

UNITEXT 140

Rocco Chirivì · Ilaria Del Corso
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Selected Exercises in Algebra

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Volume 2

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“The nice thing about mathematics is doing mathematics.”

—Pierre Deligne

To Andrea, who knows what mathematics is

Rocco

To Francesca, with a wish that she will be able to find and nurture her passions

Ilaria

To young people who already love or might come to love mathematics

Roberto

Preface

This second volume of selected exercises is meant as a companion to the first. Like its predecessor, it contains exam questions set in recent years in the context of the bachelor's degree in mathematics at the University of Pisa. The problems are accompanied by full solutions, a brief overview of the underlying theory, and a series of preliminary exercises.

Our motivation in writing this book and the objectives we hope to achieve with it remain the ones described in the preface to the first volume. Very briefly, these books stem from our conviction that, in order to study and deeply understand algebra (and, indeed, mathematics in general), attending lectures and memorising the statements and proofs of key results are not enough. What one needs to do is *apply* the theory one studies to concrete examples; in practice, use it to solve problems.

This being said, problems come in different varieties. Some require the application of rather simple procedures, or their solution can be derived immediately from definitions and theorem; others do not only require sufficient knowledge of the theory, but also some novel insight by means of which new information is unlocked.

The reader will soon realise that problems of the first type are conspicuously absent from this book; they will be forced to abandon the hope for easy solutions and instead engage with the material at a much deeper level. But *this is how it's done*: mathematics is not a novel to be passively read, but a story to be actively reinvented.

This is why we heartily recommend that the reader arm themselves with patience and do not look up the solution to an exercise before they have spent a considerable amount of time trying to come up with it themselves.

There are a couple more ideas that we would like to stress. The first is that maths enthusiasts at all levels are taken in by the *beauty* of mathematics; the second is that being enthusiastic about anything involves *curiosity* for the truth. We have tried to embody within many of the exercises in this book our personal sense of beauty and the spirit of curiosity that has always inspired us. We hope that the reader might also benefit from this aspect of our work.

As is the case for the first volume, the book is structured according to the way that algebra is taught during the first years of a mathematics degree at the University of

Pisa. When the distinction between bachelor's and master's degrees was introduced, what used to be a single algebra course was split into two parts, which are at present labelled "Arithmetic" and "Algebra 1"; these two parts perfectly correspond to the two volumes of this work.

The "Arithmetic" portion of the material, to which the first volume is devoted, is mostly concerned with basic tools such as induction, some elements of combinatorics, integers and congruences, and includes an introduction to the basic properties of fundamental algebraic structures: Abelian groups, rings, polynomials and their roots, field extensions and finite fields.

The "Algebra 1" portion covered by this second volume delves deeper into group theory, discusses commutative rings with a particular focus on unique factorisation, and deals with field extensions and the fundamentals of Galois theory.

Each part is accompanied by notes on the theory. Although exhaustive, they do not attempt to make up for an algebra textbook, as evidenced by the fact that they contain the statements of relevant results but provide no proofs. (For further information the reader may instead turn, for example, to Herstein's "Topics in Algebra", Wiley & Sons, or Artin's "Algebra", Pearson.)

This book also includes a series of preliminary exercises. These should be tackled first, since they yield results that will often be needed in subsequent problems. We also stress that none of the solutions provided require any theoretical tools other than those recalled in the introductory chapters and the preliminary exercises. Invoking more advanced theorems would simplify certain solutions and occasionally render an exercise trivial, but doing so would be contrary to the very spirit of this work.

Acknowledgements We would like to thank Filippo Callegaro for his collaboration in preparing some of the exercises and Alessandro Berarducci and Ilaria Damiani for their precious advice. We thank Dr. Francesca Bonadei and Dr. Francesca Ferrari of Springer Italia for their assistance. Finally, a special thanks to all the students who attended our lectures and busied themselves with the exercises during exams.

Updates We warmly invite readers to provide us with feedback and point out mistakes (which, in a book containing the detailed solutions to hundreds of exercises, are almost inevitably to be found) by contacting us at the addresses: `rocco.chirivi@unisalento.it` or `ilaria.delcorso@unipi.it` or `roberto.dvornicich@unipi.it`.

Updates and corrections will appear on the web page http://www.dmf.unisalento.it/~chirivi/libroEserciziAlgebra.html.

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July 2018

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Chapter 1
Theory



1.1 Groups

1.1.1 Basic Notions

In this section, we briefly recall some basic notions in group theory that were introduced in the first volume [Volume 1, Theory, Section 4].

A group is a nonempty set G equipped with an associative operation $\cdot : G \times G \longrightarrow G$, such that there exists a neutral element e_G and that each element g in the group has an inverse, denoted by g^{-1}. When the meaning is clear in context, we shall simply write e rather than e_G.

A group G is Abelian if $gh = hg$ for all g and h in G. The order $|G|$ of a group G is the number of its elements. The order $\mathrm{ord}(g)$ of an element g of G is the smallest positive integer n such that $g^n = e$, provided one exists; if no positive integer has this property, then we say that g has infinite order.

If the operation $\cdot$ of G can be restricted to a subset H (that is, if we have $\cdot : H \times H \longrightarrow H$) and if H is itself a group when equipped with this operation, we say that H is a subgroup of G. The subgroup $\{e\}$, which we sometimes denote by e to lighten notation, is called trivial; subgroups different from G are called proper. The centre $Z(G)$ of G is the set of all elements z of G such that $zg = gz$ for all $g \in G$, and is a subgroup of G.

Given a subset X of a group G, the subgroup generated by X, denoted by $\langle X \rangle$, is the intersection of all subgroups of G containing X. It is also the smallest subgroup of G that contains X. A group is called cyclic if generated by a single element.

Lagrange's Theorem states that in a finite group G the order of any subgroup divides the order G. By applying Lagrange's Theorem to the subgroup generated by an element, we obtain that the order of each element also divides the order of the group. For a subgroup H, the number $[G : H]$ of its cosets gH, with $g \in G$, that is, the cardinality of the set G/H, is called the index of H in G; if G is a finite group, then $[G : H] = |G|/|H|$.

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Recall that a subgroup K of a group G is called normal if $gKg^{-1} = K$ for all g in G. If K is a normal subgroup of G, then the set G/K of all cosets gK, where $g \in G$, is a group when endowed with the operation $gK \cdot g'K = (gg')K$; the quotient map $G \ni g \overset{\pi}{\longmapsto} gK \in G/K$ is a group homomorphism, often referred to as the quotient homomorphism . The kernel of the quotient homomorphism $G \overset{\pi}{\longrightarrow} G/K$ is of course K. Moreover, a subgroup of G is normal if and only if it is the kernel of some group homomorphism $G \longrightarrow G'$. Also note that a subgroup of index 2 is always normal; indeed, if K has index 2, then the cosets of K in G are K and $G \setminus K$.

If H and K are subgroups of a group G, then the set HK of all elements of the form hk, with $h \in H$ and $k \in K$, is a subgroup of G if and only if $HK = KH$. In particular, if one of the two subgroups is normal then this condition is satisfied. If H and K are finite subgroups, then the set HK, independently of whether or not it is a subgroup itself, has cardinality $|H||K|/|H \cap K|$.

Given a group G, an isomorphism $G \longrightarrow G$ is called an automorphism of G; the set $\mathrm{Aut}(G)$ of all automorphisms of G is itself a group, called the automorphism group of G, with the operation given by composition of maps. The group $\mathrm{Aut}(G)$ is not Abelian in general.

Given groups G and H, the Cartesian product $G \times H$ is a group with the operation $(g, h) \cdot (g', h') = (gg', hh')$. The order of an element (g, h) is the least common multiple of $\mathrm{ord}(g)$ and $\mathrm{ord}(h)$; the centre of $G \times H$ is the product of centres $Z(G) \times Z(H)$, and in particular $G \times H$ is Abelian if and only if both G and H are.

An especially important group is the symmetric group S_n, whose elements are the permutations of $\{1, 2, \ldots, n\}$ (that is, the bijective maps from $\{1, 2, \ldots, n\}$ to itself), and whose operation is the composition of maps. The group S_n has order $n!$.

Given a sequence of ℓ distinct integers $i_1, i_2, \ldots, i_\ell$ between 1 and n, the cycle $(i_1, i_2, \ldots, i_\ell)$ is the permutation σ defined as follows: for $t = 1, 2, \ldots, \ell - 1$, we have $\sigma(i_t) = i_{t+1}$; moreover, we have $\sigma(i_\ell) = i_1$; finally, for $j \in \{1, 2, \ldots, n\} \setminus \{i_1, i_2, \ldots, i_\ell\}$, we have $\sigma(j) = j$. Clearly, the name "cycle" refers to the fact that σ cyclically permutes the integers $i_1, i_2, \ldots, i_\ell$, while every other integer is kept fixed:

$$i_1 \overset{\sigma}{\longmapsto} i_2 \overset{\sigma}{\longmapsto} i_3 \overset{\sigma}{\longmapsto} \cdots \overset{\sigma}{\longmapsto} i_{\ell-1} \overset{\sigma}{\longmapsto} i_\ell \overset{\sigma}{\longmapsto} i_1$$

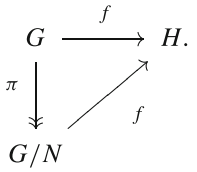
$$j \overset{\sigma}{\longmapsto} j \qquad \text{for all } j \in \{1, 2, \ldots, n\} \setminus \{i_1, i_2, \ldots, i_n\}.$$

The integer ℓ is called the length of the cycle σ; we also say that σ is an ℓ-cycle. An ℓ-cycle has order ℓ. In particular, the transposition (i, j) is a 2-cycle: it switches i and j, does not permute any other elements, and has order 2.

1.1.2 Homomorphism Theorems

There are several important results concerning group homomorphisms, one of which has already been recalled in the first volume [Volume 1, Theorem 4.18 (Fundamental Homomorphism Theorem), page 31]; for the sake of completeness, we repeat the statement here in a slightly more general form, followed by two other classical theorems about group homomorphisms.

Theorem 1.1.1 (First Homomorphism Theorem) *Given a group homomorphism $G \xrightarrow{f} H$ and a normal subgroup N of G contained in* $\mathrm{Ker}(f)$*, let $G \xrightarrow{\pi} G/N$ be the quotient homomorphism. There exists a unique homomorphism $\overline{f}$ making the following diagramme commutative:*



Moreover, the homomorphisms f and $\overline{f}$ have the same image, and $\overline{f}$ is injective if and only if $N = \mathrm{Ker}(f)$.

The homomorphism $\overline{f}$ from the First Homomorphism Theorem is said to be *induced* from the homomorphism f by passing to the quotient.

Recall that the quotient homomorphism $G \xrightarrow{\pi} G/N$ induces the one-to-one correspondence

$$G \supseteq K \longmapsto \pi(K) = KN/N \subseteq G/N$$

between subgroups of G containing N and subgroups of G/N. Via this bijection, normal subgroups of G that contain N correspond to normal subgroups of G/N.

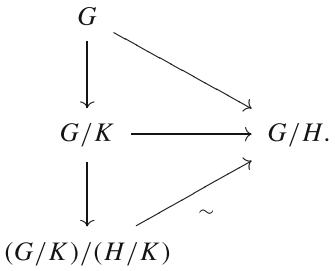
Theorem 1.1.2 (Second Homomorphism Theorem) *Given a group G, a normal subgroup K of G and a subgroup H of G, the group $H/(H \cap K)$ is isomorphic to HK/K. An isomorphism between them is induced by the projection $H \longrightarrow HK/K$, and we have the following commutative diagramme:*

$$\begin{array}{ccc} H \ni h & \longmapsto & h \in HK \\ \downarrow & & \downarrow \\ H/(H \cap K) \ni h(H \cap K) & \xrightarrow[\sim]{} & hK \in HK/K. \end{array}$$

As previously stated, if H and K are normal subgroups of G and K is contained in H then H/K is a normal subgroup of G/K. Furthermore, we have

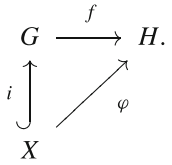
Theorem 1.1.3 (Third Homomorphism Theorem) *Suppose that K is a subgroup of H and that both K and H are normal subgroups of G; then the group $(G/K)/(H/K)$ is isomorphic to G/H.*

Indeed, the quotient homomorphism $G \longrightarrow G/H$ induces a surjective homomorphism $G/K \longrightarrow G/H$, which in turn induces an isomorphism between $(G/K)/(H/K)$ and G/H. We therefore have the following commutative diagramme:



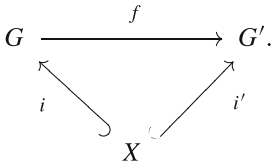
1.1.3 Free Groups

Let G be a group, X a set, and $i : X \hookrightarrow G$ an injective map. The group G is said to be *free* on X if for every group H and every map $\varphi : X \longrightarrow H$ there is a unique group homomorphism $f : G \longrightarrow H$ such that $f(i(x)) = \varphi(x)$ for all $x \in X$. What we require is, in other words, for f to make the following diagramme commutative:



The property expressed above identifies the group G up to unique isomorphism; it is in fact straightforward to prove the following

Proposition 1.1.4 *Given an injective map $i' : X \hookrightarrow G'$ and a free group G' on X, there exists a unique group isomorphism $f : G \longrightarrow G'$ making the following diagramme commutative:*



The structure of a free group on the set X can be explicitly described as follows. Let L_0 be the set of all finite strings made up of symbols of the form x or x^{-1}, where $x \in X$. On L_0, consider the operation of juxtaposing strings, which is associative and admits the empty string as a neutral element. Now let L be the quotient of L_0 by the equivalence relation generated by $s_1 x x^{-1} s_2 \sim s_1 s_2 \sim s_1 z^{-1} z s_2$ for all $x, z \in X$ and $s_1, s_2 \in L$. It is easy to prove that string juxtaposition passes to the quotient and defines a group structure on L. Equipped with the inclusion $X \ni x \overset{i}{\longmapsto} z \in L$, L is a free group on X.

Remark that we can choose as representatives for the quotient $L_0/\sim$ those strings that do not contain substrings of the form xx^{-1} or $x^{-1}x$ for any $x \in X$. We call such strings *reduced*; we shall thus think of L as the set of all reduced strings. Note that the juxtaposition of two reduced strings might yield a string that is not reduced, but one can obtain a reduced string via a finite number of cancellations of adjacent pairs of the form x, x^{-1}.

For the sake of brevity, given $n \in \mathbb{N}$ and $x \in X$, we shall from now on write x^n to denote the string $x \cdots x$ consisting of n copies of the symbol x; similarly, we write $x^{-n} = x^{-1} \cdots x^{-1}$ for n copies of x^{-1}; we thus have $x^n \cdot x^m = x^{n+m}$ for all $n, m \in \mathbb{Z}$.

As a consequence of this construction, we have

Proposition 1.1.5 *For every set X there exists a free group on X.*

In view of the results above, we will often commit a slight abuse of notation by speaking of *the* free group on a set X; when doing so, we are usually referring to the group constructed above, as the context will make clear. Loosely speaking, one may think of the free group on X as the most general group that can be built from the elements of X, treated as independent of each other.

Remark that a subset X of a group G is a set of generators for G if and only if there exists a surjective homomorphism from the free group on X to G. We say that a group is *finitely generated* if it admits a finite set of generators.

Let us introduce some examples. The free group on the empty set is the trivial group whose only element is the empty string, whereas the free group on a one-element set $X = \{x\}$ contains all strings x^n, where n ranges in $\mathbb{Z}$; it is thus clearly isomorphic to $\mathbb{Z}$.

Throughout the rest of this volume, if L is a free group on a set X with inclusion $X \overset{i}{\hookrightarrow} L$, we shall often implicitly identify X with its image $i(X)$ in L.

1.1.4 Presentations of Groups

Let X be a set and let $\mathcal{R}$ be a subset of the free group L on X; we shall think of elements of $\mathcal{R}$ as strings of symbols in $X \sqcup X^{-1}$. Denote by $G = \langle X \mid \mathcal{R} \rangle$ the quotient group of the free group L by the normal subgroup generated by $\mathcal{R}$; we shall say that X is a *set of generators* for G and that $\mathcal{R}$ is a *set of relations* defining G. The expression $\langle X \mid \mathcal{R} \rangle$ is said to be a *presentation* of the group G. Given $X = \{x_1, x_2, \ldots, x_n\}$ and $\mathcal{R} = \{r_1, r_2, \ldots, r_m\}$, we will sometimes write $\langle x_1, x_2, \ldots, x_n \mid r_1, r_2, \ldots, r_m \rangle$ for the presentation $\langle X \mid \mathcal{R} \rangle$.

We could think of the group $\langle X \mid \mathcal{R} \rangle$ as the most general group that admits the elements of X as generators and is such that $r = e$ for each $r \in \mathcal{R}$, where e is the neutral element of $\langle X \mid \mathcal{R} \rangle$. As with free groups, we shall from now on implicitly identify elements of X with their images in $\langle X \mid \mathcal{R} \rangle$.

It is clear that $\langle x \mid \varnothing \rangle$ is a presentation of the infinite cyclic group, that is, of $\mathbb{Z}$. Another simple example is the following. Given a natural number n, the group $\langle x \mid x^n \rangle$ is a cyclic group of order n and therefore isomorphic to the additive group $\mathbb{Z}/n\mathbb{Z}$ of residue classes modulo n; an explicit isomorphism is obtained by extending the assignment of the image $x \in \langle x \mid x^n \rangle$ to the generator $1 \in \mathbb{Z}/n\mathbb{Z}$.

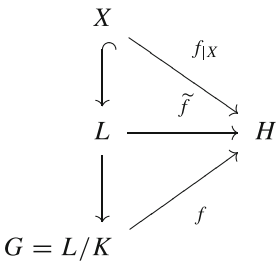
Here is another example: $\langle r, s \mid r^3, s^2, srsr \rangle$ is a presentation of S_3, the group of permutations of $\{1, 2, 3\}$. Indeed, the mapping $r \longmapsto (1, 2, 3), s \longmapsto (1, 2)$ extends to a group isomorphism between $\langle r, s \mid r^3, s^2, srsr \rangle$ and S_3. We shall occasionally write relations of the form $r_1 r_2^{-1}$ as $r_1 = r_2$; the above presentation of S_3 can thus be rewritten as $\langle r, s \mid r^3 = s^2 = e, sr = r^{-1}s \rangle$.

We should stress that $\mathcal{R}$ is *not*, in general, the set of all relations that hold between elements of X in $G = \langle X \mid \mathcal{R} \rangle$. What is true is that every relation between elements of X in G can be deduced from the relations in $\mathcal{R}$.

For example, if we consider the presentation above for S_3, we do have $(rs)^2 = e$ even though $(rs)^2$ is not an element of $\mathcal{R}$; however, using $sr = r^{-1}s$ and $s^2 = e$ we can indeed deduce $(rs)^2 = r(sr)s = r(r^{-1}s)s = (rr^{-1})s^2 = e$.

Remark that if X is a set of generators for a group G then, as already stated, there exists a surjective homomorphism f from the free group on X to G, and therefore $\langle X \mid \mathrm{Ker}(f) \rangle$ is a presentation of G. Naturally, such a presentation might be extremely redundant, given it contains *all* relations between elements of X in G. Usually, one looks for presentations with as few generators and relations as possible.

Let us now consider a homomorphism f from a group G with given presentation $\langle X \mid \mathcal{R} \rangle$ to a group H. By definition, G is the quotient of the free group L on X by the normal subgroup K of L generated by $\mathcal{R}$. Since X generates G, the homomorphism f is completely determined once the images $f(x) \in H$ of all elements $x \in X$ are given, that is, once we have the map $f_{|X} : X \longrightarrow H$. Not every choice is possible for $f_{|X}$, however; we do have to satisfy the following commutative diagramme



where $\widetilde{f}$ is the unique homomorphism induced by $f_{|X}$ on L. It follows from the First Homomorphism Theorem that, in order for $\widetilde{f}$ to pass to the quotient, it is necessary and sufficient that $K \subseteq \mathrm{Ker}(\widetilde{f})$. Since K is generated by $\mathcal{R}$, this is equivalent to the condition that for all $r \in \mathcal{R}$ we have $\widetilde{f}(r) = e_H$. Hence,

Remark 1.1.6 *Let G be a group with presentation $\langle X \mid \mathcal{R} \rangle$ and let H be a group. Given elements $h_x \in H$, one for each $x \in X$, there exists a homomorphism $f : G \longrightarrow H$ such that $f(x) = h_x$ for all $x \in X$ if and only if the following holds: for each relation $r \in \mathcal{R}$, letting $h(r)$ be the element of H obtained by replacing each generator x by h_x, we have $h(r) = e_H$. If the condition is satisfied, then f is unique and is called the homomorphism* induced *by the map $X \ni x \longmapsto h_x \in H$.*

Let us discuss a very basic example. We know that the cyclic group $\mathbb{Z}/n\mathbb{Z}$ admits $\langle x \mid x^n \rangle$ as a presentation, and that $\mathbb{Z}/n\mathbb{Z} \ni 1 \longmapsto x \in \langle x \mid x^n \rangle$ extends to an isomorphism between the two groups. If h_1 is an element of a group H, then there exists a group homomorphism from $\mathbb{Z}/n\mathbb{Z}$ to H such that $1 \longmapsto h_1$ if and only if $h(x^n) = e_H$, that is, if and only if $h_1^n = e_H$, that is to say, if and only if the order of h_1 in H divides n.

1.1.5 The Dihedral Group

Let U and V be vector spaces on a field $\mathbb{K}$. A map $f : U \longrightarrow V$ is said to be *linear* if $f(u_1 + u_2) = f(u_1) + f(u_2)$ and $f(\lambda u) = \lambda f(u)$ for all $u_1, u_2, u \in V$ and for all $\lambda \in \mathbb{K}$. If a linear map is invertible then its inverse is also linear.

The set of all invertible linear maps from V to itself, endowed with the composition of maps as its operation, is a group; we shall denote it by $\mathsf{GL}(V)$ and call it the *general linear group* of V.

When $\mathbb{K} = \mathbb{R}$ and $V = \mathbb{R}^n$, we also consider the set $\mathsf{O}(\mathbb{R}^n)$ of all linear maps that preserve distances from the origin. Such maps, which are called *orthogonal*, are always invertible, so $\mathsf{O}(\mathbb{R}^n)$ is a subgroup of $\mathsf{GL}(\mathbb{R}^n)$, called the *orthogonal group*.

Given a subset X of $\mathbb{R}^n$ we call an *isometry* of X any orthogonal map $f : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ for which $f(X) = X$. It is easy to prove that the set of isometries of X forms a subgroup of $\mathsf{O}(\mathbb{R}^n)$; it is called the *isometry group* of X and denoted by $\mathrm{Iso}(X)$.

Given an integer $n \geq 3$, the nth *dihedral group*, which we denote by D_n, is the isometry group of the regular n-gon in $\mathbb{R}^2$ centred at the origin and with a vertex at $(1, 0)$. Notice that the symmetry $\sigma : (x, y) \longmapsto (x, -y)$ and the rotation ρ by $2\pi/n$ radians are both isometries of this n-gon. The following proposition summarises the main properties of D_n.

Proposition 1.1.7 *The nth dihedral group D_n is non-Abelian, has order $2n$ and is generated by σ and ρ. These elements satisfy $\rho^n = \sigma^2 = \mathrm{Id}$ and $\sigma\rho = \rho^{-1}\sigma$. Moreover, $\langle r, s \mid r^n = s^2 = e,\ sr = r^{-1}s\rangle$ is a presentation of D_n: indeed, there is an isomorphism between the group thus presented and D_n such that $r \longmapsto \rho$, $s \longmapsto \sigma$.*

In the future, we will directly refer to $\langle r, s \mid r^n = s^2 = e,\ sr = r^{-1}s\rangle$ as the nth dihedral group: the isomorphism obtained by extending $r \longmapsto \rho$, $s \longmapsto \sigma$ is implied. When there is no risk of confusion, we will use the symbols ρ and σ in the presentation in place of r and s.

Remark that the presentation given for D_n immediately yields that D_3 is isomorphic to S_3. This is also clear for geometric reasons: an isometry of an n-gon induces a permutation of its vertices; in the case of the equilateral triangle, every permutation of its vertices can be realised with an isometry.

We use the proposition's presentation again in order to construct a homomorphism from D_n to S_n. Given $\eta = (1, 2, \ldots, n)$ and $\tau = (1, 2)$, mapping $\rho \longmapsto \eta$ and $\sigma \longmapsto \tau$ induces a homomorphism $\mathsf{D}_n \longrightarrow \mathsf{S}_n$ because the relations $\eta^n = \tau^2 = e$ and $\tau\eta = \eta^{-1}\tau$ are satisfied.

Similarly, given a divisor m of n, letting ρ' and σ' be the generators of D_m given in the proposition above, mapping $\rho \longmapsto \rho'$ and $\sigma \longmapsto \sigma'$ induces a homomorphism $\mathsf{D}_n \longrightarrow \mathsf{D}_m$, as the relations $\rho'^n = \sigma'^2 = e$ and $\sigma'\rho' = \rho'^{-1}\sigma'$ do hold in D_m.

1.1.6 Group Automorphisms

Given a subgroup H of a group G and an automorphism f of G, the image $f(H)$ is a subgroup of G. If we have $f(H) = H$ for all automorphisms $f \in \mathrm{Aut}(G)$ then we say that the subgroup H is *characteristic*. Note that $f(H)$ has the same order as H; in particular, if H is the only subgroup of G having order $\mathrm{ord}(H)$, then it is a characteristic subgroup. For example, the subgroup generated by the permutation $(1, 2, 3)$ is the only subgroup of order 3 in S_3, and is therefore characteristic.

Given a group G and one of its elements g, the map $\psi_g : h \longmapsto ghg^{-1}$ is an automorphism of G called *conjugation* by g. Automorphisms of the form ψ_g for some g in G are called *inner automorphisms*. Note that the composition of two inner automorphisms is still an inner automorphism, and in particular $\psi_g\psi_{g'} = \psi_{gg'}$. For the inverse, we clearly have $\psi_g^{-1} = \psi_{g^{-1}}$. In other words, the map $\psi : g \longmapsto \psi_g$ is a homomorphism from G to $\mathrm{Aut}(G)$; the image $\mathrm{Inn}(G)$ of this homomorphism, that is, the set of inner automorphisms, is called the *inner automorphism group* of G.

Remark that $\mathrm{Inn}(G)$ may very well be a proper subgroup of $\mathrm{Aut}(G)$. For example, if G is Abelian, then we have $\mathrm{Inn}(G) = \{\mathrm{Id}_G\}$: indeed, $\psi_g(h) = ghg^{-1} = gg^{-1}h = h$ for all $g, h \in G$. More generally, $\psi_g = \mathrm{Id}_G$ if and only if g belongs to the centre $Z(G)$ of G. We therefore have $\mathrm{Ker}(\psi) = Z(G)$, and the First Homomorphism Theorem implies

Remark 1.1.8 *The map* $G \ni g \longmapsto \psi_g \in \mathrm{Inn}(G)$ *induces an isomorphism between* $G/Z(G)$ *and* $\mathrm{Inn}(G)$.

A subgroup H of a group G is normal if and only if $\psi_g(H) = H$ for all g in G; that is, a subgroup being normal is equivalent to its being invariant under inner automorphisms. It is thus clear that all characteristic subgroups are normal, as they are invariant under all automorphisms, and therefore in particular under all inner ones.

1.1.7 Commutators and Abelianisation

Let G be a group and consider $g, h \in G$; the element $[g, h] = ghg^{-1}h^{-1}$ is called the *commutator* of g and h in G. Note that $[g, h] = e$ if and only if g and h commute. The *commutator subgroup*, or *derived group*, denoted by $[G, G]$, is the subgroup generated by all elements of the form $[g, h]$, with g and h in G. We shall sometimes alternatively denote the derived group of G by G'. Given an automorphism φ of G, we have $\varphi([g, h]) = [\varphi(g), \varphi(h)]$ for all $g, h \in G$; this immediately implies that the commutator subgroup is characteristic, and therefore normal. We can thus construct the quotient $G/[G, G]$, which is called the *Abelianisation* of G. Remark that $G/[G, G]$ is clearly Abelian, because commutators of G become trivial in $G/[G, G]$. But one can say even more:

Proposition 1.1.9 *Given a normal subgroup* H *of* G*, the group* G/H *is Abelian if and only if* $[G, G] \subseteq H$.

An equivalent way to state the preceding proposition is the following: the Abelianisation of G is the largest possible homomorphic image of G that is Abelian; in other words, K is an Abelian homomorphic image of G if and only if there exists a surjective homomorphism $G/[G, G] \longrightarrow K$.

Clearly, any Abelian group has a trivial commutator subgroup and is its own Abelianisation. As an example, let us construct the Abelianisation of D_5 using the presentation $\langle r, s \mid r^5 = s^2 = e,\ sr = r^{-1}s\rangle$. Remark that mapping $r \longmapsto 0$ and $s \longmapsto 1$ extends to a surjective homomorphism from D_5 to the Abelian group $\mathbb{Z}/2\mathbb{Z}$; therefore, $[\mathsf{D}_5, \mathsf{D}_5]$ is contained in the kernel of this homomorphism, that is, in the subgroup generated by r. Moreover, $[\mathsf{D}_5, \mathsf{D}_5]$ cannot be the trivial group, as D_5 is not Abelian. The only possibility is that $[\mathsf{D}_5, \mathsf{D}_5] = \langle r\rangle = \{e, r, \ldots, r^4\}$, so $\mathsf{D}_5/[\mathsf{D}_5, \mathsf{D}_5] \simeq \mathbb{Z}/2\mathbb{Z}$.

1.1.8 Group Actions

Given a group G and a set X, a homomorphism $G \longrightarrow \mathsf{S}(X)$ from G to the group of permutations of X is called an *action* of G on X. In other words, what an action consists of is a map $G \times X \ni (g, x) \longmapsto g \cdot x \in X$ satisfying the following requirements: $(gh) \cdot x = g \cdot (h \cdot x)$ for all g, h in G and $x \in X$, and moreover $e \cdot x = x$ for all $x \in X$.

There are innumerable examples of group actions. The *trivial action* of G on X is $G \ni g \longmapsto \mathrm{Id}_X \in \mathsf{S}(X)$. Another easy example is $\mathsf{S}(X)$ acting on X via the map $\mathrm{Id} : \mathsf{S}(X) \longrightarrow \mathsf{S}(X)$. Many more examples can be constructed as follows: given a subgroup G of $\mathsf{S}(X)$, the inclusion $G \hookrightarrow \mathsf{S}(X)$ is an action of G on X; we shall sometimes refer to it as the *natural action* of G on X, of which a common instance is the action of $\mathsf{GL}(V)$ on the vector space V. As a slightly different example, consider the set $\mathbb{C}$ of complex numbers and the map from $\mathbb{Z}/2\mathbb{Z}$ to $\mathsf{S}(\mathbb{C})$ that sends 0 to $\mathrm{Id}_{\mathbb{C}}$ and 1 to the complex conjugation map given by $a + bi \overset{\tau}{\longmapsto} a - bi$ for all a and b in $\mathbb{R}$. Since $\tau^2 = \mathrm{Id}_{\mathbb{C}}$, this provides an action of $\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{C}$.

Let G be a group acting on a set X and let x be an element of X. The subset $\mathcal{O}(x)$ of X whose elements are all those of the form $g \cdot x$ for some $g \in G$ is called the *orbit* of x in X under G. The family of orbits in X is a partition of X. Indeed, define a relation by setting $x \sim y$ if and only if there exists $g \in G$ such that $g \cdot x = y$; it is easy to show that $\sim$ is an equivalence relation and that its equivalence classes are the orbits of G in X.

For example, the natural action of $\mathsf{GL}(V)$ on V has two orbits: $\{0\}$ and $V \setminus \{0\}$. The action of $\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{C}$ by complex conjugation has two types of orbits: if z is in $\mathbb{R}$ then its orbit is $\{z\}$, whereas if z is not in $\mathbb{R}$ then $\mathcal{O}(z) = \{z, \overline{z}\}$, where $\overline{z}$ is the complex conjugate of z. The natural action of S_n on $\{1, 2, \ldots, n\}$ has but one orbit: that is, for all $1 \leq h \leq n$ we have $\mathcal{O}(h) = \{1, 2, \ldots, n\}$. Indeed, given $1 \leq h, k \leq n$, if $h = k$ then $\mathrm{Id}(h) = k$ and if $h \neq k$ then the permutation (h, k) sends h to k. If a group G acts on a set X and the action admits a unique orbit then we say that G acts *transitively* on X.

A *set of representatives* for the orbits of the action of a group G on a set X is a subset $\mathcal{R} \subseteq X$ such that for all $x \in X$ there is a unique $y \in \mathcal{R}$ such that $x \in \mathcal{O}(y)$: one constructs a set of representatives by picking an element from each orbit.

An action $\rho : G \longrightarrow \mathsf{S}(X)$ of G on X is called *faithful* if ρ is injective. A faithful action ρ sends G to an isomorphic copy of itself within $\mathsf{S}(X)$. Note that any action induces a faithful action of $G/\mathrm{Ker}(\rho)$ on X via $G/\mathrm{Ker}(\rho) \overset{\widetilde{\rho}}{\longrightarrow} \mathsf{S}(X)$, where $\widetilde{\rho}$ is the homomorphism induced by ρ on the quotient $G/Ker(\rho)$.

In order to understand an action's orbits it is often useful to first consider the question of which elements in the group fix a certain element in the set. Let G be a group acting on a set X and consider $x \in X$; the subset $\mathrm{St}(x)$ consisting of elements $g \in G$ such that $g \cdot x = x$, that is, those that "stabilise", or "fix", the element x, is called the *stabiliser* of x in G. As $g, h \in G$ fixing x implies $g^{-1}x = x$ and $(gh)x = g(hx) = gx = x$, it is clear that $\mathrm{St}(x)$ is a subgroup of G.

In the case of the action of $\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{C}$ described above, we have that if $z \in \mathbb{R}$ then $\mathrm{St}(z) = \mathbb{Z}/2\mathbb{Z}$, whereas if $z \notin \mathbb{R}$ then $\mathrm{St}(z) = \{0\}$. As this example suggests, the larger the stabiliser of an element, the smaller its orbit. This insight will be made clear by the following

Proposition 1.1.10 (Orbit-Stabiliser Theorem) *Let G be a group acting on a set X and let x be some element of X. The map*

$$G/\mathrm{St}(x) \ni g\mathrm{St}(x) \longmapsto g \cdot x \in \mathcal{O}(x)$$

is well-defined and is a bijection between the set $G/\mathrm{St}(x)$ of all cosets of $\mathrm{St}(x)$ in G, and the orbit of x in X. In particular, if G and X are both finite then $|\mathcal{O}(x)| = [G : \mathrm{St}(x)]$.

For the sake of clarity, let us explicitly remark that $\mathrm{St}(x)$ is *not* a normal subgroup of G in general; indeed, in the context of the proposition above, $G/\mathrm{St}(x)$ is to be considered as just the set of all cosets of $\mathrm{St}(x)$ in G and *not* as a group.

If y belongs to the orbit of x, that is, $y = gx$ for some g, then it is easy to show that $\mathrm{St}(y) = g\mathrm{St}(x)g^{-1}$. In other words, elements of the same orbit have conjugate stabilisers.

Let us now give an example of how one might use the Orbit-Stabiliser Theorem to compute the order of a group. Consider the cube C with vertices $(\pm 1, \pm 1, \pm 1)$ in $\mathbb{R}^3$; we wish to compute the cardinality of $G = \mathrm{Iso}(C)$. Clearly, an orthogonal map sending C to itself must permute its vertices, as those are the points of C whose distance to the origin is maximal. Let us then consider the vertex $v = (1, 1, 1)$ and its stabiliser $\mathrm{St}(v)$.

An orthogonal map that fixes v must permute the three vertices $(-1, 1, 1)$, $(1, -1, 1)$ and $(1, 1, -1)$ that are adjacent to v. One finds that rotations of $2\pi/3$ radians around the axis v, $-v$ and the symmetry $(x, y, z) \longmapsto (y, x, z)$ are elements of $\mathrm{St}(v)$; from these we obtain all 6 possible permutations of the three vertices. We therefore have $|\mathrm{St}(v)| = 6$, because the three vertices adjacent to v give a basis of $\mathbb{R}^3$. However, the action of G on the 8 vertices of C being transitive, the Orbit-Stabiliser Theorem yields $|\mathrm{Iso}(C)| = |\mathcal{O}(v)| \cdot |\mathrm{St}(v)| = 8 \cdot 6 = 48$. One could proceed in a similar way to obtain the isometry groups of other Platonic solids.

An important consequence of the Orbit-Stabiliser Theorem is the following formula, which one obtains immediately by using the fact that orbits of a group action on a set form a partition of the set itself.

Proposition 1.1.11 (Burnside's Lemma) *Given a finite group G acting on a finite set X,*

$$|X| = \sum_{x \in \mathcal{R}} [G : \mathrm{St}(x)],$$

where $\mathcal{R}$ is a set of representatives for the orbits of G in X.

From an action of G on X one can construct an action of G on the power set $\mathcal{P}(X)$ of X by setting $g \cdot Y = \{g \cdot y \mid y \in Y\}$ for all $g \in G$ and $Y \subseteq X$. For example, the stabiliser of a subset X of $\mathbb{R}^n$ for the natural action of $\mathsf{O}(\mathbb{R}^n)$ on $\mathbb{R}^n$ is the group $\mathrm{Iso}(X)$ of its isometries.

1.1.9 The Action by Conjugation

A group G acts on itself in various different ways. For example, we can consider the *action by conjugation* given by the map $G \longrightarrow \mathsf{S}(G)$ defined as

$$g \longmapsto (h \longmapsto ghg^{-1}).$$

The orbit of an element $h \in G$ according to this action is called the *conjugacy class* of h and is denoted by $\mathcal{C}\ell(h)$; it is the set of all elements of the form ghg^{-1} for some $g \in G$, that is, all elements that are *conjugates* of h.

In Abelian groups, this action is not interesting, as it is none other than the trivial action. More generally, in any group G, elements $h \in Z(G)$ have the obvious conjugacy class $\mathcal{C}\ell(h) = \{h\}$. But a more varied range of behaviour is possible outside of the group's centre: in S_3, for example, we can prove easily the existence of exactly three conjugacy classes: $\mathcal{C}\ell(e)$, $\mathcal{C}\ell((1,2))$, and $\mathcal{C}\ell((1,2,3))$.

The stabiliser of an element $h \in G$, when considering the action of the group on itself by conjugation, is called the *centraliser* of h in G and denoted by $Z(h)$. It is the subgroup of all elements $g \in G$ such that $gh = hg$, that is, all elements that commute with h. As we've already seen, if $k = ghg^{-1}$, then $Z(k) = gZ(h)g^{-1}$: the centralisers of conjugate elements are conjugate themselves. Elements in the centre admit the whole group as their centraliser.

The Orbit-Stabiliser Theorem immediately yields $|\mathcal{C}\ell(h)| = [G : Z(h)]$. The next proposition will combine this with the fact that any set of representatives for the conjugacy classes contains the centre $Z(G)$ of the group, as each element in the centre forms its own conjugacy class.

Proposition 1.1.12 (Class Equation) *For all finite groups G we have*

$$|G| = |Z(G)| + \sum_{g \in \mathcal{R} \setminus Z(G)} \frac{|G|}{|Z(g)|},$$

where $\mathcal{R}$ is a set of representatives for the conjugacy classes of G.

Let us, as an example, check the formula above for S_3. Thanks to our previous remarks, we can take $\mathcal{R} = \{e, (1,2), (1,2,3)\}$. We immediately find that $(1,2)$ only commutes with its own powers, and the same is true for $(1,2,3)$; that is, we

have $Z((1,2)) = \{e, (1,2)\}$ and $Z((1,2,3)) = \{e, (1,2,3), (1,3,2)\}$. Moreover, it is clear that the centre of S_3 is trivial. The Class Equation then gives

$$|Z(\mathsf{S}_3)| + \frac{|\mathsf{S}_3|}{|Z((1,2))|} + \frac{|\mathsf{S}_3|}{|Z((1,2,3))} = 1 + \frac{6}{2} + \frac{6}{3} = 1 + 3 + 2 = 6 = |\mathsf{S}_3|.$$

Note that a normal subgroup H of a group G entirely contains the conjugacy class of each of its elements; indeed, we could say that a subgroup is normal if and only if it's a union of conjugacy classes.

One can use conjugation to also define an action on the set of subgroups of G; indeed, if H is a subgroup of G then gHg^{-1} is also a subgroup of G and is said to be *conjugate* to H. For this action, the orbit of the subgroup H is the set of all subgroups of G that are conjugate to H, and the stabiliser of H is called *normaliser* of H; it is denoted by $N(H)$. Clearly, H is contained in its normaliser. Moreover, we can give a characterisation of the normaliser of H as the largest subgroup of G in which H is normal. Thus, H is normal in G if and only if $N(H) = G$.

The Orbit-Stabiliser Theorem immediately tells us that H has exactly $[G : N(H)]$ distinct conjugate subgroups in G.

Given a subgroup H of G, its *centraliser* $Z_G(H)$ is the subgroup of all elements g of G such that $gh = hg$ for all h in H; it is obvious that

$$Z_G(H) = \bigcap_{h \in H} Z(h).$$

1.1.10 The Action by Multiplication

Another way to have G act on itself is via left multiplication:

$$g \longmapsto (h \longmapsto gh).$$

It follows from the Cancellation Laws that this action is faithful. Consequently, we have the following

Theorem 1.1.13 (Cayley) *Any group G is isomorphic to some subgroup of a symmetric group. In particular, if G has order n then it is isomorphic to some subgroup of S_n.*

It is clear that one can construct an analogous action of G on itself using right multiplication

$$g \longmapsto (h \longmapsto hg^{-1})$$

and that the two actions have similar properties.

Letting H be a subgroup of G, left multiplication can also be used to define an action of G on the set of all cosets G/H of H in G, by setting $g \cdot (kH) = (gk)H$ for all $g, k \in G$. We thus have a homomorphism $G \to \mathsf{S}(G/H)$, which is not surjective or injective in general. In fact, one can prove that its kernel is the largest possible normal subgroup N of G that is contained H; moreover, we have $N = \cap_{g \in G} gHg^{-1}$.

1.1.11 p-Groups

Groups whose order is only divisible by a single prime p are called *p-groups*. There are several remarkable properties of p-groups, which form the first class of groups whose structure we will study in some detail. Using the Class Equation, one finds

Proposition 1.1.14 *The centre of a p-group is nontrivial.*

This is because all summands in

$$\sum_{g \in \mathcal{R} \setminus Z(G)} \frac{|G|}{|Z(g)|}$$

are divisible by p, so $|Z(g)|$ must also be divisible by p.

Let G be a p-group of order p^n; if $n = 1$, then G is of course cyclic. Suppose instead that $n = 2$ and let g be an element of G that does not belong to the centre. Then $Z(g)$ is a proper subgroup of G and must contain both $Z(G)$ and the element g; that is, it has at least $p + 1$ elements. But, since $Z(g)$ is a subgroup, this implies it must be all of G, which is a contradiction. We have thus proven the following

Remark 1.1.15 *Given a prime p, any group of order p^2 is Abelian.*

In the next proposition, we will see how a p-group admits normal subgroups of all possible orders. The proof is worked out in detail in Preliminary Exercise 15, and consists in reducing to the Abelian case by using induction together with the fact that the centre of any p-group is nontrivial.

Proposition 1.1.16 *If G is a group of order p^n, where p is prime, then it admits a chain of inclusions*

$$\{e\} = G_0 \subseteq G_1 \subseteq \cdots \subseteq G_n = G$$

consisting of normal subgroups of G whose orders are $|G_h| = p^h$, for $h = 0, 1, \ldots, n$.

Preliminary Exercises 16 and 17 present some other important properties of p-groups.

1.1.12 Permutations

Given a natural number n, we now wish to investigate the symmetric group S_n. Let us start by counting how many ℓ-cycles $(i_1, i_2, \ldots, i_\ell)$ the group contains. By definition, we have $(i_1, i_2, \ldots, i_\ell) = (i_2, i_3, \ldots, i_\ell, i_1)$; the expression $(i_1, i_2, \ldots, i_\ell)$ is therefore non-unique, as one can start with any i_k, where $k = 1, 2, \ldots, \ell$. On the other hand, any expression that is not obtained by cycling the terms in the sequence $i_1, i_2, \ldots, i_\ell$ actually represents a different permutation. Consequently, each ℓ-cycle can be expressed in exactly ℓ ways. It follows that S_n contains

$$\frac{n(n-1)(n-2)\cdots(n-\ell+1)}{\ell} = \binom{n}{\ell}(\ell-1)!$$

different cycles of length ℓ.

Note that the sequence $i_1, i_2, \ldots, i_\ell$ can be recovered from the cycle $\sigma = (i_1, i_2, \ldots, i_\ell)$, for example by taking $i_1, \sigma(i_1), \sigma^2(i_1), \ldots, \sigma^{\ell-1}(i_1)$. Moreover, if the cycles $\sigma = (i_1, i_2, \ldots, i_\ell)$ and $\tau = (j_1, j_2, \ldots, j_m)$ are disjoint, that is, if $\{i_1, i_2, \ldots, i_\ell\} \cap \{j_1, j_2, \ldots, j_m\} = \varnothing$, then σ and τ commute; this is clear from the fact that they permute disjoint sets of elements.

Given a permutation σ, consider the action of the cyclic subgroup $\langle\sigma\rangle$ of S_n on the set $\{1, 2, \ldots, n\}$ obtained by restricting the natural action of S_n. The orbits of $\langle\sigma\rangle$ partition $\{1, 2, \ldots, n\}$ into disjoint sets: clearly, the orbit of i is given by $i_1 = i$, $i_2 = \sigma(i)$, $i_3 = \sigma^2(i)$, …, $i_\ell = \sigma^{\ell-1}(i)$ if $\sigma^\ell(i) = i$. We thus find that $\sigma_{|\{i_1,i_2,\ldots,i_\ell\}} = (i_1, i_2, \ldots, i_\ell)_{|\{i_1,i_2,\ldots,i_\ell\}}$. The ℓ-cycle $(i_1, i_2, \ldots, i_\ell)$ is called *a cycle* of σ. Since the orbits form a partition, the cycles of σ are disjoint and therefore commute with each other. This discussion is summarised by the following proposition.

Proposition 1.1.17 *Given a permutation σ whose cycles are $\sigma_1, \sigma_2, \ldots, \sigma_r$,*

(i) the cycles $\sigma_1, \sigma_2, \ldots, \sigma_r$ are disjoint and commute pairwise;
(ii) we have $\sigma = \sigma_1\sigma_2\cdots\sigma_r$, that is, the permutation σ can be expressed as a product of its cycles (in a unique way up to the order of the factors);
(iii) the order of σ is the least common multiple of the orders of its cycles.

The expression $\sigma = \sigma_1\sigma_2\cdots\sigma_r$ in the proposition is called the *decomposition of σ into disjoint cycles*. Given a permutation σ, let $\sigma_1, \sigma_2, \ldots, \sigma_r$ be its cycles and let $\ell_1, \ell_2, \ldots, \ell_r$ be their lengths. Without loss of generality, thanks to the fact that the cycles of σ commute with each other, we may take $\ell_1, \ell_2, \ldots, \ell_r$ to be in decreasing order: $\ell_1 \geq \ell_2 \geq \cdots \geq \ell_r$. This sequence of lengths forms a partition of n, that is, we have $\ell_1 + \ell_2 + \cdots + \ell_r = n$. This partition is called the *cycle structure* of σ. Permutations having $\ell_1 + \ell_2 + \cdots + \ell_r = n$ as a cycle structure are sometimes referred to as $\ell_1 + \ell_2 + \cdots + \ell_r$-cycles, and 1-cycles are often omitted when using this notation. For example, the $3 + 3$-cycles in S_8 are the permutations with cycle structure $3 \geq 3 \geq 1 \geq 1$.

Proving that the conjugate of an ℓ-cycle is also an ℓ-cycle is straightforward. Indeed,

$$\tau(i_1, i_2, \ldots, i_\ell)\tau^{-1} = (\tau(i_1), \tau(i_2), \ldots, \tau(i_\ell)).$$

Remark that if $\sigma = \sigma_1\sigma_2\cdots\sigma_r$ is the decomposition of σ into disjoint cycles, then the decomposition of $\tau\sigma\tau^{-1}$ into disjoint cycles is $(\tau\sigma_1\tau^{-1})(\tau\sigma_2\tau^{-1})\cdots(\tau\sigma_r\tau^{-1})$. In particular, the cycle structure is invariant under conjugation. The converse is also true; indeed, one can prove

Proposition 1.1.18 *Two permutation are conjugates if and only if they have the same cycle structure. In particular, the conjugacy classes in S_n are in a one-to-one correspondence with the partitions of n.*

As an example, let us list all possible cycle structures of elements of S_5, or, equivalently, all conjugacy classes in S_5. We pick a single permutation to represent each class and omit 1-cycles. The partitions $5 = 4+1 = 3+2 = 3+1+1 = 2+2+1 = 2+1+1+1 = 1+1+1+1+1$ of 5 correspond to the classes of $(1,2,3,4,5)$, $(1,2,3,4)$, $(1,2,3)(4,5)$, $(1,2,3)$, $(1,2)(3,4)$, $(1,2)$, e.

Using the decomposition into disjoint cycles $\sigma = \sigma_1\sigma_2\cdots\sigma_r$, one can identify some elements belonging to the centraliser $Z(\sigma)$. The permutations $\sigma_1, \sigma_2, \ldots, \sigma_r$ do clearly commute with σ, as they commute with all of its cycles; therefore, their powers also commute with σ. Moreover, if for example $\sigma_1 = (i_1, i_2, \ldots, i_\ell)$ and $\sigma_2 = (j_1, j_2, \ldots, j_\ell)$ are cycles of the same length ℓ, then conjugating σ by the permutation $\tau = (i_1, j_1)(i_2, j_2)\cdots(i_\ell, j_\ell)$ exchanges σ_1 and σ_2 and fixes all other cycles. As a consequence, $\tau \in Z(\sigma)$. In Preliminary Exercises 25, 28 and 30 we proceed to investigate centralisers for various permutations.

Given that the conjugacy class $\mathcal{C}\ell(\sigma)$ of a permutation σ consists of all permutations with the same cycle structure, we can compute the cardinality of $\mathcal{C}\ell(\sigma)$ by just counting such permutations. The general formula one obtains this way is, however, rather complicated and not very insightful; rather than include it, we shall work through an example.

We shall compute the cardinality of the conjugacy class with cycle structure $3+3+2$ in S_{10}. Let us begin by picking the first 3-cycle, which we can do in $10\cdot 9\cdot 8/3 = 240$ ways. Next, we pick the second 3-cycle using the 7 remaining numbers in $\{1, 2, \ldots, 10\}$ that were not included in the first 3-cycle: there are $7\cdot 6\cdot 5/3 = 70$ ways of doing it. Finally, pick the 2-cycle using the 4 remaining numbers; this step can be performed in $4\cdot 3/2 = 6$ ways. The cardinality we're looking for is thus $240\cdot 70\cdot 6/2 = 50400$, where we divide the product by 2 because we should disregard the order in which the first and second 3-cycle have been chosen. In other words, the product would yield the number of expressions $\sigma_1\sigma_2\sigma_3$, where σ_1, σ_2 are 3-cycles and σ_3 is a 2-cycle; but the expressions $\sigma_1\sigma_2\sigma_3$ and $\sigma_2\sigma_1\sigma_3$ represent the same permutation, which is why we divide by 2. For more complicated cycle structures we proceed in the same way, dividing by $k!$ for each cycle length that appears k times.

We now wish to ponder the question of how to express permutations in S_n as a product of transpositions. Consider a permutation σ and let $h = \sigma(n)$; the permutation $\sigma' = (h, n)\sigma$ fixes the number n. Note that the stabiliser of n for the natural action of S_n on $\{1, 2, \ldots, n\}$ is given by the subgroup of all permutations of $\{1, 2, \ldots, n-1\}$ and is thus isomorphic to S_{n-1}. We may therefore assume that σ' is a product of transpositions and use induction on n to obtain

Proposition 1.1.19 *Each permutation in* S_n *is the product of at most* $n-1$ *transpositions.*

Note that the expression of a permutation as a product of transpositions is *not* unique in general, and that it is *not* possible to write all permutations as a product of disjoint transpositions. For instance, $(1, 2, 3) = (1, 3)(1, 2)$; however, $(1, 2, 3)$ cannot be written as a product of disjoint transpositions, as such a product would necessarily have order 2.

An alternative way to state the proposition we just proved is to say that the set of transpositions generates group of all permutations. One can extract an even smaller set of generators by considering the transpositions $t_1 = (1, 2)$, $t_2 = (2, 3)$, ..., $t_{n-1} = (n-1, n)$, called *simple* transpositions. Indeed, it is clear that, for $i < j$, setting $\eta = t_i t_{i+1} \cdots t_{j-2}$ yields $(i, j) = \eta t_{j-1} \eta^{-1}$. Simple transpositions satisfy the relations $t_i^2 = e$ for $i = 1, 2, \ldots, n-1$, as well as $t_i t_j = t_i t_j$ for i, j such that $|i - j| > 1$, and $t_i t_{i+1} t_i = t_{i+1} t_i t_{i+1}$ for $i = 1, 2, \ldots, n-2$. One can show that simple transpositions, accompanied by the relations we just listed, provide a presentation of S_n.

We shall now give a definition of the *sign* of a permutation. Given $\sigma \in \mathsf{S}_n$, consider the number

$$\prod_{1 \le i < j \le n} \frac{\sigma(j) - \sigma(i)}{j - i}.$$

Remark that, σ being a bijection of $\{1, 2, \ldots, n\}$, the factors appearing as numerators are all possible differences $j - i$, where $1 \le i < j \le n$, up to sign changes. It follows that the fraction above can only take the values $+1$ or -1; we shall denote its value by $(-1)^\sigma$ and call it the sign of σ. Furthermore, given a permutation τ in S_n, we have

$$\prod_{1 \le i < j \le n} \frac{\sigma\tau(j) - \sigma\tau(i)}{j - i} = \prod_{1 \le i < j \le n} \frac{\sigma\tau(j) - \sigma\tau(i)}{\tau(j) - \tau(i)} \prod_{1 \le i < j \le n} \frac{\tau(j) - \tau(i)}{j - i},$$

which yields $(-1)^{\sigma\tau} = (-1)^\sigma (-1)^\tau$. What this proves is that the sign map

$$\mathsf{S}_n \ni \sigma \longmapsto (-1)^\sigma \in \{\pm 1\} \simeq \mathbb{Z}/2\mathbb{Z}$$

is actually a group homomorphism. A permutation is said to be *even* if its sign is $+1$, *odd* if its sign is -1. Even permutations form the kernel of the sign homomorphism:

they are a normal subgroup of S_n (in fact, a subgroup of index 2) which we denote by A_n and call the nth *alternating group*.

A quick computation shows that all transpositions are odd. It follows that, since the sign is a homomorphism, one can determine the sign of a permutation σ by expressing it as a product of transpositions: if σ is a product of an even number of transpositions then $(-1)^\sigma = +1$, otherwise $(-1)^\sigma = -1$, which explains the use the terms "even" and "odd" in this context. Moreover, since the sign depends only on σ and not on the specific expression as a product of transpositions, we have

Proposition 1.1.20 *If $\sigma = \tau_1\tau_2\cdots\tau_r$ and $\sigma = \tau'_1\tau'_2\cdots\tau'_{r'}$ are two expressions of the same permutation σ as a product of transpositions, then r and r' have the same parity, that is, $r \equiv r' \pmod 2$.*

One can easily check that ℓ-cycles have sign $(-1)^{\ell-1}$; thus cycles of even length are odd and cycles of odd length are even.

Since A_n is a normal subgroup, it must be a union of conjugacy classes in S_n. For example, A_5 is the union of the conjugacy classes of $(1,2,3,4,5)$, $(1,2,3)$, $(1,2)(3,4)$, and e in S_5. However, it should be remarked that, in general, conjugacy classes of even sign are *not* conjugacy classes within A_n. This is already apparent for $n = 3$, where, A_3 being Abelian, the two 3-cycles $(1,2,3)$ and $(1,3,2)$ are conjugate in S_3 but not in A_3. Conjugacy classes in A_n will be described more generally in Preliminary Exercise 27.

1.1.13 Abelian Groups

Finite Abelian groups can be completely classified: they are isomorphic to direct products of cyclic groups. Thanks to this fact, one can answer many questions about finite Abelian groups by reducing to the case of cyclic groups.

Theorem 1.1.21 (Structure Theorem for Finite Abelian Groups) *Any finite Abelian group G is isomorphic to a direct product of cyclic groups $\mathbb{Z}/d_1\mathbb{Z} \times \mathbb{Z}/d_2\mathbb{Z} \times \cdots \times \mathbb{Z}/d_r\mathbb{Z}$, with $d_1 \mid d_2 \mid \cdots \mid d_r$. Moreover, the numbers $d_1, d_2, \ldots, d_r$ are uniquely determined by G.*

Given an Abelian group G, the set of its elements having order some power of p is a subgroup, called the *p-torsion* of G. It is clear that the p-torsion subgroup has order a power of p and that torsion subgroups relative to different primes only share the neutral element. This shows

Remark 1.1.22 *Any Abelian group is isomorphic to the direct product of its p-torsion subgroups.*

Consider now an Abelian group G of order p^m. By the Structure Theorem for Finite Abelian Groups, G is isomorphic to a product $\mathbb{Z}/p^{e_1}\mathbb{Z} \times \mathbb{Z}/p^{e_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{e_r}\mathbb{Z}$, where $e_r \geq \cdots \geq e_2 \geq e_1$ and $e_1 + e_2 + \cdots + e_r = m$. Thus the integers $e_r \geq \cdots \geq e_2 \geq e_1$ form a partition of m which is uniquely determined by G.

Corollary 1.1.23 *Let p be a prime and let m be a positive natural number. The number of isomorphism classes of Abelian groups of order p^m is the same as the number of partitions of m.*

One can combine the Structure Theorem for Finite Abelian Groups with the previous remark to obtain the following alternative version of the theorem.

Corollary 1.1.24 *Let G be an Abelian group of order $p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r}$, where $p_1, p_2, \ldots, p_r$ are distinct primes. The group G is isomorphic to the direct product $G_1 \times G_2 \times \cdots \times G_r$ of its p_h-torsion subgroups.*

For $h = 1, 2, \ldots, r$, the subgroup G_h, which has order $p_h^{m_h}$, is isomorphic to a direct product of the form $\mathbb{Z}/p_h^{f_{1,h}}\mathbb{Z} \times \mathbb{Z}/p_h^{f_{2,h}}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_h^{f_{r_h,h}}\mathbb{Z}$, where $f_{r_h,h} \geq \cdots \geq f_{2,h} \geq f_{1,h}$ is a partition of m_h. Moreover, all integers $f_{i,h}$, for $h = 1, 2, \ldots, r$ and $i = 1, 2, \ldots, r_h$, are uniquely determined by G.

A cyclic group has exactly one subgroup for each divisor of its order. It follows easily from the results stated in this section that an Abelian group has at least one subgroup for each divisor of its order, but it may have more than one. For instance, the group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has three subgroups of order 2.

1.1.14 Semidirect Product

As seen with the Structure Theorem for Finite Abelian Groups, direct products can be used to construct and describe a group in terms of factors that we deem easier to understand. We know that if G and H are Abelian groups, then $G \times H$ is Abelian. We shall now introduce a more general construction that equips the Cartesian product $G \times H$ of two groups G and H with a group structure; this time, it will be possible to obtain a non-Abelian group from Abelian factors.

Let G and H be groups, and let $\varphi : H \longrightarrow \mathrm{Aut}(G)$ be a homomorphism. Denote by φ_h the automorphism of G which is the image of h in $\mathrm{Aut}(G)$. It is not hard to check that setting

$$(g_1, h_1) \circ_\varphi (g_2, h_2) = (g_1 \varphi_{h_1}(g_2), h_1 h_2)$$

provides a group structure on the Cartesian product $G \times H$. We denote the ensuing group by $G \rtimes_\varphi H$ and call it the *semidirect product* of G and H with respect to φ. Note that the homomorphism φ defines an action of H on G.

When there is no ambiguity, we will simply write $\circ$ for the operation $\circ_\varphi$, or even omit the operation symbol, as is often done with groups. Similarly, we shall simply write $G \rtimes H$ rather than $G \rtimes_\varphi H$.

Remark that $G \times e_H$ is a normal subgroup of $G \rtimes H$ that is isomorphic to G, and that $e_G \times H$ is a subgroup of $G \rtimes H$ that is isomorphic to H. Moreover, in the case where φ is the trivial homomorphism $H \ni h \longmapsto \mathrm{Id}_G \in \mathrm{Aut}(G)$, the semidirect product $G \rtimes_\varphi H$ is the same as the direct product $G \times H$.

Given a group K, let G and H be two of its subgroups. Suppose that G is normal in K, that $G \cap H = \{e\}$, and that $K = G \cdot H$; then K is isomorphic to the semidirect product $G \rtimes_\varphi H$, where $\varphi_h(g) = hgh^{-1}$. In this case, H acts on G via internal automorphisms of K, by conjugation by elements of H. This way, the group K has been fully described in terms of its subgroups G and H.

Let us now see a standard application of semidirect products.

Remark 1.1.25 *Let G be a group and suppose it has order pq, where $p > q$ are primes. The group G is then isomorphic to the semidirect product $\mathbb{Z}/p\mathbb{Z} \rtimes \mathbb{Z}/q\mathbb{Z}$. Moreover, if q does not divide $p-1$, then G is actually the direct product of the two subgroups, and therefore isomorphic to $\mathbb{Z}/pq\mathbb{Z}$. On the other hand, if q does divide $p-1$, then there is a unique non-Abelian semidirect product of the two groups up to isomorphisms; in this case, in particular, there are exactly two isomorphism classes for groups of order pq.*

A proof of this result is presented in Preliminary Exercises 36, 37, 38, and 39.

1.1.15 Sylow Theorems

We have seen in Sect. 1.1.13 that an Abelian group has a subgroup for each divisor of its order; this, however, does not necessarily hold if the group is not Abelian. For instance, the group A_4, which has order 12, does *not* have any subgroups of order 6: see Preliminary Exercise 14.

A series of theorems due to the Norwegian mathematician Ludwig Sylow clarifies what one can say for general finite groups about the existence of subgroups of certain orders.

Let G be a finite group, p a prime, and suppose p^n is the largest power of p to divide the order of G. A subgroup of G of order p^n is called a *p-Sylow subgroup* of G. It is clear that, if p does not divide $|G|$, then G has no subgroups whose order is a multiple of p. Consequently, p-Sylow subgroups are only interesting for primes p that divide the order G.

Theorem 1.1.26 (First Sylow Theorem) *For each prime divisor p of the order of G these exists a p-Sylow subgroup of G. More generally, if p^n is the largest power of p that divides the order of G and if $r \leq n$, then there exists a subgroup H of G of order p^r, which is contained in a p-Sylow subgroup of G.*

Notice that any conjugate of a p-Sylow is still a p-Sylow. In particular, if there is a unique p-Sylow for some prime p, then this p-Sylow is a normal subgroup. In fact, we can say something stronger:

Theorem 1.1.27 (Second Sylow Theorem) *Given a prime p and a finite group G, all p-Sylow subgroups of G are pairwise conjugate. In particular, they are all isomorphic.*

We can deduce from this theorem that an Abelian group whose order is a multiple of p has a unique p-Sylow, which is its p-torsion subgroup. Remark that, by the Second Sylow Theorem, a p-Sylow subgroup is normal if and only if it is the only p-Sylow. In the case of a non-Abelian group, there may be multiple p-Sylow subgroups, but their number, which we denote by n_p, always satisfies certain conditions, expressed in the following theorem.

Theorem 1.1.28 (Third Sylow Theorem) *Let p be a prime, let n_p be the number of p-Sylow subgroups of a finite group G, and let P be a p-Sylow subgroup of G. The following hold:*

(i) n_p is equal to the index of the normaliser of P, and in particular n_p divides the index of P;
(ii) n_p is congruent to 1 modulo p.

When solving exercises, we will refer collectively to the theorems we just stated as "the Sylow Theorems", without further specification: which statement is being applied will be clear from the context.

1.1.16 Simple Groups

A nontrivial group G is called *simple* if its only normal subgroups are $\{e\}$ and G. Since a subgroup is normal if and only if it is the kernel of a homomorphism, a group is simple if and only if all of its nontrivial homomorphic images are isomorphic to the group itself. This motivates the choice of the term "simple", as such a group cannot be "simplified" by passing to the quotient, other than by turning it into the trivial group.

Since an Abelian group has a subgroup for each divisor of its order, and since these subgroups are all normal, it is clear that an Abelian group is simple if and only if its order is prime: the only simple Abelian groups are thus the cyclic groups of prime order.

Given our remark about groups of order pq, where p and q are distinct primes, a group of order pq cannot be simple. Neither can a group of order p^2, where p is a prime, as any such group must be Abelian. It is sometimes possible to use the Sylow Theorems in order to prove that a certain group must have a normal p-Sylow; the group is thus shown to not be simple.

Proving that a group is simple, on the other hand, is difficult in general. As we shall see in Preliminary Exercise 1.4, the alternating group A_n is simple for $n \geq 5$. This fact has deep consequences, for example concerning the solvability by radicals of polynomial equations of degree at least 5.

Any nontrivial action of a simple group is faithful. In particular, considering the action on cosets of a subgroup immediately yields

Remark 1.1.29 *If a simple group G has a proper subgroup of index r, then G is isomorphic to a subgroup of S_r, and in particular the order of G divides $r!$.*

1.2 Rings

1.2.1 Basic Notions

We briefly recall some basic notions in ring theory, a few of which we have already introduced in the first volume [Volume 1, Theory, Section 5].

A ring A is called commutative if $ab = ba$ for all $a, b \in A$ and unitary if it has a neutral element for multiplication, usually denoted by 1 and called the unity of the ring. If A and B are unitary rings, then we require that a homomorphism from A to B send the unity of A to the unity of B.

From now on, we assume every ring to be commutative and unitary, unless otherwise stated.

An ideal I in a ring is a subgroup with respect to addition that absorbs multiplication, that is, such that $ab \in I$ for all $a \in I$ and $b \in A$. Given a ring A, the sets $\{0\}$ and A are always ideals. We will often simply denote by 0 the ideal $\{0\}$, which is called *trivial*. Let us remark explicitly that in a unitary ring any nonempty subset I that is closed under addition and absorbs multiplication is an ideal: indeed, if a is an element of I then $-a = (-1) \cdot a$ is also an element of I.

An element a of a ring A is called invertible if there exists an element b such that $ab = 1$. The set of invertible elements of A is denoted by A^*.

Given an ideal I of A, we have $I = A$ if and only if $1 \in I$; an ideal I is called *proper* if $I \neq A$. A proper ideal I in a ring A is maximal if I is not strictly contained in any proper ideal J of A, that is, if I is maximal for the inclusion relation within the family of all proper ideals of A.

The intersection of a family of ideals is itself an ideal. For instance, recall that given a subset X of a ring A, the ideal (X) generated by X is the intersection of all ideals containing X; it is the smallest ideal of A that contains X. If $X = \{a_1, a_2, \ldots, a_n\}$ is finite, then the ideal generated by X is denoted by $(a_1, a_2, \ldots, a_n)$. An ideal $I = (a_1)$ generated by a single element is called *principal* and is the set $A \cdot a_1$ of all multiples of a_1 in A.

As we do for groups and normal subgroups, we can define a quotient structure of a ring by an ideal. Given a ring A and an ideal I of A, the quotient ring A/I is the set of additive cosets $a + I$, where $a \in A$, endowed with the operations induced by those of A, that is: $(a + I) + (b + I) = (a + b) + I$ and $(a + I) \cdot (b + I) = a \cdot b + I$. The natural map $A \ni a \longmapsto a + I \in A/I$ is called the *quotient homomorphism* , and its kernel is the ideal I.

Given two rings A and B, we can define the operation $(a_1, b_1) \cdot (a_2, b_2) = (a_1 a_2, b_1 b_2)$ on the Abelian group $A \times B$ and thus give $A \times B$ a ring structure. If A and B are unitary commutative rings, then $A \times B$ is also unitary and commutative. Moreover, it is easy to check that all ideals of $A \times B$ are of the form $I \times J$, where I is an ideal of A and J an ideal of B.

An element a of a ring A is called a zero divisor if there exists a nonzero element b such that $ab = 0$. An integral domain is a nonzero ring such that 0 is its only zero divisor.

1.2.2 Maximal and Prime Ideals

In order to show the existence of maximal ideals, we need the following result from logic, which is called Zorn's Lemma and is equivalent to the Axiom of Choice and to the Well-Ordering Theorem.[1]

Let X be a set endowed with a partial order $\leq$. A subset C of X is called a *chain* if for all $x, y \in C$ we have $x \leq y$ or $y \leq x$; in other words, a chain is a totally ordered subset of X. Given a subset Y of X, an *upper bound* for Y is an element $x \in X$ such that $y \leq x$ for all $y \in Y$. Finally, we say that an element $x \in X$ is *maximal* if, for each $y \in X$, $x \leq y$ implies $y = x$.

Lemma 1.2.1 (Zorn's Lemma) *Let X be a nonempty set endowed with a partial order $\leq$. If every chain of X has an upper bound, then X has a maximal element.*

Let A be a nonzero ring and let I be a proper ideal of A. Consider the set X of all proper ideals of A that contain I, endowed with the partial order given by set inclusion. The set X is nonempty, as I belongs to it, and the union of any chain of proper ideals containing I is still a proper ideal of A that contains I. We can thus apply Zorn's Lemma to deduce the existence of a maximal element in X, that is, of a maximal ideal of A that contains I. We have proven the following

Theorem 1.2.2 *Given a proper ideal I in a nonzero ring A, there exists a maximal ideal of A containing I.*

In particular, applying the theorem to the ideal 0 yields

Corollary 1.2.3 *Every nonzero ring contains at least one maximal ideal.*

By applying the theorem to the proper ideal generated by a non-invertible element, we also obtain

Corollary 1.2.4 *Given an element a that is not invertible, there exists a maximal ideal containing a.*

An ideal P of a ring A is called *prime* if $P \neq A$ and for all $x, y \in A$ such that $xy \in P$, we have $x \in P$ or $y \in P$. Remark that, as one can immediately deduce from the definition, an ideal P of a ring A is prime if and only if the quotient A/P is an integral domain. It follows that the ideal 0 is prime if and only if the ring A is an integral domain.

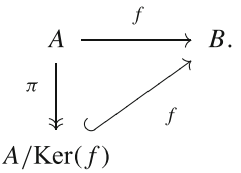
On the other hand, it is clear that an ideal M of a commutative ring A is maximal if and only if A/M is a field. Thus, since every field is an integral domain, we find that every maximal ideal is prime.

[1] See for example H. Herrlich, *"Axiom of Choice"*, Springer, 2006, Berlin and W. Hodges, *"Krull implies Zorn"*, J. London Math. Soc. 19, (1979), 285–287.

1.2.3 Quotient Rings

In the same way as normal subgroups are the kernels of group homomorphisms, ideals are the kernels of ring homomorphisms, and we have an analogue of the Fundamental Homomorphism Theorem in the context of rings.

Theorem 1.2.5 (Fundamental Homomorphism Theorem for Rings) *Let A and B be rings, let $f : A \longrightarrow B$ be a ring homomorphism and let $\pi : A \longrightarrow A/\mathrm{Ker}(f)$ be the quotient homomorphism. There exists a unique homomorphism $\overline{f}$ making the following diagramme commutative:*



In particular, if f is surjective then $A/\mathrm{Ker}(f)$ is isomorphic to B. Moreover, it is clear that the image and pre-image of a subring with respect to a homomorphism are both subrings themselves. In the context of rings, the quotient homomorphism induces a bijective correspondence between ideals; indeed, we have

Proposition 1.2.6 *Let I be an ideal in the ring A and let $\pi : A \longrightarrow A/I$ be the quotient homomorphism. The set of ideals in A/I is in bijection with the set of ideals in A that contain I. The maps $J' \longmapsto \pi^{-1}(J')$ and $J \longmapsto \pi(J) = J/I$ realise this bijection and are inverses of each other. Moreover, maximal ideals correspond to maximal ideals and prime ideals correspond to prime ideals.*

1.2.4 Ideal Operations

Given two ideals I and J in the ring A, their *sum* $I + J$ is the set of all sums $a + b$, where $a \in A$ and $b \in B$; it is easy to see that $I + J$ is an ideal of A. An equivalent definition of $I + J$ is as the ideal generated by $I \cup J$. The *product* IJ is the ideal generated by all products ab, where $a \in I$ and $b \in J$; in other words, IJ is the set of all possible sums

$$a_1b_1 + a_2b_2 + \cdots + a_nb_n,$$

where n ranges among all natural numbers, $a_1, a_2, \ldots, a_n$ are elements of I and $b_1, b_2, \ldots, b_n$ are elements of J. Remark that the product IJ is contained in the intersection ideal $I \cap J$.

Two ideals I, J in a ring A are called *coprime* if $I + J = A$. Clearly, I and J are coprime if and only if there exist $a \in I$ and $b \in J$ such that $a + b = 1$. If $I_1, I_2, \ldots, I_r$ are pairwise coprime ideals, then $I_1 I_2 \cdots I_r = I_1 \cap I_2 \cap \cdots \cap I_r$.

By an argument essentially analogous to the one that applies to congruences, replacing the hypothesis that moduli are coprime by the condition that ideals are coprime, one obtains the following

Theorem 1.2.7 (Chinese Remainder Theorem for Rings) *Let $I_1, I_2, \ldots, I_r$ be pairwise coprime ideals in a ring A, and let $I = I_1 \cap I_2 \cap \cdots \cap I_r$ be their intersection. The map*

$$A/I \ni a + I \longmapsto (a + I_1, a + I_2, \ldots, a + I_r) \in A/I_1 \times A/I_2 \times \cdots \times A/I_r$$

is a ring isomorphism.

Given two ideals I, J in a ring A, we define their *ideal quotient* $(I : J)$ as the set of elements $a \in A$ such that $aJ \subseteq I$. One can show that $(I : J)$ is itself an ideal; see for example Preliminary Exercise 48. The ideal quotient $(0 : I)$ is called the *annihilator* of I, usually denoted by $\mathrm{Ann}(I)$, and is the set of elements $a \in A$ such that $aI = 0$.

Given an ideal I, its *radical* $\sqrt{I}$ is the set of elements $a \in A$ such that $a^n \in I$ for some $n \in \mathbb{N}$. The radical of an ideal is itself an ideal, and moreover $\sqrt{I}$ is the intersection of all prime ideals containing I. These properties of the radical ideal, among others, are discussed in Preliminary Exercise 50. Remark that the radical $\sqrt{0}$ of the zero ideal, which is called the *nilradical*, is the set of nilpotent elements of A, that is, those elements $a \in A$ such that $a^n = 0$ for some $n \in \mathbb{N}$; it is the intersection of all prime ideals of A.

In the previous section, we considered a correspondence between ideals induced by the quotient homomorphism. We now discuss the behaviour of ideals when taking their images and pre-images according to general homomorphisms. Given a ring homomorphism $\varphi : A \longrightarrow B$ and an ideal I of A, the image $\varphi(I)$ is not necessarily an ideal; this leads us to define I^e, the *extension* of I with respect to φ, as the ideal of B generated by $\varphi(I)$. On the other hand, given an ideal J of B, the pre-image $I^c = \varphi^{-1}(J)$ is an ideal, called the *contraction* of J with respect to φ; note that every contraction contains the kernel $\mathrm{Ker}(\varphi)$.

Remark that $I^{ec} \supseteq I$ for all ideals I of A, and that $J^{ce} \subseteq J$ for all ideals J of B. Moreover, the contraction of a prime ideal is still a prime ideal. What is not true in general is that the extension of a prime ideal is a prime ideal: for example, (x^2) is prime in $\mathbb{Q}[x^2]$, but its extension with respect to the inclusion homomorphism is not prime in $\mathbb{Q}[x]$.

1.2.5 Fraction Fields and Localisations

Given an integral domain A, one can construct a field containing A via the following procedure.

On the Cartesian product $A \times (A \setminus \{0\})$, define the relation $\sim$ given by: $(a, b) \sim (c, d)$ if and only if $ad = bc$. It is immediate to check that $\sim$ is an equivalence relation. We denote the equivalence class of (a, b) by a/b. The sum and product

$$a/b + c/d = (ad + bc)/bd$$
$$a/b \cdot c/d = ac/bd$$

are well-defined, that is, they do not depend on the representatives a, b, c, d, but only on the classes a/b, c/d. We have thus defined two operations on the set $\mathbb{K}$ of equivalence classes of $A \times (A \setminus \{0\})$ with respect to $\sim$. When endowed with these operations, $\mathbb{K}$ is a field, which we call the *fraction field* of A.

For example, the fraction field of $\mathbb{Z}$ is $\mathbb{Q}$; this is not surprising, as the very definition of the fraction field is modelled on the standard construction of $\mathbb{Q}$ from $\mathbb{Z}$. The fraction field of the ring of polynomials $\mathbb{K}[x]$, where $\mathbb{K}$ is a field, is the field $\mathbb{K}(x)$ of *rational functions*, that is, the set of fractions $f(x)/g(x)$, where $f(x)$ and $g(x)$ are polynomials and $g(x)$ is not the zero polynomial.

The map $A \ni a \longmapsto a/1 \in \mathbb{K}$ is an injective ring homomorphism; from now on, we shall implicitly identify A with a subring of $\mathbb{K}$ via this homomorphism. We have obtained the field $\mathbb{K}$ from the integral domain A by adding the inverse of each nonzero element, which we were able to do thanks to the fact that there are no zero divisors in A. The fraction field of A is thus the smallest field that A can be embedded in.

It is sometimes useful to consider a sort of "partial" fraction field, in a sense that we will now clarify. A subset S of a ring A is called a *multiplicative set* if $1 \in S$, $0 \notin S$, and for all $s, t \in S$ we have $st \in S$. Remark that, if A is an integral domain, then the set of fractions a/s in $\mathbb{K}$ such that $a \in A$ and $s \in S$ is a subring of $\mathbb{K}$; we denote this subring by $S^{-1}A$ and call it the *localisation* of A with respect to S.

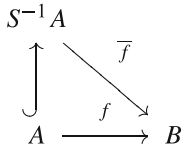
It is clear that for $S = A \setminus \{0\}$ we will again obtain the fraction field $\mathbb{K}$. In general, we have a pair of injective ring homomorphisms given by $A \ni a \longmapsto a/1 \in S^{-1}A$ and $S^{-1}A \ni a/s \longmapsto a/s \in \mathbb{K}$; in other words, by identifying A with its image in $\mathbb{K}$, we have the inclusions of rings $A \subseteq S^{-1}A \subseteq \mathbb{K}$.

As an example, consider the subset $S = \{1, 2, 4, 8, \ldots\}$ of $\mathbb{Z}$; S is a multiplicative set, and the localisation $S^{-1}\mathbb{Z}$ is the set of rational numbers whose denominator is a power of 2. Another example is the following: given a prime ideal P of a ring A, its complement $S = A \setminus P$ is a multiplicative set; the localisation $S^{-1}A$ is usually denoted by A_P and called the localisation of A at P: this concept will turn out to be of particular importance. The localisation at a prime ideal is the subject of Preliminary Exercise 53.

We now go back to a more general context. The localisation $S^{-1}A$ of an integral domain A is the smallest ring that contains A and is such that every element of S is

invertible. As partly proven in Preliminary Exercise 51, we in fact have the stronger result of

Proposition 1.2.8 *Let A be an integral domain and let S be a multiplicative set in A. Given a ring homomorphism $f : A \longrightarrow B$, there exists a homomorphism $\overline{f}$ making the diagramme*



commute if and only if $f(s)$ is invertible in B for all s in S.

If I is an ideal of A then $S^{-1}I = \{a/s \,|\, a \in A, s \in S\}$ is an ideal of $S^{-1}A$; conversely, every ideal of $S^{-1}A$ is of this form. Note that $S^{-1}I$ is the extension of I with respect to the inclusion $A \ni a \longmapsto a/1 \in S^{-1}A$, whereas the contraction of an ideal J with respect with that same homomorphism is clearly $J \cap A$. Moreover, we have $S^{-1}I = A$ if and only if $I \cap S$ is nonempty. It is easy to show that if I and J are ideals of A that are disjoint from S, then $S^{-1}I = S^{-1}J$ if and only if $I = J$. This proves the first part of the following

Proposition 1.2.9 *Ideals of A that do not intersect the multiplicative set S are in bijection with proper ideals of $S^{-1}A$ via the maps*

$$\begin{aligned} I &\longmapsto S^{-1}I \\ J \cap A &\longleftarrow\!\shortmid J. \end{aligned}$$

In particular, prime ideals correspond to prime ideals.

An immediate consequence of this proposition is the fact that all ideals of the localisation A_P of A at the prime ideal P (that is, the localisation with respect to the multiplicative set $A \setminus P$) are of the form $S^{-1}I = \{a/s \,|\, a \in I, s \notin P\}$, where I is an ideal of A contained in P. In particular, A_P admits $S^{-1}P$ as its unique maximal ideal.

1.2.6 Divisibility

Throughout this section, we assume A to be an integral domain. Given elements a and b in A, we say that a *divides* b, and write $a \,|\, b$, if there exists c in A such that $b = ac$. When $a \,|\, b$ we say that a is a *divisor* of b and that b is a *multiple* of a. Remark that the divisors of one in A are precisely the invertible elements of A. Moreover, $a \,|\, b$ if and only if $(b) \subseteq (a)$. In what follows, we will often use principal ideals of A to express divisibility conditions on elements of A.

Two elements a, b of A are said to be *associates* if there exists an invertible element $c \in A$ such that $a = cb$. Equivalently, a and b are associates if and only if a divides b and b divides a. Another equivalent condition is that the principal ideals (a) and (b) coincide.

An element d is a *greatest common divisor* of two elements a and b that are not both zero if the following conditions hold: $d \mid a$, $d \mid b$ and for all common divisors c of a and b, we have $c \mid d$. Given two elements, a greatest common divisor does not necessarily exist; but, if it does, then it is unique up to associates. In other words, if d and d' are both greatest common divisors of a and b, then d and d' are associates, that is, there is an invertible element $c \in A$ such that $d' = cd$. Since all invertible elements are associates of 1, we will sometimes say that a greatest common divisor of two elements is 1 to mean it is invertible.

Let d be a greatest common divisor of a and b; it follows from the definition that the ideal (a, b) generated by a and b is contained in the ideal generated by d. Moreover, every principal ideal (c) containing (a, b) also contains (d). On the other hand, if (a, b) is a principal ideal, then we have a result similar to the one we stated for integers [Volume 1, Proposition 3.2 (Bézout's Identity), page 16].

Proposition 1.2.10 (Bézout's Identity) *If the ideal (a, b) is a principal ideal then there exists a greatest common divisor d of a and b, and we have $(a, b) = (d)$. In particular, there exist u and v in A such that $d = ua + vb$.*

A nonzero, non-invertible element p of A is said to be *prime* if whenever $p \mid ab$, with $a, b \in A$, we have $p \mid a$ or $p \mid b$. Obviously, the prime elements in the ring $\mathbb{Z}$ are those of the form $\pm p$, where p is a prime number; the general definition of a prime in an integral domain is based on Euclid's Lemma [Volume 1, Remark 3.4 (Euclid's Lemma), page 16]. It is easy to show the following

Proposition 1.2.11 *Given an integral domain A, an element p of A is prime if and only if the principal ideal (p) is a nonzero prime ideal.*

A nonzero, non-invertible element a of A is said to be *irreducible* if whenever $a = bc$, with b and c in A, either b or c is invertible. Again, it is possible to draw a parallel between irreducible elements and certain ideals.

Proposition 1.2.12 *Given an integral domain A, an element a of A is irreducible if and only if (a) is a proper nonzero ideal that is maximal within the family of principal ideals of A.*

In $\mathbb{Z}$, an element is irreducible if and only if it is prime. It can easily be shown in general that we have

Proposition 1.2.13 *In an integral domain, every prime element is irreducible.*

The converse, however, is not true. For instance, in the integral domain $\mathbb{Z}[\sqrt{-5}]$, whose elements are the complex numbers of the form $a + b\sqrt{-5}$, where a and b are integers, we have

$$2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

in spite of the fact that 2 does not divide either of the factors of the right-hand side; so 2 is not prime in $\mathbb{Z}[\sqrt{-5}]$. But, as we show in Preliminary Exercise 1.4, the element 2 is irreducible in $\mathbb{Z}[\sqrt{-5}]$.

1.2.7 Euclidean Domains

In this section, we discuss how the concept of Euclidean Division can be generalised to rings other $\mathbb{Z}$, while maintaining its key properties.

We say that an integral domain A is a *Euclidean domain* if there is a map $d : A \setminus \{0\} \longrightarrow \mathbb{N}$, called *degree* on A, such that

(i) for all a, b in A with $b \neq 0$ there exist q, r in A such that $a = qb + r$ and either $r = 0$ or $d(r) < d(b)$,
(ii) $d(a) \leq d(ab)$ for all $a, b \in A \setminus \{0\}$.

We shall call the expression $a = qb+r$, where $r = 0$ or $d(r) < d(b)$, the *division with remainder* of a by b. The elements q and r are, respectively, the *quotient* and the *remainder* in the division $a = qb + r$. Note, however, that if the remainder is nonzero, then the quotient and remainder are *not* unique in general: we can have other valid expressions of the form $a = q'b + r'$, where $d(r') < d(b)$ and q', r' are different from q, r.

The ring of integers $\mathbb{Z}$ is a Euclidean domain with degree given by $\mathbb{Z} \setminus \{0\} \ni a \longmapsto |a| \in \mathbb{N}$; its division with remainder is obviously given by the standard Euclidean division. The ring of polynomials $\mathbb{K}[x]$, where $\mathbb{K}$ is a field, is also a Euclidean domain, with degree given by $f(x) \longmapsto \deg(f(x))$ for $f(x) \in \mathbb{K}[x]\setminus\{0\}$.

Given a field $\mathbb{K}$, a *formal power series* with coefficients in $\mathbb{K}$ is an expression of the form

$$\sum_{n\geq 0} a_n x^n$$

where $a_0, a_1, \ldots$ are elements $\mathbb{K}$. We can define the sum of two formal power series as

$$\sum_{n\geq 0} a_n x^n + \sum_{n\geq 0} b_n x^n = \sum_{n\geq 0} (a_n + b_n) x^n$$

and the product as

$$\sum_{n\geq 0} a_n x^n \cdot \sum_{n\geq 0} b_n x^n = \sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) x^n.$$

It is easy to prove that these operations endow the set $\mathbb{K}[[x]]$ of formal power series with an integral domain structure. We call the *zero series* the formal power series

all of whose coefficients are 0, which clearly is the 0 element in the ring $\mathbb{K}[[x]]$. Moreover, given a nonzero series $\sum_{n\geq 0} a_n x^n$, we define the degree

$$d\left(\sum_{n\geq 0} a_n x^n\right) = \min\{n \mid a_n \neq 0\}.$$

By inductively solving a system in an infinite number of unknowns, it is easy to check that every formal power series of degree 0 is invertible. Now, note that a nonzero formal power series $f(x)$ can be expressed as $f(x) = x^n \cdot f_0(x)$, where $n = d(f)$ and $f_0(x)$ has degree 0 and is therefore invertible. Let $f(x) = x^n f_0(x)$ and $g(x) = x^m g_0(x)$ be two formal power series, both nonzero, of degree n and m, respectively. If $n \geq m$ then $f(x) = x^{n-m} g_0(x) f_0(x)^{-1} + 0$, and in particular $f(x)$ is a multiple of $g(x)$; if $n < m$ then $f(x) = 0 \cdot g(x) + f(x)$. Remark that, since $\mathbb{K}$ is a field, we have $d(f(x) \cdot g(x)) = d(f(x)) + d(g(x))$ for all pairs of formal power series $f(x), g(x) \in \mathbb{K}[[x]] \setminus \{0\}$. It follows that d is indeed a degree on $\mathbb{K}[[x]]$, which is thus a Euclidean domain. Some additional properties of $\mathbb{K}[[x]]$ are presented in Preliminary Exercise 56.

The ring $\mathbb{Z}[i]$, that is, the set of complex numbers $a + bi$ such that $a, b \in \mathbb{Z}$, is called the ring of *Gaussian integers*. Setting

$$\mathbb{Z}[i] \ni z = a + bi \longmapsto N(z) = a^2 + b^2 \in \mathbb{N}$$

gives us a degree on the ring of Gaussian integers. Note that the degree $N(z) = |z|^2 = z \cdot \overline{z}$ of z is the square of its modulus as a complex number, which is why the map $z \longmapsto N(z)$ is often called *norm*. For more details, consult Sect. 1.2.10, which is entirely dedicated to the Gaussian integers.

By adapting the analogous proof on $\mathbb{Z}$, one finds

Proposition 1.2.14 *In a Euclidean domain, each nonzero ideal is generated by any of its elements of minimal degree.*

By applying the above proposition to the whole ring, one immediately obtains

Corollary 1.2.15 *In a Euclidean domain, the elements of minimal degree are precisely the invertible elements.*

Thus in a Euclidean domain A all ideals are principal; in particular, what we saw in Sect. 1.2.6 implies

Remark 1.2.16 *In a Euclidean domain A, every pair of elements a, b that are not both zero admits a greatest common divisor d; any such greatest common divisor is a generator for (a, b). Moreover, we have Bézout's Identity, that is, $d = ua + vb$, for some $u, v \in A$.*

An analogue of Euclid's Algorithm on $\mathbb{Z}$ [Volume 1, Proposition 3.3 (Euclid's Algorithm), page 16] enables us to explicitly compute both a greatest common divisor d for any pair of elements a, b, and a pair of elements u, v satisfying Bézout's Identity $d = ua + vb$.

1.2.8 Principal Ideal Domains

We now turn our attention to another class of rings defined by generalising properties of the ring of integers. We know that all ideals of $\mathbb{Z}$ are principal; we call any integral domain in which every ideal is principal a *principal ideal* domain.

It is clear that every Euclidean domain is a principal ideal domain by what we showed in the previous section. Let us immediately remark that the converse does not hold, that is, not all principal ideal domains are Euclidean: for instance, the ring $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$, that is, the set of complex numbers of the form $a + b\frac{1+\sqrt{-19}}{2}$, where a and b are integers, is a principal ideal domain but *not* a Euclidean domain.[2]

Some properties of rings can be expressed in a natural way in terms of chains of ideals. An ascending chain of ideals $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ in a ring A is said to be *stationary* if there is $k \in \mathbb{N}$ such that $I_k = I_{k+1} = I_{k+2} = \cdots$. The same definition applies to descending chains of ideals.

Now let $I_1 \subseteq I_2 \subseteq \cdots$ be an ascending chain of ideals in a principal ideal domain A. The union I of all ideals in the chain is still an ideal, and it is generated by a single element, say a. But then a belongs to I_h for some $h \geq 1$; in particular, the chain is stationary. We have thus proven that in a principal ideal domain every ascending chain of ideals is stationary. In general, we call a ring in which every ascending chain is stationary *Noetherian*, in honour of the German mathematician Emmy Noether. We summarise what we just found in the following

Proposition 1.2.17 *Every principal ideal domain is Noetherian.*

From what we discussed in Sect. 1.2.6 we have

Corollary 1.2.18 *Let A be a principal ideal domain. If a and b are elements of A and are not both zero, then there exists a greatest common divisor d of a and b, and d generates the ideal (a, b). Moreover, Bézout's Identity holds: that is, there exist u and v in A such that $d = ua + vb$.*

As opposed to what happens in a Euclidean domain, we cannot use Euclid's Algorithm to compute the greatest common divisor or the coefficients of Bézout's Identity in a general principal ideal domain, as we do not have the tool of division with remainder at our disposal.

An important property of principal ideal domains is given by the following

Remark 1.2.19 *In a principal ideal domain, an element is prime if and only if it is irreducible.*

The fact that every ideal is principal and the fact that prime elements are the same as irreducible elements imply

Remark 1.2.20 *In a principal ideal domain, an ideal is prime if and only if it is maximal or the zero ideal.*

[2] For a proof of this fact, see J. C. Wilson, *"A Principal Ring that is Not a Euclidean Ring"*, Mathematics Magazine 46, (1973), 34–38.

1.2.9 Unique Factorisation Domains

Throughout this section, we still assume that A is an integral domain. We say that a nonzero, non-invertible element *a factors* if there exists a *factorisation* for a, that is, if there are irreducible elements $a_1, a_2, \cdots, a_n$, with $n \geq 1$, such that $a = a_1 a_2 \cdots a_n$.

Obviously, if a is irreducible then it factors. On the other hand, if a is not irreducible, not zero and not invertible, then we can write $a = bc$ with b and c both nonzero and non-invertible; so if b and c are irreducible then we have a factorisation of a, otherwise we can start building factorisations for b and c. It is not necessarily true, however, that this procedure will end after a finite number of steps. Consider for example the ring $A = \mathbb{Q}[\sqrt[n]{x} \mid n \in \mathbb{N}]$ of polynomials in all roots of the variable x; in A, we have $x = \sqrt{x}\sqrt{x} = \sqrt{x}\sqrt[4]{x}\sqrt[4]{x} = \sqrt{x}\sqrt[4]{x}\sqrt[8]{x}\sqrt[8]{x} = \cdots$, so the factoring procedure never ends for the element x.

By using statements from Sect. 1.2.6 about divisibility, principal ideals and irreducible elements, one can show

Proposition 1.2.21 *In an integral domain the following are equivalent:*

(i) every nonzero, non-invertible element has a factorisation as a product of irreducible elements;
(ii) all ascending chains of principal ideals are stationary.

We say that an element a of A that factors *has unique factorisation* if, given two factorisations of a as a product of irreducible elements $a = a_1 a_2 \cdots a_n$ and $a = b_1 b_2 \cdots b_m$, we have $m = n$ and, up to reordering factors, a_h and b_h are associates for $h = 1, 2, \ldots, n$.

We say that A is a *unique factorisation domain* if every nonzero, non-invertible element of A has unique factorisation. While existence of a factorisation as a product of irreducible elements is tied to the properties of chains of principal ideals, uniqueness of such factorisations depends on whether irreducible elements are prime. We have in fact

Proposition 1.2.22 *An integral domain A is a unique factorisation domain if and only if the following hold:*

1. *every irreducible element is prime,*
2. *ascending chains of prime ideals are stationary.*

It follows that in a unique factorisation domain there is no difference between prime elements and irreducible elements, and thus the two terms will be used interchangeably.

By what we saw in the previous section, every principal ideal domain, and in particular every Euclidean domain, is a unique factorisation domain. The following are all examples of unique factorisation domains: the ring of integers $\mathbb{Z}$, any field, the ring of polynomials $\mathbb{K}[x]$ with coefficients in a field $\mathbb{K}$, the Gaussian integers $\mathbb{Z}[i]$.

We have seen in Sect. 1.2.6 that 2 is irreducible but not prime in $\mathbb{Z}[\sqrt{-5}]$, which is thus a ring that is not a unique factorisation domain; indeed, $2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$ are two distinct factorisations of 6 as a product of irreducible elements. The ring $\mathbb{Q}[\sqrt[n]{x} \mid x \in \mathbb{N}]$ is not a unique factorisation domain either: as an addendum to our discussion at the beginning of this section, remark that the ascending chain of principal ideals $(x) \subseteq (\sqrt{x}) \subseteq (\sqrt[4]{x}) \subseteq (\sqrt[8]{x}) \subseteq \cdots$ is not stationary.

From uniqueness it follows that, given an irreducible element p and an element $a = a_1 a_2 \cdots a_n$ with a factorisation as a product of irreducibles, the number of irreducible factors a_h, $h = 1, 2, \ldots, n$, such that a_h and p are associates only depends on a and *not* on the specific factorisation. We call this non-negative integer number the *multiplicity* of p in a. It is clear that $a = u p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r}$, where $p_1, p_2, \ldots, p_r$ are the distinct irreducible elements appearing in $a, m_1, m_2, \ldots, m_r$ are the respective multiplicities, and u is an invertible element. We shall call an expression of this form a *factorisation with multiplicity*. Moreover, remark that if an irreducible element p does not appear in a factorisation of a, then its multiplicity is zero; we can thus include p in any factorisation with multiplicity for a, by setting its exponent to zero. In particular, given two elements, we may assume that the same irreducible elements appear in their factorisations with multiplicity. From this one can prove

Remark 1.2.23 *Given two elements a, b in a unique factorisation domain that are not both zero, there exists a greatest common divisor of a and b. In particular, if $a = u p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r}$ and $b = v p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r}$ are factorisations with multiplicity, then, letting $\ell_h = \min\{m_h, n_h\}$ for $h = 1, 2, \ldots, r$, the element $p_1^{\ell_1} p_2^{\ell_2} \cdots p_r^{\ell_r}$ is a greatest common divisor for a and b.*

Thanks to the existence of greatest common divisors in unique factorisation domains, we can extend Gauss's Lemma, which was stated in the first volume for polynomials with integer coefficients [Volume 1, Lemma 5.16 (Gauss' Lemma), page 42]. Let A be a unique factorisation domain and let $f(x) = a_0 + a_1 x + \cdots + a_n x^n$ be a polynomial with coefficients in A; the *content* of $f(x)$ is a greatest common divisor of the coefficients $a_0, a_1, \ldots, a_n$. Just like the greatest common divisor, the content is defined up to associates. A polynomial is said to be *primitive* if its content is 1. With a proof entirely analogous to the one for polynomials with integer coefficients, one can show

Lemma 1.2.24 (Gauss's Lemma) *Let A be a unique factorisation domain. The product of any two primitive polynomials in $A[x]$ is a primitive polynomial.*

Gauss's Lemma comes with several important corollaries.

Corollary 1.2.25 *Let A be a unique factorisation domain. The content of the product of two polynomials in $A[x]$ is the product of their contents.*

We can relate divisibility in $A[x]$ to divisibility in $\mathbb{K}[x]$, where $\mathbb{K}$ is the fraction field of A, as explained by the next corollary.

Corollary 1.2.26 *Let A be a unique factorisation domain and let $\mathbb{K}$ be its fraction field. Consider $f(x),\ g(x) \in A[x]$; if $f(x)$ is primitive and divides $g(x)$ in $\mathbb{K}[x]$ then $f(x)$ divides $g(x)$ in $A[x]$.*

Not only does divisibility transfer from $\mathbb{K}[x]$ to $A[x]$, but the factorisations of polynomials in the two rings correspond to each other; it is sometimes the following corollary that is presented under the name of Gauss's Lemma.

Corollary 1.2.27 *Let A be a unique factorisation domain and let $\mathbb{K}$ be its fraction field. If $f(x) \in A[x]$ and $g(x), h(x) \in \mathbb{K}[x]$ are such that $f(x) = g(x)h(x)$ then there exist $g_1(x), h_1(x) \in A[x]$, where $g_1(x)$ and $g(x)$ are associates in $\mathbb{K}[x]$, and $h_1(x)$ and $h(x)$ are associates in $\mathbb{K}[x]$, such that $f(x) = g_1(x)h_1(x)$.*

Finally, it follows easily from the results recalled above that irreducibility can be tested in $A[x]$ or in $\mathbb{K}[x]$ interchangeably: indeed,

Corollary 1.2.28 *Let A be a unique factorisation domain and let $\mathbb{K}$ be its fraction field. A primitive polynomial $f(x) \in A[x]$ is irreducible in $\mathbb{K}[x]$ if and only if it is irreducible in $A[x]$.*

From the above results, we conclude that if A is a unique factorisation domain, then a polynomial in $A[x]$ is irreducible if and only if one of the following holds: the polynomial has degree 0 and is irreducible as an element of A, or the polynomial has positive degree, it is primitive and it is irreducible in $\mathbb{K}[x]$.

Since the ring of polynomials $\mathbb{K}[x]$ is a unique factorisation domain, our statements about $A[x]$ can be used to prove that the property of being a unique factorisation domain is inherited by rings of polynomials. That is,

Proposition 1.2.29 *If A is a unique factorisation domain then $A[x]$ is also a unique factorisation domain.*

For example, $\mathbb{Z}[x]$ is a unique factorisation domain. The ring $\mathbb{Z}[x]$ is an example of how not every unique factorisation domain is a principal ideal domain. Indeed, the ideal $(2, x)$ of $\mathbb{Z}[x]$ is not principal, as it is not generated by 1, which is the greatest common divisor of 2 and x; thus, $\mathbb{Z}[x]$ is a unique factorisation domain but *not* a principal ideal domain. Note that what we just said about the ideal $(2, x)$ shows how, while in a unique factorisation domain a greatest common divisor for a, b does exist, it need not be a generator of the ideal (a, b).

Obviously, repeatedly applying the previous proposition yields

Corollary 1.2.30 *If A is a unique factorisation domain and n is a positive integer, then the ring of polynomials $A[x_1, x_2, \ldots, x_n]$ in n variables is a unique factorisation domain.*

For example, given a field $\mathbb{K}$, the ring $\mathbb{K}[x, y]$ is a unique factorisation domain. Like $\mathbb{Z}[x]$, however, $\mathbb{K}[x, y]$ is not a principal ideal domain: for instance, the greatest common divisor of x and y is 1, which is not a generator for the ideal (x, y).

We can obtain a further generalisation of the last corollary by remarking that each element in the ring $A[x_1, x_2, \ldots]$ of polynomials in an infinite number of variables only involves a finite number of variables. Moreover, if a variable does

not appear in a polynomial $f(x_1, x_2, \ldots)$, then it cannot appear in any factorisation of $f(x_1, x_2, \ldots)$. This allows one to show

Corollary 1.2.31 *If A is a unique factorisation domain then the ring $A[x_1, x_2, \ldots]$ of polynomials in an infinite number of variables is also a unique factorisation domain.*

Eisenstein's Criterion [Volume 1, Proposition 5.22 (Eisenstein's Criterion), page 43] can also be generalised from $\mathbb{Z}$ to any unique factorisation domain.

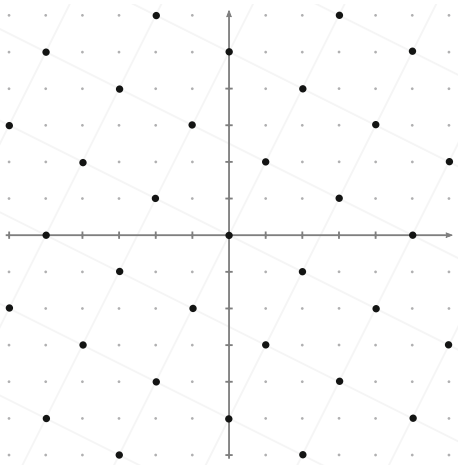
Proposition 1.2.32 (Eisenstein's Criterion) *Let $f(x) = a_0 + a_1 x + \cdots + a_n x^n$ be a polynomial with coefficients in a unique factorisation domain A. If there is a prime p of A such that p does not divide a_n, p divides $a_0, a_1, \ldots, a_{n-1}$ and p^2 does not divide a_0, then $f(x)$ is irreducible in $A[x]$ and therefore in $\mathbb{K}[x]$, where $\mathbb{K}$ is the fraction field of A.*

As an example of how Eisenstein's Criterion might be applied, consider the unique factorisation domain $A = \mathbb{K}[t]$, where $\mathbb{K}$ is a field whose characteristic is some prime p. It is clear that the element t is prime in A. Eisenstein's Criterion then implies that $x^p - t$ is irreducible in $A[x]$. Note that $x^p - t$ has a single root of multiplicity p in an algebraic closure of the fraction field $\mathbb{K}(t)$ of $\mathbb{K}[t]$; we have thus found an example of an irreducible polynomial with a multiple root.

1.2.10 Gaussian Integers

We have previously defined the ring $\mathbb{Z}[i]$ of Gaussian integers as the set of complex numbers $a + bi$ with $a, b \in \mathbb{Z}$. It is a Euclidean domain with degree given by the norm $a + bi \longmapsto N(a + bi) = a^2 + b^2$.

Via the usual identification of complex numbers with points of the plane $\mathbb{R}^2$, Gaussian integers correspond to the subset $\mathbb{Z}^2$ of points with integer coordinates, which form a lattice. Ideals of $\mathbb{Z}[i]$ are also lattices in the complex plane. Indeed, we know that every ideal is principal, and given any element $\alpha = a + bi$ in $\mathbb{Z}[i]$, the ideal $I = \mathbb{Z}[i] \cdot \alpha$ generated by α is the set of all possible linear combinations with integer coefficients of α and $i \cdot \alpha$. But, given that $i \cdot \alpha$ is obtained from α via a rotation by $\pi/2$ radians, we can immediately see that I is a square lattice. In the following figure, the dots represent the points of I for $\alpha = 1 + 2i$.



The minimal value of the norm on $\mathbb{Z}[i] \setminus \{0\}$ is 1, so the invertible elements of $\mathbb{Z}[i]$ are $\pm 1, \pm i$. We now proceed to study prime factorisations in $\mathbb{Z}[i]$; note that prime factors will be defined up to associates, that is, up to multiplication by -1, i and $-i$.

Since $\mathbb{Z} \subseteq \mathbb{Z}[i]$, we might want to ask whether a given prime number, that is, a number that is prime as an element of $\mathbb{Z}$, is still a prime in $\mathbb{Z}[i]$. The two notions are not equivalent: for instance, we have $2 = (1+i)(1-i) = -i(1+i)^2$ and also $5 = (2+i)(2-i)$, so 2 and 5 are not prime in $\mathbb{Z}[i]$, as the only invertible elements are $\pm 1, \pm i$. On the other hand, 3 remains a prime in $\mathbb{Z}[i]$.

Surprisingly, in order to check whether a prime number p remains a prime in $\mathbb{Z}[i]$, it is enough to consider the congruence class of p modulo 4. We expand on this in the following theorem, which also describes which primes can be expressed as the sum of two squares; for one way of proving it, see Preliminary Exercise 59.

Theorem 1.2.33

(i) A prime p of $\mathbb{Z}$ can be written as the sum of two squares if and only if $p \equiv 1 \pmod 4$ or $p = 2$.

(ii) For each prime p of $\mathbb{Z}$, the factorisation of p in $\mathbb{Z}[i]$ is as follows:

- *(a) if $p = 2$ then $p = (-i)(1+i)^2$ and $1+i$ is prime $\mathbb{Z}[i]$;*
- *(b) if $p \equiv 1 \pmod 4$ then $p = wz$, where w and z are not associates in $\mathbb{Z}[i]$;*
- *(c) if $p \equiv 3 \pmod 4$ then p is prime in $\mathbb{Z}[i]$.*

(iii) An element $w = a + bi \in \mathbb{Z}[i]$ is prime if and only if one of the following conditions holds: $a^2 + b^2 = p$ with p prime in $\mathbb{Z}$, in which case we necessarily have $p \equiv 1 \pmod 4$ or $p = 2$; or $a^2 + b^2 = p^2$ with $p \equiv 3 \pmod 4$, in which case w is an associate of the prime p of $\mathbb{Z}$.

It is worth noting that, thanks to the theorem above, an investigation of the arithmetic of the ring $\mathbb{Z}[i]$, which is a larger ring containing $\mathbb{Z}$, leads to the discovery of a property of $\mathbb{Z}$, namely, which primes can be written as the sum of two squares.

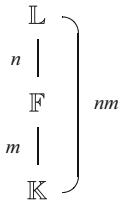
This is not an isolated instance, but an example of a very common and important paradigm in number theory.

1.3 Fields and Galois Theory

1.3.1 Basic Notions

We recall here some basic notions in field theory, many of which were already introduced in the first volume [Volume 1, Theory, Section 6].

The intersection of all subfields of a field $\mathbb{K}$ is a subfield, which is called the *prime subfield* of $\mathbb{K}$. The prime subfield of $\mathbb{K}$ is $\mathbb{Q}$ if the characteristic of $\mathbb{K}$ is 0 and $\mathbb{F}_p$, that is, the finite field with p elements, if the characteristic of $\mathbb{K}$ is p.



A field extension is a pair of fields $\mathbb{K}, \mathbb{F}$, one contained in the other: $\mathbb{K} \subseteq \mathbb{F}$. The degree $[\mathbb{F} : \mathbb{K}]$ of the extension is the dimension of $\mathbb{F}$ as a vector space over $\mathbb{K}$; the extension is said to be finite if its degree is finite and infinite otherwise. We write $\mathbb{F}/\mathbb{K}$ to indicate that $\mathbb{K} \subseteq \mathbb{F}$ is a field extension. Remark that the degree is multiplicative in towers of extensions : if the extensions $\mathbb{L}/\mathbb{F}$ and $\mathbb{F}/\mathbb{K}$ have degree n and m, respectively, then the extension $\mathbb{L}/\mathbb{K}$ has degree $[\mathbb{L} : \mathbb{K}] = [\mathbb{L} : \mathbb{F}][\mathbb{F} : \mathbb{K}] = n \cdot m$; in particular, if n or m is infinite, then the degree $[\mathbb{L} : \mathbb{K}]$ is infinite.

Given an extension $\mathbb{F}/\mathbb{K}$ and a subset X of $\mathbb{F}$, define the field $\mathbb{K}(X)$ generated by X over $\mathbb{K}$ as the intersection of all subfields of $\mathbb{F}$ that contain both $\mathbb{K}$ and X. Being an intersection of subfields, $\mathbb{K}(X)$ is a subfield itself: it is the smallest subfield of $\mathbb{F}$ that contains both $\mathbb{K}$ and X. It is easy to show that $\mathbb{K}(X)$ is the set of all ratios $f(x_1, x_2, \ldots, x_n)/g(x_1, x_2, \ldots, x_n)$ where n is a natural number, $x_1, x_2, \ldots, x_n$ are elements of X and $f(t_1, t_2, \ldots, t_n)$, $g(t_1, t_2, \ldots, t_n)$ are polynomials in the n variables $t_1, t_2, \ldots, t_n$ with coefficients in $\mathbb{K}$, with the property that $g(x_1, x_2, \ldots, x_n) \neq 0$.

Let $\mathbb{F}/\mathbb{K}$ be a field extension. Given $a \in \mathbb{F}$, we write $\mathbb{K}(a)$ for the field generated by $\{a\}$ over $\mathbb{K}$; we call extensions of this type *simple* and we say that a is a *primitive element* for the extension $\mathbb{K}(a)/\mathbb{K}$. The extension $\mathbb{F}/\mathbb{K}$ is said to be *finitely generated* if there exists a finite set X such that $\mathbb{F} = \mathbb{K}(X)$.

Given two subfields $\mathbb{K}, \mathbb{E}$ of $\mathbb{F}$, we define their composite field $\mathbb{K} \cdot \mathbb{E}$ as the field $\mathbb{K}(\mathbb{E}) = \mathbb{E}(\mathbb{K})$, that is, the smallest subfield of $\mathbb{F}$ that contains both $\mathbb{K}$ and $\mathbb{E}$.

An element a of $\mathbb{F}$ is said to be algebraic over $\mathbb{K}$ if there exists a nonzero polynomial with coefficients in $\mathbb{K}$ that vanishes at a. In other words, $a \in \mathbb{F}$ is algebraic over $\mathbb{K}$ if and only if the kernel of the map $\mathbb{K}[x] \ni f(x) \longmapsto f(a) \in \mathbb{F}$ is nontrivial. When a is algebraic, we call the monic generator of the kernel in question the minimal polynomial of a over $\mathbb{K}$ and denote it by $\mu_{a,\mathbb{K}}(x)$, or just by $\mu_a(x)$. If $a \in \mathbb{F}$ is algebraic over $\mathbb{K}$, then the extension $\mathbb{K}(a)$ has $1, a, a^2, \ldots, a^{n-1}$, with

$n = \deg(\mu_a(x))$, as a basis over $\mathbb{K}$; in particular, $[\mathbb{K}(a) : \mathbb{K}] = \deg(\mu_a(x))$. An element that is not algebraic is called transcendental.

An extension $\mathbb{F}/\mathbb{K}$ is algebraic if every element of $\mathbb{F}$ is algebraic over $\mathbb{K}$. Every finite extension is algebraic and every algebraic extension that is finitely generated is also finite. Moreover, if $\mathbb{L}/\mathbb{F}$ and $\mathbb{F}/\mathbb{K}$ are algebraic, then $\mathbb{L}/\mathbb{K}$ is also algebraic.

A field Ω is said to be algebraically closed if every non-constant polynomial in $\Omega[x]$ has a root in Ω. Clearly, Ω is algebraically closed if and only if every polynomial in $\Omega[x]$ splits as a product of linear factors. In other words, the only irreducible polynomials over an algebraically closed field are those of degree one.

An extension Ω of $\mathbb{K}$ is an algebraic closure of $\mathbb{K}$ if Ω is algebraically closed and $\Omega/\mathbb{K}$ is an algebraic extension. Recall that every field $\mathbb{K}$ admits an algebraic closure and that any two algebraic closures of $\mathbb{K}$ are isomorphic via an isomorphism that fixes every point of $\mathbb{K}$.

Given $f(x) \in \mathbb{K}[x]$ and an algebraic closure $\Omega/\mathbb{K}$, the splitting field of $f(x)$ in Ω is the field $\mathbb{K}(a_1, a_2, \ldots, a_n)$, where $a_1, a_2, \ldots, a_n$ are the roots of $f(x)$ in Ω. The splitting field is actually independent of the chosen algebraic closure: splitting fields in different algebraic closures of $\mathbb{K}$ are isomorphic to each other. The degree of the splitting field of $f(x)$ over $\mathbb{K}$ is at most $\deg(f)!$, and can in fact be shown to be a divisor $\deg(f)!$, as we shall soon see.

Analogously, we can define the splitting field of a family $\mathcal{F} \subseteq \mathbb{K}[x]$ of polynomials as the composite field in Ω of the splitting fields of the polynomials in $\mathcal{F}$. Equivalently, we can express such a splitting field as $\mathbb{K}(X)$, where X is the set of roots in Ω of the polynomials in $\mathcal{F}$.

Finally, recall that every field homomorphism is injective; we shall frequently make use of this observation, often implicitly.

1.3.2 Homomorphism Extensions

Let Ω be an algebraic closure of the field $\mathbb{K}$. An element a that is algebraic over $\mathbb{K}$ is said to be *separable* over $\mathbb{K}$ if its minimal polynomial over $\mathbb{K}$ has distinct roots in Ω. Clearly, this property does not depend on the chosen algebraic closure Ω. An algebraic element that is not separable is called *inseparable*. An algebraic extension $\mathbb{F}/\mathbb{K}$ is called *separable* if every element of $\mathbb{F}$ is separable over $\mathbb{K}$, and *inseparable* otherwise.

We recall the Derivative Criterion for Multiple Roots, which guarantees that a polynomial $f(x) \in \mathbb{K}[x]$ has a multiple root in some algebraic closure of $\mathbb{K}$ if and only if $f(x)$ and its derivative $f'(x)$ have a nontrivial greatest common divisor. In particular, an irreducible polynomial can have multiple roots only if its derivative is zero. Now, note that, over a field of characteristic zero, a polynomial that is not constant cannot have zero derivative. On the other hand, over a field $\mathbb{K}$ of characteristic $p > 0$, a polynomial $f(x)$ has zero derivative if and only if it is a polynomial in x^p; moreover, if $\mathbb{K}$ is finite then a polynomial in x^p cannot be irreducible. This proves

Proposition 1.3.1 *Let $\mathbb{F}/\mathbb{K}$ be an algebraic extension. If $\mathbb{K}$ has characteristic zero or is a finite field then $\mathbb{F}/\mathbb{K}$ is a separable extension.*

It is easy, however, to construct an inseparable extension. Indeed, as we showed in Sect. 1.2.9 using Eisenstein's Criterion, the polynomial $x^p - t$ is irreducible in the field of rational functions $\mathbb{F}_p(t)$. But, letting α be one of its roots in some algebraic extension, we have that $x^p - t = (x-\alpha)^p$ since we are in characteristic p. It is then clear that the extension $\mathbb{F}_p(\alpha)/\mathbb{F}_p(t)$ is inseparable.

From now on, we will mainly concern ourselves with extensions of fields that are either finite or of characteristic zero: we will thus always be dealing with separable extensions, unless otherwise stated.

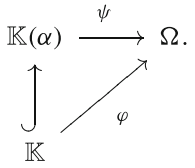
Let $\mathbb{K}$ and $\mathbb{F}$ be fields and let $\varphi : \mathbb{K} \longrightarrow \mathbb{F}$ be a field homomorphism. We can extend φ to a ring homomorphism from $\mathbb{K}[x]$ to $\mathbb{F}[x]$, which we will still denote by φ, by setting

$$\mathbb{K}[x] \ni \sum_{h=0}^{n} a_h x^h \overset{\varphi}{\longmapsto} \sum_{h=0}^{n} \varphi(a_h) x^h \in \mathbb{F}[x].$$

From now on, Ω will always denote some fixed algebraic closure of the field $\mathbb{K}$. The following, though completely trivial, is a key remark to make.

Remark 1.3.2 *Let $\alpha \in \mathbb{K}$ be a root of a polynomial $f(x) \in \mathbb{K}[x]$ and let $\varphi : \mathbb{K} \longrightarrow \Omega$ be a homomorphism. Then $\varphi(\alpha)$ is a root of the polynomial $\varphi(f(x))$.*

Now, let α be an element of Ω. We wish to determine how many ways there are to extend a homomorphism $\varphi : \mathbb{K} \longrightarrow \Omega$ to a homomorphism $\psi : \mathbb{K}(\alpha) \longrightarrow \Omega$, that is, to make a commutative diagramme



By the above remark applied to the homomorphism $\psi : \mathbb{K}(\alpha) \longrightarrow \Omega$ and to the minimal polynomial $\mu(x)$ of α over $\mathbb{K}$, we have that $\psi(\alpha)$ must be a root of the polynomial $\varphi(\mu(x))$. In fact, this condition is also sufficient: we have

Theorem 1.3.3 *Letting $\beta \in \Omega$ be a root of the polynomial $\varphi(\mu(x))$, there exists a homomorphism $\psi : \mathbb{K}(\alpha) \longrightarrow \Omega$ that extends the homomorphism $\varphi : \mathbb{K} \longrightarrow \Omega$ and is such that $\psi(\alpha) = \beta$. In particular, if α is separable over $\mathbb{K}$ then the number of distinct homomorphism that extend φ to $\mathbb{K}(\alpha)$ is* $\deg(\mu(x)) = [\mathbb{K}(\alpha) : \mathbb{K}]$.

From the fact that every finite extension can be obtained as a tower of simple extensions, we obtain

Theorem 1.3.4 *Given a finite separable extension $\mathbb{F}/\mathbb{K}$ and a homomorphism $\varphi : \mathbb{K} \longrightarrow \Omega$, there exist $[\mathbb{F} : \mathbb{K}]$ distinct extensions of φ to a homomorphism $\mathbb{F} \longrightarrow \Omega$.*

What we just obtained for extensions of homomorphisms applies in particular to extensions of the inclusion $\mathbb{K} \hookrightarrow \Omega$. A homomorphism ψ being an extension of the inclusion $\mathbb{K} \hookrightarrow \Omega$ is clearly equivalent to $\psi_{|\mathbb{K}} = \mathrm{Id}_{\mathbb{K}}$, so if $\mathbb{F}/\mathbb{K}$ is finite and $\psi : \mathbb{F} \longrightarrow \Omega$ is a homomorphism such that $\psi_{|\mathbb{K}} = \mathrm{Id}_{\mathbb{K}}$, then given $\alpha \in \mathbb{F}$ we have that $\psi(\alpha)$ is still a root of the minimal polynomial of α over $\mathbb{K}$.

In general, the roots in Ω of the minimal polynomial of α over $\mathbb{K}$ are called the *conjugates* of α over $\mathbb{K}$. Any extension of the inclusion $\mathbb{K} \hookrightarrow \Omega$ to a field containing $\mathbb{K}(\alpha)$ will thus send α to one of its conjugates.

An algebraic extension $\mathbb{F}/\mathbb{K}$ is called *normal* if every homomorphism $\varphi : \mathbb{F} \longrightarrow \Omega$ that extends the inclusion, that is, such that $\varphi_{|\mathbb{K}} = \mathrm{Id}_{\mathbb{K}}$, has the property that $\varphi(\mathbb{F}) = \mathbb{F}$. Given the discussion of the previous section, it is easy to show

Proposition 1.3.5 *An algebraic extension $\mathbb{F}/\mathbb{K}$ is normal if and only if every irreducible polynomial in $\mathbb{K}[x]$ that has a root in $\mathbb{F}$ splits as the product of linear factors in $\mathbb{F}$, that is, all of its roots are in $\mathbb{F}$.*

Using the above proposition, we obtain an important characterisation of normal extensions.

Theorem 1.3.6 *An algebraic extension $\mathbb{F}/\mathbb{K}$ is normal if and only if it is the splitting field of a family of polynomials in $\mathbb{K}[x]$. In particular, a finite extension $\mathbb{F}/\mathbb{K}$ is normal if and only if it is the splitting field of a polynomial in $\mathbb{K}[x]$.*

The extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is normal, as $\mathbb{Q}(\sqrt{2})$ is the splitting field over $\mathbb{Q}$ of the polynomial $x^2 - 2$ in $\mathbb{Q}[x]$, whereas the extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is *not* normal: the polynomial $x^3 - 2$ of $\mathbb{Q}[x]$, which we know to be irreducible by Eisenstein's Criterion, has only the root $\sqrt[3]{2}$ in $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$; its other two roots, namely, $\sqrt[3]{2}(-1 \pm \sqrt{-3})/2$, are not real numbers.

Remark that given a normal extension $\mathbb{F}/\mathbb{K}$ every homomorphism $\mathbb{F} \longrightarrow \Omega$ that extends $\mathrm{Id}_{\mathbb{K}}$ is an automorphism of $\mathbb{F}$. This remark will be crucial for the setup of Galois theory, which we will discuss in the next section.

Given an extension $\mathbb{L}/\mathbb{K}$, we can construct the smallest extension $\mathbb{F}$ of $\mathbb{L}$ that is normal over $\mathbb{K}$, called the *normal closure* of $\mathbb{L}$ over $\mathbb{K}$. Indeed, the family of all extensions of $\mathbb{L}$ that are normal over $\mathbb{K}$ is nonempty, as it contains for example the algebraic closure of $\mathbb{L}$, which is certainly normal over $\mathbb{K}$. So, since the intersection of normal extensions is normal, taking the intersection of all extensions of $\mathbb{L}$ that are normal over $\mathbb{K}$ yields the normal closure of $\mathbb{L}$. Note that if $\mathbb{L}$ is generated by elements $\alpha_1, \alpha_2, \ldots, \alpha_n$, then one can easily prove that the splitting field of the minimal polynomials of $\alpha_1, \alpha_2, \ldots, \alpha_n$ is the normal closure of $\mathbb{L}$.

If an extension $\mathbb{F}/\mathbb{K}$ is simple, that is, if there exists a primitive element $\alpha \in \mathbb{F}$, or in other words if $\mathbb{F} = \mathbb{K}(\alpha)$, then we can study the extensions of homomorphisms from $\mathbb{K}$ to $\mathbb{F}$ via the image of α. The following important result ensures that, in the case of separable extensions, we can always assume we are in this situation. Indeed, we have

Theorem 1.3.7 (Primitive Element Theorem) *Every finite separable extension is simple.*

We recall once more the fact that the condition of separability is automatically satisfied by fields of characteristic zero and finite fields; every finite extension of such fields is thus simple. For instance, the element $\sqrt{2}+\sqrt{3}$ is primitive for the extension $\mathbb{Q}(\sqrt{2},\sqrt{3})$. Working through this special case yields an argument that is actually not that far from a general proof of the theorem we just stated. For fields with an infinite number of elements, one can prove that $\mathbb{K}(\alpha,\beta) = \mathbb{K}(\alpha + a\beta)$ for some $a \in \mathbb{K}$ and conclude by induction on the number of generators of the extension.

1.3.3 Galois Correspondence

A normal separable extension of a field $\mathbb{K}$ is called a *Galois extension*. In particular, an extension of a field of characteristic zero or a finite field is Galois if and only if it is normal.

We continue to denote by Ω some fixed algebraic closure of the field $\mathbb{K}$. By what we discussed in the previous section, we know that, given a Galois extension $\mathbb{F}/\mathbb{K}$, every extension of $\mathrm{Id}_{\mathbb{K}}$ to a homomorphism $\mathbb{F} \longrightarrow \Omega$ is an automorphism of $\mathbb{F}$. As we already remarked, an automorphism of $\mathbb{F}$ that extends $\mathrm{Id}_{\mathbb{K}}$ is simply an automorphism of $\mathbb{F}$ that fixes every point of $\mathbb{K}$. Moreover, it is clear that the composition of any two such automorphism is again an extension of $\mathrm{Id}_{\mathbb{K}}$. We therefore define the *Galois group* of the extension $\mathbb{F}/\mathbb{K}$ as the set

$$\mathrm{Gal}(\mathbb{F}/\mathbb{K}) = \{\varphi : \mathbb{F} \longrightarrow \mathbb{F} \mid \varphi \text{ is an automorphism of } \mathbb{F}, \varphi_{|\mathbb{K}} = \mathrm{Id}_{\mathbb{K}}\}$$

endowed with the operation given by the composition of maps. Again thanks to the results about extensions of homomorphisms, we immediately obtain that for a finite Galois extension $\mathbb{F}/\mathbb{K}$ we have

$$|\mathrm{Gal}(\mathbb{F}/\mathbb{K})| = [\mathbb{F} : \mathbb{K}].$$

Our next objective is to state the fundamental theorem of Galois theory; before we can do that, however, we need to make some preliminary remarks. Let $\mathbb{F}/\mathbb{K}$ be some fixed Galois extension and let $\mathbb{L}$ be an intermediate extension, that is, suppose $\mathbb{K} \subseteq \mathbb{L} \subseteq \mathbb{F}$. Remark that $\mathbb{F}/\mathbb{L}$ is still Galois, as one can easily show using che characterisation of normal extensions as splitting fields. The extension $\mathbb{L}/\mathbb{K}$, however, is not Galois in general.

An automorphism of $\mathbb{F}$ that fixes every point of $\mathbb{L}$ also fixes every point of $\mathbb{K}$, so $\mathrm{Gal}(\mathbb{F}/\mathbb{L})$ is a subgroup of $\mathrm{Gal}(\mathbb{F}/\mathbb{K})$. Conversely, if H is a subgroup of $\mathrm{Gal}(\mathbb{F}/\mathbb{K})$ then the set of H-fixed points in $\mathbb{F}$, that is, the set

$$\mathbb{F}^H = \{\alpha \in \mathbb{F} \mid \varphi(\alpha) = \alpha \text{ for all } \varphi \in H\},$$

is an intermediate extension between $\mathbb{K}$ and $\mathbb{F}$.

We now introduce a series of lemmas that can be used to prove the upcoming theorem about Galois correspondence. The first ensures that, given a Galois extension $\mathbb{F}/\mathbb{K}$, the elements that are fixed by all of the Galois group are *only* those of $\mathbb{K}$; that is, we have

Lemma 1.3.8 *Letting $\mathbb{F}/\mathbb{K}$ be a Galois extension with Galois group $G = \mathrm{Gal}(\mathbb{F}/\mathbb{K})$, we have $\mathbb{F}^G = \mathbb{K}$.*

We shall see that the lemma we just stated is in fact a special case of Galois correspondence. The next lemma allows us to find a polynomial that vanishes at a given element by using the orbit of a subgroup of the Galois group. Proposition 1.3.13 will later improve on this result in the case of Galois extensions and provide a method for computing the minimal polynomial of an element.

Lemma 1.3.9 *Let $\mathbb{F}/\mathbb{K}$ be an extension and let H be a group of automorphisms of $\mathbb{F}$ that fix every point of $\mathbb{K}$. Given $\alpha \in \mathbb{F}$, the polynomial*

$$\prod_{\varphi \in H} (x - \varphi(\alpha))$$

has coefficients in $\mathbb{F}^H$ and vanishes at α.

We can now state the main result in this section, which is due to the French mathematician Évariste Galois, though the general approach that we have taken here using field extensions is due to Emil Artin.

Theorem 1.3.10 (Fundamental Theorem of Galois Theory) *Let $\mathbb{F}/\mathbb{K}$ be a finite Galois extension. There exists a one-to-one correspondence between intermediate extensions of $\mathbb{F}/\mathbb{K}$ and subgroups of $\mathrm{Gal}(\mathbb{F}/\mathbb{K})$. The two maps*

$$\begin{aligned} \mathbb{L} &\longmapsto \mathrm{Gal}(\mathbb{F}/\mathbb{L}) \\ \mathbb{F}^H &\longleftarrow\!\shortmid H \end{aligned}$$

are inverses of each other and realise this correspondence, which is called the Galois correspondence. *Moreover, if $\mathbb{L} = \mathbb{F}^H$ then we have the following: $\mathbb{L}/\mathbb{K}$ is a normal (equivalently, Galois) extension if and only if $H = \mathrm{Gal}(\mathbb{F}/\mathbb{L})$ is a normal subgroup of $\mathrm{Gal}(\mathbb{F}/\mathbb{K})$; in that case, $\mathrm{Gal}(\mathbb{L}/\mathbb{K})$ is isomorphic to $\mathrm{Gal}(\mathbb{F}/\mathbb{K})/\mathrm{Gal}(\mathbb{F}/\mathbb{L})$.*

The next proposition lists some other important properties of the Galois correspondence.

Proposition 1.3.11 *Let $\mathbb{F}/\mathbb{K}$ be a finite Galois extension and let H, H' be subgroups of the Galois group $\mathrm{Gal}(\mathbb{F}/\mathbb{K})$. We have the following:*

(i) the Galois correspondence inverts inclusions

$$H \subseteq H' \text{ if and only if } \mathbb{F}^H \supseteq \mathbb{F}^{H'},$$

(ii) the intersection of subgroups corresponds to the composite field of subfields

$$\mathbb{F}^{H\cap H'} = \mathbb{F}^H \cdot \mathbb{F}^{H'},$$

(iii) the intersection of subfields corresponds to the subgroup generated by subgroups

$$\mathbb{F}^{\langle H,H'\rangle} = \mathbb{F}^H \cap \mathbb{F}^{H'}.$$

The Galois group of the composite of two extensions $\mathbb{F}_1/\mathbb{K}$, $\mathbb{F}_2/\mathbb{K}$ that intersect in the field $\mathbb{K}$ can be expressed in terms of the Galois groups of the two extensions. In general, we have

Proposition 1.3.12 *Let $\mathbb{F}_1/\mathbb{K}$ and $\mathbb{F}_2/\mathbb{K}$ be finite Galois extensions; $\mathbb{F}_1 \cdot \mathbb{F}_2$ is itself a Galois extension of $\mathbb{K}$. The map*

$$\mathrm{Gal}(\mathbb{F}_1 \cdot \mathbb{F}_2/\mathbb{K}) \ni \sigma \longmapsto (\sigma_{|\mathbb{F}_1}, \sigma_{|\mathbb{F}_2}) \in \mathrm{Gal}(\mathbb{F}_1/\mathbb{K}) \times \mathrm{Gal}(\mathbb{F}_2/\mathbb{K})$$

is an injective homomorphism. It is surjective if and only if $\mathbb{F}_1 \cap \mathbb{F}_2 = \mathbb{K}$, in which case $[\mathbb{F}_1 \cdot \mathbb{F}_2 : \mathbb{Q}] = [\mathbb{F}_1 : \mathbb{Q}] \cdot [\mathbb{F}_2 : \mathbb{Q}]$.

We can now improve on Lemma 1.3.9 by introducing a method that allows us to compute the minimal polynomial of any given element in a Galois extension.

Proposition 1.3.13 *Let $\mathbb{F}/\mathbb{K}$ be a Galois extension and let α be an element of $\mathbb{F}$. The orbit $\mathcal{O} = \mathrm{Gal}(\mathbb{F}/\mathbb{K}) \cdot \alpha$ of α with respect to the Galois group of the extension is the set of conjugates of α, and*

$$\prod_{\beta\in\mathcal{O}} (x - \beta)$$

is the minimal polynomial of α over $\mathbb{F}$.

We now wish to apply Galois theory to the finite extensions of the field $\mathbb{F}_p$, where p is a prime. Every finite extension of $\mathbb{F}_p$ is of the form $\mathbb{F}_{p^n}$ for some positive integer n. The extension $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois, because it is separable and it is the splitting field of the polynomial $x^{p^n} - x$ over $\mathbb{F}_p$. The Frobenius endomorphism

$$\mathbb{F}_{p^n} \ni \alpha \longmapsto \alpha^p \in \mathbb{F}_{p^n}$$

is an automorphism of $\mathbb{F}_{p^n}$, and its fixed points are exactly the points of $\mathbb{F}_p$. We thus have

Remark 1.3.14 *The Galois group $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ is cyclic of order n and is generated by the Frobenius automorphism.*

Given a polynomial $f(x) \in \mathbb{Q}[x]$, let $\mathbb{F}$ be the splitting field of $f(x)$ over $\mathbb{Q}$. The extension $\mathbb{F}/\mathbb{Q}$ is clearly Galois, and we shall call *Galois group* of $f(x)$ the Galois group of the extension $\mathbb{F}/\mathbb{Q}$.

As a first simple example, consider a quadratic extension $\mathbb{F}/\mathbb{Q}$. There clearly exists $a \in \mathbb{Q} \setminus \mathbb{Q}^2$ such that $\mathbb{F} = \mathbb{Q}(\sqrt{a})$, so $\mathbb{F}/\mathbb{Q}$ is a normal extension, as $\mathbb{F}$ is the splitting field of $x^2 - a \in \mathbb{Q}[x]$. The group $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ has two elements: the identity of $\mathbb{F}$ and the automorphism $\mathbb{F} \ni u + v\sqrt{a} \longmapsto u - v\sqrt{a} \in \mathbb{F}$; in particular, the Galois group of $\mathbb{F}/\mathbb{Q}$ is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

As a second example, consider the polynomial $x^3 - 2 \in \mathbb{Q}[x]$, which is irreducible. Its splitting field is $\mathbb{F} = \mathbb{Q}(\sqrt[3]{2}, \zeta)$, where $\zeta = e^{2\pi i/3} = (-1 + \sqrt{-3})/2$ is a primitive third root of unity. The degree of $\mathbb{F}/\mathbb{Q}$ is 6, so the Galois group of $x^3 - 2$ over $\mathbb{Q}$, that is, $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$, has 6 elements. This, together with the fact that the group must permute the three conjugates $\sqrt[3]{2}$, $\sqrt[3]{2}\zeta$, $\sqrt[3]{2}\zeta^2$ of $\sqrt[3]{2}$, implies that it is isomorphic to S_3. It follows that for each permutation σ of these three elements there exists an automorphism of $\mathbb{F}$ that extends the identity of $\mathbb{Q}$ and induces σ on the set of the three roots.

In fact, we can generalise the argument above to any polynomial $f(x) \in \mathbb{Q}[x]$ of degree n and its splitting field $\mathbb{F}$ over $\mathbb{Q}$: by restricting an element of the Galois group $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ of $f(x)$, one induces a permutation of the n roots of $f(x)$ in $\mathbb{F}$. We thus have a homomorphism

$$\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \longrightarrow \mathsf{S}_n,$$

which is injective because $\mathbb{F}$ is generated by the roots of $f(x)$. If, additionally, $f(x)$ is irreducible, then by the results of the previous section we have that, given any two roots α, β of $f(x)$, there exists an extension of the identity of $\mathbb{Q}$ to an automorphism of $\mathbb{F}$ that sends α to β. This proves that the image of the homomorphism $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \longrightarrow \mathsf{S}_n$ is a group that acts transitively on $\{1, 2, \ldots, n\}$, that is, a *transitive* subgroup of S_n. We summarise this discussion in the following

Remark 1.3.15 *The Galois group of a polynomial of degree n in* $\mathbb{Q}[x]$ *is isomorphic to a subgroup of* S_n. *If the polynomial is irreducible, then the subgroup is transitive.*

We should stress the fact that the homomorphism $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \hookrightarrow \mathsf{S}_n$ actually depends on the way we have chosen to number the roots of the polynomial $f(x)$. Renumbering the roots corresponds to conjugating the image of this homomorphism in S_n.

Since the order of the Galois group of a polynomial is the degree of its splitting field, Lagrange's Theorem immediately yields

Corollary 1.3.16 *The degree of the splitting field of a polynomial* $f(x)$ *of* $\mathbb{Q}[x]$ *divides* $\deg(f)!$.

Consider now a polynomial $f(x)$ of degree three, irreducible and with rational coefficients. We wish to determine the Galois group of $f(x)$ as a subgroup G of S_3. Only $\mathsf{A}_3 \simeq \mathbb{Z}/3\mathbb{Z}$ and S_3 are transitive subgroups of S_3, so $G = \mathsf{A}_3$ or $G = \mathsf{S}_3$.

Let $\mathbb{F}$ be the splitting field of $f(x)$ over $\mathbb{Q}$ and let $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{F}$ be the three roots of $f(x)$. The extension $\mathbb{F}/\mathbb{Q}$ is Galois and thus we have the following possibilities for its degree: $[\mathbb{F} : \mathbb{Q}] = 3$ if $G = \mathsf{A}_3$, or $[\mathbb{F} : \mathbb{Q}] = 6$ if $G = \mathsf{S}_3$; our objective is to distinguish between these two alternatives. Setting $\delta = (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_2 - \alpha_3)$, define the *discriminant* of $f(x)$ as $\Delta = \delta^2$. The discriminant is invariant under permutations of the roots, and in particular under the action of G; it must therefore be a rational number. We have $\sigma(\delta) = (-1)^\sigma \delta$ and so δ is invariant under A_3 but not under S_3. It clearly follows that $G = \mathsf{A}_3$ if and only if $\Delta \in \mathbb{Q}^2$.

To compute Δ as a function of the coefficients of $f(x)$, notice that we can always rewrite the polynomial in the form $f(x) = x^3 + ax + b$: indeed, we can use the substitution $x' = x + \lambda$ for an appropriate $\lambda \in \mathbb{Q}$ in order to ensure that the term of degree 2 vanishes. Developing the product $\Delta = (\alpha_1 - \alpha_2)^2(\alpha_1 - \alpha_3)^2(\alpha_2 - \alpha_3)^2$ and using $s_1 = \alpha_1 + \alpha_2 + \alpha_3 = 0$, $s_2 = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3 = a$ and $s_3 = \alpha_1\alpha_2\alpha_3 = -b$ yields that $\Delta = -4a^3 - 27b^2$. Putting together our observations so far, we have

Proposition 1.3.17 *The discriminant of $f(x) = x^3 + ax + b$ is*

$$\Delta = -4a^3 - 27b^2.$$

If $f(x)$ is irreducible in $\mathbb{Q}[x]$ then the Galois group of $f(x)$ is isomorphic to $\mathbb{Z}/3\mathbb{Z}$ if $\Delta \in \mathbb{Q}^2$, and to S_3 if $\Delta \notin \mathbb{Q}^2$.

Note that this formula for Δ can be obtained in a more straightforward way by using the Fundamental Theorem of Symmetric Polynomials, a result which however is not central to our objectives and thus will not be covered. This theorem guarantees that Δ, being a symmetric polynomial in $\alpha_1, \alpha_2, \alpha_3$, can be expressed as a polynomial in the elementary symmetric polynomials s_1, s_2, s_3. Knowing this, it is enough to observe that Δ is homogeneous of degree 6 in $\alpha_1, \alpha_2, \alpha_3$ to obtain that it is of the form $us_2^3 + vs_3^2$ for some coefficients $u,\ v \in \mathbb{Q}$. Finally, evaluating the discriminant on, for example, the polynomials $x^3 - x$ and $x^3 - 3x + 2$, yields the values of the coefficients $u = -4$ and $v = -27$.

The irreducible polynomial $x^3 - x + 1$ has discriminant $\Delta = -4 \cdot (-1)^3 - 27 \cdot 1^3 = -23 \notin \mathbb{Q}^2$ and so its Galois group is isomorphic to S_3. The irreducible polynomial $x^3 - 7x + 7$ has discriminant $\Delta = -4 \cdot (-7)^3 - 27 \cdot 7^2 = 7^2 \in \mathbb{Q}^2$ and so its Galois group is isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

In Preliminary Exercise 1.4, we study the Galois group of a biquadratic polynomial in $\mathbb{Q}[x]$.

1.3.4 Cyclotomic Extensions

In this section, we introduce cyclotomic extensions of $\mathbb{Q}$, which are of paramount importance, for example in number theory.

Recall from the first volume that, given a positive integer n, an nth root of unity in $\mathbb{C}$ is a root of the polynomial $x^n - 1$, that is, a complex number ζ such that $\zeta^n = 1$.

The nth roots of unity are thus the elements $e^{2\pi hi/n}$, with $h = 0, 1, \ldots, n-1$. If the order of ζ is equal to n then we say that ζ is a primitive nth root of unity. Thus, an nth root of unity ζ is a primitive mth root of unity, where m is the order of ζ in $\mathbb{C}^*$; clearly, m divides n. The nth roots of unity form a cyclic group of order n in $\mathbb{C}^*$, and the primitive nth roots of unity are the possible generators for this group; they are thus given by $e^{2\pi hi/n}$, where $1 \leq h \leq n-1$ and $(h, n) = 1$, and there are $\phi(n)$ of them.

If ζ is a primitive nth root of unity, the extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ is called a *cyclotomic extension* of *order* n, or the nth cyclotomic field. Note that $\mathbb{Q}(\zeta)$ is clearly the splitting field of the polynomial $x^n - 1$ over $\mathbb{Q}$, so the extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ is Galois.

We call the polynomial

$$\Phi_n(x) = \prod_{\zeta}(x - \zeta),$$

where ζ ranges among all nth primitive roots of unity, the nth *cyclotomic polynomial*. Remark that $\Phi_n(x)$ is, by definition, a polynomial with coefficients in $\mathbb{C}$. In fact, since cyclotomic extensions are Galois and since the Galois group permutes the primitive nth roots of unity because it preserves orders, the coefficients of $\Phi_n(x)$ are actually in $\mathbb{Q}$. Using the standard argument of reducing modulo a prime, one can show

Theorem 1.3.18 (Dedekind) *For all positive integers n the cyclotomic polynomial* $\Phi_n(x)$ *is irreducible in* $\mathbb{Q}[x]$.

The cyclotomic polynomial $\Phi_n(x)$ has degree $\phi(n)$, so that given a primitive nth root of unity ζ, we have $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \phi(n)$, and the Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ has $\phi(n)$ elements.

We now wish to give an explicit description of the elements φ of the Galois group of the nth cyclotomic extension. As we remarked before, φ preserves the order of ζ in $\mathbb{C}^*$, so there is an integer a such that $\varphi(\zeta) = \zeta^a$, coprime to n and well-defined modulo n. Moreover, for each integer k, we have $\varphi(\zeta^k) = \zeta^{ak}$, and so a does not depend on the generator ζ of the extension but only on φ. We have thus shown the first part of the following

Theorem 1.3.19 (Cyclotomic Extensions of the Rationals) *Let* $\mathbb{Q}(\zeta)/\mathbb{Q}$ *be a cyclotomic extension of order n. For all* $\varphi \in \mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ *there exists* $a_\varphi \in \mathbb{Z}$ *such that* $\varphi(\omega) = \omega^{a_\varphi}$ *for every nth root of unity* ω. *Moreover, the map*

$$\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \ni \varphi \longmapsto a_\varphi \in (\mathbb{Z}/n\mathbb{Z})^*$$

is a group isomorphism. In particular, $[\mathbb{Q}(\zeta) : \mathbb{Q}] = |\mathrm{Gal}(\mathbb{Q}(\zeta) : \mathbb{Q})| = \phi(n)$.

Now let $\mathbb{Q}(\zeta)$ be a cyclotomic extension of order n and let $\mathbb{Q}(\omega)$ be a cyclotomic extension of order m. It is clear that $\zeta \cdot \omega$ is a primitive root of unity whose order is the least common multiple $[n, m]$. This proves

Remark 1.3.20 *The composite field of two cyclotomic extensions is a cyclotomic extension whose order is the least common multiple of the original orders.*

It is also clear that the intersection $\mathbb{Q}(\zeta) \cap \mathbb{Q}(\omega)$ is contained in the cyclotomic extension of order the greatest common divisor (m, n). In fact, by comparing degrees one can show

Remark 1.3.21 *The intersection of two cyclotomic extensions is a cyclotomic extension whose order is the greatest common divisor of the original orders.*

We now turn our attention to the problem of describing the real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R}$ of the nth cyclotomic field. If $n \leq 2$, then $\mathbb{Q}(\zeta) = \mathbb{Q}$, so we may assume that $n > 2$, in which case complex conjugation is an element of order 2 in $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$. We have $\zeta + \zeta^{-1} = 2\cos(2\pi/n) \in \mathbb{R}$, so $\mathbb{Q}(\zeta + \zeta^{-1}) \subseteq \mathbb{Q}(\zeta) \cap \mathbb{R}$. On the other hand, $\mathbb{Q}(\zeta)$ has degree 2 over $\mathbb{Q}(\zeta + \zeta^{-1})$ because ζ is a root of the polynomial $x^2 - (\zeta + \zeta^{-1})x + 1 \in \mathbb{Q}(\zeta + \zeta^{-1})[x]$ and because $\mathbb{Q}(\zeta)$ is not a real extension for $n > 2$. Via the Galois correspondence, we find that $\mathbb{Q}(\zeta + \zeta^{-1})$ corresponds to the subgroup of $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ generated by complex conjugation, and in particular we have

Remark 1.3.22 *If ζ is a primitive nth root of unity, then the real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R}$ is given by $\mathbb{Q}(\zeta + \zeta^{-1}) = \mathbb{Q}(\cos(2\pi/n))$. It is Galois and, if $n > 2$, its degree is $\phi(n)/2$ over $\mathbb{Q}$.*

For the sake of completeness, we conclude this section with a result we already stated in the first volume [Volume 1, Theorem 6.22, page 51] about cyclotomic extensions of a finite field $\mathbb{F}_p$.

Theorem 1.3.23 (Cyclotomic Extensions in Positive Characteristic) *Let n be a positive integer that is not divisible by the prime p. The splitting field of $x^n - 1$ over $\mathbb{F}_p$ is $\mathbb{F}_{p^r}$, where r is the order of p in the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$.*

1.3.5 Straightedge and Compass Constructions

In this section we present, for the sake of completeness, some classical result about the constructibility with straightedge and compass of certain geometric shapes and numbers. Though these will not be used in any of the exercises, they are usually a feature of algebra courses at a Bachelor's level.

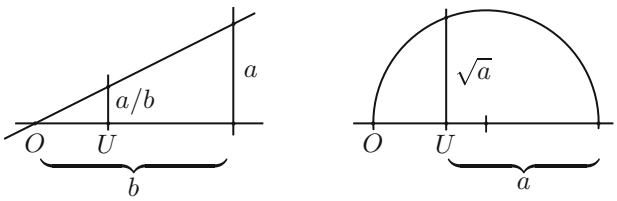
Having fixed two distinct points O and U in the plane, we define *constructible* points, lines and circles according to the following recursive rules:

(i) the points O and U are constructible,
(ii) a line through two constructible points is constructible,
(iii) a circle centred at a constructible point through a constructible point is constructible,
(iv) a point that is the intersection of two constructible lines, or of a constructible line and a constructible circle, or of two constructible circles, is constructible.

Some elementary straightedge and compass constructions are well known. It is possible to construct, for example, the perpendicular line to a constructible line through a constructible point, the axis of a segment whose endpoints are constructible, the midpoint between two constructible points, the parallel line to a constructible line through a constructible point, and a segment along a constructible line, whose length is given by the distance between two constructible points, with one prescribed constructible endpoint.

In particular, one can construct a Cartesian coordinate system with O as the origin and U as the point of coordinates $(1, 0)$. Additionally, we say that a real number a is *constructible* if there exist two constructible points at distance $|a|$. It is straightforward to prove that a point P of coordinates (x, y) is constructible if and only if x and y are constructible numbers. If a and b are constructible then $a + b$, $a - b$ are also constructible.

Thanks to the construction represented in the figure below, we find that, given constructible numbers a and $b \neq 0$, the ratio a/b is also constructible. It follows that $1/b$ and $ab = a/(1/b)$ are constructible.



Constructions for the ratio a/b and the square root $\sqrt{a}$.

We can make the point (a, b) of the plane correspond to the complex number $\alpha = a + bi$; we shall say that α is constructible if and only if both a and b are; the complex number i, for instance, is constructible. From what we showed so far we immediately obtain

Theorem 1.3.24 *The set of constructible complex numbers is a field. Moreover, if α is a constructible complex number and $\beta^2 = \alpha$ then β is also constructible.*

In fact, the theorem above does give an accurate description of which complex numbers are constructible: all constructible numbers are obtained by taking sums, differences, products, ratios and square roots. Indeed, we have

Theorem 1.3.25 *A complex number α is constructible if and only if there exists a tower of extensions*

$$\mathbb{Q} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \mathbb{K}_2 \subseteq \cdots \subseteq \mathbb{K}_n$$

such that $\alpha \in \mathbb{K}_n$ and $[\mathbb{K}_h : \mathbb{K}_{h-1}] = 2$ for $h = 1, 2, \ldots, n$.

With the notation introduced in the theorem, we have that $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides $[\mathbb{K}_n : \mathbb{Q}] = 2^n$, and therefore

Corollary 1.3.26 *If a complex number α is constructible then α is algebraic of degree some power of* 2 *over* $\mathbb{Q}$.

The converse, however, does not hold: there are algebraic numbers of degree 4 over $\mathbb{Q}$ that are not constructible. The preceding corollary is already enough to prove that the classical constructibility problems are all impossible.

Corollary 1.3.27 *Using only straightedge and compass, one cannot duplicate a cube of general side length, trisect a general angle, or construct a square whose area is equal to that of a given circle.*

Indeed, in order to duplicate a cube of side ℓ, one would need to construct a cube of side $\ell \cdot \sqrt[3]{2}$, which would only be possible if the number $\sqrt[3]{2}$ were constructible. As $\sqrt[3]{2}$ is a root of the cubic irreducible polynomial with rational coefficients $x^3 - 2$, this is clearly not the case.

There is no construction to trisect a general angle because there is no such construction for the angle $\pi/3$. Indeed, one can easily check that $\cos(\pi/9)$ is a root of an irreducible polynomial of degree 3.

The problem of constructing a square whose area is equal to that of a given circle, commonly referred to as the *problem of squaring the circle*, was proven to be impossible in 1882, when Ferdinand von Lindemann showed that the number π is transcendental, that is, not only that it does not satisfy any polynomial equation of degree a power of 2 with rational coefficients, but that it does not satisfy any rational polynomial equation of any degree.

Using Galois theory, we can prove a partial converse to the corollary stated above. The proof hinges on the fact that a group whose order is a power of 2, that is, a 2-group, has a chain of subgroups, each of index 2 in the next.

Theorem 1.3.28 *A complex number contained in a Galois extension whose degree is a power of* 2 *is constructible.*

We can apply the results we just stated to the constructibility of regular polygons inscribed in the unit circle: a regular n-gon inscribed in the unit circle is constructible if and only if a primitive nth root of unity is constructible.

A prime number p such that $p - 1$ is a power of 2 is called a *Fermat prime*.

Theorem 1.3.29 *A regular n-gon is constructible if and only if $n = 2^r p_1 p_2 \cdots p_k$ for some natural number r and some distinct Fermat primes $p_1, p_2, \ldots, p_k$.*

We briefly summarise the idea of the proof. Adding to the rationals the nth roots of unity yields a Galois extension whose degree over $\mathbb{Q}$ is $\phi(n)$; if n is as required by the theorem, the degree of the extension is a power of 2. In particular, the Galois group of the extension is a 2-group, so it has a chain of subgroups, each of index 2 in the next, which corresponds to a tower of extensions of degree 2. In particular, the nth roots of unity are constructible by Theorem 1.3.28.

For instance, the regular heptagon and nonagon are not constructible with straightedge and compass, whereas the equilateral triangle and regular pentagon are.

We conclude this section by remarking that a Fermat prime is necessarily of the form $p_m = 2^{2^m} + 1$ for some integer m. While Fermat thought that the numbers $p_0, p_1, \ldots$ were all prime, Euler found that $2^{2^5} + 1 = 641 \cdot 6700417$ and thus that p_5 is *not* prime. For $0 \leq m \leq 4$, we do obtain the prime numbers $3, 5, 17, 257, 65537$, but the existence of Fermat primes other than those is an open problem. As for an effective construction, two thousand years after the classical constructions devised by the Greeks for the equilateral triangle and the regular pentagon, it was Gauss that, in 1796, at the young age of 19, provided a construction for the regular polygon with 17 sides. The same method can be generalised to regular polygons with 257 and 65537 sides.

1.4 Preliminary Exercises

This section contains a selection of preliminary exercises. These should be read before attempting the questions from past exam papers: their statements and the techniques employed in their solutions will be instrumental to solving problems in subsequent chapters. Some exercises will reference previous ones; in particular, this will be the case for small collections of exercises that are meant to be solved sequentially and aim to show an important result by breaking down its proof into several steps.

Exercise 1 Show that any non-Abelian group with 6 elements is isomorphic to S_3.

Solution Let G be a non-Abelian group of order 6. By Cauchy's Theorem, there exist $\sigma \in G$ of order 2 and $\tau \in G$ of order 3. If these two elements commuted then $\sigma\tau$ would have order 6 and the group G would therefore be cyclic, which would contradict our hypothesis of it not being Abelian. It is easy to check that the elements

$$e, \tau, \tau^2$$
$$\sigma, \sigma\tau, \sigma\tau^2$$

are all distinct. Moreover, we find that $\tau\sigma$ must coincide with the element $\sigma\tau^2$. We have thus shown that G is isomorphic to S_3.

Exercise 2 Classify all groups of order 8 up to isomorphism.

Solution By the Structure Theorem for Finite Abelian Groups, if G is Abelian then it is isomorphic to one of the following: $\mathbb{Z}/8\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, or $(\mathbb{Z}/2\mathbb{Z})^3$. These three groups are pairwise not isomorphic: the first one is the only one with elements of order 8, the second one has elements of order 4, and the third one only contains the neutral element and elements of order 2.

Now consider the case of G not being Abelian. The group G cannot contain elements of order 8, or it would be cyclic; it also cannot only contain elements of order 1 and 2, or it would be Abelian.

It follows that G contains an element h of order 4; denoting by H the subgroup generated by h, remark that, since it has index 2, the subgroup H is normal in G.

We now consider the following cases.

(1) The set $G \setminus H$ contains an element g of order 2. Letting K be the subgroup generated by g, we obtain that $G = H \cdot K$; the action of g by conjugation on the normal subgroup H cannot be trivial, because G is not Abelian. The only possibility is therefore for the action to send h to h^{-1}; but then we have $ghg^{-1} = h^{-1}$, so G is isomorphic to D_4, the dihedral group with 8 elements.

(2) The set $G \setminus H$ only contains elements of order 4. In this case, G contains only one element of order 2, namely, h^2. For ease of notation, set $-1 = h^2$. Since it is the only element of order 2, -1 is invariant under conjugation, so it belongs to the centre G. Again for ease of notation, set $i = h$ and $-i = (-1) \cdot i = i \cdot (-1)$. We have $i^2 = h^2 = -1$ and $i^3 = i^2 \cdot i = (-1) \cdot i = -i$.

Similarly, if $j \in G \setminus H$ then j has order 4 and so j^2 has order 2, that is, $j^2 = -1$. Setting $-j = (-1) \cdot j = j \cdot (-1)$, we have $j^3 = j^2 \cdot j = (-1) \cdot j = -j$. Finally, let $k \notin H$, $k \neq \pm j$; we have $k^2 = -1$ and, setting $-k = (-1) \cdot k = k \cdot (-1)$, we obtain that $k^3 = -k$. We have obtained, in particular, the relations

$$i^2 = j^2 = k^2 = -1.$$

Now consider the product ij. By the cancellation rule, we must have $ij \neq \pm 1, \pm i, \pm j$. By exchanging k with $-k$ if necessary, we may suppose that $ij = k$. We now have that $jk = i$, as we can see by cancelling the factor i from the relation $ijk = k^2 = -1$. In an entirely analogous way, we obtain the complete set of relations

$$ij = k,\ jk = i,\ ki = j,\ ji = -k,\ kj = -i,\ ik = -j.$$

The group G is thus isomorphic to the group Q_8 of unit quaternions, which completes our list of all groups of order 8 up to isomorphism.

Exercise 3 Show that, given a finite group G and a normal subgroup of G whose order is the smallest prime that divides the order of G, the subgroup is contained in the centre of G.

Solution Let p be the smallest prime that divides $|G|$ and let H be a normal subgroup of G having order p.

Let g be an element of G and let ψ_g be the automorphism of G given by conjugation by g. We know that H is stable under ψ_g because it is normal; let $\varphi_g = \psi_{g|H} : H \longrightarrow H$ be the automorphism obtained by restricting ψ_g to H. In

order to prove that H is contained in the centre, it is enough to show that $\varphi_g = \mathrm{Id}_H$, or equivalently that, letting t be the order of φ_g in the group $\mathrm{Aut}(H)$, we have $t = 1$.

If r is the order of $g \in G$ then $\varphi_g^r = \psi_{g|H}^r = (\psi_g^r)_{|H} = (\psi_{g^r})_{|H} = (\psi_e)_{|H} = \mathrm{Id}_H$. So t divides r and thus t also divides $n = |G|$.

On the other hand, H is a group whose cardinality is a prime p, so H must be isomorphic to $\mathbb{Z}/p\mathbb{Z}$. It follows that $\mathrm{Aut}(H)$ is isomorphic to $(\mathbb{Z}/p\mathbb{Z})^*$, which is a group with $p-1$ elements. Therefore, t divides $p-1$, and in particular $t < p$.

Now assume that t is not 1 and let q be a prime dividing t. By what we have shown, q divides n and $q < p$; but this contradicts the hypothesis of minimality for p.

Exercise 4 Letting G be a finite group, show that any subgroup whose index is the smallest prime that divides the order of G is normal.

Solution Let p be the smallest prime that divides $|G|$, let H be a subgroup of index p and let X be the collection of the p left cosets of H in G.

One can immediately check that

$$G \ni g \overset{\varphi}{\longmapsto} \big(xH \longmapsto (gx)H\big) \in \mathsf{S}(X)$$

is an action of G on X.

Since the image of φ is a subgroup of $\mathsf{S}(X) \simeq \mathsf{S}_p$, the kernel K of φ is a normal subgroup of G whose index divides $p!$. Moreover, we have $K \subseteq H$: if $g \in K$ then in particular $gH = H$, so $g \in H$. It follows that p divides $[G : K]$.

However, we have $(p!, |G|) = p$ by assumption, so $[G : K] = p$ and thus $H = K$ is a normal subgroup of G.

Exercise 5 Given a prime p, consider the subgroup

$$G = \{z \in \mathbb{C}^* \mid \text{ there exists a natural number } n \text{ such that } z^{p^n} = 1\}$$

of $\mathbb{C}^*$ and let H be a proper subgroup of G. Show that H is cyclic and that G/H is isomorphic to G.

Solution Set

$$X(H) = \{n \in \mathbb{N} \mid \exists h \in H \text{ such that } \mathrm{ord}(h) = p^n\}.$$

There are two cases to consider: either there exists a maximal element N of $X(H)$ or the set $X(H)$ is unbounded. In the first case, H is contained in the subgroup of the p^Nth roots of unity, which is cyclic, and therefore H is cyclic. In fact, we can say something stronger, which we will use later: H is equal to the subgroup of the p^Nth roots of unity, because it contains a generator for it.

We now intend to show that the second case, that of $X(H)$ being unbounded, is in fact not possible, as it would yield $H = G$, which is excluded by the problem statement. Indeed, consider $z \in G$ and let p^m be the order of z. If $X(H)$ is unbounded, then there exists $h \in H$ such that $\mathrm{ord}(h) = p^n > p^m$; but then H contains the cyclic subgroup $\{z \in \mathbb{C} \mid z^{p^n} = 1\}$, and in particular $z \in H$, as claimed. This concludes the proof that $X(H)$ is bounded.

We now show that, under our assumptions, $G/H \simeq G$. We have seen that there exists a natural number N such that H coincides with the subgroup of the p^Nth roots of unity.

The map $f : G \longrightarrow G$ given by $z \longmapsto z^{p^N}$ is clearly a homomorphism whose kernel is H, so it induces an injective homomorphism $\varphi : G/H \longrightarrow G$. Now, let z be an element of G and let p^n be its order; we have $z = e^{2\pi i r/p^n}$ for some integer r such that $(r, p) = 1$. It is immediately apparent that $z = \varphi(e^{2\pi i r/p^{n+N}})$. It follows that φ is also surjective and therefore an isomorphism.

Exercise 6 Show that the group D_n is isomorphic to the semidirect product $\mathbb{Z}/n\mathbb{Z} \rtimes_i \mathbb{Z}/2\mathbb{Z}$, where $i : \mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/n\mathbb{Z})$ is given by $i(b) = (a \longmapsto (-1)^b a)$.

Solution Remark that the multiplication in the semidirect product from the problem statement is given by

$$(a_1, b_1)(a_2, b_2) = (a_1 + (-1)^{b_1} a_2, b_1 + b_2).$$

In order to prove that the mapping $f : \sigma \longmapsto (0, 1)$, $f : \tau \longmapsto (1, 0)$ extends to a homomorphism $\mathsf{D}_n \longrightarrow \mathbb{Z}/n\mathbb{Z} \rtimes_i \mathbb{Z}/2\mathbb{Z}$, we need to prove that $f(\sigma)^2 = (0, 0)$, $f(\tau)^n = (0, 0)$ and $f(\sigma) f(\tau) f(\sigma) = f(\tau)^{-1}$. We have

$$\begin{gathered}(0, 1)(0, 1) = (0, 2) = (0, 0) \\ \underbrace{(1, 0) \cdots (1, 0)}_{n \text{ times}} = (n, 0) = (0, 0) \\ (0, 1)(1, 0)(0, 1) = (0 + (-1)^1 1, 1)(0, 1) = (-1, 1)(0, 1) = (-1, 0) = (1, 0)^{-1}\end{gathered}$$

and so f does extend to a homomorphism, which we still denote by f and which is clearly surjective, because the generators $(1, 0)$ and $(0, 1)$ of $\mathbb{Z}/n\mathbb{Z} \rtimes_i \mathbb{Z}/2\mathbb{Z}$ belong to its image. But then f is also injective, as $|\mathsf{D}_n| = 2n = |\mathbb{Z}/n\mathbb{Z} \rtimes_i \mathbb{Z}/2\mathbb{Z}|$.

Exercise 7 Determine the centre of the dihedral group D_n.

Solution Let $\langle r, s \mid r^n = s^2 = e,\ sr = r^{-1}s \rangle$ be the usual presentation of D_n; from it, one immediately obtains that the elements of D_n are r^k, for $k = 0, 1, \ldots, n-1$, and $r^k s$, for $k = 0, 1, \ldots, n-1$.

An element belongs to the centre if and only if it commutes with the generators r, s. Since $r \cdot r^k s \cdot r^{-1} = r^{k+2} s$, no element of the form $r^k s$ belongs to the centre, as we are assuming that $n \geq 3$. On the other hand, elements of the form r^k certainly

commute with r, while the commutation relation for s is $r^k = s \cdot r^k \cdot s^{-1} = r^{-k}$; so r^k is in the centre if and only if $2k \equiv 0 \pmod{n}$. Now, if n is odd, then the only solution to this congruence is $k = 0$, whereas if n is even, we have the two solutions 0 and $n/2$.

We can therefore conclude that the centre of D_n is the trivial subgroup $\{e\}$ if n is odd and the subgroup $\{e, r^{n/2}\}$ if n is even.

Exercise 8 Let $K \subseteq H \subseteq G$ be groups, with K a characteristic subgroup of H and H a characteristic subgroup of G; show that K is characteristic in G. Additionally, show that if K is characteristic in H and H is normal in G then K is normal in G. Finally, provide a counterexample to the following statement: if K is normal in H and H is normal in G then K is normal in G.

Solution For the first part, we need to show that $\varphi(K) = K$ for all automorphisms φ of G. Notice that $\varphi(H) = H$, as H is characteristic in G, and so $\varphi' = \varphi_{|H}$ is clearly an automorphism of H: it is an injective homomorphism as it is the restriction of one, and it is surjective as $\varphi(H) = H$. But then $\varphi(K) = \varphi'(K) = K$, because K is characteristic in H.

For the second part, consider an element $g \in G$; since H is normal in G, conjugation by g induces by restriction an automorphism $\psi_{g|H}$ of H. Since K is characteristic in H, we have $gKg^{-1} = \psi_{g|H}(K) = K$, which proves that K is normal in G.

Finally, as a counterexample to the last statement, letting $G = \mathsf{S}_4$, consider the Klein subgroup $H = \big\{e, (1,2)(3,4), (1,3)(2,4), (1,4)(2,3)\big\}$ and the subgroup $K = \big\{e, (1,2)(3,4)\big\}$. The Klein subgroup is normal in G as it is the union of the two conjugacy classes $\mathcal{C}\ell\,(e)$ and $\mathcal{C}\ell\,\big((12)(34)\big)$ of G. Moreover, K is normal in H because H is Abelian. However, K is not normal in G: it contains a $2+2$-cycle but not all the $2+2$-cycles.

Exercise 9 Given groups H and K, consider the map

$$\mathrm{Aut}(H) \times \mathrm{Aut}(K) \ni (\varphi, \psi) \overset{f}{\longmapsto} \big((h,k) \longmapsto (\varphi(h), \psi(k))\big) \in \mathrm{Aut}(H \times K).$$

Show that f is an injective group homomorphism and that it is an isomorphism if and only if $H \times e_K$ and $e_H \times K$ are characteristic in $H \times K$.

Solution Let $(\varphi, \psi), (\varphi', \psi') \in \mathrm{Aut}(H) \times \mathrm{Aut}(K)$; for all $(h,k) \in H \times K$ we have

$$\begin{aligned} f\big((\varphi,\psi),(\varphi',\psi')\big)(h,k) &= f(\varphi\varphi', \psi\psi')(h,k) \\ &= (\varphi(\varphi'(h)), \psi(\psi'(k))) \\ &= f(\varphi,\psi)(\varphi'(h), \psi'(k)) \\ &= f(\varphi,\psi)(f(\varphi',\psi')(h,k)) \\ &= f(\varphi,\psi) f(\varphi',\psi')(h,k) \end{aligned}$$

and thus f is a homomorphism. The kernel of f is given by the set of pairs (φ, ψ) such that $f(\varphi, \psi)(h, k) = (h, k)$ for all $(h, k) \in H \times K$. By the definition of f, this is equivalent to $\varphi(h) = h$ for all $h \in H$ and $\psi(k) = k$ for all $k \in K$, that is, $\varphi = \mathrm{Id}_H$ and $\psi = \mathrm{Id}_K$. This proves that f is injective.

We now need to show that f is surjective if and only if $H \times e_K$ and $e_H \times K$ are characteristic in $H \times K$.

Suppose f is surjective. Given any automorphism Φ of $H \times K$, there exist $\varphi \in \mathrm{Aut}(H)$ and $\psi \in \mathrm{Aut}(K)$ such that $\Phi = f(\varphi, \psi)$. We therefore have

$$\Phi\big(H \times e_K\big) = \varphi(H) \times \psi(e_K) = H \times e_K,$$

which proves that $H \times e_K$ is characteristic in $H \times K$. Similarly, one shows that $e_H \times K$ is characteristic in $H \times K$.

Conversely, suppose that $H \times e_K$ and $e_H \times K$ are characteristic in $H \times K$. Given an automorphism Φ of $H \times K$ and an element $h \in H$, there exists h' in H such that $\Phi(h, e_K) = (h', e_K)$. Moreover, it is easy to prove that the map $\varphi : h \longmapsto h'$ is an automorphism of H. One analogously constructs an automorphism $\psi : k \longmapsto k'$ such that $\Phi(e_H, k) = (e_H, k')$. We obtain that

$$\begin{aligned}
\Phi(h, k) &= \Phi\big((h, e_K)(e_H, k)\big) \\
&= \Phi(h, e_K)\Phi(e_H, k) \\
&= (h', e_K)(e_H, k') \\
&= (h', k') \\
&= (\varphi(h), \psi(k)) \\
&= f(\varphi, \psi)(h, k),
\end{aligned}$$

and so $f(\varphi, \psi) = \Phi$ and the map f is surjective.

Exercise 10 Let H and K be finite groups such that $(|H|, |K|) = 1$. Prove that $\mathrm{Aut}(H \times K)$ is isomorphic to the group $\mathrm{Aut}(H) \times \mathrm{Aut}(K)$.

Solution By the previous exercise, it is enough to prove that $H \times e_k$ and $e_H \times K$ are characteristic in $H \times K$. Let $\varphi : H \times K \longrightarrow H \times K$ be an automorphism and let $h \in H$. Observe that $(h', k') = \varphi(h, e_K)$ has the same order as (h, e_K), namely, $\mathrm{ord}(h)$ in H. Since the order of (h', k') is the least common multiple of $\mathrm{ord}(h')$ and $\mathrm{ord}(k')$ and since $(|H|, |K|) = 1$, we necessarily have $\mathrm{ord}(k') = 1$, that is, $k' = e_K$. It follows that $\varphi(h, e_K) = (h', e_K) \in H \times e_K$. We have thus shown that $H \times e_K$ is characteristic in $H \times K$. The fact that $e_H \times K$ is characteristic is proven in the same way.

Exercise 11 Show that, for all positive integers n, the group $\mathrm{Aut}\big((\mathbb{Z}/p\mathbb{Z})^n\big)$ is isomorphic to the group $\mathsf{GL}_n(\mathbb{F}_p)$ of $n \times n$ invertible matrices with coefficients in the field $\mathbb{F}_p$. Compute the order of $\mathrm{Aut}\big((\mathbb{Z}/p\mathbb{Z})^n\big)$.

Solution Remark that a group automorphism of $(\mathbb{Z}/p\mathbb{Z})^n$ is the same thing as an isomorphism of the vector space $\mathbb{F}_p^n$ over the field $\mathbb{F}_p$, as the multiplication by scalars can be defined in terms of the sum of elements. This already proves the first statement about the isomorphism between $\mathrm{Aut}\big((\mathbb{Z}/p\mathbb{Z})^n\big)$ and $\mathsf{GL}_n(\mathbb{F}_p)$. We now consider the problem of computing the order of this matrix group.

Having fixed a basis $v_1, v_2, \ldots, v_n$ of $\mathbb{F}_p^n$, any linear map $f : \mathbb{F}_p^n \longrightarrow \mathbb{F}_p^n$ is uniquely determined by the images $u_1 = f(v_1), u_2 = f(v_2), \ldots, u_n = f(v_n)$. Moreover, f is an isomorphism if and only if $u_1, u_2, \ldots, u_n$ is a basis of $\mathbb{F}_p^n$. Counting the isomorphisms of $\mathbb{F}_p^n$ as a vector space over $\mathbb{F}_p$ is therefore equivalent to computing the number of bases of $\mathbb{F}_p^n$. It is clear that the vectors $u_1, u_2, \ldots, u_n$ form a basis if and only if they are linearly independent.

We start by choosing the vector u_1; the only condition on it is that it cannot be the zero vector, so it can take any value in $\mathbb{F}_p^n \setminus \{0\}$, that is, we have $p^n - 1$ possibilities for it. The second vector u_2 must not belong to the line generated by u_1, so we have $p^n - p$ possible values for it. The third vector u_3 must not belong to the plane generated by u_1, u_2, so we have $p^n - p^2$ possible values for it, and so on; if we continue in this way, we end up with $p^n - p^{k-1}$ possible values for u_k.

Choosing each vector in turn, we obtain a total of $(p^n - 1)(p^n - p)\cdots(p^n - p^{n-2})(p^n - p^{n-1})$ possibilities, which give us the cardinality of $\mathrm{Aut}((\mathbb{Z}/p\mathbb{Z})^n)$.

Exercise 12 Show that $\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$ is isomorphic to S_3.

Solution By the previous exercise, we have $|\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})| = 6$, so the group is isomorphic to $\mathbb{Z}/6\mathbb{Z}$ if Abelian, and to S_3 if not. Setting $e_1 = (1, 0)$, $e_2 = (0, 1)$, consider the homomorphisms f, g given by $f(e_1) = e_1$, $f(e_2) = e_1 + e_2$ and $g(e_1) = e_1 + e_2$, $g(e_2) = e_2$. We have $fg(e_1) = f(e_1 + e_2) = f(e_1) + f(e_2) = e_1 + e_1 + e_2 = e_2$ and $gf(e_1) = g(e_1) = e_1 + e_2$. It follows that $\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$ is not Abelian, because $fg \neq gf$. We thus conclude that it is isomorphic to S_3.

▌An alternative solution goes as follows. Any automorphism of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ permutes the three elements of order 2, which are generators for $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$; this yields an injective homomorphism from $\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$ to S_3. Since both groups have 6 elements, this homomorphism is in fact an isomorphism.▐

Exercise 13 Show that the automorphism group of D_5 has 20 elements.

Solution Let $\langle \sigma, \tau \mid \sigma^2 = \tau^5 = \sigma\tau\sigma\tau = e \rangle$ be a presentation of the dihedral group $G = \mathsf{D}_5$. Remark that any automorphism φ of G is uniquely determined by the images of σ and τ, where $\varphi(\sigma)$ must have order 2 and $\varphi(\tau)$ must have order 5. The elements of G are: the neutral element, whose order is 1, the elements $\sigma\tau^h$ for $h = 0, 1, 2, 3, 4$, of order 2, and finally τ^k for $k = 1, 2, 3, 4$, of order 5. We must therefore have $\varphi(\sigma) = \sigma\tau^h$, $\varphi(\tau) = \tau^k$ for some $0 \leq h \leq 4$ and $1 \leq k \leq 4$, as well as $\varphi(\sigma)\varphi(\tau)\varphi(\sigma)\varphi(\tau) = e$. But this does hold for all h and k as above, since $\varphi(\sigma)\varphi(\tau)\varphi(\sigma)\varphi(\tau) = \sigma\tau^h\tau^k\sigma\tau^h\tau^k = \sigma\sigma\tau^{-h-k}\tau^{h+k} = e$.

What we have shown is that, for $0 \leq h \leq 4$ and $1 \leq k \leq 4$, the mapping $\varphi(\sigma) = \sigma\tau^h$, $\varphi(\tau) = \tau^k$ extends to a homomorphism of D_5 to itself; we now

wish to prove that the homomorphism φ is in fact an automorphism, that is, it is bijective. It is enough to show that φ is surjective; in order to do so, remark that, since k and 5 are relatively prime, the element $\varphi(\tau)$ generates the subgroup $\langle\tau\rangle$; but then $\sigma = \varphi(\sigma)\tau^{-h}$ also belongs to the image of φ. The homomorphism φ is thus surjective because its image contains the generators τ, σ of D_5.

Exercise 14 Show that A_4 has no subgroups of order 6.

Solution Assume by contradiction that H is a subgroup of order 6 of A_4. By Cauchy's Theorem, H contains an element τ of order 3 and an element σ of order 2. The elements of order 3 in A_4 are the 3-cycles, so without loss of generality we may assume that $\tau = (1, 2, 3)$.

Since the subgroup generated by τ has index 2 and therefore is normal in H, the permutation σ must conjugate τ either into τ itself or into τ^{-1}. We have $\sigma\tau\sigma^{-1} = (\sigma(1), \sigma(2), \sigma(3))$, so we must have $\sigma(4) = 4$, as 4 is the only element that does not appear in τ.

However, the only elements of order 2 in A_4 are the $2 + 2$-cycles, none of which fixes 4. We conclude that such a subgroup H cannot exist.

▌Here is an alternative solution. The subgroup H is normal in A_4 and is thus a union of conjugacy classes of A_4. However, a simple computation shows that the Class Equation for A_4 is

$$12 = 1 + 3 + 4 + 4$$

and it is not possible to obtain 6 as a sum of a subset of these terms. ▐

Exercise 15 Show that any group G of order p^n, where p is a prime, contains a chain

$$\{e\} = G_0 \subseteq G_1 \subseteq \cdots \subseteq G_n = G$$

of subgroups that are normal in G and have orders given by $|G_h| = p^h$ for $h = 0, 1, \ldots, n$.

Solution For the sake of brevity, we shall call any chain with the properties described in the problem statement *normal and complete*. We procede by induction on n. If $n = 1$ then G is a group with p elements and the chain of subgroups $\{e\} = G_0 \subseteq G_1 = G$ is normal and complete.

Given $n \geq 2$, assume first that G is Abelian. By Cauchy's Theorem there exists an element g of order p; we denote by K the subgroup generated by g, which is normal because G is Abelian, and by $\overline{G}$ the quotient G/K. Since $\overline{G}$ has fewer elements than G and is still a p-group, by the induction hypothesis there exists a chain

$$\{e_{\overline{G}}\} = \overline{G}_0 \subseteq \overline{G}_1 \subseteq \cdots \subseteq \overline{G}_{n-1} = \overline{G}$$

of subgroups that are normal in $\overline{G}$, with $|\overline{G}_h| = p^h$. Now, letting $\pi : G \longrightarrow \overline{G}$ denote the quotient homomorphism, we have that $G_h = \pi^{-1}(\overline{G}_h)$ is a normal subgroup of G of order p^{h+1}. In particular, the chain of subgroups

$$\{e\} = G_0 \subseteq G_1 \subseteq \cdots \subseteq G_{n-1} = G$$

is normal and complete in G.

Now let G be a non-Abelian p-group. We know that G, being a p-group, has nontrivial centre $Z = Z(G)$. Since Z is normal in G, we may consider the quotient group $\overline{G} = G/Z$, whose cardinality is p^m for some $1 \leq m < n$. By the induction hypothesis, $\overline{G}$ has a chain

$$\{e_{\overline{G}}\} = \overline{G}_0 \subseteq \overline{G}_1 \subseteq \cdots \subseteq \overline{G}_m = \overline{G}$$

that is normal and complete. But then, letting $\pi : G \longrightarrow \overline{G}$ denote again the quotient homomorphism and setting $G_h = \pi^{-1}(\overline{G_h})$ for $h = 0, 1, \ldots, m$, the subgroups of the chain $Z = G_0 \subseteq \cdots \subseteq G_m = G$ are all normal in G. Moreover, Z is an Abelian group, and, by our previous argument, there exists a chain of subgroups $\{e\} = Z_0 \subseteq \cdots \subseteq Z_{n-m} = Z$ that is normal and complete in Z.

Remark that every subgroup of Z is normal in G. But then we can join the two chains together to obtain the chain

$$\{e\} = Z_0 \subseteq Z_1 \subseteq \cdots \subseteq Z_{n-m} = Z = G_0 \subseteq G_1 \subseteq \cdots \subseteq G_m = G,$$

which is normal and complete in G.

Exercise 16 Show that in every p-group G each nontrivial normal subgroup has nontrivial intersection with the centre of G.

Solution Let H be a normal subgroup of the p-group G. The subgroup H is a union of conjugacy classes (because it is normal) and every conjugacy class of G has either one element, which must belong to the centre of G, or a number of elements that is a multiple of p.

The subgroup H must contain the conjugacy class of the neutral element, whose cardinality is 1; since the order of H is a multiple of p, it must contain at least another $p - 1$ conjugacy classes of cardinality 1. That is, the intersection of H and the centre of G cannot only contain the neutral element.

Exercise 17 Show that the normaliser of a proper subgroup H of a p-group G must strictly contain H.

Solution First of all, remark that the statement is obvious if the p-group G is Abelian: indeed, letting $N(H)$ denote the normaliser of H, in this case we have $H \subsetneqq N(H) = G$.

We now proceed by induction on the order of G; the base case is given by $G \simeq \mathbb{Z}/p\mathbb{Z}$, which is Abelian and therefore already dealt with.

Let Z be the centre of G and let $\overline{G}$ be the quotient G/Z; denote by $\pi : G \longrightarrow \overline{G}$ the quotient homomorphism. Since the centre of a p-group cannot be trivial, the order of $\overline{G}$ is less than the order of G; moreover, we can assume that G is not Abelian, so $\overline{G}$ is still a p-group.

Now, if Z is not contained in H, then the statement is trivial, as Z is clearly contained in $N(H)$. Suppose $Z \subseteq H$ and consider $\overline{H} = \pi(H)$, which is a proper subgroup of $\overline{G}$ because H is proper in G and $Z \subseteq H$. By the induction hypothesis, $\overline{H}$ is strictly contained in $N_{\overline{G}}(\overline{H})$.

Note that $\pi^{-1}(\overline{H}) = H \cdot Z = H$ and so proving that we also have $\pi^{-1}(N_{\overline{G}}(\overline{H})) = N(H)$ would yield

$$H = \pi^{-1}(\overline{H}) \subsetneq \pi^{-1}(N_{\overline{G}}(\overline{H})) = N(H),$$

as π is surjective; this would conclude the proof of the desired statement.

In order to show that $\pi^{-1}(N_{\overline{G}}(\overline{H})) = N(H)$, it is enough to remark that we have the following chain of equivalences:

$$\begin{aligned} g \in \pi^{-1}(N_{\overline{G}}(\overline{H})) &\iff gZ \in N_{\overline{G}}(\overline{H}) \\ &\iff gZ \cdot HZ \cdot g^{-1}Z = HZ \\ &\iff gHg^{-1}Z = HZ \\ &\iff g \in N(H), \end{aligned}$$

where we have repeatedly used the assumption that $Z \subseteq H$.

Exercise 18 For each h, determine the cycle structure of σ^h, where σ is an ℓ-cycle in S_n.

Solution Letting r be the greatest common divisor of ℓ and h, we show that the permutation σ^h is the product of r disjoint ℓ/r-cycles. Without loss of generality, we may assume that $\sigma = (1, 2, 3, \ldots, \ell)$. The orbit of 1 under σ^h is $1, 1+h, 1+2h, \ldots$ and will therefore consist of t elements, where t is the smallest positive integer such that $1 + th \equiv 1 \pmod{\ell}$. We immediately find that $t = \ell/r$. It is clear that all the orbits of elements in $\{1, 2, \ldots, \ell\}$ under σ^h have the same cardinality. Moreover, these orbits determine the cycle structure of σ^h, and in particular we find that σ^h consists of r disjoint cycles of length ℓ/r.

Exercise 19 Show that the permutations $(1, 2)$ and $(1, 2, \ldots, n)$ generate all of S_n. Show that the permutations $(1, 3)$ and $(1, 2, 3, 4)$ do *not* generate S_4.

Solution Set $\sigma = (1, 2)$, $\tau = (1, 2, \ldots, n)$, and let H denote the subgroup of S_n generated by σ and τ. Remark that for $1 \leq i \leq n-1$ we have $(i, i+1) =$

$\psi_{\tau^{i-1}}\big((1,2)\big) = \tau^{i-1}\sigma\tau^{1-i} \in H$. Therefore, $H = \mathsf{S}_n$, because the simple transpositions generate all of S_n.

We now show the second statement. It is easy to check that $\sigma\tau\sigma^{-1} = \sigma^3 = \sigma^{-1}$. It follows that σ belongs to the normaliser of τ, so $\langle\sigma\rangle \cdot \langle\tau\rangle$ is a subgroup of S_4 whose order is 8, and thus cannot be S_4.

Exercise 20 Show that the centre of S_n is trivial for $n \geq 3$.

Solution Letting σ denote an element in the centre of S_n, we shall prove that $\sigma(i) = i$ for all $1 \leq i \leq n$. Having fixed i, thanks to the fact that $n \geq 3$ we can choose $1 \leq j, k \leq n$ in such a way that the integers i, j, k are all distinct. The permutation σ acts trivially by conjugation, so $\psi_\sigma\big((i,j)\big) = (i,j)$ and $\psi_\sigma\big((i,k)\big) = (i,k)$. But $\psi_\sigma\big((i,j)\big) = (\sigma(i),\sigma(j))$ and $\psi_\sigma\big((i,k)\big) = (\sigma(i),\sigma(k))$, so $\sigma(i) \in \{i,j\} \cap \{i,k\} = \{i\}$, that is, $\sigma(i) = i$.

Exercise 21 Given $n \geq 3$, show that if a subgroup of A_n contains all of the 3-cycles then it is A_n itself.

Solution Let H be a subgroup of A_n containing all of the 3-cycles. We know that a permutation σ belongs to A_n if and only if it is the product of an even number of transpositions. It follows that, if we show that for all i, j, k, t such that $i \neq j$ and $k \neq t$ we have $(i,j)(k,t) \in H$, then we obtain that $H = \mathsf{A}_n$.

Let r be the cardinality of the intersection $\{i,j\} \cap \{k,t\}$. If $r = 2$ then $(i,j) = (k,t)$ and $(i,j)(k,t) = e \in H$. If $r = 1$ then we may assume that $t \neq j = k \neq i$, which yields $(i,j)(k,t) = (i,j)(j,t) = (i,j,t) \in H$. Finally, if $r = 0$, that is, if i, j, k, t are all distinct, then $(i,j)(k,t) = (k,i,t)(i,j,k) \in H$.

Exercise 22 Given $n \geq 3$, show that if a normal subgroup H of A_n contains a 3-cycle then $H = \mathsf{A}_n$. For $n \geq 5$, prove that if a normal subgroup H of A_n contains a $2+2$-cycle then $H = \mathsf{A}_n$.

Solution First, we assume that H contains a 3-cycle, which without loss of generality we may assume to be $(1,2,3)$. We wish to prove that every 3-cycle belongs to H, which would imply that $H = \mathsf{A}_n$ by the previous exercise. To this end, let (α,β,γ) by any 3-cycle. If we were able to produce a permutation $\tau \in \mathsf{A}_n$ such that $\psi_\tau\big((1,2,3)\big) = (\alpha,\beta,\gamma)$ then we would have $(\alpha,\beta,\gamma) \in H$, because H is normal in A_n.

Let $\tau \in \mathsf{S}_n$ be any permutation such that $\psi_\tau\big((1,2,3)\big) = (\alpha,\beta,\gamma)$; such a permutation exists, because all 3-cycles are pairwise conjugate in S_n. If τ is even then we are done. If τ is odd, set $\tau' = (\beta,\gamma)\tau$ and remark that τ' is now even. Moreover, $\psi_{\tau'}\big((1,2,3)\big) = \psi_{(\beta,\gamma)}\psi_\tau\big((1,2,3)\big) = \psi_{(\beta,\gamma)}\big((\alpha,\beta,\gamma)\big) = (\alpha,\gamma,\beta)$; it follows that $(\alpha,\gamma,\beta) \in H$ and, since H is a subgroup, we have $(\alpha,\beta,\gamma) = (\alpha,\gamma,\beta)^{-1} \in H$.

Now assume that $n \geq 5$ and let H be a normal subgroup of A_n that contains a $2+2$-cycle, which we may assume to be $\tau = (1,2)(3,4)$ without loss of generality. The subgroup H, being normal, must also contain $\sigma = \psi_{(1,2)(4,5)}(\tau) = (1,2)(3,5)$. It follows that H contains $\tau\sigma = (3,5,4)$ and thus by the previous argument we have $H = \mathsf{A}_n$, as wanted.

Exercise 23 Show that $\{e\}$, A_n and S_n are the only normal subgroups of S_n for $n \geq 5$.

Solution Let $H \neq \{e\}$ be a normal subgroup of S_n. Given a transposition τ and an element $\eta \neq e$ of H, the element $\sigma_\tau = \eta(\tau\eta\tau^{-1}) = (\eta\tau\eta^{-1})\tau^{-1}$ belongs to H and is the product of two transpositions. If σ_τ is not the neutral element then it is either a 3-cycle or a $2+2$-cycle; in both cases, H contains A_n thanks to the previous exercise, and we conclude that $H = \mathsf{A}_n$ or $H = \mathsf{S}_n$.

On the other hand, if $\sigma_\tau = e$ for all transpositions τ, then η commutes with every transposition τ and therefore belongs to the centre of S_n, since the transpositions generate S_n. But this is a contradiction, as the centre is trivial and η is not the neutral element.

Exercise 24 Show that A_n is simple for all $n \geq 5$.

Solution

- First solution. Let H such that $\{e\} \neq H \neq \mathsf{A}_n$ be a normal subgroup of A_n. By the previous exercise, H is not normal in S_n; moreover, A_n is contained in the normaliser $N = N_{\mathsf{S}_n}(H)$, because H is normal in A_n. We therefore have $N = \mathsf{A}_n$, whose index in S_n is 2, and thus H has two conjugates in S_n, namely, H itself and $H' = \tau H \tau^{-1}$, where τ is any odd permutation.

  The group H' is also a normal subgroup of A_n; indeed, it is contained in A_n because it is a conjugate of H and A_n is invariant under conjugation, and it is normal in A_n because conjugating H' by elements of A_n corresponds to conjugating H by odd permutations.

  Consider now $H \cap H'$; since it is the intersection of all conjugates of H in S_n, it is a normal subgroup of S_n; moreover, it is strictly contained in A_n, so it is the trivial group $\{e\}$. On the other hand, $H \cdot H'$ is a nontrivial normal subgroup of S_n that is contained in A_n, and therefore $H \cdot H' = \mathsf{A}_n$. We have thus shown that A_n is isomorphic to $H \times H'$, because H and H' are both normal in A_n.

  Since the order of A_n is even, the order of H must also be even. We can thus find an element $\tau \in H$ of order 2, which we express as a product of disjoint 2-cycles as $\tau = \tau_1\tau_2\cdots\tau_r$. The transposition τ_1 is an odd permutation, so $\tau_1 H \tau_1^{-1} = H'$. But τ_1 commutes with τ, so $\tau = \tau_1\tau\tau_1^{-1} \in H'$, and we have thus found a nontrivial element that belongs to both H and H', yielding a contradiction.
- Second solution. Let $n \geq 5$ and let $H \neq \{e\}$ be a normal subgroup of A_n. We will break down the proof into several steps.

(1) Suppose τ is an element of H of even order m. The element $\tau' = \tau^{m/2}$ can be written as a product of disjoint 2-cycles, and in fact as a product of an even number of disjoint 2-cycles because $\tau' \in H \subseteq \mathsf{A}_n$. Suppose $\tau' = \tau_1\tau_2\cdots\tau_{2r}$, where we may assume that $\tau_1 = (1,2)$, $\tau_2 = (3,4)$. Notice that, since H is normal, the permutation $\sigma = \psi_{(2,3,4)}(\tau') = (1,3)(2,4)\tau_3\cdots\tau_{2r}$ is still an element of H. But then $\tau'\sigma = (1,4)(2,3)$ is also in H, and a previous exercise yields the desired conclusion.

(2) Now assume that H contains an element τ whose order m has a prime factor $\ell > 3$. The permutation $\tau' = \tau^{m/\ell}$ is a product of disjoint ℓ-cycles $\tau_1\tau_2\cdots\tau_r$, and we may suppose that $\tau_1 = (1,2,\ldots,\ell)$. Since $\eta = (3,4,\ldots,\ell) \in \mathsf{A}_n$, the permutation $\sigma = \psi_\eta(\tau') = (1,2,4,5,\ldots,\ell,3)\tau_2\cdots\tau_r$ also belongs to H; moreover, $\sigma^{-1} = (3,\ell,\ell-1,\ldots,5,4,2,1)\tau_2^{-1}\cdots\tau_r^{-1} \in H$. We thus have $\tau'\sigma^{-1} = (1,2,\ldots,\ell)(3,\ell,\ell-1,\ldots,5,4,2,1) = (1,4,3) \in H$, and we reach the desired conclusion thanks to a previous exercise.

(3) Finally, suppose $\tau \in H$ is of order 3^k. If $k = 1$ then H contains a 3-cycle and a previous exercise yields that $H = \mathsf{A}_n$. If $k \geq 2$, notice that $\tau' = \tau^{3^{k-1}}$ is an element of H that can be expressed as the product $\tau_1\tau_2\cdots\tau_{3^{k-1}}$ of 3^{k-1} disjoint 3–cycles, and $3^{k-1} > 1$. We may assume that $\tau_1 = (1,2,3)$ and $\tau_2 = (4,5,6)$. Remark that $\sigma = \psi_{(3,4,5)}(\tau') = (1,2,4)(5,3,6)\tau_3\cdots\tau_r$ is again an element of H, and so is $\tau'\sigma^{-1} = (1,2,3)(4,5,6)(1,4,2)(5,6,3) = (1,5,4,3,6)$. We have thus produced an element of order 5 and therefore the previous case applies.

Exercise 25 Describe the centraliser of the cycle $(1,2,\ldots,\ell)$ in S_n.

Solution Let $Z(\tau)$ be the centraliser of $\tau = (1,2,\ldots,\ell)$. Every permutation σ that fixes the numbers $1,2,\ldots,\ell$ commutes with τ. Moreover, the set of such permutations is a subgroup H of S_n that is clearly isomorphic to $\mathsf{S}_{n-\ell}$. Note that the ℓ powers of τ, which form the subgroup $K = \langle\tau\rangle$, also commute with τ. Moreover, we have $K \cap H = \{e\}$, because any element belonging to the intersection must fix $1,2,\ldots,\ell$, and the only power of τ that does is e. We have thus found $|KH| = |K||H| = \ell(n-\ell)!$ elements belonging to $Z(\tau)$.

We know that the number of ℓ-cycles in S_n is

$$\binom{n}{\ell}(\ell-1)!,$$

and that the centraliser is the stabiliser for the action of G on itself by conjugation; the set of all ℓ-cycles, that is, the conjugacy class of τ, is the orbit for this action. We therefore have

$$|Z(\tau)| = |\mathsf{S}_n|/|\mathcal{C}\ell(\tau)| = n!/(n!/\ell(n-\ell)!) = \ell(n-\ell)!$$

and so $KH = Z(\tau)$. We have thus shown that every element of $Z(\tau)$ is of the form $\tau^h\eta$ for some integer h and some permutation η that fixes $1,2,\ldots,\ell$.

Exercise 26 Determine the cardinality of the normaliser of the subgroup generated by an n-cycle in S_n.

Solution Let τ be an n-cycle in S_n. We know that the centraliser $Z(\tau)$ of τ coincides with the subgroup H generated by τ, which has order n.

A permutation σ in the normaliser $N(H)$ conjugates τ into another element of order n in H, that is, into τ^k for some k that is coprime to n. Since $H = Z(\tau) = Z(H)$ is the kernel of the action of $N(H)$ on H by conjugation, the permutation σ corresponding to some fixed k is uniquely determined up to multiplication by a power of τ.

This proves that $N(H)$ has $n \cdot \phi(n)$ elements.

Exercise 27 Let τ be an even permutation in S_n. Show that

(i) if $Z_{\mathsf{A}_n}(\tau) \neq Z_{\mathsf{S}_n}(\tau)$ then the conjugacy class $\mathcal{C}\ell_{\mathsf{A}_n}(\tau)$ of τ in A_n coincides with the conjugacy class $\mathcal{C}\ell_{\mathsf{S}_n}(\tau)$ of τ in S_n;
(ii) if $Z_{\mathsf{A}_n}(\tau) = Z_{\mathsf{S}_n}(\tau)$ then $\mathcal{C}\ell_{\mathsf{S}_n}(\tau)$ is the union of two conjugacy classes in A_n having the same cardinality.

Solution Denote by $\mathcal{C}_0 = \mathcal{C}\ell_{\mathsf{A}_n}(\tau)$ the conjugacy class of τ in A_n and by $\mathcal{C} = \mathcal{C}\ell_{\mathsf{S}_n}(\tau)$ the conjugacy class of τ in S_n. Clearly, $\mathcal{C}_0 \subseteq \mathcal{C}$ and $|\mathcal{C}_0| = |\mathsf{A}_n|/|Z_{\mathsf{A}_n}(\tau)|$, whereas $|\mathcal{C}| = |\mathsf{S}_n|/|Z_{\mathsf{S}_n}(\tau)|$. Remark that $Z_{\mathsf{A}_n}(\tau)$ is the kernel of the sign homomorphism restricted to $Z_{\mathsf{S}_n}(\tau)$, so there are two cases to consider: either ① $Z_{\mathsf{S}_n}(\tau)$ is not contained in A_n and so $Z_{\mathsf{A}_n}(\tau)$ has index 2 in $Z_{\mathsf{S}_n}(\tau)$, or ② $Z_{\mathsf{S}_n}(\tau) \subseteq \mathsf{A}_n$ and so $Z_{\mathsf{A}_n}(\tau) = Z_{\mathsf{S}_n}(\tau)$. We consider each in turn.

① The fact that $|Z_{\mathsf{A}_n}(\tau)| = |Z_{\mathsf{S}_n}(\tau)|/2$ yields $|\mathcal{C}_0| = |\mathcal{C}|$ and so $\mathcal{C} = \mathcal{C}_0$. This proves (i).

② It is clear that $|\mathcal{C}_0| = |\mathcal{C}|/2$. Given a permutation $\tau' \in \mathcal{C} \setminus \mathcal{C}_0$, we know that τ' is even thanks to τ being even, so we have $\mathcal{C} \subseteq \mathsf{A}_n$. If $\mathcal{C}_0'$ is the conjugacy class of τ' in A_n then $\mathcal{C}_0 \sqcup \mathcal{C}_0' \subseteq \mathsf{A}_n$. By applying the same arguments used for τ to the permutation τ', we find that $\mathcal{C}_0$ and $\mathcal{C}_0'$ both have $|\mathcal{C}|/2$ elements, so $\mathcal{C} = \mathcal{C}_0 \sqcup \mathcal{C}_0'$. This concludes the proof of (ii).

Exercise 28 How many conjugates does $\sigma = (1,2)(3,4)$ have in S_n for $n \geq 4$? Describe the centraliser of σ.

Solution The conjugates of $\sigma = (1,2)(3,4)$ are the $2+2$-cycles. A $2+2$-cycle is determined by two disjoint pairs of elements in $\{1, 2, \ldots, n\}$. There are $\binom{n}{2}$ ways to choose a pair of elements from the set, and $\binom{n-2}{2}$ to choose 2 elements among the remaining $n-2$. Each $2+2$-cycles can be constructed by picking the two pairs in two possible orders, so we have

$$\left|\mathcal{C}\ell\left((1,2)(3,4)\right)\right| = \frac{1}{2}\binom{n}{2}\binom{n-2}{2} = \frac{1}{8}\frac{n!}{(n-4)!}.$$

Every element of the subgroup $H = \{\tau \in \mathsf{S}_n \mid \tau(i) = i \text{ for } i = 1, 2, 3, 4\}$ commutes with σ. This subgroup has $(n-4)!$ elements and is isomorphic to S_{n-4}.

Now set $A = \{1, 2\}$, $B = \{3, 4\}$. If a permutation η fixes the numbers $4, 5, \ldots, n$ and sends A to A and B to B, or sends A to B and B to A, then it commutes with σ. Such permutations form a subgroup K with 8 elements of S_n. Moreover, we have $K \cap H = \{e\}$, so $|KH| = 8(n-4)!$.

But the cardinality of $Z(\sigma)$ is given by $\left|\mathsf{S}_n\right| / \left|\mathcal{C}\ell\left((1,2)(3,4)\right)\right| = 8(n-4)!$, so $Z(\sigma) = KH$. Finally, notice that the elements of K commute with those of H, so we can conclude that $Z(\sigma) \simeq K \times H$.

Exercise 29 Describe the normal subgroups of S_4.

Solution The conjugacy classes in S_4 are in bijection with the possible cycle structures: we have $\mathcal{C}\ell(e), \mathcal{C}\ell\left((1,2)\right), \mathcal{C}\ell\left((1,2)(3,4)\right), \mathcal{C}\ell\left((1,2,3)\right)$ and $\mathcal{C}\ell\left((1,2,3,4)\right)$, of cardinality 1, 6, 3, 8 and 6, respectively.

Any normal subgroup H of S_4 is a union of conjugacy classes and its order must divide 24. Using the cardinalities above, it is immediate to check that the possible values for $|H|$ are 1, 4, 12 and 24. We now show that each of these values corresponds to a single normal subgroup of S_4: in the first case, $H = e$; in the second, $H = \left\{e, (1,2)(3,4), (1,3)(2,4), (1,4)(2,3)\right\}$, that is, H is the Klein group; in the third, $H = \mathsf{A}_4$, and the last one yields $H = \mathsf{S}_4$.

Exercise 30

(i) Find the centraliser of $(1,2,3,4,5,6,7)(8,9,10)$ in S_{10}.
(ii) Show that S_{10} contains subgroups of order 42, none of which is Abelian.

Solution

(i) The order of the permutation $\sigma = (1,2,3,4,5,6,7)(8,9,10)$ is the least common multiple of the lengths of its disjoint cycles, namely, 21. The subgroup $H = \langle\sigma\rangle$ is contained in the centraliser $Z(\sigma)$ and thus consists of 21 elements of $Z(\sigma)$. On the other hand, the cardinality of $Z(\sigma)$ is given by $|\mathsf{S}_{10}| / |\mathcal{C}\ell(\sigma)|$ and the conjugacy class $\mathcal{C}\ell(\sigma)$ is the set of all $7+3$-cycles, whose cardinality is

$$|\mathcal{C}\ell(\sigma)| = 6! \binom{10}{7} 2! \binom{3}{3}.$$

This implies that $|Z(\sigma)| = 21$ and thus that $Z(\sigma) = H$, that is, σ only commutes with its own powers.

(ii) First of all, we show that there exists a subgroup of S_{10} with 42 elements. Letting $\tau = (8,9)$, remark that $\tau \notin H = \langle\sigma\rangle$, as 2 does not divide 21. We denote by K the subgroup $\langle\tau\rangle = \{e, \tau\}$; we have just remarked that $H \cap K = \{e\}$.

Note that we have $\tau\sigma\tau = \tau\sigma\tau^{-1} = \psi_\tau(\sigma) = (1,2,3,4,5,6,7)(9,8,10) = \sigma^8$ and so $\tau\sigma = \sigma^8\tau$. This proves that τ belongs to the normaliser of H in S_{10}, and thus that $HK = KH$. It follows that HK is a group consisting of $|HK| = |H||K|/|H \cap K| = 42$ elements.

On the other hand, an Abelian group G with $42 = 2 \cdot 3 \cdot 7$ elements must be cyclic. Indeed, Cauchy's theorem implies that it must have elements x, y, z of order 2, 3, 7, respectively. Since 2, 3, 7 are relatively prime, we immediately obtain that xyz has order 42 and is thus a generator for G.

Assume by contradiction that S_{10} contains an Abelian subgroup with 42 elements; S_{10} would then contain an element η of order 42. But the order of η is the least common multiple of the lengths of its cycles, and thus η must necessarily contain a 7-cycle, as 7 divides 42. The cycle structure of η is therefore one of the following:

$$\begin{aligned} 10 &= 7+1+1+1 \\ &= 7+2+1 \\ &= 7+3. \end{aligned}$$

By direct inspection, we find that the order of η cannot be 42.

Exercise 31 Show that if an automorphism of S_n sends transpositions to transpositions then it is an inner automorphism.

Solution Let φ be an automorphism of S_n sending transpositions to transpositions. By induction on h, where $1 \le h \le n-1$, we wish to show the following: there exists $\tau \in \mathsf{S}_n$, depending on h, such that $\psi_\tau\varphi$ sends $(t, t+1)$ to itself for $1 \le t \le h$. Remark that the case $h = n-1$ yields a permutation τ such that $\psi_\tau\varphi$ fixes every simple transposition, and therefore $\varphi = \psi_{\tau^{-1}}$, as the simple transpositions generate S_n.

We first show the base case of our induction. Letting $h = 1$ and $\varphi\big((1,2)\big) = (i,j)$, take $\tau \in \mathsf{S}_n$ such that $\tau(i) = 1$ and $\tau(j) = 2$; we have $\psi_\tau\varphi\big((1,2)\big) = \psi_\tau\big((i,j)\big) = (\tau(i), \tau(j)) = (1,2)$.

For the inductive step, consider $h \ge 2$. By the induction hypothesis, there is $\tau \in \mathsf{S}_n$ such that, letting $\varphi' = \psi_\tau\varphi$, we have $\varphi'\big((t,t+1)\big) = (t,t+1)$ for $t = 1, 2, \ldots, h-1$. Suppose $\varphi'\big((h,h+1)\big) = (i,j)$. Remark that $(h,h+1)$ commutes with $(1,2), (3,4), \ldots, (h-2,h-1)$, and moreover $(i,j) \neq (1,2), (3,4), \ldots, (h-2,h-1)$ because φ' is injective. It follows that $\{i,j\} \cap \{1,2,\ldots,h-1\} = \varnothing$. The two permutations $(h-1,h)$ and $(h,h+1)$ do not commute, so neither do $\varphi'\big((h-1,h)\big) = (h-1,h)$ and $\varphi'\big((h,h+1)\big) = (i,j)$; consequently, $|\{i,j\} \cap \{h,h-1\}| = 1$ and, by the argument above, the element they have in common can only be h. In conclusion,

we can assume without loss of generality that $i = h$ and $j \geq h + 1$, so there exists $\tau' \in \mathsf{S}_n$ such that

$$\begin{array}{c} 1 \overset{\tau'}{\longmapsto} 1 \\ 2 \longmapsto 2 \\ \vdots \\ h-1 \longmapsto h-1 \\ h \longmapsto h \\ j \longmapsto h+1. \end{array}$$

The map $\psi_{\tau'\tau}\varphi = \psi_{\tau'}\varphi'$ sends $(t, t+1)$ to itself for $t = 1, 2, \ldots, h$, which completes the proof by induction.

Exercise 32 Show that for all n other than 6 every automorphism of S_n is inner.

Solution We shall show that, if n is different from 6, then every automorphism of S_n sends transpositions to transpositions; the desired conclusion follows by the previous exercise.

Any automorphism φ preserves the orders of elements, so $\varphi\big((1,2)\big)$ must be an element of order 2; therefore, there exists $h \geq 1$ such that $\varphi\big((1,2)\big)$ is the product of h disjoint 2-cycles. It is also clear that any automorphism φ sends conjugacy classes to conjugacy classes, and in particular $\varphi(\mathcal{C}\ell(\sigma)) = \mathcal{C}\ell(\varphi(\sigma))$. It follows that the classes $\mathcal{C}\ell\left((1,2)\right)$ and $\mathcal{C}\ell\left((1,2)(3,4)\cdots(2h-1,2h)\right)$ must have the same cardinality.

In order to show that φ sends transpositions to transpositions, assume by contradiction that $h > 1$.

By comparing the relevant cardinalities, we obtain that

$$\frac{n!}{2\cdot(n-2)!} = \left|\,\mathcal{C}\ell\left((1,2)\right)\right| = \left|\,\mathcal{C}\ell\left((1,2)(3,4)\cdots(2h-1,2h)\right)\right| = \frac{1}{h!}\cdot\frac{n!}{2^h\cdot(n-2h)!}.$$

It is immediate to check that there are no solutions for $h = 2$. We can thus assume that $h \geq 3$ and rewrite the equation as

$$2^{h-1} = \binom{n-2}{h}\cdot(n-h-2)(n-h-3)\cdots(n-2h+1),$$

where, other than the binomial coefficient, the right-hand side has $h-2 \geq 1$ factors. However, since there cannot be any odd factors greater than 1, we only have the two possibilities $n-h-2 = 1$ and $h-2 = 1$. These yield the values $n = 6$, $h = 3$ and $n-h-2 = 2$, $h-2 = 2$, that is, $n = 8$, $h = 4$, which is not a solution.

Since we are assuming that $n \neq 6$, we can conclude that for $h \neq 1$ the two cardinalities are different.

[This proof implies that, for $n = 6$, any non-inner automorphism must send transpositions to $2 + 2 + 2$-cycles. It follows that for $n = 6$ we have at most $|\mathrm{Aut}(\mathsf{S}_6)/\mathrm{Inn}(\mathsf{S}_6)| = 2$. It is in fact possible to construct an automorphism of S_6 that is *not* inner, and in particular we find that the subgroup of inner automorphisms has index 2 in the group of all automorphisms of S_6.]

Exercise 33 Count the automorphisms of $\mathsf{S}_3 \times \mathbb{Z}/3\mathbb{Z}$.

Solution Set $G = \mathsf{S}_3 \times \mathbb{Z}/3\mathbb{Z}$ and let $\varphi : G \longrightarrow G$ be an automorphism. Remark that, like every automorphism, φ sends the centre $Z(G)$ of G to itself. Since $Z(G) = Z(\mathsf{S}_3) \times Z(\mathbb{Z}/3\mathbb{Z}) = e \times \mathbb{Z}/3\mathbb{Z}$, we have that $e \times \mathbb{Z}/3\mathbb{Z}$ is a characteristic subgroup of G.

Now let τ be a transposition in S_3; the element $(\tau, 0)$ has order 2, so $\varphi((\tau, 0))$ must also have order 2. But the elements of order 2 in G all belong to $\mathsf{S}_3 \times 0$, so $\varphi((\tau, 0)) \in \mathsf{S}_3 \times 0$. The transpositions generate S_3, so $\mathsf{S}_3 \times 0$ is sent to itself. It follows that $\mathsf{S}_3 \times 0$ is also characteristic.

We thus obtain that $\mathrm{Aut}(\mathsf{S}_3 \times \mathbb{Z}/3\mathbb{Z}) \simeq \mathrm{Aut}(\mathsf{S}_3) \times \mathrm{Aut}(\mathbb{Z}/3\mathbb{Z}) \simeq \mathsf{S}_3 \times (\mathbb{Z}/3\mathbb{Z})^*$, where we have used the fact that, since every automorphism of S_3 is inner by the previous exercise and since S_3 has trivial centre, we have $\mathrm{Aut}(\mathsf{S}_3) \simeq \mathsf{S}_3$. In conclusion, $|\mathrm{Aut}(\mathsf{S}_3 \times \mathbb{Z}/3\mathbb{Z})| = 12$.

[Another way to show that every automorphism of S_3 is inner is the following. The group $\mathrm{Aut}(\mathsf{S}_3)$ acts on the set of the three transpositions of S_3, and this action is faithful, as the transpositions generate S_3; in particular, $\mathrm{Aut}(\mathsf{S}_3)$ is a subgroup of S_3.]

Exercise 34 Explicitly construct a non-Abelian group of order 55.

Solution Note that $55 = 11 \cdot 5$ and that $\mathrm{Aut}(\mathbb{Z}/11\mathbb{Z}) \simeq (\mathbb{Z}/11\mathbb{Z})^*$; the element $4 \in (\mathbb{Z}/11\mathbb{Z})^*$ of order 5 corresponds to the automorphism $\mathbb{Z}/11\mathbb{Z} \ni a \longmapsto 4a \in \mathbb{Z}/11\mathbb{Z}$. The map

$$\mathbb{Z}/5\mathbb{Z} \ni b \overset{\varphi}{\longmapsto} \left(a \longmapsto 4^b a\right) \in \mathrm{Aut}(\mathbb{Z}/11\mathbb{Z})$$

is thus a homomorphism and defines the semidirect product $G = \mathbb{Z}/11\mathbb{Z} \rtimes_\varphi \mathbb{Z}/5\mathbb{Z}$. Clearly, G has 55 elements and cannot be Abelian, as $\varphi_1 \neq \mathrm{Id}_{\mathbb{Z}/11\mathbb{Z}}$. The operation of G is given by

$$(a_1, b_1) \cdot (a_2, b_2) = (a_1 + 4^{b_1} a_2, b_1 + b_2)$$

for all $(a_1, b_1),\ (a_2, b_2) \in G$.

Exercise 35 Show that the group Q_8 of unit quaternions cannot be expressed as the semidirect product of two nontrivial groups.

Solution We know that the proper nontrivial subgroups of Q_8 are the centre $Z(\mathsf{Q}_8) = \{\pm 1\}$ and the cyclic subgroups of order 4 generated by i, j and k. It

follows that $Z(\mathsf{Q}_8)$ is the only subgroup of order 2 and that it is contained in every nontrivial subgroup. So there are no two nontrivial subgroups that intersect trivially.

Exercise 36 Let K be a finite Abelian group and n an integer that is coprime to the order of K. Given a group H and a homomorphism $\varphi : K \longrightarrow \mathrm{Aut}(H)$, show that

(i) the map $K \ni b \overset{\psi}{\longmapsto} \varphi(b^n) \in \mathrm{Aut}(H)$ is a homomorphism;
(ii) the semidirect products $H \rtimes_\varphi K$ and $H \rtimes_\psi K$ are isomorphic.

Solution

(i) Since K is Abelian, the map $K \ni k \overset{p}{\longmapsto} k^n \in K$ is a homomorphism; moreover, p is injective, as n is coprime with the order of K, and so it is bijective because K is finite. The map ψ is the composition $\varphi \circ p$, so it is a homomorphism.
(ii) Consider the map

$$H \rtimes_\psi K \ni (h, k) \overset{f}{\longmapsto} (h, k^n) \in H \rtimes_\varphi K.$$

First of all, remark that f is bijective, as $f = \mathrm{Id}_H \times p$, and Id_H and p are bijections. What remains to be done is show that f is a homomorphism. We have

$$\begin{aligned} f\big((h_1, k_1) \circ_\psi (h_2, k_2)\big) &= f\big(h_1\psi(k_1)(h_2), k_1k_2\big) \\ &= f\big(h_1\varphi(k_1^n)(h_2), k_1k_2\big) \\ &= (h_1\varphi(k_1^n)(h_2), k_1^nk_2^n) \\ &= (h_1, k_1^n) \circ_\varphi (h_2, k_2^n) \\ &= f(h_1, k_1) \circ_\varphi f(h_2, k_2), \end{aligned}$$

which finally yields that f is an isomorphism.

Exercise 37 Let $p > q$ be primes. Show that every group with pq elements is isomorphic to some semidirect product $\mathbb{Z}/p\mathbb{Z} \rtimes \mathbb{Z}/q\mathbb{Z}$.

Solution Let G be a group with pq elements. By Cauchy's Theorem, there exist $x, y \in G$ of order p and q, respectively. Letting $H = \langle x \rangle$ and $K = \langle y \rangle$, notice that $H \cap K = \{e\}$ because p and q are relatively prime; it follows that $|HK| = |H||K|/|H \cap K| = pq = |G|$, so $HK = G$.

Moreover, if $H' \neq H$ were another subgroup of G of order p then we would have $|HH'| = |H||H'|/|H \cap H'| = p^2 > |G|$, which proves that H is the only subgroup of order p in G, and in particular that it is normal.

We have thus found two subgroups H and K such that $H \cap K = \{e\}$, $HK = G$ and H is normal in G. We can therefore conclude that G is isomorphic to a semidirect product of $H \simeq \mathbb{Z}/p\mathbb{Z}$ and $K \simeq \mathbb{Z}/q\mathbb{Z}$, with H normal.

Exercise 38 Let $p > q$ be primes. Show that if q does not divide $p - 1$ then a group with pq elements is cyclic.

Solution We know because of the previous exercise that a group with pq elements is isomorphic to a semidirect product $\mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/q\mathbb{Z}$ with respect to some $\varphi : \mathbb{Z}/q\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z}) \simeq (\mathbb{Z}/p\mathbb{Z})^*$. Now, the order of $\varphi(1)$ must divide both $q = \mathrm{ord}(1)$ and $p - 1 = |\mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})|$, but q does not divide $p - 1$, so $\varphi(1)$ must have order 1, that is, $\varphi(1) = \mathrm{Id}$. It follows that for every element $a \in \mathbb{Z}/q\mathbb{Z}$ we have $\varphi(a) = \mathrm{Id}$ and that the semidirect product is actually a direct product. Moreover, since p, q are relatively prime, the group $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$ is cyclic of order pq.

Exercise 39 Let $p > q$ be primes. Show that if q divides $p - 1$ then there exists a unique isomorphism class of non-Abelian groups of order pq.

Solution By a previous exercise, we know that a group with pq elements must be isomorphic to a semidirect product $\mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/q\mathbb{Z}$ for some homomorphism $\varphi : \mathbb{Z}/q\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z}) \simeq (\mathbb{Z}/p\mathbb{Z})^* \simeq \mathbb{Z}/(p-1)\mathbb{Z}$.

There exists a homomorphism φ such that $\varphi(1) \neq \mathrm{Id}$, as the order q of $1 \in \mathbb{Z}/q\mathbb{Z}$ divides the order $p - 1$ of $\mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$. In particular, such a homomorphism yields a non-Abelian semidirect product $\mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/q\mathbb{Z}$, and the image $\varphi(1)$ generates the unique subgroup with q elements in the cyclic group $\mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$. It follows that, given any homomorphism $\psi : \mathbb{Z}/q\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$, there exists a unique integer k such that $\psi(1) = \varphi(1)^k = \varphi(k)$. In conclusion, by Preliminary Exercise 36, the semidirect products of $\mathbb{Z}/p\mathbb{Z}$ and $\mathbb{Z}/q\mathbb{Z}$ with respect to φ and ψ are isomorphic, and so there is a unique isomorphism class of non-Abelian groups of order pq.

Exercise 40 Show that all groups of order 245 are Abelian.

Solution Let G be a group of order 245. By the First Sylow Theorem, G has a 7-Sylow subgroup, that is, a subgroup H of order 49. Moreover, by the Third Sylow Theorem, the number n_7 of subgroups of G having order 49 is congruent to 1 modulo 7 and divides the index $i_G(H) = 5$. The only possible value is thus $n_7 = 1$; in particular, since it is the only subgroup of order 49, H is normal in G. Notice that H is Abelian, because its order is the square of the prime number 7, and thus we have either $H \simeq \mathbb{Z}/49\mathbb{Z}$ or $H \simeq \mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$.

By Cauchy's Theorem, there exists an element of order 5 in G; let K be the subgroup it generates. Clearly, $H \cap K = \{e\}$, as any element in $K \setminus \{e\}$ has order 5, which does not divide $49 = |H|$. It follows that $|HK| = |H||K| = 245$ and thus $HK = G$. This proves that G is isomorphic to a semidirect product $H \rtimes_\varphi K$ for some homomorphism $\varphi : K \longrightarrow \mathrm{Aut}(H)$.

We now wish to prove that φ can only be the map sending each element of K to the neutral element Id_H of $\mathrm{Aut}(H)$. This implies that $G \simeq H \times K$, and thus that G is Abelian, because H and K are.

We have seen that either $H \simeq \mathbb{Z}/49\mathbb{Z}$ or $H \simeq \mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$. In the first case, $|\mathrm{Aut}(H)| = |(\mathbb{Z}/49\mathbb{Z})^*| = 42$, and in the second $|\mathrm{Aut}(H)| = (7^2 - 1)(7^2 - 7)$;

in both cases, 5 does not divide $|\mathrm{Aut}(H)|$. But, denoting by x a generator of K, the order r of $\varphi(x)$ must divide both the order of x, which is 5, and the order of $|\mathrm{Aut}(H)|$: we can therefore only have $r = 1$. It follows that $\varphi(x) = \mathrm{Id}_H$, hence every element of $K = \langle x \rangle$ is sent to Id_H by φ.

▌The fact that every subgroup with 49 elements is normal can also be shown without making use of the Third Sylow Theorem. Indeed, if there were two subgroups $H \neq H'$ with 49 elements then we would have $|H \cap H'| \leq 7$ and thus $|HH'| = |H||H'|/|H \cap H'| \geq 7^3 > 245 = |G|$.▐

Exercise 41 Show that all groups of order 255 are cyclic.

Solution Let G be a group of order 255 and let N be a 17-Sylow subgroup of G. The number n_{17} of 17-Sylow subgroups satisfies the congruence $n_{17} \equiv 1 \pmod{17}$ and divides the index $[G : N] = 15$ of N in G. The only possible value for it is thus $n_{17} = 1$, which implies that N, being the only 17-Sylow subgroup, is normal in G. It follows that, given a 5-Sylow subgroup K, the set $H = NK$ is a subgroup of G whose order is $|N||K|/|N \cap K| = 17 \cdot 5/1 = 85$.

Moreover, since 5 does not divide $16 = 17 - 1$, the group H is cyclic because of a previous exercise. By noting that the index of H is the smallest prime that divides the order of G, we obtain that H is normal thanks to another previous exercise.

By Cauchy's Theorem, there exists an element $g \in G$ of order 3; set $M = \langle g \rangle$ and notice that $H \cap M = \{e\}$ because 3 and 85 are relatively prime. This proves that $|HM| = |H||M|/|H \cap M| = 85 \cdot 3 = 255$, and so $HM = G$. The group G is thus isomorphic to a semidirect product $H \rtimes_\varphi M$, where φ is some homomorphism from K to $\mathrm{Aut}(H)$. The order of $\varphi(g)$ must divide both the order of g, which is 3, and the order of $\mathrm{Aut}(H) \simeq \mathrm{Aut}(\mathbb{Z}/85\mathbb{Z}) \simeq (\mathbb{Z}/85\mathbb{Z})^*$, which is 64. It follows that $\varphi(g) = \mathrm{Id}_H$ and thus that the semidirect product is a direct product, that is, $G \simeq \mathbb{Z}/85\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \simeq \mathbb{Z}/255\mathbb{Z}$.

Exercise 42 Given an additive subgroup I of a (possibly non-commutative) ring A, we say that I is a *bilateral ideal* of A if $aI \subseteq I$ and $Ia \subseteq I$ for all $a \in A$. Show that the ring $\mathrm{Mat}_{n\times n}(\mathbb{K})$ of $n \times n$ matrices with coefficients in the field $\mathbb{K}$ does not have any proper nontrivial bilateral ideals.

Solution Denote by $E_{h,k}$ the $n \times n$ matrix whose entries are all 0 except for the (h, k) entry, which is 1. When we multiply a matrix A by $E_{h,k}$ on the right, we obtain a matrix all of whose columns are zero save for the kth column, which coincides with the hth column of A. Analogously, multiplying A by $E_{h,k}$ on the left yields a matrix whose entries are all zero save for those of the hth row, which coincides with the kth row of A.

Now let $I \neq \{0\}$ be a bilateral ideal of $\mathrm{Mat}_{n\times n}(\mathbb{K})$ and let A be a nonzero element of I; we wish to show that $I = \mathrm{Mat}_{n\times n}(\mathbb{K})$.

If the entry (h, k) of A is nonzero and equal to a, then $A_1 = a^{-1}E_{1,h} \cdot A \cdot E_{k,1}$ is a matrix all of whose entries are zero except for the (1, 1) entry, whose value is 1. Moreover, $A_1 \in I$ as I is an ideal of $\mathrm{Mat}_{n\times n}(\mathbb{K})$.

Now, multiplying A_1 by $E_{h,1}$ on the left and by $E_{1,h}$ on the right, for $1 \leq h \leq n$, yields the matrix $A_h \in I$, whose (h,h) entry is equal to 1 whereas all other entries equal to 0. It follows that $\mathrm{Id} = A_1 + A_2 + \cdots + A_n \in I$ and thus that I is the whole ring $\mathrm{Mat}_{n\times n}(\mathbb{K})$.

Exercise 43 Let A be the subring $\mathbb{Q}[x^2, x^3]$ of $\mathbb{Q}[x]$. Determine whether or not each of the following ideals of A is prime and whether or not it is maximal: $I = (x^3 + 2)$, $J = (x^3 + x^2)$.

Solution Remark that $x^{2n} = (x^2)^n$, $x^{2n+3} = x^3(x^2)^n \in A$ for all $n \geq 0$, whereas $x \notin A$. It follows that A is the set of polynomials whose linear term is zero.

We show that J is not a prime ideal, and thus that it is not a maximal ideal. Indeed, the element $x^6 - x^4 = (x^3 - x^2)(x^3 + x^2)$ belongs to J, as $x^3 - x^2 \in A$; moreover, $x^6 - x^4 = x^2 \cdot x^2 \cdot (x^2 - 1)$ and $x^2, x^2 - 1 \in A$ but it is easy to see that $x^2, x^2 - 1 \notin J$.

We now show that I is a maximal ideal. First of all, letting $\tilde{I} \subseteq \mathbb{Q}[x]$ denote the ideal generated by $x^3 + 2$ in $\mathbb{Q}[x]$, we have $\tilde{I} \cap A = I$.

The fact that $I \subseteq \tilde{I} \cap A$ is obvious. Conversely, given $f(x) \in \tilde{I} \cap A$, there exists $g(x) \in \mathbb{Q}[x]$ such that $f(x) = (x^3 + 2)g(x) \in A$. The linear term of $f(x)$ must be zero since $f(x) \in A$, but it must also equal twice the linear term of $g(x)$. It follows that $g(x) \in A$, hence $f(x) \in I$.

Now set $\alpha = -\sqrt[3]{2} \in \mathbb{R}$ and consider the surjective homomorphism with kernel $\tilde{I}$ given by $\mathbb{Q}[x] \ni x \longmapsto \alpha \in \mathbb{Q}(\alpha)$. Letting φ be the restriction of this homomorphism to A, we have $\mathrm{Ker}(\varphi) = \tilde{I} \cap A = I$. Moreover, φ is surjective because $-x^4/2 \in A$ and $\varphi(-x^4/2) = -\alpha^4/2 = \alpha$; it therefore induces an isomorphism between A/I and the field $\mathbb{Q}(\alpha)$. In conclusion, I is a maximal ideal.

Exercise 44

(i) Show that, given any automorphism φ of the ring $\mathbb{Q}[x]$, there exist $a \in \mathbb{Q}^*$ and $b \in \mathbb{Q}$ such that $\varphi(x) = ax + b$.
(ii) Show that the group $\mathrm{Aut}(\mathbb{Q}[x])$ is isomorphic to the group $\mathbb{Q} \rtimes \mathbb{Q}^*$.

Solution

(i) Since $\varphi(1) = 1$ and thus $\varphi(a) = a$ for all $a \in \mathbb{Q}$, given $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ we have $\varphi(f(x)) = a_n(\varphi(x))^n + a_{n-1}(\varphi(x))^{n-1} + \cdots + a_1\varphi(x) + a_0 = f(\varphi(x))$. It follows that $\mathrm{Im}(\varphi) = \mathbb{Q}[\varphi(x)]$. But then the degree of $\varphi(x)$ must be 1, so there are $a \in \mathbb{Q}^*$ and $b \in \mathbb{Q}$ such that $\varphi(x) = ax + b$.
(ii) From (i), we know that every element of $\mathrm{Aut}(\mathbb{Q}[x])$ is of the form $\varphi_{a,b}$, where $\varphi_{a,b}(f(x)) = f(ax + b)$ for all $f(x) \in \mathbb{Q}[x]$. Consider the map

$$\mathrm{Aut}(\mathbb{Q}[x]) \ni \varphi_{a,b} \overset{F}{\longmapsto} (b,a)^{-1} = (-a^{-1}b, a^{-1}) \in \mathbb{Q} \rtimes \mathbb{Q}^*,$$

where the semidirect product $\mathbb{Q} \rtimes \mathbb{Q}^*$ is with respect to the homomorphism

$$\mathbb{Q}^* \ni a \longmapsto (b \longmapsto ab) \in \mathrm{Aut}(\mathbb{Q}).$$

The map F is clearly bijective; we now prove it is a group homomorphism. Indeed, we have

$$\begin{aligned}(\varphi_{a,b} \circ \varphi_{c,d})(x) &= \varphi_{a,b}(\varphi_{c,d}(x)) \\ &= \varphi_{a,b}(cx + d) \\ &= c(ax + b) + d \\ &= acx + bc + d \\ &= \varphi_{ac,bc+d}(x)\end{aligned}$$

and thus

$$\begin{aligned}F(\varphi_{a,b} \circ \varphi_{c,d}) &= F(\varphi_{ac,bc+d}) \\ &= (-a^{-1}c^{-1}(bc + d), a^{-1}c^{-1}) \\ &= (-a^{-1}b - a^{-1}c^{-1}d, a^{-1}c^{-1}) \\ &= (-a^{-1}b, a^{-1})(-c^{-1}d, c^{-1}) \\ &= F(\varphi_{a,b})F(\varphi_{c,d}).\end{aligned}$$

Exercise 45 Let A be a ring and let $P = (x)$ be a prime ideal of A. Show that if Q is a prime ideal of A that is strictly contained in P, then $Q \subseteq \cap_{n>0} P^n$.

Solution First of all, remark that x does not belong to Q: if it did, then we would have $P = (x) \subseteq Q$ and thus $Q = P$, which contradicts the fact that Q is a proper subset of P.

Given any element $y \in Q \subseteq P = (x)$, we have $y = y_1 x$ for some $y_1 \in A$. Since Q is prime, this implies that $y_1 \in Q$ or $x \in Q$; but we cannot have $x \in Q$, so we must have $y_1 \in Q$. Repeating the same argument on y_1 yields $y_2 \in Q$ such that $y_1 = y_2 x$, hence $y = y_2 x^2 \in P^2$. In the same way, there exists $y_3 \in Q$ such that $y_2 = y_3 x$ and thus $y = y_3 x^3 \in P^3$, and so on. We thus obtain that $y \in P^n$ for all n, and therefore $y \in \cap_{n>0} P^n$.

Exercise 46 Let $A = C^0[0, 1]$ denote the ring of continuous functions[3] on $[0, 1]$ and let M be a maximal ideal of A. Show that there exists $\gamma \in [0, 1]$ such that $M = \{f \in A \mid f(\gamma) = 0\}$.

Solution Consider the set $Z = \left\{\alpha \in [0, 1] \mid f(\alpha) = 0 \text{ for all } f \in M\right\} \subseteq [0, 1]$ and, for $\alpha \in [0, 1]$, define $I(\alpha) = \left\{f \in A \mid f(\alpha) = 0\right\}$. The ideal $I(\alpha)$ is proper for all α, as it does not contain the constant function $x \longmapsto 1$.

[3] This exercise and its solution require some basic notions of topology. See for example: M. Manetti, *"Topology"*, Springer, 2015.

Notice that if there exists $\gamma \in Z$, then $M \subseteq I(\gamma)$ and, since M is maximal, we must have $M = I(\gamma)$, that is, γ satisfies the problem statement. It is therefore enough to prove that Z is nonempty.

We proceed by contradiction. Assuming $Z = \varnothing$, for every $\alpha \in [0, 1]$ there exists $f_\alpha \in M$ such that $f_\alpha(\alpha) \neq 0$. Moreover, since f_α is continuous, there exists $\epsilon_\alpha > 0$ such that $f_\alpha(x) \neq 0$ for all x in the open set $U_\alpha = (\alpha - \epsilon_\alpha, \alpha + \epsilon_\alpha) \cap [0, 1]$.

The family $\{U_\alpha \mid \alpha \in [0, 1]\}$ is an open cover of $[0, 1]$. By compactness of $[0, 1]$, we know that there are $\alpha_1, \alpha_2, \cdots, \alpha_m \in [0, 1]$ such that

$$[0, 1] = U_{\alpha_1} \cup U_{\alpha_2} \cup \cdots \cup U_{\alpha_m}.$$

Consider now the continuous function $f(x) = \sum_{h=1}^{m} f_{\alpha_h}(x)^2$; we have $f(x) \in M$ because $f_{\alpha_h} \in M$ for $h = 1, \ldots, m$ and M is an ideal.

Given $x \in [0, 1]$, there exists h such that $x \in U_{\alpha_h}$. The fact that $f_{\alpha_h}(x) \neq 0$ yields that $f(x) \geq f_{\alpha_h}(x)^2 > 0$. It follows that f is a continuous function that is never zero on $[0, 1]$, so $1/f(x)$ is also a continuous function on $[0, 1]$. But then we can write $1 = (1/f(x)) \cdot f(x) \in M$, which contradicts the fact that M is a proper ideal.

Exercise 47 Find all integer solutions of the equation $x^2 - 2y^2 = \pm 1$.

Solution Let A be the ring $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\} \subseteq \mathbb{R}$. Consider the following automorphism of A

$$A \ni u = a + b\sqrt{2} \longmapsto \overline{u} = a - b\sqrt{2} \in A.$$

and define the *norm* of an element of A as

$$A \ni u = a + b\sqrt{2} \overset{N}{\longmapsto} u\overline{u} = a^2 - 2b^2 \in \mathbb{Z}.$$

Because $z \longmapsto \overline{z}$ is an automorphism, it is easy to prove that $N(uv) = N(u)N(v)$ for all $u, v \in A$. We wish to show that $A^* = \{u \in A \mid N(u) = \pm 1\}$.

But, indeed, if $u \in A^*$ then there exists $v \in A$ such that $uv = 1$. It follows that $N(u)N(v) = 1$ and, since the norm takes values in $\mathbb{Z}$, we find that $N(u) = \pm 1$. Conversely, given $u \in A$ such that $N(u) = \pm 1$, set $v = N(u)\overline{u}$. We have $uv = uN(u)\overline{u} = N(u)^2 = 1$, that is, v is an inverse of u in A and therefore $u \in A^*$.

From this description of the invertible elements of A, one immediately obtains that a pair (a, b) is a solution to the equation in the problem statement if and only if $u = a + b\sqrt{2}$ is a unit of A.

We now intend to show that we have the following group isomorphism

$$\begin{aligned} \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z} &\xrightarrow{\mu} A^* \\ (h, n) &\longmapsto (-1)^h \gamma^n \end{aligned}$$

where γ denotes the unit $1+\sqrt{2}$. It is clear that μ is a group homomorphism, and it is injective because γ is not a root of unity. In order to prove that μ is also surjective, we show that A^* does not contain any elements that are strictly greater than 1 and strictly smaller than γ.

Suppose by contradiction that $u = a+b\sqrt{2}$ is such an element. If $a < 0$ then we cannot also have $b \leq 0$, as u is positive, so $b > 0$. The fact that $u > 1$ then implies $0 > a > 1 - b\sqrt{2}$ and thus $\pm 1 = a^2 - 2b^2 < (1-b\sqrt{2})^2 - 2b^2 = 1 - 2b\sqrt{2}$. But this inequality implies that $b < 1/\sqrt{2}$ and therefore, since b is an integer, $b \leq 0$, which we have already excluded.

Now suppose that $a > 0$. We cannot have $b > 0$ or we would have $u \geq \gamma$. If $b = 0$ then $N(u) = a^2 = \pm 1$ implies $u = a = 1$, but we know that $u > 1$. The only possibility is therefore that $b < 0$. From the fact that $N(u) = u\overline{u} = \pm 1$, taking absolute values in $\mathbb{R}$, we obtain $|u||\overline{u}| = 1$; since $|u| = u > 1$, this yields $\overline{u} \leq |\overline{u}| < 1$. But then the fact that $a > 0$ and $b < 0$ implies $\overline{u} = a - \sqrt{2}b > 1$, a contradiction. This concludes the proof that A^* does not contain any elements strictly between 1 and γ.

Now consider $u \in A^*$ and suppose that $u \geq 1$. Since $\gamma > 1$, there exists n such that $1 \leq u\gamma^{-n} < \gamma$. From the fact that $u\gamma^{-n} \in A^*$, by our arguments above, we must have $u\gamma^{-n} = 1$, that is, $u = \gamma^n \in \mathrm{Im}(\mu)$.

On the other hand, if $0 < u < 1$ then $u^{-1} > 1$, so $u^{-1} \in \mathrm{Im}(\mu)$ as before, which again yields $u \in \mathrm{Im}(\mu)$. Finally, if $u < 0$ then $-u > 0$, $-u \in A^*$ and so $-u \in \mathrm{Im}(\mu)$, hence $u \in \mathrm{Im}(\mu)$.

Since we have described all elements of A^* via the isomorphism μ, we are now able to find all integer solutions of the initial equation.

▌The Diophantine equation $x^2 - 2y^2 = \pm 1$ is called *Pell's equation*. Pell's equation and some of its variants have been studied since ancient times by Greeks, Indians and Arabs, and have subsequently been a focal point in the work of Fermat, Euler and Lagrange; see for example: A. Weil, *"Basic Number Theory"*, Springer, 1967.▐

Exercise 48 Let I, J be ideals of the ring A. Show that $(I : J) = \{a \in A \mid aJ \subseteq I\}$ is an ideal of A.

Solution We obviously have $0 \cdot J = 0 \subseteq I$, so $(I : J)$ is nonempty. Given $a, b \in (I : J)$ we have $(a+b)J = aJ + bJ \subseteq I + I = I$, hence $a + b \in (I : J)$. This proves that $(I : J)$ is a subset of A that is closed under addition. Finally, for all $c \in A$ we have $(ca)J = c(aJ) \subseteq cI \subseteq I$, which proves that $(I : J)$ is an ideal of A.

Exercise 49 Let S be a multiplicative set in a ring A and let I be an ideal of A that does not intersect S. Prove that there exists a prime ideal of A that contains I and does not intersect S.

Solution Consider the family $\mathcal{F}$ of all ideals J of the ring A that contain I and do not intersect S; this family is nonempty, as $I \in \mathcal{F}$. Now let $J_1 \subseteq J_2 \subseteq \cdots$ be a

chain in $\mathcal{F}$. Remark that $J = \cup_n J_n$ is an ideal and that it contains I; moreover, $J \cap S = \cup_n (J_n \cap S) = \varnothing$. This proves that every chain of $\mathcal{F}$ has an upper bound and thus, by Zorn's Lemma, the existence of a maximal element P in $\mathcal{F}$.

By the definition of $\mathcal{F}$, we have that P contains I and does not intersect S; we now show that P is prime. Let $x, y \in A$ be such that $xy \in P$ and suppose by contradiction that $x \notin P$ and $y \notin P$. Since $P \subsetneq (x, P)$, we must have $(x, P) \notin \mathcal{F}$; but $I \subseteq (x, P)$, so we must have $S \cap (x, P) \neq \varnothing$. It follows that there exist $s_1 \in S$, $a_1 \in A$ and $p_1 \in P$ such that $s_1 = a_1 x + p_1$. An analogous argument for (y, P) yields $s_2 \in S$, $a_2 \in A$ and $p_2 \in P$ such that $s_2 = a_2 y + p_2$.

But then $s_1 s_2 = a_1 a_2 xy + a_2 y p_1 + a_1 x p_2 + p_1 p_2$ is an element of $P \cap S$, which contradicts the fact that $P \cap S = \varnothing$.

Exercise 50 Let A be a ring, $I \subseteq A$ an ideal, and $\sqrt{I} = \{x \in A \mid \text{there exists } n \in \mathbb{N} \text{ such that } x^n \in I\}$ the radical of I. Show that

(i) the set $\sqrt{I}$ is an ideal of A;
(ii) for all pairs of ideals I, J of A we have $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$;
(iii) for all pairs of ideals I, J of A we have $\sqrt{\sqrt{I} + \sqrt{J}} = \sqrt{I + J}$;
(iv) the radical $\sqrt{I}$ is the intersection of all prime ideals of A that contain I;
(v) if P is a prime ideal of A then $\sqrt{P^n} = P$ for all natural numbers $n > 0$.

Finally, for each integer m, compute $\sqrt{(m)}$ in the ring $\mathbb{Z}$.

Solution

(i) First of all, remark that $\sqrt{I}$ is nonempty, as $0 = 0^1 \in I$. Now consider $x \in \sqrt{I}$ and $a \in A$. If $x^n \in I$ then we also have $(ax)^n = a^n x^n \in I$ because I is an ideal, so $ax \in \sqrt{I}$.
Given $x, y \in \sqrt{I}$, we now show that $x + y \in \sqrt{I}$. Let $n, m \geq 0$ be such that $x^n, y^m \in I$ and let $k = n + m - 1$. We have $(x + y)^k = \sum_{h=0}^{k} \binom{k}{h} x^h y^{k-h}$. If $h \geq n$ then, since I is an ideal, $\binom{k}{h} x^h y^{k-h} \in I$, because $x^h = x^{h-n} x^n \in I$. If $h < n$ then $k - h > k - n \geq m - 1$ and thus $\binom{k}{h} x^h y^{k-h} \in I$, because $y^{k-h} = y^{k-h-m} y^m \in I$. It follows that every summand in the development of $(x+y)^k$ belongs to I, and so $(x+y)^k \in I$ because I is an ideal; consequently, we have $x + y \in \sqrt{I}$. This concludes the proof that I is an ideal of A.
(ii) First of all, we have $IJ \subseteq I \cap J$, so $\sqrt{IJ} \subseteq \sqrt{I \cap J}$.
Moreover, given $x \in \sqrt{I \cap J}$, there exists a natural number n such that $x^n \in I \cap J$, hence $x^n \in I$ and $x^n \in J$. It follows that $x \in \sqrt{I}$ and $x \in \sqrt{J}$, that is, $x \in \sqrt{I} \cap \sqrt{J}$.
Conversely, given $x \in \sqrt{I} \cap \sqrt{J}$, there exist $n, m \geq 0$ such that $x^n \in I$, $x^m \in J$, so $x^{n+m} = x^n x^m \in IJ$, hence $x \in \sqrt{IJ}$.
We have thus shown the inclusions $\sqrt{IJ} \subseteq \sqrt{I \cap J} \subseteq \sqrt{I} \cap \sqrt{J} \subseteq \sqrt{IJ}$, which imply that the three sets coincide.
(iii) One inclusion is obvious: the inclusions $I \subseteq \sqrt{I}$, $J \subseteq \sqrt{J}$ imply $\sqrt{I + J} \subseteq \sqrt{\sqrt{I} + \sqrt{J}}$.

Conversely, given $x \in \sqrt{\sqrt{I}+\sqrt{J}}$, there exist $n \geq 0$, $y \in \sqrt{I}$ and $z \in \sqrt{J}$ such that $x^n = y + z$. Moreover, there exist $h, k \geq 0$ such that $y^h \in I$ and $z^k \in J$. We set $t = h + k - 1$ and consider $x^{nt} = (y+z)^t = \sum_{j=0}^{t} \binom{t}{j} y^j z^{t-j}$. If $j < h$ then $t - j \geq k$ and so every summand of $(y+z)^t$ belongs to $I + J$. In conclusion, $x \in \sqrt{I+J}$.

(iv) Consider $x \in \sqrt{I}$. There exists an integer $n \geq 0$ such that $x^n \in I$, so, given a prime ideal P containing I, we have $x^n \in P$ and thus $x \in P$. This proves that $\sqrt{I}$ is contained in the intersection of all prime ideals containing I.
Conversely, suppose $x \notin \sqrt{I}$ and denote by S the set $\{x^n \mid n \geq 0\}$. Notice that S is a multiplicative set of A and that $S \cap I = \varnothing$ by definition of $\sqrt{I}$. By the previous exercise, there exists a prime ideal P that contains I and does not intersect S. But $x \in S$, so $x \notin P$. This proves that x does not belong to the intersection of all prime ideals containing I.

(v) Using (ii), we have $\sqrt{P^n} = \sqrt{P} \cap \cdots \cap \sqrt{P} = \sqrt{P}$, so (iv) yields $\sqrt{P} = P$ because P is a prime ideal.

In order to compute the radical of (m) in $\mathbb{Z}$, denote by $p_1, p_2, \ldots, p_n$ the distinct prime factors of m; equivalently, $(p_1), (p_2), \ldots, (p_n)$ are the prime ideals of $\mathbb{Z}$ that contain (m). From (iii), we have that $\sqrt{(m)} = (p_1) \cap (p_2) \cap \cdots \cap (p_n) = (p_1 p_2 \ldots p_n)$.

Exercise 51 Let A, B be integral domains and let $f : A \longrightarrow B$ be a ring homomorphism. Let S be a multiplicative set of A such that $f(s) \in B^*$ for all $s \in S$. Show that there exists a unique homomorphism $\tilde{f} : S^{-1}A \longrightarrow B$ that extends f and determine its kernel.

Solution We first show that, if a homomorphism $\tilde{f}$ with the required property exists, then it is unique. Indeed, given $a \in A$ and $s \in S$, we have $f(a) = \tilde{f}(a/1) = \tilde{f}(s \cdot a/s) = \tilde{f}(s)\tilde{f}(a/s) = f(s)\tilde{f}(a/s)$ and so, since $f(s) \in B^*$, we can multiply by $f(s)^{-1}$ to obtain $\tilde{f}(a/s) = f(s)^{-1} f(a)$.

We now show that such a homomorphism does exist. We have found that we necessarily have $\tilde{f}(a/s) = f(s)^{-1} f(a)$; we prove that this map is a well-defined ring homomorphism and that it extends f.

If $a/s = b/t$ for some $a, b \in A$ and $s, t \in S$, then $at = bs$ and so $f(a)f(t) = f(b)f(s)$, hence $f(s)^{-1} f(a) = f(t)^{-1} f(b)$. This proves that $\tilde{f}$ is well-defined.

Given $a, b \in A$ and $s, t \in S$ we have

$$\begin{aligned}
\tilde{f}(a/s + b/t) &= \tilde{f}\big((ta + sb)/st\big) \\
&= f(st)^{-1} f(ta + sb) \\
&= f(s)^{-1} f(t)^{-1} \big(f(t)f(a) + f(s)f(b)\big) \\
&= f(s)^{-1} f(a) + f(t)^{-1} f(b) \\
&= \tilde{f}(a/s) + \tilde{f}(b/t)
\end{aligned}$$

and also

$$\begin{aligned}\tilde{f}(a/s \cdot b/t) &= \tilde{f}\big(ab/(st)\big)\\ &= f(st)^{-1} f(ab)\\ &= f(s)^{-1} f(t)^{-1} f(a) f(b)\\ &= \tilde{f}(a/s)\tilde{f}(b/t).\end{aligned}$$

It follows that $\tilde{f}$ is a ring homomorphism. Moreover, $\tilde{f}$ extends f as $\tilde{f}(a/1) = f(1)^{-1} f(a) = 1 \cdot f(a) = f(a)$.

An element a/s belongs to the kernel of $\tilde{f}$ if and only if $f(s)^{-1} f(a) = 0$ in B. But $f(s)^{-1}$ is an invertible element of B, so $\tilde{f}(a/s) = 0$ if and only if $f(a) = 0$. We therefore have $\mathrm{Ker}(\tilde{f}) = S^{-1}\mathrm{Ker}(f)$.

Exercise 52 Show that there are no surjective homomorphism $f : \mathbb{Z}[x] \longrightarrow \mathbb{Q}$; letting S denote the set of odd integers, construct a surjective homomorphism $f : S^{-1}\mathbb{Z}[x] \longrightarrow \mathbb{Q}$.

Solution A ring homomorphism f from $\mathbb{Z}[x]$ to $\mathbb{Q}$ is uniquely determined once the image α/β of x is fixed. Indeed, note that $f(1) = 1$ implies that $f(a) = a$ for all $a \in \mathbb{Z}$. Moreover, any element of $\mathbb{Z}[x]$ is a polynomial of some degree n, that is, an expression of the form $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$, where $a_n, a_{n-1}, \ldots, a_1, a_0 \in \mathbb{Z}$, and we have $f(p(x)) = f(a_n x^n) + \cdots + f(a_1 x) + f(a_0) = a_n f(x)^n + \cdots + a_1 f(x) + a_0 = p(\alpha/\beta)$. It follows that every nonzero homomorphism f consists in evaluating polynomials at some $\alpha/\beta \in \mathbb{Q}$.

We show that f cannot be surjective. Indeed,

$$\begin{aligned}f(p(x)) &= p(\alpha/\beta)\\ &= (a_n\alpha^n + a_{n-1}\alpha^{n-1}\beta + \cdots + a_0\beta^n)/\beta^n \in \tfrac{1}{\beta^n}\mathbb{Z}.\end{aligned}$$

The image of the degree n polynomials is therefore contained $\frac{1}{\beta^n}\mathbb{Z}$, from which we deduce that $\mathrm{Im}(f) \subseteq \cup_{n\geq 0}\frac{1}{\beta^n}\mathbb{Z}$. This union is not all of $\mathbb{Q}$: given a prime p that does not divide β, the element $1/p$ does not belong to the union.

We now construct a surjective map $S^{-1}\mathbb{Z}[x] \longrightarrow \mathbb{Q}$. Define $f : \mathbb{Z}[x] \longrightarrow \mathbb{Q}$ by setting $f(p(x)) = p(1/2)$; the image of S via f is contained in $\mathbb{Q}^*$.

It is thus easy to show that the map given by $\tilde{f}(p(x)/d) = p(1/2)/d$ is a ring homomorphism. Moreover, since any a/b in $\mathbb{Q}$ can be written as $a/(2^n d)$ where d is odd, we have $a/b = f(ax^n/d)$. This proves that the homomorphism $\tilde{f}$ is surjective.

Exercise 53 Let P be a prime ideal in an integral domain A.

(i) Prove that $S = A \setminus P$ is a multiplicative set of A.
(ii) Letting A_P denote the ring of fractions $S^{-1}A$, prove that proper ideals of A_P correspond to ideals of A that are contained in P.
(iii) Show that $S^{-1}P$ is the only maximal ideal of A_P.

Solution

(i) Since P is prime, $P \neq A$ and so $1 \notin P$, hence $1 \in S$. Moreover, $0 \in P$ and so $0 \notin S$. Given $s, t \in S$, that is, $s, t \notin P$, we cannot have $st \in P$ by the definition of a prime ideal, so we have $st \in S$. We have thus shown that S is a multiplicative set of A.

(ii) In general, given a multiplicative set S of A, the correspondence $I \longrightarrow S^{-1}I$ is a bijection between the set of ideals of A that do not intersect S and the set of proper ideals of $S^{-1}A$.
The problem statement follows from the fact that, letting $S = A \setminus P$, an ideal I of A does not intersect S if and only if it is contained in P.

(iii) The statement follows immediately from (ii), as P is the largest ideal of A that is contained in P and so $S^{-1}P$ is the largest proper ideal of A_P, that is, its only maximal ideal.

Exercise 54 Let $S = \{f \in \mathbb{Q}[x] \mid f \text{ does not have any rational roots}\} \subseteq \mathbb{Q}[x]$. Prove that

(i) S is a multiplicative set of $\mathbb{Q}[x]$;
(ii) $A = S^{-1}\mathbb{Q}[x]$ is a principal ideal domain;
(iii) an ideal M of A is maximal if and only if $A/M \simeq \mathbb{Q}$.

Solution

(i) It is clear that $1 \in S$ because the constant polynomial 1 has no roots; moreover, the constant polynomial 0 has, for example, the rational root 0, so $0 \notin S$. If $f(x), g(x) \in S$ then the polynomial $f(x)g(x)$ has no rational roots because the set of its roots is the union of the set of roots of $f(x)$ and that of $g(x)$.

(ii) We know that ideals of A are of the form $S^{-1}I$, where I is an ideal of $\mathbb{Q}[x]$. But $\mathbb{Q}[x]$ is a Euclidean domain, and therefore a principal ideal domain; it follows that, if $I = f(x)$, then $S^{-1}I = S^{-1}(f(x)) = A \cdot f(x)$, and in particular is a principal ideal.

(iii) If $A/M \simeq \mathbb{Q}$ then M is maximal because $\mathbb{Q}$ is a field.
Conversely, let M be a maximal ideal of A; since A is a principal ideal domain, we have $M = A \cdot f(x)$ and, since A is a localisation of $\mathbb{Q}[x]$, we may assume that $f(x) \in \mathbb{Q}[x]$.
We now wish to prove that $f(x)$ has a rational root. If this were not the case than we would have $f(x) \in S$, so $1 = \big(1/f(x)\big) \cdot f(x) \in M$, which contradicts the fact that M, being maximal, is proper.
So let $a \in \mathbb{Q}$ be a root of $f(x)$; we have $f(x) = (x - a)g(x)$ for some $g(x) \in \mathbb{Q}[x]$ by Ruffini's Theorem. Observe that $M = (f(x)) \subseteq (x - a) \neq A$ and, since M is maximal, we have $M = (x - a)$. Consider the map $A \ni f(x)/g(x) \longmapsto f(a)/g(a) \in \mathbb{Q}$. First of all, this map is well-defined: $g(a) \neq 0$ because g has no rational roots, given that $g(x) \in S$. The map is certainly surjective and $M = (x - a) \subseteq \mathrm{Ker}(f)$. In conclusion, $M = \mathrm{Ker}(f)$ because M is maximal. We have thus shown that $A/M \simeq \mathbb{Q}$.

Exercise 55 Show that in the ring $\mathbb{Z}[\sqrt{-5}]$ the element 2 is irreducible but not prime.

Solution In $A = \mathbb{Z}[\sqrt{-5}]$ we have

$$2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

but 2 does not divide either of the factors of the right-hand side. This proves that 2 is not prime.

In order to show that 2 is irreducible, define the map $A \ni u = a + b\sqrt{-5} \longmapsto \overline{u} = a - b\sqrt{-5} \in A$ and observe that it is an automorphism of the ring A, so the map

$$A \ni u = a + b\sqrt{-5} \overset{N}{\longmapsto} u \cdot \overline{u} = a^2 + 5b^2 \in \mathbb{N}$$

is multiplicative. Moreover, if $u \in A$ is such that $N(u) = 1$, then $u = \pm 1$ and in particular u is invertible in A.

Now remark that if $2 = u \cdot v$ in A, then we must have the equalities between integers $4 = N(2) = N(u)N(v)$. It follows that the only possible values for $N(u)$ are $1,\ 2,\ 4$, and the corresponding values for $N(v)$ are $4,\ 2,\ 1$. Consequently, either u is invertible, or v is invertible, or $N(u) = N(v) = 2$. But there is no pair of integers $a,\ b$ such that $a^2 + 5b^2 = 2$, so no element u of A is such that $N(u) = 2$. This concludes the proof that 2 is irreducible in A.

Exercise 56 Let $\mathbb{K}$ be a field and let $A = \mathbb{K}[[x]]$ be the ring of formal power series in x with coefficients in $\mathbb{K}$. Show that

(i) $f(x) = \sum_{m \geq 0} a_m x^m$ is invertible in A if and only if $a_0 \neq 0$;
(ii) the inverse of $1 - x$ is $\sum_{m \geq 0} x^m$;
(iii) (x) is the only maximal ideal of A;
(iv) the ideals of A are (x^k) for $k = 0, 1, \ldots$

Solution

(i) The formal power series $f(x) = \sum_{m \geq 0} a_m x^m$ is invertible if and only if there exists a formal power series $g(x) = \sum_{n \geq 0} b_n x^n$ such that $f(x)g(x) = 1$, that is, such that $\sum_{h \geq 0} (\sum_{k=0}^{h} a_k b_{h-k}) x^h = 1$ in A. This equation in $\mathbb{K}[[x]]$ can be rewritten as a system of the following equations in $\mathbb{K}$; (E_0) : $a_0 b_0 = 1$ and (E_h) : $\sum_{k=0}^{h} a_k b_{h-k} = 0$ for $h \geq 1$.
Clearly, (E_0) implies that if $a_0 = 0$ then $f(x)$ cannot be invertible. Conversely, suppose $a_0 \neq 0$; we will prove that we can find suitable elements $b_0, b_1, b_2, \ldots$ by induction. Indeed, (E_0) yields $b_0 = a_0^{-1}$ and, assuming we have found $b_0, b_1, \ldots, b_{h-1}$, the equation (E_h) implies that $a_0 b_h + a_1 b_{h-1} + a_2 b_{h-2} + \cdots + a_{h-1} b_1 + a_h b_0 = 0$, hence $b_h = -a_0^{-1}(a_1 b_{h-1} + a_2 b_{h-2} + \cdots + a_{h-1} b_1 + a_h b_0)$. We have thus constructed $g(x)$.

(ii) We have $(1-x)\sum_{m\geq 0} x^m = \sum_{m\geq 0} x^m - \sum_{m\geq 0} x^{m+1} = 1$.
(iii) If $f(x) \notin (x)$ then $f(x) = a_0 + xg(x)$ with $a_0 \neq 0$. We have already shown that $f(x)$ is then invertible. It follows that $(x) \neq A$ is the only maximal ideal of A.
(iv) We know that A is a Euclidean domain where the degree of $\sum_{n\geq 0} a_n x^n \neq 0$ is the smallest n such that $a_n \neq 0$. In particular, A is a principal ideal domain. Let $I = (f(x))$ be an ideal and let $n = \deg(f)$. We have $f(x) = x^n \tilde{f}(x)$ for some invertible $\tilde{f}(x)$, thus $I = (f(x)) = (x^n \tilde{f}(x)) = (x^n)$.

Exercise 57 Let A be a unique factorisation domain, let x be an irreducible element of A and let $P = (x)$ be the ideal generated by x. Show that

(i) $S = A \setminus P$ is a multiplicative set of A,
(ii) $S^{-1}A$ is a Euclidean domain.

Solution

(i) Since A is a unique factorisation domain, x is prime and P is a prime ideal; it follows that $S = A \setminus P$ is a multiplicative set. From now on, we denote by A_P the localisation $S^{-1}A$.
(ii) Given an element $a/b \in A_P \setminus \{0\}$, define $\deg(a/b)$ as the exponent of x in the factorisation of a as a product of irreducible elements. We wish to show that A_P is a Euclidean domain with respect to this degree map.
First, we show that the map $\deg : A_P \setminus \{0\} \longrightarrow \mathbb{N}$ is well-defined. If $a/b = c/d$ in A_P then $ad = bc$ in A. Note that x does not divide b or d, because $b,\ d \in S$, so, thanks to the factorisation's uniqueness, the exponent of x in the factorisation of a must be the same as in the factorisation of c. We have thus shown that $\deg(a/b) = \deg(c/d)$.
It is clear that $\deg(a/b \cdot c/d) = \deg(a/b) + \deg(c/d) \geq \deg(c/d)$ for all pairs of elements $a/b,\ c/d$ in $A_P \setminus \{0\}$.
We now prove that, given a/b and $c/d \neq 0$ in A_P, there exist α and ρ in A_P, with $\rho = 0$ or $\deg(\rho) < \deg(c/d)$, such that $a/b = \alpha \cdot c/d + \rho$. Letting $n = \deg(a/b)$ and $m = \deg(c/d)$, we have $a = x^n a'$ and $c = x^m c'$, where x does not divide a' or c'. Remark that $a'/b \cdot d/c'$ is an element of A_P because x does not divide bc'. Thus, if $n \geq m$ then we set $\alpha = b/a' \cdot d/c' x^{n-m}$ and $\rho = 0$, and if $n < m$ then we set $\alpha = 0$ and $\rho = a/b$. This concludes the proof that A_P is a Euclidean domain.

Exercise 58 Let J be a nonzero ideal of the ring $\mathbb{Z}[i]$ of Gaussian integers. Show that

(i) $J \cap \mathbb{Z} \neq \{0\}$;
(ii) $\mathbb{Z}[i]/J$ is a finite ring;
(iii) if $J = (1+3i)$ then $\mathbb{Z}[i]/J$ is isomorphic to $\mathbb{Z}/10\mathbb{Z}$.

Solution

(i) Since $J \neq 0$, there exists $z = a + bi \in J$ such that $z \neq 0$, that is, such that $0 \neq a^2 + b^2 = |z|^2 = \overline{z}z \in J \cap \mathbb{Z}$.

(ii) The ring $\mathbb{Z}[i]$ is a Euclidean domain with respect to the degree given by the norm $a+bi \longmapsto N(a+bi) = |a+bi|^2 = a^2+b^2$, so in particular every ideal is principal. Consequently, there exists $w \in \mathbb{Z}[i] \setminus \{0\}$ such that $J = (w)$.
Given $z \in \mathbb{Z}[i]$, let $z = qw + r$ be the division of z by w; whether or not there is a remainder, we have $|r|^2 < |w|^2$. Remark that $z - r \in J$, so $z + J = r + J$ in $\mathbb{Z}[i]/J$. We have thus shown that every element of $\mathbb{Z}[i]/J$ is of the form $r + J$, with $|r|^2 < |w|^2$. But there is only a finite number of such elements r: indeed, if $r = a + bi$ then the integers a and b must satisfy $a^2 + b^2 < |w|^2$.

(iii) Let $f : \mathbb{Z}[i] \longrightarrow \mathbb{Z}/10\mathbb{Z}$ be the map given by $f(a + bi) = a + 3b$. Let us check that f is a ring homomorphism. We have

$$\begin{aligned} f\big((a + bi) + (c + di)\big) &= f\big((a + c) + (b + d)i\big) \\ &= a + c + 3(b + d) \\ &= (a + 3b) + (c + 3d) \\ &= f(a + bi) + f(c + di) \end{aligned}$$

and so f is a homomorphism of Abelian groups. Moreover,

$$\begin{aligned} f\big((a + bi)(c + di)\big) &= f\big((ac - bd) + (ad + bc)i\big) \\ &= (ac - bd) + 3(ad + bc) \\ &= ac + 9bd + 3ad + 3bc \\ &= (a + 3b)(c + 3d) \\ &= f(a + bi)\, f(c + di), \end{aligned}$$

so f is indeed a ring homomorphism.
Clearly, f is surjective, given that $f(1) = 1$. We now wish to show that $\mathrm{Ker}(f) = J$. We have $f(1 + 3i) = 1 + 3 \cdot 3 = 10 = 0$ and thus, since $\mathrm{Ker}(f)$ is an ideal, $J \subseteq \mathrm{Ker}(f)$.
Conversely, if $f(a + bi) = 0$ then we have $a + 3b = 0$ in $\mathbb{Z}/10\mathbb{Z}$, so $a \equiv -3b \pmod{10}$. It follows that there exists $h \in \mathbb{Z}$ such that $a = -3b + 10h$, hence $a + bi = -3b + 10h + bi = b(1 + 3i)i + (1 - 3i)(1 + 3i)h = (bi + (1 - 3i)h)(1 + 3i) \in J$.
This proves that $\mathrm{Ker}(f) = J$ and thus that $\mathbb{Z}[i]/J \simeq \mathbb{Z}/10\mathbb{Z}$.

Exercise 59

(i) Show that a prime p in $\mathbb{Z}$ is the sum of two squares if and only if $p \equiv 1 \pmod{4}$ or $p = 2$.

(ii) Show that the factorisation of a prime p of $\mathbb{Z}$ as an element of $\mathbb{Z}[i]$ is as follows:

(a) if $p = 2$ then $p = (-i)(1 + i)^2$ and $1 + i$ is prime in $\mathbb{Z}[i]$;

(b) if $p \equiv 1 \pmod 4$ then $p = wz$ for some w and z that are not associates in $\mathbb{Z}[i]$;
(c) if $p \equiv 3 \pmod 4$ then p is prime in $\mathbb{Z}[i]$.

(iii) Show that an element $w = a + bi \in \mathbb{Z}[i]$ is prime if and only if one of the following holds: $a^2 + b^2 = p$ for some prime $p \in \mathbb{Z}$, in which case $p \equiv 1 \pmod 4$ or $p = 2$; or $a^2 + b^2 = p^2$ for some prime p with $p \equiv 3 \pmod 4$, in which case w is an associate of the prime p of $\mathbb{Z}$.

Solution We know that $\mathbb{Z}[i]$ is a Euclidean domain with degree given by the norm $a+bi \longmapsto N(a+bi) = a^2+b^2$; in other words, the degree of an element is simply given by its squared norm as a complex number. It is then clear that $N(wz) = N(w)N(z)$ for all w, z in $\mathbb{Z}[i]$. Moreover, recall that the invertible elements are ± 1 and $\pm i$.

(i) The prime $2 = 1^2 + 1^2$ is clearly the sum of two squares. If p is a prime congruent to 1 modulo 4, then -1 is a square modulo p. Let $a \in \mathbb{Z}$ be such that $a^2 \equiv -1 \pmod p$, so that p divides $a^2 + 1 = (a + i)(a - i)$.
If p were prime in $\mathbb{Z}[i]$, then we would have $p \mid a + i$ or $p \mid a - i$. In fact, p would have to divide both elements, as p is real and $a - i$ is the conjugate of $a + i$; it follows that p would divide $(a + i) - (a - i) = 2i$ in $\mathbb{Z}[i]$, which is not possible because p is odd.
Consequently, p is not prime and thus not irreducible, since we are in a Euclidean domain and therefore in a unique factorisation domain. That is, we can write $p = wz$, where w and z are not invertible. By considering the norms, we find that $p^2 = N(p) = N(w)N(z)$. In the positive integers, p^2 can be expressed as the product of two factors only as $p^2 \cdot 1$, $p \cdot p$ and $1 \cdot p^2$. The first and last decompositions do not satisfy our requirements, as they would make w or z invertible. It follows that the norm of $w = u + iv$ is p, so $p = u^2 + v^2$ is a sum of two squares.
Finally, the squares of the integers modulo 4 are 0 and 1, so if $p \equiv 3 \pmod 4$ then it is not the sum of two squares: in this case, there are no elements of norm p.

(ii) First of all, remark that, if $N(w) = p$ is a prime, then w is prime. Indeed, if $w = \alpha \cdot \beta$ then we have $p = N(w) = N(\alpha)N(\beta)$, so one among α and β has norm 1 and is thus invertible.
Since $N(1 + i) = 2$, the expression $2 = (-i)(1 + i)^2$ is a factorisation of 2 in $\mathbb{Z}[i]$, which proves (a).
Thanks to the proof of (i), we know that if $p \equiv 1 \pmod 4$ then $p = wz$ with $N(w) = N(z) = p$, so the expression $p = wz$ is a factorisation of p in $\mathbb{Z}[i]$. We show that w and z are not associates.
If $w = a + bi$, then we obtain from $p = a^2 + b^2 = (a + bi)(a - bi)$ that $z = a - bi$. So, if w and z were associates, since clearly $w \neq \pm z$, we would have $w = \pm i \cdot z = \pm(b + ai)$, and thus $|a| = |b|$, so $a^2 + b^2 = 2a^2$. But this is not possible because p is odd. This concludes the proof of (b).

If $p \equiv 3 \pmod 4$, then p cannot be written as the product of two elements of norm p or it would be the sum of two squares, which is excluded by (i); it follows that p is irreducible and therefore prime, thus proving (c).

(iii) If $w \in \mathbb{Z}[i\,]$ is prime, then the ideal (w) is prime and so the ideal $(w) \cap \mathbb{Z} = (p)$ is a prime ideal of $\mathbb{Z}$. Since $(p) \subseteq (w)$, we have that w divides p in $\mathbb{Z}[i\,]$; it follows that, up to associates, every prime $a + bi$ of $\mathbb{Z}[i\,]$ is the divisor of a prime p of $\mathbb{Z}$. Going through the solution of (ii) again, we find

(a) for $p = 2$, the only prime divisor is $1 + i$, for which $a = b = 1$ and $a^2 + b^2 = 2$;
(b) for $p \equiv 1 \pmod 4$, p has two prime divisors, both of which are such that $a^2 + b^2 = p$;
(c) primes congruent to 3 modulo 4 remain prime in $\mathbb{Z}[i\,]$.

Exercise 60 Let $A = \mathbb{C}[x]$ and let G be the subgroup of the group of automorphisms of A generated by

$$f(x) \overset{\sigma}{\longmapsto} f(-x)$$
$$f(x) \overset{\tau}{\longmapsto} f(1-x).$$

Show that the subring A^G of the fixed points of G in A is $\mathbb{C}$.

Solution It is clear that $\mathbb{C}$ is contained in A^G. Conversely, let $f(x)$ be a polynomial that is fixed by G; assume by contradiction that $f(x)$ has positive degree and let α be a root of $f(x)$ in $\mathbb{C}$.

Remark that $(\tau\sigma)(f(x)) = \tau(\sigma(f(x))) = f(x-1)$, so we have $f(x) = f(x-1)$. This identity remains true if we replace x by any element of $\mathbb{C}$. In particular, replacing x by the element α yields $0 = f(\alpha) = f(\alpha - 1)$, that is, $\alpha - 1$ is also a root of $f(x)$. It follows that $f(x)$ has infinitely many roots, namely, $\alpha, \alpha - 1, \alpha - 2, \ldots$, which is clearly impossible for a nonzero polynomial.

Exercise 61 Find all fields $\mathbb{K}$ such that there exists a surjective homomorphism $\mathbb{Z}[x] \longrightarrow \mathbb{K}$.

Solution We show that the required fields are precisely the finite fields.

Let $\mathbb{K}$ be a finite field and let γ be a generator of the finite cyclic group $\mathbb{K}^*$. It is clear that the unique ring homomorphism induced by $\mathbb{Z}[x] \ni x \longmapsto \gamma \in \mathbb{K}$ is surjective.

Conversely, given a surjective homomorphism $\varphi : \mathbb{Z}[x] \longrightarrow \mathbb{K}$, its kernel M must be a maximal ideal of $\mathbb{Z}[x]$. We first show by contradiction that we cannot have $M \cap \mathbb{Z} = \{0\}$.

Suppose $M \cap \mathbb{Z} = \{0\}$; in this case, $\varphi_{|\mathbb{Z}}$ is injective, so $\mathbb{K}$ has characteristic 0, and in particular contains $\mathbb{Q}$. Then, setting $S = \{1, 2, 3, \ldots\} = \mathbb{N} \setminus \{0\}$, the image $\varphi(S)$ is contained in $\mathbb{K}^*$, so φ extends to a surjective homomorphism $\widetilde{\varphi}$ from $S^{-1}\mathbb{Z}[x] =$

$\mathbb{Q}[x]$ to $\mathbb{K}$. Now let $\widetilde{M} = \mathbb{Q}[x] \cdot f(x)$ be the kernel of this homomorphism and let $\alpha = \varphi(x) \in \mathbb{K}$. Since the image of $\widetilde{\varphi}$ is the field $\mathbb{K}$, $f(x)$ is irreducible; moreover, we can choose it to have integer coefficients and content equal to 1. With this choice of $f(x)$, Gauss's Lemma shows that M is the principal ideal generated by $f(x)$ in $\mathbb{Z}[x]$. Moreover, denoting by n the degree of $f(x)$ and by a_n its leading coefficient, for each polynomial $h(x) \in \mathbb{Z}[x]$ the Euclidean division of $h(x)$ by $f(x)$ produces a remainder $r_h(x)$ that is either zero or of degree smaller than n and has coefficients in $\frac{1}{a_n} \cdot \mathbb{Z}$. In any case,

$$\varphi(h(x)) = \widetilde{\varphi}(h(x)) = r_h(\alpha),$$

and $r_h(\alpha)$ is an element of the Abelian group generated by $1/a_n, \alpha/a_n, \ldots, \alpha^{n-1}/a_n$. In particular, given a prime p that does not divide a_n, the rational number $1/p$ is not in the image of φ, contradicting the fact that φ is surjective. This concludes the proof that $M \cap \mathbb{Z}$ is a nontrivial ideal of $\mathbb{Z}$.

Since M is maximal, it is prime, so $M \cap \mathbb{Z}$ is also prime; having just shown that it is nonzero, this implies that $M \cap \mathbb{Z} = (p)$ for some prime p. Consequently, φ passes to the quotient $(\mathbb{Z}/p\mathbb{Z})[x] = \mathbb{F}_p[x]$ and the image is $\mathbb{K} = \mathbb{F}_p[\alpha]$. This ring is a field if and only if α is algebraic over $\mathbb{F}_p$. But in this case $\mathbb{F}_p[\alpha] = \mathbb{F}_p(\alpha)$ is an extension of $\mathbb{F}_p$ whose degree is equal to that of the minimal polynomial of α over $\mathbb{F}_p$, so $\mathbb{K}$ is a finite field.

Exercise 62 Determine all subfields of the splitting field of $x^7 - 1$ over $\mathbb{Q}$.

Solution The splitting field of $x^7 - 1$ is the seventh cyclotomic extension of $\mathbb{Q}$ in $\mathbb{C}$: it is generated by a primitive seventh root of unity ζ, its degree is $\phi(7) = 6$, and its Galois group G is isomorphic to $(\mathbb{Z}/7\mathbb{Z})^*$. Each automorphism φ is uniquely determined by the image it assigns to ζ, which can be any seventh root of unity ζ^k with $1 \leq k \leq 6$. The map $\varphi \longmapsto k$ is an isomorphism between G and $(\mathbb{Z}/7\mathbb{Z})^*$. Remark that the latter is a cyclic group of order 6, generated for example by 3, and thus G is generated by the homomorphism σ that sends ζ to ζ^3.

By the Fundamental Theorem of Galois Theory, the subfields of $\mathbb{Q}(\zeta)$ correspond to the subgroups of G; other than the whole group and the trivial subgroup, we have the subgroups $G_2 = \langle \sigma^2 \rangle$ of order 3 and $G_3 = \langle \sigma^3 \rangle$ of order 2.

Denote by $\mathbb{K}_2$ the fixed subgroup of G_2 and consider the element $\alpha = \zeta + \zeta^2 + \zeta^4$ of $\mathbb{Q}(\zeta)$. Clearly, α is fixed by σ^2; moreover, G_2 has order 3, so the degree of $\mathbb{K}_2/\mathbb{Q}$, which is the index of G_2, is 2. In order to conclude that $\mathbb{K}_2 = \mathbb{Q}(\alpha)$, it is enough to observe that α cannot be rational, as $1, \zeta, \zeta^2, \ldots, \zeta^5$ is a basis of $\mathbb{Q}(\zeta)/\mathbb{Q}$. It is easy to see that α satisfies the polynomial $x^2 + x + 2$ and so $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{-7})$.

The subgroup G_3 is generated by the automorphism sending ζ to ζ^{-1}, which is the restriction to $\mathbb{Q}(\zeta)$ of conjugation on $\mathbb{C}$. The corresponding subfield $\mathbb{K}_3$ is thus the real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R} = \mathbb{Q}(\zeta + \zeta^{-1})$.

In conclusion, we have four subfields: $\mathbb{Q}(\zeta)$, which has degree 6, $\mathbb{Q}(\zeta + \zeta^{-1})$, which is real of degree 3, $\mathbb{Q}(\sqrt{-7})$, whose degree is 2, and finally $\mathbb{Q}$.

Exercise 63 Let $\mathbb{K}$ be the splitting field of $x^3 - 5$ over $\mathbb{Q}$ and let $\mathbb{F}$ be the splitting field of $x^{11} - 1$ over $\mathbb{Q}$.

(i) Determine the Galois group of $\mathbb{K}/\mathbb{Q}$ and find a primitive element for this extension.
(ii) Compute the degree of $\mathbb{K} \cdot \mathbb{F}/\mathbb{Q}$ and determine its Galois group.
(iii) Count the subextensions $\mathbb{L}$ of $\mathbb{K} \cdot \mathbb{F}$ such that $\sqrt[3]{5} \in \mathbb{L}$ and $\mathbb{L}/\mathbb{Q}$ is Galois.

Solution In what follows, we denote by ζ_n, for each natural number n, some fixed primitive nth root of unity in $\mathbb{C}$.

(i) The roots of the polynomial $x^3 - 5$ are $\sqrt[3]{5}$, $\zeta_3\sqrt[3]{5}$, $\zeta_3^2\sqrt[3]{5}$, so its splitting field is

$$\mathbb{K} = \mathbb{Q}(\sqrt[3]{5},\ \zeta_3\sqrt[3]{5},\ \zeta_3^2\sqrt[3]{5}) = \mathbb{Q}(\sqrt[3]{5},\ \zeta_3).$$

The formula for the degree of a tower of extensions yields $[\mathbb{K} : \mathbb{Q}] = [\mathbb{K} : \mathbb{Q}(\sqrt[3]{5})][\mathbb{Q}(\sqrt[3]{5}) : \mathbb{Q}]$. Now, we have $[\mathbb{Q}(\sqrt[3]{5}) : \mathbb{Q}] = 3$ because $x^3 - 5$ is irreducible, and $[\mathbb{K} : \mathbb{Q}(\sqrt[3]{5})] = 2$ because ζ_3 has degree 2 over $\mathbb{Q}$ and does not belong to $\mathbb{Q}(\sqrt[3]{5})$, which is a real field. We thus obtain that $[\mathbb{K} : \mathbb{Q}] = 6$ and therefore that the Galois group we are looking for, being the Galois group of a cubic, is isomorphic to S_3.
We now check that the element $\alpha = \zeta_3 + \sqrt[3]{5}$ of $\mathbb{K}$ generates $\mathbb{K}/\mathbb{Q}$. We can do this, for example, by showing that the orbit of α for the action of the Galois group consists of 6 elements. This implies that the degree of α over $\mathbb{Q}$ is 6 and thus that α generates the extension.
The elements of the Galois group of $\mathbb{K}/\mathbb{Q}$ permute the roots of $x^3 - 5$ and send ζ_3 to some primitive third root of unity. They are thus the automorphisms $\sigma_{h,k}$, for $h = 0, 1, 2$ and $k = 1, 2$, where $\sigma_{h,k}(\sqrt[3]{5}) = \zeta_3^h\sqrt[3]{5}$ and $\sigma_{h,k}(\zeta_3) = \zeta_3^k$. We will now determine the stabiliser of α for the action of the Galois group. We have

$$\sigma_{h,k}(\alpha) = \alpha \iff \zeta_3^k + \zeta_3^h\sqrt[3]{5} = \zeta_3 + \sqrt[3]{5} \iff (\zeta_3^h - 1)\sqrt[3]{5} = \zeta_3 - \zeta_3^k.$$

It follows that, in order for $\sigma_{h,k}$ to be in the stabiliser, we must have $h = 0$, or we would have $\sqrt[3]{5} \in \mathbb{Q}(\zeta_3)$, which contradicts the fact that $\mathbb{K} \neq \mathbb{Q}(\zeta_3)$. But then we must also have $k = 1$. The stabiliser is thus trivial, and the orbit of α has 6 elements, as claimed.

(ii) It is clear that $\mathbb{F} = \mathbb{Q}(\zeta_{11})$. Remark that $\mathbb{Q}(\zeta_{11}, \zeta_3) = \mathbb{Q}(\zeta_{33})$, as 11 and 3 are relatively prime. It follows that $\mathbb{K}\mathbb{F} = \mathbb{Q}(\zeta_{11}, \zeta_3, \sqrt[3]{5}) = \mathbb{Q}(\zeta_{33}, \sqrt[3]{5})$.
Now, the numbers $[\mathbb{Q}(\zeta_{33}) : \mathbb{Q}] = \phi(33) = 20$ and $[\mathbb{Q}(\sqrt[3]{5}) : \mathbb{Q}] = 3$ are relatively prime, so $[\mathbb{K}\mathbb{F} : \mathbb{Q}] = 60$.
Since $\mathbb{K}/\mathbb{Q}$ and $\mathbb{F}/\mathbb{Q}$ are Galois and since their intersection is $\mathbb{Q}$, there exists an isomorphism

$$G = \mathrm{Gal}(\mathbb{K}\mathbb{F}/\mathbb{Q}) \longrightarrow \mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{F}/\mathbb{Q}),$$

that is, G is isomorphic to $\mathsf{S}_3 \times (\mathbb{Z}/11\mathbb{Z})^* \simeq \mathsf{S}_3 \times \mathbb{Z}/10\mathbb{Z}$.

(iii) A normal extension $\mathbb{L}$ of $\mathbb{Q}$ containing $\sqrt[3]{5}$ must also contain the splitting field $\mathbb{K}$ of $x^3 - 5$. Using the isomorphism introduced above, we find that $\mathbb{K}$ is fixed by $H = e \times (\mathbb{Z}/11\mathbb{Z})^*$. By the Fundamental Theorem of Galois Theory, the extensions $\mathbb{L}$ we are looking for correspond to the subgroups of H that are normal in $\mathsf{S}_3 \times (\mathbb{Z}/11\mathbb{Z})^*$.
The group H is cyclic of order 10, so it has exactly 4 subgroups, each of which is characteristic. Also notice that H is the centre of $\mathsf{S}_3 \times (\mathbb{Z}/11\mathbb{Z})^*$, and in particular is characteristic. We conclude that every subgroup of H is characteristic, and therefore normal, in $\mathsf{S}_3 \times (\mathbb{Z}/11\mathbb{Z})^*$. There are thus 4 extensions satisfying the given requirements.

Exercise 64 Determine the Galois group of the polynomial $x^4 - 10x^2 + 1$ over $\mathbb{Q}$.

Solution First of all, we prove that $f(x) = x^4 - 10x^2 + 1$ is irreducible in $\mathbb{Q}[x]$. One immediately finds that the roots of $f(x)$ are the four real numbers

$$\alpha_1 = \sqrt{5 + 2\sqrt{6}},\ \alpha_2 = -\sqrt{5 + 2\sqrt{6}},\ \alpha_3 = \sqrt{5 - 2\sqrt{6}},\ \alpha_4 = -\sqrt{5 - 2\sqrt{6}}.$$

None of them is rational, or $\sqrt{6}$ would be rational. Moreover, if $f(x)$ were the product of two quadratic polynomials with rational coefficients, then one would be able to group the four roots into pairs in such a way that, denoting by β, γ two paired roots, we have $\beta\gamma,\ \beta + \gamma \in \mathbb{Q}$.

However, in order to obtain a rational product we would need to pair α_1 with α_3 or α_4; indeed, $\alpha_1^2 = -\alpha_1\alpha_2 = 5 + 2\sqrt{6} \notin \mathbb{Q}$ whereas $\alpha_1\alpha_3 = 1$ and $\alpha_1\alpha_4 = -1$. But $(\alpha_1 + \alpha_3)^2 = 12$ and $(\alpha_1 + \alpha_4)^2 = 8$, so $\alpha_1 + \alpha_3,\ \alpha_1 + \alpha_4 \notin \mathbb{Q}$. This concludes the proof that $f(x)$ is irreducible in $\mathbb{Q}[x]$.

We have also shown that $\alpha_3 = \alpha_1^{-1}$, so the splitting field of $f(x)$ over $\mathbb{Q}$ is $\mathbb{K} = \mathbb{Q}(\alpha_1)$, and in particular $[\mathbb{K} : \mathbb{Q}] = 4$. Since $\sqrt{3} = (\alpha_1 + \alpha_3)/2$ and $\sqrt{2} = (\alpha_1 + \alpha_4)/2$, the field $\mathbb{K} = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ contains three distinct quadratic subextensions, namely, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{6})$.

But then, letting G denote the Galois group of $f(x)$ over $\mathbb{Q}$, we necessarily have $G \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, because $|G| = 4$ and G must have three distinct subgroups of index 2 corresponding to the three quadratic subextensions we have found.

We can also describe explicitly each automorphism in the Galois group of $\mathbb{K}/\mathbb{Q}$. The root α_1 must be sent by each automorphism to some root of $f(x)$, so we have the four automorphism induced by the mappings $\alpha_1 \longmapsto \alpha_1$, $\alpha_1 \longmapsto -\alpha_1$, $\alpha_1 \longmapsto \alpha_1^{-1}$ and $\alpha_1 \longmapsto -\alpha_1^{-1}$.

▌The polynomial $f(x)$ is peculiar in that it is reducible modulo p for all primes p. In order to prove this, it is enough to observe that the product of two quadratic non-residues is a quadratic residue, which allows us to pair up the roots and obtain a factorisation as a product of two polynomials of degree 2 in $\mathbb{F}_p[x]$ for each prime p. In particular, it is not possible to show the irreducibility of $f(x)$ using some argument modulo p. Another example of a polynomial with this property is $x^4 + 1$.▐

Exercise 65 For each m in $\mathbb{Z}$, compute the Galois group of the polynomial $f(x) = (x^4 + 1)(x^2 - m)$ over $\mathbb{Q}$.

Solution The roots of the polynomial $x^4+1 = (x^8-1)/(x^4-1)$ are the eighth roots of unity that are not fourth roots of unity, that is, the primitive eighth roots of unity, namely, $(\pm\sqrt{2} \pm i\sqrt{2})/2$. Its splitting field over $\mathbb{Q}$ is therefore $\mathbb{K} = \mathbb{Q}((\pm\sqrt{2} \pm i\sqrt{2})/2) = \mathbb{Q}(i, \sqrt{2})$. Its degree is $[\mathbb{K} : \mathbb{Q}] = [\mathbb{K} : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 \cdot 2 = 4$ thanks to the fact that $i \notin \mathbb{Q}(\sqrt{2})$, as the latter is a real field.

Moreover, we have $\mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$: indeed, the Galois group has order 4 and it is not cyclic, as the extension has the three quadratic subextensions $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-2})$. Notice that those listed are the only quadratic subextensions of $\mathbb{K}$, as their number coincides with the number of subgroups of index 2 of the Galois group.

The roots of the polynomial $x^2 - m$ are $\pm\sqrt{m}$, so it splits as a product of linear factors in $\mathbb{Q}[x]$ if and only if m is a square: in this case, $\mathbb{K}$ is the splitting field of $f(x)$ and, as we already established, its Galois group over $\mathbb{Q}$ is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

If m is not a square, then we have $m = k^2 d$ for some squarefree $d \neq 0, 1$ and some $k \neq 0$, the splitting field of $x^2 - m$ is $\mathbb{Q}(\sqrt{d})$, and its Galois group is $\mathbb{Z}/2\mathbb{Z}$. In this case, the splitting field of the polynomial $f(x)$ is $\mathbb{F} = \mathbb{K}(\sqrt{d}) = \mathbb{Q}(i, \sqrt{2}, \sqrt{d})$ and its degree is $[\mathbb{F} : \mathbb{Q}] = [\mathbb{F} : \mathbb{K}][\mathbb{K} : \mathbb{Q}] = [\mathbb{F} : \mathbb{K}] \cdot 4$. It follows that $[\mathbb{F} : \mathbb{Q}] = 4$ if $\sqrt{d} \in \mathbb{K}$, whereas $[\mathbb{F} : \mathbb{Q}] = 8$ if $\sqrt{d} \notin \mathbb{K}$.

We have $\sqrt{d} \in \mathbb{K} = \mathbb{Q}(i, \sqrt{2})$ if and only if the field $\mathbb{Q}(\sqrt{d})$ is a quadratic subextension of $\mathbb{K}$; equivalently, since d is squarefree, if and only if $d = -1, 2$ or -2, that is, $m = -k^2, \pm 2k^2$. In this case we have $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

On the other hand, if $\sqrt{d} \notin \mathbb{K}$ then we have $\mathbb{F} = \mathbb{K}\mathbb{Q}(\sqrt{d})$ and $\mathbb{K} \cap \mathbb{Q}(\sqrt{d}) = \mathbb{Q}$, so $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \simeq \mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\sqrt{d}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

[In order to find the splitting field of $x^4 + 1$ over $\mathbb{Q}$ it's enough to remark that $x^4 + 1$ is the eighth cyclotomic polynomial. Its splitting field is therefore $\mathbb{Q}(\sqrt{2}(1 + i)/2)$, it has degree 4 over $\mathbb{Q}$, and its Galois group is isomorphic to $(\mathbb{Z}/8\mathbb{Z})^* \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.]

Exercise 66 Find all odd primes p such that the congruence class of 2 is a square modulo p.

Solution Letting p be an odd prime, given $a \in \mathbb{F}_p^*$, we recall the Legendre symbol $\left(\frac{a}{p}\right)$, defined as follows:

$$\left(\frac{a}{p}\right) = \begin{cases} +1 \text{ if } a \text{ is a square in } \mathbb{F}_p^* \\ -1 \text{ otherwise.} \end{cases}$$

First, we show that $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}}$. Indeed, letting α denote a root of the polynomial $x^2 - a$ in some fixed algebraic closure of $\mathbb{F}_p$, we have $\alpha^p = \alpha$ if and only if $\alpha \in \mathbb{F}_p$, so if and only if a is a square in $\mathbb{F}_p^*$. It follows that $a^{\frac{p-1}{2}} = \alpha^{p-1} = 1$ if and only

if a is a square, that is, if and only if $\left(\frac{a}{p}\right) = 1$. On the other hand, $a^{\frac{p-1}{2}}$ can only take the values ± 1, as it is a root of the polynomial $x^2 - 1$ in $\mathbb{F}_p$. This proves that $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}}$.

We now wish to compute the Legendre symbol $\left(\frac{2}{p}\right)$. Letting ω denote a primitive eighth root of unity in some algebraic closure of $\mathbb{F}_p$, observe that $\omega^4 = -1$ and thus $\omega^2 = -\omega^{-2}$. Setting $\alpha = \omega + \omega^{-1}$, we have $\alpha^2 = (\omega + \omega^{-1})^2 = \omega^2 + 2 + \omega^{-2} = 2$, that is, α is a square root of 2. But then $\left(\frac{2}{p}\right) = 2^{\frac{p-1}{2}} = \alpha^{p-1} = \alpha^p/\alpha$.

Now remark that p is congruent to ± 1 or ± 3 modulo 8. In the first case, $\alpha^p = \omega^p + \omega^{-p} = \omega + \omega^{-1} = \alpha$, and in the second $\alpha^p = \omega^p + \omega^{-p} = \omega^3 + \omega^{-3} = (\omega + \omega^{-1})(\omega^2 - 1 + \omega^{-2}) = -\alpha$.

We thus obtain that $\left(\frac{2}{p}\right) = +1$ if $p \equiv \pm 1 \pmod 8$ and $\left(\frac{2}{p}\right) = -1$ if $p \equiv \pm 3 \pmod 8$, which can be rewritten as the more concise expression

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

▌It immediately follows from the identity $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}}$ that the map

$$\mathbb{F}_p^* \ni a \longmapsto \left(\frac{a}{p}\right) \in \{\pm 1\}$$

is a group homomorphism. In Preliminary Exercise 18 of the First Volume, we prove that -1 is a square modulo an odd prime p if and only if p is congruent to 1 modulo 4. The statements we have shown for -1 and 2 act as supplements to the Law of Quadratic Reciprocity, proven for the first time in its complete form by Gauss in 1797, and consisting in the following statement: for all pairs of odd primes p and q, we have

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Not only is this law a deep result expressing an unexpected symmetry, but it provides the crucial tool to explicitly compute Legendre's symbol. For more information, see: J.P. Serre, *"A course in Arithmetic"*, Springer, 1996. ▐

Chapter 2
Exercises



2.1 Groups

1 Consider the group $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$.

(i) Find the number of elements of order 6 and the number of cyclic subgroups of order 6 in G.
(ii) Find all possible orders of cyclic subgroups of G.
(iii) For each positive divisor d of the order of G, construct an explicit subgroup of G having order d.

2 Let G be a group and, given $g, h \in G$, let $\psi_{g,h}$ be the map

$$G \ni x \overset{\psi_{g,h}}{\longmapsto} gxh^{-1} \in G.$$

(i) Show that $\psi_{g,h}$ is an element of the group $\mathsf{S}(G)$ of permutations of G.
(ii) Show that the map

$$G \times G \ni (g, h) \overset{\psi}{\longmapsto} \psi_{g,h} \in \mathsf{S}(G)$$

is a group homomorphism.
(iii) For which pairs (g, h) is the map $\psi_{g,h}$ a homomorphism of G?
(iv) Show that $\mathrm{Ker}(\psi)$ is isomorphic to the centre of G.

3 Given an Abelian group G, consider the maps

$$\begin{aligned} G \ni g &\overset{\psi}{\longmapsto} (g^2, g^{-1}) \in G \times G \\ G \times G \ni (g, h) &\overset{\pi}{\longmapsto} gh^2 \in G. \end{aligned}$$

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Show that

(i) ψ and π are group homomorphisms;
(ii) ψ is injective;
(iii) π is surjective;
(iv) $\mathrm{Ker}(\pi) = \mathrm{Im}(\psi)$.

4 Let G be an Abelian group and let $\sigma : G \longrightarrow G$ be a homomorphism such that $\sigma^2 = \mathrm{Id}_G$.

(i) Show that σ is an automorphism of G.
(ii) Show that $H = \{h \in G \mid \sigma(h) = h\}$ is a subgroup of G.
(iii) Show that $G/H \ni gH \overset{\tau}{\longmapsto} \sigma(g)H \in G/H$ is a well-defined map, and indeed an automorphism of G/H.
(iv) Assuming that the order of G is finite and odd, show that

$$K = \{gH \in G/H \mid \tau(gH) = gH\}$$

is the singleton $\{H\}$.

5 Let G be a group with the following property: for each finite subset S of G, the subgroup $\langle S \rangle$ generated by S is cyclic.

(i) Show that G is Abelian.
(ii) Show that G is not necessarily cyclic.

6 Show that there exists a unique non-Abelian group of order 28 with an element of order 4, up to isomorphism.

7

(i) Show that the derived subgroup $[\mathsf{S}_5, \mathsf{S}_5]$ of S_5 is A_5.
(ii) Show that, if $\mathsf{S}_5 \longrightarrow H$ is a surjective group homomorphism and H is Abelian, then $|H| \leq 2$.

8 Let G be a group of order 385 with a subgroup of order 77. Prove that there are characteristic subgroups H_1, H_2 of G such that $\{e\} \subsetneq H_1 \subsetneq H_2 \subsetneq G$.

9 Compute the number of elements and the number of subgroups of order 50 in the group $\mathbb{Z}/100\mathbb{Z} \times \mathbb{Z}/20\mathbb{Z}$.

10 Let p be a prime and G be a group with p^4 elements. Show that

(i) for all $x \in G$, the integer p^2 divides the order of the centraliser $Z(x)$ of x in G;
(ii) if G is non-Abelian, then there exists $x \in G$ such that $|Z(x)| = p^3$.

11 Let G be the multiplicative group of all real 3×3 matrices A with the following properties: all coefficients of A belong to the set $\{0, 1, -1\}$; each row and each column of A has exactly one nonzero coefficient.

Show that the set of diagonal matrices of G is a normal subgroup, and that G is isomorphic to a semidirect product $(\mathbb{Z}/2\mathbb{Z})^3 \rtimes \mathsf{S}_3$.

12 A subset S of a group G is said to be *stable under conjugation* if

$$xsx^{-1} \in S \quad \forall x \in G,\ s \in S.$$

(i) Show that for all $X \subseteq G$ the set

$$\{S \mid X \subseteq S \subseteq G,\ S \text{ is stable under conjugation}\}$$

has an element that is minimal for inclusion.

(ii) Show that if S is stable under conjugation, then the subgroup generated by S is normal.

13

(i) Describe the conjugacy classes in $\mathsf{S}_4 \times \mathbb{Z}/3\mathbb{Z}$.
(ii) Describe all possible homomorphisms from $\mathsf{S}_4 \times \mathbb{Z}/3\mathbb{Z}$ to $\mathbb{Z}/6\mathbb{Z}$.

14 Let G be a group of order $3 \cdot 5 \cdot 17$. Show that

(i) there exists a subgroup H of order 17 that is characteristic in G;
(ii) the quotient G/H is cyclic;
(iii) G is a cyclic group.

15 Determine how many permutations σ in S_{11} are such that $\sigma^5 = (1, 2)$.

16 Consider the set $X = \{1, 2, \ldots, n\}$ and define the following action of S_n on X^3:

$$\mathsf{S}_n \ni \sigma \longmapsto \big((i, j, k) \longmapsto \big(\sigma(i), \sigma(j), \sigma(k)\big)\big) \in \mathsf{S}(X^3).$$

Show that there are 5 orbits in X^3 for this action of S_n, and compute the cardinality of each.

17 Given $\sigma = (1, 2, 3, 4, 5) \in \mathsf{A}_5$, compute the cardinality of its conjugacy class $\mathcal{C}\ell_{\mathsf{A}_5}(\sigma)$ in A_5 and the cardinality of the intersection $\mathcal{C}\ell_{\mathsf{A}_5}(\sigma) \cap \langle\sigma\rangle$.

18 Let G be a finite Abelian group denoted using additive notation. We say that a subgroup H of G is *pure* if for all $n \in \mathbb{Z}$ and for all $h \in H$ the following holds:

$$\Big(\text{there exists } x \in G \text{ such that } nx = h\Big)$$

$$\text{if and only if}$$

$$\Big(\text{there exists } y \in H \text{ such that } ny = h\Big).$$

Let p be a prime divisor of $|G|$ and P be the p-torsion subgroup of G. Show that

(i) the subgroup P is a pure subgroup of G;
(ii) all subgroups of P are pure in G if and only if $px = 0$ for all $x \in P$.

19 Given a group G and an automorphism f of G, set

$$H = \big\{(x, f(x)) \in G \times G \mid x \in G\big\}.$$

(i) Show that H is a subgroup of $G \times G$.
(ii) Determine whether or not H is normal.
(iii) Find the centraliser of H in $G \times G$.

20

(i) Show that the subgroup H of S_7 generated by all 7-cycles is normal.
(ii) Show that $H = \mathsf{A}_7$.

21 Show that a p-group G is non-cyclic if and only if there exists a normal subgroup N of G such that G/N is isomorphic to $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

22 Let G be a finite group and $\varphi : G \longrightarrow G$ a homomorphism.

(i) Show that there is an integer $N \geq 0$ such that $\varphi^n(G) = \varphi^N(G)$ for all $n \geq N$.
(ii) Show that G is isomorphic to the semidirect product of its subgroups $\mathrm{Ker}(\varphi^N)$ and $\mathrm{Im}(\varphi^N)$.
(iii) Provide an example showing that the subgroup $\mathrm{Im}(\varphi^N)$ is not necessarily normal in G.

23

(i) Give a description of the centraliser of $\sigma = (1, 2, 3, \ldots, 11)$ and of the normaliser of the group generated by σ in the permutation group S_{11}.
(ii) Show that A_{11} has subgroups of order 55 but no subgroups of order 110.

24 Let G be a group and H be a proper subgroup of G that is maximal with respect to set inclusion.

(i) Show that if H is the only maximal proper subgroup then H is normal in G.
(ii) Show that if H does not contain the centre of G then H is normal in G.
(iii) Provide an example of a group G and a maximal proper subgroup that is not normal in G.

25

(i) Show that S_5 has no Abelian subgroup of order 8 or of order 10.
(ii) List the possible orders of Abelian subgroups of S_5.

26 Let m, n be positive integers.

(i) Show that the group $\mathrm{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ is isomorphic to the group $\mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z})$.
(ii) Show that $\mathrm{Hom}(\mathsf{S}_n, \mathbb{Z}/m\mathbb{Z})$ has at most two elements.

27 Find all permutations σ in S_6 such that $\sigma^4 = (1, 2, 3)$.

28 Let p be a prime and G a group of order p^4 whose centre has order p^2. Find the number of conjugacy classes in G and determine the group of internal automorphisms of G.

29 Let G be a group, H a characteristic subgroup of G, and $\pi : G \longrightarrow G/H$ the quotient homomorphism.

(i) Show that for all $\varphi \in \mathrm{Aut}(G)$ there exists a unique element $\overline{\varphi} \in \mathrm{Aut}(G/H)$ such that $\pi \circ \varphi = \overline{\varphi} \circ \pi$. Additionally, show that the map

$$\mathrm{Aut}(G) \ni \varphi \overset{F}{\longmapsto} \overline{\varphi} \in \mathrm{Aut}(G/H)$$

is a homomorphism.
(ii) Show that, in the case where $H = Z(G)$, the kernel of F is the centraliser in $\mathrm{Aut}(G)$ of the subgroup of internal automorphisms.
(iii) Provide an example where, setting $H = Z(G)$, the homomorphism F is surjective, and an example where it isn't.

30 Let G be a finite Abelian group and let $G(d)$ be its subgroup $\{x \in G \mid dx = 0\}$.

(i) Show that if $G(p)$ is cyclic for all primes p dividing the order of G, then G is cyclic.
(ii) Prove or disprove the following: if G has order m^2 and $G(p) \simeq \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ for all primes p dividing m, then $G \simeq \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.
(iii) Show that if G has order m^2 and for all $d < m$ such that $d \mid m$ we have $G(d) \simeq \mathbb{Z}/d\mathbb{Z} \times \mathbb{Z}/d\mathbb{Z}$, then $G \simeq \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.

31 Given a prime p, describe the conjugacy classes of the group $\mathbb{Z}/p\mathbb{Z} \rtimes_\varphi (\mathbb{Z}/p\mathbb{Z})^*$, where φ is the homomorphism $(\mathbb{Z}/p\mathbb{Z})^* \ni a \overset{\varphi}{\longmapsto} (n \longmapsto an) \in \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$. Additionally, find the centre of the group and the centraliser of $(1, 1)$.

32 Let G be a group with centre Z and let

$$K = \big\{\varphi \in \mathrm{Aut}(G) \mid \varphi(gZ) = g\varphi(Z) \text{ for all } g \in G\big\}.$$

(i) Show that K is a normal subgroup of $\mathrm{Aut}(G)$.
(ii) Find K in the case where $G = \mathsf{D}_6$.

33 Find the number of homomorphisms from $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ to S_7.

34 Let $G = \mathbb{Z}/13\mathbb{Z} \times \mathbb{Z}/13\mathbb{Z}$, $H = \mathbb{Z}/13\mathbb{Z}$ and $K = \mathbb{Z}/4\mathbb{Z}$. Construct explicit examples of non-Abelian semidirect products $G \rtimes H$ and $G \rtimes K$.

35 Show that in any p-group the number of non-normal subgroups is divisible by p.

36 Find all primes p for which the equation

$$\sigma^p = (1, 2, \cdots, p)(p + 1, p + 2, \cdots, 2p), \quad \sigma \in \mathsf{S}_{2p}$$

admits a solution, and for each of these primes find all solutions.

37 Show that any finite non-Abelian group whose centre is nontrivial has at least 4 distinct conjugacy classes.

38 Let A and B be groups of order m and n, respectively. Show that a necessary and sufficient condition for all the subgroups of the direct product $A \times B$ to be of the form $A' \times B'$, where A' is a subgroup of A and B' is a subgroup of B, is that $(m, n) = 1$.

39 Let $G = (\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}) \rtimes_\varphi \mathbb{Z}/3\mathbb{Z}$, where

$$\varphi : Z/3\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z})$$

is defined by

$$\begin{cases} \varphi_0(x, y, z) = (x, y, z) \\ \varphi_1(x, y, z) = (y, z, x) \\ \varphi_2(x, y, z) = (z, x, y). \end{cases}$$

(i) Find the centre of G.
(ii) Find the number of elements of order 3 in G.

40 Let $p < q$ and r be prime numbers, let G be a group of order pq and let H be the group $\mathbb{Z}/r\mathbb{Z}$. Find all homomorphisms from G to H and all homomorphisms from H to G.

41 Let $G = \mathsf{SL}_2(\mathbb{Z})$ be the multiplicative group of 2×2 matrices with coefficients in $\mathbb{Z}$ that have determinant 1, and denote by $\mathcal{H}$ the subset of the complex numbers consisting of those whose imaginary part is positive.

(i) Show that setting

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}$$

defines an action of G on $\mathcal{H}$.
(ii) Show that this action passes to the quotient L of G by the subgroup $\{\pm I\}$, where I is the identity matrix.
(iii) Show that for all integers a, b, c, d such that $ad - bc = 1$ the element

$$\frac{ai + b}{ci + d}$$

of $\mathcal{H}$ has a nontrivial stabiliser in L.

42 Let G be a group, set $K = G \times G$, and consider the subgroup $H = \{(g, g) \mid g \in G\}$ of K. For each $k \in K$, define the *double coset* of k with respect to H as the set

$$HkH = \{hkh' | h, h' \in H\}.$$

(i) Show that for $k, k' \in K$ we have either $\big(HkH\big) \cap \big(Hk'H\big) = \varnothing$ or $HkH = Hk'H$.
(ii) Show that the set of all double cosets of elements in K with respect to H is in bijection with the set of conjugacy classes of G.

43 Find all finite Abelian groups of odd order that have exactly 10 subgroups, including the group itself and the subgroup consisting of only the neutral element.

44 Let H and K be groups, let $\varphi : K \longrightarrow \mathrm{Aut}(H)$ be a group homomorphism and $G = H \rtimes_\varphi K$ the corresponding semidirect product.

(i) Given an automorphism ω of H, show that

$$G \ni (h, k) \longmapsto (\omega(h), k) \in G$$

is an automorphism of G if and only if ω belongs to the centraliser of $\mathrm{Im}(\varphi)$ in $\mathrm{Aut}(H)$.

(ii) Given an automorphism ϵ of K, show that

$$G \ni (h, k) \longmapsto (h, \epsilon(k)) \in G$$

is an automorphism of G if and only if the following conditions hold: $\epsilon(\mathrm{Ker}(\varphi)) = \mathrm{Ker}(\varphi)$, and the map induced on the quotient $k\,\mathrm{Ker}(\varphi) \longmapsto \epsilon(k)\,\mathrm{Ker}(\varphi)$ is the identity of $K/\mathrm{Ker}(\varphi)$.

45

(i) Find an element σ in the group $\mathsf{S}(\mathbb{N})$ of permutations of $\mathbb{N}$ such that $Z(\sigma) \subsetneqq Z(\sigma^n)$ for all $n > 1$.

(ii) Let $\mathsf{S}_0(\mathbb{N})$ be the subgroup of $\mathsf{S}(\mathbb{N})$ consisting of the permutations σ such that $\sigma(k) = k$ for all but a finite number of $k \in \mathbb{N}$. Show that $\mathsf{S}_0(\mathbb{N})$ is a normal subgroup of $\mathsf{S}(\mathbb{N})$ and that the quotient $\mathsf{S}(\mathbb{N})/\mathsf{S}_0(\mathbb{N})$ is still an infinite group.

46 Let σ be a 19-cycle in S_{19}.

(i) Compute the cardinality of the normaliser of the subgroup $\langle\sigma\rangle$ in S_{19}.

(ii) Show that the normaliser of $\langle\sigma\rangle$ in A_{19} is different from that in S_{19}.

47 Let G be the group $\mathsf{S}_3 \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.

(i) Prove that the subgroups $\mathsf{S}_3 \times \{0\} \times \{0\}$ and $\{e\} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ are characteristic in G.

(ii) Compute the cardinality of the set $\mathrm{Aut}(G)$.

48 Show that the only permutation σ in S_5 such that

$$\begin{cases} \sigma^2 = (1,2)\sigma(1,2) \\ \sigma^3 = (2,3)\sigma(2,3) \end{cases}$$

is the identity.

49 Let p be a prime and H a subgroup of order p of the group $\mathsf{GL}_2(\mathbb{F}_p)$ of 2×2 invertible matrices with coefficients in $\mathbb{F}_p$. Denote by V the vector space $\mathbb{F}_p^2$ and consider the natural action of H on V, which sends $A \in H$ to the linear map $V \ni v \longmapsto A \cdot v \in V$.

Determine the number of orbits in V and the cardinality of each.

50 Let G be a group with centre Z and let S be a non abelian simple subgroup of G. Show that, if G is isomorphic to $S \times Z$, then S is characteristic in G.

51 Find all positive integers n such that the group S_n has a subgroup of order 21.

52

(i) For each prime p, explicitly construct a non-Abelian group G of order p^3 containing an element of order p^2.
(ii) Find all normal subgroups of the group constructed in (i).

53 Given three natural numbers $1 \leq h \leq k \leq n$ and the permutations

$$\sigma = (1, 2, \ldots, h)$$
$$\tau = (k, k+1, \ldots, n),$$

let G be the subgroup of S_n generated by σ and τ. Find

(i) the smallest normal subgroup of S_n containing G for $n \geq 5$,
(ii) the centraliser of G in S_n,
(iii) the orbits of G in $\{1, 2, \ldots, n\}$.

54 Find all possible Abelian groups G of order 1.000.000 with the following property: if H and K are subgroups of G having the same cardinality, then H and K are isomorphic.

55 Given groups G and H, an action $\varphi : G \longrightarrow \mathsf{S}(H)$ of G on H is said to be an action *by group automorphisms* if $\mathrm{Im}(\varphi)$ is contained in $\mathrm{Aut}(H)$.

(i) Given $G = \mathbb{Z}/7\mathbb{Z}$ and $H = (\mathbb{Z}/2\mathbb{Z})^3$, show that there is a nontrivial action of G on H by group automorphisms.
(ii) Show that any such action has two orbits in H.

56

(i) Let G be an Abelian group, let H be a group whose centre is trivial, and let $\varphi : H \longrightarrow \mathrm{Aut}(G)$ be a homomorphism. Show that the centre of the group $G \rtimes_\varphi H$ is $G_0 \times e_H$, where $G_0 = \left\{g \in G \mid \varphi_h(g) = g \ \forall h \in H\right\}$.
(ii) For $n \geq 3$, construct a semidirect product $(\mathbb{Z}/2\mathbb{Z})^n \rtimes S_n$ which is not direct and determine its centre.

57

(i) Find the minimum n such that there is an injective homomorphism from D_5 to S_n.
(ii) Find the minimum n such that there is an injective homomorphism from D_7 to A_n.

58 Let G be a finite Abelian group with the following property: for each positive integer n, the equation $x^n = e$ has at most n solutions in G. Show that G is cyclic.

59 Let G be a group and N a cyclic normal subgroup of G. Show that the centraliser $Z_G(N)$ of N in G is a normal subgroup of G and that $G/Z_G(N)$ is Abelian.

60

(i) Let G be the semidirect product $\mathbb{Z}/11\mathbb{Z} \rtimes (\mathbb{Z}/11\mathbb{Z})^*$ with respect to the homomorphism

$$(\mathbb{Z}/11\mathbb{Z})^* \ni a \longmapsto (n \longmapsto an) \in \mathrm{Aut}(\mathbb{Z}/11\mathbb{Z}).$$

Show that there exists an injective homomorphism from G to S_{11}.

(ii) Given an odd prime p, show that there is no injective homomorphism from $\mathbb{Z}/p\mathbb{Z} \times (\mathbb{Z}/p\mathbb{Z})^*$ to S_p.

61 Let G be the group $\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/49\mathbb{Z}$ and let H be its subgroup $\mathbb{Z}/7\mathbb{Z} \times \{0\}$. Determine the number of automorphisms φ of G for which there exists an automorphism $\widetilde{\varphi}$ of G/H such that $\widetilde{\varphi}(g + H) = \varphi(g) + H$ for all g in G.

62

(i) Prove that any group of order n is isomorphic to a subgroup of A_{n+2}.

(ii) Prove that A_{35} contains a subgroup isomorphic to D_{35}.

63 Let p be a prime, let $G = \mathbb{Z}/p^2\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, and let

$$A = \big\{\varphi \in \mathrm{Aut}(G) \,|\, \varphi(0, 1) = (0, 1)\big\}.$$

Compute the order of the subgroup A and its index in $\mathrm{Aut}(G)$.

64 Let σ be an ℓ-cycle in S_n. For which integers m do we have $Z(\sigma^m) = Z(\sigma)$?

65 Show that any p-group with a single subgroup of index p is cyclic.

66 Letting X be the set $\big\{\langle\sigma\rangle \,|\, \sigma \text{ is an } n\text{-cycle in } \mathsf{S}_n\big\}$, show that

(i) the group S_n acts on X by conjugation;

(ii) for even n, the action of A_n on X is transitive;

(iii) for any $n \geq 5$, the action of S_n on X is faithful.

67 Construct an example of a group of order 18 having a non-normal subgroup, a non-characteristic normal subgroup, and a nontrivial characteristic subgroup.

68 Let $p > 2$ be a prime and let G be the group $\mathbb{Z}/p\mathbb{Z} \rtimes (\mathbb{Z}/p\mathbb{Z})^*$ with operation

$$(a, \alpha)(b, \beta) = (a + \alpha b, \alpha\beta) \text{ for all } a, b \in \mathbb{Z}/p\mathbb{Z},\ \alpha, \beta \in (\mathbb{Z}/p\mathbb{Z})^*.$$

Describe all homomorphisms from G to $\mathbb{C}^*$.

69 Let $\Omega = \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n \mid x_i \neq x_j \text{ for all } i \neq j\}$. For each $\sigma \in \mathsf{S}_n$ and $x = (x_1, \ldots, x_n) \in \Omega$, set

$$\sigma \cdot x = (x_{\sigma^{-1}(1)}, x_{\sigma^{-1}(2)}, \ldots, x_{\sigma^{-1}(n)}).$$

(i) Show that the map defined above is an action of S_n on Ω.
(ii) Compute the cardinality of the orbit of each element in Ω.
(iii) A subset D of Ω is called a *fundamental domain* if for all $x \in \Omega$ there is a unique $y \in D$ that belongs to the orbit of x under S_n. Give an example of a fundamental domain.

70 We say that G *has the finite index property* if all nontrivial subgroups of G have finite index.

(i) Suppose G has the finite index property let K be a homomorphic image of G. Show that if K is infinite, then it is isomorphic to G.
(ii) Show that if H is a subgroup of G and G has the finite index property, then H also has the finite index property.
(iii) Describe all Abelian groups having the finite index property.

71

(i) Find the number of solutions of the equation $\sigma^2 = (1, 2)$ in S_{10}.
(ii) Find the number of solutions of the equation $\sigma^2 = (1, 2)(3, 4)$ in S_{10}.

72 Let G be the group $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/15\mathbb{Z}$.

(i) Compute the number of elements in G of each possible order.
(ii) Determine all homomorphisms from G to $\mathbb{Z}/10\mathbb{Z}$.

73 Compute the number of conjugacy classes in the group

$$\{f : \mathbb{Z}/7\mathbb{Z} \longrightarrow \mathbb{Z}/7\mathbb{Z} \mid f(x) = ax + b, a \in (\mathbb{Z}/7\mathbb{Z})^*, b \in \mathbb{Z}/7\mathbb{Z}\}.$$

74

(i) Determine all groups of order $5^2 \cdot 13$ up to isomorphism.
(ii) Show that there exists a non-Abelian group of order $5^2 \cdot 11$.

75 Let X be the set of conjugacy classes of the dihedral group D_8 and, for each automorphism φ of D_8, define

$$X \ni C \overset{F_\varphi}{\longmapsto} \varphi(C) \in X.$$

Show that F_φ is well-defined, that it is a permutation of X, and that the map

$$\mathrm{Aut}(\mathsf{D}_8) \ni \varphi \overset{F}{\longmapsto} F_\varphi \in \mathsf{S}(X)$$

is an action of the group $\mathrm{Aut}(\mathsf{D}_8)$ on the set X. Additionally, determine the orbits for this action.

76 Find all prime numbers p such that there exist at least three groups, not pairwise isomorphic, of order $25p$.

77 Describe all homomorphisms $\varphi : \mathsf{A}_4 \longrightarrow \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

78 Find the number of isomorphism classes of groups of order 52.

79 Determine the number of subgroups of order 6 of S_5 by conjugacy class.

80 Let p and q be distinct odd primes and let G be a group of order p^3q.

(i) Show that the order of the centre of G cannot be equal to q.
(ii) Show that G is not simple.

81

(i) Show that S_7 contains a subgroup isomorphic to D_{12}.
(ii) Show that A_7 does not contain any subgroups isomorphic to D_{12}.

82 Let $\varphi : \mathbb{Z}/3\mathbb{Z} \longrightarrow \mathrm{Aut}\left((\mathbb{Z}/7\mathbb{Z})^3\right)$ be the homomorphism

$$\begin{cases} \varphi_0(x, y, z) = (x, y, z) \\ \varphi_1(x, y, z) = (y, z, x) \\ \varphi_2(x, y, z) = (z, x, y) \end{cases}$$

and let $\mathcal{H}$ be the set of all subgroups of order 7 in $(\mathbb{Z}/7\mathbb{Z})^3$. Show that setting

$$a \longmapsto \left(H \longmapsto \varphi_a(H)\right)$$

defines an action of $\mathbb{Z}/3\mathbb{Z}$ on $\mathcal{H}$ and find the number of its orbits.

83 Let p be a prime number and let $G = (\mathbb{Z}/p\mathbb{Z})^3 \rtimes_\varphi \mathsf{S}_3$, where

$$\varphi_\sigma(x_1, x_2, x_3) = (x_{\sigma^{-1}(1)}, x_{\sigma^{-1}(2)}, x_{\sigma^{-1}(3)})$$

for all $(x_1, x_2, x_3) \in (\mathbb{Z}/p\mathbb{Z})^3$ and $\sigma \in \mathsf{S}_3$.

(i) Find the centre of G.
(ii) Find all normal subgroups of G contained in $(\mathbb{Z}/p\mathbb{Z})^3 \times \{e\}$.

84 Let G be a group of order 21 generated by x and y, with $\mathrm{ord}(x) = 7$, $\mathrm{ord}(y) = 3$ and $yxy^{-1} = x^2$.

(i) Show that

$$\langle \widetilde{x}, \widetilde{y} \,|\, \widetilde{x}^7 = \widetilde{y}^3 = e, \widetilde{y}\widetilde{x}\widetilde{y}^{-1} = \widetilde{x}^2 \rangle$$

is a presentation of G.
(ii) Compute the order of $\mathrm{Aut}(G)$.
(iii) Show that $\{\varphi \in \mathrm{Aut}(G) \,|\, \varphi(x) = x\}$ is a normal subgroup of $\mathrm{Aut}(G)$ and compute its order.
(iv) Determine whether or not all of the automorphisms of G are inner automorphisms.

85 Let $G = \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(i) Find the number of subgroups of G of each possible order.
(ii) Show that all subgroups of G of order 4 are characteristic.

86

(i) Determine the minimum n such that S_n has a subgroup isomorphic to D_{15}.
(ii) Determine the minimum n such that A_n has a subgroup isomorphic to D_{15}.

87 Find the number of solutions σ in S_{10} for the equation $\sigma^4 = (1, 2, 3)(4, 5, 6)$.

88 Classify the groups of order 2013 up to isomorphism.

89 Show that the group $\mathrm{Aut}(\mathsf{S}_3 \times \mathsf{S}_3)$ is isomorphic to a semidirect product of $\mathsf{S}_3 \times \mathsf{S}_3$ and $\mathbb{Z}/2\mathbb{Z}$.

90 Let $\sigma = (1, 2)(3, 4)(5, 6, 7)$ and let $\tau = (1, 2)(8, 9, 10)$. Determine the centraliser of σ and the centraliser of $\langle \sigma, \tau \rangle$ in S_{10}.

91 Classify all groups of order 20 up to isomorphism.

92

(i) Letting A, B, C be Abelian groups, show that there is a group isomorphism between $\mathrm{Hom}(A, C) \oplus \mathrm{Hom}(B, C)$ and $\mathrm{Hom}(A \oplus B, C)$.
(ii) Letting G be an Abelian group of order n, show that G is isomorphic to $\mathrm{Hom}(G, \mathbb{Z}/n\mathbb{Z})$.

93 Determine all isomorphism classes of subgroups of order 8 in S_6.

94 Show that every group of order p^4 has an Abelian subgroup of order p^3.

95 Find the number of subgroups of order 10 in S_7 according to their isomorphism class.

96

(i) Let p be a prime and let G be the group $\mathbb{Z}/p^{a_1}\mathbb{Z} \times \mathbb{Z}/p^{a_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{a_r}\mathbb{Z}$, where $a_1 \geq a_2 \geq \cdots \geq a_r$ are positive integers. Compute the number of automorphisms φ of G such that $\varphi(h) = h$ for all elements h of G whose first coordinate is zero.
(ii) Show that, given an Abelian group G with p^n elements, p^{n-1} divides the order of $\mathrm{Aut}(G)$.

97 Let G be a group of order 120.

(i) Show that if G is simple then it is isomorphic to a subgroup of A_6;
(ii) deduce that G cannot be simple.

98 Find the number of permutations σ in S_7 such that the order of σ^2 is odd.

99 Let G be a group of order pqr, where p, q, r are distinct primes.

(i) Show that G has a normal subgroup whose order is prime.
(ii) Show that G has a normal subgroup whose index is prime.

100 Let p be a prime, let $n \geq 1$ be a natural number, and let G be a group of order p^n. Show that the number of elements of order p in G is congruent to -1 modulo p.

101 Compute the cardinalities of the centralisers in S_{10} and in A_{10} of each of the powers of $(1, 2, 3, 4, 5, 6, 7, 8)$.

102 Find all primes p for which there exists a non-Abelian group of order $125p$ containing an Abelian subgroup of order 125.

103 Find the minimum n such that

(i) the group S_n contains a subgroup with 360 elements;
(ii) the group S_n contains a cyclic subgroup with 360 elements.

104 Given a prime p and an injective homomorphism $\varphi : \mathbb{Z}/(p-1)\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$, let $G = \mathbb{Z}/p\mathbb{Z} \rtimes_\varphi \mathbb{Z}/(p-1)\mathbb{Z}$, and let d be a divisor of $p-1$.

(i) Show that all subgroups of order d in G are cyclic.
(ii) Show that, if H and K are distinct subgroups of order d in G, then $H \cap K = \{e\}$.

105 Find the minimum positive integer n such that

(i) the group S_n contains a subgroup of order 36;
(ii) the group S_n contains a subgroup of order 72;
(iii) the group S_n contains a subgroup of order 144.

106 Let G be an Abelian group of order $2^n 5^m$, where m, n are positive integer. Show that the following are equivalent:

(i) G is a cyclic group;
(ii) $\{g \in G \mid 10g = 0\}$ is cyclic;
(iii) the subgroups $\{2g \mid g \in G\}$ and $\{5g \mid g \in G\}$ are both cyclic.

107 Consider the permutations $\tau = (1, 2)(3, 4, 5)$ and $\sigma = (5, 6, 7)$ in S_9. Describe the centralisers of τ and $\tau\sigma$ and the normalisers of the subgroups generated by τ and $\tau\sigma$, and in particular determine their orders.

108 For each prime p, find all pairs of positive integers (a, b) such that the group $\mathbb{Z}/p^a\mathbb{Z} \times \mathbb{Z}/p^b\mathbb{Z}$ has a characteristic subgroup of order p.

109 Find all possible orders for the centre of a group of order 75.

110 Let G be a finite group acting on a set $\mathcal{B} = \{e_1, \cdots, e_n\}$ by permuting its elements, and let V be the vector space on $\mathbb{C}$ for which $\mathcal{B}$ is a basis. Consider the homomorphism $G \longrightarrow \mathsf{GL}(V)$ induced by the action of G on elements of the basis $\mathcal{B}$ of V. Show that the dimension of the vector subspace on which G acts trivially is equal to the number of orbits of G in $\mathcal{B}$.

111 Show that a group of order 870 cannot be simple.

112 Find the minimum positive integer n such that S_n contains a subgroup isomorphic to the group Q_8 of unit quaternions.

113 Show that

(i) the group S_7 contains a subgroup isomorphic to D_{12};
(ii) the group A_7 contains a subgroup isomorphic to D_6;
(iii) the group A_7 does not contain a subgroup isomorphic to D_{12}.

114 Let G be a finite group of order $p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$, where $p_1, p_2, \ldots, p_n$ are distinct primes; for $h = 1, 2, \ldots, n$, let P_h be some p_h-Sylow subgroup of G. Given a subgroup K of G, we denote by $N(K)$ its normaliser.

(i) Show that $N(N(P_h)) = N(P_h)$ for $h = 1, 2, \ldots, n$.
(ii) Show that $G \simeq P_1 \times P_2 \times \cdots \times P_n$ if and only if $N(K) \neq K$ for all subgroups K such that $K \neq G$.

115

(i) Let p be a prime and let G be a p-group acting on a finite set X. Show that if $|X|$ is not divisible by p then there is an element $x \in X$ such that $g \cdot x = x$ for all $g \in G$.
(ii) Let V be a finite-dimensional vector space on the field $\mathbb{F}_p$ and let G be a p-group in $\mathsf{GL}(V)$. Show that there is a nonzero vector $v \in V$ such that $g \cdot v = v$ for all $g \in G$.

116 Find the number of elements of order 18 and the number of subgroups of order 18 in the group $\mathbb{Z}/36\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z}$.

117 Let p be a prime and let G be a group of order $p^3 + p^2$. Show that the intersection of all p-Sylow subgroups of G is nontrivial.

118

(i) Solve the equation $\sigma^3 = (1, 2)(3, 4)(5, 6)$ in S_6.
(ii) Determine the structure of the centraliser of $(1, 2)(3, 4)(5, 6)$ in S_6.

119 A subgroup H of a group G is said to be *subnormal* in G if there exists a finite sequence $H = H_0 \subseteq H_1 \subseteq \ldots \subseteq H_n = G$ of subgroups of G, each of them normal in the next. Show the following statements:

(i) if G is a p-group, then all of its subgroups are subnormal;
(ii) a subnormal group H of a group G is not necessarily normal in G;
(iii) if H is subnormal in G and P is a p-Sylow subgroup of G, then $P \cap H$ is a p-Sylow subgroup of H.

120 Show that a group of order 300 cannot be simple.

121 Let p be a prime and let τ be the product of three disjoint p-cycles in the group S_{3p}.

(i) Determine, as a function of p, the number of solutions $\sigma \in \mathsf{S}_{3p}$ to the equation $\sigma^p = \tau$.
(ii) Show that, for $p \geq 3$, there is a subgroup of S_{3p} containing τ that is isomorphic to the dihedral group D_p.

122 Let G be a group of order 1045.

(i) Show that G has a unique 19-Sylow subgroup, which is contained in its centre. Additionally, determine all possible values for the cardinality of the centre of G.
(ii) Show that there is a nontrivial homomorphism $G \longrightarrow \mathbb{Z}/154\mathbb{Z}$ if and only if G is cyclic.

123 Let σ be the product of two 3-cycles and two 5-cycles in S_{16}, all pairwise disjoint.

(i) Express the centraliser of σ as a semidirect product.
(ii) Show that the centraliser of σ contains a subgroup isomorphic to D_{15}.

124

(i) Show that a group of order p^2q^2, where p, q are distinct primes, cannot be simple.
(ii) Let $n(p, q)$ be the number of isomorphism classes of groups with p^2q^2 elements; determine the minimum possible value of $n(p, q)$, where p and q are distinct primes.

125 Show that the automorphism group of $\mathsf{A}_4 \times \mathbb{Z}/2\mathbb{Z}$ is isomorphic to S_4.

126 Let G be a group of order 399.

(i) Show that G is isomorphic to a semidirect product of cyclic groups.
(ii) What are the possible values for the order of the centre of G? For each possible value, provide an example.

127 For each $n \geq 3$, determine the smallest normal subgroup of S_n that contains an n-cycle.

128 Show that any group of order $5^2 \cdot 7 \cdot 17$ is Abelian.

129 Let $\sigma = (1, 2, 3, 4)$, $\tau = (2, 4)(5, 6)$, and let H be the subgroup of S_6 they generate.

(i) Determine the order of H and the order of the centraliser of H in S_6.
(ii) Determine the normaliser of H and show that H is contained in a unique 2-Sylow subgroup of S_6.

130 Let G be a finite group and let G' be its commutator subgroup.

(i) Show that if M is a maximal subgroup of G then

$$Z(G) \subseteq M \text{ or } G' \subseteq M.$$

(ii) Provide an example of a non-Abelian group G and a maximal subgroup M of G such that $G' \subseteq M$ and $Z(G) \not\subseteq M$.

131

(i) Show that $\mathsf{SL}_2(\mathbb{F}_5)$ contains a subgroup isomorphic to Q_8.
(ii) Show that S_5 is not isomorphic to $\mathsf{SL}_2(\mathbb{F}_5)$.

132 The exponent of a finite group G is the smallest positive integer d such that $g^d = 1$ for all g in G.

(i) Show that the exponent of G is equal to the order of G if and only if all Sylow subgroups of G are cyclic.
(ii) Show that if G is Abelian then the exponent of G is equal to the maximal order of an element of G.

133 Prove that the possible orders for an Abelian subgroup of S_7 are 12 and all positive integers less than 11.

134 A group G is *hypercyclic* if all of its Sylow subgroups are cyclic. Show that if G is hypercyclic then

(i) all subgroups and quotients of G are hypercyclic;
(ii) given any prime p and natural number r, all subgroups of G of order p^r are pairwise conjugate;
(iii) for each normal subgroup N of G and p-Sylow subgroup P of G, the order of $|N \cap P|$ is the greatest common divisor of $|N|$ and $|P|$.

135

(i) Show that D_{15} has at least a subgroup of order d for each divisor d of 30.
(ii) Find all divisors d of 30 such that D_{15} has a unique subgroup of order d.

2.2 Rings

136 Find the number of zero divisors and the number of invertible elements in the ring $\mathbb{K}[x]/\big((x^2-2)(x^3-2)\big)$ for $\mathbb{K} = \mathbb{F}_3$ and for $\mathbb{K} = \mathbb{F}_7$.

137 Consider the polynomials in $\mathbb{C}[x]$ recursively defined by

$$p_0(x) = 1$$
$$p_{n+1}(x) = (x-1)\,p_n(\zeta x)$$

where $\zeta \in \mathbb{C}$ is a primitive third root of unity.

(i) Show that $p_{3n}(x) = (x^3-1)^n$ for all $n \geq 0$.
(ii) For what $n \in \mathbb{N}$ do we have $p_n(x) \in \mathbb{Z}[x]$?

138 Let A be a ring with the following property: for each $x \in A$ there is $n > 1$ (depending on x) such that $x^n = x$. Show that an ideal of A is prime if and only if it is maximal.

139 Consider the ideals $I = (5, x)$ and $J = (25, x)$ in $\mathbb{Z}[x]$. Show that

(i) I is maximal;
(ii) J is *primary*, that is, it has the following property: given $f(x), g(x) \in \mathbb{Z}[x]$, if $f(x)g(x) \in J$ then $f(x) \in J$ or $g(x)^n \in J$ for some natural number n;
(iii) the radical of J is equal to I.

140 Let $A = \mathbb{Q}[x]$ be the ring of polynomials with coefficients in $\mathbb{Q}$ and let $S_1 = A \setminus \{0\}$ and $S_2 = A \setminus (x)$. Show that S_1 and S_2 are multiplicative sets in A and that $S_1^{-1}A$ is not isomorphic to $S_2^{-1}A$.

141 Consider the rings $A = \mathbb{Z}[i\,]/(2)$ and $B = \mathbb{Z}[i\,]/(3)$.

(i) Show that A is not an integral domain.
(ii) Are the rings $A[x]$ and $B[x]$ isomorphic?

142 Prove that $A = \{m/n \in \mathbb{Q} \mid n \text{ odd}\}$ is a subring of $\mathbb{Q}$ and that, given an ideal I of A, I is generated by 2^k for some natural number k. Find the fraction field of A.

143 Determine the unit group of the ring $\mathbb{Z}[\sqrt{-3}]$.

144 Let A be the ring $\mathbb{Q}[t, t^{-1}]$ and let σ be the automorphism

$$A \ni f(t) \longmapsto f(t^{-1}) \in A.$$

Show that the set of fixed points $A^\sigma = \{f \in A \mid \sigma(f) = f\}$ is the ring of polynomials $\mathbb{Q}[t + t^{-1}]$.

145 Let A be a commutative ring and let I, J, K be three ideals of A.

(i) Show that, if $I + J + K = A$, then $I^n + J^n + K^n = A$ for all $n \geq 1$.
(ii) Show that, if $I + J = J + K = K + I = A$, then $IJ + JK + KI = A$.

146 Let A be an integral domain and let $\mathbb{K}$ be its fraction field. We say that A is *integrally closed* is it has the following property: if an element α of $\mathbb{K}$ is a root of a monic polynomial with coefficients in A, then α belongs to A.

(i) Show that every unique factorisation domain is integrally closed.
(ii) Show that the rings $\mathbb{Z}[\sqrt{4n+1}]$, where n is an integer and $4n + 1$ is not a square in $\mathbb{Z}$, are not unique factorisation domains.

147 Given an ideal I of a ring A, define $\mathcal{V}(I)$ as the set of prime ideals of A that contain I. Given two ideals I, J of A, show that

(i) there is an ideal M of A such that $\mathcal{V}(I) \cup \mathcal{V}(J) = \mathcal{V}(M)$,
(ii) there is an ideal N of A such that $\mathcal{V}(I) \cap \mathcal{V}(J) = \mathcal{V}(N)$.

Is it true that $\mathcal{V}(I) = \mathcal{V}(J)$ if and only if $I = J$?

148 Let A be an integral domain and let S be a multiplicative set in A. Show that if an ideal I is maximal within the family of ideals of A that do not intersect S, then I is a prime ideal of A.

149 Show that the radical of (x^2, y^2) is a prime ideal in the ring $\mathbb{Q}[x, y]$.

150 Let A be the ring $\mathbb{Q}[x, y]$ and let I be the ideal generated by the polynomial $xy - 1$. Determine the set of all ring homomorphisms from A/I to $\mathbb{Q}$.

151 Let a be an element of a ring A and let I be an ideal of A. Setting $\mathcal{J} = \{f(x) \in A[x] \mid f(a) \in I\}$, show that

(i) the set $\mathcal{J}$ is an ideal of $A[x]$;
(ii) I is a prime ideal of A if and only if $\mathcal{J}$ is a prime ideal of $A[x]$;
(iii) if $A = \mathbb{Z}$, $I = (5)$ and $a = 1$, then $\mathcal{J} = (5, x - 1)$.

152 Let $S = \mathbb{Z} \setminus 2\mathbb{Z}$ and let $A = \mathbb{Z}[i]$ be the ring of Gaussian integers.

(i) Find all invertible elements in $S^{-1}A$.
(ii) Show that all ideals of $S^{-1}A$ are of the form $S^{-1}A \cdot (1 + i)^k$, for $k \geq 0$.

153 Let A be a ring, let I be an ideal of A and let $I[x]$ be the set of polynomials in $A[x]$ whose coefficients are in I.

(i) Show that $I[x]$ is an ideal of $A[x]$;
(ii) show that $I[x]$ is the ideal of $A[x]$ generated by I;
(iii) is it true that if I is prime then $I[x]$ is prime?

154 Let A be the ring $\mathbb{Z} \times \mathbb{Z}$.

(i) Show that all ideals of A are principal.
(ii) Find all prime ideals and all maximal ideals of A.

155 Let A be a ring and let $A[x]$ be the ring of polynomials in one variable with coefficients in A. Show that if $A[x]$ is a principal ideal domain then A is a field.

156 Determine for which primes p and for which elements $a \in \mathbb{F}_p^*$ the rings $\mathbb{F}_p[x]/(x^2 - a)$ and $\mathbb{F}_p[x]/(x^2 + a)$ are isomorphic.

157 Let A be the subring $\{a + b\sqrt{7} \mid a, b \in \mathbb{Z}\}$ of $\mathbb{R}$; consider the map $N : A \to \mathbb{Z}$ given by $N(a + b\sqrt{7}) = (a + b\sqrt{7})(a - b\sqrt{7})$ for all $a, b \in \mathbb{Z}$. Show that

(i) for all $u,\ v \in A$ we have $N(uv) = N(u)N(v)$;
(ii) the element u is invertible in A if and only if $N(u) = \pm 1$;
(iii) a prime number $p \in \mathbb{Z}$ is reducible in A if and only if there is $u \in A$ such that $N(u) = \pm p$;
(iv) 2 is reducible in A and 5 is irreducible in A.

158 Show that in the ring $\mathbb{Z}/m\mathbb{Z}$, where m is a positive integer, an element a is nilpotent if and only if $1 - ab$ is invertible for all $b \in \mathbb{Z}/m\mathbb{Z}$.

159 Let A be a unique factorisation domain, let x be a prime element of A and let S be the multiplicative set $A \setminus (x)$. Show that

(i) the localisation $S^{-1}A$ is a principal ideal domain;
(ii) the intersection of all nonzero ideals of $S^{-1}A$ is $\{0\}$.

160 Determine whether each of the ideals $(x^2 + y^2 - 1)$ and $(x^2 - 3, y^2 - x)$ of $\mathbb{Q}[x, y]$ is prime, and whether it is maximal.

161 Given a positive integer m, consider the multiplicative sets $S = \{m^k \mid k \in \mathbb{N}\}$ and $T = \{a \in \mathbb{Z} \mid (a, m) = 1\}$ in $\mathbb{Z}[x]$.

(i) Show that there is no surjective homomorphism $S^{-1}\mathbb{Z}[x] \longrightarrow \mathbb{Q}$.
(ii) Provide an example of a surjective homomorphism $T^{-1}\mathbb{Z}[x] \longrightarrow \mathbb{Q}$.

162 Let A be a principal ideal domain. Show that if B is an integral domain and $\varphi : A \longrightarrow B$ is a surjective homomorphism, then φ is an isomorphism or B is a field.

163 Let A be a principal ideal domain that is not a field and let $I \neq \{0\}$ be a proper ideal of A. For each of the following families of ideals of A, determine whether they are always finite, always infinite, or they can be either:

(i) $\{I + J \mid J \text{ ideal of } A\}$,
(ii) $\{I \cap J \mid J \text{ ideal of } A\}$,
(iii) $\{J \mid I + J = L\}$, where L is some fixed ideal containing I.

164 Let A be a ring whose every ideal other than A itself is prime. Show that A is a field.

165

(i) Describe the set of all integers a such that the ideal $(11, x^2 + a)$ is prime in $\mathbb{Z}[x]$.
(ii) Check that $S = A \setminus (11, x^2 + 3)$ is a multiplicative set in $\mathbb{Z}[x]$ and determine for which values of the parameter $\lambda \in \mathbb{Z}$ the polynomial $x^4 + \lambda x^2 + 5$ is invertible in $S^{-1}\mathbb{Z}[x]$.

166 Let A be an integral domain and let $\mathcal{I}$ be the family of all nonzero ideals of A. Define the following relation on $\mathcal{I}$:

$I \sim J$ if and only if there exist nonzero elements $a, b \in A$ such that $aI = bJ$.

Show that

(i) the relation $\sim$ is an equivalence relation on $\mathcal{I}$;
(ii) the ring A is a principal ideal domain if and only if for all $I, J \in \mathcal{I}$ we have $I \sim J$.

167 Let A, B be rings and let $\varphi : A \to B$ be a ring homomorphism. Prove or disprove the following statements.

(i) If P_1, P_2 are prime ideals of A then $P_1 + P_2$ is a prime ideal of A.
(ii) If the radical $\sqrt{I}$ of an ideal I is prime then I is a prime ideal.
(iii) If Q is a prime ideal of B then $\varphi^{-1}(Q)$ is a prime ideal of A.
(iv) If Q is a maximal ideal of B then $\varphi^{-1}(Q)$ is a maximal ideal of A.

168 A ring is said to be *local* if it has a unique maximal ideal. Show that

(i) if $\varphi : A \longrightarrow B$ is a ring homomorphism and A is local, then $\varphi(A)$ is local;
(ii) the ring A is local if and only if $A \setminus A^*$ is an ideal of A.

169 Let $\mathbb{K}$ be a field and let $\mathbb{K}(x)$ be the ring of rational functions over $\mathbb{K}$. Given a nonzero element $\rho(x)$ of $\mathbb{K}(x)$, define $\operatorname{ord}(\rho(x))$ as the integer v such that $\rho(x) = x^v f(x)/g(x)$ with $f(x)$ and $g(x)$ polynomials in $\mathbb{K}[x]$ that do not vanish at 0.

(i) Show that

$$A = \left\{\rho(x) \in \mathbb{K}(x) \setminus \{0\} \mid \operatorname{ord}(\rho(x)) \geq 0\right\} \bigcup \{0\}$$

is a subring of $\mathbb{K}(x)$ and find its invertible elements.
(ii) Show that there is a multiplicative set S of $\mathbb{K}[x]$ such that $S^{-1}\mathbb{K}[x] = A$ as a subset of $\mathbb{K}(x)$.

170 Find the number of invertible elements and the number of zero divisors in the ring

$$\mathbb{F}_5[x]/(x^6 + 2x^5 + x + 2) \times \mathbb{F}_7[x]/(x^6 + 2x^5 + x + 2);$$

additionally, find all prime ideals.

171 Let S be the multiplicative set $\left\{n \in \mathbb{Z} \mid n \equiv 1 \pmod{30}\right\}$ of $\mathbb{Z}$ and let $A = S^{-1}\mathbb{Z}$.

(i) Find all maximal ideals of A.
(ii) Letting $\mathbb{Z} \ni x \overset{i}{\longmapsto} x/1 \in A$ be the canonical inclusion homomorphism, determine $i^{-1}(A^*)$.

172 Let $\mathbb{K}$ be a field, let $f(t), g(t) \in \mathbb{K}[t]$ be two non-constant polynomials and let I be the ideal $(f(x), g(y))$ in $\mathbb{K}[x, y]$.

(i) Show that I is not a principal ideal.
(ii) Find the dimension of $\mathbb{K}[x, y]/I$ as a vector space over $\mathbb{K}$.

173 Given a formal power series $f(x) = \sum_{n\geq 0} a_n x^n$ in $\mathbb{K}[[x]]$, where $\mathbb{K}$ is a field, denote by $Df(x)$ the formal power series $\sum_{n\geq 0}(n+1)a_{n+1}x^n$.

(i) Show that for all $f(x), g(x) \in \mathbb{K}[[x]]$ we have $D(f(x)g(x)) = (Df(x))g(x) + f(x)(Dg(x))$.
(ii) Show that the system

$$\begin{cases} D(f(x)) = f(x) \\ f(0) = 1 \end{cases}$$

has a unique solution if $\mathrm{Char}(\mathbb{K}) = 0$ and has no solutions if $\mathrm{Char}(\mathbb{K}) > 0$.

174 Let A be the ring $\mathbb{Z}^n$, where n is a positive integer. Show that there are finitely many automorphisms of A and determine their number.

175

(i) Show that there is no homomorphism $\varphi : \mathbb{Z}[x^2, x^3] \longrightarrow \mathbb{Q}$ such that $\varphi(x^2) = 1/3$;
(ii) find all homomorphisms $\varphi : \mathbb{Z}[x^2, x^3] \longrightarrow \mathbb{Q}$ such that $\varphi(x^2) = 1/4$;
(iii) describe a system of generators for $\mathrm{Ker}(\varphi)$ for each homomorphism φ as above.

176

(i) Check that

$$S = \left\{a_0 + a_1 x + \cdots + a_n x^n \in \mathbb{Z}[x] \,|\, a_o + a_1 + \cdots + a_n \text{ is odd}\right\}$$

is a multiplicative set in $\mathbb{Z}[x]$;
(ii) show that the ring $S^{-1}\mathbb{Z}[x]$ has infinitely many prime ideals;
(iii) show that the ring $\mathsf{S}^{-1}\mathbb{Z}[x]$ has a unique maximal ideal.

177 Let A be a ring.

(i) Is every ideal that is maximal within the family of proper principal ideals a prime ideal of A?
(ii) For each $x \in A$, let $\mathrm{Ann}(x)$ denote the ideal $\{a \in A \,|\, ax = 0\}$. Is it true that every ideal that is maximal within the family of ideals of the form $\mathrm{Ann}(x)$ with $x \in A \setminus \{0\}$ is a prime ideal of A?

178 Let A be a unique factorisation domain with exactly three prime ideals, all of them principal: $\{0\}$, P and Q. Show that

(i) for all pairs (m, n) of positive integers, $P^m + Q^n = A$;
(ii) the ring A is a principal ideal domain.

179 Let the complex numbers ζ_5 and ζ_8 be a primitive fifth root and a primitive eighth root of unity, respectively.

(i) Determine an ideal I of $\mathbb{Z}[x]$ such that $\mathbb{Z}[\zeta_5] \simeq \mathbb{Z}[x]/I$;
(ii) show that the ring $\mathbb{Z}[\zeta_5]/(11)$ is not a field;
(iii) show that the two rings $\mathbb{Z}[\zeta_5]/(11)$ and $\mathbb{Z}[\zeta_8]/(11)$ are not isomorphic.

180

(i) Show that $\mathbb{Q}[x, y]$ has infinitely many prime ideals that are not maximal.
(ii) Determine the set of non-constant polynomials $f(x, y)$ such that

$$(f(x, y)) \neq (f(x, y), x) \neq \mathbb{Q}[x, y].$$

181 Show that for any field $\mathbb{K}$ the ring $\mathbb{K}[x, y]/(x^m, y^n)$ has a unique prime ideal if and only if the natural numbers m, n satisfy $mn > 0$.

182 Let $\mathbb{K}$ be a field and let A be a ring containing $\mathbb{K}$. Assume that A is finite-dimensional as a vector space over $\mathbb{K}$ and prove that

(i) every element $a \in A$ is the root of a nonzero polynomial in $\mathbb{K}[x]$;
(ii) every prime ideal of A is maximal.

183 Let $\Phi_n(t) \in \mathbb{Z}[t]$ be the nth cyclotomic polynomial, let p be an odd prime and let $\Psi_n(t)$ be the reduction of $\Phi_n(t)$ modulo p.

(i) Show that $\mathbb{F}_p[t]/(\Psi_{p-1}(t))$ is isomorphic to $\mathbb{F}_p^d$, where $d = \phi(p - 1)$.
(ii) Show that there are nonzero nilpotent elements in $\mathbb{F}_p[t]/(\Psi_p(t))$.

184 Letting A be a ring and I, J be ideals of A, we recall the definition of the ideal $(I : J) = \{a \in A \mid aJ \subseteq I\}$.

(i) Given a prime ideal P of A, determine $(P : J)$ as J varies among all ideals of A.
(ii) Suppose $A = \mathbb{Z}[i]$; determine $((18 + 6i) : (10))$.
(iii) Provide an example of a ring A and elements x, y of A such that the ideal $((x) : (y))$ is *not* principal.

185 Let A be the subring of $\mathbb{Q}(x)$ defined as

$$A = \Big\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in \mathbb{Q}[x],\ g(0)g(-1) \neq 0 \Big\}.$$

(i) Determine all invertible elements of A.
(ii) Show that A is a principal ideal domain.
(iii) Determine all prime ideals of A.

186 Given a ring A, show that $f(x) \in A[x]$ is nilpotent if and only if all of its coefficients are nilpotent elements of A.

187 Given a ring A, denote by $J(A)$ the intersection of all its maximal ideals.

(i) Show that $x \in J(A)$ if and only if $1 + xy$ is invertible in A for all $y \in A$.
(ii) Let $\varphi : A \longrightarrow B$ be a surjective ring homomorphism. Show that $\varphi(J(A)) \subseteq J(B)$.

188 Determine the invertible elements and the prime ideals of the ring

$$\Big\{ \frac{\alpha}{\beta} \,\Big|\, \alpha, \beta \in \mathbb{Z}[i\,],\ (\beta, 3 + i) = 1 \Big\}.$$

189 Given a subring A of a ring R, define the set

$$\mathfrak{f} = \{a \in R \mid aR \subseteq A\}.$$

(i) Show that $\mathfrak{f}$ is both an ideal of R and an ideal of A.
(ii) Show that $\mathfrak{f}$ is the largest ideal of A that is also an ideal of R.
(iii) Determine a generator of $\mathfrak{f}$ as an ideal of R in the case where $A = \mathbb{Z}[\sqrt{-3}]$ and $R = \mathbb{Z}[(-1 + \sqrt{-3})/2]$.

190 Let A be a ring and let $N = \sqrt{0}$ be its nilradical. Show that

(i) if N is finitely generated then there exists $k \in \mathbb{N}$ such that $N^k = \{0\}$;
(ii) an element a is invertible in A if and only if its class $a + N$ is invertible in A/N.

191

(i) Show that the ring $A = \mathbb{Z}[\,(1 + \sqrt{-7})/2\,]$ is a Euclidean domain with degree

$$A \ni a + b\frac{1 + \sqrt{-7}}{2} \ \overset{d}{\longmapsto}\ a^2 + ab + 2b^2 \in \mathbb{N}.$$

(ii) Determine the factorisation of 10 as a product of irreducible elements in A.

192 Let A be a ring all of whose ideals are principal.

(i) Show that if A is an integral domain and I is an ideal of A such that $I^2 = I$, then $I = 0$ or $I = A$.
(ii) Suppose the maximal ideals of A are $M_1, M_2, \ldots, M_n$ (in particular, they are finitely many). Show that every ideal I of $A/(M_1 \cdot M_2 \cdots M_n)$ is such that $I^2 = I$.

193

(i) Show that if p is a prime number and $a \in \mathbb{Z}$ then $(p, x-a)$ is a maximal ideal of $\mathbb{Z}[x]$.
(ii) Show that, for each monic irreducible polynomial $f(x) \in \mathbb{Z}[x]$, the ideal $(f(x))$ is not maximal.

194 Let A be a unique factorisation domain.

(i) Show that every prime ideal P of A is generated by the set

$$\mathcal{P} = \{\pi \in P \mid \pi \text{ is a prime element of } A\}.$$

(ii) Show that if A has finitely many prime elements and is such that for each prime π in A the ideal (π) is maximal, then the sum of two principal ideals is principal.

195 Let A be a ring and let D be the set of its zero divisors.

(i) Show that D can be expressed as the union of a family of ideals.
(ii) Show that if D is an ideal, then it is prime.
(iii) Show that if D is an ideal and A is finite, then all elements of D are nilpotent.

196 Let A be an integral domain, let S be a multiplicative set of A and let $S^{-1}A$ be the corresponding ring of fractions. Show that

(i) if I is an ideal of A then $S^{-1}\sqrt{I} = \sqrt{S^{-1}I}$;
(ii) if A is a unique factorisation domain then

$$\mathrm{MCD}_{S^{-1}A}\big(a/1, b/1\big) = \mathrm{MCD}_A(a, b)/1$$

up to associates, for all $a, b \in A$ that are not both zero.

197 Consider the ideals $I = (5 + 14i)$ and $J = (-4 + 7i)$ of the ring $\mathbb{Z}[i]$.

(i) Find a generator for $I \cap J$ and one for $I + J$.
(ii) Determine all prime ideals of $\mathbb{Z}[i]/(I + J)$.

198 Given a principal ideal domain A, show that every maximal ideal of $A[x]$ has a set of at most two generators.

199 Let A be an integral domain. A multiplicative set S in A is called *saturated* if for all $s \in S$ and $d \in A$ such that $d \mid s$, we have $d \in S$.

(i) Show that, given a multiplicative set S in A, there exists the smallest saturated multiplicative set of A containing S, which we call the *saturation* of S.
(ii) Show that two multiplicative sets S and T in A have the same saturation if and only if $S^{-1}A = T^{-1}A$, seen as subsets of the fraction field of A.

200 Let A be a ring and let N be the ideal of nilpotent elements of A. Show that the following are equivalent:

(a) the ring A has a unique prime ideal,
(b) every element of A is either invertible or nilpotent,
(c) the quotient A/N is a field.

201 Given a Gaussian integer $a + bi$, let $A = \mathbb{Z}[a + bi]$ be the subring of $\mathbb{Z}[i]$ generated by $a + bi$. Consider the ideal

$$\mathfrak{f} = \{z \in A \mid \mathbb{Z}[i] \cdot z \subseteq A\}$$

of A. Determine the index of $\mathfrak{f}$ in A.

202 Consider the ring of Gaussian integers $A = \mathbb{Z}[i]$ and let N be the usual norm given by $N(a + bi) = a^2 + b^2$.

(i) Show that if I is the ideal generated by $a + bi$ then $|A/I| = N(a + bi)$.
(ii) Find the number of ideals I of A whose index is 100.

203 A proper ideal Q of a ring A is said to be *primary* if it has the following property: if $ab \in Q$ and $a \notin Q$ then $b^n \in Q$ for some natural number n.

(i) Show that an ideal Q of A is primary if and only if every zero divisor in A/Q is nilpotent.
(ii) Determine all primary ideals of $\mathbb{Z}[i]$.
(iii) Determine all primary ideals of $\mathbb{Z} \times \mathbb{Z}$.

204 Let A be the ring $\mathbb{Z}[\sqrt{2}]$.

(i) Show that the ideal (5) is prime in A.
(ii) Determine how many ideals of A contain 7.
(iii) Show that there are two distinct prime ideals P, Q in A such that $P \cap \mathbb{Z} = Q \cap \mathbb{Z}$.

205 Let p be a prime and let

$$A = \left\{(a_1, a_2, a_3, \dots) \mid a_n \in \mathbb{Z}/p^n\mathbb{Z},\ a_{n+1} \equiv a_n \pmod{p^n} \text{ for all } n \geq 1\right\}.$$

The set A, endowed with component-wise operations, is a unitary commutative ring.

(i) What are the invertible elements of A?
(ii) Show that A has a unique maximal ideal and that it is principal.
(iii) Show that every nonzero ideal of A is a power of the unique maximal ideal.

206 Let A be a unique factorisation domain, let P be a prime ideal of A and let $S = A \setminus P$. Show that $S^{-1}A$ is a unique factorisation domain.

207 Let A be the ring $\mathbb{Z}[\sqrt{13}]$.

(i) Check that $18 + 5\sqrt{13}$ is invertible in A and that A^* is infinite.
(ii) Check that the elements 2 and $3 + \sqrt{13}$ are irreducible in A.
(iii) Show that the ring A is not a unique factorisation domain.

208 Let $\mathbb{K}_1, \mathbb{K}_2, \dots, \mathbb{K}_n$ be fields and set $A = \mathbb{K}_1 \times \mathbb{K}_2 \times \cdots \times \mathbb{K}_n$.

(i) Show that every ideal of A is principal.
(ii) Find the number of prime ideals of A.

209 Let X be an infinite set, let $\mathbb{K}$ be a field and let A be the ring of maps $f : X \longrightarrow \mathbb{K}$, with the following operations: $(f+g)(x) = f(x)+g(x)$ and $(fg)(x) = f(x)g(x)$. Show that A does contain non-principal ideals.

2.3 Fields and Galois Theory

210

(i) Compute the minimal polynomial $f(x)$ of $\sqrt{5}+i$ over $\mathbb{Q}$ and check that $f(x)$ has integer coefficients.
(ii) Compute the degree of the splitting field of $f(x)$ over $\mathbb{Q}$, over $\mathbb{Q}(i)$ and over $\mathbb{F}_5$.

211 Compute the degree of the splitting field of the polynomial $x^4 - 4x^2 + 16$ over $\mathbb{Q}$, over $\mathbb{F}_{13}$ and over $\mathbb{F}_5$.

212 Determine the degree of the splitting field of the polynomial $x^4 + x^2 + 1$ over $\mathbb{Q}$, over $\mathbb{F}_3$ and over $\mathbb{F}_7$.

213 Given a natural number n, let ζ and η be two distinct primitive nth roots of unity in $\mathbb{C}$. Show that $\zeta - \eta$ is not a rational number.

214 Let $\mathbb{E}$, $\mathbb{F}$ and $\mathbb{K}$ be the splitting fields over $\mathbb{Q}$ of the polynomials x^2-3, x^3-2 and $(x^2-3)(x^3-2)$, respectively.

(i) Describe the subextensions of $\mathbb{F}$ that are normal over $\mathbb{Q}$.
(ii) Show that $\mathbb{E} \cap \mathbb{F} = \mathbb{Q}$.
(iii) Compute the degree of $\mathbb{K}$ over $\mathbb{Q}$.

215 Let ζ be a primitive pth root of unity in $\mathbb{C}$, where p is an odd prime, and let G be the Galois group of the extension $\mathbb{Q}(\zeta)/\mathbb{Q}$. Show that we have

$$\sum_{\varphi \in G} \varphi(\zeta) = -1, \qquad \prod_{\varphi \in G} \varphi(\zeta) = 1.$$

216 For each $n \in \mathbb{N}$, find the degree of the splitting field of $x^4 - n$ over $\mathbb{Q}$.

217 Find all subextensions of the twelfth cyclotomic field over $\mathbb{Q}$.

218 Find the smallest positive integer n such that the cyclotomic extension of order n over $\mathbb{Q}$ has a subfield whose Galois group over $\mathbb{Q}$ is isomorphic to $\mathbb{Z}/7\mathbb{Z}$.

219 Determine the degree of the splitting field over $\mathbb{Q}$ of the polynomial $(x^2+1)^2+1$.

220 Let $f(x)$ be the polynomial $(x^2-2)(x^2-3)(x^2-6)$.

(i) Compute the Galois group of $f(x)$ over $\mathbb{Q}$.
(ii) Show that $f(x)$ has a root in $\mathbb{F}_p$ for every prime p.

221 Show that $\mathbb{K} = \mathbb{Q}(i, \sqrt{3}, \sqrt[3]{3})$ is a normal extension of $\mathbb{Q}$ and compute $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$. Additionally, determine the number of subextensions of $\mathbb{K}$ of degree 6 over $\mathbb{Q}$.

222 Letting $\mathbb{F}$ be the field $\mathbb{Q}(\sqrt[5]{5})$, find an extension $\mathbb{K}/\mathbb{F}$ of degree 4 such that $\mathbb{K}$ is normal over $\mathbb{Q}$. Show that $\mathbb{K}$ is unique and has a unique subextension of degree 4 over $\mathbb{Q}$.

223 Letting ζ denote a primitive fifth root of unity in $\mathbb{C}$, compute the degree and the Galois group of the extension $\mathbb{Q}(\sqrt{n}, \zeta)/\mathbb{Q}$ for each positive integer n.

224 Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial and let $\mathbb{K}$ be its splitting field. Find all possible values for the degree of $f(x)$ under the assumption that $[\mathbb{K} : \mathbb{Q}] = 8$. For each possible degree, provide an example of a polynomial $f(x)$ with the required properties.

225 Given the polynomial $(x^4-x^2+1)(x^2-3)$, determine the degree of its splitting field and its Galois group over $\mathbb{Q}$ and over $\mathbb{F}_7$.

226 Compute the degree of the splitting field and the Galois group of the polynomial $x^4 - 5x^2 + 9$ over $\mathbb{Q}$ and over $\mathbb{F}_{11}$.

227 Let $f(x)$ be the polynomial $(x^8 - 1)(x^3 - 1)$.

(i) For which primes p does the polynomial $f(x)$ split as a product of linear factors over $\mathbb{F}_p$?
(ii) Compute the Galois group of $f(x)$ over $\mathbb{Q}$.

228 Given the polynomial $x^6 - 7x^4 + 3x^2 + 3$, compute the degree of its splitting field over $\mathbb{Q}$ and over $\mathbb{F}_{13}$.

229 Let α and β be algebraic elements over a field $\mathbb{K}$ and let $f(x)$ and $g(x)$ be their respective minimal polynomials over $\mathbb{K}$. Show that

(i) $f(x)$ is irreducible over $\mathbb{K}(\beta)$ if and only if $g(x)$ is irreducible over $\mathbb{K}(\alpha)$;
(ii) if $(\deg(f(x)), \deg(g(x))) = 1$ then $f(x)$ is irreducible over $\mathbb{K}(\beta)$.

230 Compute the degree of the splitting field of $x^7 - 2$ over $\mathbb{Q}$ and over $\mathbb{F}_5$, and determine whether or not the corresponding Galois groups are Abelian.

231 Letting $\mathbb{F}$ and $\mathbb{K}$ be the splitting fields over $\mathbb{Q}$ of the polynomials $x^4 - 2$ and $x^6 - 2$, respectively, determine the degree of $\mathbb{F} \cap \mathbb{K}$ over $\mathbb{Q}$.

232 Determine all degree 4 subfields of the splitting field over $\mathbb{Q}$ of the polynomial $(x^5 - 2)(x^2 - 5)$.

233 Let ζ be a third root of unity in $\mathbb{C}$, let $\mathbb{K} = \mathbb{Q}(\zeta)$, and denote by $\mathbb{F}$ the splitting field of the polynomial $x^{11} - 3$ over $\mathbb{K}$.

(i) Determine $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$.
(ii) Show that for all $\alpha \in \mathbb{F}$ we have $\mathbb{K}(\alpha^3) = \mathbb{K}(\alpha)$.

234 Let $\zeta \in \mathbb{C}$ be a primitive 13th root of unity and let $\alpha = \zeta + \zeta^3 + \zeta^9$. Letting $\mathbb{F} = \mathbb{Q}(\alpha)$, determine the Galois group of the extension $\mathbb{F}/\mathbb{Q}$ and whether or not $\mathbb{F}$ is a subfield of $\mathbb{R}$.

235 Let $\mathbb{K}$ be the splitting field of the polynomial $(x^3 - 3)(x^5 - 5)$ over $\mathbb{Q}$.

(i) Determine $[\mathbb{K} : \mathbb{Q}]$.
(ii) Show that for all pairs of nonzero rational numbers a, b we have $\mathbb{Q}(a\sqrt[3]{3} + b\sqrt[5]{5}) = \mathbb{Q}(\sqrt[3]{3}, \sqrt[5]{5})$.

236 Let ζ be a primitive seventh root of unity in $\mathbb{C}$.

(i) Show that $\mathbb{Q}(\zeta + \zeta^2 + \zeta^4) = \mathbb{Q}(\sqrt{-7})$.
(ii) Show that $\mathbb{Q}(\zeta + \zeta^{-1}, \sqrt{-7}) = \mathbb{Q}(\zeta)$.

237 Let ζ_3 and ζ_5 in $\mathbb{C}$ be a primitive third root of unity and a primitive fifth root of unity, respectively. Given a prime number p, let $\mathbb{K}(p) = \mathbb{Q}(\zeta_3, \zeta_5, \sqrt{p})$. Determine $\mathrm{Gal}(\mathbb{K}(p)/\mathbb{Q})$ and find the number of subfields $\mathbb{E} \subseteq \mathbb{K}(p)$ such that $[\mathbb{E} : \mathbb{Q}] = 2$ as a function of p.

238 Let $\mathbb{K}$ be the splitting field of the polynomial $x^7 - 2$ over $\mathbb{Q}$.

(i) Determine the Galois group of $\mathbb{K}$ over $\mathbb{Q}$.
(ii) Show that there exists a unique subextension $\mathbb{F}$ of $\mathbb{K}$ such that $[\mathbb{F} : \mathbb{Q}] = 3$ and describe it.

239 Let a be an integer and let $\mathbb{K}$ be the field $\mathbb{Q}(\sqrt{6}, \sqrt{10}, \sqrt{a})$.

(i) Compute the Galois group of $\mathbb{K}/\mathbb{Q}$ as a function of a.
(ii) Find all integers d such that there exists an irreducible polynomial with rational coefficients having degree d and a root in $\mathbb{K}$.
(iii) Find all integers d such that there exists an irreducible polynomial with rational coefficients that has degree d and splits as a product of linear factors in $\mathbb{K}$.

240 Let a, b be nonzero integers and let $\mathbb{K} = \mathbb{Q}(\sqrt{a}, \sqrt[3]{b})$.

(i) Determine the degree of $\mathbb{K}$ over $\mathbb{Q}$ as a function of a and b.
(ii) Determine all subextensions of $\mathbb{K}$ over $\mathbb{Q}$.

241 Let $\mathbb{F}$ be the splitting field of $(x^3 - 7)(x^2 - 3)$ over $\mathbb{Q}$.

(i) Determine the Galois group of $\mathbb{F}/\mathbb{Q}$.
(ii) Determine all subextensions of $\mathbb{F}$ that are not normal over $\mathbb{Q}$.

242 Let $\mathbb{K}$ be a finite field and let α, β be algebraic numbers over $\mathbb{K}$ with $[\mathbb{K}(\alpha) : \mathbb{K}] = 5$ and $[\mathbb{K}(\beta) : \mathbb{K}] = 4$. Show that $[\mathbb{K}(\alpha\beta) : \mathbb{K}] = 20$.

243 Let ζ be a primitive 25th root of unity in $\mathbb{C}$.

(i) Letting $\alpha = \zeta^7 + \zeta + \zeta^{-1} + \zeta^{-7}$, show that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 5$.
(ii) Determine all subfields of $\mathbb{Q}(\zeta)$.

244 Let $f(x)$ be an irreducible polynomial of degree 6 with rational coefficients. Determine all the ways that $f(x)$ may factor over $\mathbb{Q}(\sqrt{2})$ and provide an example for each.

245 Let ζ_7 be a primitive seventh root of unity in $\mathbb{C}$ and let $\alpha = \zeta_7 + \zeta_7^2 + \zeta_7^4$.

(i) Show that $\mathbb{Q}(\alpha)$ has degree 2 over $\mathbb{Q}$.
(ii) Letting ζ_5 denote a primitive fifth root of unity in $\mathbb{C}$, compute the Galois group of $\mathbb{Q}(\alpha, \zeta_5)/\mathbb{Q}$.

246 Let n be a positive integer and let ζ be a primitive nth root of unity in $\mathbb{C}$. Let $\mathbb{E}$ be a normal extension of $\mathbb{Q}(\zeta)$, let $\sigma \in \mathrm{Gal}(\mathbb{E}/\mathbb{Q}(\zeta))$ and $\alpha \in \mathbb{E}$ be such that $\sigma(\alpha) = \zeta\alpha$.

(i) Show that n divides $[\mathbb{E} : \mathbb{Q}(\zeta_n)]$.
(ii) Show that there exists a field $\mathbb{F}$ strictly contained in $\mathbb{E}$ such that $\alpha^n \in \mathbb{F}$.

247 Given positive integers n, m, set $\alpha_n = \sqrt[n]{5}$ and let ζ_m be a primitive mth root of unity in $\mathbb{C}$.

(i) For each n, compute the degree of $\mathbb{Q}(\alpha_n)$ over $\mathbb{Q}(\zeta_5)$.
(ii) Determine all pairs (n, m) such that α_n is an element of $\mathbb{Q}(\zeta_m)$.

248 Are there a field $\mathbb{K}$ of characteristic zero and an irreducible polynomial $f(x) \in \mathbb{K}[x]$ of degree 5 such that, letting $\mathbb{L}$ be the splitting field of $f(x)$ over $\mathbb{K}$, we have $[\mathbb{L} : \mathbb{K}] = 5$? What about $[\mathbb{L} : \mathbb{K}] = 10$? What about $[\mathbb{L} : \mathbb{K}] = 15$?

249 Let $\mathbb{K}$ be a finite field or a field of characteristic zero, let $f(x) \in \mathbb{K}[x]$ be an irreducible polynomial of degree 4, and let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be its roots in an algebraic closure of $\mathbb{K}$. Set $\beta = \alpha_1\alpha_2 + \alpha_3\alpha_4$, $\gamma = \alpha_1\alpha_3 + \alpha_2\alpha_4$, $\delta = \alpha_1\alpha_4 + \alpha_2\alpha_3$ and $\mathbb{L} = \mathbb{K}(\beta, \gamma, \delta)$. Show that

(i) $\mathbb{L}$ is a normal extension of $\mathbb{K}$;
(ii) the minimal polynomial of β over $\mathbb{K}$ has degree at most 3;
(iii) if $\mathbb{K}$ is a finite field then $[\mathbb{L} : \mathbb{K}] = 2$.

250 Let ζ be a primitive 24th root of unity in $\mathbb{C}$ and let $\mathbb{L} = \mathbb{Q}(\zeta)$.

(i) Find the number of quadratic subextensions of $\mathbb{L}$ and the number of subextensions of $\mathbb{L}$ whose degree is 4 over $\mathbb{Q}$.
(ii) Show that the subextensions of $\mathbb{L}$ whose degree is 4 are normal over $\mathbb{Q}$ and that their Galois groups are pairwise isomorphic.
(iii) Find a primitive element for each subextension of degree 4.

251 Determine the Galois group of the polynomial $(x^4 + 5x^2 + 5)(x^2 - a)$ over $\mathbb{Q}$ as a function of the integer a.

252 Determine all quadratic subextensions of the splitting field of $x^6 - 2$ over $\mathbb{Q}$.

253 For each integer a, determine the Galois group of the polynomial $x^4 - 2$ over $\mathbb{Q}(\sqrt{a})$.

254 Given a prime number p, let $\mathbb{K}_5$, $\mathbb{K}_7$ and $\mathbb{K}_{35}$ be the splitting fields over $\mathbb{Q}$ of the polynomials $x^5 - p$, $x^7 - p$ and $x^{35} - p$, respectively. Show that

(i) $\mathbb{K}_5 \cdot \mathbb{K}_7 = \mathbb{K}_{35}$;
(ii) $\mathbb{K}_5 \cap \mathbb{K}_7 = \mathbb{Q}$.

255 Determine the Galois groups of $x^4 - 49$ and $(x^7 - 1)(x^4 - 49)$ over $\mathbb{Q}$.

256 For each positive integer m, determine the degree of the splitting field of $x^5 - m$ over $\mathbb{Q}$ and over $\mathbb{F}_{19}$.

257 Let $\mathbb{K} = \mathbb{Q}(\sqrt[4]{2}, \sqrt[3]{2})$.

(i) Determine the degree of $\mathbb{K}$ over $\mathbb{Q}$.
(ii) Let $\mathbb{L}$ be the smallest extension of $\mathbb{K}$ that is normal over $\mathbb{Q}$; determine $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$.
(iii) Determine the subgroup of $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ that fixes the field $\mathbb{K}$.
(iv) Find the number of Galois subextensions of $\mathbb{L}$ that have degree 4 over $\mathbb{Q}$.

258 Let ζ be a primitive 15th root of unity in $\mathbb{C}$. Find the number of subextensions of $\mathbb{Q}(\zeta)$ that have degree 2 over $\mathbb{Q}$, and express each of them as $\mathbb{Q}(\sqrt{m})$ for some squarefree integer m.

259

(i) Determine the Galois group over $\mathbb{Q}$ of the polynomial $f(x) = x^4 - 12$.
(ii) Determine all possible degrees of the splitting field of $f(x)$ over a finite extension $\mathbb{K}$ of $\mathbb{Q}$ and provide an example for each possible degree.

260 For each integer a, determine the Galois group of the polynomial $(x^4 - 3)(x^2 - a)$ over $\mathbb{Q}$ and over $\mathbb{Q}(\sqrt{2})$.

261 Letting ζ be a primitive 36th root of unity in $\mathbb{C}$, determine all subextensions of $\mathbb{Q}(\zeta)$, and for each give a set of generators over $\mathbb{Q}$.

262 Letting ζ be a primitive 11th root of unity, determine the Galois group of $\mathbb{Q}(\zeta, \sqrt{11})/\mathbb{Q}$ and all its subextensions.

263 Let $\mathbb{E}$ be the splitting field of $x^6 - 2$ over $\mathbb{Q}$.

(i) Determine $[\mathbb{E} : \mathbb{Q}]$.
(ii) Show that the Galois group of $\mathbb{E}/\mathbb{Q}$ is not Abelian and has a cyclic subgroup of order 6.
(iii) Show that $\mathbb{E}$ contains a subextension that is normal over $\mathbb{Q}$ and whose Galois group is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

264 Show that $\mathbb{Q}(\sqrt{2}, \sqrt{-3}, \sqrt[3]{5})$ is a Galois extension of $\mathbb{Q}$, compute its Galois group and find all of its subextensions that are normal over $\mathbb{Q}$.

265 Let $\mathbb{E}$ be the splitting field and G the Galois group of the polynomial $x^8 - 2$ over $\mathbb{Q}$.

(i) Describe a set of generators for $\mathbb{E}$ over $\mathbb{Q}$ and compute the degree of the extension.
(ii) Set $\alpha = \sqrt[8]{2}$ and $\zeta = \sqrt{2}(1+i)/2$. Show that G contains an automorphism θ such that

$$\theta(\alpha) = \zeta\alpha,\ \theta(i) = i$$

and an automorphism σ such that

$$\sigma(\alpha) = \alpha,\ \sigma(i) = -i.$$

(iii) Show that G is not isomorphic to a dihedral group.
(iv) Determine the subfields that are fixed by the following subgroups of G: $\langle\theta\rangle$, $\langle\theta^2\rangle$, $\langle\theta^4\rangle$, $\langle\sigma\rangle$, $\langle\theta^4, \sigma\rangle$.

266

(i) Determine the degree of the splitting field $\mathbb{F}$ of $(x^4 - 3)(x^4 - 12)$ over $\mathbb{Q}$.
(ii) Determine all subfields of $\mathbb{F}$ that are quadratic extensions of $\mathbb{Q}$.

267 Describe the Galois groups of the polynomials $(x^2 + 3)(x^3 - 3x + 1)$ and $(x^2 + 3)(x^3 - 5)$ over $\mathbb{Q}$.

268

(i) For each integer a, describe the Galois group and the splitting field $\mathbb{K}$ of $(x^5 + 1)(x^2 - a)$ over $\mathbb{Q}$.
(ii) For $a = 7$, list all subfields of $\mathbb{K}$ of degree 2 over $\mathbb{Q}$.

269 Let ζ be a primitive twelfth root of unity in $\mathbb{C}$ and let $\mathbb{L} = \mathbb{Q}(\zeta, \sqrt[3]{2})$.

(i) Show that $\mathbb{L}$ is a Galois extension of $\mathbb{Q}$ and determine its degree.
(ii) Letting ω denote a primitive third root of unity in $\mathbb{C}$, compute the Galois group of $\mathbb{L}$ over $\mathbb{Q}(\omega)$.
(iii) Compute the Galois group of $\mathbb{L}$ over $\mathbb{Q}$ and provide a list of generators for it.

270 Let $\mathbb{L}$ be the splitting field of the polynomial $(x^3 - 3)(x^4 - 3)$.

(i) Determine the degree of $\mathbb{L}$ over $\mathbb{Q}$.
(ii) Show that there exists an element of $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ that fixes $\mathbb{Q}(i, \sqrt[3]{3})$ and sends $\sqrt[4]{3}$ to $i\sqrt[4]{3}$.
(iii) Describe the subfields of $\mathbb{L}$ that have degree 4 over $\mathbb{Q}$.

271

(i) Compute the Galois group of $x^5 - 2$ over $\mathbb{Q}$.
(ii) Compute the degree of the splitting field of $(x^2 - p)(x^5 - 2)$ over $\mathbb{Q}$ as a function of the prime number p.

272 Let $\zeta \in \mathbb{C}$ be a primitive ninth root of unity and set $\alpha = \zeta^3$, $\beta = \zeta + \zeta^{-1}$.

(i) Show that $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta)$ are the only extensions $\mathbb{K}$ of $\mathbb{Q}$ such that $\mathbb{Q} \subsetneq \mathbb{K} \subsetneq \mathbb{Q}(\zeta)$.
(ii) Determine the Galois group of $\mathbb{Q}(\zeta, i)$ over $\mathbb{Q}(\alpha)$ and over $\mathbb{Q}(\beta)$.
(iii) Find all subextensions of $\mathbb{Q}(\zeta, i)$ whose degree over $\mathbb{Q}$ is prime.

273

(i) Compute the degree of the splitting field $\mathbb{E}$ of $x^4 - 2x^2 - 10$ over $\mathbb{Q}$.
(ii) Determine the Galois group of $\mathbb{E}/\mathbb{Q}$.
(iii) Find the number of subextensions of $\mathbb{E}/\mathbb{Q}$ and describe explicitly those that are Galois over $\mathbb{Q}$.

274 Let a be a positive integer and let $\mathbb{K}$ be the splitting field over $\mathbb{Q}$ of the polynomial $(x^{11} - 1)(x^3 - a)$.

(i) Determine the Galois group of $\mathbb{K}$ over $\mathbb{Q}$.
(ii) Find the number of subextensions $\mathbb{E}$ of $\mathbb{K}$ that are normal over $\mathbb{Q}$ and such that $\mathrm{Gal}(\mathbb{E}/\mathbb{Q})$ is Abelian.

275

(i) Compute the degree of the splitting field $\mathbb{K}$ of the polynomial $x^4 - 2x^2 - 2$ over $\mathbb{Q}$.
(ii) Determine the group $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ up to isomorphism.
(iii) Find generators for the each subextension of $\mathbb{K}/\mathbb{Q}$ and determine whether or not it is Galois over $\mathbb{Q}$.

276 Consider the field extension $\mathbb{C}(t) \subseteq \mathbb{C}(x)$ with $t = x^3 + x^{-3}$.

(i) Determine the degree of the extension.
(ii) Show that the extension is Galois and determine its Galois group.
(iii) Find all proper subextensions and compute a primitive element for each.

277 Let $\alpha = \sqrt[3]{3} + \sqrt{5}$.

(i) Show that $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt[3]{3}, \sqrt{5})$.
(ii) Determine the splitting field $\mathbb{K}$ and the Galois group of the minimal polynomial of α over $\mathbb{Q}$.
(iii) Describe the subextensions of $\mathbb{K}$ that are Galois over $\mathbb{Q}$ and have an Abelian Galois group.

278 Determine the degree of the splitting field $\mathbb{K}$ of $x^5 - 5$ over $\mathbb{Q}$. Additionally, determine all subextensions of $\mathbb{K}/\mathbb{Q}$ and specify which ones are normal over $\mathbb{Q}$.

279 Determine the Galois group of the smallest extension of $\mathbb{Q}(\sqrt[3]{2}, \sqrt{2})$ that is normal over $\mathbb{Q}$.

280 Determine all subextensions $\mathbb{K}$ of the 35th cyclotomic field $\mathbb{L}$ over $\mathbb{Q}$ such that $[\mathbb{L} : \mathbb{K}] = 2$.

281 Compute the Galois group of the polynomial $x^4 + 9$ over $\mathbb{Q}$.

Chapter 3
Solutions



3.1 Groups

1

(i) The group G is isomorphic to $G_2 \times G_3$, where $G_2 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $G_3 = \mathbb{Z}/3\mathbb{Z}$. An element of G has order 6 if and only if its projection onto G_2 has order 2 and its projection onto G_3 has order 3.
The number of elements in G_2 that yield the neutral element when multiplied by 2 is $2 \cdot 2 \cdot 2 = 8$; among these, all have order 2 except for the neutral element, so there are 7 elements of order 2. Similarly, there are $3 - 1 = 2$ elements of order 3 in G_3. It follows that there are $7 \cdot 2 = 14$ elements of order 6 in G, and therefore $14/\phi(6) = 14/2 = 7$ cyclic subgroups of order 6.

(ii) The set of possible orders of cyclic subgroups of G is also the set of possible orders of elements of G. We use again the isomorphism between G and $G_2 \times G_3$; this time, we need to determine the least common multiples of all possible orders of elements in G_2 and G_3. An element in G_2 can have order 1, 2 or 4, and an element in G_3 can have order 1 or 3, and there is an element for each of these possible orders. The least common multiple can therefore take the values 1, 2, 4, 3, 6, and 12.

(iii) The positive divisors of $48 = |G|$ are 1, 2, 3, 4, 6, 8, 12, 16, 24, and 48. We already constructed in (ii) subgroups of order 1, 2, 3, 4, 6, and 12. As a subgroup of order 8, consider $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \{0\}$; the subgroup $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \{0, 3\}$ has order 16, the subgroup $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \{0, 2, 4\}$ has order 24, and G itself is a subgroup of order 48.

2

(i) We show that $\psi_{g,h}$ is a bijection. Indeed, assuming $\psi_{g,h}(x) = \psi_{g,h}(y)$, we have $gxh^{-1} = gyh^{-1}$, which, by the cancellation laws, implies $x = y$. This proves that the map $\psi_{g,h}$ is injective. Moreover, given $y \in G$, setting $x =$

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$g^{-1}yh$ gives $\psi_{g,h}(x) = gxh^{-1} = gg^{-1}yhh^{-1} = y$; the map is thus also surjective.

(ii) Let g_1, h_1, g_2, h_2 be elements of G. for all $x \in G$ we have

$$\begin{aligned}\big(\psi(g_1,h_1)\circ\psi(g_2,h_2)\big)(x) &= \psi_{g_1,h_1}\big(\psi_{g_2,h_2}(x)\big)\\ &= \psi_{g_1,h_1}(g_2xh_2^{-1})\\ &= g_1g_2xh_2^{-1}h_1^{-1}\\ &= \psi(g_1g_2,h_1h_2)(x)\\ &= \psi\big((g_1,h_1)(g_2,h_2)\big)(x),\end{aligned}$$

and thus $\psi(g_1,h_1)\circ\psi(g_2,h_2) = \psi\big((g_1,h_1)(g_2,h_2)\big)$.

(iii) A necessary condition is that $\psi_{g,h}(e) = e$, that is, that $gh^{-1} = e$, which is equivalent to $g = h$. But $\psi_{g,g}$ represents conjugation by g, which is an automorphism of G, so the condition that $g = h$ is also sufficient.

(iv) An element (g,h) belongs to the kernel of ψ if and only if $\psi_{g,h}(x) = x$ for all $x \in G$. Setting $x = e$, that gives $h = g$, and the condition that $\psi_{g,h}(x) = x$ becomes $gx = xg$ for all $x \in G$. The kernel therefore consists of the pairs (g,g) such that $g \in Z(G)$, and the map $Z(G) \ni g \longmapsto (g,g) \in \mathrm{Ker}(\psi)$ is an isomorphism.

3

(i) By the fact that G is Abelian, for all $g, h \in G$ we have

$$\begin{aligned}\psi(gh) &= \big((gh)^2,(gh)^{-1}\big)\\ &= (g^2h^2,g^{-1}h^{-1})\\ &= (g^2,g^{-1})(h^2,h^{-1})\\ &= \psi(g)\psi(h),\end{aligned}$$

so ψ is a group homomorphism.
Moreover, for all $(g_1,h_1),(g_2,h_2) \in G\times G$ we have

$$\begin{aligned}\pi\big((g_1,h_1)(g_2,h_2)\big) &= \pi(g_1g_2,h_1h_2)\\ &= (g_1g_2)(h_1h_2)^2\\ &= g_1g_2h_1^2h_2^2\\ &= (g_1h_1^2)(g_2h_2^2)\\ &= \pi(g_1,h_1)\pi(g_2,h_2),\end{aligned}$$

so π is also a group homomorphism.

(ii) Suppose $g \in G$ is such that $\psi(g) = e_{G\times G} = (e,e)$. This implies $g^2 = e$ and $g^{-1} = e$, which yields $g = e$. We have thus shown that ψ is injective.

(iii) Given $g \in G$, we have $\pi(g,e) = ge^2 = g$, which proves that π is surjective.

(iv) First of all, $(\pi\circ\psi)(g) = \pi(g^2,g^{-1}) = g^2(g^{-1})^2 = e$, so we have $\mathrm{Im}(\psi) \subseteq \mathrm{Ker}(\pi)$. Moreover, if $\pi(g,h) = e$ then $gh^2 = e$, hence $g = h^{-2}$. But then,

$(g,h) = (h^{-2},h) = ((h^{-1})^2,(h^{-1})^{-1}) = \psi(h^{-1})$, and therefore we also have $\mathrm{Ker}(\pi) \subseteq \mathrm{Im}(\psi)$.

What we have just shown can also be expressed by saying that the sequence of groups and homomorphisms

$$0 \longrightarrow G \xrightarrow{\psi} G \times G \xrightarrow{\pi} G \longrightarrow 0$$

is *exact*.

4

(i) The homomorphism σ is clearly an automorphism, as $\sigma^{-1} = \sigma$.

(ii) First, remark that H is nonempty, as $e \in H$. Now consider $h_1, h_2 \in H$; we have $\sigma(h_1h_2) = \sigma(h_1)\sigma(h_2) = h_1h_2$, so H is closed by taking products. For $h \in H$, we have $\sigma(h^{-1}) = (\sigma(h))^{-1} = h^{-1}$, that is, $h^{-1} \in H$. This shows that H is a subgroup of G.

(iii) Since G is Abelian, H is a normal subgroup of G. Consider the quotient homomorphism $\pi : G \longrightarrow G/H$. The map $\tilde{\sigma} = \pi \circ \sigma : G \longrightarrow G/H$ is a surjective homomorphism, as it is the composition of two surjective homomorphisms. The kernel of $\tilde{\sigma}$ is equal to H: indeed, we have $\sigma(h) \in H$ if and only if $h \in H$.
By the First Homomorphism Theorem, $\tilde{\sigma}$ induces an isomorphism $\tau : G/H \longrightarrow G/H$ defined by $\tau(gH) = \pi \circ \sigma(g) = \sigma(g)H$.

(iv) The fact that H belongs to K is clear; indeed, $\tau(H) = H$ as τ is a homomorphism and $H = e_{G/H}$.
On the other hand, if $gH \in K$ then the definition of K implies that $\tau(gH) = gH$, that is, $\sigma(g)H = gH$. In order for this to hold we must have $h \in H$ such that $\sigma(g) = gh$. Applying σ to this equality yields $g = \sigma(g)h$, where we have used that $h \in H$. But then, $h = g^{-1}\sigma(g)$ and $h = \sigma(g)^{-1}g$, from which $h = h^{-1}$, that is, $h^2 = e$. We have found that the order of h in G must be either 1 or 2; however, this order cannot be 2, because the order of G is odd. The only remaining possibility is $h = e$, that is, $\sigma(g) = g$, so $g \in H$ and $gH = H$.

5

(i) We wish to show that G is Abelian, that is, that no matter how we choose g, h in G, they commute. The subset $S = \{g,h\}$ is finite, and so the subgroup $\langle S \rangle$ it generates is cyclic, and in particular it is Abelian. But, since g and h belong to the group $\langle S \rangle$, this means that $gh = hg$ in $\langle S \rangle$, so the same holds in G.

(ii) We show that $\mathbb{Q}$ does have the desired property, but that it is not cyclic.
Let $S = \{n_1/d_1, n_2/d_2, \ldots, n_r/d_r\}$ be any subset of $\mathbb{Q}$. Let d be the least common multiple of $d_1, d_2, \ldots, d_r$, and suppose that $d = c_1d_1 = c_2d_2 = \cdots = c_rd_r$.
Consider the subgroup $H = \langle 1/d \rangle$ in $\mathbb{Q}$, which is obviously cyclic. Now, since $n_1/d_1 = c_1n_1 \cdot 1/d$, $n_2/d_2 = c_2n_2 \cdot 1/d, \ldots, n_r/d_r = c_rn_r \cdot 1/d$, the subgroup

$\langle S\rangle$ is contained in H, so $\langle S\rangle$ is also cyclic, as it is a subgroup of a cyclic group. This proves that $\mathbb{Q}$ has the required property.
We now show that $\mathbb{Q}$ is not cyclic. Assume by contradiction that $\mathbb{Q} = \langle n/d\rangle$ and let p be a prime that does not divide d. Since $1/p$ is an element of $\mathbb{Q}$, we must have $1/p = c \cdot n/d$ for some $c \in \mathbb{Z}$. However, this yields $d = cnp$, which, since p does not divide d, is not satisfied for any integer c.

6 Let G be a non-Abelian group of order 28; by Cauchy's theorem, it has an element h of order 7, whose generated subgroup we shall denote by H.

Now let K be another subgroup of order 7 in G. Either H and K have trivial intersection, or they coincide, as each of their elements other than the identity is itself a generator (because the order of H and K is prime). If $K \neq H$ then the set HK contains 49 elements, which is not possible because G has only 28. We have thus shown that H is the only subgroup of G with 7 elements, and in particular H is normal in G (any conjugate of H is a subgroup of order 7, so coincides with H).

By assumption, G contains an element g of order 4, which generates a subgroup we hereby denote by L. We must have $H \cap L = \{e\}$, as the order of any element in the intersection must divide both 4 and 7, therefore also $1 = (4, 7)$. It follows that $HL = G$. We thus obtain that G is isomorphic to a semidirect product $H \rtimes L$. More precisely, since $H \simeq \mathbb{Z}/7\mathbb{Z}$ and $L \simeq \mathbb{Z}/4\mathbb{Z}$, G must be isomorphic to $\mathbb{Z}/7\mathbb{Z} \rtimes_\varphi \mathbb{Z}/4\mathbb{Z}$ with respect to some homomorphism $\varphi : \mathbb{Z}/4\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/7\mathbb{Z}) \simeq (\mathbb{Z}/7\mathbb{Z})^* \simeq \mathbb{Z}/6\mathbb{Z}$.

Note that the order of $\varphi(1)$ must divide 4, which is the order of 1 in $\mathbb{Z}/4\mathbb{Z}$. So the order of $\varphi(1)$ can only be 1 or 2, as $\mathbb{Z}/6\mathbb{Z}$ does not contain any elements of order 4. Moreover, if this order were 1 then every element of $\mathbb{Z}/4\mathbb{Z}$ would be sent to 1, resulting in an Abelian group G. The only remaining possibility is for $\varphi(1)$ to be an element of order 2 in $\mathbb{Z}/6\mathbb{Z}$. But the only such element is 3, so we necessarily have $\varphi(1) = 3$, and the homomorphism φ is uniquely determined, which in turn completely determines the structure of G, as it is isomorphic to $\mathbb{Z}/7\mathbb{Z} \rtimes_\varphi \mathbb{Z}/4\mathbb{Z}$.

7

(i) The subgroup $[\mathsf{S}_5, \mathsf{S}_5]$ is generated by the elements $[\sigma, \tau] = \sigma\tau\sigma^{-1}\tau^{-1}$, for σ and τ in S_5. Now, since each permutation has the same sign as its inverse, $[\sigma, \tau] \in \mathsf{A}_5$; consequently, $[\mathsf{S}_5, \mathsf{S}_5] \subseteq A_5$. Moreover, the derived subgroup is normal, and $[\mathsf{S}_5, \mathsf{S}_5]$ cannot be the trivial group as it contains, for example, the element $\big[(1, 2), (2, 3)\big] = (1, 3, 2)$. Since A_5 is simple, the only possibility is $[\mathsf{S}_5, \mathsf{S}_5] = \mathsf{A}_5$.

(ii) If H is a homomorphic image of S_5 and an Abelian group, then $[\mathsf{S}_5, \mathsf{S}_5] = \mathsf{A}_5$ is contained in the kernel of the homomorphism. But then H is a homomorphic image of $\mathsf{S}_5/\mathsf{A}_5 \simeq \mathbb{Z}/2\mathbb{Z}$, which implies the desired result.

⟦ The same conclusion holds in general if we consider S_n rather than S_5. The case $n = 2$ is straightforward; for $n \geq 3$, one can use the same arguments as above, with some small provisions for the case $3 \leq n \leq 4$, in which A_n is not simple, but the normal subgroup $[\mathsf{S}_n, \mathsf{S}_n]$ contains the 3–cycle $(1, 3, 2)$ and therefore all of A_n. ⟧

8 Let H_2 be the subgroup of order 77 of G that we are given by assumption. Let us prove that H_2 is characteristic in G. Given any subgroup K of G with 77 elements, consider $K \cap H_2$: this is a subgroup of H_2 whose order d must be a divisor of 77; the set KH_2 has $77^2/d$ elements. Now, if K did not coincide with H_2, then we would have $d \leq 11$, and therefore $|KH_2| \geq 7^2 \cdot 11 > |G|$, contradiction. This shows that H_2 is the only subgroup of G having exactly 77 elements, so H_2 is characteristic.

By Cauchy's theorem, H_2 has an element h of order 11. In a similar way as before, one can show that the subgroup H_1 generated by h is characteristic in H_2. But then H_1 is also characteristic in G, thanks to the fact that H_2 is characteristic in G, as we've just shown.

9 The group $\mathbb{Z}/100\mathbb{Z} \times \mathbb{Z}/20\mathbb{Z}$ is isomorphic to $G = \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/25\mathbb{Z}$. The number of elements of order $50 = 2 \cdot 5^2$ that G contains is therefore $(2^2 - 1)(5^3 - 5^2) = 300$.

Now, let H be a subgroup of G of order 50. We must have either $H \simeq \mathbb{Z}/50\mathbb{Z}$ or $H \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$: the two are mutually exclusive, as the first group is cyclic and the second is not. The number of subgroups of the first type (that is, cyclic subgroups of order 50) is $300/\phi(50) = 300/20 = 15$.

A subgroup H of the second type contains 24 elements of order 5, which is exactly the number of elements of order 5 contained in G. Consequently, $H = H_1 \cdot H_2 \simeq H_1 \times H_2$, where H_1 is a subgroup of order 2 in G and $H_2 = \{x \in G \mid 5x = 0\}$. There are 3 subgroup of this type, as many as the elements of order 2 in G.

In conclusion, we have 18 subgroups of order 50.

10

(i) If G is Abelian then the result holds trivially. We can thus assume that the centre Z is a proper subgroup of G.
If $x \in Z$, then $Z(x) = G$ and therefore p^2 divides $|Z(x)| = p^4$. If $x \notin Z$, then the fact that both Z and $\langle x \rangle$ are contained in $Z(x)$ implies that $Z(x)$ has more elements than the centre, which in a p-group is nontrivial; therefore, $|Z(x)| \geq p^2$.

(ii) Remark that Z cannot have order p^3, or G/Z would be cyclic and G would be Abelian; it follows that Z has either order p^2 or p.
In the first case, we can take $x \notin Z$ and obtain that $Z, \langle x \rangle \subseteq Z(x)$ so that $|Z(x)| > p^2$; since $|Z(x)| < p^4$ because $x \notin Z$, we have $|Z(x)| = p^3$.
In the second case, we write the class equation

$$p^4 = p + \sum_{x \in \mathcal{R}} p^4/|Z(x)|,$$

where $\mathcal{R}$ is a set of representatives for the conjugacy classes of elements outside Z. If $|Z(x)| \leq p^2$ then p^2 divides $p^4/|Z(x)|$; this cannot be the case for all $x \in \mathcal{R}$, or p^2 would divide p.
We therefore have some $x \in \mathcal{R}$ such that $|Z(x)| \geq p^3$; thus, because $x \notin Z$, we have $|Z(x)| = p^3$.

11 Set

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

and let V_1, V_2, V_3 be the vector subspaces of $\mathbb{R}^3$ generated by e_1, e_2, e_3, respectively. The group G acts on the set $X = \{V_1, V_2, V_3\}$ via the homomorphism $f : G \longrightarrow \mathsf{S}(X) \simeq \mathsf{S}_3$ defined by $f(A)(V_i) = A \cdot V_i$, for $i = 1, 2, 3$. The kernel of the homomorphism f is clearly the set D of diagonal matrices of G, which is therefore a normal subgroup of G.

Moreover, elements of D are of the form

$$\begin{pmatrix} \pm 1 & & \\ & \pm 1 & \\ & & \pm 1 \end{pmatrix}$$

and thus, by the definition of matrix multiplication, D is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$.

Now, let K be the subgroup of matrices of G all of whose nonzero coefficients are equal to 1. We obviously have $D \cap K = \{\mathrm{Id}\}$ and so $f_{|K}$ is injective. Since K has 3! elements, $f_{|K}$ is also surjective, and therefore K is isomorphic to S_3.

It is easy to check that $|G| = 48$: there are 6 possibilities for the first column of a matrix $A \in G$; given the first column, there are 4 possibilities for the second and, given the first two, 2 possibilities for the third. It follows that $G = DK$, from which we conclude that G is isomorphic to a semidirect product $D \rtimes K$ and thus also to $(\mathbb{Z}/2\mathbb{Z})^3 \rtimes \mathsf{S}_3$.

12

(i) Given a subset S of G which is stable under conjugation and an element $s \in S$, we have $xsx^{-1} \in S$ for all $x \in G$ and therefore $\mathcal{C}\ell(s) \subseteq S$. Let now $X \subseteq G$ and set $\overline{X} = \cup_{g \in X} \mathcal{C}\ell(g)$.
If S is stable under conjugation and contains X, then for all $g \in X$ we have $\mathcal{C}\ell(g) \subseteq S$. This implies $\overline{X} \subseteq S$, so $\overline{X}$ is minimal (for inclusion) among sets containing X that are stable under conjugation.

(ii) Let S be a set that is stable under conjugation and let $\langle S \rangle$ be the subgroup generated by S. In order to show that $\langle S \rangle$ is a normal subgroup, that is, that $x\langle S \rangle x^{-1} \subseteq \langle S \rangle$ for all $x \in G$, it is enough to check for each $x \in G$ that $xSx^{-1} \subseteq \langle S \rangle$, as $x\langle S \rangle x^{-1}$ is generated by xSx^{-1}. And indeed, the fact that $xSx^{-1} \subseteq S$ follows from S being stable by conjugation: as a consequence, we also have $xSx^{-1} \subseteq \langle S \rangle$.

13

(i) We begin by showing that for all groups H and K, setting $G = H \times K$, we have $\mathcal{C}\ell_G(h,k) = \mathcal{C}\ell_H(h) \times \mathcal{C}\ell_K(k)$. Indeed,

$$\begin{aligned}\mathcal{C}\ell_G(h,k) &= \left\{(h_1,k_1)(h,k)(h_1,k_1)^{-1} \mid h_1 \in H,\ k_1 \in K\right\}\\ &= \left\{(h_1hh_1^{-1}, k_1kk_1^{-1}) \mid h_1 \in H,\ k_1 \in K\right\}\\ &= \mathcal{C}\ell_H(h) \times \mathcal{C}\ell_K(k).\end{aligned}$$

In our case, the conjugacy classes in S_4 are parametrised by the possible cycle structures, that is, by the partitions of 4, and the conjugacy classes in $\mathbb{Z}/3\mathbb{Z}$ are parametrised by the elements of the group itself. It follows that conjugacy classes in $\mathsf{S}_4 \times \mathbb{Z}/3\mathbb{Z}$ are Cartesian products $\mathcal{C} \times \{a\}$, where $\mathcal{C}$ is the set of permutations in S_4 having some fixed cycle structure, and a is an element of $\mathbb{Z}/3\mathbb{Z}$.

(ii) It is easy to show that homomorphisms from the direct product $\mathsf{S}_4 \times \mathbb{Z}/3\mathbb{Z}$ to the group $\mathbb{Z}/6\mathbb{Z}$ are of the form $\alpha \times \beta$ for some homomorphisms $\alpha : \mathsf{S}_4 \longrightarrow \mathbb{Z}/6\mathbb{Z}$ and $\beta : \mathbb{Z}/3\mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z}$, where $(\alpha \times \beta)(h,k) = \alpha(h) + \beta(k)$.
What we need to describe is every possible homomorphism $\alpha : \mathsf{S}_4 \longrightarrow \mathbb{Z}/6\mathbb{Z}$ and $\beta : \mathbb{Z}/3\mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z}$.
The kernel K of the homomorphism α must be a normal subgroup of S_4 such that S_4/K is Abelian. Listing all possible normal subgroups of S_4, we find that the only ones with this property are A_4 and S_4. Consequently, the kernel of α contains A_4.
It follows that α induces a homomorphism $\tilde{\alpha} : \mathbb{Z}/2\mathbb{Z} \simeq \mathsf{S}_4/\mathsf{A}_4 \longrightarrow \mathbb{Z}/6\mathbb{Z}$ by passing to the quotient. There are exactly two homomorphisms from $\mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}/6\mathbb{Z}$, which in turn correspond to the following homomorphisms from S_4 to $\mathbb{Z}/6\mathbb{Z}$: one is given by $\alpha_1(\sigma) = 0$ for all $\sigma \in \mathsf{S}_4$, and the other is given by $\alpha_2(\sigma) = 0$ for $\sigma \in \mathsf{A}_4$, $\alpha_2(\sigma) = 3$ for $\sigma \notin \mathsf{A}_4$.
In order to complete our description of the homomorphisms from $\mathsf{S}_4 \times \mathbb{Z}/3\mathbb{Z}$ to $\mathbb{Z}/6\mathbb{Z}$, all that is left to do is to remark that there are three homomorphisms from $\mathbb{Z}/3\mathbb{Z}$ to $\mathbb{Z}/6\mathbb{Z}$, each one induced by $1 \longmapsto 2a$, with $a = 0, 1, 2$.

14

(i) By Cauchy's theorem, there exists a subgroup H of order 17 in G. Were we to show that H is the only subgroup of order 17, we would have that H is characteristic in G. Now, suppose K is a subgroup of order 17 that does not coincide with H; we then have $H \cap K = \{e\}$, so $|HK| = 17^2 > |G|$, contradiction.

(ii) It is clear that $|G/H| = 15$ and, since 3 does not divide $4 = 5 - 1$, the group G/H is Abelian and therefore cyclic.

(iii) We know that there is a subgroup of G/H of order 5, and since G/H is Abelian this subgroup is normal. It follows that its pre-image in G is a normal subgroup L of G of order 85. Moreover, again by Cauchy's theorem, there exists a subgroup M of order 3 in G. Naturally, $L \cap M = \{e\}$, since the orders

of the two groups are relatively prime. We can thus express G as the semidirect product of L and M, with L being normal.
Note that a group of order $85 = 5 \cdot 17$ is necessarily cyclic, as 5 does not divide $16 = 17 - 1$, and in particular $\mathrm{Aut}(L)$ is a group of order $\phi(85) = 4 \cdot 16 = 2^6$. However, $3 = |M|$ and 2^6 are relatively prime, so the only possible homomorphism between these two groups is the trivial homomorphism $M \ni g \longmapsto \mathrm{Id}_L \in \mathrm{Aut}(L)$. This implies that the semidirect product of L and M is actually a direct product, and therefore G is cyclic.

15 Let σ be a solution of the equation $\sigma^5 = (1, 2)$ in S_{11}. Notice that we must have $\sigma^{10} = (1, 2)^2 = e$, so we only have to look for solutions among permutations whose order divides 10. Clearly, the order of a solution cannot be 1 or 5, otherwise we would have $\sigma^5 = e \neq (12)$.

If σ has order 2 then we must have $\sigma = \sigma^5 = (1, 2)$: there is exactly one such solution.

Suppose now that σ has order 10. Clearly, σ cannot have a 10-cycle in its decomposition as a product of disjoint cycles: when composed with itself 5 times, such a cycle would yield 5 disjoint cycles of length 2, which contradicts the assumption that $\sigma^5 = (1, 2)$.

The cycle structure of σ must therefore contain at least one 2-cycle and at least one 5-cycle, so in fact exactly one 5-cycle, since the sum of the lengths of nontrivial cycles in the decomposition can be at most 11. Suppose $\sigma = \tau_1 \cdots \tau_r \eta$, where $\tau_1, \ldots, \tau_r$ are 2-cycles and η is a 5-cycle.

We have $\sigma^5 = \tau_1 \cdots \tau_r$, which yields $r = 1$ and $\tau_1 = (1, 2)$. In conclusion, the solutions of this second type are all the permutations of the form $\sigma = (1, 2)\eta$, where η is a 5-cycle disjoint from $(1, 2)$. There are

$$\binom{9}{5} \cdot 4! = 3024$$

such permutations, yielding a total of 3025 solutions.

16 We define the following subsets of X^3:

$$\begin{aligned}
X_1 &= \{(i, i, i) \mid 1 \leq i \leq n\}, \\
X_2 &= \{(i, i, j) \mid 1 \leq i, i \leq n,\ i \neq j\}, \\
X_3 &= \{(i, j, i) \mid 1 \leq i, j \leq n,\ i \neq j\}, \\
X_4 &= \{(j, i, i) \mid 1 \leq i, j \leq n,\ i \neq j\}, \\
X_5 &= \{(i, j, k) \mid 1 \leq i, j, k \leq n,\ i \neq j,\ i \neq k,\ j \neq k\},
\end{aligned}$$

and show that they are the orbits for the given action of S_n on X^3.

The fact that each of these sets is invariant under the action of S_n is clear: if an element (x_1, x_2, x_3) has some equal coordinates, then for each σ in S_n the same coordinates (and only they) coincide in $(\sigma(x_1), \sigma(x_2), \sigma(x_3))$.

Vice-versa, given k distinct integers $i_1, \dots, i_k$ in the set $\{1, 2, \dots, n\}$, and another k distinct integers $j_1, \dots, j_k$ in $\{1, 2, \dots, n\}$, there is a permutation σ in S_n sending i_h to j_h for $h = 1, \dots, k$. Using the above for $k = 1, 2, 3$, we find that S_n acts transitively on each of the sets listed.

This proves that $X_1, \dots, X_5$ are the orbits of the action of S_n on X^3. It is clear that $|X_1| = n$, $|X_2| = |X_3| = |X_4| = n(n-1)$, $|X_5| = n(n-1)(n-2)$.

17 The conjugacy class $\mathcal{C}\ell_{\mathsf{S}_5}(\sigma)$, that is, the set of all 5-cycles in S_5, consists of 24 elements. Consequently, the centraliser $Z_{\mathsf{S}_5}(\sigma)$ has order $5!/24 = 5$, and therefore coincides with $\langle\sigma\rangle$. Since the latter is contained in A_5, we have $Z_{\mathsf{A}_5}(\sigma) = Z_{\mathsf{S}_5}(\sigma)$, and in particular $|\,\mathcal{C}\ell_{\mathsf{A}_5}(\sigma)| = [\mathsf{A}_5 : Z_{\mathsf{A}_5}(\sigma)] = 60/5 = 12$.

In order to compute the cardinality of the desired intersection, we first make a general remark. Given $\tau \in \mathsf{S}_5$, the conjugate $\psi_\tau(\sigma) = \tau\sigma\tau^{-1}$ of σ only depends on the class of τ modulo the centraliser $Z_{\mathsf{S}_5}(\sigma)$. However, as noted above, $Z_{\mathsf{S}_5}(\sigma)$ is contained in A_5; it follows that, if $\tau, \tau' \in \mathsf{S}_5$ are such that $\psi_\tau(\sigma) = \psi_{\tau'}(\sigma)$, then τ and τ' have the same sign.

We know that σ cannot be a conjugate of $e \in \langle\sigma\rangle$, whereas it is obviously a conjugate of σ itself. Notice that $\psi_{(2,3,5,4)}(\sigma) = \sigma^2$, which, $(2, 3, 5, 4)$ being odd, implies that σ^2 is not a conjugate of σ in A_5, because of our general remark. Similarly, $\sigma^3 \notin \mathcal{C}\ell_{\mathsf{A}_5}(\sigma)$ as $\psi_{(2,4,5,3)}(\sigma) = \sigma^3$. We instead have $\psi_{(2,5)(3,4)}(\sigma) = \sigma^{-1}$, and so $\sigma^{-1} \in \mathcal{C}\ell_{\mathsf{A}_5}(\sigma)$, because $(2, 5)(3, 4)$ is even. We can therefore conclude that $\mathcal{C}\ell_{\mathsf{A}_5}(\sigma) \cap \langle\sigma\rangle = \{\sigma, \sigma^{-1}\}$ has cardinality 2.

18 Given an integer n, let μ_n be the map $G \ni g \longmapsto ng \in G$.

(i) Let $n = |G|$ and let p^m be the largest power of p that divides n, which is the order of P; finally, set $q = n/p^m$. The map $\mu_q : G \longrightarrow G$ has image P and is an isomorphism when restricted to P, as q and the order of P are coprime. Consider $x \in G$ and $h \in P$ such that $nx = h$. Applying μ_q yields $n\mu_q(x) = \mu_q(h) \in P$, and therefore, by applying $\mu_{q|P}^{-1}$, we obtain $ny = h$, where $y = \mu_{q|P}^{-1}(\mu_q(x)) \in P$. This shows that P is pure.

(ii) It is easy to show that, since P is pure in G, a subgroup H of P is pure in G if and only if it is pure in P. We can thus suppose that $G = P$, that is, that G has order p^m.
Suppose there is an element g in G of order p^t for some $t > 1$, and consider the subgroup $H = \langle pg\rangle$ generated by pg, which has order p^{t-1}. Remark that, setting $h = pg \in H$, $n = p$ and $x = g$, we have $nx = h$; however, there can be no element y in H such that $ny = h$, or H would have order at least p^t. This proves that, if every subset of G is pure, then $px = 0$ for all $x \in G$.
On the other hand, suppose $px = 0$ for all x in G. Let n be an integer, H a subgroup of G, and h an element of H for which there exists $x \in G$ such that $nx = h$. If $h = 0$, then we have $ny = h$ for $y = 0 \in H$. If $h \neq 0$, then $nx = h$ implies that n and p must be coprime, so if $m \in \mathbb{N}$ is such that $mn \equiv 1 \pmod{p}$, then $x = mh \in H$. This proves that H is pure.

19

(i) First of all, note that we have $(e, e) = (e, f(e)) \in H$, so H is nonempty. Given elements $(x, f(x)), (y, f(y)) \in H$ (where x, y are elements of G), we have $(x, f(x))(y, f(y)) = (xy, f(x)f(y)) = (xy, f(xy)) \in H$. Moreover, $(x, f(x))(x^{-1}, f(x^{-1})) = (e, e)$, so H does contain the inverse of each of its elements.
Therefore, H is a subgroup.

(ii) The subgroup H is not necessarily normal. For example, set $G = \mathsf{S}_3$ and $f = \mathrm{Id}$, which yields $H = \{(\sigma, \sigma) \mid \sigma \in \mathsf{S}_3\}$. For $g = \big((1, 2), (1, 2)\big) \in H$ and $h = \big(e, (1, 3)\big) \in G \times G$, we have $hgh^{-1} = \big((1, 2), (2, 3)\big) \notin H$.

(iii) An element (a, b) of $G \times G$ belongs to the centraliser of H if and only if $(a, b)(x, f(x))(a^{-1}, b^{-1}) = (x, f(x))$ for all $x \in G$. Equivalently, the system

$$\begin{cases} axa^{-1} = x \\ bf(x)b^{-1} = f(x) \end{cases}$$

must be satisfied for all $x \in G$. In particular, the first equation implies that a must belong to the centre Z of G.
The condition that $bf(x)b^{-1} = f(x)$ for all $x \in G$ is equivalent to $byb^{-1} = y$ for all y in G, because f is surjective. Therefore, we also must have $b \in Z$. With this, we have established that the centraliser $Z(H)$ of H is contained in $Z \times Z$.
On the other hand, the fact that $Z \times Z \subseteq Z(H)$ is clear because $f_{|Z} : Z \longrightarrow Z$ is an automorphism; therefore, $Z(H) = Z \times Z$.

20

(i) First of all, note that if σ is a 7-cycle then σ^{-1} is also a 7-cycle. As a consequence, we can say that every element of H is a finite product of 7-cycles.
We know that the set of all 7-cycles is stable under conjugation, and is in fact a conjugacy class. Given a product $h = \sigma_1\sigma_2 \cdots \sigma_n$ of 7-cycles and an element τ in S_7 we thus have that $\tau h\tau^{-1} = (\tau\sigma_1\tau^{-1})(\tau\sigma_2\tau^{-1}) \cdots (\tau\sigma_n\tau^{-1})$ is also a product of 7-cycles, which implies $\tau h\tau^{-1} \in H$. This proves that H is a normal subgroup of S_7.

(ii) Since 7-cycles are even permutations, we have $H \subseteq \mathsf{A}_7$. Also note that $(1, 2, 3, 4, 5, 6, 7)(1, 2, 7, 6, 5, 4, 3) = (1, 3, 2) \in H$, and that H is normal in S_7 and thus a normal subgroup of A_7. But any normal subgroup of A_7 containing a 3-cycle must be all of A_7.

21 Remark that any quotient of a cyclic group is cyclic. Consequently, if G is cyclic, then there is no normal subgroup N of G such that $G/N \simeq \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, since the latter is not a cyclic group. We have thus proven one of the two implications.

Let us now show that G being non-cyclic implies the existence of a subgroup N with the required properties. We proceed by induction on the order of G; but first,

we treat the case where G is Abelian, which incidentally takes care of the induction base case $|G| = p^2$.

Since G is a p-group, its order is p^n for some n; assuming it's Abelian, by the Structure Theorem for Finite Abelian Groups there is an isomorphism f between G and $\mathbb{Z}/p^{n_1}\mathbb{Z} \times \mathbb{Z}/p^{n_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{n_r}\mathbb{Z}$, where $n_1, n_2, \ldots, n_r \geq 1$ are integers such that $n_1 + n_2 + \cdots + n_r = n$. Moreover, G being non-cyclic implies $r \geq 2$. Setting

$$N = f^{-1}(\mathbb{Z}/p^{n_1-1}\mathbb{Z} \times \mathbb{Z}/p^{n_2-1}\mathbb{Z} \times \mathbb{Z}/p^{n_3}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{n_r}\mathbb{Z}),$$

the group G/N is clearly isomorphic $Z/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, as wanted.

We can now assume that G is not Abelian.

We know that the centre of a p-group is nontrivial, and therefore that the group $G' = G/Z(G)$ is a p-group whose order is strictly less than the order of G; it is also non-cyclic, because G is not Abelian. By induction, we can assume there is a subgroup N' of G' such that G'/N' is isomorphic $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$. But then we can set $N = \pi^{-1}(N')$, where $\pi : G \longrightarrow G/Z(G) = G'$ is the quotient homomorphism, and we have

$$G/N \simeq \big(G/Z(G)\big)/\big(N/Z(G)\big) = G'/N' \simeq \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$$

as wanted.

22

(i) Remark that $\varphi^{n+1}(G) \subseteq \varphi^n(G)$ for all $n \geq 0$, so the sequence of integers $(|\varphi^n(G)|)_n$ is non-increasing, bounded above by $|G| < \infty$, and bounded below by 1, given that e belongs to $\varphi^n(G)$ for all n. The sequence is therefore eventually constant, that is, there exists $N \geq 0$ such that $|\varphi^n(G)| = |\varphi^N(G)|$ for all $n \geq N$. This implies that $\varphi^n(G) = \varphi^N(G)$ for all $n \geq N$.

(ii) Set $\psi = \varphi^N$. It is clear that $\mathrm{Ker}(\psi)$ is a normal subgroup of G and that $|\mathrm{Ker}(\psi)| \cdot |\mathrm{Im}(\psi)| = |G|$ by the First Homomorphism Theorem applied to $\psi : G \longrightarrow G$. What remains to be proven is that $\mathrm{Ker}(\psi) \cap \mathrm{Im}(\psi) = \{e\}$, which would imply that G is a semidirect product of its subgroups $\mathrm{Ker}(\psi)$ and $\mathrm{Im}(\psi)$. But the map $\psi_{|\mathrm{Im}(\psi)}$ is injective as $\mathrm{Im}(\psi^2) = \mathrm{Im}(\varphi^{2N}) = \mathrm{Im}(\varphi^N) = \mathrm{Im}(\psi)$, so $\mathrm{Ker}(\psi)$ and $\mathrm{Im}(\psi)$ intersect trivially.

(iii) Let φ be the composition

$$\mathsf{S}_3 \xrightarrow{\pi} \mathsf{S}_3/\mathsf{A}_3 \xrightarrow{i} \mathsf{S}_3,$$

where π is the quotient homomorphism and i is the inclusion of $\mathsf{S}_3/\mathsf{A}_3 \simeq \mathbb{Z}/2\mathbb{Z}$ into $\langle(1,2)\rangle \subseteq \mathsf{S}_3$. In other words, set

$$\sigma \xmapsto{\varphi} \begin{cases} e & \text{if } \sigma \in \mathsf{A}_3, \\ (1,2) & \text{if } \sigma \notin \mathsf{A}_3. \end{cases}$$

It is clear that we have $N = 1$ in this case, and that $\mathrm{Im}(\varphi^N) = \{e, (1,2)\}$ is not a normal subgroup of S_3.

23

(i) Let $H = \langle\sigma\rangle$ be the subgroup generated by σ in S_{11}.
The centraliser $Z(\sigma)$ of σ in S_{11} contains H and its cardinality satisfies $|Z(\sigma)| = |\mathsf{S}_{11}|/|\mathcal{C}\ell(\sigma)|$. Since the conjugacy class $\mathcal{C}\ell(\sigma)$ is the set of all 11-cycles, we find that $|Z(\sigma)| = 11!/10! = 11$ and therefore $Z(\sigma) = H$.
The normaliser $N(H)$ of H in S_{11} acts on H by conjugation, so we have a homomorphism $N(H) \longrightarrow \mathrm{Aut}(H)$. The group H is cyclic, and thus its automorphisms are given by $\sigma \longmapsto \sigma^j$, where j and 11 are relatively prime. In particular, σ^j has order 11 and is thus an 11-cycle in S_{11}, that is, an element of $\mathcal{C}\ell(\sigma)$. This implies that there exists $\tau \in N(H)$ such that $\tau(\sigma) = \sigma^j$, so $N(H) \longrightarrow \mathrm{Aut}(H)$ is surjective. As for its kernel, it is $Z_{\mathsf{S}_{11}}(H) = Z(\sigma) = H$, hence $N(H)$ has order $|H| \cdot |\mathrm{Aut}(H)| = 11 \cdot \phi(11) = 110$.
Denote by τ the 10-cycle $(2, 3, 5, 9, 6, 11, 10, 8, 4, 7)$; we have $\tau\sigma\tau^{-1} = \sigma^2$ and therefore $\tau \in N(H)$. Set $K = \langle\tau\rangle$ and remark that $H \cap K = \{e\}$, because the orders of the two subgroups are coprime; it follows that the set HK has cardinality 110 and therefore coincides with $N(H)$. We can thus conclude that each element of $N(H)$ can be expressed as $\sigma^h\tau^k$, where $0 \le h \le 10$ and $0 \le k \le 9$.

(ii) Let M be the subgroup $N(H) \cap \mathsf{A}_{11}$ of A_{11}. The elements of M are the even permutations of the form $\eta = \sigma^h\tau^k$, where $0 \le h \le 10$ and $0 \le k \le 9$. On the other hand, since σ is an even permutation and τ is odd, the sign of η is $(-1)^k$. We have thus shown that $M = \{\sigma^h\tau^k \mid k \text{ is even}\}$, and conclude that $|M| = 55$. So A_{11} does have subgroups of order 55, as exemplified by M.
We now prove that there is no subgroup of order 110 in A_{11}. Suppose by contradiction that L is one such subgroup; by Cauchy's Theorem, L has an element of order 11, that is, an 11-cycle. Without loss of generality, we can take this element to be σ. As before, let $H = \langle\sigma\rangle$.
If $P \neq H$ were another subgroup of order 11 in L, we would have $P \cap H = \{e\}$ and therefore the subset PH of L would have cardinality $121 > 110 = |L|$. This shows that H is the only subgroup of L of order 11, and in particular H is normal in L. But then $L \subseteq N(H)$, and $L \subseteq \mathsf{A}_{11}$ yields $L \subseteq N(H) \cap A_{11} = M$. This is a contradiction because M has order 55.

24

(i) Conjugation $\psi_g : G \longrightarrow G$ by an element g of G is a bijective map which, if applied to subsets, preserves inclusions. We therefore have that $\psi_g(H)$ is still a maximal proper subgroup of G, and thus $\psi_g(H) = H$. This shows that H is normal in G.

(ii) Denote by $N(H)$ the normaliser of H and by $Z(G)$ the centre of G. We have $H, Z(G) \subseteq N(H)$, so $N(H)$ must strictly contain H, since H does not contain $Z(G)$. But H being maximal then implies $N(H) = G$; that is, H is normal in G.

(iii) Set $G = S_3$ and let H be the subgroup generated by $(1, 2)$. The subgroup H is clearly proper, maximal, and not normal.

25 We first make a general remark. The possible cycle structures in S_5, with the corresponding orders and cardinalities of centralisers, are

g	$\operatorname{ord}(g)$	$\lvert Z(g)\rvert$
e	1	120
$(1, 2)$	2	12
$(1, 2)(3, 4)$	2	8
$(1, 2, 3)$	3	6
$(1, 2, 3)(45)$	6	6
$(1, 2, 3, 4)$	4	4
$(1, 2, 3, 4, 5)$	5	5.

Given an Abelian subgroup G of S_5, for all elements g in G, and in particular for $g \neq e$, we have that G is a subgroup of $Z(g)$, as all elements of G commute with g.

(i) By what we have seen above, the only possibility up to conjugation for an Abelian subgroup of order 8 is $Z\big((1, 2)(3, 4)\big)$. This group, however, is not Abelian, as it contains the permutations $(1, 2)$ and $(1, 3)(2, 4)$, which do not commute.
An Abelian subgroup of order 10 cannot exist because of the fact that 10 does not divide the cardinality of any of the centralisers, except for $Z(e)$.

(ii) It is clear that Abelian subgroups of order 1, 2, 3, 4, 5, and 6 do exist, as we have elements of those orders. By the previous part of the solution, the only other possible order is 12, and, without loss of generality, we can assume a subgroup of order 12 is $Z\big((1, 2)\big)$. But the latter subgroup is not Abelian, as it contains all permutations of the elements 3, 4, 5.

26

(i) Letting d be the greatest common divisor of m and n, we intend to show that $\operatorname{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ is isomorphic to $\mathbb{Z}/d\mathbb{Z}$, which is equivalent to the desired result.
An element $\varphi \in \operatorname{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ is completely determined by the choice of $x = \varphi(1)$ in $\mathbb{Z}/n\mathbb{Z}$. We know that $\operatorname{ord}(x)$ divides both m and n, so it has to divide d.
Moreover, given $x \in \mathbb{Z}/n\mathbb{Z}$ such that $\operatorname{ord}(x)$ divides d, there is a homomorphism φ such that $\varphi(1) = x$. But the set of all elements in $\mathbb{Z}/n\mathbb{Z}$ whose order divides d forms the one subgroup G of order d in $\mathbb{Z}/n\mathbb{Z}$, which is itself cyclic. This shows that the map

$$\operatorname{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \ni \varphi \longmapsto x = \varphi(1) \in G \simeq \mathbb{Z}/d\mathbb{Z}$$

is a bijection. By the definition of the group operation in $\mathrm{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$, we have $(\varphi_1 + \varphi_2)(1) = \varphi_1(1) + \varphi_2(1)$, so the above map is also a homomorphism, and therefore an isomorphism.

(ii) Let φ be a homomorphism from S_n to $\mathbb{Z}/m\mathbb{Z}$. We wish to show that A_n is contained in $\mathrm{Ker}(\varphi)$.
This is clear for $n \leq 2$, so assume $n \geq 3$. Since $\mathbb{Z}/m\mathbb{Z}$ is Abelian, the subgroup $[\mathsf{S}_n, \mathsf{S}_n]$ of commutators of S_n is contained in $\mathrm{Ker}(\varphi)$. But we have $(1, 2, 3) = (2, 3)(1, 2)(2, 3)(1, 2) \in [\mathsf{S}_n, \mathsf{S}_n]$, so the kernel $\mathrm{Ker}(\varphi)$ is a normal subgroup containing a 3-cycle, and therefore it contains A_n.
As a consequence, φ induces, by passing to the quotient, a homomorphism $\overline{\varphi}$ between $\mathsf{S}_n/\mathsf{A}_n \simeq \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z}$, which satisfies $\varphi = \overline{\varphi} \circ \pi$, where $\pi : \mathsf{S}_n \longrightarrow \mathsf{S}_n/\mathsf{A}_n$ is the quotient homomorphism.
The desired result follows from the fact that $\mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \simeq \mathbb{Z}/(2, m)\mathbb{Z}$ (as proven in (i)), and in particular $|\mathrm{Hom}(S_n, \mathbb{Z}/m\mathbb{Z})| = (2, m)$.
[We actually have $[\mathsf{S}_n, \mathsf{S}_n] = \mathsf{A}_n$, since for $n \geq 3$ the group S_n is not Abelian and A_n has index 2.]

27 The fact that $\sigma^4 = (1, 2, 3)$ implies that $\sigma^{12} = e$. The decomposition of σ into disjoint cycles can therefore only consist of 2, 3, 4, and 6-cycles.

Note, however, that a 6-cycle elevated to the fourth power yields two disjoint 3-cycles, so no 6-cycle can appear in the decomposition of σ. Moreover, given a 3-cycle η appearing in the decomposition of σ, the same η must also appear in the decomposition of σ^4; the only possibility is therefore $\eta = (1, 2, 3)$. On the other hand, it is clear that such a 3-cycle must actually appear, since the 2-cycles and 4-cycles disappear when elevated to the fourth power. But then it is impossible to have a 4-cycle and we have at most one 2-cycle, because the sum of the lengths of nontrivial cycles must be less than or equal to 6.

In conclusion, the possibilities for σ are $(1, 2, 3)$, $(1, 2, 3)(4, 5)$, $(1, 2, 3)(4, 6)$, and $(1, 2, 3)(5, 6)$.

28 For each $g \in G$, the conjugacy class of g has exactly $[G : Z(g)]$ elements, where $Z(g)$ is the centraliser of g in G.

Evidently, $Z(g) = G$ if and only if g belongs to the centre $Z(G)$ of G, so there are exactly p^2 conjugacy classes consisting of single elements.

Now, let g be an element outside $Z(G)$. Since $Z(g)$ contains both g and $Z(G)$, the centraliser $Z(g)$ is a subgroup of G which strictly contains $Z(G)$ but is not G itself. It follows that $Z(g)$ has exactly p^3 elements and that the conjugacy class of g has p elements.

The $p^4 - p^2$ elements in $G \setminus Z(G)$ are therefore partitioned into $p^3 - p$ conjugacy classes, each with exactly p elements. Consequently, the total number of conjugacy classes is $p^3 + p^2 - p$.

From the isomorphism $\mathrm{Inn}(G) \simeq G/Z(G)$, we obtain that $\mathrm{Inn}(G)$ has $p^4/p^2 = p^2$ elements. Note that the quotient $G/Z(G)$ is not cyclic because G is not Abelian, and so $\mathrm{Inn}(G)$ is isomorphic to $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

29

(i) Given an automorphism φ of G we wish to construct the automorphism $\overline{\varphi}$ of G/H that makes the following diagramme commutative:

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & G \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ G/H & \overset{\overline{\varphi}}{\dashrightarrow} & G/H. \end{array}$$

The map $\pi \circ \varphi : G \longrightarrow G/H$ is a surjective homomorphism, since it is the composition of surjective homomorphisms. Its kernel is the set $\{g \in G \mid \varphi(g) \in H\}$. Since H is a characteristic subgroup of G, we have that $\varphi(g) \in H$ if and only if $g \in H$, so the kernel is H itself.
By the First Homomorphism Theorem, $\pi \circ \varphi$ induces a group isomorphism $\overline{\varphi} : G/H \longrightarrow G/H$ with the required properties, and such an isomorphism is unique.
Moreover, because of what we remarked,

$$\begin{aligned} F(\varphi \circ \psi)(xH) &= (\varphi \circ \psi)(x)H \\ &= \varphi(\psi(x))H \\ &= F(\varphi)(\psi(x)H) \\ &= F(\varphi)\big(F(\psi)(xH)\big) \\ &= \big(F(\varphi) \circ F(\psi)\big)(xH), \end{aligned}$$

so F is a homomorphism.

(ii) Let now $H = Z(G)$; for each $g \in G$, set ψ_g to be the automorphism of G given by conjugation by g.
We show that $\mathrm{Ker}(F) \subseteq Z_{\mathrm{Aut}(G)}(\mathrm{Inn}(G))$. Indeed, given $\varphi \in \mathrm{Ker}(F)$ and $g \in G$, we have $\varphi(g)Z(G) = gZ(G)$, so there exists $z_g \in Z(G)$ such that $\varphi(g) = gz_g = z_g g$. It follows that for all $x,\ g \in G$ we have

$$\begin{aligned} \psi_g \circ \varphi \circ \psi_g^{-1}(x) &= \psi_g \circ \varphi(g^{-1}xg) \\ &= g\varphi(g)^{-1}\varphi(x)\varphi(g)g^{-1} \\ &= z_g^{-1}\varphi(x)z_g \\ &= \varphi(x), \end{aligned}$$

so φ commutes with all internal automorphisms of G.
We now proceed to show that, conversely, we have $Z_{\mathrm{Aut}(G)}(\mathrm{Inn}(G)) \subseteq \mathrm{Ker}(F)$. Suppose φ commutes with all internal automorphisms of G. This means that for all $x,\ g \in G$ we have $(\psi_g \circ \varphi)(x) = (\varphi \circ \psi_g)(x)$, that is,

$$g\varphi(x)g^{-1} = \varphi(g)\varphi(x)\varphi(g)^{-1},$$

which implies that $g^{-1}\varphi(g)$ commutes with $\varphi(x)$ for all $x,\ g \in G$. Since φ is an automorphism and therefore a surjective map, this yields $g^{-1}\varphi(g) \in Z(G)$ for all g, that is, $\varphi(g)Z(G) = gZ(G)$ for all g. In conclusion, $F(\varphi) = \mathrm{Id}$.

(iii) If G is Abelian then $G/Z(G)$ consists of a single element, so F is surjective. As an example of a non-surjective F, consider the dihedral group D_4.
First, we show that $\mathrm{Aut}(\mathsf{D}_4)$ has order 8. Indeed, consider the action of $\mathrm{Aut}(\mathsf{D}_4)$ on D_4, and note we'll be using the habitual presentation $\langle r, s \mid r^4 = s^2 = e, sr = r^{-1}s\rangle$ for D_4. Remark that $\mathrm{Aut}(D_4)$ has a subgroup H of order 4 which consists of the automorphisms fixing the rotation r and sending s to one of the four possible symmetries. The stabiliser of r is precisely H, and the orbit of r consists of the two elements r and r^{-1}, because those are the only elements of order 4 and the map sending r to r^{-1} and s to s is an automorphism. From the Orbit-Stabiliser Relation we obtain that $\mathrm{Aut}(\mathsf{D}_4)$ has 8 elements, as claimed.
The centre $Z(\mathsf{D}_4)$ of D_4 is $\{e, r^2\}$, and $\mathsf{D}_4/Z(\mathsf{D}_4)$ is isomorphic $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$. Now, $\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$ is isomorphic to S_3, so it has 6 elements, and thus the homomorphism F, which is defined between a group with 8 elements and a group with 6 elements, cannot be surjective.

30 For a direct product $H \times K$, the subgroup $(H \times K)(p)$ is isomorphic to $H(p) \times K(p)$. This will turn out to be very useful in view of the fact that, by the Structure Theorem for Finite Abelian groups, G is isomorphic to a direct product of the form $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}$, where $n_1 \geq 1$ and $n_i \mid n_{i+1}$ for $i = 1, \ldots, r-1$.

(i) Clearly, we can assume G is not the trivial group. Given a prime p dividing n_1, we have $G(p) \simeq (\mathbb{Z}/p\mathbb{Z})^r$; since the latter is required to be cyclic, we obtain $r = 1$, so $G \simeq \mathbb{Z}/n_1\mathbb{Z}$ is cyclic.

(ii) The statement is false. Indeed, set $m = p^2$ and $G = \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p^3\mathbb{Z}$. In this case, $G(p) \simeq \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, whereas G is clearly not isomorphic to $(\mathbb{Z}/p^2\mathbb{Z})^2$.

(iii) With a similar argument as that of (i) applied to a prime p dividing n_1, one can prove that we must have $r = 2$, that is, $G \simeq \mathbb{Z}/n_1\mathbb{Z}\times\mathbb{Z}/n_2\mathbb{Z}$ with $n_1 \mid n_2$. What is left is to establish is that $n_1 = n_2$.
Suppose by contradiction $n_1 \neq n_2$, and let p^a and p^b be the largest powers of p dividing n_1 and n_2, respectively, for some prime p for which $a < b$. Remark that there exists an integer c such that $a + b = 2c$ because the order of G is the square m^2; so $a < b$ implies $a + 1 < b$ and therefore $a + 1 \leq c$, that is, p^{a+1} divides m.
In order to conclude, consider $G(p^{a+1})$: by hypothesis, this group is isomorphic to $\mathbb{Z}/p^{a+1}\mathbb{Z} \times \mathbb{Z}/p^{a+1}\mathbb{Z}$; but, since p^a is the largest power of p dividing n_1, we have $G(p^{a+1}) \simeq \mathbb{Z}/p^a\mathbb{Z}\times\mathbb{Z}/p^{a+1}\mathbb{Z}$. This is the contradiction we were looking for, and thus we have $n_1 = n_2 = m$.

31 Denote by G the group given in the statement, and remark that the operation of G is given by

$$(n, a) \circ (m, b) = (n + am, ab)$$

so that $(0, 1)$ is the neutral element, $(-a^{-1}n, a^{-1})$ is the inverse of (n, a), and conjugation of an element by another is given by the expression

$$(n, a)(m, b)(n, a)^{-1} = ((1 - b)n + am, b).$$

Clearly, $(0, 1)$ is only conjugate to itself, whereas

$$\mathcal{C}\ell\left((1, 1)\right) = \left\{(m, 1) \,|\, 0 < m < p\right\},$$

and for $b \neq 1$ we have

$$\mathcal{C}\ell\left((0, b)\right) = \left\{(m, b) \,|\, 0 \leq m < p\right\}.$$

Since every element of G belongs to one of the classes listed above, this concludes our description of the conjugacy classes of G.

An element (n, a) belongs to the centre of G if and only if $(n, a)(m, b) = (m, b)(n, a)$ for all $(m, b) \in G$, which is equivalent to having $n + am = m + bn$ for all $0 \leq m < p$ and $1 \leq b < p$. In particular, setting $b = 1$ yields $am = m$ for all m, and therefore $a = 1$, which in turn implies that $n = nb$ for all b, that is, $n = 0$. We can therefore conclude that the centre of G is trivial.

A similar computation yields that the centraliser of $(1, 1)$ consists of all elements of the form $(m, 1)$ with $0 \leq m < p$.

▌Another way to find the centre of G is to view it as the set of elements whose conjugacy class has cardinality one, which implies (by examination of the list of conjugacy classes) that $Z(G) = \{(0, 1)\}$. ▐

32

(i) Note that the centre is a characteristic group and therefore for all $\varphi \in \mathrm{Aut}(G)$ we have $\varphi(Z) = Z$. It follows that $K = \{\varphi \in \mathrm{Aut}(G) \mid \varphi(gZ) = gZ \text{ for all } g \in G\}$. Moreover, given $\varphi \in K$, $\psi \in \mathrm{Aut}(K)$ and any element g in G, we have

$$\begin{aligned}(\psi\varphi\psi^{-1})(gZ) &= \psi(\varphi(\psi^{-1}(gZ))) \\ &= \psi(\varphi(\psi^{-1}(g)Z)) \\ &= \psi(\psi^{-1}(g)Z) \\ &= gZ,\end{aligned}$$

which proves that K is a normal subgroup of $\mathrm{Aut}(G)$.

(ii) Consider the usual presentation of D_6 with generators r, s and relations $r^6 = s^2 = e$, $sr = r^{-1}s$. The centre of D_6 is $Z = \{e, r^3\}$. Any automorphism of D_6

sends r to either r or r^5 and s to sr^k, where $k = 0, 1, \ldots, 5$, because the order of elements must be preserved.
Now, let φ be an element of K. The condition that $\varphi(rZ) = rZ$ implies that $\varphi(r) \in r\{e, r^3\}$, and therefore by what we just observed $\varphi(r) = r$. Moreover, $\varphi(sZ) = sZ$ yields $\varphi(s) \in s\{e, r^3\}$, so either $\varphi(s) = s$ or $\varphi(s) = sr^3$.
In the first case, $\varphi = \mathrm{Id}_{\mathsf{D}_6}$, which is of course an element of K. The second case does also correspond to an automorphism, as the mappings $r \longmapsto r$, $s \longmapsto sr^3$ are compatible with the relations defining D_6, so they induce a homomorphism of D_6, which we again denote by φ. We have

$$\varphi(r^k Z) = \varphi(r)^k Z = r^k Z$$
$$\varphi(sr^k Z) = \varphi(s)\varphi(r)^k Z = sr^3 r^k Z = sr^k r^3 Z = sr^k Z$$

so φ is also an element of K. In conclusion, $K = \{\mathrm{Id}_{\mathsf{D}_6}, \varphi\} \simeq \mathbb{Z}/2\mathbb{Z}$.

33 We have $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}/10\mathbb{Z}$, so the homomorphisms we're looking for correspond to elements of S_7 whose order divides 10, because each homomorphism from $\mathbb{Z}/10\mathbb{Z}$ extends $1 \longmapsto \sigma$ for some $\sigma \in \mathsf{S}_7$. What we need to do is count elements of order 1, 2, 5 or 10 in S_7.

Clearly, e is the only element of order 1; the elements of order 2 are the 2-cycles, the $2+2$-cycles and the $2+2+2$-cycles. The number of 2-cycles is $\binom{7}{2}$, the number of $2+2$-cycles is $\binom{7}{2}\binom{5}{2}/2$ and the number of $2+2+2$-cycles is $\binom{7}{2}\binom{5}{2}\binom{3}{2}/6$.

The elements of order 5 are the 5-cycles, which are $\binom{7}{5}4!$. The elements of order 10 are given by a 5-cycle times a 2-cycle disjoint from it, so they are as many as the 5-cycles.

The number of homomorphisms from $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ to S_7 is the sum of the cardinalities found so far.

34 We know that $\mathrm{Aut}(G)$ is isomorphic to $\mathsf{GL}_2(\mathbb{F}_{13})$; it follows that constructing the required non-Abelian group structures amounts to constructing nontrivial homomorphisms $\varphi : H \longrightarrow \mathsf{GL}_2(\mathbb{F}_{13})$ and $\psi : K \longrightarrow \mathsf{GL}_2(\mathbb{F}_{13})$. Moreover, H and K are both cyclic groups, so the problem is equivalent to finding an element of order 13 in $\mathsf{GL}_2(\mathbb{F}_{13})$ (for the first example), and one of order 4 or 2 (for the second example).

Consider

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

We have that $A(x, y) = (x + y, y)$ for all $(x, y) \in G$, so $A^k(x, y) = (x + ky, y)$ for all $k \in \mathbb{Z}$, and A is thus an element of $\mathsf{GL}_2(\mathbb{F}_{13})$ having order 13. Clearly, $B^2 = 1$, so B has order 2.

The corresponding semidirect products are the following: we have $G \rtimes_\varphi H$, where

$$\begin{aligned}\big((x,y),u\big) \circ_\varphi \big((z,t),v\big) &= \big((x,y) + \varphi_u(z,t), u+v\big)\\ &= \big((x,y) + A^u(z,t), u+v\big)\\ &= \big((x,y) + (z+ut,t), u+v\big)\\ &= \big((x+z+ut, y+t), u+v\big),\end{aligned}$$

and $G \rtimes_\psi K$, where

$$\begin{aligned}\big((x,y),a\big) \circ_\psi \big((z,t),b\big) &= \big((x,y) + \psi_a(z,t), a+b\big)\\ &= \big((x,y) + B^a(z,t), a+b\big)\\ &= \big((x,y) + ((-1)^a z, (-1)^a t), a+b\big)\\ &= \big((x+(-1)^a z, y+(-1)^a t), a+b\big).\end{aligned}$$

For the first case, one can show that all non-Abelian semidirect products are actually isomorphic; for the second, one could construct a semidirect product using a matrix of order 4, which would yield a group not isomorphic to the one given above.

35 If H is a non-normal subgroup of G, then for all $g \in G$ the subgroup gHg^{-1} is also not normal. Moreover, the number of subgroups of G that are conjugate to H is given by the index of $N(H)$. This index is certainly strictly greater than 1, because H is not normal; it must therefore be divisible by p.

By partitioning the set $\mathcal{N}$ of non-normal subgroups into conjugacy classes, we find that, since the cardinality of each class is divisible by p, the cardinality of $\mathcal{N}$ must also be divisible by p.

36 Letting $\sigma \in \mathsf{S}_{2p}$ be a solution of the equation

$$\sigma^p = (1, 2, \cdots, p)(p+1, p+2, \cdots, 2p),$$

note that we have $\sigma^{p^2} = e$. The order of the permutation σ can therefore only be 1, p or p^2; consequently, the decomposition of σ into disjoint cycles can only contain p-cycles and p^2-cycles. If there were no p^2-cycles in the decomposition, then we would have $\sigma^p = e$, which is not compatible with the equation, so σ must contain at least a p^2-cycle.

Elevating a p^2-cycle to the pth power yields p disjoint cycles of length p. Given the equation σ needs to satisfy, we must have $p = 2$; the equation therefore becomes $\sigma^2 = (1,2)(3,4)$ in S_4.

By what we just showed, the solutions of the equation are $\sigma = (1,3,2,4)$ and $\sigma = (1,4,2,3)$.

We could alternatively have obtained that $p = 2$ by noticing that σ having a p^2-cycle implies $p^2 \leq 2p$, that is, $p = 2$.

37 Let Z be the centre of a group G with the required properties. By assumption, we have $\{e\} \subsetneq Z \subsetneq G$. It is clear that, for each element h in Z, we have $\mathcal{C}\ell(h) = \{h\}$, and that the conjugacy class of each element outside Z does not intersect Z.

Assume Z has at least three elements; in this case, we have at least three conjugacy classes within the centre and at least one outside, so the statement holds.

We can therefore assume that Z has exactly 2 elements. If the elements outside the centre are not all pairwise conjugate, then we end up with at least four conjugacy classes again. We can thus make the additional assumption that, letting g be an element outside the centre, we have $\mathcal{C}\ell(g) = G \setminus Z$. We will show this leads to a contradiction.

Remark that $Z, \{g\} \subseteq Z(g)$, so $|Z(g)| \geq 3$; this implies $|G| - 2 = |\mathcal{C}\ell(g)| = |G|/|Z(g)| \leq |G|/3$, and therefore $|G| \leq 3$. But then G would be Abelian, contradicting our hypothesis.

38 Suppose m and n are coprime natural numbers and let G be a subgroup of $A \times B$. Denote by $\pi_A : A \times B \longrightarrow A$ and $\pi_B : A \times B \longrightarrow B$ the projections, and set $A' = \pi_A(G)$, $B' = \pi_B(G)$, so A' and B' are subgroups of A and B, respectively. We wish to show that $G = A' \times B'$.

Given $g = (a, b) \in G$, we have $a = \pi_A(g) \in A'$ and $b = \pi_B(g) \in B'$, so $g \in A' \times B'$. This proves that $G \subseteq A' \times B'$.

Moreover, given $g = (a, b) \in G$, setting $r = o_A(a)$ and $s = o_B(b)$, we must have $(r, s) = 1$ because r is a divisor of m, s is a divisor of n, and $(m, n) = 1$. It follows that a and a^s generate the same subgroup of A, and thus $(a, e_B) \in \langle (a^s, e_B) \rangle = \langle g^s \rangle \subseteq G$; similarly, $(e_A, b) \in G$.

Now, given $a \in A'$ and $b \in B'$, we wish to show that $(a, b) \in G$. The definition of A' and B' implies the existence of $g_1 = (a, b_1) \in G$ and $g_2 = (a_2, b) \in G$, for some $b_1 \in B$ and $a_2 \in A$. But, by our argument from before, $(a, b_1) \in G$ yields $(a, e_B) \in G$, and similarly $(a_2, b) \in G$ implies $(e_A, b) \in G$. We therefore have $(a, b) = (a, e_B)(e_A, b) \in G$. We have thus shown $G = A' \times B'$.

Now assume all subgroups of G are of the form $A' \times B'$, where A' is a subgroup of A and B' a subgroup of B, and suppose by contradiction there is a prime p dividing both m and n. Cauchy's Theorem yields elements $a \in A$ and $b \in B$ having order p. Let $g = (a, b)$ and $G = \langle g \rangle$; we have $\mathrm{ord}(g) = p$, so $|G| = p$. By assumption, we must have A', B' as above for which $G = A' \times B'$, and in particular $p = |A'| \times |B'|$. But then $A' = \{e_A\}$ or $B' = \{e_B\}$; however, since G is not contained in either $e_A \times B$ or $A \times e_B$, this is a contradiction.

39 Let $V = (\mathbb{Z}/3\mathbb{Z})^3$ be a vector space of dimension 3 over the field $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$, and let A be the following matrix of order 3 in $\mathsf{GL}_3(\mathbb{F}_3)$:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

The group structure of $G = V \rtimes_\varphi \mathbb{Z}/3\mathbb{Z}$ is then given by

$$(u, n)(v, m) = (u + A^n v, n + m).$$

(i) Let (u, n) be an element in the centre $Z(G)$ of G. Since (u, n) commutes with all elements, and in particular with those of the form $(v, 0)$, we have

$$(u + A^n v, n) = (u, n)(v, 0) = (v, 0)(u, n) = (v + u, n)$$

for all $v \in V$; that is, $A^n v = v$ for all $v \in V$, or equivalently $A^n = \mathrm{Id}_V$. This implies that $n = 0$.
Moreover, the fact that $(u, 0) \in Z(G)$ implies

$$(u, 1) = (u, 0)(0, 1) = (0, 1)(u, 0) = (Au, 1)$$

and thus $u = Au$. This yields $u = Au = A^2 u$, that is, $u = \lambda u_0$ where $u_0 = (1, 1, 1)$ and $\lambda \in \mathbb{F}_3$.
Conversely, let us now prove that $(u_0, 0) \in Z(G)$. Indeed, given $(v, m) \in G$, we have $(u_0, 0)(v, m) = (u_0 + v, m)$, and also $(v, m)(u_0, 0) = (v + A^m u_0, m) = (v + u_0, m)$.
We can therefore conclude that $Z(G) = \langle (u_0, 0) \rangle$, so it is isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

(ii) Given an element $g = (u, n) \in G \setminus \{e\}$, we have $g^3 = (u + A^n u + A^{2n} u, 0)$.
If $n = 0$, then $g^3 = (u + u + u, 0) = (3u, 0) = (0, 0)$, and therefore g either has order 1 (if $u = 0$) or order 3 (if $u \neq 0$). Thus, there are $|V| - 1 = 26$ elements (u, n) of order 3 such that $n = 0$.
If $n \neq 0$ then the element g has order 3 if and only if $(\mathrm{Id}_V + A^n + A^{2n})u = 0$. For $n = 1$ and $n = 2$ we have

$$\mathrm{Id}_V + A^n + A^{2n} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

so g has order 3 if and only if $u = (x, y, z)$ with $x + y + z = 0$. So there are $2 \cdot 3^2 = 18$ elements (u, n) of order 3 such that $n \neq 0$.
This brings us to a total of 44 elements of order 3 in G.

40 By Cauchy's Theorem there exist $x, y \in G$ of order p and q, respectively; in particular, G is generated by x and y. Let $\varphi : G \longrightarrow H$ be a homomorphism; remark that $\mathrm{ord}(\varphi(x))$ divides $\mathrm{ord}(x) = p$ and also $|H| = r$, so $\mathrm{ord}(\varphi(x))$ divides the greatest common divisor (p, r), and similarly $\mathrm{ord}(\varphi(y))$ divides (q, r).

Therefore, if $r \notin \{p, q\}$ then we must have $\mathrm{ord}(\varphi(x)) = \mathrm{ord}(\varphi(y)) = 1$, so the only homomorphism from G to H is the null homomorphism given by $g \longmapsto 0$ for all $g \in G$.

A similar argument yields that, for a homomorphism $\psi : H \longrightarrow G$, the order of $\psi(1)$ must divide (r, pq), and thus if $r \notin \{p, q\}$ then the only possible homomorphism is given by $a \longmapsto e_G$ for all $a \in \mathbb{Z}/r\mathbb{Z}$.

We can therefore assume that $r = p$ or $r = q$. Consider the first case; for the moment, we additionally assume that G is cyclic, that is, $H = \mathbb{Z}/p\mathbb{Z}$ and $G \simeq \mathbb{Z}/pq\mathbb{Z}$. In order to define a homomorphism from G, all that is needed is to assign an image to its generator 1, which can be done in p different ways: the homomorphisms from G to H are those induced by $1 \longmapsto a$ for $0 \leq a \leq p-1$.

Similarly, the homomorphisms from H to G are extensions of the mapping $1 \longmapsto bq$ for $0 \leq b \leq p-1$, as the image of $1 \in H$ must be an element of order 1 or p in G.

The case where G is cyclic and $r = q$ is entirely analogous, so we now consider the case where G is not cyclic: p divides $q-1$ and G is isomorphic to $\mathbb{Z}/q\mathbb{Z} \rtimes \mathbb{Z}/p\mathbb{Z}$, unique up to isomorphism.

We first assume that $r = p$. Letting $\varphi : G \longrightarrow H$ be a homomorphism, note that $\varphi(y) = 1$, so φ passes to the quotient $G/(\mathbb{Z}/q\mathbb{Z}) \simeq \mathbb{Z}/p\mathbb{Z}$. It follows that φ is completely determined by $\varphi(x)$, so there are p possible homomorphisms.

Conversely, defining a homomorphism $\psi : \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathbb{Z}/q\mathbb{Z} \rtimes \mathbb{Z}/p\mathbb{Z}$ amounts to selecting the image of 1 among elements (u, v) in G of order 1 or p. Setting $\psi(1) = (0, 0)$ yields the null homomorphism. Note that each element $g = (u, v)$ with $v \neq 0$ has order p: for each integer k we have $g^k = (u_k, kv)$, with $u_k \in \mathbb{Z}/q\mathbb{Z}$, so the order of g is divisible by p and cannot be pq because G is not cyclic.

There are therefore $1 + q(p-1)$ homomorphisms from $\mathbb{Z}/p\mathbb{Z}$ to $\mathbb{Z}/q\mathbb{Z} \rtimes \mathbb{Z}/p\mathbb{Z}$.

The last remaining case is $r = q$. We have $x \in \mathrm{Ker}(\varphi)$ for all $\varphi : G \longrightarrow \mathbb{Z}/q\mathbb{Z}$, so $\langle x \rangle \subseteq \mathrm{Ker}(\varphi)$. But $\langle x \rangle$ is not a normal subgroup of G, since otherwise G would be Abelian, as $0 \times \mathbb{Z}/q\mathbb{Z}$ is a normal subgroup. So $|\mathrm{Ker}(\varphi)| > p$, which yields $|\mathrm{Ker}(\varphi)| = pq$, that is, φ is the null homomorphism.

Conversely, a homomorphism from $\mathbb{Z}/q\mathbb{Z}$ to $\mathbb{Z}/q\mathbb{Z} \rtimes \mathbb{Z}/p\mathbb{Z}$ is defined by mapping 1 to an element of order 1 or q in $\mathbb{Z}/q\mathbb{Z} \rtimes \mathbb{Z}/p\mathbb{Z}$, that is, an element of the form $(u, 0)$ with $0 \leq u \leq q-1$ by our argument above.

41

(i) First of all, we show that the imaginary part of

$$z' = \frac{az+b}{cz+d}$$

is positive for all integers a, b, c, d such that $ad - bc = 1$ and for all complex numbers z whose imaginary part is positive. Indeed,

$$z' = \frac{(az+b)(c\overline{z}+d)}{|cz+d|^2} = \frac{ac|z|^2 + adz + bc\overline{z} + bd}{|cz+d|^2},$$

so the imaginary part of z' is given by

$$\operatorname{Im} z' = \frac{ad - bc}{|cz + d|^2} \cdot \operatorname{Im} z = \frac{\operatorname{Im} z}{|cz + d|^2}.$$

It follows that the sign of the imaginary part of z does not change when the transformation given in the problem statement is applied.
Moreover, using the definition, it is easy to check that $(AB)(z) = A(B(z))$ for all $A, B \in \mathsf{SL}_2(\mathbb{Z})$ and for all $z \in \mathcal{H}$.

(ii) This is clear, as

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \cdot z = \frac{-z + 0}{0 \cdot z - 1} = z.$$

(iii) Remark that the elements we are considering are all part of the orbit of $i \in \mathcal{H}$ for the action of L. The stabilisers of such elements are therefore subgroups of $\mathsf{SL}_2(\mathbb{Z})$ that are conjugate to the stabiliser of the element i; hence, it is enough to show that the stabiliser of i is nontrivial. But, indeed,

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot i = \frac{0 \cdot i + 1}{-1 \cdot i + 0} = \frac{1}{-i} = i,$$

and the matrix above is not the neutral element in L as it is neither I nor $-I$.

42

(i) Let $k, k' \in K$ be such that $\big(HkH\big) \cap \big(Hk'H\big) \neq \varnothing$ and let x be an element in the intersection, so that we have $x = h_1 k h_2 = h_3 k' h_4$ for some $h_1, h_2, h_3, h_4 \in H$. By rearranging, we obtain $k = h_1^{-1} h_3 k' h_4 h_2^{-1}$, hence, since H is a subgroup,

$$HkH = Hh_1^{-1}h_3k'h_4h_2^{-1}H = \big(Hh_1^{-1}h_3\big)k'\big(h_4h_2^{-1}H\big) = Hk'H.$$

(ii) For $x, y \in G$, let $D_{x,y} = H(x, y)H$ be the double coset of $(x, y) \in K$ with respect to H, and let $\mathcal{D}$ be the set of all such double cosets. Moreover, let $\mathcal{C}\ell(t) = \{gtg^{-1} \mid g \in G\}$ be the conjugacy class of $t \in G$ in G and let $\mathcal{C}$ be the set of all conjugacy classes of G. We wish to construct a bijection from $\mathcal{D}$ to $\mathcal{C}$.
First, we show that $D_{x,y} = D_{u,v}$ if and only if xy^{-1} is conjugate to uv^{-1} in G. Indeed, if $D_{x,y} = D_{u,v}$, then there exist $g, g' \in G$ such that $(x, y) = (g, g)(u, v)(g', g')$, and thus $x = gug'$, $y = gvg'$. It follows that

$$xy^{-1} = gug'g'^{-1}vg^{-1} = guv^{-1}g^{-1},$$

which proves that xy^{-1} is conjugate to uv^{-1}.

As for the other implication, assume that xy^{-1} is conjugate to uv^{-1}, and let $g \in G$ be such that $xy^{-1} = guv^{-1}g^{-1}$. For all $g' \in G$ we have $xy^{-1} = gug'g'^{-1}v^{-1}g^{-1}$; choosing g' such that $gug' = x$, we also have $g'^{-1}v^{-1}g^{-1} = y^{-1}$, that is, $gvg' = y$. It follows that $(x, y) = (g, g)(u, v)(g', g')$, and therefore $D_{x,y} = D_{u,v}$.
We can thus conclude that the map

$$\mathcal{D} \ni D_{x,y} \longmapsto \mathcal{C}\ell(xy^{-1}) \in \mathcal{C}$$

is well-defined and injective. But it is also surjective, because the conjugacy class $\mathcal{C}\ell(t)$ can be obtained as the image of $D_{t,e}$.

43 A finite Abelian group G is the direct product of its p-torsion subgroups, which are Abelian p-groups, and a subgroup of G is a direct product of subgroups of these factors. Therefore, a group with the desired properties must either be a p-group with 10 subgroups, or the product of a p-group with 5 subgroups and a q-group with 2 subgroups. We proceed to investigate which Abelian p-groups have exactly 2, 5, 10 subgroups.
(1) A p-group has exactly 2 subgroups—the trivial group and the group itself—if and only if it is isomorphic to $\mathbb{Z}/p\mathbb{Z}$.
(2) If a p-group is cyclic of order p^n, then it has exactly $n + 1$ subgroups, as many as the divisors of p^n. Thus a cyclic p-group has exactly 5 subgroups if and only if $n = 4$.
If a p-group is not cyclic then it contains a subgroup isomorphic to $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$. This has two obvious subgroups, and additionally $(p^2 - 1)/(p - 1) = p + 1$ subgroups of order p, so $p + 3$ subgroups in total. It follows that the number of subgroups of such a group cannot be equal to 5, since by assumption $p > 2$.
(3) By our previous arguments, $\mathbb{Z}/p^9\mathbb{Z}$ and $\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$ have exactly 10 subgroups; they are the only cyclic group and the only group of order p^2 with this property, respectively. We now check whether there are other possibilities.
If a non-cyclic p-group has at least p^3 elements, then it contains either a group isomorphic to $\mathbb{Z}/p^2\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ or a group isomorphic to $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$. The first of these two groups has $(p^2 - 1)/(p - 1) = p + 1$ subgroups of order p, one non-cyclic subgroup and $(p^3 - p^2)/(p^2 - p) = p$ cyclic subgroups of order p^2, so $2p + 4$ subgroups in total, including the obvious ones. This case is therefore possible only if $p = 3$ and the group is isomorphic $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. The second groups contains $(p^3 - 1)/(p - 1) = p^2 + p + 1 \geq 3^2 + 3 + 1 = 13$ subgroups of order p and is therefore not to be considered.

In conclusion, the groups with the desired properties are the cyclic groups of order p^9, p^4q, and the two groups $\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$, $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.

▌If one also considered groups of even order, they would have to add to the list the groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$; these are obtained with essentially the same proof as above, but excluding the existence of other possibilities would be rather more laborious. ▐

44 We recall the definition of the group structure for a semidirect product: given (h_1, k_1) and (h_2, k_2) in G, we have $(h_1, k_1)(h_2, k_2) = (h_1\varphi_{k_1}(h_2), k_1k_2)$, where φ_{k_1} is the image $\varphi(k_1)$ of k_1 via φ.

(i) First of all, note that $\alpha : (h, k) \longmapsto (\omega(h), k)$ is definitely bijective, since ω is an automorphism of H. So α is an automorphism of G if and only if for all $h_1, h_2 \in H$ and $k_1, k_2 \in K$ we have

$$\alpha(h_1, k_1)\alpha(h_2, k_2) = \alpha(h_1\varphi_{k_1}(h_2), k_1k_2),$$

which is equivalent to

$$\big(\omega(h_1), k_1\big)\big(\omega(h_2), k_2\big) = \big(\omega(h_1\varphi_{k_1}(h_2)), k_1k_2\big).$$

This yields

$$\omega(h_1)\varphi_{k_1}(\omega(h_2)) = \omega(h_1)\omega(\varphi_{k_1}(h_2)),$$

and, by simplifying, $(\varphi_{k_1} \circ \omega)(h_2) = (\omega \circ \varphi_{k_1})(h_2)$ for all h_2 and k_1. It follows that α is an automorphism if and only if for all $k_1 \in K$ we have that $\varphi_{k_1} \circ \omega$ and $\omega \circ \varphi_{k_1}$ coincide as automorphisms of H. That is, if and only if in the group $\mathrm{Aut}(H)$ the automorphism ω commutes with all automorphisms of the form φ_{k_1} for k_1 in K. This is equivalent to saying that $\omega \in Z_{\mathrm{Aut}(H)}(\mathrm{Im}(\varphi))$.

(ii) By an argument similar to the above, using the same notation, we have that $(h, k) \longmapsto (h, \epsilon(k))$ is an automorphism of G if and only if

$$\big(h_1, \epsilon(k_1)\big)\big(h_2, \epsilon(k_2)\big) = \big(h_1\varphi_{k_1}(h_2), \epsilon(k_1k_2)\big),$$

and thus if and only if

$$\varphi_{\epsilon(k_1)}(h_2) = \varphi_{k_1}(h_2)$$

for all h_2 and k_1. The condition is equivalent to having $\varphi_{\epsilon(k_1)} = \varphi_{k_1}$ in $\mathrm{Aut}(H)$, for all $k_1 \in K$. Now, since $\varphi : K \longrightarrow \mathrm{Aut}(H)$ is a homomorphism, we can rewrite this as $k_1^{-1}\epsilon(k_1) \in \mathrm{Ker}(\varphi)$ for all $k_1 \in K$.
Assume the above does hold. In particular, if $k_1 \in \mathrm{Ker}(\varphi)$ then $\epsilon(k_1) \in \mathrm{Ker}(\varphi)$, which is the first of two conditions in the problem statement. Moreover, this makes it possible to pass to the quotient and define the map $\epsilon' : k_1 \mathrm{Ker}(\varphi) \longmapsto \epsilon(k_1)\mathrm{Ker}(\varphi)$. Using again the assumption about ϵ, we obtain $\epsilon' = \mathrm{Id}_{K/\mathrm{Ker}(\varphi)}$ and thus the second condition in the problem statement. Conversely, assuming the two conditions in the problem statement, we have $\epsilon(k_1)\mathrm{Ker}(\varphi) = k_1\mathrm{Ker}(\varphi)$ for all $k_1 \in K$. In particular, $\epsilon(k_1) \in k_1\mathrm{Ker}(\varphi)$, and therefore $\varphi_{\epsilon(k_1)} = \varphi_{k_1}$. This implies that $(h, k) \longmapsto (h, \epsilon(k))$ is an automorphism, as shown above.

45

(i) Let σ be the permutation

$$(1)(2,3)(4,5,6)(7,8,9,10)\cdots$$

The inclusion $Z(\sigma) \subseteq Z(\sigma^n)$ holds in general for all n and for all permutations σ. Let us show that, given $n > 1$, there is a permutation τ that commutes with σ^n but not with σ.
The permutation σ contains an n-cycle $(\ell, \ell+1, \cdots, \ell+n-1)$ for some $\ell \in \mathbb{N}$. Set $\tau = (\ell, \ell+1)$, and remark that σ and τ both send the set $I = \{\ell, \ell+1, \cdots, \ell+n-1\}$ to itself. Moreover, σ^n has each element of I as a fixed point, and thus τ and σ^n commute. On the other hand, $(\tau\sigma\tau^{-1})_{|I} = (\ell+1, \ell, \ell+2, \ell+3, \ldots, \ell+n-1) \neq \sigma_{|I}$, so τ does not commute with σ.

(ii) Consider $\omega \in \mathsf{S}_0(\mathbb{N})$ and let ϵ be any element of $\mathsf{S}(\mathbb{N})$. Letting $\{a_1, a_2, \ldots, a_n\}$ be the subset of all elements that are not fixed by ω, we have that the set $\{\epsilon(a_1), \epsilon(a_2), \ldots, \epsilon(a_n)\}$ consists exactly of the elements not fixed by $\epsilon\omega\epsilon^{-1}$. This implies that the latter permutation belongs to $\mathsf{S}_0(\mathbb{N})$, so $\mathsf{S}_0(\mathbb{N})$ is a normal subgroup of $\mathsf{S}(\mathbb{N})$.
In order to show that $\mathsf{S}(\mathbb{N})/\mathsf{S}_0(\mathbb{N})$ is an infinite group, we will prove that the permutation σ defined in part (i) of the solution has infinite order in $\mathsf{S}(\mathbb{N})/\mathsf{S}_0(\mathbb{N})$. To this end, we intend to show that σ^n is not a permutation of a finite number of elements for any natural number n.
Indeed, having fixed n, there are infinitely many natural numbers coprime with n, so there are infinitely many disjoint cycles of length coprime with n in σ. Now, σ^n does not fix any of the infinitely many integers appearing in these cycles, so it is not an element of $\mathsf{S}_0(\mathbb{N})$.

46

(i) The normaliser of the subgroup G generated by σ is

$$N(G) = \left\{\tau \in \mathsf{S}_{19} \,|\, \text{there exists } 0 \leq k \leq 18 \text{ such that } \tau\sigma\tau^{-1} = \sigma^k\right\}.$$

Remark that orders of elements are preserved under conjugation, so we cannot have $k = 0$ in the above formula. Therefore, letting N_k be the set of permutations $\tau \in \mathsf{S}_{19}$ such that $\tau\sigma\tau^{-1} = \sigma^k$, we have

$$|N(G)| = \sum_{k=1}^{18} |N_k|$$

Moreover, for each $1 \leq k \leq 18$, σ^k is still a 19-cycle, so there exists an element $\tau_k \in N_k$, and thus $N_k = \tau_k Z(\sigma)$, hence $|N_k| = |Z(\sigma)|$.
By recalling that $|Z(\sigma)| = |\mathsf{S}_{19}|/|\mathcal{C}\ell(\sigma)| = 19!/18! = 19$, we obtain $|N(G)| = 18 \cdot 19$.

(ii) The normaliser of G in A_{19} is the intersection of $N(G)$ with A_{19}. Up to conjugation, we can assume $\sigma = (1, 2, 3, \ldots, 18, 19)$. Consider the element

$$\tau = (1, 19)(2, 18)(3, 17)(4, 16)(5, 15)(6, 14)(7, 13)(8, 12)(9, 11).$$

Clearly, $\tau\sigma\tau^{-1} = (19, 18, 17, \ldots, 3, 2, 1) = \sigma^{-1}$, and moreover τ is an odd permutation. This proves that the two normalisers do not coincide.

47

(i) Let $H = \mathsf{S}_3 \times 0 \times 0$ and $K = e \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.
The elements of order 2 in G are all those of the form $(\tau, 0, 0)$ with τ a transposition in S_3. Any automorphism of G will send the set of these elements to itself, and, since this set generates $\mathsf{S}_3 \times 0 \times 0$, this implies that H is characteristic.
We know that the centre of a direct product is the product of the centres: $Z(\mathsf{S}_3 \times \mathbb{Z}/3\mathbb{Z}/ \times \mathbb{Z}/3\mathbb{Z}) = Z(\mathsf{S}_3) \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} = e \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} = K$.
So K is also characteristic, as it is the centre of G.

(ii) Since G is the product of two characteristic subgroups, we immediately have $|\operatorname{Aut}(G)| = |\operatorname{Aut}(\mathsf{S}_3)| \cdot |\operatorname{Aut}(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z})|$.
Note that an automorphism of S_3 must permute the 3 elements of order 2 that generate S_3, so the automorphisms of S_3 are at most 6. Moreover, since the centre of S_3 is trivial, we have 6 distinct automorphisms of S_3 given by its inner automorphisms, so $|\operatorname{Aut}(\mathsf{S}_3)| = 6$.
We have that $\operatorname{Aut}(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z})$ is isomorphic to $\mathsf{GL}_2(\mathbb{F}_3)$, so $|\operatorname{Aut}(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z})| = (3^2 - 1)(3^2 - 3) = 48$.
In conclusion, $|\operatorname{Aut}(G)| = 6 \cdot 48$.

48 Suppose $\sigma \in \mathsf{S}_5$ satisfies the conditions $\sigma^2 = (1, 2)\sigma(1, 2)$ and $\sigma^3 = (2, 3)\sigma(2, 3)$. Note that this implies that σ^2 and σ^3 are conjugates of σ; in particular, σ^2 and σ^3 have the same order as σ, so this order is coprime with 6.

By considering the possible cycle structures of permutations in S_5 and the corresponding orders, we find that the identity and the 5-cycles are the only permutations whose order is coprime with 6. It is clear that the identity does satisfy the required conditions. We can thus assume that σ is a 5-cycle and prove that this yields a contradiction.

Using the fact that σ has order 5, we have

$$\sigma = \sigma^6 = (\sigma^2)^3 = ((1, 2)\sigma(1, 2))^3 = (1, 2)\sigma^3(1, 2) = (1, 2)(2, 3)\sigma(2, 3)(1, 2).$$

Consequently, $(1, 2, 3) = (1, 2)(2, 3)$ is an element in the centraliser $Z(\sigma)$ of σ in S_5. But the conjugacy class of σ is the set of 5-cycles, so it has 4! elements, which implies that $Z(\sigma)$ has 5 elements and cannot therefore contain any 3-cycles.

49 The cardinality of the orbits must divide p and therefore be either p or 1. Letting a and b be the number of orbits with p elements and the number of orbits with one

element, respectively, we have $|V| = p^2 = ap + b$, so b must be divisible by p. Moreover, we have $b > 0$ since $\{0\}$ is an orbit.

The set of fixed points for the action of H is clearly a vector subspace of V; consequently, we have $b = p^k$, where $k = 1$ or $k = 2$. If we had $k = 2$, then all points of V would be fixed by the action of H; but this cannot be, as H has order p and must therefore contain elements other than the identity. We can thus conclude that $k = 1$, that is, $b = p$.

It follows that $a = p - 1$ and that there are $a + b = p - 1 + p = 2p - 1$ orbits in V.

50 Clearly, it is enough to deal with the case of $S \neq \{e_G\}$, as $\{e_G\}$ is a characteristic subgroup.

Assume that the group G is isomorphic to $S \times Z$; what we wish to prove is that, given any automorphism $\varphi : S \times Z \longrightarrow S \times Z$, we have $\varphi(S \times e_Z) = S \times e_Z$. Consider the composition

$$S \overset{i}{\hookrightarrow} S \times Z \overset{\varphi}{\longrightarrow} S \times Z \overset{\pi}{\longrightarrow} Z, \qquad S \overset{\psi}{\longrightarrow} Z$$

of the inclusion $i : s \longmapsto (s, e_Z)$ with φ and with the projection $\pi : (s, z) \longmapsto z$. Since it is a composition of homomorphisms, ψ is a homomorphism from S to Z. Its kernel $\mathrm{Ker}(\psi)$ is a normal subgroup of S; but S is simple, so there are only two possibilities: either $\mathrm{Ker}(\psi) = \{e_S\}$ or $\mathrm{Ker}(\psi) = S$.

In the first case, $\psi : S \longrightarrow Z$ is an injection of S into Z, so the fact that Z is Abelian implies that S is as well; but S is not Abelian by hypothesis. In the second case, we have $\pi \circ \varphi(s, e_Z) = e_Z$ for all $s \in S$, so $\varphi(s, e_Z) \in S \times e_Z$ for all $s \in S$. In other words, S is characteristic.

51 If n is strictly less than 7 then S_n contains no elements of order 7, whereas any group of order 21 does by Cauchy's Theorem; it follows that, for n strictly less than 7, S_n does not contain any subgroups of order 21.

We now show that S_7 does contain a subgroup of order 21, and thus that all groups S_n with n at least 7 do, as they have a subgroup isomorphic to S_7.

Let σ be a 7-cycle in S_7 and set $H = \langle\sigma\rangle$. We know that the normaliser $N(H)$ of H has order $\phi(7) \cdot |Z(\sigma)| = 6 \cdot 7!/6! = 42$, so it must contain an element τ of order 3; let $K = \langle\tau\rangle$. Since $K \subseteq N(H)$, we have $HK = KH$, and so $G = HK$ is a subgroup of S_7. Moreover, the order of G is $|H||K|/|H \cap K| = 21$, as $H \cap K = \{e\}$.

▌Given that the 7-Sylow H in a group of order 21 must be normal, any subgroup of S_n of order 21 must be contained in the normaliser of H; it is therefore natural to look for the element τ of order 3 within $N(H)$.▐

52

(i) Setting $\alpha = 1 + p \in \mathbb{Z}/p^2\mathbb{Z}$, note that $\alpha^p \equiv 1 \pmod{p^2}$, so α has order p in $(\mathbb{Z}/p^2\mathbb{Z})^*$. The map $\mathbb{Z}/p^2\mathbb{Z} \ni a \overset{\varphi}{\longmapsto} \alpha a \in \mathbb{Z}/p^2\mathbb{Z}$ is thus an automorphism of $\mathbb{Z}/p^2\mathbb{Z}$ having order p. We can therefore consider the homomorphism $\mathbb{Z}/p\mathbb{Z} \ni 1 \overset{\psi}{\longmapsto} \varphi \in \mathrm{Aut}(\mathbb{Z}/p^2\mathbb{Z})$ and construct the corresponding semidirect product $G = \mathbb{Z}/p^2\mathbb{Z} \rtimes_\psi \mathbb{Z}/p\mathbb{Z}$, which clearly has the required properties.

(ii) We consider the group G as previously constructed and denote by h the element $(1, 0)$. First of all, we show that the centre Z of G is $\langle h^p \rangle = \langle (p, 0) \rangle$. We know that Z is nontrivial because G is a p-group; so Z has cardinality either p or p^2, given that G is not Abelian. However, if the cardinality of Z were equal to p^2, then G/Z would have cardinality p and would therefore by cyclic, so G would be Abelian, contradiction. In conclusion, Z has p elements. Set $k = (0, 1)$ and notice that by definition of the group structure on G we have $khk^{-1} = (0, 1)(1, 0)(0, 1)^{-1} = (1 + p, 0)$. It follows that $k(h^p)k^{-1} = (khk^{-1})^p = (1 + p, 0)^p = (p(1 + p), 0) = (p, 0) = h^p$. This shows that h^p commutes with k; moreover, we clearly have that h^p commutes with h, so h^p belongs to the centre of G. We conclude that $Z = \langle h^p \rangle$, since h^p and Z both have order p.

We know that any nontrivial normal subgroup of a p-group has nontrivial intersection with the group's centre. In fact, since the centre of G has order p, every nontrivial normal subgroup of G actually contains the centre.

The set of nontrivial normal subgroups of G is therefore in bijection with the set of normal subgroups of G/Z. Since G/Z is a group of order p^2 and it is not cyclic, we conclude that G/Z is isomorphic to $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, so the bijection is given by the quotient homomorphism

$$G = \mathbb{Z}/p^2\mathbb{Z} \rtimes_\psi \mathbb{Z}/p\mathbb{Z} \ni (a, b) \overset{\pi}{\longmapsto} (a, b) \in \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \simeq G/Z.$$

All subgroups of G/Z are normal; they are: $\{e_{G/Z}\} = \{(0, 0)\}$, G/Z, $\langle (1, 0) \rangle$, and additionally all subgroups of the form $\langle (a, 1) \rangle$ where a varies between 0 and $p - 1$.

Since $\pi(h) = (1, 0)$ and $\pi(k) = (0, 1)$, the nontrivial normal subgroups of G are the following: Z, G, the subgroup generated by $h = (1, 0)$, and the subgroups generated by h and $(a, 1)$, where a ranges from 0 to $p - 1$.

53

(i) We know that, for $n \geq 5$, the normal subgroups of S_n are $\{e\}$, A_n and S_n. What we need to do is determine which of those is the smallest subgroup containing G.

The answer is the trivial subgroup $\{e\}$ if and only if $h = 1$ and $k = n$.

Assuming this is not the case, the desired subgroup is A_n if and only if the generators σ, τ of G are both even permutations, that is, if and only if $(-1)^{h-1} = (-1)^{n-k} = 1$, which is equivalent to h being odd and $n - k$ even.

In all other cases, the smallest subgroup containing G is S_n.

(ii) The centraliser $Z_{\mathsf{S}_n}(G)$ of G in S_n is the intersection $Z(\sigma) \cap Z(\tau)$ of the centralisers of the generators of G. Since σ and τ are cycles, we know the following facts: every element of $Z(\sigma)$ is of the form $\sigma^u \eta$ for some integer u and some permutation η fixing each point in $\{1, 2, \ldots, h\}$; every element of $Z(\tau)$ is of the form $\tau^v \epsilon$, where v is an integer and ϵ is a permutation fixing each point in $\{k, k+1, \ldots, n\}$. We consider two cases.
① Suppose $h < k$. In this case, $Z(\sigma) \cap Z(\tau)$ is the set of elements of the form $\sigma^u \delta \tau^v$, where u and v are integers which we can take to satisfy $0 \leq u \leq h-1$ and $0 \leq v \leq n-k$, and δ is a permutation fixing each point in $\{1, 2, \ldots, h, k, k+1, \ldots n\}$, that is, a permutation of $\{h+1, h+2, \ldots, k-2, k-1\}$.
② Suppose $h = k$. Any permutation α that commutes with σ must send h to an element of $\{1, 2, \ldots, h\}$; if α additionally commutes with τ, then it sends k to an element of $\{k, k+1, \ldots, n\}$; so $h = k$ must be fixed by α. This forces α to fix all elements in $\{1, \ldots, h\}$ because of the description given for $Z(\sigma)$, and similarly it must fix all elements of $\{k, k+1, \ldots, n\}$ because of the description given for $Z(\tau)$. In conclusion, α must fix all elements in $\{1, 2, \ldots, n\}$ because $h = k$, so $Z(G)$ is the trivial subgroup of S_n.

(iii) From the discussion of the previous part we glean the following.
① If $h < k$ then the orbits are $\{1, 2, \ldots, h\}$, $\{k, k+1, \ldots, n\}$ and $\{a\}$, where a ranges between $h+1$ and $k-1$.
② If $h = k$ then there is a single orbit, namely, $\{1, 2, \ldots, n\}$.

54 Let p be a prime and let $G(p)$ be the p-torsion subgroup of G. All subgroups of G, including the group itself, are isomorphic to the direct product of their p-torsion subgroups, so we can deal with each component separately.

Let $G(p) = \mathbb{Z}/p^{\alpha_1}\mathbb{Z} \times \mathbb{Z}/p^{\alpha_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{\alpha_r}\mathbb{Z}$ be a decomposition of $G(p)$ into cyclic subgroups. Remark that, if $r > 1$, then there exists a subgroup H of $G(p)$ that is isomorphic to $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$. Moreover, as soon as there is some $\alpha_i > 1$ we also have a subgroup K of $G(p)$ that is isomorphic to $\mathbb{Z}/p^2\mathbb{Z}$. Since in this case H and K would have the same cardinality but not be isomorphic, we deduce that we must have $r = 1$ or $\alpha_1 = \alpha_2 = \cdots = \alpha_r = 1$. This condition is necessary for the desired property to hold: we now prove that it is also sufficient.

Indeed, if $r = 1$ then $G(p) = \mathbb{Z}/p^{\alpha_1}\mathbb{Z}$ is a cyclic group, and therefore has a single subgroup for each possible divisor of its order: subgroups of the same order are isomorphic because they actually coincide.

If $\alpha_1 = \alpha_2 = \cdots = \alpha_r = 1$, then $G(p) = \mathbb{Z}/p\mathbb{Z} \times \cdots \times \mathbb{Z}/p\mathbb{Z}$ is a vector space of dimension r on $\mathbb{F}_p$. Two subgroups of the same order are thus vector subspaces of the same dimension, and therefore isomorphic.

In conclusion, adding the condition that $|G| = 10^6$, the 2-torsion component can be either $(\mathbb{Z}/2\mathbb{Z})^6$ or $\mathbb{Z}/2^6\mathbb{Z}$ and, similarly, the 5-torsion can be either $(\mathbb{Z}/5\mathbb{Z})^6$ or $\mathbb{Z}/5^6\mathbb{Z}$. So we have precisely the following four possibilities for the group G: $(\mathbb{Z}/2\mathbb{Z})^6 \times (\mathbb{Z}/5\mathbb{Z})^6$, $\mathbb{Z}/2^6\mathbb{Z} \times (\mathbb{Z}/5\mathbb{Z})^6$, $(\mathbb{Z}/2\mathbb{Z})^6 \times \mathbb{Z}/5^6\mathbb{Z}$ and $\mathbb{Z}/2^6\mathbb{Z} \times \mathbb{Z}/5^6\mathbb{Z}$.

55

(i) We know that $\mathrm{Aut}\left((\mathbb{Z}/2\mathbb{Z})^3\right)$ is isomorphic to the matrix group $\mathsf{GL}_3(\mathbb{F}_2)$ and has order $7 \cdot 6 \cdot 4$. Thus, by Cauchy's Theorem, there is an element φ of order 7 in $\mathrm{Aut}\left((\mathbb{Z}/2\mathbb{Z})^3\right)$. The mapping $\mathbb{Z}/7\mathbb{Z} \ni 1 \longmapsto \varphi \in \mathrm{Aut}((\mathbb{Z}/2\mathbb{Z})^3)$ extends to a nontrivial homomorphism and therefore provides an example of a nontrivial action by group automorphisms.

(ii) Let $\psi : \mathbb{Z}/7\mathbb{Z} \longrightarrow \mathrm{Aut}\left((\mathbb{Z}/2\mathbb{Z})^3\right)$ be a nontrivial action by group automorphisms. We have that $\{0\}$ must be an orbit in $(\mathbb{Z}/2\mathbb{Z})^3$ for such an action, because any automorphism will send 0 to itself. The remaining 7 elements of $(\mathbb{Z}/2\mathbb{Z})^3$ are thus permuted among themselves, and in particular we can think of $\psi(1)$ as an element of S_7 of order 7, as the action is nontrivial. But the only elements of order 7 in S_7 are the 7-cycles, so $\psi(1)$ acts transitively on $(\mathbb{Z}/2\mathbb{Z})^3 \setminus \{0\}$. We have thus shown that

$$(\mathbb{Z}/2\mathbb{Z})^3 = \{0\} \bigsqcup \left((\mathbb{Z}/2\mathbb{Z})^3 \setminus \{0\}\right)$$

is the partition into orbits for the action ψ.

56

(i) An element (g, h) belongs to the centre of $G \rtimes_\varphi H$ if and only if $(g, h)(f, k) = (f, k)(g, h)$ for all $f \in G$ and $k \in H$. Assume this is the case; by computing the two products and considering the second coordinate, we obtain that $hk = kh$. This implies that h belongs to the centre of H, so $h = e_H$ since the centre of H is trivial.

Now consider the first coordinate; we must have $g + g' = g' + \varphi_k(g)$ for all $g' \in G$ e $k \in H$. Using the fact that G is Abelian, we obtain that (g, h) is in the centre if and only if $h = e_H$ and $\varphi_k(g) = g$ for all $k \in H$, precisely as claimed in the problem statement.

(ii) We will think of $V = (\mathbb{Z}/2\mathbb{Z})^n$ as an n-dimensional vector space on $\mathbb{F}_2$ and define the following elements: $e_1 = (1, 0, \ldots, 0)$, $e_2 = (0, 1, 0, \ldots, 0)$, $\ldots$, $e_n = (0, \ldots, 0, 1)$. These vectors form a basis of V.

Given an element $\sigma \in \mathsf{S}_n$, consider the corresponding linear map f_σ whose restriction on the basis vectors is the permutation $e_i \longmapsto e_{\sigma(i)}$ for $i = 1, \ldots, n$. We obtain a homomorphism

$$\mathsf{S}_n \ni \sigma \overset{\varphi}{\longmapsto} f_\sigma \in \mathsf{GL}_n(\mathbb{F}_2) \simeq \mathrm{Aut}\left((\mathbb{Z}/2\mathbb{Z})^n\right)$$

and the corresponding semidirect product $(\mathbb{Z}/2\mathbb{Z})^n \rtimes_\varphi \mathsf{S}_n$.

In order to determine the centre of this group, we can use the result obtained in the previous part of the solution, since $(\mathbb{Z}/2\mathbb{Z})^n$ is Abelian and the centre of S_n is trivial for $n \geq 3$. What we need to do is find all elements $v \in V$ such that $f_\sigma(v) = v$ for all $\sigma \in \mathsf{S}_n$.

As the action of S_n permutes the coordinates of v, in order for v to be invariant under all permutations, its coordinates must all be equal; that is, v must be

either 0 or $e_1 + e_2 + \cdots + e_n$. It is easy to check that each of these two vectors is fixed for the action of S_n.
We have thus found that the centre of the semidirect product we constructed is $\{0, e_1+e_2+\cdots+e_n\} \times \mathrm{Id}$; since it has two elements, it is obviously isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

57

(i) If an injective homomorphism from D_5 to S_n exists, then we must have $n \geq 5$, as D_5 has an element of order 5, so S_n must also have one. Moreover, we can think of D_5 as the group of Euclidean plane isometries sending a regular pentagon to itself; we can thus see D_5 as a subgroup of the permutation group on the vertices of the pentagon, so we have an injective homomorphism from D_5 to S_5.
This yields that the required minimum is $n = 5$.

(ii) With a similar argument as for the previous part, we obtain that $n \geq 7$ because D_7 has an element of order 7; moreover, as above, we have an injective homomorphism from D_7 to S_7 obtained by interpreting elements of D_7 as permutations of the vertices of a 7-sided polygon.
Let us fix some notation: let the rotation ρ and the symmetry σ be generators of D_7. We also number the vertices of the 7-sided polygon in such a way that the homomorphism discussed above sends D_7 to the subgroup generated by the 7-cycle $(1, 2, 3, \ldots, 7)$ and the permutation $(2, 7)(3, 6)(4, 5)$, which we take to be the image of ρ and the image of σ, respectively. It is now clear that the mapping

$$\rho \longmapsto (1, 2, 3, \ldots, 7), \quad \sigma \longmapsto (2, 7)(3, 6)(4, 5)(8, 9)$$

extends to an injective homomorphism from D_7 to A_9. This proves that $n \leq 9$.
We now wish to show that for $n = 7$ and $n = 8$ no injective homomorphism exists from D_7 to A_n, and thus conclude that $n = 9$.
For $n \in \{7, 8\}$, the image of ρ via any homomorphism must be a 7-cycle, which we can assume to be $(1, 2, 3, \ldots, 7)$, up to conjugation. We know that $\sigma\rho\sigma^{-1} = \rho^{-1}$, so the image of σ must be a permutation η such that $\eta(1, 2, 3, \ldots, 7)\eta^{-1} = (7, 6, 5, \ldots, 1)$; for example, it could be the permutation $\eta_0 = (1, 7)(2, 6)(3, 5)$.
The condition given for η implies that η must be in the coset $\eta_0 Z\big((1, 2, 3, \ldots, 7)\big)$, as its action by conjugation on the 7-cycle is fixed. By computing the cardinality of the centraliser $Z\big((1, 2, 3, \ldots, 7)\big)$, we immediately find that this centraliser must coincide with the subgroup generated by the 7-cycle itself for $n = 7$ and $n = 8$. In particular, $\eta \in \eta_0 \mathsf{A}_n$, because 7-cycles are even and therefore $\langle(1, 2, 3, \ldots, 7)\rangle$ is a subgroup of A_n. But then $\eta \notin \mathsf{A}_n$, as η_0 is odd. In particular, there are no injective homomorphisms sending D_7 to A_n for $n = 7$ or $n = 8$.

58

- We first give an arithmetic solution. For each positive divisor r of n, let d_r be the number of elements in G whose order is equal to r. By assumption, either $d_r = 0$ or, given a solution y of order r of the equation $y^r = 1$, all solutions of this equation are contained in the cyclic subgroup generated by y. As a consequence, d_r can only be 0 or $\phi(r)$.

  But, since $\sum_{r\,|\,n} \phi(r) = n$, we have that d_r must always be equal to $\phi(r)$. In particular, $d_n > 0$, so G is cyclic.
- We now present a second solution. First of all, we show that the following simple property holds for any finite Abelian group G: letting n be the maximum possible order of an element in G, we have $y^n = 1$ for all $y \in G$.

  Equivalently, we are claiming that $\mathrm{ord}(y) \,|\, n$ for all $y \in G$. Fix an element y of G and a prime p; let p^α be the largest power of p that divides n, and let p^β be the largest power of p that divides $r = \mathrm{ord}(y)$. Our claim is equivalent to the fact that $\beta \leq \alpha$.

  Now let z be an element of order n; note that the element $z_1 = z^{p^\alpha}$ has order n/p^α and the element $y_1 = y^{r/p^\beta}$ has order p^β. By the definitions of α and β, we have that n/p^α and p^β are coprime; the element $z_1 y_1$ has therefore order $np^{\beta-\alpha}$. But then $\beta \leq \alpha$, or $z_1 y_1$ would have order strictly greater than n.

  We now have all the tools to solve the original problem: every element of G is a solution of $x^n = 1$, so by the assumed property there are at most n elements in G. But $\langle z \rangle$ is a group with n elements, because z has order n. So G coincides with $\langle z \rangle$ and is thus cyclic.
- Finally, here is a very simple solution using the Structure Theorem for Finite Abelian Groups. The latter implies that a non-cyclic finite Abelian group contains a product of the form $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ for some p; but then the equation $x^p = e$ has at least p^2 solutions.

59 For each $g \in G$, denote by ψ_g the automorphism of G given by conjugation by g. Since N is normal in G, the restriction $\psi_{g|N}$ is an automorphism of N. So $g \longmapsto \psi_{g|N}$ defines a homomorphism $G \longrightarrow \mathrm{Aut}(N)$, which we denote by ψ.

The kernel of ψ consists of the elements of G that act trivially by conjugation on N, so $\mathrm{Ker}(\psi) = Z_G(N)$; this shows that $Z_G(N)$ is a normal subgroup of G.

Moreover, by the First Homomorphism Theorem, the group $G/Z_G(N)$ is isomorphic to a subgroup of $\mathrm{Aut}(N)$. Now, since N is cyclic, $\mathrm{Aut}(N)$ is an Abelian group, so $G/Z_G(N)$ must be as well.

60

(i) We can express the operation on G as

$$(n, a)(m, b) = (n + am, ab),$$

for all $n, m \in \mathbb{Z}/11\mathbb{Z}$ and $a, b \in (\mathbb{Z}/11\mathbb{Z})^*$.
Since $2^5 = 32$ is congruent to -1 modulo 11, 2 is a generator for $(\mathbb{Z}/11\mathbb{Z})^*$, so $e_1 = (1, 1)$ and $e_2 = (0, 2)$ are generators for G of order 11 and 10,

respectively. In particular, each element of G can be written as $(n, 2^\alpha)$, where $0 \le n \le 10$ and $0 \le \alpha \le 9$. Moreover, e_2 acts by conjugation on e_1 in the following way:

$$\begin{aligned} e_2 e_1 e_2^{-1} &= (0,2)(1,1)(0,2)^{-1} \\ &= (0,2)(1,1)(0,2^{-1}) \\ &= (2,2)(0,2^{-1}) \\ &= (2,1) \\ &= 2e_1. \end{aligned}$$

Any injective homomorphism into S_{11} must send e_1 to a permutation of order 11, which we can assume without loss of generality to be $\sigma = (1, 2, \ldots, 11)$, and it must send e_2 to a permutation τ of order 10. Additionally, the two permutations must satisfy $\tau\sigma\tau^{-1} = \sigma^2$. We write this condition as $(\tau(1), \tau(2), \ldots, \tau(11)) = (1, 3, 5, 7, 9, 11, 2, 4, 6, 8, 10)$.
The permutation

$$\begin{aligned} \tau &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ 1 & 3 & 5 & 7 & 9 & 11 & 2 & 4 & 6 & 8 & 10 \end{pmatrix} \\ &= (2, 3, 5, 9, 6, 11, 10, 8, 4, 7) \end{aligned}$$

has order 10 and satisfies the required condition. We now show that setting

$$G \ni (n, 2^\alpha) \overset{F}{\longmapsto} \sigma^n \tau^\alpha \in \mathsf{S}_{11}$$

yields an injective homomorphism.
First of all, F is well-defined, as σ has order 11 and τ has order 10. Moreover, $\tau\sigma = \sigma^2\tau$ so $\tau^\alpha\sigma = \sigma^{2^\alpha}\tau^\alpha$ and $\tau^\alpha\sigma^b = \sigma^{2^\alpha b}\tau^\alpha$. Therefore,

$$\begin{aligned} F(a, 2^\alpha)F(b, 2^\beta) &= \sigma^a \tau^\alpha \sigma^b \tau^\beta \\ &= \sigma^a \sigma^{2^\alpha b} \tau^\alpha \tau^\beta \\ &= \sigma^{a + 2^\alpha b} \tau^{\alpha+\beta} \\ &= F(a + 2^\alpha b, 2^{\alpha+\beta}) \\ &= F\big((a, 2^\alpha)(b, 2^\beta)\big), \end{aligned}$$

which proves that F is a homomorphism. Finally, the fact that $\sigma, \tau \in \mathrm{Im}(F)$ implies that 11, 10 divide $|\mathrm{Im}(F)|$, and so $|\mathrm{Im}(F)| = 11 \cdot 10$ and thus F is injective.

(ii) Suppose by contradiction that F is an injective homomorphism from $\mathbb{Z}/p\mathbb{Z} \times (\mathbb{Z}/p\mathbb{Z})^*$ to S_p. We must have that $\sigma = F(1, 0)$ is a p-cycle in S_p, so its centraliser $Z(\sigma)$ is $\langle\sigma\rangle$. In particular, $Z(\sigma)$ does not contain any elements of order $p - 1$. But, given a generator a of $(\mathbb{Z}/p\mathbb{Z})^*$, the element $\tau = F(0, a)$ must have order $p - 1$ and must commute with σ, that is, we must have $\tau \in Z(\sigma)$, which yields a contradiction.

61 First of all, we show that an automorphism φ has the required property if and only if $\varphi(H) = H$.

Indeed, if φ does have the property, then $H = 0+H = \widetilde{\varphi}(0+H) = \widetilde{\varphi}(h+H) = \varphi(h) + H$ for all $h \in H$, so $\varphi(h) \in H$ for all $h \in H$. Since $\varphi(H)$ and H are both finite of the same cardinality, this yields $\varphi(H) = H$.

Conversely, suppose $\varphi \in \mathrm{Aut}(G)$ is such that $\varphi(H) = H$. Then the map $g + H \longmapsto \varphi(g) + H$ is a well-defined homomorphism $\widetilde{\varphi}$ from G/H to itself. But, since φ is an automorphism, $\widetilde{\varphi}$ is also an automorphism.

We shall therefore count the automorphisms φ such that $\varphi(\mathbb{Z}/7\mathbb{Z} \times 0) = \mathbb{Z}/7\mathbb{Z} \times 0$. Such an automorphism φ must satisfy $\varphi(1, 0) = (u, 0)$ for some $u \neq 0$ in $\mathbb{Z}/7\mathbb{Z}$, and $\varphi(0, 1) = (v, c)$ for some $v, c \in \mathbb{Z}/7\mathbb{Z}$ such that $7 \nmid c$, because $\varphi(0, 1)$ must have order 49. It follows that $\varphi(n, a) = n\varphi(1, 0) + a\varphi(0, 1) = (nu + av, ac)$.

We now show that, for any u, v and c as above, the map $\varphi(n, a) = (nu + av, ac)$ is an automorphism of G; the fact that it satisfies the required property is obvious.

The map φ is well-defined, as $(n, a) = (m, b)$ implies $7 \mid n - m$ and $49 \mid a - b$, that is, $n = 7h + m$ and $a = 49k + b$ for some $h, k \in \mathbb{Z}$, so $(nu + av, ac) = ((7h + m)u + (49k + b)v, (49k + b)c) = (mu + bv, bc)$.

Moreover, we have $\varphi(n + m, a + b) = ((n + m)u + (a + b)v, (a + b)c) = (nu + av, ac) + (mu + bv, bc) = \varphi(n, a) + \varphi(m, b)$, so φ is a homomorphism.

Finally, φ is surjective: $\varphi(H) = H$ has order 7 and, letting K be the subgroup $0 \times \mathbb{Z}/49\mathbb{Z}$, the subgroup $\varphi(K)$ has order 49; moreover, $H \cap \varphi(K) = \{(0, 0)\}$. It follows that $\mathrm{Im}(\varphi)$ contains $H\varphi(K)$, which has cardinality $7 \cdot 49 = |G|$.

We can thus conclude that the automorphisms with the desired property are as many as the possible choices for u, v and c, that is, $6 \cdot 7 \cdot \varphi(49) = 6^2 \cdot 7^2$.

62

(i) Consider the map φ

$$\begin{aligned} \mathsf{S}_n &\longrightarrow \mathsf{S}_{n+2} \\ \sigma &\longmapsto \begin{cases} \sigma & \text{if } \sigma \text{ is even} \\ \sigma \circ (n+1, n+2) & \text{if } \sigma \text{ is odd,} \end{cases} \end{aligned}$$

where we interpret σ as an element of S_{n+2} that permutes $1, 2, \ldots, n$.
From the fact that every permutation $\sigma \in \mathsf{S}_n$ commutes with $(n + 1, n + 2)$ when seen as an element of S_{n+2}, it is easy to show that φ is a homomorphism. Moreover, φ is injective, as $\varphi(\sigma)_{|\{1,2,\ldots,n\}} = \sigma$. Finally, the image of φ is contained in A_{n+2} because of the way φ was constructed.
Now let G be a group of order n. By Cayley's Theorem, we know that S_n contains a subgroup H isomorphic to G; but then $\varphi(H)$ is a subgroup of A_{n+2} isomorphic to G.

(ii) In order to construct a subgroup of A_{35} isomorphic to D_{35}, we need to find elements $\rho, \sigma \in \mathsf{A}_{35}$ such that $\mathrm{ord}(\rho) = 35$, $\mathrm{ord}(\sigma) = 2$ and $\sigma\rho\sigma^{-1} = \rho^{-1}$.

Set $\rho = (1, 2, \ldots, 7)(8, 9, \ldots, 12)$, which is an element of order 35 in A_{35}. We have $\rho^{-1} = (7, 6, \ldots, 1)(12, 11, \ldots, 8)$, so the permutation

$$\sigma = (1, 7)(2, 6)(3, 5)(8, 12)(9, 11)(13, 14)$$

is even, has order 2 and satisfies $\sigma\rho\sigma^{-1} = \psi_\sigma(\rho) = \rho^{-1}$. The permutations ρ and σ have the required properties and generate a subgroup of A_{35} isomorphic to D_{35}.

63 In order to compute the cardinality of A, we must determine how many automorphisms φ of G can be constructed by imposing the condition $\varphi(0, 1) = (0, 1)$.

Any automorphism with this property is uniquely determined by the image of $(1, 0)$, which we will call (a, b). The order of $(1, 0)$ is p^2, so the order of (a, b) must also be p^2. This condition is equivalent to $a \not\equiv 0 \pmod{p}$, so we have $p(p - 1)$ possibilities for (a, b). We show that for each of these possible values of (a, b), the mapping $(1, 0) \longmapsto (a, b)$, $(0, 1) \longmapsto (0, 1)$ extends to an automorphism of G; that is, we show that $\varphi(x, y) = (ax, bx + y)$ is a well-defined, injective homomorphism, which implies it is also surjective.

If $x' \equiv x \pmod{p^2}$ and $y' \equiv y \pmod{p}$ then $(ax', bx' + y') = (ax, bx + y)$, so φ is well-defined. If we have $\varphi(x, y) = (0, 0)$, that is, $(ax, bx + y) = (0, 0)$, then $x = 0$ because a and p are coprime, and thus we also have $y = 0$. This shows that φ is injective.

The map φ is a homomorphism because it is well-defined and obtained by appropriately extending a map on the generators. We can thus conclude that $|A| = p(p - 1)$.

In order to compute the index of A in $\mathrm{Aut}(G)$, we proceed as follows. The group $\mathrm{Aut}(G)$ acts on G and the subgroup A is the stabiliser of the element $(0, 1)$ for this action. So the cardinality of the orbit of $(0, 1)$ is $[\mathrm{Aut}(G) : A] = |\mathrm{Aut}(G)|/|A|$.

Let us investigate the possible images of the element $(0, 1)$ via an automorphism of G. The order of $(0, 1)$ is p, so its image must also have order p. Remark that the subgroup $\langle (p, 0) \rangle$ is equal to $p \cdot G$ and is therefore characteristic; this implies that the image of $(0, 1)$ cannot have 0 as its second coordinate.

Consider the mapping $(1, 0) \longmapsto (1, 0)$, $(0, 1) \longmapsto (pa, b)$, where $b \neq 0$; we show it can be extended to an automorphism of G. Let $\varphi(x, y) = (x + pay, by)$.

First of all, we show that φ is well-defined in the same way as above. Moreover, φ is injective: if $(x + pay, by) = (0, 0)$ then $y = 0$ because $b \neq 0$, and so $x = 0$. Finally, φ is a homomorphism by construction.

The orbit of $(0, 1)$ therefore consists of all elements (pa, b) with $b \neq 0$, and has $p(p - 1)$ elements. In conclusion, $[\mathrm{Aut}(G) : A] = p(p - 1)$.

64 We wish to show that the two centralisers coincide if and only if m and ℓ are coprime.

Indeed, assume $(m, \ell) = 1$; then the order of σ^m is ℓ, so σ^m generates $\langle \sigma \rangle$, and therefore $Z(\sigma^m) = Z(\langle \sigma^m \rangle) = Z(\langle \sigma \rangle) = Z(\sigma)$.

Conversely, suppose $(m, \ell) = d > 1$. We then know that σ^m has a decomposition into disjoint cycles of the form $\tau_1 \cdots \tau_d$, where $\tau_1, \ldots, \tau_d$ have length ℓ/d. It follows that, for example, τ_1 commutes with σ^m; we will now show it does not commute with σ.

We know that elements in the centraliser of σ are of the form $\sigma^t \eta$, where t is a natural number and η a permutation disjoint from σ. If we had $\tau_1 = \sigma^t \eta$ then we would have $\eta = e$, as τ_1 only moves elements that are actually moved by σ; we would therefore have $\tau_1 = \sigma^t$ for some t. but the left-hand side consists of a single cycle, so t and ℓ must be relatively prime, or σ^t would consist in more than one cycle. But then σ^t is still an ℓ-cycle, so it can't be equal to the ℓ/d-cycle τ_1.

65 Let G be a p-group with a single subgroup of index p. First, we show that G is cyclic under the additional assumption that it is Abelian. Suppose by contradiction G is Abelian and not cyclic. Then, by the Structure Theorem for Finite Abelian Groups, G is isomorphic to a direct product $G_1 \times G_2$ for some nontrivial p-groups G_1, G_2. We know that a p-group has subgroups of all possible orders, so in particular G_1 and G_2 have subgroups of index p, which we denote by H_1 and H_2, respectively. But then $H_1 \times G_2$ and $G_1 \times H_2$ are distinct subgroups of index p in G. This concludes the proof in the Abelian case.

Now suppose that G is a non-Abelian p-group of minimal order among those with a single subgroup of index p; we show that the existence of G yields a contradiction. We know that the centre Z of G is nontrivial, so G/Z has order strictly less than the order of G and can't be the trivial group, as G is not Abelian. It follows that G/Z is a p-group, and thus has a subgroup K of index p.

Letting $\pi : G \longrightarrow G/Z$ be the quotient homomorphism, the subgroup $\pi^{-1}(K)$, which has index p in G, must be the only subgroup of index p in G. So $K = \pi(\pi^{-1}(K))$ is the only subgroup of index p in G/Z. But then G/Z also has the property from the problem statement and has order strictly less than the order of G, so the group G/Z is Abelian by minimality of G. This, however, implies by our previous argument that G/Z is cyclic and thus G must be Abelian, contradicting our assumptions.

66

(i) For all $\tau \in \mathsf{S}_n$, we have $\tau\langle\sigma\rangle\tau^{-1} = \langle\tau\sigma\tau^{-1}\rangle$; moreover, if σ is an n-cycle then $\tau\sigma\tau^{-1}$ is as well. This shows that S_n acts on X by conjugation.

(ii) We have to show that for every pair σ_1, σ_2 of n-cycles there is an element $\tau \in A_n$ such that $\tau\langle\sigma_1\rangle\tau^{-1} = \langle\sigma_2\rangle$.
We know that σ_1 and σ_2 are conjugate via some $\eta \in \mathsf{S}_n$, because they have the same cycle structure. If η is even, then $\tau = \eta$ will satisfy the required condition. Suppose η is odd instead.
The centraliser of the n-cycle σ_1 in S_n coincides with the subgroup generated by σ_1. It is therefore not contained in A_n, because n is even and σ_1 is odd. Consider now $\epsilon \in Z_{\mathsf{S}_n}(\sigma_1) \setminus \mathsf{A}_n$ and set $\tau = \eta\epsilon$.

The permutation τ is even, as both η and ϵ are odd. Moreover, $\tau\sigma_1\tau^{-1} = \eta\epsilon\sigma_1\epsilon^{-1}\eta^{-1} = \eta\sigma_1\eta^{-1} = \sigma_2$, so τ is an element of A_n that satisfies the required condition.

(iii) Let $\varphi : \mathsf{S}_n \longrightarrow \mathsf{S}(X)$ be the action of S_n on X by conjugation. The kernel of φ is a normal subgroup of S_n, so for $n \geq 5$ it can only be $\{e\}$, A_n or S_n. Consequently, if the kernel were nontrivial, there would be at most 2 elements in each orbit for the action of S_n on X.
However, we have seen in our proof of (i) that X is a single orbit for the action of S_n, and it contains $(n-1)!$ elements, which implies that the kernel is trivial for $n \geq 5$.

67 Let $N = \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}3\mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$ and consider the group $G = N \rtimes M$, where the semidirect product is taken with respect to the action of M on N given by $1 \cdot (a, b) = (-a, -b)$ for all $(a, b) \in N$. In the discussion that follows, we will identify N with the subgroup $N \times 0$ of G and M with the subgroup $(0, 0) \times M$ of G.

The subgroup M cannot be normal, as the action of M on N is nontrivial. On the other hand, N is a characteristic subgroup, because it is the only one of its order: if there were another subgroup N' of order 9, then the set $N \cdot N'$ would have at least $9 \cdot 9/3 > 18$ elements.

We now show that $L = \mathbb{Z}/3\mathbb{Z} \times 0 \times 0$ is a normal subgroup of G that is not characteristic. Clearly, N is contained in the normaliser of L, as N is Abelian and L is a subgroup of N. Moreover, $1 \cdot (a, 0) = (-a, 0)$, so M is also contained in the normaliser of L. It follows that all of G is contained in the normaliser, so L is normal.

In order to show that L is not characteristic, we first prove that the map $\varphi : (a, b, c) \longmapsto (b, a, c)$ is an automorphism of G. Indeed,

$$\begin{aligned}\varphi\big((a_1, b_1, c_1)(a_2, b_2, c_2)\big) &= \varphi\big(a_1 + (-1)^{c_1}a_2, b_1 + (-1)^{c_1}b_2, c_1 + c_2\big)\\ &= \big(b_1 + (-1)^{c_1}b_2, a_1 + (-1)^{c_1}a_2, c_1 + c_2\big)\\ &= (b_1, a_1, c_1)(b_2, a_2, c_2)\\ &= \varphi(a_1, b_1, c_1)\varphi(a_2, b_2, c_2).\end{aligned}$$

Since the automorphism φ does not send L to itself, L is not characteristic.

68 Let $\chi : G \longrightarrow \mathbb{C}^*$ be a homomorphism. Since $\mathbb{C}^*$ is an Abelian group, the derived group G' of G must be contained in $\mathrm{Ker}(\chi)$. In particular,

$$\begin{aligned}\big[(1, 1), (0, -1)\big] &= (1, 1)(0, -1)(1, 1)^{-1}(0, -1)^{-1}\\ &= (1, 1)(0, -1)(-1, 1)(0, -1)\\ &= (1, -1)(-1, -1)\\ &= (2, 1)\end{aligned}$$

belongs to the kernel of χ. But, since p is an odd prime, the element $(2, 1)$ generates the subgroup $H = \mathbb{Z}/p\mathbb{Z} \times 1$, so H is contained in the kernel of χ. This implies that the homomorphism χ passes to the quotient $G/H \simeq (\mathbb{Z}/p\mathbb{Z})^*$.

Conversely, any homomorphism from $(\mathbb{Z}/p\mathbb{Z})^*$ to $\mathbb{C}^*$ induces, by composition with the quotient homomorphism $G \longrightarrow G/H$, a homomorphism from G to $\mathbb{C}^*$. We have thus reduced the problem to that of describing all homomorphisms from the cyclic group $(\mathbb{Z}/p\mathbb{Z})^* \simeq \mathbb{Z}/(p-1)\mathbb{Z}$ to $\mathbb{C}^*$. It is clear that those are obtained by sending a generator of $\mathbb{Z}/(p-1)\mathbb{Z}$ to a $(p-1)$th root of unity in $\mathbb{C}$. In particular, there are $p-1$ such homomorphisms.

69

(i) We need to show that the map $\sigma \longmapsto f_\sigma$, where $f_\sigma(x) = \sigma \cdot x$, is a homomorphism from S_n to the group of permutations of Ω.
First of all, remark that if the coordinates of $x \in \mathbb{R}^n$ are all distinct, that is, if x is an element of Ω, then σx is also in Ω, as the coordinates of σx are the same as those of x, just permuted by σ. We have thus shown that f_σ is a map from Ω to Ω.
Fix $\sigma, \tau \in \mathsf{S}_n$ and $x = (x_1, \dots, x_n) \in \Omega$. Let $j_1, \dots, j_n$ be such that $\tau^{-1}(h) = j_h$ for $h = 1, \dots, n$. We then have

$$\begin{aligned}\sigma(\tau x) &= \sigma(x_{j_1}, \dots, x_{j_n}) \\ &= (x_{j_{\sigma^{-1}(1)}}, x_{j_{\sigma^{-1}(2)}}, \dots, x_{j_{\sigma^{-1}(n)}}).\end{aligned}$$

Now, since for $h = 1, \dots, n$ we have $j_{\sigma^{-1}(h)} = \tau^{-1}(\sigma^{-1}(h)) = (\sigma\tau)^{-1}(h)$, we obtain that $\sigma(\tau x) = (\sigma\tau)(x)$.
In particular, for all $\sigma \in \mathsf{S}_n$ we can deduce that $f_{\sigma^{-1}}$ is the inverse of f_σ, so f_σ is a permutation of Ω. This concludes the proof that the map $\sigma \longmapsto f_\sigma$ is an action.
⟦ The group S_n permutes vector coordinates, which is what forces the use of σ^{-1} in the definition. In a sense that could be made precise, this action is the *dual* of the permutation action on a basis of $\mathbb{R}^n$. ⟧

(ii) Note each element of Ω has a trivial stabiliser, as its coordinates are all distinct and thus it cannot be fixed by any permutation other than the identity. So every element of Ω has an orbit consisting of $|\mathsf{S}_n| = n!$ elements.

(iii) We show that

$$D = \left\{(x_1, x_2, \dots, x_n) \mid x_1 < x_2 < \cdots < x_n\right\} \subseteq \Omega$$

is a fundamental domain.
Indeed, remark that, given any element x of Ω, since its coordinates are distinct, we can rearrange them into increasing order via a permutation; this shows that there exists $\sigma \in \mathsf{S}_n$ such that $\sigma x \in D$.

Moreover, any element $\sigma \neq e$ of S_n will permute the coordinates of any $x \in D$ and send it to a vector whose coordinates are not in increasing order, that is, a vector not in D. This shows that elements of D are pairwise not conjugate.
Putting this two observations together yields that for each element of Ω there is a unique element of D that is conjugate to it; that is, D is a fundamental domain.

70

(i) By the First Homomorphism Theorem, K is isomorphic to a quotient G/H of G. Since K is infinite and its order must coincide with the index of H, we find that H has infinite index in G. But then, by the finite index property, H must be the trivial subgroup $\{e_G\}$, and K is thus isomorphic to G.

(ii) Given a subgroup $K \neq \{e_G\}$ of H, K is also a subgroup of G, so, by the finite index property, K has a finite number of cosets in G. Since every coset in H is also a coset in G, the cosets in H are also finitely many.
This proves that H has the finite index property.

(iii) Clearly, every finite group has the finite index property. Assume G has infinite order; we will show that G is isomorphic to $\mathbb{Z}$.
Any element of G having finite order generates a finite subgroup of G, which has infinite index in G. This implies that there are no elements of finite order G, save for the neutral element.
Given $x \in G \setminus \{0\}$, define the integer i_x as the index $[G : \langle x \rangle]$, which we know to be finite. We consider $x \in G$ such that the index i_x is minimal. Suppose by contradiction that $i_x > 1$ and construct $y \notin \langle x \rangle$. Remark that $G/\langle x \rangle$ is a finite group and, letting m be the order of $y + \langle x \rangle$ in $G/\langle x \rangle$, we have $my = nx$ for some integer n. Necessarily $(m, n) = 1$, because if there were $d > 1$ such that $d \mid (m, n)$, then we would have

$$d\left(\frac{m}{d}x - \frac{n}{d}y\right) = 0,$$

that is, the element $mx/d - ny/d \neq 0$ would have finite order. But then we can find integers s, t such that $ms + nt = 1$.
Letting $z = sx + ty$, it is easy to check that $mz = x$ and therefore, using the fact that $m > 1$ and that x has infinite order, the subgroup generated by z is a proper subgroup of the one generated by x, that is, $i_z < i_x$, which is a contradiction. It follows that $i_x = 1$ and thus that G is cyclic, so isomorphic to $\mathbb{Z}$.

71

(i) Remark that σ^2 is an even permutation, so no σ can be such that σ^2 is equal to $(1, 2)$, which is an odd permutation; hence the equation has no solutions.

(ii) The fact that $\sigma^4 = e$ implies that the permutation σ is a product of disjoint 4-cycles and 2-cycles. Moreover, each 4-cycle becomes a pair of disjoint 2-cycles when squared, so σ can have, and in fact must have, a single 4-cycle.

There are only two possibilities for the 4-cycle in σ, namely, $\tau_1 = (1, 3, 2, 4)$ and $\tau_2 = (1, 4, 2, 3)$. We can then multiply either τ_1 or τ_2 by any number of 2-cycles, provided they are mutually disjoint and disjoint from the chosen 4-cycle; but, since σ is an element of S_{10}, the maximum number of 2-cycles that can fit is actually three.

We now count the solutions according to how many 2-cycles appear in the decomposition of σ: we have the two solutions τ_1, τ_2, with no additional 2-cycles; we obtain $2 \cdot \binom{6}{2} = 30$ solutions by adding a single 2-cycle; we obtain $2 \cdot \binom{6}{2} \cdot \binom{4}{2} \cdot \frac{1}{2} = 90$ by adding two, and finally $2 \cdot \binom{6}{2} \cdot \binom{4}{2} \cdot \binom{2}{2} \cdot \frac{1}{6} = 30$ by adding three.

In total, we have 152 solutions.

72

(i) The order of an element (x, y) in G is the least common multiple of the orders of x in $\mathbb{Z}/3\mathbb{Z}$ and y in $\mathbb{Z}/15\mathbb{Z}$; the values it can take are thus 1, 3, 5 and 15. Clearly, $(0, 0)$ is the only element of order 1.

An element $(x, y) \neq (0, 0)$ has order 3 if and only if y belongs to the unique subgroup of order 3 in $\mathbb{Z}/15\mathbb{Z}$, so there are $3 \cdot 3 - 1 = 8$ elements of order 3 in G.

An element of order 5 is of the form $(0, y)$, where $y \neq 0$ has order 5 in $\mathbb{Z}/15\mathbb{Z}$, and thus there are 4 such elements.

Finally, an element (x, y) has order 15 if and only if one of the following holds: y has order 15 in $\mathbb{Z}/15\mathbb{Z}$, or $x \neq 0$ and y has order 5 in $\mathbb{Z}/5\mathbb{Z}$. There are $3 \cdot \varphi(15) = 24$ elements of the first type and $2 \cdot 4 = 8$ of the second, so 32 elements of order 15 in G.

(ii) Let $\varphi : G \longrightarrow \mathbb{Z}/10\mathbb{Z}$ be a homomorphism and set $e_1 = (1, 0)$, $e_2 = (0, 1)$. Note that $\operatorname{ord}(e_1) = 3$ and $\operatorname{ord}(e_2) = 15$, so, since for all g in G we must have $o(\varphi(g)) \mid (\operatorname{ord}(g), 10)$, it follows that $\operatorname{ord}(\varphi(e_1)) = 1$ and either $\operatorname{ord}(\varphi(e_2)) = 1$ or $\operatorname{ord}(\varphi(e_2)) = 5$.

In the first case, we have $\varphi(e_1) = \varphi(e_2) = 0$, so φ is the null homomorphism. For the second case, remark that $\mathbb{Z}/10\mathbb{Z}$ has four elements of order 5, namely, the residue classes modulo 10 of the form $2n$, where $1 \leq n \leq 4$; so we have $\varphi(e_2) = 2n$ for some $1 \leq n \leq 4$. From the fact that φ is a homomorphism we deduce that we must have $\varphi(a, b) = a\varphi(e_1) + b\varphi(e_2) = 2bn$.

It is straightforward to show that for each n such a map φ is indeed a homomorphism:

$$\begin{aligned} \varphi\big((a, b) + (c, d)\big) &= \varphi(a + c, b + d) \\ &= 2n(b + d) \\ &= 2nb + 2nd \\ &= \varphi(a, b) + \varphi(c, d). \end{aligned}$$

We have thus determined all possible homomorphisms from G to $\mathbb{Z}/10\mathbb{Z}$.

73 Let G be the group in the problem statement, and let $f(x) = ax + b$ be an element of G; first, we compute the cardinality of the centraliser $Z(f)$ of f in G.

Given $g = cx + d \in Z(f)$, since $(f \circ g)(x) = a(cx + d) + b = acx + ad + b$ and $(g \circ f)(x) = c(ax + b) + d = acx + bc + d$, we must have $ad + b = bc + d$, that is, $(1 - a)d = (1 - c)b$ in $\mathbb{Z}/7\mathbb{Z}$. We consider several cases:
① If $a = 1$ and $b = 0$ then $Z(f) = G$. ② If $a = 1$ and $b \neq 0$ then $Z(f) = \{x + d \mid d \in \mathbb{Z}/7\mathbb{Z}\}$. ③ Finally, if $a \neq 1$ then $Z(f) = \{cx + (1-a)^{-1}(1-c)b \mid c \in (\mathbb{Z}/7\mathbb{Z})^*\}$.

The cardinality of the conjugacy class of f in G is given by the index of the centraliser of f, and thus there is a single conjugacy class consisting of exactly one element. Elements of the form $x + b$ with $b \neq 0$ are all pairwise conjugate; this follows from the fact that they are the only elements whose centraliser has cardinality 7, that the cardinality of the centraliser is invariant under conjugation, and that there are $6 = 42/7$ of them.

But then elements of the form $ax + b$ with $a \neq 1$ must also have conjugates on the same form; there are 35 of them and they must be grouped into conjugacy classes with $42/6 = 7$ elements, so they form 5 separate conjugacy classes.

In conclusion, G has $1 + 1 + 5 = 7$ conjugacy classes.

74

(i) Let n_5 be the number of 5-Sylow subgroups of a group G of order $5^2 \cdot 13$. We know n_5 must be congruent to 1 modulo 5 and must divide 13, so we must have $n_5 = 1$. Similarly, let n_{13} be the number of 13-Sylow subgroups; n_{13} must be congruent to 1 modulo 13 and divide 5^2, so we also have $n_{13} = 1$.
Given a 5-Sylow subgroup H and a 13-Sylow subgroup K, both H and K are normal in G and they have order 5^2 and 13, respectively, so $G \simeq H \times K$.
Moreover, H is Abelian, as its order is a prime number squared, and K is also Abelian because it is cyclic. In conclusion, G is Abelian and therefore can only be one of two groups, namely, $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/13\mathbb{Z}$ and $\mathbb{Z}/25\mathbb{Z} \times \mathbb{Z}/13\mathbb{Z}$.

(ii) Remark that $\mathrm{Aut}(\mathbb{Z}/11\mathbb{Z})$ is isomorphic to $(\mathbb{Z}/11\mathbb{Z})^*$, which is a cyclic group with 10 elements. The element 4 has order 5 in $(\mathbb{Z}/11\mathbb{Z})^*$, as $4^5 = 2^{10} \equiv 1 \pmod{11}$, so the automorphism $\mathbb{Z}/11\mathbb{Z} \ni a \overset{\alpha}{\longmapsto} 4a \in \mathbb{Z}/11\mathbb{Z}$ has order 5 in $\mathrm{Aut}(\mathbb{Z}/11\mathbb{Z})$.
It follows that the map $\mathbb{Z}/5\mathbb{Z} \ni b \overset{\varphi}{\longmapsto} \alpha^b \in \mathrm{Aut}(\mathbb{Z}/11\mathbb{Z})$ is well-defined and nontrivial, and so the group $(\mathbb{Z}/11\mathbb{Z} \rtimes_\varphi \mathbb{Z}/5\mathbb{Z}) \times \mathbb{Z}/5\mathbb{Z}$ is not Abelian and has order $5^2 \cdot 11$.

75 We show that for any $\varphi \in \mathrm{Aut}(\mathsf{D}_8)$ the map F_φ is a permutation of X. Indeed, letting $C \in X$ be the conjugacy class of some element $x \in \mathsf{D}_8$, we have $\varphi(C) = \{\varphi(g)\varphi(x)\varphi(g)^{-1} \mid g \in \mathsf{D}_8\}$; since φ is surjective, $\varphi(C)$ is the conjugacy class of $\varphi(x)$. Moreover, again because φ is surjective, the map F_φ is also surjective, and similarly the fact that φ is injective implies that F_φ is injective.

Now let us show that F defines an action, that is, that F is a homomorphism. Given automorphisms $\varphi,\ \psi$ of D_8 and given a conjugacy class C, we have

$$\begin{aligned} F_{\varphi\circ\psi}(C) &= (\varphi\circ\psi)(C) \\ &= \varphi(\psi(C)) \\ &= F_\varphi(F_\psi(C)) \\ &= (F_\varphi\circ F_\psi)(C). \end{aligned}$$

With the usual presentation of D_8 as generated by $r,\ s$ with relations $r^8 = s^2 = e$ and $sr = r^{-1}s$, the conjugacy classes of D_8 are

$$\begin{aligned} R_0 &= \{e\}, \\ R_1 &= \{r, r^{-1}\}, \\ R_2 &= \{r^2, r^{-2}\}, \\ R_3 &= \{r^3, r^{-3}\}, \\ R_4 &= \{r^4\}, \\ S_0 &= \{s, sr^2, sr^4, sr^{-2}\}, \\ S_1 &= \{sr, sr^3, sr^{-3}, sr^{-1}\}. \end{aligned}$$

Any automorphism must send R_0 to itself, as it fixes the neutral element. But then it sends R_4 to R_4, as R_4 is the only other conjugacy class of cardinality one. Moreover, R_2 must also be sent to itself, as it can be characterised as the set of elements of order 4. As for R_1 and R_3, which consist of all elements of order 8, those can be exchanged via the automorphism that maps $r \longmapsto r^3$, $s \longmapsto s$. Similarly, S_0 and S_1, consisting of all elements of order 2 outside the centre $\{e, r^4\}$, can be exchanged via the automorphism mapping $r \longmapsto r$, $r \longmapsto rs$.

In conclusion, the relevant action of $\mathrm{Aut}(\mathsf{D}_8)$ on X has the following orbits: $\{R_0\}$, $\{R_2\}$, $\{R_4\}$,$\{R_1, R_3\}$,$\{S_0, S_1\}$.

▌The results obtained in the first part of the solution hold for any group, and not just for D_8: the same proof applies in general.▐

76 We intend to show that the primes with the required property are 2, 3, 5, and all primes congruent to 1 modulo 5.

Clearly, we always have at least the two non-isomorphic Abelian groups $\mathbb{Z}/5\mathbb{Z}\times\mathbb{Z}/5\mathbb{Z}\times\mathbb{Z}/p\mathbb{Z}$ and $\mathbb{Z}/25\mathbb{Z}\times\mathbb{Z}/p\mathbb{Z}$ that have order $25p$. If we could construct a non-Abelian group of the same order, then p would have the required property.

Let us begin by making a remark that will be used several times. Given a group G and a positive divisor n of the order of $\mathrm{Aut}(G)$, by Cauchy's theorem there is an element φ of $\mathrm{Aut}(G)$ having order n, and we can construct the semidirect product $G\rtimes\mathbb{Z}/n\mathbb{Z}$ with respect to the homomorphism that extends $1 \longmapsto \varphi$. Since this homomorphism is nontrivial, the semidirect product is not Abelian.

Because $\mathrm{Aut}(\mathbb{Z}/5\mathbb{Z}\times\mathbb{Z}/5\mathbb{Z}) \simeq \mathsf{GL}_2(\mathbb{F}_5)$ has cardinality $(5^2-1)(5^2-5) = 2^5\cdot 3\cdot 5$, by the previous remark the primes 2, 3 and 5 have the required property. On the other hand, if $p \equiv 1 \pmod 5$ then 5 divides $p-1$, which is the order

of $\mathrm{Aut}(\mathbb{Z}/p\mathbb{Z}) \simeq (\mathbb{Z}/p\mathbb{Z})^*$. So we can construct the non-Abelian group $\mathbb{Z}/5\mathbb{Z} \times (\mathbb{Z}/p\mathbb{Z} \rtimes \mathbb{Z}/5\mathbb{Z})$ and thus p still has the required property.

Now let G be any group of order $25p$; we wish to show that if p is not 2, 3 or 5 and is not congruent to 1 modulo 5, then G is Abelian. This establishes our initial claim, as the only Abelian groups of order $25p$ are, for $p \neq 5$, the two listed in the beginning of the solution.

Let n_5 be the number of 5-Sylow subgroups in G. We have $n_5 \equiv 1 \pmod 5$ and n_5 divides p, so either $n_5 = 1$ or $n_5 = p$. The second case would imply $p \equiv 1 \pmod 5$, which we have excluded, so the only possibility is $n_5 = 1$. We have thus shown that there is only one subgroup H of order 25 in G, and in particular H is normal.

Let n_p be the number of p-Sylow subgroups of G. We have $n_p \equiv 1 \pmod p$ and n_p divides 25, so the possible values for n_p are 1, 5 and 25. If $n_p = 5$ then $5 \equiv 1 \pmod p$ yields $p = 2$, which we have excluded; analogously, if $n_p = 25$ then $25 \equiv 1 \pmod p$ yields $p = 2$ or $p = 3$, both excluded. The remaining case is $n_p = 1$: there exists a single subgroup K of order p in G, and in particular K is normal.

It follows that $G \simeq H \times K$, and since $|H| = 25$ and $|K| = p$, both H and K are Abelian groups. We conclude that G is Abelian, as claimed.

77 The group $\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is Abelian, so every homomorphism φ from A_4 to this group induces a homomorphism from the Abelianisation $\mathsf{A}_4/\mathsf{A}_4'$, where A_4' is the derived subgroup of A_4.

Remark that

$$\big[(1,2,3),(1,2,4)\big] = (1,2,3)(1,2,4)(1,2,3)^{-1}(1,2,4)^{-1} = (1,2)(3,4) \in \mathsf{A}_4';$$

since the derived subgroup is normal, this implies that all $2+2$-cycles belong to A_4', as they are pairwise conjugate in A_4. Thus, A_4' contains the Klein subgroup

$$K = \{e, (1,2)(3,4), (1,3)(2,4), (1,4)(2,3)\};$$

moreover, the quotient A_4/K has cardinality 3 and is therefore isomorphic to $\mathbb{Z}/3\mathbb{Z}$, which is Abelian. This shows that we also have the opposite inclusion of A_4' in K, so in fact $\mathsf{A}_4' = K$.

Consequently, every homomorphism $\varphi : \mathsf{A}_4 \longrightarrow \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ passes to the quotient $\overline{\varphi} : \mathsf{A}_4/K \simeq \mathbb{Z}/3\mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Finally, note that the cycle $(1,2,3)$ is not in K, so we can pick $(1,2,3)K$ as a generator of the quotient A_4/K.

The element $(1,2,3)K$ has order 3 in A_4/K, so $\varphi(1,2,3) = \overline{\varphi}((1,2,3)K)$ must have order 1 or 3 in $\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, that is, $\varphi(1,2,3) = (2a, 0)$ for some $a \in \{0,1,2\}$. We thus have 3 homomorphisms in total: they are determined by the image of $(1,2,3)$ and clearly send K to $(0,0)$.

78 Let G be a group of order 52. If G is Abelian then it is one of the two groups $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/13\mathbb{Z} \simeq \mathbb{Z}/52\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/13\mathbb{Z}$, up to isomorphism. From now on, assume G is not Abelian.

Let n be the number of 13-Sylow subgroups of G; we have $n \equiv 1 \pmod{13}$ and n must divide 4, so the only possible value for it is $n = 1$, that is, there is a unique 13-Sylow subgroup K, which is normal. Naturally, $K \simeq \mathbb{Z}/13\mathbb{Z}$.

Now assume there are no elements of order 4 in G and let H be some fixed 2-Sylow subgroup; the group H must be isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Given that $K \cap H = e$ and $|H||K| = 52$, the group G is isomorphic to a semidirect product $K \rtimes H \simeq \mathbb{Z}/13\mathbb{Z} \rtimes_\varphi (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$. Since we are assuming that G is not Abelian, the map $\varphi : \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/13\mathbb{Z}) \simeq (\mathbb{Z}/13\mathbb{Z})^* \simeq \mathbb{Z}/12\mathbb{Z}$ defining the semidirect product is nontrivial, that is, it does not send all elements to $0 \in \mathbb{Z}/12\mathbb{Z}$.

The only element of order 2 in $\mathbb{Z}/12\mathbb{Z}$ is 6; we therefore have $\varphi(1, 0) = 6$, or $\varphi(0, 1) = 6$, or both. In any case, the kernel of φ is nontrivial (if $\varphi(1, 0) = 6$ and $\varphi(0, 1) = 6$, then $\varphi(1, 1) = 6 + 6 = 0$). The group G is thus isomorphic to $(K \rtimes \mathbb{Z}/2\mathbb{Z}) \times \mathbb{Z}/2\mathbb{Z} \simeq \mathsf{D}_{13} \times \mathbb{Z}/2\mathbb{Z}$, where the last isomorphism is because of the fact that the semidirect product is uniquely determined, as $1 \in \mathbb{Z}/2\mathbb{Z}$ must be sent to $6 \in \mathbb{Z}/12\mathbb{Z}$. We have thus obtained that there is a unique non-Abelian group G of order 52 with no elements of order 4, up to isomorphism.

Now assume G has an element of order 4. The 2-Sylow subgroup H is then isomorphic to $\mathbb{Z}/4\mathbb{Z}$, so this time G is the semidirect product $K \rtimes H \simeq \mathbb{Z}/13\mathbb{Z} \rtimes_\varphi \mathbb{Z}/4\mathbb{Z}$, where $1 \in \mathbb{Z}/4\mathbb{Z}$ can be sent to 3, 6 or 9 in $\mathbb{Z}/12\mathbb{Z} \simeq \mathrm{Aut}(\mathbb{Z}/13\mathbb{Z})$ via the map φ that defines the semidirect product.

The choices of 3 and $9 = -3$ yield isomorphic groups, as we can see by exchanging h with h^{-1} in H. We shall therefore only consider the cases corresponding to 3 and 6.

We want to show that the two possibilities yield groups that are not isomorphic. Now, an element $(e, h) \in K \rtimes_\varphi H = G$ belongs to the centre of G if and only if $h \in \mathrm{Ker}(\varphi)$. But if $\varphi(1) = 3$ the kernel is trivial, whereas if $\varphi(1) = 6$ the element 2 of $\mathbb{Z}/4\mathbb{Z}$ belongs to the kernel.

In conclusion, there are 5 isomorphism classes of groups with 52 elements.

79 We shall first consider cyclic subgroups of order 6. These are generated by an element of order 6, so by a permutation whose cycle structure is $(a, b, c)(d, e)$. There are $\binom{5}{3} \cdot 2 = 20$ permutations of this form, all pairwise conjugate; consequently, the subgroups they generate are also pairwise conjugate. Since every cyclic subgroup of order 6 contains two permutations of this form, the conjugacy class of all cyclic subgroups of order 6 contains $20/2 = 10$ elements.

Subgroups that are isomorphic to S_3 contain a permutation of order 3, that is, a cycle (a, b, c), and three permutations of order 2, which must belong to the normaliser of (a, b, c), but not to its centraliser, otherwise they would generate a cyclic group, which would thus be Abelian. There are two possible choices for the three permutations:

1. (a, b), $(a, b)(a, b, c) = (b, c)$ and $(a, b)(a, c, b) = (a, c)$;
2. $(a, b)(d, e)$, $(a, b)(d, e)(a, b, c) = (b, c)(d, e)$ and $(a, b)(d, e)(a, c, b) = (a, c)(d, e)$.

Case (1) yields the subgroup that permutes a, b, c while fixing d, e; there are $\binom{5}{3} = 10$ such subgroups, all pairwise conjugate.

Case (2) still yields subgroups isomorphic to S_3, but these do not fix any elements. Again there are 10 subgroups of this type, one for each choice of the set $\{a, b, c\}$, and they are all pairwise conjugate. Indeed, given

$$\sigma = \begin{pmatrix} a & b & c & d & e \\ a' & b' & c' & d' & e' \end{pmatrix}$$

we have $\sigma\langle(a, b, c), (a, b)(d, e)\rangle\sigma^{-1} = \langle(a', b', c'), (a', b')(d', e')\rangle$.

In conclusion, there are $10 + 10 + 10 = 30$ subgroups of order 6 partitioned into three conjugacy classes, each with 10 elements.

80

(i) Assume by contradiction that the cardinality of the centre is q, that is, that the centre is the q-Sylow subgroup Q, which is therefore unique and normal. Now let Z_P be the centre of the p-Sylow subgroup P, which we know to be nontrivial. Each element in Z_P commutes with all elements in P and with all elements in Q, so with $PQ = G$, that is, Z_P is contained in the centre of G. But this is a contradiction, as we have assumed the centre to have cardinality q, whereas the order of Z_p is a power of p and cannot divide q.

(ii) If $p > q$, then $q \not\equiv 1 \pmod{p}$, so there is a unique p-Sylow subgroup, as by the Sylow Theorems the number of such subgroups must be a divisor of q and also be congruent to 1 modulo p. Naturally, being unique, the p-Sylow subgroup is normal, so in this case G is not simple.
Now assume $p < q$ and let Q be a q-Sylow subgroup. If Q is not a normal subgroup of G, then the number n of subgroups that are conjugate to Q must be a divisor of p^3 different from 1 and must also be congruent to 1 modulo q. We cannot have $n = p$, as $p < q + 1$, and $q + 1$ is the smallest number greater than 1 and congruent to 1 modulo q.
We also cannot have $n = p^2$, as $p^2 \equiv 1 \pmod{q}$ would imply $q \mid p^2 - 1 = (p - 1)(p + 1)$, rendered impossible by the condition $q > p + 1$.
Finally, suppose $n = p^3$. In this case we have p^3 distinct subgroups of order q which have pairwise trivial intersections. These provide a set X of $p^3(q-1) = p^3q - p^3$ elements of order q. It follows that every p-Sylow subgroup must be contained in $G \setminus X$; but $|G \setminus X| = p^3$ is also the cardinality of any p-Sylow subgroup, which must then coincide with $G \setminus X$. Therefore, there exists a unique p-Sylow subgroup, which must be normal: again, G is not simple.

81

(i) The element $r = (1, 2, 3)(4, 5, 6, 7)$ of S_7 has order 12, while the element $s = (1, 2)(4, 7)(5, 6)$ of S_7 has order 2 and is such that $sr = r^{-1}s$. It follows that the subgroup $\langle r, s\rangle$ of S_7 is isomorphic to D_{12}.

(ii) A subgroup isomorphic to D_{12} must contain an element of order 12. The elements of order 12 in S_7 must be products of two mutually disjoint cycles,

one of length 3 and one of length 4: this is the only possibility, given that the order of a permutation is the least common multiple of the lengths of the cycles appearing in its decomposition into disjoint cycles, which implies our permutation must have a cycle whose length is a multiple of 3 and one whose length is a multiple of 4.
However, any such permutation is odd, so there are no subgroups of A_7 that are isomorphic to D_{12}.

82 For all $a \in \mathbb{Z}/3\mathbb{Z}$, φ_a induces a permutation of the elements of $\mathcal{H}$: indeed, an automorphism will send a subgroup of order 7 to another subgroup of order 7; the induced map is injective because the automorphism is injective, and it is therefore surjective because $\mathcal{H}$ is finite.

Note that every $H \in \mathcal{H}$ is cyclic; setting $\sigma = (1, 2, 3) \in \mathsf{S}_3$ and letting $H = \langle (x_1, x_2, x_3) \rangle$, for all $a, b \in \mathbb{Z}/3\mathbb{Z}$ we have

$$\begin{aligned}
\varphi_{a+b}(H) &= \varphi_{a+b}\big(\langle (x_1, x_2, x_3) \rangle\big) \\
&= \langle \big(x_{\sigma^{a+b}(1)}, x_{\sigma^{a+b}(2)}, x_{\sigma^{a+b}(3)}\big) \rangle \\
&= \varphi_a\big(\langle x_{\sigma^b(1)}, x_{\sigma^b(2)}, x_{\sigma^b(3)} \rangle\big) \\
&= (\varphi_a \circ \varphi_b)\big(\langle (x_1, x_2, x_3) \rangle\big) \\
&= (\varphi_a \circ \varphi_b)(H),
\end{aligned}$$

so φ induces a homomorphism from $\mathbb{Z}/3\mathbb{Z}$ to $\mathsf{S}(\mathcal{H})$. This concludes the proof that we have constructed an action on $\mathcal{H}$.

Every subgroup of order 7 has exactly $\varphi(7) = 6$ elements of order 7, and any two distinct such subgroups intersect trivially; there are thus $(7^3 - 1)/(7 - 1) = 57$ elements in $\mathcal{H}$. The acting group has order 3, so each orbit has either 1 or 3 elements.

An orbit contains a single element if and only if $\langle (x, y, z) \rangle = \langle \varphi_1(x, y, z) \rangle$, that is, if and only if $(y, z, x) = a(x, y, z)$ for some $a \in (\mathbb{Z}/7\mathbb{Z})^*$. But since $\varphi_1^3 = \mathrm{Id}_{\mathcal{H}}$, this implies $a^3 = 1$, that is, a must be 1, 2 or 4. These three possibilities yield the subgroups $\langle (1, 1, 1) \rangle$, $\langle (1, 2, 4) \rangle$, $\langle (1, 4, 2) \rangle$. The other 54 subgroups are partitioned into 18 orbits, each with 3 elements, so the total number of orbits is $3 + 18 = 21$.
▌Since $(\mathbb{Z}/7\mathbb{Z})^3$ is a vector space of dimension 3 on $\mathbb{F}_7$, what the problem is asking us to investigate is the action induced by the permutations of coordinates on the lines in this vector space. The linear map φ_1 is a rotation with axis $\mathbb{F}_7 \cdot (1, 1, 1)$, which has 3 poles that correspond to third roots of unity in $\mathbb{F}_7$.▐

83

(i) We show that $Z(G) = \big\{((a, a, a), e) \mid a \in \mathbb{Z}/p\mathbb{Z}\big\}$.
First of all observe that, if $((x_1, x_2, x_3), \sigma)$ belongs to the centre of G, then σ is in the centre of S_3, as the second coordinate of the product of two elements is equal to the product of the elements' second coordinates, by definition of the semidirect product. We thus have that σ is the neutral element, because the centre of S_3 is trivial.

Moreover,

$$((x_1, x_2, x_3), e) \cdot ((0, 0, 0), \sigma) = ((x_1, x_2, x_3), \sigma),$$

whereas

$$((0, 0, 0), \sigma) \cdot ((x_1, x_2, x_3), e) = ((x_{\sigma^{-1}(1)}, x_{\sigma^{-1}(2)}, x_{\sigma^{-1}(3)}), \sigma),$$

so if $((x_1, x_2, x_3), e)$ belongs to the centre of G then we must have $x_1 = x_2 = x_3$. Conversely, elements of the form $((a, a, a), e)$ do commute both with elements of the form $((x_1, x_2, x_3), e)$, because $(\mathbb{Z}/p\mathbb{Z})^3$ is Abelian, and with elements of the form $((0, 0, 0), \sigma)$ because of the computation above, so our claim is proven.

(ii) We identify $(\mathbb{Z}/p\mathbb{Z})^3$ with the subgroup $(\mathbb{Z}/p\mathbb{Z})^3 \times e$ and remark that the two obvious subgroups, that is, the trivial group and all of $(\mathbb{Z}/p\mathbb{Z})^3$, are certainly normal.
The group $(\mathbb{Z}/p\mathbb{Z})^3$ is a vector space of dimension 3 on $\mathbb{F}_p$, so its subgroups are its vector subspaces. A subgroup of G is normal if and only if, in its interpretation as a vector subspace of $(\mathbb{Z}/p\mathbb{Z})^3$, it is invariant under permutations of the coordinates.
In particular, a subgroup of order p, which corresponds to a line on $\mathbb{F}_p$, is generated by a nonzero vector (x_1, x_2, x_3), and thus is normal if and only if $x_1 = x_2 = x_3$. This case yields the centre of G.
We now consider a subgroup of order p^2, that is, a subspace of dimension 2. Its elements are the solutions of a nontrivial equation of the form $ax_1 + bx_2 + cx_3 = 0$. If the subgroup is normal, then this equation must be invariant under permutations of the coordinates, so it must be equivalent to $x_1 + x_2 + x_3 = 0$. This yields a single normal subgroup of order p^2 in G, which is $\{((x_1, x_2, x_3), e) \mid x_1 + x_2 + x_3 = 0\}$.

84

(i) Let G' be the group with the given presentation. The mapping $\widetilde{x} \longmapsto x$, $\widetilde{y} \longmapsto y$ extends to a homomorphism from G' to G, as the relations defining G' are satisfied by x and y in G. This homomorphism is surjective, as the generators x and y of G belong to its image. Moreover, it is clear that every element of G' can be expressed, using the relations from the presentation, as $\widetilde{x}^a \widetilde{y}^b$ with $0 \leq a \leq 6$ and $0 \leq b \leq 2$; so G' has at most 21 elements. But G has exactly 21 elements and thus the existence of a surjective homomorphism $G' \longrightarrow G$ implies that G' also has exactly 21 elements; the homomorphism is thus injective, and in fact an isomorphism.

(ii) By the Sylow Theorems, the number of 7-Sylow subgroups of G is congruent to 1 modulo 7 and divides $21/7 = 3$; there is thus a single 7-Sylow subgroup, which clearly is the subgroup generated x, and is thus characteristic. An element $x^a y^b$ has order 1 if $a = b = 0$ and order 7 if $a \neq 0$ and $b = 0$; otherwise, that is, if $b \neq 0$, it has order 3.

Now let $\overline{x} = x^u$ for some $1 \leq u \leq 6$, and $\overline{y} = x^a y^b$ for some $0 \leq a \leq 6$ and $1 \leq b \leq 2$. The conditions on u, a and b are equivalent to setting the order of $\overline{x}$ to 7 and the order of $\overline{y}$ to 3. In order for the mapping $x \longmapsto \overline{x}$, $y \longmapsto \overline{y}$ to extend to a homomorphism of G to itself, a necessary and sufficient condition is that, additionally, $\overline{y}\,\overline{x} = \overline{x}^2\,\overline{y}$. This yields

$$\overline{y}\,\overline{x} = x^a y^b x^u = x^{a+2^b u} y^b$$
$$\overline{x}^2\,\overline{y} = x^{a+2u} y^b$$

and thus the condition that $b \equiv 1 \pmod 3$, as 2 has order 3 in $(\mathbb{Z}/7\mathbb{Z})^*$; consequently, $b = 1$.

Remark that every homomorphism constructed as above is actually surjective, as the image contains the element $\overline{x}$ of order 7 and the element $\overline{y}$ of order 3, so has order divisible by 21, and therefore is G itself. We have thus constructed all possible automorphisms, which implies $|\operatorname{Aut}(G)| = 6 \cdot 7 = 42$, because there are 6 possible choices for u, 7 choices for a and one choice for b.

(iii) As shown above, the subgroup $\langle x \rangle$ is characteristic and there is a homomorphism $\operatorname{Aut}(H) \ni \varphi \longmapsto \varphi_{|\langle x \rangle} \in \operatorname{Aut}(\langle x \rangle)$. The kernel of this homomorphism is the subgroup $\{\varphi \in \operatorname{Aut}(G) \mid \varphi(x) = x\}$, which is thus a normal subgroup. Its order is 7 because its elements are precisely the automorphisms listed in (ii) such that $u = 1$.

(iv) Not all automorphisms are inner automorphisms, as $42 = |\operatorname{Aut}(G)| > |G| = 21$ whereas $\operatorname{Inn}(G)$ is a quotient of G.

The group G in the problem statement is isomorphic to a non-Abelian semidirect product $\mathbb{Z}/7\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$; this explicit construction of the group could be used to provide an alternative solution that is essentially equivalent to the one given.

85

(i) The order of G is 16, so its subgroups have order 1, 2, 4, 8 or 16; clearly, $\{(0,0)\}$ is the only subgroup of order 1 and G is the only subgroup of order 16. In what follows, we will repeatedly use the Structure Theorem for Finite Abelian Groups in order to categorise subgroups according to their isomorphism class.

There are as many subgroups of order 2 as elements of order 2, that is, elements of $4\mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ other than $(0,0)$; there are thus 3 such subgroups.

A subgroup of order 4 is either cyclic or isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. To obtain the number of subgroups of the first type, first we count elements of order 4: they are contained in $2\mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, so there are 4 of them; we then divide the result by $\varphi(4) = 2$, as each cyclic group of order 4 contains 2 elements of order 4. We have found that there are 2 cyclic subgroups of order 4 in G.

There is only one subgroup of the second type because G has exactly 3 elements of order 2, as many as $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

A cyclic group of order 8 contains $\varphi(8) = 4$ elements of order 8 and is generated by any one of those elements. In G, elements of order 8 are those

whose first coordinate is ± 1 or ± 3, so there are 8 of them; this implies that there are 2 cyclic subgroups of order 8.
Any group of order 8 that is isomorphic to $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ only has elements whose order divides 4. Elements of G with this property form the subgroup $2\mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, which is isomorphic to $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and is thus the only subgroup of this type.
Finally, G has no subgroups of order 8 isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$, because it does not contain 7 elements of order 2.

(ii) We have seen that G has a unique subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, which is therefore characteristic. The cyclic subgroups of order 4 are $\langle (2,0) \rangle$, $\langle (2,1) \rangle$. Moreover, $\langle (2,0) \rangle = 2G$, which is characteristic because for all $\varphi \in \mathrm{Aut}(G)$ we have $\varphi(2G) = 2\varphi(G) = 2G$. Consequently, $\langle (2,1) \rangle$ is also characteristic, because it cannot have any image via an automorphism but itself.

86

(i) The group S_n has a subgroup isomorphic to D_{15} if and only if there exist $\rho, \sigma \in \mathsf{S}_n$ such that $\mathrm{ord}(\rho) = 15$, $\mathrm{ord}(\sigma) = 2$, and $\sigma\rho = \rho^{-1}\sigma$. Indeed, with the usual presentation $\langle r, s \mid r^{15} = s^2 = e,\ sr = r^{-1}s \rangle$ for D_{15}, it's immediate to see that these conditions are necessary, and if elements satisfying them exist, then the mapping $r \longmapsto \rho$, $s \longmapsto \sigma$ extends to an isomorphism $\mathsf{D}_{15} \longrightarrow \langle \rho, \sigma \rangle \subseteq \mathsf{S}_n$.
The requirement for S_n to contain an element of order 15 yields $n \geq 8$, as the order of a permutation is the least common multiple of the orders of its disjoint cycles.
We can then check that $n = 8$ is enough by choosing ρ to be the permutation $(1,2,3,4,5)(6,7,8)$ and choosing σ to be the $2+2+2$-cycle $(1,5)(2,4)(6,8)$, which has order 2 and sends ρ to ρ^{-1} via conjugation.

(ii) As before, we must have $n \geq 8$ in order for S_n to have a permutation ρ of order 15. However, this time we show that A_8 does *not* have any subgroups isomorphic to D_{15}.
Up to renumbering, we can assume

$$\rho = (1,2,3,4,5)(6,7,8)$$

because the only elements of order 15 in S_8 are the $5+3$-cycles. We are thus looking for a permutation σ of order 2 such that $\sigma\rho\sigma^{-1} = \rho^{-1}$. This fixes the action by conjugation of σ on ρ, and the set of solutions for this equation is the coset $(1,5)(2,4)(6,8)Z(\rho)$ of the centraliser of ρ in S_n.
But the conjugacy class of ρ has

$$\binom{8}{5} \cdot 4! \cdot \binom{3}{3} \cdot 2! = 8!/15$$

elements, so $|Z(\rho)| = 15$, hence $Z(\rho) = \langle\rho\rangle$. Therefore, since ρ is even and $(1,5)(2,4)(6,8)$ is odd, the coset $(1,5)(2,4)(6,8)Z(\rho)$ contains only odd permutations. There is thus no $\sigma \in \mathsf{A}_8$ that satisfies the required condition, and so A_8 does not have any subgroups isomorphic to D_{15}.
We reach the same conclusion for $n = 9$, as the cycle structure of ρ must still be $5+3$, the centraliser of a $5+3$-cycle is still generated by the cycle itself, and thus there is no even permutation σ satisfying $\sigma\rho\sigma^{-1} = \rho^{-1}$.
On the other hand, for $n = 10$ we can choose $\rho = (1,2,3,4,5)(6,7,8)$ as above, and $\sigma = (1,5)(2,4)(6,8)(9,10)$. We conclude that the required minimum is 10.

87 From the relation $3 = \mathrm{ord}(\sigma^4) = \mathrm{ord}(\sigma)/(\mathrm{ord}(\sigma), 4)$, we obtain that

$$3 \mid \mathrm{ord}(\sigma) \mid 12,$$

so the order of σ can be 3, 6 or 12. This implies that in the decomposition of σ into disjoint cycles only 2, 3, 4 and 6-cycles can appear, given that there are no 12-cycles in S_{10}; moreover, the decomposition must have at least a 3-cycle or at least a 6-cycle. We consider two cases.
(1) If σ has a 6-cycle, say (a,b,c,d,e,f), then we must have $\sigma = (a,b,c,d,e,f)\tau$, where $\tau \in \mathsf{S}(\{7,8,9,10\}) \simeq \mathsf{S}_4$ is disjoint from the 6-cycle, $\tau^4 = e$ and

$$(a,b,c,d,e,f)^2 = (a,c,e)(b,d,f) = (1,2,3)(4,5,6).$$

Note that the only permutations in S_4 whose order is not a divisor of 4 are the 3-cycles, so there are $24 - 8 = 16$ possibilities for τ.

Moreover, we can assume without loss of generality that $a = 1$, and so $c = 2$ and $e = 3$, whereas for b we have the three possible values 4, 5 and 6, and d, f are uniquely determined given b. We thus have 3 possibilities for the 6-cycle in σ, and therefore $3 \cdot 16 = 48$ possibilities in total.
(2) If σ has no 6-cycles then $\sigma = (1,2,3)(4,5,6)\tau$, where $\tau \in \mathsf{S}(\{7,8,9,10\}) \simeq \mathsf{S}_4$ is disjoint from the two 3-cycles and such that $\tau^4 = e$. Therefore, by the same argument as above, we have 16 possibilities for τ.

In conclusion, we have a total of $48 + 16 = 64$ solutions.

88 Let G be a group of order 2013, where we can factor 2013 as $3 \cdot 11 \cdot 61$, and as usual denote by n_p the number p-Sylow subgroups of G. By the Sylow Theorems, we have $n_{11} \equiv 1 \pmod{11}$ and n_{11} divides $3 \cdot 61$, so $n_{11} = 1$. Analogous conditions for n_{61} also imply $n_{61} = 1$. On the other hand, $n_3 \equiv 1 \pmod 3$ and $n_3 \mid 11 \cdot 61$, so either $n_3 = 1$ or $n_3 = 61$. We consider two cases.
(1) If $n_3 = 1$ then there are a unique 3-Sylow subgroup, a unique 11-Sylow subgroup and a unique 61-Sylow subgroup, and each of these subgroups is cyclic. It follows that the three subgroups are normal and G is isomorphic to their direct product, that is, to $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/11\mathbb{Z} \times \mathbb{Z}/61\mathbb{Z}$, and thus to a cyclic group of order 2013.

(2) If $n_3 = 61$ then G is not Abelian, as the 3-Sylow subgroups are pairwise conjugate and distinct. We still have a unique 11-Sylow subgroup, isomorphic to $\mathbb{Z}/11\mathbb{Z}$ and normal in G, and a unique 61-Sylow subgroup, isomorphic to $\mathbb{Z}/61\mathbb{Z}$ and also normal in G. Now, given a 3-Sylow subgroup H, it acts on the 11-Sylow subgroup and on the 61-Sylow subgroup. But $\mathrm{Aut}(\mathbb{Z}/11\mathbb{Z}) \simeq (\mathbb{Z}/11\mathbb{Z})^*$ is a cyclic group of order 10, so H acts trivially on the 11-Sylow subgroup, because 3 does not divide 10.

From this we obtain that G is isomorphic to $\mathbb{Z}/11\mathbb{Z} \times (\mathbb{Z}/61\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z})$. But now the second factor is a group of order $3 \cdot 61$, not Abelian because of our previous remark, and we know that there is a unique isomorphism class of non-Abelian groups of this order.

In conclusion, a group of order 2013 is either cyclic, that is, isomorphic to $\mathbb{Z}/2013\mathbb{Z}$, or isomorphic to the unique non-Abelian product $\mathbb{Z}/11\mathbb{Z} \times (\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/61\mathbb{Z})$.

[Though it is not required, one could easily prove that $(-1+\sqrt{-3})/2 = 13$ is a primitive cubic root of 1 in $\mathbb{F}_{61}$, and so the mapping $1 \longmapsto (n \longmapsto 13n)$ can be extended to a homomorphism $\mathbb{Z}/3\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/61\mathbb{Z})$ in order to construct a non-Abelian semidirect product $\mathbb{Z}/61\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$.]

89 Note that the normal subgroup $\mathrm{Inn}(\mathsf{S}_3)$ of $\mathrm{Aut}(\mathsf{S}_3)$ is isomorphic to S_3, as the centre of S_3 is trivial; in fact, $\mathrm{Inn}(\mathsf{S}_3)$ coincides with $\mathrm{Aut}(\mathsf{S}_3)$, since each automorphism of S_3 induces a permutation of the set of the three 2-cycles, which are generators for S_3. We thus have that $H = \mathrm{Inn}(\mathsf{S}_3) \times \mathrm{Inn}(\mathsf{S}_3)$ is a subgroup of $\mathrm{Aut}(\mathsf{S}_3 \times \mathsf{S}_3)$ isomorphic to $\mathsf{S}_3 \times \mathsf{S}_3$.

Now let s be the automorphism $(\sigma, \tau) \longmapsto (\tau, \sigma)$ of $\mathsf{S}_3 \times \mathsf{S}_3$; clearly, $K = \langle s \rangle$ has order 2 and intersects H only in the identity.

We now wish to show that $KH = \mathrm{Aut}(\mathsf{S}_3 \times \mathsf{S}_3)$. This will imply that, in particular, H has index 2 and is thus normal, hence $\mathrm{Aut}(\mathsf{S}_3 \times \mathsf{S}_3) \simeq H \rtimes K \simeq (\mathsf{S}_3 \times \mathsf{S}_3) \rtimes \mathbb{Z}/2\mathbb{Z}$ as required.

Consider $\varphi \in \mathrm{Aut}(\mathsf{S}_3 \times \mathsf{S}_3)$. The elements of order 6 in $\mathsf{S}_3 \times \mathsf{S}_3$ are the pairs consisting of a 2-cycle and a 3-cycle, in some order. Therefore, given a 2-cycle σ and a 3-cycle τ, one of $\varphi(\sigma, \tau)$ and $(s \circ \varphi)(\sigma, \tau)$ is still a pair of the form (σ', τ'), where σ' is a 2-cycle and τ' a 3-cycle. Given what we intend to prove, we can assume that we are in the first case and show that this implies $\varphi \in H$.

Indeed, we have $\varphi(\sigma, e) = \varphi((\sigma, \tau)^3) = (\varphi(\sigma, \tau))^3 = (\sigma', \tau')^3 = (\sigma', e)$. But, as recalled above, the 2-cycles are generators for S_3, so $\varphi(\mathsf{S}_3 \times e) = \mathsf{S}_3 \times e$. Similarly, $\varphi(e \times \mathsf{S}_3) = e \times \mathsf{S}_3$.

Now let π_1, π_2 be the projections of $\mathsf{S}_3 \times \mathsf{S}_3$ onto the two factors and consider the maps $\varphi_1 : \eta \longmapsto \pi_1(\varphi(\eta, e))$ and $\varphi_2 : \eta \longmapsto \pi_2(\varphi(e, \eta))$. What we just proved implies that $\varphi = (\varphi_1, \varphi_2) \in \mathrm{Aut}(\mathsf{S}_3) \times \mathrm{Aut}(\mathsf{S}_3) = \mathrm{Inn}(\mathsf{S}_3) \times \mathrm{Inn}(\mathsf{S}_3) = H$.

90 Denote by $Z(\sigma)$ the centraliser of σ and by $\mathcal{C}\ell(\sigma)$ the conjugacy class of σ in S_{10}, that is, the set of all $2+2+3$-cycles. We have

$$|Z(\sigma)| = \frac{|\mathsf{S}_{10}|}{|\mathcal{C}\ell(\sigma)|} = \frac{10!}{\binom{10}{2} \cdot \binom{8}{2} \cdot \frac{1}{2} \cdot \binom{6}{3} 2!} = 144,$$

because $Z(\sigma)$ is the stabiliser for the action of S_{10} on itself by conjugation, and $\mathcal{C}\ell(\sigma)$ the corresponding orbit.

We know that all cycles of σ and all permutations disjoint from σ commute with σ because they commute with each of its cycles. It follows that $(1,2)$, $(3,4)$, $(5,6,7)$ and $\mathsf{S}(\{8,9,10\})$ are contained in $Z(\sigma)$. Moreover, the listed permutations commute with each other, so the group they generate has cardinality $2\cdot 2\cdot 3\cdot 3! = 72$ and thus has index 2 in $Z(\sigma)$.

The permutation $(1,3)(2,4)$, which is not in the subgroup of $Z(\sigma)$ we just constructed, nevertheless belongs to $Z(\sigma)$, as its action on σ by conjugation consists in exchanging $(1,2)$ and $(3,4)$ and fixing $(5,6,7)$. Thus, $Z(\sigma)$ is generated by $(1,2)$, $(3,4)$, $(1,3)(2,4)$, $(5,6,7)$ and $\mathsf{S}(\{8,9,10\})$. Remark that

$$Z(\sigma) = \big(Z(\sigma)\cap \mathsf{S}(\{1,2,3,4\})\big)\cdot\big(Z(\sigma)\cap \mathsf{S}(\{5,6,7\})\big)\cdot \mathsf{S}(\{8,9,10\}),$$

where the three factors commute with each other; the first is generated by $(1,2)$, $(3,4)$ and $(1,3)(2,4)$, while the second is generated by $(5,6,7)$.

Now let $H = \langle\sigma,\tau\rangle$; clearly, $Z_{\mathsf{S}_{10}}(H) = Z(\sigma)\cap Z(\tau)$, so what we need to establish is which permutations among those we found to be in $Z(\sigma)$ commute with τ. Among the permutations in $Z(\sigma)\cap\mathsf{S}(\{1,2,3,4\})$, only those in the subgroup of order 4 generated by $(1,2)$ and $(3,4)$ commute with τ; all three permutations in $Z(\sigma)\cap\mathsf{S}(\{5,6,7)\})$ commute with τ; finally, only the three permutations generated by $(8,9,10)$ belong to $\mathsf{S}(\{8,9,10\})\cap Z(\tau)$.

In conclusion, $Z_{\mathsf{S}_{10}}(H)$ is generated by $(1,2)$, $(3,4)$, $(5,6,7)$ and $(8,9,10)$.

[Even though this is not required, we can immediately see from the arguments above that $Z(\sigma)\simeq \mathsf{D}_4\times\mathbb{Z}/3\mathbb{Z}\times\mathsf{S}_3$, as $(1,2)$, $(3,4)$ and $(1,2,3,4)$ generate a subgroup of $\mathsf{S}(\{1,2,3,4\})\simeq\mathsf{S}_4$ isomorphic to D_4. Analogously, $Z(\langle\sigma,\tau\rangle)$ is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^2\times(\mathbb{Z}/3\mathbb{Z})^2$.]

91 By the Structure Theorem for Finite Abelian Groups, there are exactly two Abelian groups of order 20: the cyclic group $\mathbb{Z}/4\mathbb{Z}\times\mathbb{Z}/5\mathbb{Z}\simeq\mathbb{Z}/20\mathbb{Z}$, and the non-cyclic group $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/5\mathbb{Z}$. We can thus assume from now on that G is not Abelian.

Thanks to the Sylow Theorems, we know that the number of 5-Sylow subgroups of G must be congruent to 1 modulo 5 and divide 4; the only possibility is thus for the 5-Sylow subgroup to be unique and therefore normal; clearly, it is isomorphic to $\mathbb{Z}/5\mathbb{Z}$.

The 2-Sylow subgroups of G can be isomorphic to $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$: we treat these two cases separately.

① Suppose that the 2-Sylow subgroups are cyclic. The group G is then isomorphic to the semidirect product $\mathbb{Z}/5\mathbb{Z}\rtimes_\varphi\mathbb{Z}/4\mathbb{Z}$ with respect to some homomorphism $\varphi:\mathbb{Z}/4\mathbb{Z}\longrightarrow\mathrm{Aut}(\mathbb{Z}/5\mathbb{Z})\simeq(\mathbb{Z}/5\mathbb{Z})^*$. We have four semidirect products G_a, for $a\in(\mathbb{Z}/5\mathbb{Z})^*$, defined by the homomorphisms φ_a induced by $\mathbb{Z}/4\mathbb{Z}\ni 1\longmapsto(n\longmapsto an)\in\mathrm{Aut}(\mathbb{Z}/5\mathbb{Z})$. The automorphism φ_1 is the identity and so the group G_1 is Abelian, which we had excluded; we can therefore assume that $a\neq 1$.

The operation in G_a is given by

$$(n, u)(m, v) = (n + a^u m, u + v);$$

it follows by some simple computations that the centres of G_2 and G_3 are trivial, whereas $(0, 2) \in Z(G_4)$. This shows that G_4 is not isomorphic to G_1 or G_3.

By using the fact that $3 = 2^{-1}$ in $(\mathbb{Z}/5\mathbb{Z})^*$, it is immediate to show that the mapping $(1, 0) \longmapsto (1, 0), (0, 1) \longmapsto (0, -1)$ induces an isomorphism $G_2 \longrightarrow G_3$.

We can thus conclude that there are 2 isomorphism classes of non-Abelian groups of order 20 whose 2-Sylow subgroups are cyclic.

(2) Now assume that the 2-Sylow subgroups are isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and let $\varphi : \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/5\mathbb{Z})$ be a nontrivial homomorphism. This homomorphism will send exactly two of the three elements of order 2 in $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ to -1, which is the only element of order 2 in $(\mathbb{Z}/5\mathbb{Z})^*$. But we know that automorphisms of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ transitively permute elements of order 2, so there is a unique isomorphism class for a non-Abelian semidirect product $\mathbb{Z}/5\mathbb{Z} \rtimes (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$.

In conclusion, we have five groups of order 20 up to isomorphism: one is cyclic, one is Abelian but not cyclic, and three are not Abelian.

92

(i) Given two homomorphisms $\varphi \in \mathrm{Hom}(A, C)$ and $\psi \in \mathrm{Hom}(B, C)$, one can immediately check that the map

$$A \oplus B \ni (x, y) \overset{\varphi\oplus\psi}{\longmapsto} \varphi(x) + \psi(y) \in C$$

is a homomorphism. We can thus consider the map

$$\mathrm{Hom}(A, C) \oplus \mathrm{Hom}(B, C) \ni (\varphi, \psi) \overset{\alpha}{\longmapsto} \varphi \oplus \psi \in \mathrm{Hom}(A \oplus B, C)$$

and show that α is also a homomorphism. Indeed, for all $\varphi, \varphi' \in \mathrm{Hom}(A, C)$, $\psi, \psi' \in \mathrm{Hom}(B, C)$ and $a \in A$, $b \in B$ we have

$$\begin{aligned}
\alpha\big((\varphi, \psi) + (\varphi', \psi')\big)(a, b) &= \alpha(\varphi + \varphi', \psi + \psi')(a, b) \\
&= (\varphi + \varphi')(a) + (\psi + \psi')(b) \\
&= \varphi(a) + \varphi'(a) + \psi(b) + \psi'(b) \\
&= \varphi(a) + \psi(b) + \varphi'(a) + \psi'(b) \\
&= \alpha(\varphi, \psi)(a, b) + \alpha(\varphi', \psi')(a, b) \\
&= \big(\alpha(\varphi, \psi) + \alpha(\varphi', \psi')\big)(a, b),
\end{aligned}$$

so $\alpha\big((\varphi, \psi) + (\varphi', \psi')\big) = \alpha(\varphi, \psi) + \alpha(\varphi', \psi')$.

The kernel of α is given by the pairs (φ, ψ) such that $\varphi(a) + \psi(b) = 0$ for all $a \in A$ and $b \in B$. But then choosing $b = 0$ yields $\varphi(a) = 0$ for all $a \in A$, that is, $\varphi = 0$, and analogously $\psi = 0$. This proves that α is injective.
Moreover, α is surjective, as given $\gamma \in \mathrm{Hom}(A \oplus B, C)$ we have $\gamma(a, b) = \gamma\big((a, 0) + (0, b)\big) = \gamma(a, 0) + \gamma(0, b)$, so $\gamma = \alpha(\varphi, \psi)$, where φ and ψ are given by $\varphi(a) = \gamma(a, 0)$ for all $a \in A$, and $\varphi(b) = \gamma(0, b)$ for all $b \in B$.
We can thus conclude that α is a group isomorphism between $\mathrm{Hom}(A, C) \oplus \mathrm{Hom}(B, C)$ and $\mathrm{Hom}(A \oplus B, C)$.

(ii) By the Structure Theorem for Finite Abelian Groups, G is isomorphic to $\oplus_{i=1}^{r} \mathbb{Z}/d_i\mathbb{Z}$ for some divisors $d_1, d_2, \ldots, d_r$ of n, so by (i) we have

$$\mathrm{Hom}(G, \mathbb{Z}/n\mathbb{Z}) \simeq \oplus_{i=1}^{r} \mathrm{Hom}(\mathbb{Z}/d_i\mathbb{Z}, \mathbb{Z}/n\mathbb{Z});$$

it is therefore enough to show that for $d \mid n$ we have $\mathrm{Hom}(\mathbb{Z}/d\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \simeq \mathbb{Z}/d\mathbb{Z}$.
We know that a homomorphism from $\mathbb{Z}/d\mathbb{Z}$ is uniquely determined by the image of 1, which can be any element of $\mathbb{Z}/n\mathbb{Z}$ whose order divides d. It follows that $|\mathrm{Hom}(\mathbb{Z}/d\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})| = d$ and that the homomorphism group is cyclic, as the map $\varphi \longmapsto \varphi(1)$ is an injection into the cyclic group $\mathbb{Z}/n\mathbb{Z}$. This shows the desired statement.

93 We know that a group of order 8 is isomorphic to one of the following groups: $\mathbb{Z}/8\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, the dihedral group D_4, or the group Q_8 of unit quaternions. We now determine which of these are isomorphic to some subgroup of S_6.

It is clear that S_6 cannot have any subgroups isomorphic to $\mathbb{Z}/8\mathbb{Z}$, as S_6 has no elements of order 8.

The permutations $(1, 2, 3, 4)$ and $(5, 6)$ commute, so the subgroup they generate is isomorphic to $\langle(1, 2, 3, 4)\rangle \times \langle(5, 6)\rangle$ and thus to $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Analogously, the subgroup of S_6 generated by $(1, 2)$, $(3, 4)$ and $(5, 6)$ is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$.

It is also clear that D_4 injects into S_4: isometries of the square induce permutations of its four vertices; this yields a homomorphism $\mathsf{D}_4 \longrightarrow \mathsf{S}_4$, clearly injective because $\mathbb{R}^2$ is generated by the vertices of the square as a vector space on $\mathbb{R}$.

We now show that S_6 has no subgroups isomorphic to the group Q_8. We know that $\mathsf{Q}_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, where $i^4 = j^4 = k^4 = 1$ and $i^2 = j^2 = k^2 = -1$. If S_6 had a subgroup isomorphic to Q_8, this would contain 3 permutations σ, τ, ρ which would be images of i, j, k, respectively, and therefore have order 4; the subgroups generated by each of them would have to be different, and we would have $\sigma^2 = \tau^2 = \rho^2 = \gamma$ for some γ of order 2. The only permutations of order 2 are products of transpositions; among these, the only ones that are squares are those whose cycle structure is $2 + 2$, as they have to be even. Up to conjugation, we can thus assume that $\gamma = (1, 2)(3, 4)$.

However, the equation $x^2 = (1, 2)(3, 4)$ in S_6 only has the following solutions: $(1, 3, 2, 4)$ and its inverse, and $(1, 3, 2, 4)(5, 6)$ and its inverse. It follows that there

are no 3 permutations σ, τ, ρ in S_6 satisfying the required constraints, so S_6 has no subgroups isomorphic to Q_8.

94 Let G be a group of order p^4. The centre $Z(G)$ of G is nontrivial because G is a p-group; additionally, it cannot have order p^3, or $G/Z(G)$ would be cyclic, which is not possible. Therefore, $Z(G)$ can have order p^4, p^2 or p. We consider each case separately.

(1) If $|Z(G)| = p^4$, then the group G is Abelian, so it has a subgroup of order d for each divisor d of p^4; in particular, it has at least a subgroup of order p^3, which is obviously Abelian.

(2) Let $|Z(G)| = p^2$ and let g be an element that is not in $Z(G)$. The centraliser $Z(g)$ contains both g and $Z(G)$, but cannot be all of G as $g \notin Z(G)$. It follows that $|Z(g)| = p^3$, and this subgroup is definitely Abelian, as its centre contains both g and $Z(G)$.

(3) Finally, assume that $|Z(G)| = p$. By the Class Equation, we have $|G| = |Z(G)| + \sum_{g\in\mathcal{R}} |G|/|Z(g)|$, where $\mathcal{R}$ is a system of representatives for the conjugacy classes of elements outside the centre. Remark that for each $g \in \mathcal{R}$ we have $|Z(g)| > p$, as $Z(g)$ contains both $Z(G)$ and g, which is not in $Z(G)$; moreover, the fact that $g \notin Z(G)$ also implies $|Z(g)| < p^4$. We can thus have either $|Z(g)| = p^2$ or $|Z(g)| = p^3$.

Now, if we had $|Z(g)| = p^2$ for all $g \in \mathcal{R}$, then we would have $|G|/|Z(g)| = p^2$ for each summand, and the formula would give $p^4 = p + |\mathcal{R}|p^2$, which is clearly impossible as it would imply that p^2 divides p.

There is therefore an element g such that $Z(g)$ has order p^3. The centraliser $Z(g)$ is Abelian, as the centre of a non-Abelian p-group of order p^3 would have order p, whereas the centre of $Z(g)$ contains both $Z(G)$ and g.

95 A group of order $10 = 2\cdot 5$ can be either cyclic or isomorphic to D_5; we proceed to count subgroups of each type.

A cyclic group of order 10 is generated by any one of its $\phi(10) = 4$ elements of order 10; the number of such subgroups in S_7 is therefore given by the number of elements of order 10, divided by 4. But an element of S_7 has order 10 if and only if it is a $5 + 2$-cycles; the number of such elements is thus $\binom{7}{5} \cdot 4! \cdot \binom{2}{2} = 504$, and the number of cyclic subgroups of order 10 is $504/4 = 126$.

We now wish to count subgroups isomorphic to D_5 in S_7. Each such subgroup contains a unique subgroup K of order 5, which is necessarily cyclic and generated by a 5-cycle of S_7, as the only elements of order 5 in S_7 are the 5-cycles. The number of cyclic subgroups of order 5 is thus given by the number of 5-cycles (which is the same as the number of $5 + 2$-cycles), divided by $\phi(5) = 4$, so we end up with 126 choices for K. We now show that, for any given choice of K, there are exactly two subgroups isomorphic to D_5 that contain K; we will then conclude that the number of subgroups isomorphic to D_5 in S_7 is $126 \cdot 2 = 252$.

Up to conjugation, we can assume $K = \langle\sigma\rangle$, where $\sigma = (1, 2, 3, 4, 5)$. If G is a subgroup isomorphic to D_5 that contains K, then G must have exactly 5 elements τ of order 2 such that $\tau\sigma\tau^{-1} = \sigma^{-1}$, and in fact G is generated by K and by any

one of these elements τ. Consequently, we can obtain the number of subgroups G by counting the elements τ as described and dividing the result by 5.

Let τ be an element of order 2 such that $\tau\sigma\tau^{-1} = \sigma^{-1}$. Clearly, $\tau_1 = (2,5)(3,4)$ satisfies these conditions, which implies that τ is an element of the coset $\tau_1 Z(\sigma)$, because its action on σ by conjugation is fixed. The centraliser $|Z(\sigma)|$ has order $|\mathsf{S}_7|/|\mathcal{C}\ell(\sigma)| = 10$, so there are at most 2 subgroups G satisfying our requirements. On the other hand, the permutation $\tau_2 = (2,5)(3,4)(6,7)$, which does satisfy the constraints, generates with K a subgroup G different from the one generated by K and τ_1, as the latter subgroup is contained in $\mathsf{S}(\{1,2,3,4,5\})$ and the former is not. In conclusion, there are exactly 2 subgroups G for each K, as claimed.

96

(i) Let H be the subgroup of G given by the elements whose first coordinate is null, and let $e_1 = (1,0,\ldots,0) \in G$. It is clear that each $x \in G$ can be expressed in a unique way as $x = ue_1 + h$, where $u \in \mathbb{Z}/p^{a_1}\mathbb{Z}$ and $h \in H$, that is, G is isomorphic to $\mathbb{Z}/p^{a_1}\mathbb{Z} \times H$.

Denote by A the set of automorphisms φ such that $\varphi_{|H} = \mathrm{Id}_H$. Each element φ of A is uniquely determined by $\varphi(e_1)$. Fix $\varphi(e_1) = ve_1 + k$ for some $v \in \mathbb{Z}/p^{a_1}\mathbb{Z}$ and $k \in H$; if we had $v \in p\mathbb{Z}/p^{a_1}\mathbb{Z}$ then every element in the image of φ would have its first coordinate in $p\mathbb{Z}/p^{a_1}\mathbb{Z}$, so φ would not be surjective, which would contradict its being an automorphism.

On the other hand, we now prove that for any choice of $v \in (\mathbb{Z}/p^{a_1}\mathbb{Z})^*$ and $k \in H$, the map $\varphi : ue_1 + h \longmapsto u(ve_1 + k) + h$ is an automorphism of G, and clearly it is an element of A; this implies that $|A| = |(\mathbb{Z}/p^{a_1}\mathbb{Z})^*| \cdot |H| = (p-1)p^{a_1-1}p^{a_2}\cdots p^{a_r} = (p-1)p^{n-1}$.

It is easy to show that φ is well-defined and a homomorphism, as it is constructed by appropriately extending the mappings $e_1 \longmapsto ve_1 + k$ and $h \longmapsto h$, for $h \in H$. Moreover, $ve_1 = \varphi(e_1 - k)$, so the element e_1, which belongs to the subgroup generated by ve_1 since v is invertible in $\mathbb{Z}/p^{a_1}\mathbb{Z}$, is in the image of φ. But then, since $H = \varphi(H)$ is also contained in the image, φ is surjective, and therefore also injective, because G is finite.

(ii) The set A discussed in (i) is clearly a subgroup of $\mathrm{Aut}(G)$, so p^{n-1} divides $|\mathrm{Aut}(G)|$ because p^{n-1} divides $|A|$.

97

(i) We factor 120 as $2^3 \cdot 3 \cdot 5$. By the Sylow Theorems there is a 5-Sylow subgroup H of G. If H is normal, then G is not simple; otherwise, the number of its conjugates is a divisor of $2^3 \cdot 3 = 24$ that is strictly greater than 1 and congruent to 1 modulo 5; the only possibility is for this number to be equal to 6.

As a consequence, the normaliser $N(H)$, which is the stabiliser of H for the action by conjugation, has index 6 in G. We now let G act on the left cosets of $N(H)$ by left multiplication. As is the case with all left multiplication actions on left cosets, this action is nontrivial because it is transitive.

But the action's kernel is a normal subgroup of G, and since the homomorphism is nontrivial and G has no normal subgroups other than $G, \{e\}$, the

kernel must be equal to $\{e\}$. In other words, G is isomorphic to a subgroup of the permutations of 6 elements, corresponding to the 6 left cosets of $N(H)$. If the image of the action were a subgroup of S_6 that is not contained in A_6, then its intersection with A_6 would be a normal subgroup of the image, which is isomorphic to G, contradiction. So the image is contained in A_6 and G is isomorphic to a subgroup of A_6.

(ii) We reach the desired conclusion from the fact that A_6 has no subgroups of order 120. Indeed, we have $|\mathsf{A}_6| = 360$, so any subgroup of A_6 of order 120 would have index 3. But if a group has a subgroup K of index k, then there is a normal subgroup contained in K whose index is a divisor of $k!$. In this case, this would imply the existence of a normal subgroup of A_6 whose index is 3 or 6. But A_6 is simple, so it has no normal proper nontrivial subgroups.

98 As is the case with elements of any group, given a permutation σ of order m, the permutation σ^2 still has order m if m is odd, and it has order $m/2$ if m is even. It is easy to check that the square of a k-cycle, where $k \leq 7$, has even order if and only if $k = 4$. Given that the order of a permutation is the least common multiple of the orders of its cycles, the permutations whose square has even order are exactly the products of a 4-cycle with any permutation of the remaining 3 elements. There are thus

$$\binom{7}{4} \cdot 3! \cdot 3! = 1260$$

of them. The permutations we wish to enumerate are those in the complement, so they are

$$7! - 1260 = 3780.$$

99

(i) Assume by symmetry that $p < q < r$. By the Sylow Theorems, which we shall repeatedly use in this solution, if an r-Sylow subgroup of G is not normal, then the number of its conjugates is strictly more than 1, divides pq and is congruent to 1 modulo r. The only possibility is thus for this number to be pq, as long as we also have $pq \equiv 1 \pmod{r}$. In this case, there are $pq(r-1)$ elements of order r in G, so there are at most pq elements whose order is not a multiple of r. Consider a q-Sylow subgroup: if this were not normal either, it would have at least $q+1$ conjugates, as their number would have to be congruent to 1 modulo q, and therefore G would have at least $(q+1)(q-1) > pq$ elements of order q, which is a contradiction.
Consequently, G has a normal r-Sylow subgroup or a normal q-Sylow subgroup.

(ii) Up to renaming the three primes, let H be a normal p-Sylow subgroup of G and let $\pi : G \longrightarrow G/H$ be the quotient homomorphism. The order of G/H is the product of the two distinct primes q, r. Suppose by symmetry

$q < r$; the group G/H has a normal subgroup $\overline{K}$ of order r, so by the bijective correspondence between subgroups induced by the quotient homomorphism, $\pi^{-1}(\overline{K})$ is a normal subgroup of G whose index is q.

100 Letting A be the set of elements of order p in G, we partition A into conjugacy classes. The cardinality of the conjugacy class of an element $a \in A$ is 1 if $a \in Z(G)$, and a multiple of p if $a \notin Z(G)$.

The problem is thus equivalent to showing that the number of elements of A that belong to $Z(G)$, which is a nontrivial subgroup, is congruent to -1 modulo p. Since $Z(G)$ is Abelian, the set $H = \{x \in Z(G) \mid x^p = e\}$ is a subgroup of $Z(G)$ whose order is a divisor p^k of $|Z(G)|$, and so in particular a divisor of $|G| = p^n$; moreover, this divisor cannot be 1 because of Cauchy's Theorem. The subgroup H consists of all elements of order p in $Z(G)$, plus the identity. The elements of order p in H are therefore $p^k - 1$, which is a number congruent to -1 modulo p.

101 The conjugacy class $\mathcal{C}\ell(\sigma)$ of a permutation σ in S_{10} is the set of permutations having the same cycle structure as σ; for the centraliser $Z(\sigma)$, we have $|Z(\sigma)| = 10!/|\mathcal{C}\ell(\sigma)|$. Letting $Z_0(\sigma)$ be the centraliser of σ in A_{10}, we have $Z_0(\sigma) = Z(\sigma) \cap \mathsf{A}_{10}$, so if $Z(\sigma) \subseteq \mathsf{A}_{10}$ then $|Z_0(\sigma)| = |Z(\sigma)|$, and otherwise $|Z_0(\sigma)| = |Z(\sigma)|/2$.

Now, letting $\tau = (1, 2, 3, 4, 5, 6, 7, 8)$, note that $(9, 10) \in Z(\tau^k)$ for all k, because $(9, 10)$ is disjoint from τ; in particular, $Z(\tau^k)$ is not contained in A_{10} and therefore $|Z_0(\tau^k)| = |Z(\tau^k)|/2$. We thus only need to compute the cardinality of $Z(\tau^k)$ for each k; we do so by considering the following cases.

(1) If k and 8 are coprime then the permutation τ^k is still an 8-cycle, so

$$\left|\mathcal{C}\ell(\tau^k)\right| = \binom{10}{8} \cdot 7! = \frac{10!}{16}$$

and therefore $|Z(\tau^k)| = 16$.

(2) If $(k, 8) = 2$ then τ^k is a $4 + 4$-cycle and we have

$$\left|\mathcal{C}\ell(\tau^k)\right| = \binom{10}{4} \cdot 3! \cdot \binom{6}{4} \cdot 3! \cdot \frac{1}{2} = \frac{10!}{64},$$

so $|Z(\tau^k)| = 64$.

(3) If $(k, 8) = 4$ then τ^k is a $2 + 2 + 2 + 2$-cycle, so

$$\left|\mathcal{C}\ell(\tau^k)\right| = \binom{10}{2} \cdot \binom{8}{2} \cdot \binom{6}{2} \cdot \binom{4}{2} \cdot \frac{1}{4!} = \frac{10!}{768},$$

hence $|Z(\tau^k)| = 768$.

(4) Finally, if $(k, 8) = 8$ then $\tau^k = e$ and clearly $Z(\tau^k) = \mathsf{S}_{10}$ has cardinality 10!.

102 We wish to show that the primes satisfying the requirement are 2, 3, 5 and all primes congruent to 1 modulo 5.

We know that the automorphism group of $(\mathbb{Z}/5\mathbb{Z})^3$ is isomorphic to $\mathsf{GL}_3(\mathbb{F}_5)$, so it has order $(5^3-1)(5^3-5)(5^3-5^2)$, which is divisible by 2, 3, 5 and 31. It follows by Cauchy's Theorem that there exists an element of order p in $\mathrm{Aut}((\mathbb{Z}/5\mathbb{Z})^3)$ for each of the primes 2, 3, 5 and 31, and so for those primes there is a nontrivial homomorphism $\mathbb{Z}/p\mathbb{Z} \longrightarrow \mathrm{Aut}((\mathbb{Z}/5\mathbb{Z})^3)$. We can use these homomorphisms to construct semidirect products $(\mathbb{Z}/5\mathbb{Z})^3 \rtimes \mathbb{Z}/p\mathbb{Z}$ with the required properties. This proves that 2, 3, 5 and 31 satisfy the condition in the problem statement.

Now let p be a prime congruent to 1 modulo 5. Remark that the order of $\mathrm{Aut}(\mathbb{Z}/p\mathbb{Z}) \simeq (\mathbb{Z}/p\mathbb{Z})^*$ is divisible by 5, so we can construct a nontrivial homomorphism $\mathbb{Z}/125\mathbb{Z} \longrightarrow \mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$ by extending the mapping $1 \longmapsto \varphi$, where φ is an element of order 5 in $\mathrm{Aut}(\mathbb{Z}/p\mathbb{Z})$, whose existence is guaranteed by Cauchy's Theorem. As before, the ensuing semidirect product $\mathbb{Z}/p\mathbb{Z} \rtimes \mathbb{Z}/125\mathbb{Z}$ has an Abelian subgroup of order 125, has order $125p$ and is not Abelian; it follows that all primes p congruent to 1 modulo 5 satisfy the requirement.

We will now show that no primes other than those already considered satisfy the condition. Indeed, let G be a non-Abelian group of order $125p$ with an Abelian subgroup of order 125, and suppose $p \neq 5$. The Sylow Theorems imply that the number n_p of subgroups of order p in G is congruent to 1 modulo p and divides 125. We have the following possibilities: either $n_p = 125$, so $p = 2$, 31, or $n_p = 25$, so $p = 2$, 3, or $n_p = 5$, so $p = 2$, or $n_p = 1$. We can without loss of generality assume that $n_p = 1$, that is, that there is a unique p-Sylow subgroup H, which is therefore normal.

Now consider the number n_5 of 5-Sylow subgroups; n_5 divides p, so we have but the two possibilities $n_5 = 1$ and $n_5 = p$. In the first case the unique 5-Sylow subgroup K, of order 125, must be normal, so G is isomorphic to the Abelian group $H \times K$, which contradicts the hypothesis that G is not Abelian. This implies that $n_5 = p \equiv 1 \pmod 5$, as wanted.

▌Note that the prime 31 appeared in two separate constructions: indeed, we have the two distinct groups $(\mathbb{Z}/5\mathbb{Z})^3 \rtimes \mathbb{Z}/31\mathbb{Z}$ and $\mathbb{Z}/31\mathbb{Z} \rtimes \mathbb{Z}/125\mathbb{Z}$.▐

103

(i) A necessary condition is for 360 to divide $n!$, which holds for $n \geq 6$. For $n = 6$, the group S_6 contains A_6, which has 360 elements. So $n = 6$ is the required minimum.

(ii) First of all, remark that there is a cyclic group of order 360 in S_{22}; for instance, consider the group generated by the permutations $(1, 2, 3, 4, 5)$, $(6, 7, 8, 9, 10, 11, 12, 13)$ and $(14, 15, 16, 17, 18, 19, 20, 21, 22)$. Indeed, these three cycles, whose orders are 5, 8 and 9, respectively, commute with each other and thus generate a group isomorphic to $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}$, whose order is $5 \cdot 8 \cdot 9 = 360$.

We now show that if H is a cyclic group of order 360 contained in S_n then $n \geq 22$, which proves that the required minimum is 22. A cyclic group is generated by an element σ, and its order is the least common multiple of the lengths of the disjoint cycles in σ. For the subgroup to have order 360, at least one cycle of σ must have length a multiple of 5, at least one must have length

a multiple of 8, and at least one a multiple of 9. If a single cycle's length is a multiple of two of the numbers 5, 8 and 9, then the cycle's length is at least $40 = 5 \cdot 8$. It follows that the sum of the lengths of the cycles of σ is in any case at least $5 + 8 + 9 = 22$.

104

(i) Letting L be a subgroup of order d in G, consider $\pi_{|L} : L \longrightarrow G/(\mathbb{Z}/p\mathbb{Z} \times 0)$, where π is the quotient homomorphism from G to $G/(\mathbb{Z}/p\mathbb{Z} \times 0)$. The order of the kernel of $\pi_{|L}$ divides both d and p, so the kernel is trivial. It follows that L is isomorphic to a subgroup of the cyclic group $G/(\mathbb{Z}/p\mathbb{Z} \times 0) \simeq \mathbb{Z}/(p-1)\mathbb{Z}$, so it is itself cyclic.

(ii) Suppose by contradiction that H and K are two distinct subgroups of order d whose intersection $H \cap K$ contains an element $g \neq e$, and let $Z(g)$ be the centraliser of g. The order of g is a divisor of d and is thus coprime with p, so g does not belong to $\mathbb{Z}/p\mathbb{Z} \times 0$.
We now wish to prove that no element of $\mathbb{Z}/p\mathbb{Z} \times 0$ commutes with g. Indeed, letting $g = (u, k)$ and $h = (v, 0)$, we have $hgh^{-1} = (v + u - \varphi_k(v), k)$, so if h did commute with g we would have $\varphi_k(v) = v$; but if $v \neq 0$ then h generates $Z/p\mathbb{Z} \times 0$, so we would have $\varphi_k = \mathrm{Id}$, whereas φ is assumed to be injective. This concludes the proof that $Z(g) \cap (\mathbb{Z}/p\mathbb{Z} \times 0) = \{e\}$.
It follows that the order of $Z(g)$ is a divisor of $p - 1$, as all elements of order p in G belong to $\mathbb{Z}/p\mathbb{Z} \times 0$, because this is the unique p-Sylow subgroup of G. By (i), $Z(g)$ is cyclic, but this yields a contradiction: $Z(g)$ contains H and K, as they are cyclic and in particular Abelian; however, they are distinct subgroups with the same order, whereas a cyclic group has a unique subgroup of each possible order.

105

(i) A necessary condition in order for S_n to contain a subgroup of order 36 is that 36 divides $n!$. The smallest n satisfying this condition is $n = 6$. On the other hand, S_6 contains the subgroup $\mathsf{S}(\{1, 2, 3\}) \cdot \mathsf{S}(\{4, 5, 6\})$, which is isomorphic to $\mathsf{S}_3 \times \mathsf{S}_3$, so the required minimum is indeed $n = 6$.

(ii) Again, $n = 6$ is the smallest n such that 72 divides $n!$. Now let $\sigma = (1, 4)(2, 5)(3, 6)$ and let K be the subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z}$ that is generated by σ in S_6. Moreover, let H be $\mathsf{S}(\{1, 2, 3\}) \cdot \mathsf{S}(\{4, 5, 6\})$, which is a subgroup with 36 elements. Remark that conjugation by $\sigma = (1, 4)(2, 5)(3, 6)$ exchanges the two subgroups $\mathsf{S}(\{1, 2, 3\})$ and $\mathsf{S}(\{4, 5, 6\})$, so $\sigma \in N(H)$ and thus HK is a subgroup of S_6 whose cardinality is

$$\frac{|H| \cdot |K|}{|H \cap K|} = \frac{36 \cdot 2}{1} = 72.$$

The required minimum is, again, $n = 6$.

(iii) As before, $n = 6$ is the smallest value of n for which 144 divides $n!$; however, S_6 does not contain a subgroup of order 144. Indeed, assume by contradiction

that H is such a subgroup. The index of H in S_6 is 5, and S_6 acts on the set S_6/H of the 5 cosets of H by left multiplication. This action is transitive, hence nontrivial, and its kernel K is contained in H. But then K is a normal subgroup of S_6 whose index is a multiple of 5 and a divisor of 5!, which is not possible because the only proper normal subgroups of S_6 are A_6 and $\{e\}$.
On the other hand, S_7 contains the subgroup $\mathsf{S}(\{1,2,3,4\})\cdot\mathsf{S}(\{5,6,7\})$, which is isomorphic to $\mathsf{S}_4 \times \mathsf{S}_3$ and has order $4! \cdot 3! = 144$.
The required minimum is thus $n = 7$.

106 It is clear that (i) implies both (ii) and (iii), as all subgroups of a cyclic group are cyclic.

We now show that, conversely, (ii) implies (i). Assume by contradiction that (ii) holds but G is not cyclic. By the Structure Theorem for Finite Abelian Groups, G is isomorphic to the direct product of its subgroups $G(2)$ and $G(5)$ of 2-torsion and 5-torsion, whose orders are 2^n and 5^m, respectively.

By our assumption, at least one of these two subgroups is not cyclic, and thus is the direct product of at least two nontrivial cyclic subgroups. Assuming $G(2)$ is not cyclic, it must contain at least two different cyclic subgroups of order 2, which are contained in $\{g \in G \mid 10g = 0\}$; but this is a cyclic group by assumption, and thus it cannot contain two distinct subgroups of the same order. A similar argument applies if $G(5)$ is not cyclic.

Finally, we show that (iii) implies (i). Assume that the subgroups $G_2 = \{2g \mid g \in G\}$ and $G_5 = \{5g \mid g \in G\}$ are cyclic. Notice that G_2 is the image of the homomorphism $\varphi_2 : x \longmapsto 2x$ of G, and, analogously, G_5 is the image $\varphi_5 : x \longmapsto 5x$.

Now, the 2-torsion subgroup $G(2)$ is characteristic, so $\varphi_5(G(2)) \subseteq G(2)$. Moreover, $\varphi_{5|G(2)}$ is injective because the order of $G(2)$ is a power of 2 and therefore coprime with 5. The fact that $G(2)$ is finite then implies that $\varphi_{5|G(2)}$ is also surjective, that is, $G(2) = \varphi_5(G(2)) \subseteq G_5$. It follows that $G(2)$ is cyclic, as it is a subgroup of the cyclic group G_5.

Similarly, $G(5)$ is cyclic, so G is isomorphic to a direct product of cyclic groups whose orders are coprime, so it is cyclic itself.

107 We first consider the permutation τ. Its conjugacy class is the set of $2+3$-cycles, of which there are $\binom{9}{2} \cdot \binom{7}{3} \cdot 2$; its centraliser $Z(\tau)$ has thus order

$$\frac{9!}{\binom{9}{2} \cdot \binom{7}{3} \cdot 2} = 2 \cdot 3 \cdot 4!.$$

The cycles $(1,2)$, $(3,4,5)$ and the permutations of $\{6,7,8,9\}$ commute with τ and generate the subgroup

$$\langle (1,2) \rangle \cdot \langle (3,4,5) \rangle \cdot \mathsf{S}(\{6,7,8,9\}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathsf{S}_4$$

of order $2 \cdot 3 \cdot 4!$, which is thus the centraliser of τ.

Any element of the normaliser $N(\langle\tau\rangle)$ of $\langle\tau\rangle$ will conjugate τ into one of its powers having the same cycle structure as τ, so either τ or $\tau^5 = \tau^{-1}$. The permutation $(4,5)$ conjugates τ into $\tau^5 = (1,2)(3,5,4)$, so the cosets of $Z(\tau)$ in $N(\langle\tau\rangle)$ are $Z(\tau)$ and $(4,5)Z(\tau)$. In particular, the normaliser is given by

$$\langle(1,2)\rangle \cdot \langle(4,5),(3,4,5)\rangle \cdot \mathsf{S}(\{6,7,8,9\}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathsf{S}_3 \times \mathsf{S}_4$$

and has order $2 \cdot 3! \cdot 4! = 2^5 \cdot 3^2$.

The permutation $\tau\sigma = (1,2)(3,4,5,6,7)$ has $\binom{9}{2} \cdot \binom{7}{5} \cdot 4!$ conjugates in S_9. Its centraliser thus consists of

$$\frac{9!}{\binom{9}{2}\binom{7}{5}4!} = 20$$

elements. The 2-cycle $(1,2)$, the 5-cycle $(3,4,5,6,7)$ and the 2-cycle $(8,9)$ all commute with $\tau\sigma$. These elements generate the group

$$\langle(1,2)\rangle \cdot \langle(3,4,5,6,7)\rangle \cdot \langle(8,9)\rangle$$

which is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and has order 20; it is therefore the centraliser of $\tau\sigma$.

The permutation $\tau\sigma$ has order 10, and the powers of $\tau\sigma$ that are conjugate to $\tau\sigma$ itself are those of the form $(\tau\sigma)^k$ with $(k,10) = 1$, that is, with k taking values in the cyclic group $(\mathbb{Z}/10\mathbb{Z})^*$, which has order 4 and is generated by 3. The permutation $(4,6,7,5)$ conjugates $\tau\sigma$ into $(\tau\sigma)^3 = (1,2)(3,6,4,7,5)$. Consequently, the normaliser of $\langle\tau\sigma\rangle$ is given by

$$\langle(12)\rangle \cdot \langle(4,6,7,5),(3,4,5,6,7)\rangle \cdot \langle(8,9)\rangle.$$

Since $(4,6,7,5)$ sends the 5-cycle $(3,4,5,6,7)$ to $(3,4,5,6,7)^3$, the second factor in the product above is isomorphic to $\mathbb{Z}/5\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$, where $\mathbb{Z}/4\mathbb{Z}$ acts on $\mathbb{Z}/5\mathbb{Z}$ via the homomorphism induced by $\mathbb{Z}/4\mathbb{Z} \ni 1 \longmapsto (a \longmapsto 3a) \in \mathrm{Aut}(\mathbb{Z}/5\mathbb{Z})$. In particular, this second factor has 20 elements, and thus the normaliser has order $2 \cdot 20 \cdot 2 = 2^4 \cdot 5$.

108 Let $G = \mathbb{Z}/p^a\mathbb{Z} \times \mathbb{Z}/p^b\mathbb{Z}$ be the group in the problem statement. First, we consider the case where $a \neq b$; by symmetry, we can assume $a > b$. In this case, the set $p^{a-1}G$ is a characteristic subgroup, and we have

$$p^{a-1}(\mathbb{Z}/p^a\mathbb{Z} \times \mathbb{Z}/p^b\mathbb{Z}) = p^{a-1}\mathbb{Z}/p^a\mathbb{Z} \times 0 \simeq \mathbb{Z}/p\mathbb{Z},$$

so G has a characteristic subgroup of order p.

Now consider the case where $a = b$: we show that G has no characteristic subgroups of order p. Assume by contradiction that H is such a subgroup; H

is generated by some element g of order p, which is thus of the form $g = (p^{a-1}r, p^{a-1}s)$, where at least one of r and s is not congruent to zero modulo p.

Assume by symmetry that $r \not\equiv 0 \pmod{p}$, and consider the automorphism φ of G defined by

$$G \ni (x, y) \stackrel{\varphi}{\longmapsto} (x, x + y) \in G.$$

We have $\varphi(g) = \varphi(p^{a-1}r, p^{a-1}s) = (p^{a-1}r, p^{a-1}(r + s))$ and we find that $\varphi(g) \notin H$: indeed, if $\varphi(g)$ belonged to H, this would imply that $\varphi(g) = cg$ for some integer c; looking at the first coordinate, we would have $cr \equiv r \pmod{p}$, that is, $c \equiv 1 \pmod{p}$, while for the second coordinate we would have $cs \equiv r+s \pmod{p}$, hence $c \not\equiv 1 \pmod{p}$. This contradiction proves that there is no characteristic subgroup H of order p.

109 Let G be a group of order $75 = 3 \cdot 5^2$. If the group is Abelian then obviously $Z(G) = G$, so 75 is a possible order for the centre; we will spend the rest of the solution showing that 1 is the only other possible order.

Assume G is not Abelian; the 5-Sylow subgroup $G(5)$ is unique, because by the Sylow Theorems the number of 5-Sylow subgroups is congruent to 1 modulo 5 and divides 3. In particular, $G(5)$ is normal and, having order 5^2, must be Abelian and isomorphic to either $\mathbb{Z}/25\mathbb{Z}$ or $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$.

Remark that G is the semidirect product of $G(3) \simeq \mathbb{Z}/3\mathbb{Z}$ and $G(5)$ with respect to some nontrivial homomorphism $\varphi : G(3) \longrightarrow \mathrm{Aut}(G(5))$. If the subgroup $G(5)$ were cyclic, 3 would not divide the order of $\mathrm{Aut}(\mathbb{Z}/25\mathbb{Z}) \simeq (\mathbb{Z}/25)^*$, which would be 20, so the only homomorphism from $Z/3\mathbb{Z}$ to $\mathrm{Aut}(\mathbb{Z}/25\mathbb{Z})$ would be the trivial one. We therefore have $G(5) \simeq \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$, a vector space of dimension 2 on $\mathbb{F}_5$, and $\mathrm{Aut}(G(5)) \simeq \mathsf{GL}_2(\mathbb{F}_5)$. In order to simplify notation we shall from now on identify G with $\mathbb{F}_5^2 \rtimes \mathbb{Z}/3\mathbb{Z}$, and thus $G(5)$ with the subgroup $\mathbb{F}_5^2 \times 0$ and $G(3)$ with $0 \times \mathbb{Z}/3\mathbb{Z}$.

Consider the generator $g = (0, 1)$ of $G(3)$ and let $f = \varphi_g : \mathbb{F}_5^2 \longmapsto \mathbb{F}_5^2$. An element $h = (v, 0) \neq (0, 0)$ commutes with g if and only if $(f(v), 0) = ghg^{-1} = h = (v, 0)$, that is, if and only if v is an eigenvector for f with eigenvalue 1. We now show that f actually does not have 1 among its eigenvalues.

Indeed, the minimal polynomial $\mu(t)$ of f has degree at most 2, because it divides the characteristic polynomial of f; moreover, the fact that $f^3 = \mathrm{Id}$ implies that $\mu(t)$ also divides $t^3 - 1$, so it has simple roots on every field of characteristic different from 3, and in particular on $\mathbb{F}_5$. But then, if 1 were an eigenvalue of f, there would also have to be a different eigenvalue, as we cannot have $\mu(t) = t - 1$, because f is not the identity. But 1 is the only cubic root of unity in $\mathbb{F}_5$, as 3 does not divide $|\mathbb{F}_5^*|$.

The above shows that no element in $G(5) \setminus (0, 0)$ commutes with g. But then, no element of the form $(v, \pm 1) = (v, 0)g^{\pm 1}$ with $v \neq (0, 0)$, that is, no element in $G \setminus G(3)$, can commute with g. We have thus proven that the centre is trivial.

110 Let $\mathcal{B} = \mathcal{B}_1 \sqcup \mathcal{B}_2 \sqcup \cdots \sqcup \mathcal{B}_r$ be the partition of $\mathcal{B}$ into G-orbits, let V_h be the vector subspace of V generated by $\mathcal{B}_h$ for $h = 1, \ldots, r$, and let $V = V_1 \oplus \cdots \oplus V_r$ be the corresponding decomposition of V as a direct sum.

Denote by V^G the vector subspace given by all vectors v such that $g \cdot v = v$ for all $g \in G$, which we call *invariant* vectors. What we need to show is that $\dim V^G = r$. Notice that if $v = v_1 + \cdots + v_r$, where $v_h \in V_h$ for $h = 1, \ldots, r$, then $g \cdot v = g \cdot v_1 + \cdots + g \cdot v_r$ and $g \cdot v_h \in V_h$, since $\mathcal{B}_h$ is an orbit and thus V_h is stable for the action of G. It follows that $g \cdot v = v$ if and only if $g \cdot v_h = v_h$ for $h = 1, \ldots, r$. We can thus reduce the problem to the case $r = 1$ and show that for any transitive action we have $\dim V^G = 1$.

Assume from now on that $r = 1$. Clearly, $e_1 + \cdots + e_n$ is invariant and nonzero, so $\dim V^G \geq 1$. On the other hand, the image of the linear map

$$V \ni v \overset{\pi}{\longmapsto} \frac{1}{|G|} \sum_{g \in G} g \cdot v \in V$$

is clearly contained in V^G, and moreover we have $\pi(v) = v$ if v is invariant. Letting $v = a_1 e_1 + \cdots + a_n e_n$, remark that

$$\begin{aligned} \pi(v) &= \sum_{h=1}^{n} a_h \pi(e_h) \\ &= \frac{1}{|G|} \sum_{h=1}^{n} a_h \Big(\sum_{g \in G} g \cdot e_h \Big) \\ &= \frac{1}{|G|} \sum_{h=1}^{n} a_h v_0 \\ &= \Big(\frac{1}{|G|} \sum_{h=1}^{n} a_h \Big) v_0, \end{aligned}$$

where we have used transitivity of the action to express $\sum_{g \in G} g \cdot e_h$ as some vector v_0 of V that does *not* depend on h. Now, if v is invariant then $v = \pi(v) \in \mathbb{C} \cdot v_0$, and so $\dim V^G \leq 1$.

111 First, factor the number 870 as $2 \cdot 3 \cdot 5 \cdot 29$.

If a group of order 870 were simple, none of its Sylow subgroups would be normal. In particular, letting n_p be the number of p-Sylow subgroups, which is such that $n_p \equiv 1 \pmod{p}$ and $n_p \mid 870$ by the Sylow Theorems, we would have

$$n_2 \geq 3, n_3 \geq 10, n_5 \geq 6, n_{29} \geq 30.$$

Since any two distinct subgroups of order p intersect trivially, a group G having n_p subgroups of order p must contain $n_p(p-1)$ elements of order p.

By summing the number of elements of order p in G for $p = 2, 3, 5$ and 29, we find at least

$$3 + 20 + 24 + 840 = 887 > 870$$

different elements, which is a contradiction. Hence G cannot be simple.

112 The group Q_8 has order 8, so it is isomorphic to a subgroup of S_8 via the action by left multiplication on the set of its elements.

Suppose Q_8 acts on a set X of cardinality at most 7. Without loss of generality, we can assume $|X| = 7$, that is, we have a homomorphism $\mathsf{Q}_8 \longrightarrow \mathsf{S}_7$. We intend to show that such a homomorphism cannot be injective.

For each element $x \in X$, the orbit of x has at most 7 elements, so the index of the stabiliser of x in Q_8 is at most 7:

$$\big[\mathsf{Q}_8 : \mathrm{St}(x)\big] = |\,\mathcal{O}(x)| \leq 7.$$

In particular, the stabiliser of x cannot be trivial, so it is a nontrivial subgroup of S_7 whose order divides 8: that is, the order of $|\,\mathrm{St}(x)|$ is 2, 4 or 8.

Every group of order 4 or 8 contains a subgroup of order 2, and $Z(\mathsf{Q}_8) = \{\pm 1\}$ is the only subgroup of Q_8 that has order 2. It follows that $Z(\mathsf{Q}_8)$ is contained in the stabiliser of every element, and so it is contained in the kernel of the action, and the homomorphism from Q_8 to S_7 is not injective. We conclude that the minimum n is precisely 8.

113

(i) The permutation $\sigma = (1, 2, 3)(4, 5, 6, 7)$ in S_7 is the product of a 3-cycle and a disjoint 4-cycle, so it has order $3 \cdot 4 = 12$. The inverse of σ is the permutation $\sigma^{-1} = (1, 3, 2)(4, 7, 6, 5)$, so σ and σ^{-1} are conjugate via the permutation $\tau = (2, 3)(5, 7)$, whose order is 2. We therefore have

$$\sigma^{-1} = \tau\sigma\tau$$

and the subgroup of S_7 generated by σ and τ is isomorphic to the dihedral group D_{12} via the isomorphism $r \longmapsto \sigma$, $s \longmapsto \tau$, where we are using the standard presentation $\langle r, s \,|\, r^{12} = s^2 = e,\ sr = r^{-1}s\rangle$.

(ii) We consider the same permutations σ and τ as in (i); it is clear that σ^2 and τ generate a group isomorphic to D_6, that is, isomorphic to the group generated by r^2 and s in D_{12}. Moreover, τ is even and σ^2 is also even because it is the square of some permutation, so the subgroup $\langle\sigma^2, \tau\rangle$ is contained in A_7.

(iii) We show that A_7 cannot contain an element of order 12. If it did, then A_7 would also contain an element of order 3 and an element of order 4 that commute, given by the fourth and third powers of the element of order 12.
However, an element of order 4 in A_7 must be the product of two disjoint cycles, a 4-cycle, which is an odd permutation, and a 2-cycle. An element of

order 3 in A_7 is the product of one or two disjoint 3-cycles and, in order for it to commute with the element of order 4, the 3-cycles must be disjoint from the 4-cycle and the 2-cycle. We would thus have to permute at least 9 elements in total, which is of course not possible in A_7.

114

(i) For each subgroup H of G we have $H \subseteq N(H)$, hence $N(P_h) \subseteq N(N(P_h))$, which proves one inclusion.
As for the other, remark that P_h is a characteristic subgroup of $N(P_h)$; indeed, it is normal in $N(P_h)$ and thus the only p_h-Sylow subgroup of $N(P_h)$. If $g \in N(N(P_h))$ then $gN(P_h)g^{-1} \subseteq N(P_h)$. It follows that $N(P_h) \ni x \longmapsto gxg^{-1} \in N(P_h)$ is an automorphism of $N(P_h)$, and so it sends the characteristic group P_h to itself. In other words, $g \in N(P_h)$.

(ii) Suppose G is isomorphic to $P_1 \times \cdots \times P_n$ and let K be a subgroup of G of order $p_1^{b_1} \cdots p_n^{b_n}$ for some $b_1, \ldots, b_n \geq 0$. The order of the projection K_n of K on P_h divides $p_h^{b_h}$ for each h. On the other hand, it is clear that $K \subseteq K_1 \times \cdots \times K_n$, and by comparing orders we obtain that $K = K_1 \times \cdots \times K_n$. We have thus shown that every subgroup K of G is of the form $K_1 \times \cdots \times K_n$.
The normaliser of K is therefore $N(K_1) \times \cdots \times N(K_n)$ in $P_1 \times \cdots \times P_n$. If $K \neq G$ then at least one of the factors K_h is different from P_h, so, as we know must be the case in the p_h-group P_h, $N(K_h) \supsetneq K_h$. It follows that K is strictly contained in $N(K)$ in G.
Conversely, assume that for every subgroup $K \neq G$ we have $N(K) \neq K$. In particular, this holds for the subgroups $N(P_h)$ of G, each of which must not coincide with the corresponding Sylow subgroup P_h. Thanks to (i), this implies $N(P_h) = G$, that is, all Sylow subgroups are normal in G, and thus $G \simeq P_1 \times \cdots \times P_n$.

115

(i) The cardinality of each of the orbits for the action of G on X must divide the order of G, and thus be a power of p. Moreover, the cardinality of X is the sum of the cardinalities of the orbits. If all orbits had cardinality divisible by p, then the cardinality of X would also have to be a multiple of p, against our hypothesis. So there is $x \in X$ whose orbit has cardinality $p^0 = 1$, and thus $g \cdot x = x$ for all $g \in G$.

(ii) In this case, G acts on the set V. As before, the cardinalities of orbits are all powers of p. Remark that for all $g \in G$ we have $g(0) = 0$, that is, the orbit of the zero vector consists of a single element. If the cardinalities of all other orbits were divisible by p, the cardinality of V would be congruent to 1 modulo p, which is not possible. Consequently, there exists a vector $v \in V \setminus \{0\}$ whose orbit has a single element, that is, such that $g \cdot v = v$ for all $g \in G$.

116 The group G in the problem statement is isomorphic to

$$\big(\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}\big) \times \big(\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}\big),$$

that is, to the direct product of its 2-torsion subgroup $G(2) = \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ and its 3-torsion subgroup $G(3) = \mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Any element of order $18 = 2 \cdot 3^2$ is thus of the form (g_2, g_3), where $g_2 \in G(2)$ has order 2 and $g_3 \in G(3)$ has order 9.

There are $2^2 - 1 = 3$ elements of order 2 in $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. The elements of order 9 in $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ are those whose order is not less than or equal to 3, so there are $3^3 - 3^2 = 18$ of them. It follows that G contains $3 \cdot 18 = 54$ elements of order 18.

Since G is Abelian, any subgroup of G can be uniquely expressed as a product of its intersections with the torsion subgroups. In order to count subgroups of order 18 in G, we can thus count subgroups of order 2 in $G(2)$ and subgroups of order 9 in $G(3)$.

Each subgroup of order 2 in $G(2)$ contains a unique element of order 2 which is a generator for it, so there are as many subgroups as elements of order 2, that is, there are 3.

A subgroup of order 9 in $G(3)$ is isomorphic to either $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ or $\mathbb{Z}/9\mathbb{Z}$. In the first case, the subgroup contains 8 elements of order 3; since $G(3)$ contains exactly 8 elements of order 3, it has exactly one subgroup of this type. In the second case, the subgroup contains $\phi(9) = 6$ elements of order 9, each of which is a generator. The group $G(3)$ contains 18 elements of order 9, so it has $18/6 = 3$ subgroups of this second type, for a total of $1 + 3 = 4$ subgroups of order 9.

In conclusion, G contains $3 \cdot 4 = 12$ subgroups of order 18.

117 If G has a normal p-Sylow subgroup P, then it is unique, and therefore the intersection of all p-Sylow subgroups obviously coincides with P which is not the trivial group.

Assume now that the p-Sylow subgroups are not normal. Given two of them, say P_1 and P_2, let $Q = P_1 \cap P_2$ be their intersection. First of all, remark that $Q \neq \{e\}$, or the set $P_1 P_2$ would have p^4 elements, and could not be contained in G. Since $P_1 \neq P_2$, it follows that $|Q| = p$, because P_1 and P_2 each have p^2 elements.

The fact that a group of order p^2 is necessarily Abelian implies that the centraliser of Q contains both P_1 and P_2, so it has at least $|P_1 P_2| = p^3$ elements. On the other hand, $p^3 > |G|/2$, so the centraliser of Q coincides with G, and Q is therefore contained in the centre of G.

Now consider the fact that, given $g \in G$, $Q = gQg^{-1} \subseteq gP_1g^{-1}$. Since p-Sylow subgroups are pairwise conjugate, it follows that Q is contained in their intersection, which is thus nontrivial.

▌In the case of a non-Abelian group G, we could approach the problem by this alternative route. The action of G on the set of the $p + 1$ cosets of a given p-Sylow subgroup given by left multiplication cannot be injective, as the order of G is not a divisor of $(p + 1)!$. But the kernel of this action is precisely the intersection of all conjugates of the given p-Sylow subgroup, that is, the intersection of all p-Sylow subgroups of G.▐

118

(i) The equality $\sigma^3 = (1, 2)(3, 4)(5, 6)$ implies $\sigma^6 = e$, so the order of σ must divide 6; moreover, it cannot be 1 or 3, or we would have $\sigma^3 = e$. The possible

orders for σ are thus 2 and 6. The only solution σ whose order is 2 is given by $\sigma = \sigma^3 = (1,2)(3,4)(5,6)$.
The elements of order 6 in S_6 are the 6-cycles and the $2+3$-cycles, but the latter cannot be solutions of the equation because they have a fixed point. Let σ be a 6-cycle of S_6, which we write in the form $(1,a,b,c,d,e)$; we have $\sigma^3 = (1,c)(a,d)(b,e)$, so in order for σ to be a solution we must have $c = 2$ and either $\{a,d\} = \{3,4\}$ and $\{b,e\} = \{5,6\}$, or $\{a,d\} = \{5,6\}$ and $\{b,e\} = \{3,4\}$. We thus find 8 solutions: $(1,3,5,2,4,6)$, $(1,4,5,2,3,6)$, $(1,3,6,2,4,5)$, $(1,4,6,2,3,5)$, $(1,5,3,2,6,4)$, $(1,6,3,2,5,4)$, $(1,5,4,2,6,3)$ and $(1,6,4,2,5,3)$.
The equation has a total of 9 solutions in S_6.

(ii) Denote by τ the permutation $(1,2)(3,4)(5,6)$. Its conjugacy class is the set of all $2+2+2$-cycles, whose cardinality is

$$\binom{6}{2}\binom{4}{2}\binom{2}{2}\frac{1}{3!} = 15;$$

it follows that the centraliser $Z(\tau)$ has order $6!/15 = 48$.
The disjoint cycles $(1,2)$, $(3,4)$ and $(5,6)$ generate a subgroup H isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$ that is necessarily contained in $Z(\tau)$. Moreover, the permutations $(1,3,5)(2,4,6)$ and $(1,3)(2,4)$ permute the three 2-cycles by conjugation, so the subgroup K they generate is also contained in $Z(\tau)$. Note that K is clearly isomorphic to S_3. Finally, $H \cap K$ is trivial, as there is an element of H that fixes each of the three 2-cycles, whereas the only element of K that does is the neutral element. We thus have $|HK| = 48$ and so $Z(\tau) = HK$.
We obtain the structure of $Z(\tau)$ by remarking that H is the kernel of the action by conjugation of $Z(\tau)$ on the three 2-cycles, so

$$Z(\tau) \simeq (\mathbb{Z}/2\mathbb{Z})^3 \rtimes \mathsf{S}_3.$$

119

(i) Let H be a subgroup of a p-group G. We know that if $H \neq G$ then H is strictly contained in its normaliser $N(H)$, and it is a normal subgroup of $N(H)$ by definition. If $N(H)$ does not coincide with G, we can repeat the same argument with $N(H)$ instead of H. By iterating this construction, we reach G in a finite number of steps, as the index of the subgroup under consideration strictly decreases at each step.

(ii) Consider the group D_4 with its presentation $\langle r, s \mid r^4 = s^2 = e,\ sr = r^{-1}s\rangle$; it is a 2-group of order 8. The subgroup H generated by s is subnormal by our argument of (i), but it cannot be normal, or D_4 would be isomorphic to $\langle r\rangle \times H$ and thus Abelian: indeed, $\langle r\rangle$ is normal because its index is 2, and it clearly does not contain s, or D_4 would have order 4.

▌In order to show that H is not normal, one could alternatively remark that a subgroup of order 2 is normal if and only if it is contained in the centre. This is not the case here, as $sr \neq rs$.▐

(iii) In this solution, we will apply the Sylow Theorems repeatedly without mentioning them every time. We first show the required statement under the stronger assumption that H is normal in G. Let P be a p-Sylow subgroup of G and let Q be a p-Sylow subgroup of H containing $P \cap H$; such a subgroup Q exists, as the order of $P \cap H$ is a power of p.
The p-Sylow subgroups of G are pairwise conjugate, so there exists an element $g \in G$ such that Q is contained in gPg^{-1}. It follows that $Q \subseteq (gPg^{-1}) \cap H$, and so $Q = (gPg^{-1}) \cap H$, as $(gPg^{-1}) \cap H$ is a subgroup of H whose order is a power of p and contains the p-Sylow subgroup Q of H. But then $Q = (gPg^{-1}) \cap H = (gPg^{-1}) \cap (gHg^{-1}) = g(P \cap H)g^{-1}$, because we have assumed H to be normal in G. This proves that Q and $P \cap H$ have the same cardinality, so $P \cap H$ is a p-Sylow subgroup of H, as claimed.
Now let H be a subnormal subgroup of G; we proceed by induction on the length of the chain of subgroups $H_0 = H \subseteq H_1 \subseteq \cdots \subseteq H_n = G$, each of them normal in the next. If $n = 0$ then $H = G$ and the statement is obviously true. Suppose $n \geq 1$; remark that, by induction, $P' = P \cap H_1$ is a p-Sylow subgroup of H_1. But H is normal in H_1, so $P' \cap H = P \cap H$ is a p-Sylow subgroup of H by our previous argument.

120 Let G be a group of order $300 = 2^2 \cdot 3 \cdot 5^2$. By the Sylow Theorems, the number n_5 of 5-Sylow subgroups in G is congruent to 1 modulo 5 and divides $12 = 300/5^2$. The only possible values are therefore $n_5 = 1$ and $n_5 = 6$.

If $n_5 = 1$, then we have a unique 5-Sylow subgroup, which is thus normal, so G is not simple. Assume that $n_5 = 6$; let P be some 5-Sylow subgroup and let $N(P)$ be its normaliser. We know that the index of $N(P)$ in G coincides with the number of 5-Sylow subgroups, so there are exactly 6 cosets in $G/N(P)$.

The group G acts on the cosets of $N(P)$ by left multiplication, as $g \cdot hN(P) = (gh)N(P)$, so there is a homomorphism

$$\varphi : G \longrightarrow \mathsf{S}\big(G/N(P)\big) \simeq \mathsf{S}_6$$

whose kernel is nontrivial because $|G| = 300$ does not divide $|\mathsf{S}_6| = 720$. It follows that in this case G has a normal subgroup given by the nontrivial kernel of φ, so it is not simple.

121

(i) We know that the nth power of any m-cycle is the product of d disjoint cycles, each of length m/d, where d is the greatest common divisor of m and n.
Now, if σ^p is the product of three disjoing p-cycles, then the remark we just made implies that the all cycles of σ must have length divisible by p. However, the pth power of a p-cycle is trivial, so σ cannot contain any p-cycles, as σ^p does not fix any elements. But then σ cannot contain any $2p$-cycles: if it did,

the fact that it must permute all elements in $\{1, 2, \ldots, 3p\}$ would imply that it also contains a p-cycle. The only remaining possibility is for σ to be a $3p$-cycle; but then its pth power is a product of p disjoint 3-cycles. In conclusion, the equation $\sigma^p = \tau$ admits solutions if and only if $p = 3$.
Fix $p = 3$. Up to renumbering elements, we can assume $\tau = (1, 2, 3)(4, 5, 6)(7, 8, 9)$. Denoting the 9-cycle σ as $(1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3)$, we have $\sigma^3 = (1, a_2, a_3)(b_1, b_2, b_3)(c_1, c_2, c_3)$, so $a_2 = 2, a_3 = 3$. We have six choices for b_1, that is, 4, 5, 6, 7, 8 and 9, and once b_1 is fixed we can deduce b_2 and b_3, as they are the other two elements in the 3-cycle of b_1, in order. At this point, three possibilities remain for c_1: they are the elements of the 3-cycle which has not been chosen, and once c_1 is fixed, c_2 and c_3 can be deduced.
It follows that we have 18 solutions for $p = 3$ and no solutions for $p \neq 3$.

(ii) Up to renumbering integers between 1 and $3p$, we can assume

$$\tau = (1, 2, \ldots, p)(p + 1, \ldots, 2p)(2p + 1, \ldots, 3p).$$

Remark that a p-cycle $(a_1, a_2, \ldots, a_p)$ is conjugated into its inverse by the permutation

$$(a_1, a_p)\,(a_2, a_{p-1}) \cdots (a_{(p-1)/2}, a_{(p+1)/2})$$

whose order is 2. In particular, τ is conjugated into its inverse by a permutation η of order 2. Thus, by considering the presentation of D_p given by $\langle r, s \mid r^p = s^2 = e,\ sr = r^{-1}s\rangle$, we can construct an injective homomorphism that maps $r \longmapsto \tau$ and $s \longmapsto \eta$. The image of this homomorphism is a subgroup of S_{3p} satisfying the required properties.

122

(i) Factor 1045 as $5 \cdot 11 \cdot 19$; let P, Q and R be a 19-Sylow, an 11-Sylow and a 5-Sylow subgroup of G, respectively. The Sylow Theorems imply that P is the only 19-Sylow subgroup, as the number n_{19} of 19-Sylow subgroups is a divisor of 55 that is congruent to 1 modulo 19, so $n_{19} = 1$.
It follows that $H = PQ$ and $K = PR$ are subgroups of G, as P being unique implies its being normal. Moreover, H has order $11 \cdot 19$ and K has order $5 \cdot 19$, so they are both cyclic, as 11 and 5 are not divisors of $19 - 1$. But then every element of P, which clearly commutes with all other elements of P because P has prime order and is thus cyclic, commutes both with all elements of Q and with all elements of R; so P commutes with the group generated by P, Q and R, that is, with G, and is thus contained in the centre.
If P is a proper subgroup of the centre of G, then $G/Z(G)$ has order 1, 5 or 11, so $G/Z(G)$ is cyclic, and thus the only possibility is that $Z(G) = G$. It follows that either $Z(G)$ has order 1045, or it coincides with P and has order 19.

In order to show that this second case is possible, remark that the group of automorphisms of $\mathbb{Z}/209\mathbb{Z}$ has order $\phi(209) = 18 \cdot 10$, so it contains an element φ of order 5 by Cauchy's Theorem. We can thus construct a non-Abelian semidirect product $\mathbb{Z}/209\mathbb{Z} \rtimes \mathbb{Z}/5\mathbb{Z}$ with respect to the homomorphism induced by $\mathbb{Z}/5\mathbb{Z} \ni 1 \longmapsto \varphi \in \mathrm{Aut}(\mathbb{Z}/209\mathbb{Z})$. Since $\mathbb{Z}/209\mathbb{Z} \rtimes \mathbb{Z}/5\mathbb{Z}$ is not Abelian, its centre has order 19 by our previous argument.

(ii) We have $154 = 2 \cdot 7 \cdot 11$, so if G is cyclic then we can define a nontrivial homomorphism by mapping a generator of G to an element of order 11 in $\mathbb{Z}/154\mathbb{Z}$, for example to the class of 14.
Conversely, given a homomorphism $G \longrightarrow \mathbb{Z}/154\mathbb{Z}$ with kernel K, the quotient G/K is isomorphic to a subgroup of $\mathbb{Z}/154\mathbb{Z}$ and, given that its order must divide both $|G|$ and 154, the homomorphism being nontrivial implies that $|G/K|$ is equal to 11.
The normal subgroup K of G has thus order $5 \cdot 19$ and, using again the fact that 5 does not divide $19 - 1$, we obtain that K is cyclic. Letting Q still denote an 11-Sylow subgroup of G, it follows that G is isomorphic to some semidirect product $K \rtimes_\psi Q$, where ψ is a homomorphism from Q to $\mathrm{Aut}(K)$. Now, since Q has order 11, which is coprime with the order $\phi(5 \cdot 19) = 4 \cdot 18$ of $\mathrm{Aut}(K)$, the only possibility is for ψ to be trivial, that is, G is isomorphic to $K \times Q \simeq \mathbb{Z}/95\mathbb{Z} \times \mathbb{Z}/11\mathbb{Z} \simeq \mathbb{Z}/1045\mathbb{Z}$.

123

(i) Up to renumbering the integers $1, 2, \ldots, 16$, we can assume $\sigma = \tau_1 \cdot \tau_2 \cdot \tau_3 \cdot \tau_4$, where $\tau_1 = (1, 2, 3)$, $\tau_2 = (4, 5, 6)$, $\tau_3 = (7, 8, 9, 10, 11)$, $\tau_4 = (12, 13, 14, 15, 16)$.
The centraliser of σ clearly contains the elements τ_1, τ_2, τ_3 and τ_4, which commute with each other and generate a subgroup N of S_{16} isomorphic to $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$. The centraliser also contains $\eta_1 = (1, 4)(2, 5)(3, 6)$ and $\eta_2 = (7, 12)(8, 13)(9, 14)(10, 15)(11, 16)$: indeed, η_1 conjugates τ_1 into τ_2 and fixes τ_3 and τ_4, whereas η_2 fixes τ_1 and τ_2 while conjugating τ_3 into τ_4. It follows that the subgroup H generated by η_1, η_2 normalises N. Moreover, H is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ because η_1 and η_2 commute.
Remark that $N \cap H = \{e\}$ because the two subgroups' orders are coprime. It follows that $N \cdot H$ is a subgroup of S_{16} isomorphic to the semidirect product $N \rtimes H$.
We now show that the centraliser $Z(\sigma)$ is the subgroup $N \cdot H$. We have already found that $N \cdot H$ is contained in the centraliser of σ because N and H are. Moreover, the order of $N \cdot H$ is $2^2 3^2 5^2$, and the conjugacy class of σ in S_{16} is the set of all $3 + 3 + 5 + 5$-cycles, whose cardinality is

$$\frac{1}{2}\binom{16}{5}4!\binom{11}{5}4!\frac{1}{2}\binom{6}{3}2!\binom{3}{3}2! = \frac{16!}{2^2 3^2 5^2};$$

in other words, $Z(\sigma)$ has the same cardinality as $N \cdot H$, and therefore the two subgroups coincide. In particular, $Z(\sigma)$ is isomorphic to the semidirect product

$$\big(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}\big) \rtimes \big(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}\big)$$

with respect to the homomorphism induced by the mappings

$$(1,0) \longmapsto \big((a,b,c,d) \longmapsto (b,a,c,d)\big),$$
$$(0,1) \longmapsto \big((a,b,c,d) \longmapsto (a,b,d,c)\big).$$

(ii) The element $\rho = \tau_1 \tau_2^{-1} \tau_3 \tau_4^{-1}$ has order 15 and belongs to $Z(\sigma)$. The element $\tau = \eta_1 \eta_2$ also belongs to the centraliser; its order is 2 and by the relations considered in (i) it is such that $\tau \rho \tau^{-1} = \rho^{-1}$. Letting $\langle r, s \,|\, r^{15} = s^2 = e,\ sr = r^{-1}s \rangle$ be the usual presentation of D_{15}, we can extend the mapping $r \longmapsto \rho$, $s \longmapsto \tau$ to a homomorphism from D_{15} to the subgroup of $Z(\sigma)$ generated by ρ and τ, which is in fact an isomorphism between D_{15} and a subgroup of $Z(\sigma)$.

124

(i) From now on, we assume by symmetry that $p < q$. The number n_q of q-Sylow subgroups of G is congruent to 1 modulo q and divides p^2, so the fact that $p < q$ implies that either $n_q = 1$ or $n_q = p^2$. If $n_q = 1$ then the q-Sylow subgroup is unique and therefore normal, and so G is not simple. On the other hand, if $n_q = p^2$, then q divides $p^2 - 1 = (p-1)(p+1)$ and, since $p < q$, the only possibility is for q to divide $p + 1$; but, using again the fact that $p < q$, this yields that $p = 2$ and $q = 3$.
Even in this case, however, G is not simple: G has four 3-Sylow subgroups and acts on them by conjugation via an action $\varphi : G \longrightarrow \mathsf{S}_4$. If G were simple φ would be injective, which is not possible because $|G| = 4 \cdot 9 = 36$ whereas $|\mathsf{S}_4| = 24$.

(ii) For all distinct primes p and q we have the four different Abelian groups $(\mathbb{Z}/p\mathbb{Z})^2 \times (\mathbb{Z}/q\mathbb{Z})^2$, $\mathbb{Z}/p^2Z \times (\mathbb{Z}/q\mathbb{Z})^2$, $(\mathbb{Z}/p\mathbb{Z})^2 \times \mathbb{Z}/q^2\mathbb{Z}$ and $\mathbb{Z}/p^2\mathbb{Z} \times \mathbb{Z}/q^2\mathbb{Z}$, so $n(p,q) \geq 4$.
We shall now prove that for $p = 5$ and $q = 7$ there are no non-Abelian groups of order p^2q^2, which yields that the desired minimum is precisely 4. Let G be a group of order $5^2 \cdot 7^2$. By our arguments of (i), since we are not in the case $p = 2$ and $q = 3$, there is a unique 7-Sylow subgroup Q, which is thus normal; in particular, G is isomorphic to a semidirect group $Q \rtimes P$, where P is a 5-Sylow subgroup.
Moreover, P and Q both have order the square of a prime, so they are both Abelian. The group Q is isomorphic to either $(\mathbb{Z}/7\mathbb{Z})^2$ or $\mathbb{Z}/7^2\mathbb{Z}$; in any case, 5 does not divide the order of $\mathrm{Aut}(Q)$, which is $|\mathsf{GL}_2(\mathbb{F}_7)| = (7^2-1)(7^2-7)$ if Q is not cyclic and $|(\mathbb{Z}/7^2\mathbb{Z})^*| = \phi(7^2) = 6 \cdot 7$ otherwise. It follows that the

only homomorphism $P \longrightarrow \mathrm{Aut}(Q)$ is the trivial one, and G is thus Abelian because it is the direct product of two Abelian groups.

125 First of all, remark that the subgroups $e \times \mathbb{Z}/2\mathbb{Z}$ and $\mathsf{A}_4 \times 0$ are characteristic in $G = \mathsf{A}_4 \times \mathbb{Z}/2\mathbb{Z}$. Indeed, the former is the centre of G, as $Z(G) = Z(\mathsf{A}_4) \times Z(\mathbb{Z}/2\mathbb{Z}) = e \times \mathbb{Z}/2\mathbb{Z}$, so it is characteristic. Moreover, the only elements of order 3 in G are of the form $(\tau, 0)$, where τ is a 3-cycle, so they are permuted by automorphisms of G; since A_4 is generated by the 3-cycles, the subgroup $\mathsf{A}_4 \times 0$ is characteristic in G.

It follows that

$$\mathrm{Aut}(G) = \mathrm{Aut}(\mathsf{A}_4 \times \mathbb{Z}/2\mathbb{Z}) \simeq \mathrm{Aut}(\mathsf{A}_4) \times \mathrm{Aut}(\mathbb{Z}/2\mathbb{Z}) \simeq \mathrm{Aut}(\mathsf{A}_4) \times \{\mathrm{Id}_{\mathbb{Z}/2\mathbb{Z}}\},$$

and what we need to show in order to reach the desired conclusion is that $\mathrm{Aut}(\mathsf{A}_4)$ is isomorphic to S_4.

Let $\varphi : \mathsf{S}_4 \longrightarrow \mathrm{Aut}(\mathsf{A}_4)$ be the action of S_4 on its normal subgroup A_4 by conjugation. The group A_4 has trivial centre, so $\mathsf{A}_4 \simeq \mathrm{Inn}(\mathsf{A}_4) = \varphi(\mathsf{A}_4)$, which is a subset of $\varphi(\mathsf{S}_4)$, the latter being isomorphic to S_4/K, where $K = \mathrm{Ker}(\varphi)$. The kernel K has thus at most two elements; but S_4 does not have any normal subgroups of order 2, so the only possibility is for K to be trivial and for $\mathrm{Aut}(\mathsf{A}_4)$ to contain a subgroup isomorphic to S_4.

On the other hand, A_4 is generated by $(1, 2, 3)$ and $(1, 2)(3, 4)$. Indeed, $\langle (1, 2, 3), (1, 2)(3, 4) \rangle$ is a subgroup whose order is a multiple of 6, because it contains an element of order 2 and an element of order 3. Moreover, if its order were exactly 6, then it would be normal in A_4 as its index would be 2; but both $(1, 2, 3)$ and $(1, 2)(3, 4)$ have 4 conjugates in A_4, so the total number of conjugates is more than 6, contradiction. We thus find that the subgroup in question has order 12 and in fact coincides with A_4. It follows that any automorphism of A_4 is uniquely determined by the images of $(1, 2, 3)$ and $(1, 2)(3, 4)$. Since A_4 contains eight 3-cycles and three $2 + 2$-cycles, we find that A_4 has at most 24 automorphisms. It follows that $\mathrm{Aut}(\mathsf{A}_4)$ is isomorphic to S_4, as claimed.

126

(i) We factor 339 as $3 \cdot 7 \cdot 19$. By the Sylow Theorems, the number of 19-Sylow subgroups of G is congruent to 1 modulo 19 and divides 21, so there is a unique 19-Sylow subgroup P which is therefore normal.
Let Q be a 7-Sylow subgroup; since P is normal and $P \cap Q = \{e\}$ (as the orders of the two subgroups are coprime), the set $N = PQ$ is a subgroup of G of order 133 and is a semidirect product of P and Q. Moreover, P is cyclic, so its automorphism group is isomorphic to $(\mathbb{Z}/19\mathbb{Z})^*$, whose order $\phi(19) = 18$ is not divisible by 7. It is thus clear that there are no nontrivial homomorphisms from Q to $\mathrm{Aut}(P)$, and so N is the direct product of P and Q, that is, a cyclic group of order $19 \cdot 7$.
The index of N in G is 3, which is the smallest prime divisor of the order of G; it follows that N is normal in G.

Now let R be a 3-Sylow subgroup; we have $R \cap N = \{e\}$ because 3 and $|N| = 19 \cdot 7$ are coprime. Consequently, $G = NR$ and G is isomorphic to a semidirect product $N \rtimes R \simeq \mathbb{Z}/133\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$ of cyclic groups.

(ii) Let $\varphi : R \longrightarrow \mathrm{Aut}(N)$ be the homomorphism induced by conjugation in G. The group N is cyclic of order 133, so $\mathrm{Aut}(N)$ is isomorphic to $(\mathbb{Z}/7\mathbb{Z})^* \times (\mathbb{Z}/19\mathbb{Z})^*$. By fixing a generator for R, we can think of φ as the homomorphism from $\mathbb{Z}/3\mathbb{Z}$ that extends $1 \longmapsto (a, b)$ for some $a \in (\mathbb{Z}/7\mathbb{Z})^*$ and $b \in (\mathbb{Z}/19\mathbb{Z})^*$.

The possible values for a, that is, the elements of $(\mathbb{Z}/7\mathbb{Z})^*$ whose order divides 3, are 1, 2 and 4; the possible values for b in $(\mathbb{Z}/19\mathbb{Z})^*$ are 1, 7 and 11. The group G is Abelian if and only if $(a, b) = (1, 1)$, in which case the centre is all of G. Note that in all other cases we may assume that the centre does not contain any elements of order 3; indeed, if such an element were to be found, we could express the 3-Sylow subgroup R as the subgroup generated by this element of order 3, and thus have $(a, b) = (1, 1)$, which implies G is Abelian.

If $a = 1, b \neq 1$, then the group G is not Abelian, but R commutes with Q and thus Q is in the centre of G. Since the quotient by the centre cannot be cyclic and thus the centre cannot have prime index in G, the centre of G must be Q, which is a group of order 7.

If $a \neq 1$ and $b = 1$, then the group G is not Abelian, but R commutes with P, which is thus contained in the centre of G. Again, since the quotient by the centre cannot be cyclic, the centre cannot have prime index in G, and thus the centre of G must be P, which is a group of order 19.

If $a \neq 1$ and $b \neq 1$ then the centre cannot contain elements of order 7 or of order 19. Moreover, as remarked above, we can assume that the centre does not contain any elements of order 3. We conclude that the centre of G must be trivial.

In conclusion, the possible values for the order of the centre are 399, 7 and 19, with the respective example constructed above by choosing the appropriate values of (a, b).

127 Denote by σ an n-cycle and by H the smallest normal subgroup of S_n containing σ. Remark that $H \cap \mathsf{A}_n$ is a normal subgroup of A_n; if $n \geq 5$ then we know that A_n is a simple group, so given that $H \cap \mathsf{A}_n$ contains the element $\sigma^2 \neq e$, we have $H \cap \mathsf{A}_n = \mathsf{A}_n$.

It follows that, for $n \geq 5$, the subgroup H must contain both σ and A_n, and thus we have to consider two cases: if n is odd, in which case σ is an even permutation, then H is the subgroup A_n; if n is even and thus σ odd, then the subgroup H, which contains both A_n and σ, is all of S_n.

For $n = 3$ it is clear that $H = \mathsf{A}_3$. For $n = 4$, the subgroup H must contain all 4-cycles, so it is not contained in A_4, and it also contains all products of 4-cycles, including for instance $(1, 2, 3, 4)(1, 3, 2, 4) = (1, 4, 2)$. But then H is a normal subgroup containing A_4 (because it contains a 3-cycle), and thus $H = \mathsf{S}_4$.

128 Let G be a group of order $5^2 \cdot 7 \cdot 17$. By the Sylow Theorems, the number of 5-Sylow subgroups of G must be congruent to 1 modulo 5 and divide $7 \cdot 17$. But

none among the divisors 7, 17, $7 \cdot 17$ is congruent to 1 modulo 5, so there is a unique 5-Sylow subgroup P, which is therefore normal in G.

The order of P is the square of a prime, so P is Abelian and isomorphic to either $(\mathbb{Z}/5\mathbb{Z})^2$ or $\mathbb{Z}/25\mathbb{Z}$. The automorphism group of P is thus isomorphic to either $\mathsf{GL}_2(\mathbb{F}_5)$, whose order is $(5^2-1)(5^2-5)$, or to $(\mathbb{Z}/25\mathbb{Z})^*$, whose order is $\phi(25) = 20$. Whatever the case, neither 7 nor 17 divides $|\operatorname{Aut}(P)|$, so P is contained in the centre of G, as it is centralised by all Sylow subgroups.

Now, since P is contained in the centre $Z(G)$, the quotient $G/Z(G)$ is isomorphic to a quotient of G/P, that is, to a quotient of a group of order $7 \cdot 17$, which is cyclic because 7 does not divide $17-1$. It follows that $G/Z(G)$ is also cyclic, so G is Abelian.

129

(i) First of all, remark that $\tau\sigma\tau^{-1} = \sigma^{-1}$, so τ is an element of the normaliser $N(\langle\sigma\rangle)$ of the subgroup of order 4 generated by σ; it follows that $\langle\sigma\rangle \cdot \langle\tau\rangle$ is a group, and that it coincides with H. Moreover, $\langle\sigma\rangle \cap \langle\tau\rangle = \{e\}$ because τ does not belong to $\langle\sigma\rangle$, so the order of H is $4 \cdot 2 = 8$.
Clearly, the centraliser $Z_{\mathsf{S}_6}(H)$ is the intersection of the two centralisers $Z(\sigma)$ and $Z(\tau)$. We shall first identify elements of $Z(\sigma)$, and for each check whether it commutes with τ. The conjugacy class of the 4-cycle σ consists of all 4-cycles and thus has cardinality $\binom{6}{4} \cdot 3!$; it follows that the order of $Z(\sigma)$ is

$$\frac{6!}{\binom{6}{4} \cdot 3!} = 8.$$

Clearly, $\sigma, (5,6) \in Z(\sigma)$ and, since σ and $(5,6)$ commute, $\langle\sigma, (5,6)\rangle$ has order 8, so we have $Z(\sigma) = \langle\sigma, (5,6)\rangle$. Note that σ does not commute with τ but σ^2 and $(5,6)$ do; this implies that $Z_{\mathsf{S}_6}(H) = \langle(1,3)(2,4), (5,6)\rangle$ and therefore its order is 4.

(ii) For each γ in the normaliser $N(H)$ of H, the permutation $\gamma\sigma\gamma^{-1}$ is either σ or σ^{-1}, as σ and σ^{-1} are the only elements of order 4 in H. It follows that $N(H)$ is contained in $N(\langle\sigma\rangle) = Z(\langle\sigma\rangle) \cup \tau Z(\langle\sigma\rangle)$, whose order is 16. This in turn implies that $N(H)$, which obviously contains H, has order 8 if it coincides with H and order 16 if it doesn't, in which case it coincides with $N(\langle\sigma\rangle)$. Since $Z_{\mathsf{S}_6}(H)$ is contained in $N(H)$, the permutation $(5,6)$ is in $N(H)$; however, $(5,6)$ does not belong to H, so $N(H)$ is generated by σ, τ and $(5,6)$, and its order is 16.
This implies that $N(H)$ is the only 2-Sylow subgroup of S_6 that contains H: indeed, given a 2-Sylow subgroup P containing H, the index of H in P must be 2, so H is normal in P, but then $P \subseteq N(H)$, and since both groups have order 16 they must coincide.

130

(i) Let M be a maximal subgroup of G. As for any subgroup, the set $Z(G) \cdot M$ is still a subgroup of G, because the centre is normal in G. In particular, the fact that $M \subseteq Z(G) \cdot M \subseteq G$ yields that either $M = Z(G) \cdot M$ or $Z(G) \cdot M = G$. Clearly, $M = Z(G) \cdot M$ if and only if $Z(G) \subseteq M$; thus, if $Z(G) \not\subseteq M$, then $Z(G) \cdot M = G$. We wish to show that in this case G' is contained in M: what we need to do is show that for all $g, h \in G$ the commutator of g and h belongs to M. Since $G = Z(G) \cdot M$ we have $g = zm$ and $h = wn$ for some $z, w \in Z(G)$ and $m, n \in M$; therefore, $ghg^{-1}h^{-1} = zmwn(zm)^{-1}(wn)^{-1} = mnm^{-1}n^{-1} \in M$.

(ii) Letting $G = \mathsf{S}_3 \times \mathbb{Z}/2\mathbb{Z}$, we have $Z(G) = e \times \mathbb{Z}/2\mathbb{Z}$ and $G' = (\mathsf{S}_3)' \times (\mathbb{Z}/2\mathbb{Z})' = \mathsf{A}_3 \times 0$. The subgroup $M = \mathsf{S}_3 \times 0$, which is maximal because the order of $G/M \simeq \mathbb{Z}/2\mathbb{Z}$ is prime, satisfies the required properties.

131

(i) Consider the following matrices in $\mathsf{SL}_2(\mathbb{F}_5)$:

$$A_i = \begin{pmatrix} 2 & \\ & -2 \end{pmatrix}, \quad A_j = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}, \quad A_k = \begin{pmatrix} & -2 \\ -2 & \end{pmatrix};$$

we show that mapping $i \longmapsto A_i$, $j \longmapsto A_j$ and $k \longmapsto A_k$ extends to an injective homomorphism from Q_8 to $\mathsf{SL}_2(\mathbb{F}_5)$.
First of all, $A_k = A_i \cdot A_j$ and $A_i^2 = A_j^2 = A_k^2 = -\mathrm{Id}$, so the squares are elements in the centre of $\mathsf{SL}_2(\mathbb{F}_5)$. Moreover, we have $A_j \cdot A_i = -A_k$; the other commutation relations that define Q_8 follow. We thus have a homomorphism $\varphi : \mathsf{Q}_8 \longrightarrow \mathsf{SL}_2(\mathbb{F}_5)$ whose image is given by the 8 distinct matrices $\pm \mathrm{Id}$, $\pm A_i$, $\pm A_j$ and $\pm A_k$, so the homomorphism is injective, because Q_8 also has order 8.

(ii) The group S_5 has order 120, so its 2-Sylow subgroups have order 8. In particular, they are all conjugate to the subgroup generated by $(1, 2)$ and $(1, 2, 3, 4)$, which is easily proven to be isomorphic to D_4.
The group $\mathsf{SL}_2(\mathbb{F}_5)$ has order $24 \cdot 20/4 = 120$, so its 2-Sylow subgroups also have order 8. We have shown in (i) that a subgroup of $\mathsf{SL}_2(\mathbb{F}_5)$ is isomorphic to Q_8.
But the groups D_4 and Q_8 are not isomorphic: the only element of order 2 in Q_8 is -1, whereas D_4 has 5 different elements of order 2. It follows that S_5 and $\mathsf{SL}_2(\mathbb{F}_5)$ cannot be isomorphic, because their 2-Sylow subgroups are not.

132

(i) Assume that every Sylow subgroup of G is cyclic. The exponent of G is then clearly divisible by the cardinality of each Sylow subgroup, but the least common multiple of these cardinalities is the order of G, which is therefore a divisor of the exponent of G. On the other hand, the exponent is a divisor of $|G|$, as $g^{|G|} = e$ for all $g \in G$. We have thus shown that the exponent of G is the same as its order.

Conversely, assume that the exponent of G coincides with the order of G; we wish to show that every p-Sylow subgroup of G is cyclic. Note that the exponent of G is the least common multiple of the orders of elements of G. It follows that, for each prime p, if p^a divides the exponent of G, then there is an element g in G whose order is a multiple of p^a, so $g^{\mathrm{ord}(g)/p^a}$ is an element of order p^a. In particular, if p^a is the order of a p-Sylow subgroup, we find that G contains an element of order p^a, so there is a cyclic p-Sylow subgroup; but then, since they are all conjugate to each other, all p-Sylow subgroups are cyclic.

(ii) The order of an element (g_1, g_2) of a direct product $G_1 \times G_2$ is the least common multiple of the orders of g_1 and g_2; from this it follows that the exponent of $G_1 \times G_2$ is the least common multiple of the exponents of G_1 and G_2, and also that the maximal order of an element in $G_1 \times G_2$ is the least common multiple of the maximal orders in G_1 and G_2.
In particular, if the order of G_1 and the order of G_2 are coprime, then both the exponent and the maximal order of an element in $G_1 \times G_2$ are given by the product of the corresponding quantities for G_1 and G_2.
We now remark that, by the Structure Theorem for Finite Abelian Groups, G is the direct product of its p-torsion subgroups, that is, its p-Sylow subgroups, with p ranging among all primes that divide $|G|$. Each factor is isomorphic to a direct product $\mathbb{Z}/p^{a_1}\mathbb{Z} \times \mathbb{Z}/p^{a_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{a_r}\mathbb{Z}$ for some positive integers $a_1 \geq a_2 \geq \cdots \geq a_r$, and both its maximal order and its exponent are equal to p^{a_1}. It now follows from our argument above that the the exponent of G is equal to the maximal order of an element in G.

133 We list all possible orders of elements of S_7 according to their cycle decomposition

g	$\mathrm{ord}(g)$
e	1
$(1,2)$	2
$(1,2)(3,4)$	2
$(1,2)(3,4)(5,6)$	2
$(1,2,3)$	3
$(1,2,3)(4,5)$	6
$(1,2,3)(4,5)(6,7)$	6
$(1,2,3)(4,5,6)$	3
$(1,2,3,4)$	4
$(1,2,3,4)(5,6)$	4
$(1,2,3,4)(5,6,7)$	12
$(1,2,3,4,5)$	5
$(1,2,3,4,5)(6,7)$	10
$(1,2,3,4,5,6)$	6
$(1,2,3,4,5,6,7)$	7

and find that there exist cyclic, and therefore Abelian, subgroups of S_7 of order 1, 2, 3, 4, 5, 6, 7, 10 and 12. Moreover, the permutations $(1, 2, 3, 4)$ and $(5, 6)$ commute, so they generate the Abelian group $\langle (1, 2, 3, 4)\rangle \cdot \langle (5, 6)\rangle$ of order 8. Similarly, the subgroup generated by $(1, 2, 3)$ and $(4, 5, 6)$ is Abelian and equal to $\langle (1, 2, 3)\rangle \cdot \langle (4, 5, 6)\rangle$, so its order is 9. We have thus constructed an Abelian subgroup for each order up to 12 except for 11, as requested.

Conversely, we now show that there are no Abelian subgroups whose order is not one of those covered above. In particular, we need to exclude the existence of subgroups of order given by a divisor of $7! = 2^4 \cdot 3^2 \cdot 5 \cdot 7$ that is strictly greater than 12.

Any subgroup H of S_7 whose order is a multiple of 7 contains a 7-cycle σ, as the 7-cycles are the only elements of order 7 in S_7. It follows that if H is Abelian then it must be contained in $Z(\sigma)$, whose order is

$$\frac{7!}{|\mathcal{C}\ell(\sigma)|} = \frac{7!}{\binom{7}{7} \cdot 6!} = 7$$

and so we must have $H = \langle \sigma \rangle$, and the order must be 7. Similarly, if 5 divides the order of an Abelian subgroup K of S_7, then the subgroup must contain a 5-cycle τ and be contained $Z(\tau)$, but the order of $Z(\tau)$ is

$$\frac{7!}{|\mathcal{C}\ell(\tau)|} = \frac{7!}{\binom{7}{5} \cdot 4!} = 10,$$

so K must have order 5 or 10.

We now show that S_7 has no Abelian subgroups of order 16, which will imply that it has no Abelian groups whose order is a multiple of 16, as an Abelian group has at least a subgroup for each divisor of its order. Now, a subgroup of order 16 is a 2-Sylow subgroup of S_7, so it is enough to show that the 2-Sylow subgroups, which are all pairwise conjugate and therefore isomorphic, are not Abelian. Indeed, S_7 contains a subgroup isomorphic to D_4, for example the one generated by $(1, 2, 3, 4)$ and $(2, 3)$, which has order 8 and is not Abelian, but, since its order is a power of 2, is contained in some 2-Sylow subgroup, which therefore cannot be Abelian.

What remains to be done is to exclude the possible orders $2 \cdot 3^2$, $2^2 \cdot 3^2$, $2^3 \cdot 3$ and $2^3 \cdot 3^2$, and it is enough to exclude $18 = 2 \cdot 3^2$ and $24 = 2^3 \cdot 3$, as the last two orders are their multiples.

By the same argument as above, an Abelian subgroup of order 18 would contain a permutation of order 2, that is, a 2-cycle, a $2 + 2$-cycle or a $2 + 2 + 2$-cycle, and would have to be contained in the centraliser of such an element of order 2. But the orders of the centralisers of elements with these cycle structures are

$$\frac{7!}{\binom{7}{2}} = 2 \cdot 5!, \quad \frac{7!}{\binom{7}{2}\binom{5}{2} \cdot \frac{1}{2}} = 4 \cdot 3!, \quad \frac{7!}{\binom{7}{2}\binom{5}{2}\binom{3}{2} \cdot \frac{1}{3!}} = 8 \cdot 3!;$$

none is divisible by 9. This proves that there are no Abelian subgroups of order 18.

Finally, an Abelian subgroup of order 24 contains a 6-cycle or a $3+2$-cycle and must be contained in the centraliser of such an element. But the centralisers of such elements have orders

$$\frac{7!}{\binom{7}{6}\cdot 5!} = 6, \qquad \frac{7!}{\binom{7}{3}\cdot 2!\cdot\binom{4}{2}} = 12$$

and neither possible value is divisible by 24; it follows that S_7 has no Abelian subgroups of order 24.

134

(i) Let H be a subgroup of G and let P be a Sylow subgroup of H. Since its order is a prime power, P is contained in a Sylow subgroup of G, and thus it is cyclic as every subgroup of a cyclic group is cyclic. We have thus shown that H is hypercyclic.
Let G/N be a quotient of G and let $\pi : G \longmapsto G/N$ be the quotient homomorphism; let p be a prime and suppose p^a, p^b are the greatest powers of p dividing $|G|$ and $|N|$, respectively. Given a p-Sylow subgroup $\widetilde{P}$ of G/N, we have $|\widetilde{P}| = p^{a-b}$ and its pre-image $H = \pi^{-1}(\widetilde{P})$ has cardinality $p^{a-b}\cdot|N|$, so H contains a p-Sylow subgroup P of G.
We will now show that $\pi(P) = \widetilde{P}$. Indeed, we have $|\pi(P)| = |PN/N| = |P/N\cap P| = |P|/|N\cap P|$ and, assuming $|N| = p^b\cdot r$ where p does not divide r, it is clear that the order of $N\cap P$ divides $(p^a, p^b\cdot r) = p^b$. This implies that the order of $\pi(P)$ is a multiple of p^{a-b}, which is the order of $\widetilde{P}$, so, since $\pi(P)\subseteq\pi(H) = \widetilde{P}$, we obtain that $\pi(P) = \widetilde{P}$.
We can now conclude that $\widetilde{P}$, being the image of the cyclic subgroup P, is itself cyclic, and therefore G/N is hypercyclic.

(ii) Let H_1 and H_2 be subgroups of G whose order is p^r, and let P_1, P_2 be p-Sylow subgroups containing H_1 and H_2, respectively. The subgroups P_1 and P_2, being p-Sylow subgroups, are conjugate. Moreover, their being cyclic implies that each contains a unique subgroup of order p^r. It follows that conjugating P_1 into P_2 must also send H_1 to H_2.

(iii) By comparing orders, we find that the equation is satisfied if and only if $N\cap P$ is a p-Sylow subgroup of N. We show that this is always the case.
Let K be a p-Sylow subgroup of N. By (ii), K is conjugate to a subgroup K' of P. Since N is normal, K' is also a subgroup of N, so $N\cap P$ contains K'.
Now, K' is also a p-Sylow subgroup of N, again because N is normal, and thus we have the inclusion $N\cap P\subseteq K'$ because $N\cap P$ is a p-group and its cardinality cannot be greater than that of the p-Sylow subgroup K'.
We have thus shown that $N\cap P = K'$, and in particular $N\cap P$ is a p-Sylow subgroup of N.

135

(i) The divisors of 30 are 1, 2, 3, 5, 6, 10, 15 and 30. Obviously, $\{e\}$ has order 1 and D_{15} has order 30; moreover, by Cauchy's Theorem, D_{15} has subgroups of order 2, 3 and 5.
Note that D_{15} is isomorphic to the semidirect product $\mathbb{Z}/15\mathbb{Z} \rtimes \mathbb{Z}_2 \simeq (\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}) \rtimes \mathbb{Z}/2\mathbb{Z}$, where the generator of $\mathbb{Z}/2\mathbb{Z}$ acts by conjugation on both $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/5\mathbb{Z}$ via $n \longmapsto -n$.
It is then clear that the subsets $(\mathbb{Z}/3\mathbb{Z} \times 0) \rtimes \mathbb{Z}/2\mathbb{Z}$, $(0 \times \mathbb{Z}/5\mathbb{Z}) \rtimes \mathbb{Z}/2\mathbb{Z}$ and $(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}) \rtimes 0$ are subgroups of $(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}) \rtimes \mathbb{Z}/2\mathbb{Z}$ of order 6, 10 and 15, respectively; thus D_{15} does contain subgroups of these three orders.

(ii) Clearly, the subgroups of order 1 and 30 are unique.
Moreover, D_{15} contains 15 symmetries, that is, 15 elements of order 2, and the subgroup consisting of the remaining rotations has an element of order 1, $2 = \phi(3)$ elements of order 3, $4 = \phi(5)$ elements of order 5, $8 = \phi(15)$ elements of order 15.
It follows that D_{15} contains a unique subgroup of order 3, a unique subgroup of order 5 and a unique subgroup of order 15, as a group of order n, for $n = 3, 5, 15$, is cyclic and contains $\phi(n)$ elements of order n.
We will now show that the only values of n for which there is a unique subgroup of order n are 1, 3, 5, 15 and 30; in other words, that for possible even orders n other than 30 there are multiple subgroups of order n. We do this by showing that a subgroup H with such an order cannot be normal.
Since the order of H is even, H contains an element s of order 2. The centraliser of s also has order 2, because s does not commute with any element of order 3 or 5, as these elements belong to the rotation subgroup and the centre of the group is trivial. It follows that the number of conjugates of s is $[\mathsf{D}_{15} : Z(s)] = 15$, that is, that all elements of order 2 are pairwise conjugate.
Assume by contradiction that H is normal; it would then contain all elements of order 2, but these generate D_{15}, so we would have $H = \mathsf{D}_{15}$, contradicting the requirement that H be a proper subgroup.

3.2 Rings

136 Set $p(x) = (x^2 - 2)(x^3 - 2)$ and denote by A the quotient ring $\mathbb{K}[x]/(p(x))$. An element $f(x) + \big(p(x)\big)$ of A is a zero divisor if and only if $f(x)$ and $p(x)$ are not relatively prime, whereas $f(x) + \big(p(x)\big)$ is invertible if and only if $f(x)$ and $p(x)$ are relatively prime.

As we can choose $f(x)$ to have degree less than $5 = \deg(p(x))$, we find that A has $N = p^5$ elements, where $p = 3$ if $\mathbb{K} = \mathbb{F}_3$ and $p = 7$ if $\mathbb{K} = \mathbb{F}_7$. Letting D be the number of zero divisors in A, the difference $N - D$ is the number of invertible elements, so all we need to do is find the number of zero divisors in A. In order to obtain it, we start by writing $p(x)$ as a product of irreducible polynomials.

First, consider the case $\mathbb{K} = \mathbb{F}_3$. The polynomial $x^2 - 2$ is irreducible, because 2 is not a square in $\mathbb{F}_3$, whereas we have $x^3 - 2 = (x - 2)^3$. Thus $f(x)$ and $p(x)$ are not relatively prime if and only if $f(x)$ is divisible by $x^2 - 2$ or $x - 2$.

In the first case, we have $f(x) = (x^2 - 2)g(x)$ where $g(x)$ is one of 3^3 polynomials of degree at most 2 in $\mathbb{F}_3[x]$. In the second case, $f(x) = (x - 2)h(x)$ where $h(x)$ is a polynomial of degree at most 3: there are 3^4 such polynomials in $\mathbb{F}_3[x]$. If $f(x)$ is a multiple of both x^2-2 and $x-2$, then $f(x) = (x^2-2)(x-2)k(x)$ where $k(x)$ is a polynomial of degree at most 1, and therefore can take 3^2 values. The Inclusion-Exclusion Principle then yields $D = 3^4 + 3^3 - 3^2 = 99$, which implies that the number of invertible elements in A is $3^5 - 99 = 144$.

We now consider the case $\mathbb{K} = \mathbb{F}_7$. The polynomial $x^3 - 2$ is irreducible because it has no roots, whereas $x^2 - 2 = (x+3)(x-3)$. We proceed as above by remarking that $f(x)$ and $p(x)$ are not relatively prime if and only if $f(x)$ is divisible by $x^3 - 2$ or $x + 3$ or $x - 3$. In the first case, $f(x) = (x^3 - 2)g(x)$ where $g(x)$ is one of 7^2 polynomials of degree at most 1. In the second case, $f(x) = (x + 3)h(x)$ where $h(x)$ is one of 7^4 polynomials of degree at most 3; the third case yields again 7^4 polynomials $f(x)$ that are divisible by $x - 3$.

If $f(x)$ is a multiple of both $x^3 - 2$ and $x + 3$, then $f(x) = (x^3 - 2)(x + 3)k(x)$ where $k(x)$ has degree 0, that is, it is a constant: there are 7 such polynomials. The same is true for multiples of both $x^3 - 2$ and $x - 3$. If $f(x)$ is divisible by $x + 3$ and $x - 3$ then $f(x) = (x + 3)(x - 3)l(x)$ where $l(x)$ has degree at most 2, so $f(x)$ is one of 7^3 polynomials. Finally, since the degree of $f(x)$ is at most 4, if it is a multiple of $x^3 - 2$, $x + 3$ and $x - 3$ then it is the constant polynomial 0.

It follows that $D = 7^2 + 7^4 + 7^4 - 7 - 7 - 7^3 + 1 = 4495$ and that the number of invertible elements is $7^5 - 4495 = 12312$.

137

(i) We proceed by induction on n. The base case $n = 0$ is trivial: $p_0(x) = 1 = (x^3 - 1)^0$. Assume the assertion holds for n, that is, assume that $p_{3n}(x) = (x^3 - 1)^n$; we will deduce that the formula also hold for $n + 1$.
Indeed, using the recursion defining our polynomials three times, we have

$$\begin{aligned} p_{3n+3}(x) &= (x - 1)\,p_{3n+2}(\zeta x) \\ &= (x - 1)(\zeta x - 1)\,p_{3n+1}(\zeta^2 x) \\ &= (x - 1)(\zeta x - 1)(\zeta^2 x - 1)\,p_{3n}(x) \\ &= (x - 1)(x - \zeta)(x - \zeta^2)\,p_{3n}(x) \\ &= (x^3 - 1)\,p_{3n}(x) \\ &= (x^3 - 1)(x^3 - 1)^n \\ &= (x^3 - 1)^{n+1}. \end{aligned}$$

(ii) By what we proved in (i), we have $p_n(x) \in \mathbb{Z}[x]$ if $n \equiv 0 \pmod 3$. If $n \equiv 1 \pmod 3$, that is, $n = 3m + 1$ for some m, then $p_n(x) = (x - 1)\,p_{3m}(\zeta x) = (x - 1)((\zeta x)^3 - 1)^m = (x - 1)(x^3 - 1)^m \in \mathbb{Z}[x]$. Finally, if $n \equiv 2 \pmod 3$,

that is, if $n = 3m + 2$ for some m, then $p_n(x) = (x-1)p_{3m+1}(\zeta x) = (x-1)(\zeta x-1)(x^3-1)^m$. The latter polynomial is not in $\mathbb{Z}[x]$, as the coefficient of its monomial of maximal degree is ζ.

138 The fact that a maximal ideal is prime is true in general, so we show the converse.

Let P be a prime ideal of A and let x be an element of A that is not in P. If we show that the ideal (x, P), that is, the ideal generated by P and x, coincides with the ring A itself, we obtain as a consequence that P is a maximal ideal.

We know that there is $n > 1$ such that $x^n = x$, and so $x(x^{n-1} - 1) = 0 \in P$. Since P is a prime ideal and $x \notin P$, we have $y = x^{n-1} - 1 \in P$. It follows that $1 = x^{n-1} - y \in (x, P)$, which implies $(x, P) = A$ as wanted.

139

(i) It is clear that $\mathbb{Z}[x]/I \simeq \mathbb{Z}/5\mathbb{Z}$ is a field, so I is maximal.

(ii) We have $\mathbb{Z}[x]/J \simeq \mathbb{Z}/25\mathbb{Z}$, and in the latter ring all zero divisors, that is, 0 and 5, are nilpotent. It follows easily that J is primary: if $f(x)g(x) \in J$ and $f(x) \notin J$, then $f(x)g(x) = 0$ in $\mathbb{Z}[x]/J$ and $f(x) \neq 0$ in $\mathbb{Z}[x]/J$, that is, $g(x)$ is a zero divisor in $\mathbb{Z}[x]/J$. By our observation above, $g(x)$ is thus nilpotent in $\mathbb{Z}[x]/J$, that is, $g(x)^n \in J$ for some n.

(iii) Suppose $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_rx^r$ is an element of $\sqrt{J}$. We then have $f(x)^n \in J$ for some $n \in \mathbb{N}$, and $f(x)^n = a_0^n + xg(x)$ for some polynomial $g(x)$, so 25 must divide a_0^n. It follows that 5 must divide a_0, so $f(x) \in I$.
Conversely, suppose $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_rx^r$ is an element of I. Then 5 divides a_0, so 25 divides the constant term of $f(x)^2$, that is, $f(x)^2 \in J$, and thus $f(x) \in \sqrt{J}$.

140 The ring A is an integral domain, so $\{0\}$ is a prime ideal of A; the ideal (x) is also prime in A. It follows that S_1 and S_2 are multiplicative sets, as they are complements of prime ideals.

The two localisations are not isomorphic, as $S_1^{-1}A$ is a field, and in particular it is the field of rational functions in the variable x, whereas $S_2^{-1}(x) \neq \{0\}$ is a maximal ideal of $S_2^{-1}A$, so $S_2^{-1}A$ is not a field.

141

(i) A is not an integral domain, as $(1 + i)^2 = -2i$ in $\mathbb{Z}[i]$, so $1 + i$ is a zero divisor in the quotient $A = \mathbb{Z}[i]/(2)$.

(ii) We show that 3 is prime in $\mathbb{Z}[i]$. Let $u = a + bi$, $v = c + di$ be elements of $\mathbb{Z}[i]$ such that $3 = u \cdot v$; we wish to prove that at least one of u and v is invertible.
Note that $9 = |u|^2|v|^2 = (a^2 + b^2)(c^2 + d^2)$. If we had $a^2 + b^2,\ c^2 + d^2 > 1$ then we would necessarily have $a^2 + b^2 = 3 = c^2 + d^2$, but these equations have no solutions in $\mathbb{Z}$. It follows that one of the elements u, v must have norm 1, that is, it must be invertible. This shows that 3 is irreducible; the fact

that $\mathbb{Z}[i]$ is a Euclidean domain and thus a unique factorisation domain then implies that 3 is also prime.
We therefore have that B is a field and that $B[x]$ is an integral domain. We conclude that the two rings of polynomials are not isomorphic, as $A[x]$ is not an integral domain because A is not.

142 The fact that A is a subring of $\mathbb{Q}$ is proven in the obvious way. Letting S be the subset of $\mathbb{Z}$ consisting of the odd integers, S is a multiplicative set in $\mathbb{Z}$, and we have $A \simeq S^{-1}\mathbb{Z}$. The ideals of A are thus of the form $S^{-1}I$, where I is an ideal of $\mathbb{Z}$. This shows that all ideals of A are of the form $S^{-1}(2^k \cdot d)$, where d is an odd integer; but $d \in S$, so $S^{-1}(2^k \cdot d) = S^{-1}\mathbb{Z} \cdot (2^k)$. Finally, it is clear that the fraction field of A is $\mathbb{Q}$.

143 If the element $u = a + b\sqrt{-3}$, where $a, b \in \mathbb{Z}$, is invertible, then

$$u^{-1} = \frac{1}{a + b\sqrt{-3}} = \frac{a}{a^2 + 3b^2} - \frac{b}{a^2 + 3b^2}\sqrt{-3}.$$

Since 1 and $\sqrt{-3}$ are linearly independent over $\mathbb{Q}$, we have that $u^{-1} \in \mathbb{Z}[\sqrt{-3}]$ if and only if the rational numbers

$$\frac{a}{a^2 + 3b^2}, \quad \frac{b}{a^2 + 3b^2}$$

are actually integers. But if $b \neq 0$ then $a^2 + 3b^2 > |b|$, so the second number above cannot be an integer. It follows that $b = 0$, which yields that a^2 divides a, and so $a = \pm 1$. We have thus shown that $\mathbb{Z}[\sqrt{-3}]^* = \{\pm 1\}$ is a group isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

144 Set $x = t + t^{-1}$. It is easy to show that A^σ is a subring of A using the fact that σ is a ring automorphism. It then follows from $\sigma(x) = x$ that $\mathbb{Q}[x] \subseteq A^\sigma$.

Conversely, let $f = \sum_{|n| \leq N} a_n t^n$, where N is some natural number, be an element of A, and suppose that $\sigma(f) = f$. We have $\sigma(f) = \sum_{|n| \leq N} a_{-n} t^n$, so we obtain that $a_{-n} = a_n$, that is, $f = \sum_{n=0}^{N} a_n (t^n + t^{-n})$. We now wish to prove that such an element must be a polynomial in x.

We proceed by induction on N. If $N = 0$ then $f = a_0$ and the statement is obvious. Now assume the statement holds for $N - 1$ and consider the element $f' = f - a_N x^N = f - a_N (t + t^{-1})^N$. It is clear that f' is still fixed by σ and that it only features powers of t and t^{-1} with exponents smaller than N. It follows by the induction hypothesis that f' is a polynomial in x, so $f = f' + a_N x^N$ is as well.

145

(i) If $I + J + K = A$, then there are $i \in I$, $j \in J$ and $k \in K$ such that $1 = i + j + k$. It follows that

$$\begin{aligned} 1 &= 1^{3n} \\ &= (i + j + k)^{3n} \\ &= \sum \alpha_{a,b,c} i^a j^b k^c \end{aligned}$$

where the sum is taken over all natural numbers a, b, c such that $a+b+c = 3n$, and the numbers $\alpha_{a,b,c}$ are integer coefficients. If we have $a, b, c < n$ then $a + b + c < 3n$, so each term actually appearing in the sum has at least one among a, b and c that is greater than or equal to n. It follows that each summand $\alpha_{a,b,c} i^a j^b k^c$ belongs to I^n, J^n or K^n. We conclude that the sum, which is equal to 1, belongs to $I^n + J^n + K^n$.

(ii) The assumptions on I, J and K imply that there are $i_1, i_2 \in I$, $j_1, j_3 \in J$ and $k_2, k_3 \in K$ such that $1 = i_1 + j_1 = i_2 + k_2 = j_3 + k_3$, so we have $1 = (i_1 + j_1)(i_2 + k_2)(j_3 + k_3)$. Developing the product yields 8 summands, each of which is the product of three terms, not all three belonging to the same ideal I, J or K. In other words, each summand belongs to IJ, IK or JK. But then the sum, which is 1, belongs to $IJ + IK + JK$.

146

(i) Let A be a unique factorisation domain, let $\mathbb{K}$ be its fraction field and let $\alpha \in \mathbb{K}$ be a root of the monic polynomial $p(x) = x^n + a_1 x^{n-1} + \cdots + a_{n-1}x + a_n$ with coefficients in A. We want to show that $\alpha \in A$.
We can write $\alpha = a/b$ for some elements a and b of A that are relatively prime. The fact that $p(\alpha) = 0$ implies

$$\begin{aligned} a^n + a_1 a^{n-1} b + \cdots + a_{n-1} a b^{n-1} + b^n &= 0, \\ b(a_1 a^{n-1} + \cdots + a_{n-1} a b^{n-2} + b^{n-1}) &= -a^n. \end{aligned}$$

Now, given a prime π of A that divides b, the second equation yields that π must divide $-a^n$, so it must divide a. But a and b are relatively prime, so they cannot share a common prime divisor π, and thus b is invertible in A. This implies that $\alpha = a/b = a \cdot b^{-1} \in A$.

(ii) Let n be an integer such that $4n + 1$ is not a square in $\mathbb{Z}$, let $A = \mathbb{Z}[\sqrt{4n+1}]$ and let $\mathbb{K}$ be the fraction field of A.
The element $\alpha = \big(1 + \sqrt{4n+1}\big)/2$ of $\mathbb{K}$ is a root of the monic polynomial $x^2 - x - n$ in $A[x]$.
Since $4n + 1$ is not a square in $\mathbb{Z}$, the elements 1 and $\sqrt{4n+1}$ are a basis of $\mathbb{K}$ over $\mathbb{Q}$. Thus

$$\alpha = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \sqrt{4n+1}$$

is the unique expression of α as a linear combination of the basis elements over $\mathbb{Q}$, so α is not an element of $A = \{a + b\sqrt{4n+1} \mid a, b \in \mathbb{Z}\}$. This proves that A is not integrally closed, which by (i) implies that it is not a unique factorisation domain.

147

(i) We show that $M = I \cap J$ has the desired property.

We first prove that $\mathcal{V}(I) \cup \mathcal{V}(J) \subseteq \mathcal{V}(M)$. Indeed, given an element P of $\mathcal{V}(I)$, we have $I \subseteq P$ and thus $M \subseteq P$. It follows from the definition that $P \in \mathcal{V}(M)$. The argument for an ideal P in $\mathcal{V}(J)$ is entirely analogous.
We now prove the opposite inclusion. Given $P \in \mathcal{V}(M)$, we have $M = I \cap J \subseteq P$. If $I \subseteq P$ does not hold, then there is an element $x \in I \setminus P$. Now, let y be an element of J; we have $xy \in IJ \subseteq I \cap J$, so $xy \in P$; but $x \notin P$ and thus $y \in P$ because P is prime. This shows that $J \subseteq P$, hence $P \in \mathcal{V}(J)$.

(ii) We can take $N = I + J$. Indeed, $P \in \mathcal{V}(I) \cap \mathcal{V}(J)$ if and only if $I, J \subseteq P$, that is, if and only if $N = I + J \subseteq P$, which is equivalent to $P \in \mathcal{V}(N)$.

In order to show that $\mathcal{V}(I) = \mathcal{V}(J)$ does not imply $I = J$, consider $A = \mathbb{Z}$, $I = (2)$, $J = (4)$. It is easy to see that $\mathcal{V}(I) = \{(2)\} = \mathcal{V}(J)$, whereas $I \neq J$.

148 Given an ideal I with the property from the problem statement, consider the ideal $S^{-1}I$ of $S^{-1}A$. This is a proper ideal, as I does not intersect S, and it is a maximal ideal of $S^{-1}A$, since every ideal of $S^{-1}A$ is of the form $S^{-1}J$ for some ideal J of A. In particular, $S^{-1}I$ is a prime ideal of $S^{-1}A$, so $S^{-1}I \cap A$ is a prime ideal of A.

If we show that $S^{-1}I \cap A = I$, then we are done. It is clear that $I \subseteq J = S^{-1}I \cap A$. Moreover, J does not intersect S, or $S^{-1}I = S^{-1}J$ would not be a proper ideal, and the property in the problem statement implies $I = J$.

149 Let I be the ideal (x^2, y^2) of $\mathbb{Q}[x, y]$ and let $J = \sqrt{I}$; note that $x^2, y^2 \in I$ implies $x, y \in J$, and thus $(x, y) \subseteq J$. Moreover, for each natural number h, the element $1^h = 1$ does not belong to I, so $1 \notin J$. Thus, using the fact that $(x, y) \subseteq J$, we find that all polynomials in J have constant term equal to zero.

This proves that $J = (x, y)$, which implies that $A/J \simeq \mathbb{Q}$, so J is maximal, and in particular prime in $\mathbb{Q}[x, y]$.

150 The ring homomorphisms from A/I to $\mathbb{Q}$ are in bijection with the ring homomorphisms from A to $\mathbb{Q}$ whose kernel contains I. Let $\varphi : A \longrightarrow \mathbb{Q}$ be one such homomorphism.

The homomorphism φ is completely determined by the images of x and y; suppose $\varphi(x) = \alpha$ and $\varphi(y) = \beta$, where $\alpha, \beta \in \mathbb{Q}$. The condition that $I \subseteq \mathrm{Ker}(\varphi)$ is equivalent to $xy - 1 \in \mathrm{Ker}(\varphi)$, as $xy - 1$ generates I.

We have $\varphi(xy - 1) = \varphi(x)\varphi(y) - 1 = \alpha\beta - 1$, so requiring that I be contained in the kernel of φ is equivalent to asking that $\alpha \neq 0$ and $\beta = \alpha^{-1}$.

What we have shown is that the desired homomorphisms are $\varphi_\alpha : A/I \longrightarrow \mathbb{Q}$ for α in $\mathbb{Q}^*$, where φ_α is the unique homomorphism such that $\varphi_\alpha(x + I) = \alpha$ and $\varphi_\alpha(y + I) = \alpha^{-1}$.

151

(i) Consider the valuation homomorphism $v : A[x] \longrightarrow A$ given by $v(f(x)) = f(a)$. Since v is a ring homomorphism and $v^{-1}(I) = \mathcal{J}$, we have that $\mathcal{J}$ is an ideal of $A[x]$.

(ii) The valuation homomorphism defined in (i) is surjective, because $A[x]$ contains A in the form of constant polynomials. In particular, v induces a

one-to-one correspondence between ideals of $A[x]$ that contain $\mathrm{Ker}(v) = A[x] \cdot (x - a)$ and ideals of A, wherein prime ideals correspond to prime ideals. But $\mathcal{J}$ corresponds to I, so $\mathcal{J}$ is prime in $A[x]$ if and only if I is prime in A.

(iii) It is clear that $(5, x - 1) \subseteq \mathcal{J}$. Conversely, let $f(x) \in \mathbb{Z}[x]$ be such that $f(1) \in (5)$, that is, $f(1) = 5h$ for some $h \in \mathbb{Z}$, and set $g(x) = f(x) - 5h$. We have $g(1) = 0$, so there is $q(x) \in \mathbb{Q}[x]$ such that $g(x) = q(x)(x - 1)$. By Gauss's Lemma, $q(x) \in \mathbb{Z}[x]$, because $x - 1$ is a primitive polynomial. It follows that $f(x) = q(x)(x-1)+5h \in (5, x-1)$, hence the desired inclusion.

152

(i) We show that an element z/d of $S^{-1}A$, where $z \in A$ and d is an odd integer, is invertible if and only if the norm of z, that is, the integer $N(z) = z\overline{z}$, is odd. Indeed, if $N(z)$ is odd then $d\overline{z}/N(z)$ is an element of $S^{-1}A$; moreover, $(z/d) \cdot (d\overline{z}/N(z)) = 1$ and thus $d\overline{z}/N(z)$ is the inverse of z/d.
Conversely, if an element $w \in A$ and an odd integer c are such that w/c is the inverse of z/d, then we have $(z/d) \cdot (w/c) = 1$ and so $zw = dc$, hence $N(z)N(w) = d^2c^2$. Since d, c are both odd, it follows that $N(z)$ is also odd.

(ii) We need to show that every ideal of $S^{-1}A$ is generated by $(1 + i)^k$ for some natural number k. We know that A is a principal ideal domain, so any ideal I of $S^{-1}A$ is generated by some element $z = a + bi$ of A. Denote by $r = N(z) = z\overline{z} = a^2 + b^2$ the norm of z, and remark that $r \in I \cap \mathbb{Z}$.
We consider several cases. If r is odd, then $I = S^{-1}A$ by (i).
Suppose r is even. If $4 \nmid r$ then $r = 2d$, where d is odd. Since d is invertible in $S^{-1}A$ and $r \in I$, we have $2 \in I$. Moreover, the fact that r is even and $4 \nmid r$ yields that a, b are both odd, thus of the form $a = 2a' + 1$ and $b = 2b' + 1$. It follows that $z = a + bi = 2(a' + b'i) + (1 + i)$, and thus $1 + i \in I$. Also note that $2 = -i(1 + i)^2$, so $z = -i(1 + i)^2(a' + b'i) + 1 + i \in (1 + i)$. We can therefore conclude that $I = S^{-1}A \cdot (1 + i)$.
Now suppose $r = 2^h d$, where d is odd and $h \geq 2$. In this case, a, b are both even, of the form $a = 2a'$ and $b = 2b'$. Thus $z = 2(a' + b'i) = -i(1 + i)^2(a' + b'i)$, which implies that $I = (z) = (1 + i)^2(a' + b'i)$ because $-i$ is invertible. Set $z' = a' + b'i$ and remark that $N(z') = 2^{h-2}d$.
By iterating this argument a finite number t of times, we reduce the problem to one of the cases above, in which $4 \nmid r$: we thus find $I = S^{-1}A \cdot (1 + i)^{2t}$ or $I = S^{-1}A \cdot (1 + i)^{2t+1}$, as wanted.

153

(i) Denote by $\pi : A \longrightarrow A/I$ the quotient homomorphism and let $\widetilde{\pi} : A[x] \longrightarrow (A/I)[x]$ be the ring homomorphism given by $\widetilde{\pi}_{|A} = \pi$ and $\widetilde{\pi}(x) = x$.
By definition, we have $\mathrm{Ker}(\widetilde{\pi}) = I[x]$, so $I[x]$ is an ideal of $A[x]$.

(ii) Let $\mathcal{I}$ be the ideal of $A[x]$ generated by I. Clearly, the fact that $I \subseteq I[x]$ implies that $\mathcal{I} \subseteq I[x]$. Conversely, consider $f(x) = \sum_{h=0}^{n} a_h x^h \in I[x]$; since $a_h \in I$ for all h, we have $a_h x^h \in \mathcal{I}$ for all h, and so $f(x) \in \mathcal{I}$.

(iii) Assume I is a prime ideal of A; then A/I is an integral domain, so $(A/I)[x]$ is also an integral domain. Remark that $A[x]/I[x]$ can be embedded into $(A/I)[x]$, as $I[x] = \mathrm{Ker}(\widetilde{\pi})$. We thus conclude that $A[x]/I[x]$ is also an integral domain, and so $I[x]$ is a prime ideal of $A[x]$.

154 We know that every ideal in the ring $\mathbb{Z} \times \mathbb{Z}$ is the direct product $I \times J$ of two ideals I, J of $\mathbb{Z}$.

(i) Let $I \times J$ be an ideal of $\mathbb{Z} \times \mathbb{Z}$. Since $\mathbb{Z}$ is a principal ideal domain, there are $a, b \in \mathbb{Z}$ such that $I = \mathbb{Z}\cdot a$ and $J = \mathbb{Z}\cdot b$. Therefore, $I \times J = (\mathbb{Z}\cdot a) \times (\mathbb{Z}\cdot b) = \mathbb{Z} \times \mathbb{Z} \cdot (a, b)$, and so it is principal.

(ii) The ideal $I \times J$ is prime if and only if $(\mathbb{Z} \times \mathbb{Z})/(I \times J) \simeq \mathbb{Z}/I \times \mathbb{Z}/J$ is an integral domain. On the other hand, a direct product of rings is an integral domain if and only if one of the factors is the zero ring and the other is an integral domain. Indeed, if both factors were nonzero, the product would have zero divisors, as $(1, 0) \cdot (0, 1) = (0, 0)$. It follows that $I \times J$ is prime if and only if it is of the form $\mathbb{Z} \times (p)$ or $(p) \times \mathbb{Z}$, where p is a prime number or 0. Similarly, $\mathbb{Z}/I \times \mathbb{Z}/J$ is a field if and only if one of the two factors is zero and the other is a field. It follows that $I \times J$ is maximal if and only if it is prime and not equal to $\mathbb{Z} \times 0$ or $0 \times \mathbb{Z}$.

155 Since A is isomorphic to $A[x]/(x)$, it is enough to show that $I = (x)$ is a maximal ideal. Suppose $I \subseteq J \subseteq A[x]$ for some ideal J. By assumption, we know that there is $f(x) \in A[x]$ such that $J = (f(x))$.

Note that A is an integral domain because $A[x]$ is. In particular, the degree of a product of polynomials is the sum of the degrees of the factors. the fact that $(x) \subseteq (f(x))$ implies the existence of $g(x) \in A[x]$ such that $x = g(x)f(x)$. By comparing degrees, we find that $g(x)$ has degree 0 and $f(x)$ has degree 1, or $g(x)$ has degree 1 and $f(x)$ has degree 0.

In the first case, $g(x) = a \in A$ and $f(x) = bx + c$, with $b, c \in A$, so $x = a(bx + c)$. This yields $ab = 1$, and in particular that b is invertible and $c = 0$, so we have $I = (x) = (bx) = (f(x)) = J$. In the second case, $g(x) = ax + b$, with $a, b \in A$, and $f(x) = c \in A$, so we have $x = (ax + b)c$. Thus $ac = 1$, and in particular c is invertible and $J = (f(x)) = (c) = A[x]$. This proves that I is a maximal ideal.

156 If $p = 2$ then $x^2 - a$ and $x^2 + a$ are the same polynomial, so the two quotient rings under consideration actually coincide.

Assume from now on that $p \neq 2$. Let $f(x)$ be the polynomial $x^2 - a$ or the polynomial $x^2 + a$. By checking the derivative, we find that $f(x)$ has no multiple roots, because $a \neq 0$. Consequently, $f(x)$ is either irreducible or the product of two distinct factors of degree 1.

If $f(x)$ is irreducible then $\mathbb{F}_p[x]/(f(x)) \simeq \mathbb{F}_{p^2}$, whereas if $f(x) = f_1(x)f_2(x)$ with $f_1(x) \neq f_2(x)$, the Chinese Remainder Theorem for Rings implies that $\mathbb{F}_p[x]/(f(x)) \simeq \mathbb{F}_p[x]/(f_1(x)) \times \mathbb{F}_p[x]/(f_2(x)) \simeq \mathbb{F}_p \times \mathbb{F}_p$.

The two quotient rings are thus isomorphic if and only if the two polynomials have the same type of factorisation. Since $x^2 - a$ is irreducible if and only if

$\sqrt{a} \in \mathbb{F}_p$ and $x^2 + a$ is irreducible if and only if $\sqrt{-a} \in \mathbb{F}_p$, the two quotients are isomorphic if and only if -1 is a square in $\mathbb{F}_p^*$, which is equivalent to p being congruent to 1 modulo 4.

157

(i) Given $u = a+b\sqrt{7}$, set $\overline{u} = a-b\sqrt{7}$. The elements $u = a+b\sqrt{7}$, $v = c+d\sqrt{7}$ in A satisfy $u \cdot v = (ac + 7bd) + (ad + bc)\sqrt{7}$, so $\overline{(u \cdot v)} = (ac + 7bd) - (ad + bc)\sqrt{7} = (a - b\sqrt{7})(c - d\sqrt{7}) = \overline{u} \cdot \overline{v}$. We have thus shown that the map $u \longmapsto \overline{u}$ is multiplicative.
It follows that $N(uv) = u\,v\,\overline{u}\,\overline{v} = u\,v\,\overline{u}\,\overline{v} = u\,\overline{u}\,v\,\overline{v} = N(u)N(v)$, so the map N is also multiplicative.

(ii) Suppose u is invertible, that is, suppose we have $v \in A$ such that $uv = 1$. Then $N(u)N(v) = N(uv) = N(1) = 1$, so $N(u) = \pm 1$ because $N(u)$ and $N(v)$ are integers.
Conversely, if $N(u) = \pm 1$ then $u\overline{u} = N(u) = \pm 1$, hence $u \cdot (\pm\overline{u}) = 1$, and thus u is invertible.

(iii) Suppose p is reducible in A, that is, assume $p = uv$ where u, v are not invertible. We have $p^2 = N(p) = N(u)N(v)$, so $N(u) = N(v) = \pm p$ because $|N(u)|,\ |N(v)| \neq 1$, as u and v are not invertible.
Conversely, if there is $u \in A$ such that $N(u) = \pm p$ then $p = \pm u\overline{u}$, so p is reducible in A, because u and $\overline{u}$ are not invertible, as $|N(u)| = |N(\overline{u})| = p > 1$.

(iv) The prime 2 is reducible, as $N(3+\sqrt{7}) = 2$. We now show that 5 is irreducible by proving that there is no $u \in A$ such that $N(u) = \pm 5$.
Indeed, finding such a u is equivalent to finding integer solutions a, b to the equation $a^2 - 7b^2 = \pm 5$, which we can reduce modulo 5 to obtain $a^2 - 2b^2 = 0$. Since 2 is not a square modulo 5, the only potential solution is given by $a = b = 0$ modulo 5, that is, $a = 5a_1$ and $b = 5b_1$ for some $a_1,\ b_1 \in \mathbb{Z}$. But $a_1,\ b_1$ must satisfy $5^2a_1^2 - 7 \cdot 5^2b_1^2 = 5$, which is impossible because the left-hand side is congruent to 0 modulo 25, whereas the right-hand side is not.

158 First of all, remark that a is nilpotent if and only if a is divisible for all primes p that divide m.

Now, if a is divisible for every prime p that divides m, then $1 - ab$ is coprime to every such p for all integers b, so $1 - ab$ is invertible.

Conversely, suppose a is not divisible by some prime p that divides m. Then a is invertible in $\mathbb{Z}/p\mathbb{Z}$, so there exists b such that $1 - ab$ is divisible by p. It follows that the element $1 - ab$ is not invertible in $\mathbb{Z}/m\mathbb{Z}$.

159

(i) All ideals of $S^{-1}A$ are of the form $S^{-1}I$, where I is an ideal of A. Note that, since A is a unique factorisation domain, for each $y \in A \setminus \{0\}$ there is a well-defined integer n_y such that x^{n_y} divides y and x^{n_y+1} does not divide y. In particular, every element $y \neq 0$ in A can be expressed as $y = ux^{n_y}$ with

$u \in S$. Moreover, if $I \neq 0$ then we can define the non-negative integer n_I as $\min\{n_y \mid y \in I \setminus \{0\}\}$.
Concerning the ideal $S^{-1}I$ of $S^{-1}A$, we have

$$S^{-1}I = \big(y \mid y \in I \setminus \{0\}\big)_{S^{-1}A} = \big(x^{n_y} \mid y \in I \setminus \{0\}\big)_{S^{-1}A} = S^{-1}A \cdot x^{n_I}.$$

This proves that every nonzero ideal in $S^{-1}A$ is principal, and in fact of the form $S^{-1}A \cdot x^n$ for some $n \geq 0$.

(ii) The intersection of all nonzero ideals is still an ideal. If such an intersection were nonzero, by our argument in (i) it would be the ideal of $S^{-1}A$ generated by x^n for some non-negative integer n. But this is not possible, as $S^{-1}A \cdot x^{n+1}$ is a nonzero ideal that is strictly contained in $S^{-1}A \cdot x^n$.

160 Remark that $x^2+y^2-1 = x^2+(y-1)(y+1)$, so the ideal $I = (x^2+y^2-1)$ is contained in the ideal $(x, y+1) \neq \mathbb{Q}[x, y]$, which implies that I is not a maximal ideal.

We now show that I is a prime ideal. Indeed, $\mathbb{Q}[x, y]$ is a unique factorisation domain, so it is enough to prove that x^2+y^2-1 is irreducible. Let $\psi : \mathbb{Q}[x, y] \longrightarrow \mathbb{Q}[x]$ be the unique homomorphism such that $\psi(x) = x$ and $\psi(y) = 2$; we have $\psi(x^2 + y^2 - 1) = x^2 + 3$, which is an irreducible polynomial in $\mathbb{Q}[x]$, and this implies that $x^2 + y^2 - 1$ is irreducible in $\mathbb{Q}[x, y]$.

Now consider the ideal $J = (x^2 - 3, y^2 - x)$; we shall prove it is a maximal ideal. Indeed, let $\alpha = \sqrt[4]{3} \in \mathbb{R}$ and let $\varphi : \mathbb{Q}[x, y] \longrightarrow \mathbb{Q}[\alpha] = \mathbb{Q}(\alpha)$ be the unique homomorphism such that $\varphi(x) = \alpha^2$ and $\varphi(y) = \alpha$. Clearly, φ is surjective, because α is in its image. Showing that $\mathrm{Ker}(\varphi) = J$ would allow us to conclude that J is a maximal ideal, because $\mathbb{Q}(\alpha)$ is a field.

Since $\varphi(x^2 - 3) = \alpha^2 - 3 = (\sqrt[4]{3})^2 - 3 = 0$ and $\varphi(y^2 - x) = \alpha^2 - \alpha^2 = 0$, we find that $J \subseteq \mathrm{Ker}(\varphi)$. We now proceed to show the opposite inclusion.

Let $f(x, y)$ be a polynomial in $\mathbb{Q}[x, y]$, which we write as $f(x, y) = a + bx + cy + dxy + g(x, y)$ for some a, b, c and d in $\mathbb{Q}$, and $g(x, y) \in J$. If $\varphi(f(x, y)) = 0$ then, since $J \subseteq \mathrm{Ker}(\varphi)$, we obtain that $0 = \varphi(f(x, y)) = a + b\alpha + c\alpha^2 + d\alpha^3$. But $t^4 - 3$ is the minimal polynomial of α over $\mathbb{Q}$, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$, and in particular the elements $1, \alpha, \alpha^2, \alpha^3$ are linearly independent over $\mathbb{Q}$. It follows that $a = b = c = d = 0$ and that $f(x, y) = g(x, y) \in J$. We have thus shown that $\mathrm{Ker}(\varphi) = J$.

161 Let R be a multiplicative set in $\mathbb{Z}$ and let $\varphi : R^{-1}\mathbb{Z}[x] \longrightarrow \mathbb{Q}$ be a ring homomorphism. If $\varphi \neq 0$ then $\varphi(1) = 1$, so $\varphi(r) = r$ for all $r \in R$ and $\varphi(1/r) = 1/r$ for all $r \in R$. The homomorphism φ is therefore completely determined once we fix the image $\varphi(x) = a/b \in \mathbb{Q}$ of x.

(i) Let $\varphi : S^{-1}\mathbb{Z}[x] \longrightarrow \mathbb{Q}$ be a homomorphism. By our previous argument, we have $\varphi(S^{-1}\mathbb{Z}) = \mathbb{Z}[1/m]$, and moreover $\varphi(S^{-1}\mathbb{Z}[x]) = \mathbb{Z}[1/m, a/b]$, where $\varphi(x) = a/b \in \mathbb{Q}$. It is then clear that, given a prime p that does not divide m or b, the rational number $1/p$ does not belong to the image of φ.

(ii) We intend to show that the unique homomorphism $\varphi : T^{-1}\mathbb{Z}[x] \longrightarrow \mathbb{Q}$ such that $\varphi(x) = 1/m$ is surjective. Indeed, since $\mathbb{Z} \subseteq \mathrm{Im}(\varphi)$ it is enough to prove that $1/p \in \mathrm{Im}(\varphi)$ for all primes p.
Now, if a prime p does not divide m then $p \in T$ and so $1/p \in T^{-1}\mathbb{Z}[x]$ and $\varphi(1/p) = 1/p$. On the other hand, if p divides m then $a = m/p$ is an integer and we have $\varphi(ax) = (m/p) \cdot (1/m) = 1/p$.

162 Assume that φ is not an isomorphism; we shall prove that B is a field. We know that the ideal $I = \mathrm{Ker}(\varphi)$ is nonzero because the surjective homomorphism φ is not an isomorphism.

Clearly, $B \simeq A/I$, so I is a prime ideal because B is an integral domain. But a nonzero prime ideal in a principal ideal domain is maximal, so B is a field.

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(i) We show that the family $X = \{I + J \mid J \text{ ideal of } A\}$ is always finite. First of all, remark that X is the family of all ideals containing I. Suppose $I = (a)$ where $a = x_1 \cdots x_n$ and $x_1, \ldots, x_n$ are irreducible; then the ideals that contain I are generated by divisors of a, that is, by the products of elements in the set $\{x_1, \ldots, x_n\}$. Consequently, they form a finite family.

(ii) Let $I = (a)$ be as above; we show that the sequence $I^n = (a^n)$ for $n = 1, 2, 3, \ldots$ is a strictly decreasing sequence of ideals in $Y = \{I \cap J \mid J \text{ ideal of } A\}$. The fact that such ideals belong to Y is obvious, as they are contained in I.
Moreover, if we had $(a^n) = (a^{n+1})$ then there would exist an element $b \in A$ such that $a^n = ba^{n+1}$, and we would have $a^n(1-ba) = 0$. But A is an integral domain and I is a nonzero ideal, so this would imply $ba = 1$, that is, a would be invertible in A. This contradicts the fact that I is a prime ideal of A.
We have thus shown that the family Y is always infinite.

(iii) We now show that $Z = \{J \mid I + J = L\}$ can be either finite or infinite.
Let $A = \mathbb{Z}$, $I = L = (2)$. For all ideals of the form $J = (2n)$ we have $I + J = (2) + (2n) = (2) = L$, so the family Z is infinite in this case.
Now let $A = \mathbb{Q}[[t]]$ be the ring of formal power series in the variable t over $\mathbb{Q}$, and let $I = (t^2)$, $L = (t)$. Since every ideal of A is of the form $J = (t^n)$ for some n, we have $I + J = L$ if and only if $(t^2) + (t^n) = (t)$. But $(t^2) + (t^n) = (t^{\min(2,n)})$, so only $n = 1$ yields an ideal of Z; indeed, we have $Z = \{(t)\}$.

164 The ring A is an integral domain, as the ideal 0 is prime. Given $a \in A$ such that $a \neq 0$, we now show that a is invertible. Consider the ideal (a^2). Either $(a^2) = (1)$, in which case there exists $b \in A$ such that $a^2b = a(ab) = 1$, and so a is invertible, or (a^2) is a prime ideal. In the latter case, the fact that $a \cdot a \in (a^2)$ yields $a \in (a^2)$, that is, there exists $c \in A$ such that $a = ca^2$, so $a(1-ac) = 0$. Since A is an integral domain and $a \neq 0$, we obtain $ac = 1$, and so a is invertible.

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(i) Let J be the ideal generated by 11 in $\mathbb{Z}[x]$ and let I be the ideal $(11, x^2 + a)$. The quotient $\mathbb{Z}[x]/I$ is isomorphic to $(\mathbb{Z}[x]/J)/(I/J) \simeq \mathbb{F}_{11}[x]/(x^2 + a)$, so

I is a prime ideal if and only if the polynomial $x^2 + a$ is irreducible over the field $\mathbb{F}_{11}$, that is, if and only if $-a$ is not a square modulo 11. By computing the squares of all elements in $\mathbb{F}_{11}$, we find that I is irreducible for all $a \in \mathbb{Z}$ congruent to -2, 1, 3, 4 or 5 modulo 11.

(ii) What we showed in (i) implies that $P = (11, x^2 + 3)$ is a prime ideal, so $S = \mathbb{Z}[x] \setminus P$ is a multiplicative set. We now prove that the polynomial $f_\lambda = x^4 + \lambda x^2 + 5$ is invertible if and only if $f_\lambda \notin P$.
Indeed, if $f_\lambda \notin P$ then $f_\lambda \in S$ and so $1/f_\lambda \in S^{-1}\mathbb{Z}[x]$. Conversely, if f_λ is invertible then there are $a \in \mathbb{Z}[x]$ and $s \notin P$ such that $f_\lambda \cdot a/s = 1$, that is, such that $s = a \cdot f_\lambda$ in $\mathbb{Z}[x]$. So $f_\lambda \notin P$, or we would have $s \in P$.
Dividing f_λ by $x^2 + 3$ yields $f_\lambda = (x^2 + \lambda - 3)(x^2 + 3) + 14 - 3\lambda$, so $f_\lambda \notin P$ if and only if $14 - 3\lambda \notin P$. The latter being a polynomial of degree 0, we have $14 - 3\lambda \notin P$ if and only if $3(1 - \lambda) \neq 0$ in $\mathbb{F}_{11}$.
In conclusion, f_λ is invertible in $S^{-1}\mathbb{Z}[x]$ if and only if λ is not congruent to 1 modulo 11.

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(i) By taking $a = b = 1$ we find that $I \sim I$ for all $I \in \mathcal{I}$. Moreover, exchanging a and b yields that $I \sim J$ if and only if $J \sim I$.
Now assume that $I \sim J$ and $J \sim K$. Let $a, b, c, d \in A \setminus \{0\}$ be such that $aI = bJ$ and $cJ = dK$. We have $acI = c(aI) = c(bJ) = b(cJ) = bdK$, so $I \sim K$ because $ac \neq 0$ and $bd \neq 0$ thanks to A being an integral domain. This proves that $\sim$ is an equivalence relation.

(ii) Suppose A is a principal ideal domain and let $I = (x)$ and $J = (y)$ be two elements of $\mathcal{I}$. We have $yI = y(x) = (xy) = x(y) = xJ$, so $I \sim J$.
Conversely, suppose that all elements in $\mathcal{I}$ are in relation with each other; we will show that A is a principal ideal domain. Let I be a nonzero ideal of A; by assumption, we have $I \sim (1)$, so there are $a, b \in A \setminus \{0\}$ such that $aI = b(1) = (b)$.
This condition implies that for every $x \in I$ there is $c_x \in A$ such that $ax = c_x b$, and also that there is $z \in I$ such that $b = az$. We now prove that I is generated by z.
Since $z \in I$, it is clear that $(z) \subseteq I$. Conversely, for each $x \in I$ we have $ax = c_x b = c_x az$, so $a(x - c_x z) = 0$, which, since A is an integral domain and $a \neq 0$, yields $x = c_x z$, that is, $x \in (z)$. This concludes the proof that $I = (z)$ is a principal ideal.

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(i) The statement is false. For example, take $A = \mathbb{Z}$, $P_1 = (2)$ and $P_2 = (3)$; we have that $P_1 + P_2 = (1)$ is not prime in $\mathbb{Z}$.

(ii) The statement is false. If we take $A = \mathbb{Z}$, $I = (4)$, then $\sqrt{I} = (2)$ is prime but I is not.

(iii) The statement is true. Indeed, the fact that Q is prime in B implies that $Q \neq B$ and thus $1_B \notin Q$; it follows that $1_A \notin \varphi^{-1}(Q)$, and so $\varphi^{-1}(Q)$ is a proper ideal of A. Moreover, if $xy \in \varphi^{-1}(Q)$ then $\varphi(xy) = \varphi(x)\varphi(y) \in Q$, and

the fact that Q is prime yields $\varphi(x) \in Q$ or $\varphi(y) \in Q$. We conclude that $x \in \varphi^{-1}(Q)$ or $y \in \varphi^{-1}(Q)$, which proves that $\varphi^{-1}(Q)$ is prime in A.

(iv) The statement is false. Consider $A = \mathbb{Z}$, $B = \mathbb{Q}$ and set φ to be the inclusion of $\mathbb{Z}$ in $\mathbb{Q}$. The ideal 0 is maximal in $\mathbb{Q}$, as $\mathbb{Q}$ is a field, but $\varphi^{-1}(0) = 0$ is not maximal in $\mathbb{Z}$ because, for example, $0 \subsetneq (2) \subsetneq \mathbb{Z}$.

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(i) The ring $B' = \varphi(A)$ is isomorphic to $A/\operatorname{Ker}(\varphi)$ and φ induces a one-to-one correspondence between ideals of A that contain $\operatorname{Ker}(\varphi)$ and ideals of B', which pairs maximal ideals with maximal ideals. It follows that if an ideal $\varphi(M)$ of B' is maximal, then M is maximal in A and is thus the unique maximal ideal of A; so $\varphi(M)$ is the unique maximal ideal of B', and B' is local.

(ii) First, assume that $M = A \setminus A^*$ is an ideal of A; we show that M is the unique maximal ideal of A.

Let I be an ideal of A such that $M \subseteq I \subseteq A$. We either have $I = M$ or $x \in I$ for some $x \notin M$; if the latter were the case, we would have $x \in A^*$, so $I = A$. This proves that M is a maximal ideal.

Moreover, any maximal ideal N cannot contain any invertible elements (otherwise $N = A$), so $N \subseteq M$ and therefore $N = M$, because N is maximal. We have thus shown that M is the unique maximal ideal of A, that is, that A is local.

Conversely, assume that A is local and that M is its unique maximal ideal. We show that $A \setminus A^* = M$ and so in particular $A \setminus A^*$ is an ideal of A. Of course, M does not contain any invertible elements, because $M \neq A$, and so $M \subseteq A \setminus A^*$.

Now consider $x \in A \setminus A^*$; we have $(x) \neq A$, so there exists a maximal ideal containing (x). But M is the unique maximal ideal of A, so $(x) \subseteq M$. In particular, $x \in M$.

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(i) We have $0 \in A$ by definition and $1 \in A$ because $\operatorname{ord}(1) = 0$. Given $\rho_1(x) = x^u f_1(x)/g_1(x)$, $\rho_2(x) = x^v f_2(x)/g_2(x)$ of order u, $v \geq 0$, respectively, with $f_1(0), g_1(0), f_2(0), g_2(0) \neq 0$, if $u \geq v$ then we have

$$\rho_1(x) + \rho_2(x) = x^v \cdot \frac{x^{u-v} g_2(x) f_1(x) + g_1(x) f_2(x)}{g_1(x) g_2(x)}.$$

The denominator of the fraction above does not vanish at 0, so $\operatorname{ord}(\rho_1(x) + \rho_2(x)) \geq v \geq 0$. Moreover, it is clear that $\operatorname{ord}(-\rho(x)) = \operatorname{ord}(\rho(x))$ for all $\rho(x) \in \mathbb{K}(x)$, and in particular if $\operatorname{ord}(\rho(x)) \geq 0$ then $\operatorname{ord}(-\rho(x)) \geq 0$. This proves that A is an additive subgroup. Using the same notation as above, we also have

$$\rho_1(x)\rho_2(x) = x^{u+v} \cdot \frac{f_1(x) f_2(x)}{g_1(x) g_2(x)},$$

and thus $\operatorname{ord}(\rho_1(x)\rho_2(x)) = \operatorname{ord}(\rho_1(x)) + \operatorname{ord}(\rho_2(x))$, so $\rho_1(x)\rho_2(x) \in A$. This concludes the proof that A is a subring of $\mathbb{K}(x)$.

Now let $\rho_1(x)$ be an element of A with inverse $\rho_2(x)$. By our previous arguments, $0 = \operatorname{ord}(1) = \operatorname{ord}(\rho_1(x)\rho_2(x)) = \operatorname{ord}(\rho_1(x)) + \operatorname{ord}(\rho_2(x))$, which is a sum of non-negative integers, and thus $\operatorname{ord}(\rho_1(x)) = \operatorname{ord}(\rho_2(x)) = 0$. Conversely, if $\operatorname{ord}(\rho(x)) = 0$ then $\rho(x) = f(x)/g(x)$ with $f(0), g(0) \neq 0$, so $g(x)/f(x) \in A$.

We have shown that A^* is the set of rational functions of order 0.

(ii) Letting $S = \{g(x) \in \mathbb{K}[x] \mid g(0) \neq 0\}$, we begin by showing that S is a multiplicative set. Indeed, $0 \notin S$, $1 \in S$, and if $g_1(x)$ and $g_2(x)$ do not vanish at 0 then neither does $g_1(x)g_2(x)$.

We now prove that $S^{-1}\mathbb{K}[x] = A$. If a rational function $g(x)$ does not vanish at 0, then its order is not positive: we have $\operatorname{ord}(g(x)^{-1}) \geq 0$ and $1/g(x) \in A$; this proves that $S^{-1}\mathbb{K}[x]$ is contained in A. Conversely, if a rational function $\rho(x) = x^v f(x)/g(x)$, with $f(0), g(0) \neq 0$, is an element of A, then $v \geq 0$ and so $\rho(x) = (x^v f(x))/g(x)$ is an element of $S^{-1}\mathbb{K}[x]$ because $g(0) \neq 0$.

170 In $\mathbb{Z}[x]$, we have $x^6 + 2x^5 + x + 2 = (x+2)(x^5+1)$. Factoring over $\mathbb{F}_5$ yields $f_1(x) = (x+2)(x^5+1) = (x+2)(x+1)^5$. Over $\mathbb{F}_7$, we have $f_2(x) = (x+2)(x^5+1) = (x+2)(x+1)(x^4+x^3+x^2+x+1)$; the final three factors are irreducible because $\mathbb{F}_7$ does not contain any primitive fifth roots of unities, as $\mathbb{F}_7^* \simeq \mathbb{Z}/6\mathbb{Z}$ has no elements of order 5.

The ring $A = \mathbb{F}_5[x]/(f_1(x)) \times \mathbb{F}_7[x]/(f_2(x))$ is finite, so each of its elements is either invertible or a zero divisor. It is thus enough to compute the number n of zero divisors: the ring will have $5^6 \cdot 7^6 - n$ invertible elements. An element (a, b) is a zero divisor if and only if one of the coordinates is a zero divisor. We thus proceed as follows, using the Inclusion-Exclusion Principle: letting n_1 be the number of zero divisors in $\mathbb{F}_5[x]/(f_1(x))$ and n_2 the number of zero divisors in $\mathbb{F}_7[x]/(f_2(x))$, in the ring A we have $n = 7^6 n_1 + 5^6 n_2 - n_1 n_2$ zero divisors.

Let us now find n_1. A zero divisor in $\mathbb{F}_5[x]/(f_1(x))$ is the coset of a polynomial of degree less than 6 that is not coprime to $f_1(x)$, that is, sharing a common irreducible factor with $f_1(x)$. We thus find 5^5 multiples of $x+2$ and 5^5 multiples of $x+1$; this yields a total of $2 \cdot 5^5$ polynomials, where the multiples of $(x+2)(x+1)$ have been counted twice. In conclusion, $n_1 = 2 \cdot 5^5 - 5^4$.

The computation of n_2 is analogous. We have the multiples of $x+2$, the multiples of $x+1$ and the multiples of $x^4 + x^3 + \cdots + x + 1$. Using the Inclusion-Exclusion Principle, we obtain $n_2 = 7^5 + 7^5 + 7^2 - (7^4 + 7 + 7) + 1$.

The prime ideals of A are either of the form $P \times \mathbb{F}_7[x]/(f_2(x))$, with P prime in $\mathbb{F}_5[x]/(f_1(x))$, or of the form $\mathbb{F}_5[x]/(f_1(x)) \times Q$, with Q prime in $\mathbb{F}_7[x]/(f_2(x))$.

In the quotient ring $\mathbb{F}_5[x]/(f_1(x))$, prime ideals are generated by irreducible factors of $f_1(x)$: we have thus the prime ideals $P = (x+1)$ and $P = (x+2)$. Similarly, in $\mathbb{F}_7[x]/(f_2(x))$ we have $Q = (x+2)$, $Q = (x+1)$ and $Q = (x^4 + x^3 + \cdots + x + 1)$.

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(i) First of all, remark that an element of A is invertible if and only if it is of the form d/s with $(d, 30) = 1$ and $s \in S$. Indeed, for every d that is coprime to 30 there exists an integer e such that $de \equiv 1 (\mathrm{mod}\, 30)$, so that $r = de \in S$; it follows that $se/r \in A$ and that $d/s \cdot se/r = dse/sr = 1$. Conversely, if an element d/s is invertible then there exists $e/r \in A$ such that $de/sr = 1$, which yields $de = sr \in S$, and so $de \equiv 1 (\mathrm{mod}\, 30)$ and d must be coprime to 30.
All ideals of A are of the form $S^{-1}(n)$, where n is an integer. Moreover, if $n = md$ where d and 30 are relatively prime, we have $S^{-1}(n) = S^{-1}(m)$, because $d/1$ is invertible in A by our argument above.
Given integers m and n such that $m \mid n$, we have $(n) \subseteq (m)$, so $S^{-1}(n) \subseteq S^{-1}(m)$. In order for $S^{-1}(n)$ to be maximal, n must be a prime dividing 30. That is, $n \in M = \{2, 3, 5\}$.
Given $p \in M$, if we had $S^{-1}(p) = A$ then $p/1$ would be invertible in A, but p and 30 are not relatively prime. It follows that all ideals of the form $S^{-1}(p)$, for $p \in M$, are different from A.
Now let $q \in M$ be different from p. If we had $p \in S^{-1}(q)$ then we would have $p = (n/s) \cdot q$ for some $n \in \mathbb{Z}$ and $s \in S$. This would yield $ps = nq$, and so q would divide s, as $q \neq p$. But this is not possible, because $s \equiv 1 (\mathrm{mod}\, 30)$ and $q \mid 30$.
We have thus shown that $S^{-1}(2)$, $S^{-1}(3)$ and $S^{-1}(5)$ are the maximal ideals of $S^{-1}\mathbb{Z}$.

(ii) An element of $i^{-1}(A^*)$ is an integer d such that $d/1$ is invertible in A; we have already seen that such integers d are those that are coprime to 30.

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(i) Denote by $\deg_x$ the degree of a polynomial in $A = \mathbb{K}[x, y]$ with respect to the variable x, that is, seen as a polynomial with coefficients in $\mathbb{K}[y]$; similarly, denote by $\deg_y$ the degree with respect to the variable y. Since $\mathbb{K}$ is a field and therefore an integral domain, we have $\deg_x(u(x, y) \cdot v(x, y)) = \deg_x(u(x, y)) + \deg_x(v(x, y))$ for all pairs of polynomials $u(x, y), v(x, y) \in A$. The same holds for $\deg_y$.
Now assume by contradiction that I is generated by $h(x, y) \in A$. From the fact that $f(x) \in I$, it follows that there is $u(x, y) \in A$ such that $f(x) = u(x, y) \cdot h(x, y)$. Consequently, $0 = \deg_y(f(x)) = \deg_y(u(x, y)) + \deg_y(h(x, y))$, hence $\deg_y(h(x, y)) = 0$, that is, only the variable x appears in the polynomial $h(x, y)$. Similarly, $g(y) \in I$ yields that $\deg_x(h(x, y)) = 0$. In conclusion, $h(x, y) = \lambda \in \mathbb{K}$. But then either $\lambda = 0$ and $(h(x, y)) = 0 \neq I$, or $\lambda \neq 0$ and $(h(x, y)) = A$. It is thus enough to show that $I \neq A$.
Indeed, $I = A$ if and only if $1 \in I$, that is, if and only if there exist polynomials $a(x, y)$, $b(x, y)$ such that $1 = a(x, y)f(x) + b(x, y)g(y)$. Suppose by contradiction this is the case and let α be a root of the non-constant polynomial

$f(t)$ in an algebraic closure $\overline{\mathbb{K}}$ of $\mathbb{K}$ and, similarly, let β be a root of $g(t)$ in $\overline{\mathbb{K}}$. If we replace x by α and y by β, we find

$$1 = a(\alpha, \beta) f(\alpha) + b(\alpha, \beta) g(\beta) = 0$$

which is clearly impossible.

(ii) Letting $m = \deg(f(t))$ and $n = \deg(g(t))$, we show that $\mathbb{K}[x, y]/I$ has dimension mn over $\mathbb{K}$ by proving that the elements $x^h y^k + I$, with $0 \le h \le n-1$ and $0 \le k \le m-1$, are a basis over $\mathbb{K}$.

If the expression for $u(x, y) \in \mathbb{K}[x, y]$ includes a monomial $x^r y^s$ with $r \ge n$ or $s \ge m$, then we can use the fact that $f(x) \in I$ and $g(y) \in I$ to replace $x^r y^s$ with monomials of smaller degree, up to adding elements of I. This proves that the elements $x^h y^k$, with $0 \le h \le n-1$ and $0 \le k \le m-1$, form a system of generators for $\mathbb{K}[x, y]/I$ as a vector space over $\mathbb{K}$.

We now prove that the monomials $x^h y^k + I$, with $0 \le h \le n-1$ and $0 \le k \le m-1$, are linearly independent over $\mathbb{K}$ in $\mathbb{K}[x, y]/I$. Indeed, let $\ell(x, y) = \sum_{h,k} \lambda_{h,k} x^h y^k$ be linear combination that is zero in $\mathbb{K}[x, y]/I$, that is, a linear combination yielding an element of I: there exist two polynomials $u(x, y)$ and $v(x, y)$ such that $\ell(x, y) = u(x, y) f(x) + v(x, y) g(y)$.

If $u(x, y)$ features a term whose degree with respect to y is at least $m = \deg_y(g(y))$, we can replace it with terms of lower degree provided we add a multiple of $g(y)$ to $v(x, y)g(y)$, that is, we can replace $v(x, y)$ by some $v'(x, y)$ in order to balance out the desired change in the polynomial $u(x, y)$. By repeating this process, we can eliminate from $u(x, y)$ all terms whose degree with respect to y is at least m and obtain a new polynomial $u'(x, y)$ with $\deg_y(u'(x, y)) < m$.

Consider now the polynomial $\ell(x, y) = u'(x, y) f(x) + v'(x, y) g(y)$: the terms of highest degree with respect to y all appear in the second product $v'(x, y)g(y)$ if $v'(x, y) \neq 0$. But if this were the case, then we would have $\deg_y(\ell(x, y)) \ge \deg_y(g(y)) = m$, which contradicts the fact that $\ell(x, y)$ only has monomials with powers of y whose exponents are less than m. This proves that $v'(x, y) = 0$.

We must therefore have $\ell(x, y) = u'(x, y) f(x)$; this implies that, if $u'(x, y) \neq 0$, then $\deg_x(\ell(x, y)) \ge \deg_x(f(x)) = n$, again impossible because $\ell(x, y)$ only has monomials whose degree in x is less than n. So we also have $u'(x, y) = 0$.

It follows that $\ell(x, y) = 0$ in $\mathbb{K}[x, y]$, which means that $\lambda_{h,k} = 0$ for all h and k. This proves linear independence, as wanted.

▌The formula for the dimension immediately follows from the isomorphism between $\mathbb{K}[x, y]/I$ and the tensor product $\mathbb{K}[x]/(f(x)) \otimes_{\mathbb{K}} \mathbb{K}[y]/(f(y))$, which is however beyond the scope of this book.▐

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(i) Given a formal power series $f(x) = \sum a_n x^n \in \mathbb{K}[[x]]$, let $\big[f(x)\big]_n = a_n$ denote its nth coefficient. For any formal power series $g(x) = \sum b_n x^n$ with coefficients in $\mathbb{K}$, we have

$$\begin{aligned}\big[(Df(x))g(x) + f(x)(Dg(x))\big]_n &= \sum_{h+k=n} \big[Df(x)\big]_h \big[g(x)\big]_k + \\ &\quad + \sum_{h+k=n} \big[f(x)\big]_h \big[Dg(x)\big]_k \\ &= \sum_{h+k=n} (h+1)a_{h+1}b_k + a_h(k+1)b_{k+1} \\ &= (n+1)\sum_{h=0}^{n+1} a_h b_{n+1-k} \\ &= (n+1)\big[f(x)g(x)\big]_{n+1} \\ &= \big[D(f(x)g(x))\big]_n\end{aligned}$$

which proves that $D(f(x)g(x)) = (Df(x))g(x) + f(x)Dg(x)$ in $\mathbb{K}[[x]]$.

(ii) Let $f(x) = \sum a_n x^n$ be a solution to the system $Df(x) = f(x)$, $f(0) = 1$. From $Df(x) = \sum_{n\geq 0}(n+1)a_{n+1}x^n = \sum_{n\geq 0} a_n x^n$ we obtain that $(n+1)a_{n+1} = a_n$ for all $n \geq 0$. Together with the condition that $f(0) = a_0 = 1$, this implies that $a_n = 1/n!$ if $\mathrm{Char}(\mathbb{K}) = 0$. In this case, we have a unique solution.

Now assume that $\mathrm{Char}(\mathbb{K}) = p > 0$. In this case, we have $0 = pa_p = a_{p-1}$, so $a_{p-1} = 0$. It follows that $0 = (p-1)a_{p-1} = a_{p-2}$, and we can repeat this argument to obtain that $a_0 = 0$, which contradicts $a_0 = 1$. Thus, the system has no solutions.

174 For $1 \leq i \leq n$, let $e_i = (0, \ldots, 0, 1, 0 \ldots, 0)$ be the element whose coordinates are all zero save for the ith coordinate, which is equal to 1. We wish to prove that if $\varphi : A \longrightarrow A$ is an automorphism, then φ permutes the elements $e_1, \ldots, e_n$.

For $1 \leq i \leq n$, set $\epsilon_i = \varphi(e_i)$. Remark that if $i \neq j$ then $e_i \cdot e_j = (0, \ldots, 0)$, so $\epsilon_i \cdot \epsilon_j = (0, \ldots, 0)$; this implies that, for $1 \leq k \leq n$, the set $\{\epsilon_1, \ldots, \epsilon_n\}$ contains only one element whose kth coordinate is nonzero. On the other hand, every element $\epsilon_1, \ldots, \epsilon_n$ has at least one nonzero coordinate because the map φ is injective. In conclusion, each of $\epsilon_1, \ldots, \epsilon_n$ has exactly one nonzero coordinate.

This implies that there exists a permutation $\sigma \in \mathsf{S}_n$ such that $\epsilon_i = c_i e_{\sigma(i)}$, with $c_i \in \mathbb{Z}$, for all $1 \leq i \leq n$. But then the fact that $e_i^2 = e_i$ yields $c_i^2 = c_i$, and so $c_i = 1$ because $\epsilon_i \neq 0$. We have thus shown that $\varphi(e_i) = e_{\sigma(i)}$.

The fact that $e_1, \ldots, e_n$ is a set of generators for A as a group implies that there exists at most one automorphism such that $\varphi(e_1), \ldots, \varphi(e_n)$ take certain assigned values. Thus, by the argument above, there is at most one automorphism for each permutation σ, that is, $|\operatorname{Aut}(A)| \leq n!$.

In fact, for each permutation $\sigma \in S_n$, the map

$$A \ni (a_1, a_2, \ldots, a_n) \longmapsto (a_{\sigma(1)}, a_{\sigma(2)}, \ldots, a_{\sigma(n)}) \in A$$

is actually an automorphism, so $|\operatorname{Aut}(\mathbb{Z}^n)| = n!$.

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(i) If such a homomorphism existed, it would satisfy $\varphi(x^3)^2 = \varphi(x^6) = \varphi(x^2)^3 = 1/27$, so $\varphi(x^3)$ would be a rational number whose square is $1/27$, and such a rational number clearly does not exist.

(ii) For the sake of brevity, we denote by A the ring $\mathbb{Z}[x^2, x^3]$. Let $\varphi : A \longrightarrow \mathbb{Q}$ be a homomorphism such that $\varphi(x^2) = 1/4$. By the argument used in (i), we have $\varphi(x^3) = \pm 1/8$. We have found a necessary condition on φ; we will now show that both possibilities for $\varphi(x^3)$ are actually possible.
Remark that A is a subring of $\mathbb{Z}[x]$, and from $\mathbb{Z}[x]$ we do have the homomorphisms $\psi_h : \mathbb{Z}[x] \longrightarrow \mathbb{Q}$ given by $\psi_h(x) = (-1)^h 1/2$, for $h = 0, 1$. The restrictions of the homomorphisms ψ_0, ψ_1 to A are precisely the two homomorphisms from A to $\mathbb{Q}$ that we wished to construct.

(iii) Using the same notation as in (ii), we want to find generators for the kernel of $\varphi_h = \psi_{h|A}$, for $h = 0, 1$. By definition, $\operatorname{Ker}(\varphi_h) = \operatorname{Ker}(\psi_h) \cap A$, so we first consider the kernel of ψ_h.
It is clear that $2x - 1$ belongs to the kernel of ψ_0. Conversely, if $f(x)$ is such that $\psi(f(x)) = 0$ then $1/2$ is a root of $f(x)$ in $\mathbb{Q}$, because the homomorphism ψ_0 yields the value of a polynomial at $1/2$. It follows that $x - 1/2$ divides $f(x)$ in $\mathbb{Q}[x]$, that is, the primitive polynomial $2x - 1$ of $\mathbb{Z}[x]$ divides the polynomial $f(x)$ of $\mathbb{Z}[x]$ in $\mathbb{Q}[x]$. By Gauss's Lemma, this implies that $2x - 1$ also divides $f(x)$ in $\mathbb{Z}[x]$, so $f(x) \in \mathbb{Z}[x] \cdot (2x - 1)$. One can prove that $\operatorname{Ker}(\psi_1) = \mathbb{Z}[x] \cdot (2x + 1)$ in the same way.
We now show that the three polynomials $4x^2 - 1$, $2x^3 - x^2$, $2x^4 - x^3$ generate the ideal $\operatorname{Ker}(\varphi_0) = (2x - 1) \cap A$ of A.
The three polynomials are elements of $(2x - 1) \cap A$, as they belong to A and vanish at $1/2$. Conversely, an element $f(x)$ of $(2x - 1) \cap A$ can be expressed as $f(x) = (2x - 1)(k_0 + k_1 x + k_2 x^2 + k_3 x^3 + k_4 x^4 + \cdots + k_n x^n)$, where $k_0, \ldots, k_n \in \mathbb{Z}$, and also $k_1 = 2k_0$ because $f(x)$, being an element of A, has no linear term. So $f(x) = k_0(2x-1)(2x+1) + k_2(2x^3 - x^2) + k_3(2x^4 - x^3) + \cdots + k_n(2x^{n+1} - x^n)$. Now, for $r \geq 4$ we have $x^{r-2} \in A$, and $2x^{r+1} - x^r = x^{r-2}(2x^3 - x^2)$, so each summand belongs to the ideal of A generated by the three polynomials we are considering. This concludes the proof that such polynomials are generators.
Analogously, one finds that $4x^2 - 1$, $2x^3 + x^2$ and $2x^4 + x^3$ are generators for $\operatorname{Ker}(\varphi_1)$.

176

(i) Remark that a polynomial $h(x)$ is an element of S if and only if $h(1)$ is odd. It follows easily that S is a multiplicative set; indeed, we clearly have that $0 \notin S$,

$1 \in S$ and, given elements $f(x)$, $g(x)$ of S, the integer $f(1)g(1)$ is the product of two odd integers, so it is odd.

(ii) Consider $f_p(x) = x + p$, where p is an odd prime. The polynomial $f_p(x)$ is irreducible in $\mathbb{Z}[x]$ and $f_p(x) \notin S$ because $f_p(1) = 1 + p$ is even. It follows that the ideal $I_p = (f_p(x))$ of $\mathbb{Z}[x]$ is prime and does not intersect S, as for each $g(x) \in \mathbb{Z}[x]$ the product $g(1)f_p(1)$ is even, because $f_p(1)$ is even.
Moreover, $f_p(x)$ is the only polynomial of minimal degree in I_p; so the ideals I_p, where p ranges among all odd primes, are pairwise distinct.
We conclude that the ideals $S^{-1}I_p$ are prime in $S^{-1}\mathbb{Z}[x]$ and they are distinct thanks to the correspondence between ideals of A and prime ideals of $\mathbb{Z}[x]$ that do not intersect S.

(iii) We know that proper ideals of A are of the form $S^{-1}I$, where I is some ideal of $\mathbb{Z}[x]$ that does not intersect S; that is, $I \subseteq A \setminus S$.
Thus if we prove that $M = \mathbb{Z}[x] \setminus S$ is an ideal of $\mathbb{Z}[x]$ then the desired result follows. But M is the kernel of the composition of surjective homomorphisms

$$\begin{array}{ccccc} \mathbb{Z}[x] & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}/2\mathbb{Z} \\ f(x) & \longmapsto & f(1) & & \\ & & a & \longmapsto & a + 2\mathbb{Z} \end{array}$$

and thus it is an ideal.

177

(i) We show that an ideal (x) is maximal within the family of proper principal ideals if and only if x is an irreducible element.
Indeed, (x) being maximal among principal ideals is equivalent to the fact that if $x = ab$ then either $(x) = (a)$, so b is invertible, or $(a) = A$, so a is invertible, which is precisely the definition of being irreducible.
On the other hand, we know that a principal ideal is prime if and only if it is generated by a prime element.
The statement under consideration is thus equivalent to every irreducible element being prime, and so it is false in general: for instance, in $\mathbb{Z}[\sqrt{-5}]$, which is not a unique factorisation domain, the element 2 is irreducible but not prime.

(ii) Let $M = \mathrm{Ann}(x)$ be a maximal element in the family defined in the problem statement. We now prove that M is a prime ideal of A.
Let $y, z \in A$ be such that $yz \in M$, that is, $yz \cdot x = 0$, and assume that $z \notin M$, that is, $z \cdot x \neq 0$. We shall prove that $y \in M$.
Remark that if $t \in \mathrm{Ann}(x)$ then $t \cdot zx = z(tx) = 0$, so $t \in \mathrm{Ann}(zx)$. This shows that $\mathrm{Ann}(x) \subseteq \mathrm{Ann}(zx)$, where $\mathrm{Ann}(zx)$ belongs to the family of ideals under consideration because we have assumed $zx \neq 0$.

But $M = \mathrm{Ann}(x)$ is maximal among annihilators of nonzero elements, so we have $\mathrm{Ann}(x) = \mathrm{Ann}(zx)$. Since $y \cdot zx = yz \cdot x = 0$ we have $y \in \mathrm{Ann}(zx) = \mathrm{Ann}(x) = M$, as wanted.

178

(i) Let $p, q \in A$ be such that $P = (p)$ and $Q = (q)$. A maximal ideal M of A is necessarily prime, so we can only have $M = P$ or $M = Q$, whereas the ideal 0 is not maximal, as it is properly contained, for example, in the ideal P.
Now assume by contradiction that $P^m + Q^n \neq A$ for some positive integers m, n. It follows that $P^m + Q^n$ is contained in a maximal ideal of A; by symmetry we may assume that $P^m + Q^n \subseteq P$. Since obviously $P^m \subseteq P$, this yields $Q^n = (q^n) \subseteq P = (p)$, that is, that p divides q^n. But, since p is prime, this in turn implies that p divides q and thus $P = (p) = (q) = Q$, because q is irreducible. This is clearly a contradiction.

(ii) We keep denoting by p and q two generators for P and Q, respectively. Remark that p and q are the only irreducible elements in A, as all irreducible elements in a unique factorisation domain are prime, so they generate prime ideals. It follows that every element of A has a factorisation of the form $up^c q^d$, where c, d are natural numbers and u is a unit of the ring A.
We show that if $I \neq 0$ is an ideal of A then it is of the form $I = (p^r q^s)$, where $p^r q^s$ is a greatest common divisor of the elements in I. In other words, r and s are the maximum exponents of p and q, respectively, such that all elements of I are divisible by p^r and q^s.
From the definition of r and s we have that, given any $x \in I$, the element x is divisible by $p^r q^s$, hence $I \subseteq (p^r q^s)$.
As for the opposite inclusion, again by the definition of r and s, we know that there are $a, b \in I$ such that p^r divides a, p^{r+1} does not divide a, q^s divides b, and q^{s+1} does not divide b. Then, up to multiplication by invertible elements, we may assume that $a = p^r q^{s_1}$ and $b = p^{r_1} q^s$, where $s_1 \geq s$, $r_1 \geq r$ are natural numbers.
Now, if $r_1 = r$ or $s_1 = s$, then we find that $p^r q^s \in I$, and thus $(p^r q^s) \subseteq I$. Suppose instead that $r_1 = r + m$, $s_1 = s + n$ with $m, n > 0$. By (i), there exist $x, y \in A$ such that $xp^m + yq^n = 1$; multiplying by $p^r q^s$ yields $xb + ya = p^r q^s$, and thus $p^r q^s \in I$.

179

(i) It is known that $f(x) = x^4 + x^3 + x^2 + x + 1$ is the minimal polynomial of ζ_5 over $\mathbb{Q}$. We proceed to show that $\mathbb{Z}[x]/(f(x)) \simeq \mathbb{Z}[\zeta_5]$. Consider the homomorphism $\mathbb{Z}[x] \ni h(x) \longmapsto \varphi(h(x)) = h(\zeta_5) \in \mathbb{Z}[\zeta_5]$, which is clearly surjective by the definition of $\mathbb{Z}[\zeta_5]$. We have $\varphi(f(x)) = f(\zeta_5) = 0$, so $(f(x)) \subseteq \mathrm{Ker}(\varphi)$.
Conversely, consider $h(x) \in \mathrm{Ker}(\varphi)$. We have $h(\zeta_5) = 0$ and so in particular $h(x)$ is divisible by $f(x)$ in $\mathbb{Q}[x]$. But then $h(x)$ is also divisible by $f(x)$ in $\mathbb{Z}[x]$ by Gauss's Lemma, that is, $h(x) \in (f(x))$.

(ii) Thanks to (i), we have

$$\mathbb{Z}[\zeta_5]/(11) \simeq \mathbb{Z}[x]/(f(x), 11) \simeq (\mathbb{Z}/11\mathbb{Z})[x]/(f(x)) \simeq \mathbb{F}_{11}[x]/(f(x)).$$

But $f(x)$ is a product of linear factors in $\mathbb{F}_{11}[x]$, as 5 divides the order of $\mathbb{F}_{11}^*$. It follows that $\mathbb{Z}[\zeta_5]/(11)$ is isomorphic to $\mathbb{F}_{11}^4$, which is not a field.

(iii) Since x^4+1 is the eighth cyclotomic polynomial over $\mathbb{Q}[x]$, a similar argument to that of (i) yields that $\mathbb{Z}[\zeta_8] \simeq \mathbb{Z}[x]/(x^4+1)$ and thus $\mathbb{Z}[\zeta_8]/(11) \simeq \mathbb{F}_{11}[x]/(x^4+1)$.
The roots of x^4+1 are all primitive eighth roots of unity. Since the order of $\mathbb{F}_{11}^*$ is not divisible by 8, x^4+1 has no roots in $\mathbb{F}_{11}$, so $\mathbb{F}_{11}[x]/(x^4+1) \not\simeq \mathbb{F}_{11}^4$.
▌The eighth roots of unity belong to $\mathbb{F}_{11^2}$, because $8 \mid 11^2 - 1 = 120$, so the polynomial x^4+1 is the product of two irreducible factors of degree two in $\mathbb{F}_{11}[x]$; it follows that $\mathbb{F}_{11}[x]/(x^4+1) \simeq (\mathbb{F}_{11^2})^2$. One can also check that the actual factorisation of x^4+1 in $\mathbb{F}_{11}[x]$ is $x^4+1 = (x^2-3x-1)(x^2+3x-1)$.▐

180

(i) Since $\mathbb{Q}[x, y]$ is a unique factorisation domain, every irreducible element is prime and thus generates a prime ideal. We will show that, for example, the ideals $I_a = (x + ay)$, where $a \in \mathbb{Q}$, which are clearly prime because any polynomial of degree one is irreducible, are not maximal and are all distinct.
Indeed, if $I_a = I_b$ then $x + ay = c(x + by)$, where c is some constant. But equating the coefficients of the linear terms yields $c = 1$, so $a = b$. Moreover, $I_a \subsetneq (x, y)$, as $x + ay$ does not divide y, so I_a is not maximal.

(ii) First of all, note that $(f(x, y)) = (f(x, y), x)$ if and only if $x \in (f(x, y))$, that is, if and only if $f(x, y) \mid x$. The only divisors of x, up to multiplication by invertible elements, are 1 and x, and $f(x, y)$ is not constant by hypothesis, so the equality holds if and only if $f(x, y) = cx$, where c is some nonzero constant.
We write the polynomial $f(x, y)$ in the form $f(x, y) = g(y) + xh(x, y)$ and proceed to determine which cases lead to $(f(x, y), x) = \mathbb{Q}[x, y]$.
We have

$$(f(x, y), x) = (g(y) + xh(x, y), x) = (g(y), x),$$

so if $g(y)$ is a nonzero constant then $(f(x, y), x) = \mathbb{Q}[x, y]$. On the other hand, if $(f(x, y), x) = \mathbb{Q}[x, y]$ then there exist $a(x, y)$, $b(x, y)$ in $\mathbb{Q}[x, y]$ such that $a(x, y)g(y) + b(x, y)x = 1$ and, letting $x = 0$, we have $a(0, y)g(y) = 1$. So $g(y)$ is a nonzero constant.
In conclusion, the required polynomials are those of the form $f(x, y) = c + xh(x, y)$, where c is a nonzero constant and $h(x, y)$ is a polynomial in $\mathbb{Q}[x, y]$.

181 Let A denote the ring $\mathbb{K}[x, y]$ and I its ideal (x^m, y^n). If $mn = 0$ then $m = 0$ or $n = 0$, so $1 \in I$, and thus the ring $A/I = 0$ has no prime ideals.

Assume $m, n > 0$ and let $\overline{P}$ be a prime ideal of A/I. Letting $\pi : A \longrightarrow A/I$ be the quotient homomorphism, remark that $P = \pi^{-1}(\overline{P})$ is a prime ideal of A and contains I. In particular, $x^m \in P$ and $m > 0$, so we also have $x \in P$ and, by the same argument, $y \in P$. This proves that $(x, y) \subseteq P$, which yields $P = (x, y)$ as $A/(x, y) \simeq \mathbb{K}$ and so (x, y) is maximal. It follows that $\overline{P} = \pi(P) = (x, y)/I$ is the unique prime ideal of A/I.

182

(i) Let n be the dimension of A as a vector space over $\mathbb{K}$ and consider an element $a \in A$. The $n+1$ elements $1, a, a^2, \ldots, a^n$ of A cannot be linearly independent over $\mathbb{K}$, as $n + 1 > \dim_{\mathbb{K}}(A)$. Consequently, there exist some coefficients $\lambda_0, \lambda_1, \ldots, \lambda_n \in \mathbb{K}$, not all of them zero, such that

$$\lambda_n a^n + \cdots + \lambda_1 a + \lambda_0 = 0.$$

The polynomial $f(x) = \lambda_n x^n + \cdots + \lambda_1 x + \lambda_0 \in \mathbb{K}[x]$ is thus nonzero and has a as a root.

(ii) Letting P be a prime ideal of A, consider the quotient $B = A/P$. We know that B is an integral domain and that it is a finite-dimensional vector space over $\mathbb{K}$, because A is finite-dimensional. We wish to show that B is a field, which will imply that P is a maximal ideal.
Indeed, given a nonzero element $b \in B$, consider the $\mathbb{K}$–linear map $B \ni c \longmapsto f(c) = bc \in B$. Since $b \neq 0$ and B is an integral domain, we immediately obtain that $\mathrm{Ker}(f) = 0$, that is, f is injective. But then f is also surjective, as B is finite-dimensional over $\mathbb{K}$. In particular, there exists $c \in B$ such that $f(c) = bc = 1$, that is, b is invertible.

183

(i) The polynomial $\Phi_{p-1}(t)$ has degree $r = \phi(p-1)$ and is a divisor of $t^{p-1} - 1$. Moreover, thanks to Fermat's Little Theorem, in $\mathbb{F}_p(t)$ we have $t^{p-1} - 1 = (t-1)(t-2)\cdots(t-p+1)$ and so $\Psi_{p-1}(t) = (t-a_1)\cdots(t-a_r)$ for some $a_1, \ldots, a_r$. Since the a_i's are all different, the factors $t - a_i$ are relatively prime. On the other hand, the Chinese Remainder Theorem for Rings yields the isomorphism

$$\mathbb{F}_p[t]/\big((t-a_1)\cdots(t-a_r)\big) \simeq \mathbb{F}_p[t]/(t-a_1) \times \cdots \times \mathbb{F}_p[t]/(t-a_r) \simeq \mathbb{F}_p^r.$$

(ii) The polynomial $\Phi_p(t)$ has degree $p-1$ and is a divisor of $t^p - 1$. The latter polynomial factors in $\mathbb{F}_p[t]$ as $(t-1)^p$. It follows that the factorisation of $\Psi_p(t)$ is $(t-1)^{p-1}$, and thus the class of $t-1$ is a nonzero nilpotent element in $\mathbb{F}_p[t]/\big((t-1)^{p-1}\big)$.

184

(i) It is clear that $P \subseteq (P : J)$. Suppose that some element $x \in (P : J)$ does not belong to P. For all $a \in J$ we have $xa \in P$, which, since P is prime, yields

that $a \in P$; so $J \subseteq P$. In conclusion: if $J \not\subseteq P$ then $(I : J) = P$, whereas if $J \subseteq P$, we clearly have $1 \in (P : J)$ and thus $(P : J) = A$.

(ii) By definition,

$$D = ((18 + 6i) : (10)) = \left\{ \alpha \in \mathbb{Z}[i] \,|\, 10\alpha \in (6(3 + i)) \right\},$$

so D is the set of elements α such that $6(3 + i)$ divides 10α. Since $\mathbb{Z}[i]$ is a principal ideal domain, and in particular a unique factorisation domain, this divisibility condition is equivalent to $3(3 + i)$ dividing 5α.
In $\mathbb{Z}[i]$, the prime 5 factors as the product of irreducibles $(2 + i)(2 - i)$, whereas 3 is itself irreducible, and finally $3 + i = (1 + i)(2 - i)$, where $1 + i$ and $2 - i$ are irreducible. It follows that the condition on α is equivalent to $3(1 + i) \,|\, (2 + i)\alpha$.
Now, 3, $1 + i$ and $2 + i$ are irreducible and they are not associates, so $\alpha \in D$ if and only if $3(1 + i) \,|\, \alpha$, that is, $D = (3 + 3i)$.

(iii) Let $A = \mathbb{Z}[\sqrt{-5}]$ and set $x = 2$, $y = 1 + \sqrt{-5}$. We wish to prove that $D = ((2) : (1 + \sqrt{-5}))$ is not principal in A.
We begin by checking that $D = (2, 1 + \sqrt{-5})$. Indeed, the definition yields that $\alpha = a + b\sqrt{-5} \in D$ if and only if

$$(a + b\sqrt{-5})(1 + \sqrt{-5}) = (a - 5b) + (a + b)\sqrt{-5} \in (2),$$

that is, if and only if $a - 5b$ and $a + b$ are even; this is obviously equivalent to $a + b$ being even. It is then clear that $(2, 1 + \sqrt{-5}) \subseteq D$.
Conversely, if $a + b$ is even then we have the following possibilities: a is even, so b is also even and $a + b\sqrt{-5} \in (2)$, or a is odd, so b is also odd and therefore $a + b\sqrt{-5} = \big((a - 1) + (b - 1)\sqrt{-5}\big) + (1 + \sqrt{-5})$, with $\big((a - 1) + (b - 1)\sqrt{-5}\big) \in (2)$ and $a + bi \in (2, 1 + \sqrt{-5})$. This completes the proof that $D = (2, 1 + \sqrt{-5})$.
We now show that the ideal $(2, 1 + \sqrt{-5})$ is not principal in A. First of all, note that 2 is irreducible: indeed, if we consider as usual the norm

$$A \ni a + b\sqrt{-5} \overset{N}{\longmapsto} (a + b\sqrt{-5})(a - b\sqrt{-5}) = a^2 + 5b^2 \in \mathbb{N},$$

we immediately find that the invertible elements of A are those of norm 1. Now, $N(2) = 4$ and, if $u + v\sqrt{-5}$ is a divisor of 2 that is not invertible and is not an associate of 2, then it must necessarily have norm 2. But the equation $u^2 + 5v^2 = 2$ clearly has no solutions in $\mathbb{Z}$; this shows that 2 is irreducible.
It follows that the ideal (2) is maximal among principal ideals, and since $(2) \subsetneq (2, 1 + \sqrt{-5}) \subsetneq A$, the ideal $(2, 1 + \sqrt{-5})$ cannot be principal.

185

(i) An element $f(x)/g(x) \in A$ that is reduced to lowest terms is invertible in A if and only if its inverse $g(x)/f(x)$ in the field $\mathbb{Q}(x)$ belongs to A; that is, if and

only if the denominator $f(x)$ does not vanish at 0 or -1, which is equivalent to $f(0)f(-1) \neq 0$.

(ii) The ring A is clearly an integral domain, because it is a subring of the field $\mathbb{Q}(x)$. Since $\mathbb{Q}[x]$ is a unique factorisation domain, every polynomial $f(x) \in \mathbb{Q}[x] \setminus \{0\}$ has a unique expression of the form $f(x) = x^a(x+1)^b u(x)$ with $a, b \in \mathbb{N}$ and $u(x) \in \mathbb{Q}[x]$ such that $u(0)u(-1) \neq 0$. It follows that for every element $f(x)/g(x) \in A \setminus \{0\}$ we have

$$\frac{f(x)}{g(x)} = x^a(x+1)^b \frac{u(x)}{g(x)}$$

where $a, b \in \mathbb{N}$ are uniquely determined and $u(x)/g(x)$ is invertible in A by (i). So every element of A is an associate of a polynomial of the form $x^a(x+1)^b$.

Let $I \subseteq A$ a nonzero ideal; we have just established that I is generated by a family of polynomials of the form $x^{a_r}(x+1)^{b_r}$ for some a_r, b_r in $\mathbb{N}$, where r ranges within some set of indices. Let $x^a(x+1)^b$ be a greatest common divisor in $\mathbb{Q}[x]$ of the polynomials $x^{a_r}(x+1)^{b_r}$: such a greatest common divisor does exist and is a finite linear combination over $\mathbb{Q}[x]$ of polynomials of the form $x^{a_r}(x+1)^{b_r}$, because $\mathbb{Q}[x]$ is a Euclidean domain. This implies that I is generated by $x^a(x+1)^b$. We have thus shown that A is a principal ideal domain.

(iii) Since A is a principal ideal domain by (ii), the prime ideals of A are 0 and the maximal ideals, which are those generated by irreducible elements of A. We have seen that every element of A is an associate of a polynomial of the form $x^a(x+1)^b$, where a, b are natural numbers, and that the elements x and $x+1$ are not invertible, so x and $x+1$ are precisely the irreducible elements of A. In conclusion, the prime ideals of A are 0, (x) and $(x+1)$.

186 Recall that the set of nilpotent elements in a given ring is an ideal, as it is the nilradical of the ring.

Let $N(A)$ and $N(A[x])$ be the nilradicals of the rings A and $A[x]$, respectively. First of all, note that $N(A) \subseteq N(A[x])$, and thus we can refer to nilpotent elements without specifying the ring, with no risk of ambiguities.

Let $f(x) = \sum_{h=0}^{n} a_h x^h$ be a polynomial whose coefficient are nilpotent, that is, such that $a_h \in N(A) \subseteq N(A[x])$ for all $h = 0, 1, \ldots, n$; the polynomial $f(x)$ is also nilpotent, because $N(A[x])$ is an ideal.

Conversely, assume that $f(x) = \sum_{h=0}^{n} a_h x^h$ is nilpotent; we show by induction on n that its coefficients must be as well.

For $n = 0$, the statement is clearly true. Suppose $n \geq 1$; by definition of a nilpotent element, there exists $k > 0$ such that $f(x)^k = 0$. The highest degree term in $f(x)^k$ is $a_n^k x^{nk}$, which implies that $a_n^k = 0$, that is, a_n is nilpotent. So $a_n x^n$ is also nilpotent, and $g(x) = f(x) - a_n x^n$ is as well because of the fact that $N(A[x])$ is an ideal. But the degree of $g(x)$ is strictly less than n, so its coefficients $a_0, a_1, \ldots, a_{n-1}$ are all nilpotent by the induction hypothesis.

187

(i) Consider $x \in J(A)$ and $y \in A$. Given a maximal ideal M of A, remark that $x \in M$ and so $xy \in M$; but then $1+xy \notin M$, or we would have $1 \in M$, which is impossible. So $1 + xy$ does not belong to any maximal ideal, which implies that it is invertible.
Conversely, suppose that $x \in A$ is such that $1 + xy$ is invertible for all $y \in A$. Given a maximal ideal M of A, we have $(M, x) \neq A$: if this were not the case, there would exist $a \in M$ and $y \in A$ such that $1 = a + xy$, which would yield $1 + x(-y) = a \in M$, and so the element $1 + x(-y)$, since it belongs to the maximal ideal M, would not be invertible, contradicting our assumption. So (M, x) is a proper ideal containing the maximal ideal M, which can only be the case if $x \in M$. Thus the element x belongs to every maximal ideal of A, that is, it belongs to $J(A)$.

(ii) Letting $x \in J(A)$, in order to show that $\varphi(x) \in J(B)$, we can use (i) and prove that $1 + \varphi(x)b$ is invertible in B for all $b \in B$. Since φ is surjective, we have $b = \varphi(a)$ for some $a \in A$; thus, $1 + \varphi(x)b = \varphi(1 + xa)$ is the image of an invertible element in A and is thus invertible in B.

▌The ideal $J(A)$ is called the *Jacobson radical* of the ring A.▐

188 Let γ denote the element $3 + i$ of $\mathbb{Z}[i]$ and let A be the ring in the problem statement. First of all, remark that $S = \{\beta \in \mathbb{Z}[i] \,|\, (\beta, \gamma) = 1\}$ is a multiplicative set in $\mathbb{Z}[i]$. Indeed, this follows from $\mathbb{Z}[i]$ being a Euclidean domain, and thus a unique factorisation domain: if β_1 and β_2 are not divisible by any prime π that divides γ, then neither is $\beta_1\beta_2$.

Moreover, it is clear from the definition that $A = S^{-1}\mathbb{Z}[i]$ as subrings of the quotient field of $\mathbb{Z}[i]$.

If $a = \alpha/\beta$ is invertible in A, that is, if there exists $b = \alpha'/\beta' \in A$ such that $ab = 1$, then $\alpha\alpha' = \beta\beta' \in S$, so α and γ are relatively prime, that is, $\alpha \in S$. Conversely, it is clear that if $\alpha \in S$ then a is invertible. We have thus shown that A^* is the set of elements α/β with α and β coprime to γ.

Obviously, 0 is a prime ideal of A because A is an integral domain. Moreover, the nonzero prime ideals of the localisation $S^{-1}\mathbb{Z}[i]$ are the ideals $S^{-1}P$, where P is a nonzero prime ideal of $\mathbb{Z}[i]$ that does not intersect S. As mentioned before, $\mathbb{Z}[i]$ is a Euclidean domain, so $P = (\pi)$ is prime if and only if π is prime, or, equivalently, irreducible. Since the factorisation of $\gamma = 3+i$ as a product of irreducible elements in $\mathbb{Z}[i]$ is $\gamma = (1 + i)(2 - i)$, we obtain that $(\pi) \cap S = \varnothing$ if and only if, up to associates, $\pi = 1 + i$ or $\pi = 2 - i$.

We can thus conclude that the prime ideals of A are 0, $(1 + i)$ and $(2 - i)$.

189

(i) First of all, we show that $\mathfrak{f}$ is contained in A: if $a \in \mathfrak{f}$ then $a = a \cdot 1 \in aR \subseteq A$. Clearly, it is now enough to prove that $\mathfrak{f}$ is an ideal of R to obtain the desired result.

If $a, b \in \mathfrak{f}$ then $(a+b)R \subseteq aR + bR \subseteq A + A = A$, so $a + b \in \mathfrak{f}$. Moreover, if $a \in \mathfrak{f}$ and $b \in R$ then $(ab)R = a(bR) \subseteq aR \subseteq A$ and $ab \in \mathfrak{f}$. This proves that $\mathfrak{f}$ is an ideal of R.

(ii) Let $I \subseteq A$ be an ideal of R; we have $IR \subseteq I \subseteq A$, that is, $I \subseteq \mathfrak{f}$. So $\mathfrak{f}$ contains all ideals of R that are contained in A and, since it is itself contained in A, it is the largest ideal of R with this property.

(iii) For the sake of brevity, set $\zeta = (-1 + \sqrt{-3})/2$. Observe that $2\zeta = -1 + \sqrt{-3} \in A$ and thus $R \cdot 2 \subseteq \mathfrak{f}$ because $\mathfrak{f}$ is an ideal of R.
Moreover, if $x + y\sqrt{-3} \in \mathfrak{f}$ then $(x + y\sqrt{-3})\zeta \in (x + y\sqrt{-3})R \subseteq A$, which yields

$$(x + y\sqrt{-3})\zeta = \frac{-x - 3y}{2} + \frac{x - y}{2}\sqrt{-3} \in A.$$

What we obtained is clearly equivalent to $x + y$ being even. Conversely, a multiple of 2 in R is an element of the form $2(u + v\zeta)$, where u and v are integers. But we have

$$2(u + v\zeta) = 2(u - \frac{v}{2} + \frac{v}{2}\sqrt{-3}) = (2u - v) + v\sqrt{-3},$$

so being a multiple of 2 is equivalent to being an element of $\mathfrak{f}$. We have thus shown that $\mathfrak{f}$ is generated by 2 as an ideal of R.

[The ideal $\mathfrak{f}$ is called the *conductor* of A in R.]

190

(i) Assume $N = (a_1, a_2, \ldots, a_m)$ is finitely generated and let $k_1, k_2, \ldots, k_m$ be such that $a_h^{k_h} = 0$ for $h = 1, 2, \ldots, m$. Letting $k = k_1 + k_2 + \cdots + k_m$, remark that N^k is generated by the products $a_1^{r_1} a_2^{r_2} \cdots a_m^{r_m}$ with $r_1 + r_2 + \cdots + r_m = k$. Now, if we had $r_h < k_h$ for all h then we would have $r_1 + r_2 + \cdots + r_m < k$. So at least one of the exponents, say r_t, is such that $r_t \geq k_t$, hence $a_t^{r_t} = 0$ and thus $a_1^{r_1} a_2^{r_2} \cdots a_m^{r_m} = 0$. This proves that $N^k = 0$.

(ii) If $ab = 1$ in A then $(a + N)(b + N) = 1 + N$, so $a + N$ is invertible in A/N because $1 + N$ is the one of A/N.
Conversely, if $a \in A$ is such that $(a + N)(b + N) = 1 + N$ for some $b \in A$, then there exists $x \in N$ such that $ab = 1 + x$. The element x is nilpotent, so there is $n > 0$ such that $x^n = 0$ and we can obviously assume n to be odd. Thus

$$1 = 1 + x^n = (1 + x)(1 - x + x^2 + \cdots \pm x^{n-1})$$

and so, letting c be the second factor of the right-hand side, we have $a(bc) = (ab)c = (1 + x)c = 1$, that is, a is invertible in A.

191

(i) By definition, A is a subring of $\mathbb{C}$, so it is an integral domain. Setting $\gamma = (1+\sqrt{-7})/2$ for ease of notation, remark that every element α of A can be expressed in a unique way as $a+b\gamma$, where a and b are integers. Moreover, we have

$$|\alpha|^2 = \alpha\cdot\overline{\alpha} = \big(a+b\frac{1+\sqrt{-7}}{2}\big)\cdot\big(a+b\frac{1-\sqrt{-7}}{2}\big) = a^2+ab+2b^2 = d(\alpha),$$

so the degree of α is its square norm as a complex number. In particular, the degree is multiplicative, and the fact that it takes natural values implies that $d(\alpha\cdot\beta) = d(\alpha)d(\beta) \geq d(\alpha)$ for all $\alpha,\beta \in A\setminus\{0\}$.
We still need to show that we can perform divisions with remainder. Fix some $\beta = a+b\gamma \in A\setminus\{0\}$ and let $\Lambda = (\beta) = \mathbb{Z}[\gamma]\cdot\beta$ be the ideal generated by β in A. Clearly, every element of Λ is a linear combination of β and $\beta' = \gamma\beta$ with integer coefficients.
If we think of β and β' as vectors in the complex plane, we find that their lengths are $|\beta|$ and $|\beta'| = |\gamma||\beta| = \sqrt{2}|\beta|$, and that the cosine of the angle between them is $\sqrt{2}/4$; it follows that $|\beta'-\beta| = \sqrt{2}|\beta|$. Thus, we can see the ideal Λ as a set of points in $\mathbb{C}$ that partition the plane into congruent isosceles triangles with sides of length $|\beta|$, $\sqrt{2}|\beta|$ and $\sqrt{2}|\beta|$.
It is easy to compute the radius of the circumcircle of one of these triangles, which is $2|\beta|/\sqrt{7}$. This radius is less than $|\beta|$, and for every point inside a triangle the minimal distance to one of the vertices is at most equal to the radius of the circumcircle, so we have the following: for each point α of $\mathbb{C}$ there exists a point $\alpha' = \delta\cdot\beta$ of Λ such that $|\alpha-\alpha'| \leq |\beta|$. In particular, this is true for the elements α of A, and, letting $\epsilon = \alpha-\alpha'$, we have $\alpha = \delta\beta+\epsilon$ with $d(\epsilon) \leq d(\beta)$. We have thus constructed the division with remainder of α by β, where the (possible) remainder is ϵ.

(ii) Let us see whether the factorisation $10 = 2\cdot 5$ in $\mathbb{Z}$ can be made finer in A.
Note that the minimal value of the degree of nonzero elements is 1, so elements of degree 1 are invertible in A. Moreover, since the degree is multiplicative, elements whose degree is prime in $\mathbb{Z}$ are necessarily irreducible.
In particular, the factorisation

$$2 = \gamma\cdot(1-\gamma) = \frac{1+\sqrt{-7}}{2}\cdot\frac{1-\sqrt{-7}}{2}$$

consists of irreducible factors, because their degree is 2.
Now, the degree of 5 is 25, so a nontrivial factorisation in A would involve elements of degree 5; we show that 5 is irreducible by proving that there are no elements of degree 5 in A. Indeed, if

$$a+b\gamma = \big(a+\frac{b}{2}\big)+\frac{b}{2}\sqrt{-7}$$

has square norm equal to 5, then $(2a+b)^2+7b^2=20$ and so $|b|<2$. But for $b=0,\pm1$, the integer $20-7b^2$ is not a square; so the equation has no integer solutions for a, b, and thus the ring has no elements of degree 5.
In conclusion, the factorisation of 10 in A is given by

$$10=5\cdot\gamma\cdot(1-\gamma).$$

192

(i) Suppose $I=(x)$; the fact that $I=I^2$ is clearly equivalent to $x\in(x^2)$. Now, if $x\in(x^2)$, then there exists $y\in A$ such that $x=x^2y$, which yields $x=x(xy)$, that is, $x(xy-1)=0$. But A is an integral domain, so either $x=0$ and thus $I=0$, or $xy-1=0$ and thus x is invertible, so $I=A$.

(ii) Any two distinct maximal ideals M_i and M_j are clearly such that $M_i+M_j=A$. So, by the Chinese Remainder Theorem for Rings, $A/(M_1\cdot M_2\cdots M_n)$ is isomorphic to $A/M_1\times A/M_2\times\cdots\times A/M_n$.
The ideals of a direct product of rings are direct products of ideals; in this case, since A/M_h is a field for each h, each factor of a direct product of ideals is either the zero ideal or the whole ring.
In other words, any ideal I of the quotient is generated by some $(\epsilon_1,\epsilon_2,\dots,\epsilon_n)$, where $\epsilon_h=0$ or $\epsilon_h=1$ for each h. In any case,

$$I^2=(\epsilon_1,\epsilon_2,\dots,\epsilon_n)^2=(\epsilon_1^2,\epsilon_2^2,\dots,\epsilon_n^2)=(\epsilon_1,\epsilon_2,\dots,\epsilon_n)=I.$$

193

(i) Consider the ring homomorphism

$$\mathbb{Z}[x]\ni f(x)\overset{\varphi}{\longmapsto} f(a)\in\mathbb{Z}/p\mathbb{Z}$$

that sends a polynomial to the class modulo p of its value at a. If we show that φ is surjective and that its kernel is $(p,x-a)$, then we obtain that the ideal $(p,x-a)$ is prime in $\mathbb{Z}[x]$ because $\mathbb{Z}/p\mathbb{Z}$ is a field.
Clearly, φ is surjective, because the image of the subset of constant polynomials is all of $\mathbb{Z}/p\mathbb{Z}$. Moreover, p and $x-a$ are obviously in $\mathrm{Ker}(\varphi)$, so $(p,x-a)\subseteq\mathrm{Ker}(\varphi)$.
In order to show the opposite inclusion, let $h(x)\in\mathbb{Z}[x]$ be such that $\varphi(h(x))=0$. We have $h(a)\in p\mathbb{Z}$, that is, there exists $r\in\mathbb{Z}$ such that $h(a)=rp$. But then the polynomial $h(x)-rp$ has a as a root, and so, by Ruffini's Theorem, it is divisible by $x-a$ in $\mathbb{Q}[x]$, and thus in $\mathbb{Z}[x]$ by Gauss's Lemma. Consequently, there exists $k(x)\in\mathbb{Z}[x]$ such that $h(x)-rp=k(x)(x-a)$ or, in other words, $h(x)=rp+k(x)(x-a)\in(p,x-a)$. This concludes the proof that $\mathrm{Ker}(\varphi)=(p,x-a)$.

(ii) Fix some prime number p and denote by $\overline{g(x)}$ the reduction modulo p of the polynomial $g(x)$ of $\mathbb{Z}[x]$. Consider the ring homomorphism

$$\mathbb{Z}[x] \ni g(x) \overset{\psi}{\longmapsto} \overline{g(x)} + (\overline{f(x)}) \in \mathbb{F}_p[x]/(\overline{f(x)}).$$

It is easy to show, by an argument similar to the one used for (i), that $\mathrm{Ker}(\psi)$ is the ideal $(p, f(x))$ of $\mathbb{Z}[x]$. Now, $f(x)$ is monic and thus the degree of $\overline{f(x)}$ is the same as the degree of $f(x)$; in particular, $\mathbb{F}_p[x]/(\overline{f(x)})$ is not the zero ring. Therefore, since ψ is surjective, the ring $\mathbb{Z}[x]/(p, f(x))$ is not the zero ring either, so $(p, f(x))$ is a proper ideal of $\mathbb{Z}[x]$.
This is enough to establish that $(f(x))$ is not maximal, as it is contained in $(p, f(x))$ and the two ideals are not equal because $(f(x))$ does not contain any constant polynomials.

194

(i) Obviously, we have $(\mathcal{P}) \subseteq P$, as $\mathcal{P} \subseteq P$. Consider now $x \in P$ and assume that its factorisation in A is $x = \pi_1^{a_1}\pi_2^{a_2}\cdots\pi_r^{a_r}$, where $\pi_1, \pi_2, \ldots, \pi_r$ are prime elements of A. Since P is prime and $\pi_1^{a_1}\pi_2^{a_2}\cdots\pi_r^{a_r} \in P$, at least one of the prime factors $\pi_1, \pi_2, \ldots, \pi_r$ belongs to P, so it belongs to $\mathcal{P}$ by definition. It follows that $x \in (\mathcal{P})$.

(ii) Let $\pi_1, \pi_2, \ldots, \pi_r$ be the prime elements of A, let $I = (x)$, $J = (y)$ be two principal ideals and let $x = \pi_1^{e_1}\pi_2^{e_2}\cdots\pi_r^{e_r}$, $y = \pi_1^{f_1}\pi_2^{f_2}\cdots\pi_r^{f_r}$ be the factorisations of x and y. The greatest common divisor of x and y is $z = \pi_1^{d_1}\pi_2^{d_2}\cdots\pi_r^{d_r}$ with $d_h = \min\{e_h, d_h\}$ for $h = 1, 2, \ldots, r$.
Clearly, $x, y \in (z)$ and so $I + J \subseteq (z)$. In order to show the opposite inclusion, consider the following systems of congruences in A:

$$\begin{cases} a \equiv \alpha_1 \pmod{\pi_1} \\ a \equiv \alpha_2 \pmod{\pi_2} \\ \quad\vdots \\ a \equiv \alpha_r \pmod{\pi_r} \end{cases}, \qquad \begin{cases} b \equiv \beta_1 \pmod{\pi_1} \\ b \equiv \beta_2 \pmod{\pi_2} \\ \quad\vdots \\ b \equiv \beta_r \pmod{\pi_r} \end{cases}$$

where, for $h = 1, 2, \ldots, r$, we set $\alpha_h = 1$, $\beta_h = 0$ if $a_h \leq b_h$ (in which case $d_h = a_h$), and $\alpha_h = 0$, $\beta_h = 1$ if $a_h > b_h$ (in which case $d_h = b_h$). The two systems each have a solution by the Chinese Remainder Theorem for Rings, because the ideals generated by $\pi_1, \pi_2, \ldots, \pi_r$ are maximal and pairwise distinct, so they are pairwise relatively prime.
Letting a and b be solutions of the two systems, $\pi_h^{d_h}$ divides $ax + by$ but $\pi_h^{d_h+1}$ does not divide $ax + by$. It follows that $ax + by = uz$, where u is an invertible element of A. This proves that $z \in I + J$ and so that $I + J = (z)$ is principal.

195

(i) For $a \in A \setminus \{0\}$, let I_a be the ideal $\{b \in A \mid ab = 0\} = ((b) : 0)$. It is clear that every element of I_a is a zero divisor because $a \neq 0$; conversely, every zero divisor belongs to some I_a by definition. This proves that

$$D = \bigcup\nolimits_{b \in A \setminus \{0\}} I_a .$$

(ii) Suppose D is an ideal and assume that $ab \in D$, that is, there is some $c \neq 0$ such that $abc = 0$. If $bc = 0$ then $b \in D$, whereas if $bc \neq 0$ then $a \in D$. We have thus shown D to be a prime ideal.

(iii) Let A be a finite ring and let a be an element of the ideal D. The powers of a cannot be all different, so there exist positive integers n, m such that $a^n = a^{n+m}$, hence $a^n(a^m - 1) = 0$. If $a^n = 0$, then a is nilpotent. Assume by contradiction that $a^n \neq 0$, so $a^m - 1 \in D$. Since $a \in D$, we also have $a^m \in D$. But D is an ideal, so this implies that $1 = a^m - (a^m - 1) \in D$, that is, 1 is a zero divisor, which is clearly a contradiction.

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(i) Consider $x/s \in S^{-1}\sqrt{I}$; we have $x \in \sqrt{I}$, so there exists $n \in \mathbb{N}$ such that $x^n \in I$. It follows that $(x/s)^n = x^n/s^n \in S^{-1}I$, so $x/s \in \sqrt{S^{-1}I}$.
Conversely, consider $x/s \in \sqrt{S^{-1}I}$; we have that $(x/s)^n = x^n/s^n \in S^{-1}I$ for some $n \in \mathbb{N}$, so $x^n/s^n = y/t$ where $y \in I$ and $t \in S$, which yields $x^n t = s^n y \in I$. But then $(xt)^n = x^n t \cdot t^{n-1} \in I$ and $x/s = (xt)/(st)$ has a numerator in $\sqrt{I}$ and a denominator in S, that is, $x/s \in S^{-1}\sqrt{I}$.

(ii) From now on, for ease of notation, we will identify $a \in A$ with $a/1 \in S^{-1}A$, thanks to the injective ring homomorphism $A \ni a \longmapsto a/1 \in S^{-1}A$.
If $a = 0$, which implies $b \neq 0$, then the statement is obviously true; we therefore assume $a, b \neq 0$.
Every nonzero element in a unique factorisation domain can be written as a product of irreducible (or, equivalently, prime) elements. We distinguish two types of irreducible elements of A: (1) those that do not divide any element of S and (2) those that divide some element of S.
We now prove that irreducible elements of type (1) remain irreducible in $S^{-1}A$. Indeed, suppose π is of this type and divides $(a/s) \cdot (b/t)$, that is, $\pi \cdot c/u = (a/s) \cdot (b/t)$ for some $c/u \in S^{-1}A$. Then π divides abu in A. Using the fact that π is prime in A and does not divide u in A, we obtain that π divides ab in A; therefore, π divides a, in which case π divides a/s in $S^{-1}A$, or π divides b, in which case π divides b/t in $S^{-1}A$.
As for irreducible elements of type (2), they become invertible in $S^{-1}A$: indeed, if π is an irreducible element of this type, that is, if it divides s in A for some $s \in S$, then there exists $a \in A$ such that $\pi a = s$ and so $\pi \cdot (a/s) = 1$.

We now write factorisations for a and b, distinguishing between the two types of irreducible factors and using null exponents where needed:

$$a = u \cdot \pi_1^{a_1} \cdots \pi_h^{a_h} \cdot \eta_1^{b_1} \cdots \eta_k^{b_k}, \quad b = v \cdot \pi_1^{c_1} \cdots \pi_h^{c_h} \cdot \eta_1^{d_1} \cdots \eta_k^{d_k},$$

where $u, v \in A^*$, $\pi_1, \ldots, \pi_h$ are irreducible of type (1), $\eta_1, \ldots, \eta_k$ are irreducible of type (2), and $a_1, \ldots, a_h, b_1, \ldots, b_k, c_1, \ldots, c_h, d_1, \ldots, d_k \geq 0$. Define $e_j = \min\{a_j, c_j\}$ for $j = 1, \ldots, h$, and $f_j = \min\{b_j, d_j\}$ for $j = 1, \ldots, k$. We have

$$\mathrm{MCD}_A(a, b) = \pi_1^{e_1} \cdots \pi_h^{e_h} \cdot \eta_1^{f_1} \cdots \eta_k^{f_k} = w \cdot \pi_1^{e_1} \cdots \pi_h^{e_h}$$

where $w = \eta_1^{f_1} \cdots \eta_k^{f_k}$ is invertible in $S^{-1}A$, while

$$\mathrm{MCD}_{S^{-1}A}(a, b) = \pi_1^{e_1} \cdots \pi_h^{e_h}.$$

This proves the desired result.

197

(i) Denote by $N(a + bi) = a^2 + b^2$ the usual norm, which is a degree making $\mathbb{Z}[i]$ into a Euclidean domain.
Letting $z = 5 + 14i$ and $w = -4 + 7i$, remark that $N(z) = 221 = 13 \cdot 17$ and $N(w) = 65 = 5 \cdot 13$, where the prime numbers 5, 13 and 17 are all congruent to 1 mod 4. In $\mathbb{Z}[i]$ we thus have that z factors as the product of two primes, one of degree 13 and the other of degree 17, and w factors as the product of two primes, one of degree 13 and the other of degree 5.
By solving the Diophantine equation $a^2 + b^2 = 13$, we immediately find the integer solutions $a = 2$ and $b = \pm 13$, up to a change of sign. That is, up to invertible factors, the Gaussian integers having norm 13 are $2 + 3i, 2 - 3i$. Indeed, we find the factorisations $z = 5+14i = (2+3i)(4+i)$, $w = -4+7i = (2 + 3i)(1 + 2i)$.
Now, $\mathbb{Z}[i]$ is a principal ideal domain, so $I \cap J$ is generated by the least common multiple of z and w, that is, by $(2 + 3i)(1 + 2i)(4 + i) = -23 + 24i$, and $I + J$ is generated by the greatest common divisor of z and w, that is, by $2 + 3i$.

(ii) The Gaussian integer $2+3i$ is prime, so the quotient $\mathbb{Z}[i]/(I+J) = \mathbb{Z}[i]/(2+3i)$ is a field and its only prime ideal is 0.

198 Let M be a maximal ideal of $A[x]$ and let $P = A \cap M = i^{-1}(M)$, where $i : A \longrightarrow A[x]$ is the natural inclusion. Since P is the contraction of a prime ideal, it is a prime ideal of A. We consider the following two cases, according to whether or not P is zero.

(1) The ideal P is nonzero.

Letting P^e be the ideal generated by P in $A[x]$, remark that $P^e \subseteq M$. The quotient homomorphism

$$A[x] \ni f(x) \longmapsto f(x) + P^e \in A[x]/P^e \simeq (A/P)[x]$$

induces a one-to-one correspondence between ideals of $A[x]$ containing P^e and ideals of $(A/P)[x]$. Let J be the ideal of $(A/P)[x]$ corresponding to M in $A[x]$.

Since in a principal ideal domain every nonzero prime ideal is maximal, A/P is a field, so the ideals of $(A/P)[x]$ are all principal. In particular, there exists $f(x) \in A[x]$ whose reduction modulo P generates J in $(A/P)[x]$.

It is now immediate to check that $M = (\pi, f(x))$, where π is a generator of P in A.

(2) The ideal P is zero.

Letting $\mathbb{K}$ be the fraction field of A, consider the ideal M^e generated by M in $\mathbb{K}[x]$; clearly, $M^e = \mathbb{K} \cdot M$. Let $f(x)$ be a monic generator of M^e in $\mathbb{K}[x]$. If c is the greatest common divisor of the denominators of the coefficients of $f(x)$, the polynomial $f_0(x) = c \cdot f(x)$ is in $A[x]$ and is primitive. Clearly, $f_0(x) \in M^e \cap A[x]$; if we show that $M^e \cap A[x] = M$ then we have $f_0(x) \in M$.

It is clear that $M \subseteq M^e \cap A[x]$ and, since M is a maximal ideal of $A[x]$, we either have $M^e \cap A[x] = M$ or $M^e \cap A[x] = (1)$. In the second case, we would have $1 \in M^e$, and so there would exist $a/b \in \mathbb{K}$, with $a, b \in A$ and $b \neq 0$, and $g(x) \in M$ such that $(a/b) \cdot g(x) = 1$. This would imply that $a \cdot g(x) = b$ in A; but then $b \in M \cap A[x] = P = 0$, which is a contradiction. We have thus shown that $M^e \cap A[x] = M$.

Now, the fact that $f_0(x) \in M$ yields $(f_0(x)) \subseteq M$. Conversely, if $h(x) \in M$ then the primitive polynomial $f_0(x) \in A[x]$ divides the polynomial $h(x)$ in $\mathbb{K}[x]$ and, by Gauss's Lemma, $f_0(x)$ divides $h(x)$ in $A[x]$, that is, $h(x) \in (f_0(x))$.

In conclusion, in this case the ideal $M = (f_0(x))$ is principal.

[Not that (2) can actually happen: let S be the set of odd integers, $A = S^{-1}\mathbb{Z}$ and $M = (1+2x) \subseteq A[x]$; the ideal M is maximal and $M \cap A[x] = 0$.]

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(i) Consider the set

$$S^+ = \{t \in A \setminus 0 \mid \text{there exists } s \in S \text{ such that } t \text{ divides } s\};$$

we show that S^+ is the smallest saturated multiplicative set of A that contains S.

First of all, it is clear that $s \mid s$ for all $s \in S$, so $S \subseteq S^+$. Now, $0 \notin S^+$, whereas $1 \in S^+$ as $1 \in S$; given $t_1, t_2 \in S^+$, there exist $s_1, s_2 \in S$ such that $t_1 \mid s_1$ and $t_2 \mid s_2$, so $t_1 t_2 \mid s_1 s_2 \in S$, which yields $t_1 t_2 \in S^+$. We have thus shown S^+ to be a multiplicative set.

Moreover, S^+ is saturated: if $d \mid t \in S^+$ then there is $s \in S$ such that $t \mid s$, so d divides s by transitivity, and thus $d \in S^+$. Finally, given a saturated

multiplicative set T that contains S, it must contain all divisors of elements of S by definition, so it must contain S^+.

(ii) We show that, letting S^+ be the saturation of S, we have $S^{-1}A = S^{+,-1}A$. Clearly, the fact that $S \subseteq S^+$ implies $S^{-1}A \subseteq S^{+,-1}A$. Conversely, if $a/t \in S^{+,-1}A$, then there exists $s \in S$ such that $t \mid s$ and, letting $s = tu$, we have $a/t = au/s \in S^{-1}A$.
This proves that if S and T have the same saturation, then $S^{-1}A = T^{-1}A$.
Conversely, we now show that the saturation S^+ of S in A is the set of elements of A that are invertible in $S^{-1}A$, that is, $S^+ = (S^{-1}A)^* \cap A$.
Indeed, given $t \in S^+$, there exist $a \in A$ and $s \in S$ such that $ta = s$, hence $t \cdot (a/s) = 1$ and so t is invertible in $S^{-1}A$. On the other hand, if $t \in A$ is invertible in $S^{-1}A$, then there exist $a \in A$ and $s \in S$ such that $t \cdot (a/s) = 1$, so $ta = s$ and, since S^+ is saturated, $t \in S^+$.
It is now clear that if $S^{-1}A = T^{-1}A$ then $S^+ = (S^{-1}A)^* \cap A = (T^{-1}A)^* \cap A = T^+$.

200 First of all, we recall that N is the intersection of all prime ideals of A.

We first prove that (a) implies (b). If A has a unique prime ideal P, then $N = P$. Since every maximal ideal is also a prime ideal, A has a unique maximal ideal, which is therefore equal to N, so A is a local ring with maximal ideal N. In a local ring, every non-invertible element belongs to the maximal ideal, so every non-invertible element of A belongs to N and is thus nilpotent.

We now prove that (b) implies (c). Consider the quotient homomorphism $\pi : A \longrightarrow A/N$, let $\overline{x}$ be a nonzero element in A/N, and let $x \in A$ be such that $\pi(x) = \overline{x}$. The element x does not belong to N, so there is an element $y \in A$ such that $xy = 1$. But then, setting $\overline{y} = \pi(y)$, we have $\overline{x} \cdot \overline{y} = \overline{xy} = \overline{1}$, so $\overline{x}$ is invertible.

Finally, let us show that (c) implies (a). If A/N is a field, then N is maximal. But, since N is contained in every prime ideal P, by maximality of N every prime ideal P must be equal to N, so A has a unique prime ideal, which is N itself.

201 First of all, remark that $A = \mathbb{Z}[bi]$, as $a \in \mathbb{Z}$, so A is the set $\{r + sbi \mid r, s \in \mathbb{Z}\}$. The ideal $\mathfrak{f}$ consists of all elements $h + kbi$ such that $(h + kbi)(x + yi) \in A$ for all $x, y \in \mathbb{Z}$.

Since we have

$$(h + kbi)(x + yi) = (hx - kby) + (hy + kbx)i,$$

the element $h + kbi$ is in $\mathfrak{f}$ if and only if $hy + kbx$ is divisible by b for all $x, y, \in \mathbb{Z}$, that is, if and only if h is divisible by b, say $h = bl$.

Therefore, $\mathfrak{f} = b\{l + ki \mid l, k \in \mathbb{Z}\}$ and $[A : \mathfrak{f}] = b$.

202

(i) Let $n = N(a + bi) = a^2 + b^2$.
First, assume that a, b are relatively prime in $\mathbb{Z}$. We intend to construct a homomorphism from A/I to $\mathbb{Z}/n\mathbb{Z}$. We begin by observing that b and n are

relatively prime: indeed, a common prime divisor of b and n would also divide a. Let then c be an integer such that $bc \equiv 1 \pmod{n}$ and set

$$A \ni x + yi \overset{\varphi}{\longmapsto} x - acy \in \mathbb{Z}/n\mathbb{Z}.$$

Since $a^2 + b^2 \equiv 0 \pmod{n}$, we have $(-ac)^2 \equiv -1 \pmod{n}$ and so φ is a ring homomorphism, which is obviously surjective and whose kernel contains $a + bi$.
We will prove that actually $\mathrm{Ker}(\varphi) = I$ and so $A/I \simeq \mathbb{Z}/n\mathbb{Z}$ and $|A/I| = n = N(a+bi)$. Indeed, if $\varphi(x+yi) = 0$ then we have $x - acy = hn$ for some integer h, and so

$$\begin{aligned} x + yi &= acy + yi + hn \\ &= cay + cbyi + h'n \\ &= \big(cy + h'(a - bi)\big)(a + bi) \end{aligned}$$

for some $h' \in A$, which yields $x + yi \in I$.
If a and b are not relatively prime in $\mathbb{Z}$, let m be their greatest common divisor; we have $a = a'm$, $b = b'm$, with a' and b' relatively prime. The index of I in $I' = (a' + b'i)$ is $m^2 = N(m)$ and we have $[A : I] = [A : I'][I' : I] = N(a' + b'i)m^2 = N(a + bi)$.

(ii) The ring A is a principal ideal domain, so every ideal I of A is of the form $I = (a + bi)$ for some $a + bi \in A$. We are thus looking for the ideals that are generated by some $a + bi$ such that $N(a + bi) = 100 = 2^2 \cdot 5^2$.
The norm of a Gaussian integer is the product of the norms of its prime factors. Up to associates, the only prime whose norm is 2 is $1 + i$. The norm of a prime of A can be p^2, where p is a prime of $\mathbb{Z}$, only if p is congruent to 3 modulo 4; but 5 is congruent to 1 modulo 4, so there are no primes of norm 25. The only primes of norm 5 are $2 \pm i$.
It follows that the ideals of index 100 in A are:

$$\big((1+i)^2(2+i)^2\big), \quad \big((1+i)^2(2-i)^2\big) \quad \text{and} \quad \big((1+i)^2 5\big).$$

These three ideals are all different because they are generated by different products of primes and A is a unique factorisation domain.

203

(i) Let $A \ni x \longmapsto \overline{x} \in A/Q$ be the quotient homomorphism.
Assume that Q is primary and let $\overline{b}$ be a zero divisor in A/Q, so that there exists $a \in A \setminus Q$ such that $\overline{a} \cdot \overline{b} = \overline{ab} = \overline{0}$. This implies that $ab \in Q$ and thus, since Q is primary, that $b^n \in Q$ for some n. In other words, $\overline{b}^n = \overline{0}$, that is, $\overline{b}$ is nilpotent.
Now suppose every zero divisor in A/Q is nilpotent; we show that Q is primary. Consider $a, b \in A$ such that $ab \in Q$ and $a \notin Q$. We have $\overline{a} \neq \overline{0}$

and $\overline{a} \cdot \overline{b} = \overline{0}$, that is, $\overline{b}$ is a zero divisor in A/Q; it follows that it is also nilpotent, that is, there exists n such that $\overline{b}^n = \overline{b^n} = \overline{0}$, and therefore $b^n \in Q$. This shows that Q is primary.

(ii) It is immediate to see that 0 is a primary ideal of $\mathbb{Z}[i]$, as obviously $\mathbb{Z}[i]/0 \simeq \mathbb{Z}[i]$ and the ring $\mathbb{Z}[i]$, being a subset of $\mathbb{C}$, does not contain any zero divisors. Recall that $\mathbb{Z}[i]$ is a Euclidean domain; we wish to show that $Q = (z) \neq 0$ is primary if and only if z is a power of some irreducible element of $\mathbb{Z}[i]$.
Indeed, if $z = \pi^n$, where π is irreducible in $\mathbb{Z}[i]$, then for all $x, y \in \mathbb{Z}[i]$ such that $xy \in Q$, if $x \notin Q$ then π^n does not divide x but π^n divides xy; it follows that π divides y and thus π^n divides y^n, that is, $y^n \in Q$. This proves that Q is primary.
Conversely, let Q be a primary ideal and assume by contradiction that it is generated by $z = z_1 \cdot z_2$, where z_1 and z_2 are relatively prime and are not invertible. We have $z = z_1 \cdot z_2 \in Q$, $z_1 \notin Q$ but we also have $z_2^n \notin Q$, because any irreducible divisor of z_1 does not divide z_2^n for any n; this clearly contradicts the definition of a primary ideal.

(iii) The ideals of a direct product of rings are direct products of ideals of the factors, so the ideals of $\mathbb{Z} \times \mathbb{Z}$ are of the form $m\mathbb{Z} \times n\mathbb{Z}$, with corresponding quotient $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.
If both factors are nontrivial, then the resulting ideal is not primary: the element $(\overline{1}, \overline{0})$, for example, is a zero divisor that is not nilpotent. Let us consider ideals of the form $m\mathbb{Z} \times \mathbb{Z}$ or $\mathbb{Z} \times m\mathbb{Z}$. The corresponding quotient is isomorphic to $\mathbb{Z}/m\mathbb{Z}$, whose zero divisors are the divisors of m. It follows that the zero divisors are all nilpotent if and only if $m = 0$ or m is a prime power.
In conclusion, the primary ideals of $\mathbb{Z} \times \mathbb{Z}$ are products of $\mathbb{Z}$ and some ideal of the form $\mathbb{Z} \cdot p^k$, where p is a prime and k is a natural number, or $\mathbb{Z} \times 0$.

204

(i) One can easily show that

$$A \ni z = a + b\sqrt{2} \longmapsto \overline{z} = a - b\sqrt{2} \in A$$

is an automorphism of A; in particular, the map

$$\mathbb{Z}[\sqrt{2}] \ni z = a + b\sqrt{2} \overset{N}{\longmapsto} z \cdot \overline{z} = a^2 - 2b^2 \in \mathbb{Z}$$

is multiplicative.
If $z, w \in A$ are such that $zw \in (5)$, then 5 divides $N(zw)$, so, up to exchanging z and w, we may assume that 5 divides $N(z)$, that is, $z = a + \sqrt{2}b$ with $a^2 - 2b^2 \equiv 0 \pmod 5$. Now, if b were not congruent to 0 modulo 5, then we would have $(a/b)^2 \equiv 2 \pmod 5$; however, the only squares modulo 5 are $0, \pm 1$, which proves that 5 divides b, and so 5 also divides a and $z \in (5)$. We have thus proven that (5) is prime in A.

(ii) Any ideal I of A that contains 7 necessarily contains all of the ideal (7). Ideals of A containing 7 are thus in a one-to-one correspondence with ideals of $A/(7)$. Now, $A/(7)$ is isomorphic to $\mathbb{Z}[x]/(7, x^2-2) \simeq \mathbb{F}_7[x](x^2-2)$ and, since $x^2-2 = (x-3)(x+3)$ in $\mathbb{F}_7[x]$, the Chinese Remainder Theorem for Rings yields

$$A/(7) \simeq \mathbb{F}_7[x]/(x-3) \times \mathbb{F}_7[x]/(x+3) \simeq \mathbb{F}_7 \times \mathbb{F}_7.$$

The ideals of a product are product of ideals of the factors; in this case, the factors are fields, and so they only have the two obvious ideals. We thus find 4 ideals in total in the quotient, and so there are 4 ideals of A that contain 7.

(iii) Denote by P and Q the two ideals from (ii) that yield the quotients $\mathbb{F}_7 \times 0$ and $0 \times \mathbb{F}_7$: these are maximal because quotienting by them gives the field $\mathbb{F}_7$, and so both P and Q are prime. Their intersections with $\mathbb{Z}$ are prime ideals containing $\mathbb{Z} \cdot 7$, so both intersections are equal to $\mathbb{Z} \cdot 7$.

205

(i) The one of the ring A is clearly $(1, 1, 1, \ldots)$. An element $\alpha = (a_n)_n \in A$ is invertible if and only if there exists $\beta = (b_n)_n \in A$ such that $a_n b_n \equiv 1 \pmod{p^n}$ for all $n \geq 1$. We now show that the invertible elements α are exactly those such that $a_1 \not\equiv 0 \pmod{p}$.
Indeed, if $(a_n)_n$ is invertible, then $a_1 b_1 \equiv 1 \pmod{p}$, so $a_1 \not\equiv 0 \pmod{p}$. Conversely, if $a_1 \not\equiv 0$ then, since for all $n \geq 1$ we have $a_n \equiv a_1 \not\equiv 0 \pmod{p}$, the class a_n is invertible in $\mathbb{Z}/p^n\mathbb{Z}$; moreover, $a_{n+1} \equiv a_n \pmod{p^n}$ implies that $a_{n+1}^{-1} \equiv a_n^{-1} \pmod{p^n}$ for all n, and thus if we set $b_n \equiv a_n^{-1} \pmod{p^n}$, we find that the element $(b_n)_n$ is in A and is the inverse of α.

(ii) A ring has a unique maximal ideal if and only if $M = A \setminus A^*$ is an ideal. By (i), we have

$$M = \left\{(a_n)_n \in A \,|\, a_1 \equiv 0 \pmod{p}\right\}$$

and this set is indeed an ideal, as it is the kernel of the homomorphism $A \ni (a_n)_n \longmapsto a_1 \in \mathbb{Z}/p\mathbb{Z}$.
We now prove that the element $\pi = (0, p, p, p \ldots) \in M$ is a generator for M. Indeed, consider $\alpha = (a_n)_n \in M$; we have $a_1 = 0$ and for all $n \geq 2$ there exists $b_n \in \mathbb{Z}/p^n\mathbb{Z}$ such that $a_n \equiv p b_n \pmod{p^n}$. Given b_2, for all $n \geq 2$ we can obviously sum a multiple of p^{n-1} to b_n in order to also have $b_n \equiv b_{n-1} \pmod{p^{n-1}}$, that is, to have $\beta = (b_n)_n \in A$. We obtain $\alpha = \beta \cdot \pi$, that is, $\alpha \in (\pi)$. This proves that $M = (\pi)$ is a principal ideal.

(iii) Let $M = (\pi)$ be the unique maximal ideal found in (ii) and let I be a nonzero ideal of A.
Since for $k \geq 0$ the element π^k has its first k coordinates equal to zero and a nonzero $(k+1)$th coordinate, there exists k such that $I \subseteq M^k = (\pi^k)$ and $I \not\subseteq M^{k+1}$, as I is nonzero.

So there exists an element $(a_n)_n \in I$ such that $a_n \equiv 0 \pmod{p^n}$ for $n \leq k$ and $a_{k+j} \not\equiv 0 \pmod{p^{k+j}}$ for all $j \geq 1$. By the same argument used above, we can construct $\beta = (b_n)_n \in A^*$ such that $\alpha = \pi^k \beta$, that is, $\pi^k \in (\alpha)$ and so $M^k \subseteq I$. Therefore, $I = M^k$, as wanted.

▌The ring in the problem statement is usually denoted by $\mathbb{Z}_p$ in the literature and is called the ring of *p-adic integers*.▐

206 The ring $A_P = S^{-1}A$ is an integral domain because it is contained in the fraction field of A. First of all, we proceed to characterise invertible elements and irreducible elements of A_P.

We prove that $a/s \in A_P$ is invertible if and only if $a \in S$. Indeed, if $a \in S$, then $s/a \in A_P$ is clearly the inverse of a/s. Conversely, if a/s is invertible then there exists $b/t \in A_P$ such that $(a/s) \cdot (b/t) = 1$, so $ab = st \in S$, that is, $ab \notin P$, hence $a \notin P$ because P is an ideal of A.

We now prove that $a/s \in A_P$ is irreducible if and only if $a = u\pi$ in A, where $u \in S$ and $\pi \in P$ is irreducible in A.

Let $a = u\pi$ with $u \in S$ and $\pi \in P$ irreducible, and assume that $a/s = (b/t) \cdot (c/v)$ in A_P; we then have $\pi utv = bcs$ and so π divides bcs in A. Since π is irreducible and thus prime, in A we have $\pi \mid b$ or $\pi \mid c$, as clearly π cannot divide s because $\pi \in P$ whereas $s \in S = A \setminus P$. If $b = d\pi$ then we obtain $utv = dcs$, which yields $c \notin P$, and so c/s is invertible in A_P. A similar argument applies if π divides c. We have thus shown that $a = u\pi$ is irreducible in A_P.

Conversely, assume that a/s is irreducible in A_P and let $a = \pi_1 \cdots \pi_k$ be a factorisation as a product of irreducible elements of A, with $\pi_j \in P$ for $1 \leq j \leq r$ and $\pi_j \notin P$ for $r < j \leq k$. By our arguments so far, π_j is irreducible in A_P for $1 \leq j \leq r$ and invertible for $r < j \leq k$. It is thus clear that $r = 1$ and thus that every irreducible element in A_P is of the form we claimed.

Let us now prove that every non-invertible element of A_P can be written in a unique way as a product of irreducible elements. Let $a/s \in A_P$ be a non-invertible element, and in particular such that $a \in P$. As explained above, if in A we have $a = \pi_1 \cdots \pi_r t$ with $t \notin P$, $\pi_j \in P$ irreducible in A for $j = 1, \ldots, r$, then $a/s = (t/s) \cdot \pi_1 \cdots \pi_r$ is a factorisation in A_P as a product of irreducible elements, with t/s invertible.

Now suppose that an element of A_P has two distinct factorisations. Multiplying by invertible elements yields an equality of the form: $v \cdot \pi_1 \cdots \pi_r = u \cdot \eta_1 \cdots \eta_s$ with $\pi_j, \eta_k \in P$ irreducible in A for $j = 1, \ldots r$ and for $k = 1, \ldots s$, and $u, v \notin P$. So, letting $v = \rho_1 \cdots \rho_l$ and $u = \epsilon_1 \ldots \epsilon_h$ be the factorisations of u and v, we have in A the equality $\rho_1 \cdots \rho_l \cdot \pi_1 \cdots \pi_r = \epsilon_1 \cdots \epsilon_h \cdot \eta_1 \cdots \eta_r$. By uniqueness of the factorisation in A, the factors appearing on the left and right are pairwise associates, but then the π_j's and the η_k's must be associates, as they are the factors belonging to P. It follows that the factorisation is also unique in A_P.

207

(i) Consider the automorphism of A given by

$$A \ni u = a + b\sqrt{13} \longmapsto \overline{u} = a - b\sqrt{13} \in A.$$

and define the norm

$$A \ni u = a + b\sqrt{13} \longmapsto N(u) = u \cdot \overline{u} = a^2 - 13b^2 \in \mathbb{Z}.$$

The norm is multiplicative, so every $w \in A$ of norm ± 1 is invertible, as $w \cdot (N(w)\overline{w}) = N(w)^2 = 1$ and so $w^{-1} = N(w)\overline{w} \in A$ is the inverse of w.
But then the element $w = 18 + 15\sqrt{13}$ is invertible in A because $N(18 + 5\sqrt{13}) = -1$.
Moreover, w is a real number greater than 1, so all powers of w are distinct. It follows that the elements of the form $\pm w^n$ for $n \in \mathbb{Z}$ are all distinct and invertible, and thus that A^* is infinite.

(ii) Note that $N(2) = 4$ and $N(3 + \sqrt{13}) = -4$, so if 2 or $3 + \sqrt{13}$ were reducible there would exist some element of A whose norm is 2. We show that A contains no elements of norm 2.
Let $u = a + b\sqrt{13}$ and assume $N(u)$ is even. In particular, a and b are both even or both odd, and so $a^2 \equiv b^2 \pmod 4$; it follows that $N(u) = a^2 - 13b^2 \equiv a^2 - b^2 \equiv 0 \pmod 4$. Being divisible by 4, the norm of u cannot be equal to 2.

(iii) We have the equality

$$4 = 2^2 = (3 + \sqrt{13})(-3 + \sqrt{13}).$$

By (ii), all factors in the two expressions are irreducible, and 2 is not an associate of $3 + \sqrt{13}$. Indeed, all associates of 2 in A are of the form $u = a + b\sqrt{13}$ with a and b even. The two factorisations are thus distinct, which implies that A is not a unique factorisation domain.

208 Since the ideals of a field $\mathbb{K}$ are 0 and $\mathbb{K}$, the ideals of A are $I_1 \times I_2 \times \cdots \times I_n$ where for each h we have $I_h = 0$ or $I_h = \mathbb{K}_h$.

(i) It is then clear that the ideals of A are all of the form $A \cdot x$, where $x = (u_1, u_2, \ldots, u_n)$ with $u_h \in \{0, 1\}$ for each h.

(ii) An ideal $I = A \cdot (u_1, \ldots, u_h)$ is prime if and only if A/I is an integral domain. But $A/I \simeq \mathbb{K}_{h_1} \times \mathbb{K}_{h_2} \times \cdots \times \mathbb{K}_{h_r}$, where $\{h_1, \ldots, h_r\} = \{h \mid u_h = 0\}$. So I is a prime ideal if and only if $r = 1$. In conclusion, there are n prime ideals of A, namely: $0 \times \mathbb{K}_2 \times \cdots \times \mathbb{K}_n$, $\mathbb{K}_1 \times 0 \times \mathbb{K}_2 \times \cdots \times \mathbb{K}_n$, …, $\mathbb{K}_1 \times \cdots \times \mathbb{K}_{n-1} \times 0$.

209 Given a map $f : X \longrightarrow \mathbb{K}$, denote by $\mathrm{supp}(f)$ the *support* of f, that is, the set of all $x \in X$ such that $f(x) \neq 0$. It is clear that $\mathrm{supp}(f + g) \subseteq \mathrm{supp}(f) \cup \mathrm{supp}(g)$ and $\mathrm{supp}(f \cdot g) = \mathrm{supp}(f) \cap \mathrm{supp}(g)$. This implies that the set I of maps with finite support is an ideal of A.

We now prove that I is not a principal ideal. Given $f \in I$, since X is infinite and the support of f is finite, there exists $x_0 \in X$ such that $f(x_0) = 0$. Every $g \in (f)$ is such that $g(x_0) = 0$ and so f cannot generate I, as the map

$$X \ni x \longmapsto \begin{cases} 1 \text{ if } x = x_0 \\ 0 \text{ otherwise} \end{cases}$$

is in I.

3.3 Fields and Galois theory

210

(i) Set $\alpha = \sqrt{5} + i$ and remark that α is a root of the polynomial with rational coefficients

$$\begin{aligned} g(x) &= \big(x - (\sqrt{5} + i)\big)\big(x - (\sqrt{5} - i)\big)\big(x - (-\sqrt{5} + i)\big)\big(x - (-\sqrt{5} - i)\big) \\ &= x^4 - 8x^2 + 36. \end{aligned}$$

The polynomial $g(x)$ is thus a multiple in $\mathbb{Q}[x]$ of the minimal polynomial $f(x)$ of α over $\mathbb{Q}$. But then, since $g(x)$ only has non-real roots, $f(x)$ can only have degree 2 or 4. In fact, if the degree of $f(x)$ were 2, then it would have two roots that are complex conjugates, namely, α and $\overline{\alpha} = \sqrt{5} - i$; in this case, the coefficient of the linear term of $f(x)$ would be $-(\alpha + \overline{\alpha}) = -2\sqrt{5}$, which is not rational. The only remaining possibility is for $f(x)$ to have degree 4 and thus to be equal to $g(x)$, which in particular has integer coefficients.

(ii) We show that $\mathbb{K} = \mathbb{Q}(\alpha)$ is the splitting field of $f(x)$ over $\mathbb{Q}$. Clearly, $\mathbb{K}$ is contained in the splitting field because α is a root of $f(x)$. Moreover,

$$\alpha^{-1} = \frac{\sqrt{5} - i}{6}$$

is also an element of $\mathbb{K}$, and $\sqrt{5} = (\alpha + 6\alpha^{-1})/2$ and $i = (\alpha - 6\alpha^{-1})/2$ are in $\mathbb{K}$ as well. It follows that $\mathbb{K} = \mathbb{Q}(\sqrt{5}, i)$ and thus, since it contains all four roots, $\mathbb{K}$ is the splitting field of $f(x)$. In particular, the degree of the splitting field of $f(x)$ over $\mathbb{Q}$ is 4.
Since $\mathbb{Q}(i)$ is a subfield of $\mathbb{K}$, the splitting field of $f(x)$ over $\mathbb{Q}(i)$ is again $\mathbb{K} = \mathbb{Q}(i)(\sqrt{5})$, and for the degree we have $[\mathbb{K} : \mathbb{Q}(i)] = [\mathbb{K} : \mathbb{Q}]/[\mathbb{Q}(i) : \mathbb{Q}] = 2$.
Over $\mathbb{F}_5$ we have $f(x) = x^4 - 8x^2 + 36 = x^4 + 2x^2 + 1 = (x^2 + 1)^2 = (x - 2)^2(x + 2)^2$; the splitting field is thus $\mathbb{F}_5$ itself.

211 Setting $p(x) = x^4 - 4x^2 + 16$ and $x = 2y$, we have $p(x) = (2y)^4 - 4(2y)^2 + 16 = 16(y^4 - y^2 + 1)$, and so $p(x)$ and $f(y) = y^4 - y^2 + 1$ have the same splitting field. Remark that in every field $\mathbb{K}$ we have

$$f(y)(y^2 + 1)(y^6 - 1) = y^{12} - 1,$$

which implies that every root of $f(y)$ is a twelfth root of unity. Also note that the sixth roots of unity are roots of the polynomial $y^6 - 1$ and that, if $\mathbb{K}$ has characteristic different from 2 and 3, then the primitive fourth roots of unity are the roots of $y^2 + 1$, and thus the roots of $f(y)$ are exactly the primitive twelfth roots of unity.

It is then clear that the splitting field over $\mathbb{Q}$ is the twelfth cyclotomic field, whose degree is $\phi(12) = 4$. On the other hand, thanks to the Theorem of Cyclotomic Extensions in Positive Characteristic, the degree over $\mathbb{F}_p$ is the order of p in $(\mathbb{Z}/12\mathbb{Z})^*$ for $p \neq 2,\ 3$; in particular, the degree is 1 over $\mathbb{F}_{13}$ and 2 over $\mathbb{F}_5$.

▌One could observe directly that $\mathbb{F}_{13}^* \simeq \mathbb{Z}/12\mathbb{Z}$ and so $\mathbb{F}_{13}$ already contains all twelfth roots of unity. Analogously, 12 divides $|\mathbb{F}_{25}^*| = 24$ but not $|\mathbb{F}_5^*| = 4$, so $\mathbb{F}_{25}$ contains all twelfth roots of unity and $\mathbb{F}_5$ does not.▐

212 Setting $f(x) = x^4 + x^2 + 1$, observe that multiplying $f(x)$ by $x^2 - 1$ yields $x^6 - 1$. In particular, the splitting fields of $f(x)$ and $x^6 - 1$ are the same over every field $\mathbb{K}$, as the roots ± 1 of $x^2 - 1$ automatically belong to $\mathbb{K}$.

Over $\mathbb{Q}$, the roots of the polynomial $f(x)$ are the primitive sixth roots of unity, so its splitting field is the sixth cyclotomic field, which has degree $\phi(6) = 2$.

Over $\mathbb{F}_3$, we have $x^6 - 1 = (x^3 - 1)(x^3 + 1) = (x - 1)^3(x + 1)^3$, so the degree of the splitting field is 1.

As for $\mathbb{F}_7$, remark that $\mathbb{F}_7^*$ has order 6, so $a^6 = 1$ for all elements $a \in \mathbb{F}_7$. It follows that all six elements of $\mathbb{F}_7^*$ are roots of $f(x)$ and thus the splitting field obviously has degree 1.

213 Let $\mathbb{Q}(\zeta) = \mathbb{Q}(\eta)$ be the nth cyclotomic extension of $\mathbb{C}$ and assume by contradiction that $\eta = \zeta + a$ for some $a \in \mathbb{Q}$. The action of the Galois group G of $\mathbb{Q}(\zeta)/\mathbb{Q}$ is transitive on primitive nth roots of unity, so there exists $\sigma \in G$ such that $\sigma(\zeta) = \eta = \zeta + a$.

It is easy to show by induction that $\sigma^h(\zeta) = \zeta + ha$ for all $h \geq 1$; in particular, when r is equal to the order of σ in the finite group G, we get $\zeta = \sigma^r(\zeta) = \eta + ra$ and thus necessarily $r = 0$, which gives a contradiction.

214 We denote the three polynomials in the problem statement by $e(x) = x^2 - 3$, $f(x) = x^3 - 2$ and $k(x) = (x^2 - 3)(x^3 - 2)$.

(i) By the Fundamental Theorem of Galois Theory, the subextensions of $\mathbb{F}$ that are normal over $\mathbb{Q}$ are in bijection with the normal subgroups of $G = \mathrm{Gal}(\mathbb{F}/\mathbb{Q})$, a group which we now set out to determine.
The roots of $f(x)$ are $\alpha = \sqrt[3]{2}$, $\zeta\alpha$ and $\zeta^2\alpha$, where ζ is a primitive third root of unity in $\mathbb{C}$. The polynomial $f(x)$ is irreducible by Eisenstein's Criterion, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(\alpha) \subseteq \mathbb{F}$, the degree $[\mathbb{F} : \mathbb{Q}]$ must be divisible by 3; moreover, $\mathbb{Q}(\alpha) \neq \mathbb{F}$ as $\mathbb{Q}(\alpha) \subseteq \mathbb{R}$ whereas $\zeta\alpha \notin \mathbb{R}$. In particular, $\mathbb{Q}(\alpha)/\mathbb{Q}$

is not normal, and the fact that $[\mathbb{F} : \mathbb{Q}] \leq 3! = 6$ yields $[\mathbb{F} : \mathbb{Q}] = 6$, so G is isomorphic to S_3.
We know that the only normal subgroup of S_3 is A_3, so there exists a unique normal subextension of $\mathbb{F}$, which is clearly $\mathbb{Q}(\zeta)$, as $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 2 = [\mathsf{S}_3 : \mathsf{A}_3]$.

(ii) The splitting field of $e(x)$ is $\mathbb{E} = \mathbb{Q}(\sqrt{3})$. The extension $\mathbb{E} \cap \mathbb{F}$ is normal over $\mathbb{Q}$ because it is the intersection of two normal extensions, and its degree is at most 2. Tanks to (i), $\mathbb{E} \cap \mathbb{F}$ can only be $\mathbb{Q}(\zeta)$ or $\mathbb{Q}$, and the fact that $\mathbb{E}$ is a real extension immediately yields $\mathbb{E} \cap \mathbb{F} = \mathbb{Q}$.

(iii) It follows easily from $k(x) = e(x)f(x)$ that $\mathbb{K} = \mathbb{E} \cdot \mathbb{F}$. Moreover, since the degree of $\mathbb{E}$ over $\mathbb{Q}$ is 2 and since $\mathbb{E} \cap \mathbb{F} = \mathbb{Q}$, the degree of $\mathbb{K} = \mathbb{F}(\sqrt{3})$ over $\mathbb{F}$ is 2. So $[\mathbb{K} : \mathbb{Q}] = [\mathbb{K} : \mathbb{F}][\mathbb{F} : \mathbb{Q}] = 12$.

▌In part (i) we could have immediately concluded that $G \simeq \mathsf{S}_3$ by observing that the discriminant $-4 \cdot 0^3 - 27 \cdot 2^2$ of $f(x)$ is not a square in $\mathbb{Q}$.▐

215 We know that in a Galois extension, and in particular in $\mathbb{Q}(\zeta)/\mathbb{Q}$, the set of roots of the minimal polynomial of an element is the orbit of the element for the action of the Galois group. Since the cyclotomic polynomial $\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + x + 1$ is irreducible over $\mathbb{Q}$, the orbit of ζ for the action of G is exactly the set of primitive pth roots of unity. Moreover, ζ is a generator of $\mathbb{Q}(\zeta)$, so its stabiliser in G is trivial. It is thus clear that

$$\sum_{\varphi \in G} \varphi(\zeta), \qquad \prod_{\varphi \in G} \varphi(\zeta)$$

are the sum and product of the roots of $\Phi_p(x)$, respectively, and the two formulas in the problem statement follow.

216 Let $\mathbb{K}$ be the splitting field of $x^4 - n$ over $\mathbb{Q}$. First of all, observe that if $n = 0$ then $\mathbb{K} = \mathbb{Q}$. From now on, we assume n to be a positive integer.

Setting $\alpha = \sqrt[4]{n} \in \mathbb{R}^+$, the roots of $x^4 - n$ are α, $i\alpha$, $-\alpha$, $-i\alpha$, and thus $\mathbb{K} = \mathbb{Q}(\alpha, i)$. We have $i \notin \mathbb{Q}(\alpha) \subseteq \mathbb{R}$, so $[\mathbb{K} : \mathbb{Q}(\alpha)] > 1$. It follows that $[\mathbb{K} : \mathbb{Q}(\alpha)] = 2$, as $[\mathbb{Q}(i) : \mathbb{Q}] = 2$ implies $[\mathbb{K} : \mathbb{Q}(\alpha)] \leq 2$. What we need to do is compute $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ for each $n \in \mathbb{N}$. We consider the following cases.

① If $n = m^4$ is a fourth power then $\alpha = m \in \mathbb{N}$, hence $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 1$ and $[\mathbb{K} : \mathbb{Q}] = 2$.

② If $n = m^2$ is a square but not a fourth power, that is, $m \notin \mathbb{N}^2$, then $\alpha = \sqrt{m} \notin \mathbb{Q}$, and so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2$ and $[\mathbb{K} : \mathbb{Q}] = 4$.

③ Finally, if n is not a square, suppose $n = r^4 m$ where m is free from fourth powers; m is not a square and $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt[4]{m})$, so we can assume without loss of generality that n is free from fourth powers. Now, there must exist a prime p such that either p divides n and p^2 does not divide n, or p^3 divides n and p^4 does not divide n.

In the first case, $x^4 - n$ is irreducible by Eisenstein's Criterion, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$ and $[\mathbb{K} : \mathbb{Q}] = 8$.

In the second case, remark that $1/\alpha = \sqrt[4]{n^3}/n$ and thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\beta)$ with $\beta = \sqrt[4]{n^3}$. We can write $n^3 = r^4 m$ where m is squarefree and divisible by p but not by p^2; we thus reduce the problem to the previous case and find that $[\mathbb{K} : \mathbb{Q}] = 8$.

217 Let ζ be a primitive twelfth root of unity in $\mathbb{C}$ and let $\mathbb{L} = \mathbb{Q}(\zeta)$ be the cyclotomic extension of order 12 over $\mathbb{Q}$. We know that the Galois group of $\mathbb{L}/\mathbb{Q}$ is isomorphic to $(\mathbb{Z}/12\mathbb{Z})^* \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. By the Fundamental Theorem of Galois Theory, there are 5 subfields of $\mathbb{L}$ corresponding to the 5 subgroups of $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$.

The fourth cyclotomic field $\mathbb{K}_1 = \mathbb{Q}(\sqrt{-1})$ and the third cyclotomic field $\mathbb{K}_2 = \mathbb{Q}(\sqrt{-3})$ are proper subextensions of $\mathbb{L}$. It follows that $\mathbb{K}_3 = \mathbb{Q}(\sqrt{3})$ is also a proper subextension of $\mathbb{L}$.

Remark that $\mathbb{K}_1$, $\mathbb{K}_2$ and $\mathbb{K}_3$ are quadratic extensions and that they are distinct, as none of the products $(-1)\cdot 3$, $(-1)\cdot(-3)$ and $3\cdot(-3)$ is a square in $\mathbb{Q}$. This implies that the subextensions of $\mathbb{L}$ are: $\mathbb{L}$, $\mathbb{K}_1$, $\mathbb{K}_2$, $\mathbb{K}_3$ and $\mathbb{Q}$.

218 We know that the Galois group G of the nth cyclotomic field over $\mathbb{Q}$ is isomorphic to $G = (\mathbb{Z}/n\mathbb{Z})^*$, which is an Abelian group of order $\phi(n)$. By the Fundamental Theorem of Galois Theory, using the fact that G is Abelian and that 7 is prime, the existence of a subfield with the required property is equivalent to the existence of a subgroup of G whose index is 7.

In particular, 7 must divide the order of G. If $n = p_1^{e_1}\cdots p_r^{e_r}$ is the prime factorisation of n, then $\phi(n) = p_1^{e_1-1}(p_1-1)\cdots p_r^{e_r-1}(p_r-1)$. It follows that 7 divides $\phi(n)$ if and only if 7^2 divides n or there is a prime p that divides n and is such that 7 divides $p-1$. In the first case, we have $n \geq 49$, whereas 29 is the smallest prime for which the second condition holds. So $n = 29$ is the smallest positive integer such that 7 divides $\phi(n)$.

Moreover, we have $(\mathbb{Z}/29\mathbb{Z})^* \simeq \mathbb{Z}/28\mathbb{Z}$, and thus, since the latter group is cyclic, $(\mathbb{Z}/29\mathbb{Z})^*$ has a subgroup of order 7, as required. In conclusion, $n = 29$ is the required minimum.

219 The roots of $f(x) = (x^2+1)^2 + 1 = x^4 + 2x^2 + 2$ are $\pm\alpha_1$, $\pm\alpha_2$, where $\alpha_1 = \sqrt{-1+i}$ and $\alpha_2 = \sqrt{-1-i}$; moreover, $f(x)$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion. It is then clear that, letting $\mathbb{K}$ be the splitting field, we have $\mathbb{K} = \mathbb{Q}(\alpha_1, \alpha_2)$, and 4 divides $[\mathbb{K} : \mathbb{Q}]$.

Remark that $\alpha_1\alpha_2 = \sqrt{2}$ and therefore $\mathbb{K} = \mathbb{Q}(\alpha_1, \sqrt{2})$. Since $\sqrt{2}$ is a root of $x^2 - 2 \in \mathbb{Q}(\alpha_1)[x]$, the degree of $\mathbb{K}$ over $\mathbb{Q}(\alpha_1)$ can be either 2 or 1, corresponding to $[\mathbb{K} : \mathbb{Q}] = 8$ or $[\mathbb{K} : \mathbb{Q}] = 4$. We wish to exclude the second possibility, thus proving that $[\mathbb{K} : \mathbb{Q}] = 8$.

If we had $[\mathbb{K} : \mathbb{Q}] = 4$ then we would have $\mathbb{K} = \mathbb{Q}(\alpha_1) = \mathbb{Q}(\alpha_2)$. Recalling the values we computed for α_1 and α_2, it is clear that these two fields are quadratic extensions of $\mathbb{Q}(i)$. But then they could only be equal if $(-1+i)(-1-i) = 2$ were a square in $\mathbb{Q}(i)$, which is not the case: since 2 is a positive real, it would have to be a square in $\mathbb{Q}$, but $\sqrt{2}$ is irrational.

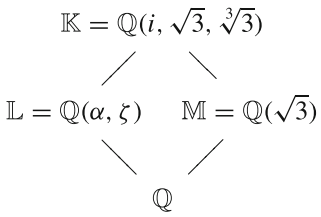
220

(i) Setting $\alpha = \sqrt{2}$, $\beta = \sqrt{3}$, the roots of $f(x)$ are $\pm\alpha$, $\pm\beta \pm \alpha\beta$. Now let $\mathbb{K} = \mathbb{Q}(\alpha, \beta)$ be the splitting field of $f(x)$ over $\mathbb{Q}$ and let $G = \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$. The two quadratic extensions $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta)$ are distinct, as $2 \cdot 3 = 6$ is not a square in $\mathbb{Q}$. So $[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}] = [\mathbb{Q}(\alpha, \beta) : \mathbb{Q}(\alpha)][\mathbb{Q}(\alpha) : \mathbb{Q}] = 2 \cdot 2 = 4$. The extensions $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta)$ of $\mathbb{Q}$ are normal, and in particular Galois, as they are quadratic; moreover, $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\beta) = \mathbb{Q}$ because the intersection has degree less than 2. It follows that $G \simeq \mathrm{Gal}(\mathbb{Q}(\alpha)/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\beta)/\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(ii) Since 0 is a root of $f(x)$ in $\mathbb{F}_2$ and in $\mathbb{F}_3$, we may assume that p is not 2 or 3. Denote by Q the subgroup of index 2 of $\mathbb{F}_p^*$ given by nonzero squares. If 2 and 3 are not in Q, that is, if $(x^2 - 2)(x^2 - 3)$ has no roots in $\mathbb{F}_p$, then $2 \cdot 3 \in Q$ and so $x^2 - 6$ has a root in $\mathbb{F}_p$. In particular, $f(x)$ always has a root in $\mathbb{F}_p$.

221 Denote by α the real number $\sqrt[3]{3}$ and by ζ the primitive third root of unity $(-1 + i\sqrt{3})/2$ in $\mathbb{C}$. Any automorphism of $\mathbb{K}/\mathbb{Q}$ must send i to $\pm i$, $\sqrt{3}$ to $\pm\sqrt{3}$ and α to $\zeta^k\alpha$, with $k = 0$, 1 or 2, because i is a root of $x^2 + 1$, $\sqrt{3}$ is a root of $x^2 - 3$ and α is a root of $x^3 - 3$. It follows that $\mathbb{K}$ is a normal extension of $\mathbb{Q}$, since $\zeta \in \mathbb{F} = \mathbb{Q}(i, \sqrt{3}) \subseteq \mathbb{K}$.

It is also clear that $[\mathbb{F} : \mathbb{Q}] = 4$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, and so $12 \mid [\mathbb{F} : \mathbb{Q}]$. From the fact that $\mathbb{K} = \mathbb{F}(\alpha)$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, we obtain $[\mathbb{K} : \mathbb{F}] \leq 3$, and thus finally $[\mathbb{K} : \mathbb{Q}] = 12$.



By our arguments so far, $\mathbb{L} = \mathbb{Q}(\alpha, \zeta)$ is the splitting field of $x^3 - 3$, so $\mathrm{Gal}(\mathbb{L}/\mathbb{Q}) \simeq \mathsf{S}_3$ because $[\mathbb{L} : \mathbb{Q}] = 6$ and $\mathbb{Q}(\alpha)$ is a subextension of $\mathbb{L}$ that is not normal. Setting $\mathbb{M} = \mathbb{Q}(\sqrt{3})$, we clearly have $\mathrm{Gal}(\mathbb{M}/\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z}$. Remark that $\mathbb{K} = \mathbb{L} \cdot \mathbb{M}$ and also $\mathbb{L} \cap \mathbb{M} = \mathbb{Q}$, because we would otherwise have $\mathbb{L} \cap \mathbb{M} = \mathbb{M}$, which would yield $\mathbb{K} = \mathbb{L}$, contradicting the fact that $[\mathbb{K} : \mathbb{Q}] = 12$ and $[\mathbb{L} : \mathbb{Q}] = 6$.

It is thus clear that $\mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \simeq \mathsf{S}_3 \times \mathbb{Z}/2\mathbb{Z}$.

By the Fundamental Theorem of Galois Theory, the subextensions of $\mathbb{K}$ of degree 6 over $\mathbb{Q}$ correspond to subgroups of order 2 of G, that is, to elements of order 2. All we need is thus to count elements of order 2 in G, and it is easy to see from the expression of G as a direct product that there are 7 such elements.

▌The fact that the Galois group of $\mathbb{L}$ over $\mathbb{Q}$ is isomorphic to S_3 can alternatively be deduced by observing that the discriminant $-4 \cdot 0^3 - 27 \cdot 3^2$ of $x^3 - 2$ is not a square in $\mathbb{Q}$.▐

222 Let $\mathbb{K}$ be the splitting field of the polynomial $x^5 - 5$ over $\mathbb{Q}$ and let ζ be a primitive fifth root of unity in $\mathbb{C}$. Since $\mathbb{K} = \mathbb{Q}(\sqrt[5]{5}, \zeta)$, the field $\mathbb{F}$ is a subextension of $\mathbb{K}$; clearly, $\mathbb{K}$ is normal over $\mathbb{Q}$ as it is a splitting field.

Let $\mathbb{E}$ be the fifth cyclotomic field $\mathbb{Q}(\zeta)$; notice that $\mathbb{K}$ is the composite field of $\mathbb{F}$ and $\mathbb{E}$, and so $[\mathbb{K} : \mathbb{Q}] = 4 \cdot 5 = 20$ as the degrees of $\mathbb{F}$ and $\mathbb{E}$ over $\mathbb{Q}$ are relatively

prime. In particular, $[\mathbb{K} : \mathbb{F}] = [\mathbb{K} : \mathbb{Q}]/[\mathbb{F} : \mathbb{Q}] = 4$. Thus the extension $\mathbb{K}/\mathbb{F}$ does have the required properties.

Now, let $\mathbb{K}'$ be a normal extension of $\mathbb{Q}$ containing $\mathbb{F}$ such that $[\mathbb{K}' : \mathbb{F}] = 4$. The polynomial $x^5 - 5 \in \mathbb{Q}[x]$ has a root in $\mathbb{K}'$ and it is irreducible over $\mathbb{Q}$, so it must split as a product of linear factors in $\mathbb{K}'$ because $\mathbb{K}'$ is normal over $\mathbb{Q}$. But then $\mathbb{K} \subseteq \mathbb{K}'$ and the equality $[\mathbb{K}' : \mathbb{F}] = 4 = [\mathbb{K} : \mathbb{F}]$ implies that $\mathbb{K}' = \mathbb{K}$. We have thus shown that the extension $\mathbb{K}$ is the only one that satisfies the requirements.

By the Fundamental Theorem of Galois Theory, a subextension of $\mathbb{K}$ of degree 4 over $\mathbb{Q}$ corresponds to a subgroup H of order 5 in $G = \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$, that is, to a 5-Sylow subgroup of G. However, letting n_5 denote the number of 5-Sylow subgroups of G, we have $n_5 \equiv 1 \pmod 5$ and $n_5 \,|\, 4 = [G : H]$, so necessarily $n_5 = 1$. This implies that H is the unique 5-Sylow subgroup of G, and thus that there is a unique subextension of $\mathbb{K}/\mathbb{Q}$ of degree 4, which is given by the field $\mathbb{E} = \mathbb{Q}(\zeta)$ we introduced earlier.

223 Define the extensions $\mathbb{E} = \mathbb{Q}(\sqrt{n})$, $\mathbb{F} = \mathbb{Q}(\zeta)$ and $\mathbb{K} = \mathbb{Q}(\sqrt{n}, \zeta) = \mathbb{E} \cdot \mathbb{F}$. The fifth cyclotomic field $\mathbb{F}$ has Galois group over $\mathbb{Q}$ isomorphic to $(\mathbb{Z}/5\mathbb{Z})^*$ and thus to $\mathbb{Z}/4\mathbb{Z}$. Moreover, we can clearly assume n to be squarefree.

If $n = 1$ then $\mathbb{E} = \mathbb{Q}$, $\mathbb{K} = \mathbb{F}$, hence $[\mathbb{K} : \mathbb{Q}] = 4$ and $\mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \simeq \mathbb{Z}/4\mathbb{Z}$.

Suppose $n \neq 1$. Since $[\mathbb{E} : \mathbb{Q}] = 2$, the degree $[\mathbb{K} : \mathbb{F}] = [\mathbb{F}(\sqrt{n}) : \mathbb{F}]$ is equal to either 1 or 2, depending on whether or not we have $\mathbb{E} \subseteq \mathbb{F}$. In the first case, we have again that $\mathbb{K} = \mathbb{F}$; in the second, that is, if $\mathbb{E} \not\subseteq \mathbb{F}$, then the fact that $\mathbb{E}$ and $\mathbb{F}$ are normal extensions of $\mathbb{Q}$ that intersect in $\mathbb{Q}$ itself implies that $\mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \simeq \mathrm{Gal}(\mathbb{E}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ and $[\mathbb{K} : \mathbb{Q}] = 8$.

What is left to do is determine for which n we have $\mathbb{E} \subseteq \mathbb{F}$. By the Fundamental Theorem of Galois Theory, $\mathbb{F}$ has a unique quadratic subfield, as $\mathbb{Z}/4\mathbb{Z}$ has only one subgroup of index 2, and this field is $\mathbb{Q}(\alpha)$, where $\alpha = \zeta + \zeta^{-1}$. Indeed, from $\zeta^4 + \zeta^3 + \zeta^2 + \zeta + 1 = 0$, dividing by ζ^2 yields $\zeta^2 + \zeta + 1 + \zeta^{-1} + \zeta^{-2} = 0$, that is, $\alpha^2 + \alpha - 1 = 0$, and we obtain $\alpha = (-1 + \sqrt{5})/2$.

In conclusion, $\mathbb{E} \subseteq \mathbb{F}$ if and only if $n = 5$.

224 Denote by n the degree of $f(x)$; first of all, remark that we cannot have $n = 1$ or $n = 2$, as this would imply $[\mathbb{K} : \mathbb{Q}] = 1$ or $[\mathbb{K} : \mathbb{Q}] = 2$.

Now let $\alpha \in \mathbb{C}$ be a root of $f(x)$. From the inclusion $\mathbb{Q}(\alpha) \subseteq \mathbb{K}$ and the fact that $f(x)$ is irreducible we obtain that $n = [\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides $8 = [\mathbb{K} : \mathbb{Q}]$. We can thus either have $n = 4$ or $n = 8$, and we will now set out to prove that both cases are actually possible.

Consider the polynomial $f(x) = x^4 - 2$, which is irreducible by Eisenstein's Criterion. Its splitting field is $\mathbb{K} = \mathbb{Q}(\sqrt[4]{2}, i)$. Since $\mathbb{Q}(\sqrt[4]{2})$ has degree 4 and $\mathbb{Q}(i)$ has degree 2 over $\mathbb{Q}$, and since the intersection of the two extensions is $\mathbb{Q}$ because the first one is real, we immediately obtain that $\mathbb{K} = \mathbb{Q}(\sqrt[4]{2})(i)$ has degree 8 over $\mathbb{Q}$, as wanted.

The case $n = 8$ is also possible, as we can see for example from the polynomial $f(x) = x^8 + 1$. Indeed, we have $f(x)(x^8 - 1) = x^{16} - 1$, and the roots of $f(x)$ are thus the sixteenth roots of unity that are not eighth roots of unity. It follows

that $f(x)$ is the cyclotomic polynomial of order 16 and its splitting field has degree $\phi(16) = 8$ over $\mathbb{Q}$.

225 Remark that $(x^4+x^2+1)(x^2-1) = x^6-1$ and that $(x^4-x^2+1)(x^4+x^2+1)(x^4-1) = x^{12}-1$. It follows that, in any field of characteristic different from 2 and 3, the roots of $f(x) = x^4 - x^2 + 1$ are the primitive twelfth roots of unity; in other words, $f(x)$ is the twelfth cyclotomic polynomial.

Letting $\mathbb{K}$ be the splitting field of $f(x)$ over $\mathbb{Q}$ and $\zeta \in \mathbb{C}$ the primitive twelfth root of unity $\zeta = (\sqrt{3}+i)/2$, we have $\mathbb{Q}(\zeta) \subseteq \mathbb{K}$. Moreover, the fact that $\zeta + \zeta^{-1} = \sqrt{3}$ yields that in fact $\mathbb{K} = \mathbb{Q}(\zeta)$ is the twelfth cyclotomic field. So $[\mathbb{K} : \mathbb{Q}] = \phi(12) = 4$ and the Galois group we are investigating is isomorphic to $(\mathbb{Z}/12\mathbb{Z})^* \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Thanks to the Theorem of Cyclotomic Extensions in Positive Characteristic, the degree of the cyclotomic extension of order twelve over $\mathbb{F}_7$ is given by the order of 7 in $(\mathbb{Z}/12\mathbb{Z})^*$, which is 2, as one can immediately check. It is then clear that the splitting field is $\mathbb{F}_{49}$, as $x^2 - 3$ splits as a product of linear factors over this field. The required degree is thus 2 and the Galois group is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

226 Let $\mathbb{K}$ be the splitting field of $f(x) = x^4 - 5x^2 + 9$ over $\mathbb{Q}$. Since $f(x) = (x^2+3)^2 - 11x^2 = (x^2 - \sqrt{11}x + 3)(x^2 + \sqrt{11}x + 3)$, the roots of $f(x)$ are $(\pm\sqrt{11} \pm i)/2$, and therefore $\mathbb{K} = \mathbb{Q}(\sqrt{11}, i)$.

Remark that 11, -1 and $(-1)\cdot 11$ are not elements of $\mathbb{Q}^2$, so $\mathbb{K}$ has degree 4 over $\mathbb{Q}$ and contains three different quadratic subextensions. This implies that the Galois group of $\mathbb{K}$ over $\mathbb{Q}$ is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

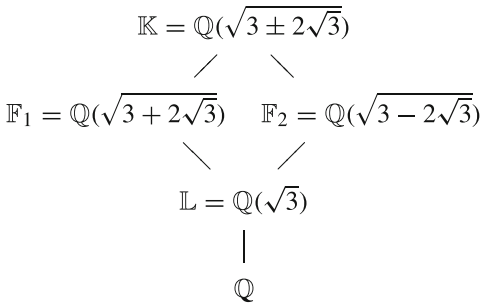
Over $\mathbb{F}_{11}$, we have $f(x) = (x^2+3)^2$, and we can check that -3 is not a square in $\mathbb{F}_{11}$. It follows that $\mathbb{F}_{11^2}$ is the splitting field of $f(x)$ and that the Galois group is $\mathbb{Z}/2\mathbb{Z}$.

227

(i) By the Theorem of Cyclotomic Extensions in Positive Characteristic, we have the following: assuming p does not divide n, the polynomial $x^n - 1$ splits as a product of linear factors over $\mathbb{F}_p$ if and only if $p \equiv 1 \pmod{n}$. We consider separately the cases $p = 2$, $p = 3$ and $p > 3$.
For $p = 2$, the polynomial $x^3 - 1$ does not split as a product of linear factors over $\mathbb{F}_2$, as $2 \not\equiv 1 \pmod 3$, so neither does the polynomial $f(x)$. Similarly, for $p = 3$, the polynomial $f(x)$ does not split as a product of linear factors because $3 \not\equiv 1 \pmod 8$.
For $p > 3$, the conditions are that $p \equiv 1 \pmod 8$ and $p \equiv 1 \pmod 3$: $f(x)$ splits as a product of linear factors if and only if $p \equiv 1 \pmod{24}$.
[For example, for $p = 73$ the polynomial splits as a product of linear factors; in fact, there exist infinitely many primes congruent to 1 modulo 24.]

(ii) Denote by ζ_n the primitive nth root of unity $e^{2\pi i/n} \in \mathbb{C}$. The splitting field $\mathbb{K}$ of $f(x)$ over $\mathbb{Q}$ is $\mathbb{Q}(\zeta_8, \zeta_3)$. Moreover, since 8 and 3 are coprime, $\mathbb{K} = \mathbb{Q}(\zeta_{24})$ is the 24th cyclotomic field. We therefore have $[\mathbb{K} : \mathbb{Q}] = \phi(24) = 8$, and $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $(\mathbb{Z}/24\mathbb{Z})^* \simeq (\mathbb{Z}/2\mathbb{Z})^3$.

228 By noticing that 1 and -1 are roots of the polynomial $f(x) = x^6 - 7x^4 + 3x^2 + 3$, we obtain its factorisation $f(x) = (x-1)(x+1)(x^4 - 6x^2 - 3)$. The roots of the biquadratic factor, which is irreducible by Eisenstein's Criterion, are $\pm\sqrt{3 \pm 2\sqrt{3}}$.



Letting $\mathbb{K}$ be the splitting field of $f(x)$ over $\mathbb{Q}$, consider its subextensions $\mathbb{L} = \mathbb{Q}(\sqrt{3})$ and $\mathbb{F}_1 = \mathbb{L}(\sqrt{3+2\sqrt{3}})$, $\mathbb{F}_2 = \mathbb{L}(\sqrt{3-2\sqrt{3}})$.
The two quadratic extensions $\mathbb{F}_1/\mathbb{L}$ and $\mathbb{F}_2/\mathbb{L}$ are different because $(3+2\sqrt{3})(3-2\sqrt{3}) = -3 \notin \mathbb{L}^2$, as $\mathbb{L}$ is a real extension of $\mathbb{Q}$. It follows that $[\mathbb{K} : \mathbb{Q}] = 8$.

Over $\mathbb{F}_{13}$, we have $8 + (-2) = 6$ and $8 \cdot (-2) = -3$, so $x^4 - 6x^2 - 3 = (x^2 - 8)(x^2 + 2)$. We can check directly that 2 is not a square in $\mathbb{F}_{13}$. The splitting field has thus degree 2 over $\mathbb{F}_{13}$.

229 Let n and m be the degrees of the polynomials $f(x)$ and $g(x)$, respectively. We have $[\mathbb{K}(\alpha) : \mathbb{K}] = n$ and $[\mathbb{K}(\beta) : \mathbb{K}] = m$.

(i) The polynomial $g(x)$ is irreducible over $\mathbb{K}(\alpha)$ if and only if $[\mathbb{K}(\alpha)(\beta) : \mathbb{K}(\alpha)] = m$, so if and only if $[\mathbb{K}(\alpha, \beta) : \mathbb{K}] = [\mathbb{K}(\alpha)(\beta) : \mathbb{K}(\alpha)][\mathbb{K}(\alpha) : \mathbb{K}] = mn$.
On the other hand, $[\mathbb{K}(\alpha, \beta) : \mathbb{K}] = [\mathbb{K}(\beta)(\alpha) : \mathbb{K}(\beta)][\mathbb{K}(\beta) : \mathbb{K}] = mn$ if and only if $[\mathbb{K}(\beta)(\alpha) : \mathbb{K}(\beta)] = n$, that is, if and only if $f(x)$ is irreducible over $\mathbb{K}(\beta)$.

(ii) If $(n, m) = 1$ then the least common multiple mn divides $[\mathbb{K}(\alpha, \beta) : \mathbb{K}] \leq mn$, hence $[\mathbb{K}(\alpha, \beta) : \mathbb{K}] = mn$. By what we remarked in (i), this is equivalent to the irreducibility of the polynomial $f(x)$ over $\mathbb{K}(\beta)$.

230 Letting $f(x) = x^7 - 2$, consider the problem over $\mathbb{Q}$. Denote by α the real number $\sqrt[7]{2}$ and by $\zeta \in \mathbb{C}$ a primitive seventh root of unity; the roots of $f(x)$ are α, $\alpha\zeta$, $\alpha\zeta^2, \ldots, \alpha\zeta^6$.

The splitting field $\mathbb{K} = \mathbb{Q}(\alpha, \zeta)$ of $f(x)$ thus contains the subfields $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\zeta)$, whose degrees are 7 and $\phi(7) = 6$, respectively. Since the degrees are relatively prime, we have $[\mathbb{K} : \mathbb{Q}] = 42$.

Remark that $\mathbb{Q}(\alpha) \subseteq \mathbb{R}$ is a proper subfield of $\mathbb{K}$ that contains only one root of $f(x)$, so it is not a normal extension of $\mathbb{Q}$. It follows that the Galois group $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is not Abelian, or all of its subgroups would be normal and thus, by the Fundamental Theorem of Galois Theory, every subextension would be normal.

Over $\mathbb{F}_5$, we know that the Galois group is necessarily Abelian, and in fact cyclic, as is the case for every finite Galois extension of a finite field. As for the degree of the splitting field $\mathbb{K}$, remark that $2 = (-2)^7$, so $\mathbb{K}$ can be obtained by simply adding a primitive seventh root of unity to $\mathbb{F}_5$.

Thanks to the Theorem of Cyclotomic Extensions in Positive Characteristic, the cyclotomic extension of order 7 over $\mathbb{F}_5$ has degree given by the order k of 5 in $(\mathbb{Z}/7\mathbb{Z})^*$, that is, the minimum k such that 7 divides $5^k - 1$. It is easy to check that this minimum is 6; this proves that $[\mathbb{K} : \mathbb{F}_5] = 6$.

231 Set $\alpha = \sqrt[4]{2}$, $\beta = \sqrt[6]{2}$, and let ζ be the primitive sixth root of unity $(1 + \sqrt{-3})/2 \in \mathbb{C}$. We have that $\mathbb{F} = \mathbb{Q}(\alpha, i)$, $\mathbb{K} = \mathbb{Q}(\beta, \zeta)$ and that $[\mathbb{F} : \mathbb{Q}] = 8$, $[\mathbb{K} : \mathbb{Q}] = 12$; this is because $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta)$ are real extensions, whereas $\mathbb{Q}(i)$ and $\mathbb{Q}(\zeta)$ are not real and have degree 2 over $\mathbb{Q}$.

First of all, remark that $\alpha^2 = \beta^3 = \sqrt{2}$, so $\mathbb{Q}(\sqrt{2})$ is a subextension of $\mathbb{F} \cap \mathbb{K}$. It follows that, since $r = [\mathbb{F} \cap \mathbb{K} : \mathbb{Q}]$ must divide $[\mathbb{F} : \mathbb{Q}]$ and $[\mathbb{K} : \mathbb{Q}]$, we have either $r = 2$ or $r = 4$.

The group $G = \mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ is a subgroup of order 8 of S_4, so it is a 2-Sylow subgroup of S_4 and is thus isomorphic to D_4.

Assume by contradiction that $r = 4$. By the Fundamental Theorem of Galois Theory, the extension $\mathbb{F} \cap \mathbb{K}$ of $\mathbb{Q}$ corresponds to a normal subgroup H of D_4 that has index 4. Since G/H has order 4, it must be an Abelian group, and so H must contain the commutator subgroup of D_4.

On the other hand, the normal subgroup $K = \langle r^2 \rangle$ is the commutator subgroup of D_4: it must contain the commutator subgroup because D_4/K, which also has 4 elements, is Abelian; moreover, K has only 2 elements and the commutator subgroup cannot be trivial because D_4 is not Abelian. Thus we have $H = K$, because both groups have two elements. It follows that $\mathbb{F} \cap \mathbb{K}$ is the only normal subfield of $\mathbb{F}$ that has degree 4 over $\mathbb{Q}$.

Now, $\mathbb{Q}(\sqrt{2}, i)$ is a subextension of $\mathbb{F}$ that has degree 4 and is normal, as it is the splitting field of $(x^2 - 2)(x^2 + 1)$. We therefore have $\mathbb{F} \cap \mathbb{K} = \mathbb{Q}(\sqrt{2}, i)$, and in particular $i \in \mathbb{K}$, which implies that $\mathbb{Q}(\sqrt{2}, \sqrt{-3}, i)$ is a subextension of $\mathbb{K}$. But this is a contradiction, as $\mathbb{Q}(\sqrt{2}, \sqrt{-3}, i)$ has degree 8, which is not a divisor of $12 = [\mathbb{K} : \mathbb{Q}]$.

We have thus shown that $r = 2$, that is, $\mathbb{F} \cap \mathbb{K} = \mathbb{Q}(\sqrt{2})$.

232 Set $\alpha = \sqrt[5]{2}$ and denote by ζ some primitive fifth root of unity in $\mathbb{C}$. The roots of $f(x) = (x^5 - 2)(x^2 - 5)$ are $\alpha\zeta^h$, for $h = 0, 1, \ldots, 4$, and $\pm\sqrt{5}$. The element $\zeta + \zeta^{-1}$ of the fifth cyclotomic field $\mathbb{Q}(\zeta)$ is a root of the polynomial $x^2 + x - 1$, whose discriminant is 5. It follows that $\sqrt{5} \in \mathbb{Q}(\zeta)$ and that the splitting field of $f(x)$ is $\mathbb{K} = \mathbb{Q}(\alpha, \zeta)$.

The field $\mathbb{K}$ thus contains the subfields $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\zeta)$, whose respective degrees are 5 (since $x^5 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion) and $\phi(5) = 4$. The degrees being relatively prime yields that $[\mathbb{K} : \mathbb{Q}] = 20$.

A group of order 20, such as the Galois group of $\mathbb{K}/\mathbb{Q}$, has a unique 5-Sylow subgroup: indeed, the number n_5 of such subgroups must satisfy both $n_5 \equiv 1 \pmod 5$ and $n_5 \mid 4 = 20/5$. By the Fundamental Theorem of Galois Theory, it follows that $\mathbb{K}$ has a unique subfield of degree 4 over $\mathbb{Q}$.

But the subfield $\mathbb{Q}(\zeta)$ has degree 4, so it is the unique subfield in question.

233

(i) Let ω be a primitive eleventh root of unity in $\mathbb{C}$ and set $\beta = \sqrt[11]{3}$. Letting $\mathbb{E}$ denote the splitting field of $x^{11} - 3$ over $\mathbb{Q}$, we have that $\mathbb{E} = \mathbb{Q}(\beta, \omega)$, because the roots of the polynomial are $\beta\omega^k$ for $0 \leq k \leq 10$. Moreover, $[\mathbb{Q}(\omega) : \mathbb{Q}] = \phi(11) = 10$ and $\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \simeq (\mathbb{Z}/11\mathbb{Z})^*$, because $\mathbb{Q}(\omega)$ is a cyclotomic extension; we also have that $[\mathbb{Q}(\beta) : \mathbb{Q}] = 11$, as $x^{11} - 3$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion. It follows that $[\mathbb{E} : \mathbb{Q}] = 110$ because 10 and 11 are relatively prime.
An automorphism φ of $\mathbb{E}/\mathbb{Q}$ sends the root β to a root of $x^{11} - 3$, that is, to $\beta\omega^h$ for some h which is unique modulo 11, and it sends ω to a primitive 11th root of unity, that is, to ω^k for some k which is both unique modulo 11 and coprime to 11. We can thus encode φ by the pair (h, k), which immediately yields a group isomorphism from $\mathrm{Gal}(\mathbb{E}/\mathbb{Q})$ to $\mathbb{Z}/11\mathbb{Z} \rtimes (\mathbb{Z}/11\mathbb{Z})^*$, where the semidirect product is with respect to the action $(\mathbb{Z}/11\mathbb{Z})^* \ni a \longmapsto (k \longmapsto ak) \in \mathrm{Aut}(\mathbb{Z}/11\mathbb{Z})$.
The splitting field $\mathbb{F}$ of $x^{11} - 3$ over $\mathbb{K}$ is the composite field $\mathbb{E} \cdot \mathbb{K} = \mathbb{E}(\zeta)$. Moreover, assuming by contradiction that $\zeta \in \mathbb{E}$, we obtain that $\omega\zeta \in \mathbb{E}$; but $\omega\zeta$ is a 33rd primitive root of unity, so $20 = \phi(33)$ must divide $[\mathbb{E} : \mathbb{Q}] = 110$, which is not the case. We therefore have $[\mathbb{F} : \mathbb{Q}] = 220$ and $\mathbb{E} \cap \mathbb{K} = \mathbb{Q}$.
The two extensions $\mathbb{E}/\mathbb{Q}$ and $\mathbb{K}/\mathbb{Q}$ are Galois (they are splitting fields) and their intersection is $\mathbb{Q}$, so $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{E}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ and thus, by our argument above, to $(\mathbb{Z}/11\mathbb{Z} \rtimes (\mathbb{Z}/11\mathbb{Z})^*) \times \mathbb{Z}/2\mathbb{Z}$.

(ii) Let α be an element of $\mathbb{F}$; remark that α is a root of the polynomial $x^3 - \alpha^3 \in \mathbb{K}(\alpha^3)[x]$. If such a polynomial were irreducible in $\mathbb{K}(\alpha^3)$, then we would have $[\mathbb{K}(\alpha) : \mathbb{K}(\alpha^3)] = 3$, and so 3 would have to be a divisor of $[\mathbb{F} : \mathbb{K}]$, since we have the tower of extensions $\mathbb{K} \subseteq \mathbb{K}(\alpha^3) \subseteq \mathbb{K}(\alpha) \subseteq \mathbb{F}$. But this is a contradiction, because our arguments from (i) imply that $[\mathbb{F} : \mathbb{K}] = 110$.
It follows that $x^3 - \alpha^3$ cannot be irreducible over $\mathbb{K}(\alpha^3)$, so it must have a root in this field. But its roots are $\alpha, \zeta\alpha, \zeta^2\alpha$ and, since $\zeta \in \mathbb{K}$, we obtain that $\alpha \in \mathbb{K}(\alpha^3)$. In other words, $\mathbb{K}(\alpha) = \mathbb{K}(\alpha^3)$.

234 In order to determine the Galois group of $\mathbb{F}/\mathbb{Q}$, consider the larger extension $\mathbb{K} = \mathbb{Q}(\zeta)$, that is, the cyclotomic extension of order 13 over $\mathbb{Q}$ within the field $\mathbb{C}$. The group $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $(\mathbb{Z}/13\mathbb{Z})^*$, which is the cyclic group of order 12; we denote by σ_h, where $h \in (\mathbb{Z}/13\mathbb{Z})^*$, the element of $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ such that $\sigma_h(\zeta) = \zeta^h$.

Note that $[\mathbb{K} : \mathbb{Q}] = 12$, so the elements $1, \zeta, \ldots, \zeta^{11}$ are a basis of the extension $\mathbb{K}/\mathbb{Q}$; moreover, by multiplying each of these elements by ζ, we obtain that the elements $\zeta, \zeta^2, \ldots, \zeta^{12}$ are also a basis.

We now wish to determine the subgroup of $(\mathbb{Z}/13\mathbb{Z})^*$ that corresponds to $\mathbb{F}$ via the Galois correspondence. If an element σ_h of the Galois group fixes α, then

$$\zeta + \zeta^3 + \zeta^9 = \alpha = \sigma_h(\alpha) = \zeta^h + \zeta^{3h} + \zeta^{9h}.$$

Using the fact that $\zeta, \zeta^2, \ldots, \zeta^{12}$ is a basis, we obtain that $\zeta^h = \zeta$ or $\zeta^h = \zeta^3$ or $\zeta^h = \zeta^9$, that is, $h = 1$ or $h = 3$ or $h = 9$. If $h = 1$ then we have $\sigma_h = \sigma_1 = \mathrm{Id}_{\mathbb{K}}$, which clearly fixes α. For $h = 3$, the element $\sigma_3(\alpha) = \zeta^3 + \zeta^9 + \zeta^{27} = \zeta^3 + \zeta^9 + \zeta = \alpha$ fixes α. Since the elements that fix α form a subgroup, we also have that $\sigma_9 = \sigma_3^2$ fixes α.

We can thus conclude that the elements of $\mathrm{Gal}(K/\mathbb{Q})$ that keep α fixed form the subgroup $\langle \sigma_3 \rangle$ of order 3. It follows that the Galois group of $\mathbb{F}$ over $\mathbb{Q}$ is isomorphic to $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})/\langle \sigma_3 \rangle$, and thus to the cyclic group with 4 elements.

Denote by τ the automorphism of $\mathbb{C}$ given by complex conjugation; notice that $\alpha \in \mathbb{R}$ if and only if $\tau(\alpha) = \alpha$. Consider now the action of σ_{-1} on $\mathbb{F}$: we have $\sigma_{-1}(\zeta) = \zeta^{-1} = \tau(\zeta)$, so, since both τ and σ_{-1} are field automorphisms, we have that $\sigma_{-1} = \tau_{|\mathbb{K}}$.

We conclude that $\alpha \notin \mathbb{R}$, as σ_{-1} does not fix α because -1 is not 1, 3 or 9 in $\mathbb{Z}/13\mathbb{Z}$. Thus $\mathbb{F}$ is not a subfield of $\mathbb{R}$.

235

(i) Let $\zeta \in \mathbb{C}$ be a primitive fifteenth root of unity; remark that

$$\mathbb{Q}(\sqrt[3]{3}),\ \mathbb{Q}(\sqrt[5]{5}),\ \mathbb{Q}(\zeta) \subseteq \mathbb{K},$$

and in fact $\mathbb{K} = \mathbb{Q}(\sqrt[3]{3}, \sqrt[5]{5}, \zeta)$. Moreover, since the degrees $[\mathbb{Q}(\sqrt[3]{3}) : \mathbb{Q}] = 3$, $[\mathbb{Q}(\sqrt[5]{5}) : \mathbb{Q}] = 5$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \varphi(15) = 8$ are relatively prime, we have $[\mathbb{K} : \mathbb{Q}] = 3 \cdot 5 \cdot 8$.

(ii) Since $\mathbb{K}$ is a splitting field, the extension $\mathbb{K}/\mathbb{Q}$ is Galois; we denote by G its Galois group. The roots of a polynomial with coefficients in $\mathbb{Q}$ are permuted by every element of the Galois group, so $\sqrt[3]{3}$ can only be sent to $\sqrt[3]{3}\zeta^{5h}$ for some integer h such that $0 \le h \le 2$, and similarly $\sqrt[5]{5}$ is sent to $\sqrt[5]{5}\zeta^{3k}$ for some integer k such that $0 \le k \le 4$, while ζ is sent to ζ^m for some integer m such that $1 \le m \le 14$ and $(m, 15) = 1$.

By comparing the number of ensuing possibilities to the order of G, which is $[\mathbb{K} : \mathbb{Q}] = 3 \cdot 5 \cdot 8$, we find that each choice of images does correspond to an automorphism. We can thus denote elements of G by $\varphi_{h,k,m}$, where the triple of indices refers to the images chosen for the elements discussed above. In the next part of the solution, we will use the fact that the automorphisms $\varphi_{h,k,1}$ form a subgroup of G isomorphic to $\mathbb{Z}/15\mathbb{Z}$: this follows from the fact, which one can easily check, that $\varphi_{h,k,1} \circ \varphi_{h',k',1} = \varphi_{h+h',k+k',1}$, where $h + h'$ is taken modulo 3 and $k + k'$ modulo 5.

Having fixed some $a, b \in \mathbb{Q}^*$, let $\alpha = a\sqrt[3]{3} + b\sqrt[5]{5}$. First of all, we obviously have $\mathbb{Q}(\alpha) \subseteq \mathbb{Q}(\sqrt[3]{3}, \sqrt[5]{5})$. In order to prove the converse, we will show that the subgroup of G corresponding to the subextension $\mathbb{Q}(\alpha)$ of $\mathbb{K}$ fixes $\sqrt[3]{3}$ and $\sqrt[5]{5}$, and so $\mathbb{Q}(\alpha)$ must contain the extension $\mathbb{Q}(\sqrt[3]{3}, \sqrt[5]{5})$, which establishes the desired equality.

We now show that only elements of the form $\varphi_{0,0,m}$ may fix α, thus proving our claim. Indeed, let $\varphi_{h,k,m}$ be an element that fixes α; we have $\varphi_{h,k,1}(\alpha) = \varphi_{h,k,m}(\alpha) = \alpha$, that is, $\varphi_{h,k,1}$ must also fix α.

But we have already remarked that automorphisms with $m = 1$ form a subgroup of G isomorphic to $\mathbb{Z}/15\mathbb{Z}$; if any of them were to fix α, its powers would also fix α, and unless $h = k = 0$ among them would be an element of the form $\varphi_{1,0,1}$ or $\varphi_{0,1,1}$. But this is a contradiction, because the image of α via either of these automorphisms is not real, whereas α itself is a real number.

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(i) Letting $\alpha = \zeta + \zeta^2 + \zeta^4$, we have $\alpha^2 = \zeta^2 + \zeta^4 + \zeta + 2\zeta^3 + 2\zeta^5 + 2\zeta^6$, and thus $\alpha^2 + \alpha + 2 = 2\zeta^6 + \cdots + 2\zeta + 2 = 0$. This yields that α is one of the two roots $(-1 \pm \sqrt{-7})/2$ of the polynomial $x^2 + x + 2$. We conclude that $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{-7})$, as wanted.

(ii) We know that $\mathbb{Q}(\zeta)$ has degree 2 over the real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R} = \mathbb{Q}(\zeta + \zeta^{-1})$. Therefore, $[\mathbb{Q}(\zeta + \zeta^{-1}) : \mathbb{Q}] = 3$, which we can combine with the fact that $[\mathbb{Q}(\sqrt{-7}) : \mathbb{Q}] = 2$ to obtain that 6 divides the degree of $\mathbb{Q}(\zeta + \zeta^{-1}, \sqrt{-7})$ over $\mathbb{Q}$. The desired result follows.

237 Since 3 and 5 are relatively prime, $\mathbb{L} = \mathbb{Q}(\zeta_3, \zeta_5)$ is the cyclotomic extension of order 15 of $\mathbb{Q}$ in $\mathbb{C}$, namely, $\mathbb{Q}(\zeta_{15})$. The Galois group of this extension is $(\mathbb{Z}/15\mathbb{Z})^* \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, and so there are exactly three quadratic subextensions of $\mathbb{Q}(\zeta_{15})$ over $\mathbb{Q}$, corresponding to the three subgroups of index 2 in the Galois group.

Obviously, we have $\sqrt{-3} \in \mathbb{Q}(\zeta_3) \subseteq \mathbb{L}$. Setting $\alpha = \zeta_5 + \zeta_5^{-1}$, the subextension $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta_5) \cap \mathbb{R}$ is another quadratic extension of $\mathbb{Q}$. Moreover, one can immediately show that $\alpha^2 + \alpha - 1 = 0$, and so $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{5})$. We have thus found the quadratic subextensions $\mathbb{Q}(\sqrt{d}) \subseteq \mathbb{L}$ with $d = -3$ or $d = 5$ or $d = -15$.

Consider now the field $\mathbb{K}(p)$. First of all, remark that $\mathbb{K}(p)/\mathbb{Q}$ is a Galois extension, since $\mathbb{K}(p)$ is the splitting field of $(x^2 - p)(x^{15} - 1)$. If $p = 5$ then $\mathbb{K}(p) = \mathbb{L}$, so we already know the Galois group and the number of quadratic subextensions.

If $p \neq 5$, then the quadratic field $\mathbb{Q}(\sqrt{p})$ has intersection with $\mathbb{L}$ equal to $\mathbb{Q}$, so $[\mathbb{K}(p) : \mathbb{Q}] = 16$ and the Galois group of $\mathbb{K}(p)$ is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. Moreover, $\mathbb{K}(p)$ contains 7 quadratic subfields, which correspond to the subgroups of index 2 in $\mathrm{Gal}(\mathbb{K}(p)/\mathbb{Q})$, or equivalently in $2\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, which is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$.

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(i) Let ζ be a primitive seventh root of unity in $\mathbb{C}$ and let $\alpha = \sqrt[7]{2}$. We have $\mathbb{K} = \mathbb{Q}(\alpha, \zeta)$ because the roots of the polynomial $x^7 - 2$ are $\alpha\zeta^k$ for $0 \leq k \leq 6$. Moreover, $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \phi(7) = 6$ and $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \simeq (\mathbb{Z}/7\mathbb{Z})^*$, because $\mathbb{Q}(\zeta)$ is a cyclotomic extension, while $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 7$ because $x^7 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion. Since 6 and 7 are relatively prime, it follows that $[\mathbb{E} : \mathbb{Q}] = 42$.
An automorphism φ of $\mathbb{E}/\mathbb{Q}$ sends the root α to some root of $x^7 - 2$, that is, to $\alpha\zeta^h$ for some h that is well-defined modulo 7, and it sends ζ to a primitive seventh root of unity, that is, to ζ^k for some k that is well-defined modulo 7 and coprime to 7. The automorphism φ can thus be encoded by the pair (h, k); in

fact, it is easy to prove that this gives an isomorphism between $\mathrm{Gal}(\mathbb{E}/\mathbb{Q})$ and $\mathbb{Z}/7\mathbb{Z} \rtimes (\mathbb{Z}/7\mathbb{Z})^*$, where the semidirect product is with respect to the action $(\mathbb{Z}/7\mathbb{Z})^* \ni a \longmapsto (k \longmapsto ak) \in \mathrm{Aut}(\mathbb{Z}/7\mathbb{Z})$.

(ii) By the Fundamental Theorem of Galois Theory, a subfield $\mathbb{F}$ of $\mathbb{K}$ such that $[\mathbb{F} : \mathbb{Q}] = 3$ corresponds to a subgroup H of order 14 in the group $G = \mathrm{Gal}(\mathbb{E}/\mathbb{Q})$. In particular, by Cauchy's Theorem, H must contain a 7-Sylow subgroup H' of G. Letting n_7 be the number of 7-Sylow subgroups of G, we have $n_7 \equiv 1 \pmod 7$ and $n_7 \mid 6 = [G : H']$, so $n_7 = 1$, that is, H' is the unique 7-Sylow subgroup of G. But then, again by the Fundamental Theorem of Galois Theory, we have that $\mathbb{Q}(\zeta)$, whose degree over $\mathbb{Q}$ is 6, is the unique subfield of degree 6 and corresponds to H'; moreover, since H contains H', $\mathbb{F}$ must be contained in $\mathbb{Q}(\zeta)$.
The Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ is cyclic of order 6, so it contains a unique subgroup of index 3. But we know that the real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R} = \mathbb{Q}(\zeta + \zeta^{-1})$ has degree $6/2 = 3$ over $\mathbb{Q}$, and thus we conclude that $\mathbb{F} = \mathbb{Q}(\zeta + \zeta^{-1})$ is the unique subextension of $\mathbb{K}$ of degree 3 over $\mathbb{Q}$.

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(i) Letting $\mathbb{E} = \mathbb{Q}(\sqrt{6}, \sqrt{10})$, consider its subextensions $\mathbb{Q}(\sqrt{6})$ and $\mathbb{Q}(\sqrt{10})$. These are quadratic extensions whose intersection is $\mathbb{Q}$, as $6 \cdot 10$ is not a square in $\mathbb{Q}$; it follows that $\mathbb{E}$ has degree 4 over $\mathbb{Q}$ and that its Galois group is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.
We have that $\mathbb{K} = \mathbb{E}(\sqrt{a})$, so $\mathbb{E}/\mathbb{Q}$ has degree 1 if $\mathbb{Q}(\sqrt{a}) \subseteq \mathbb{E}$ and degree 2 otherwise. By the Fundamental Theorem of Galois Theory, we know that quadratic subextensions of $\mathbb{E}$ correspond to subgroups of index 2 in $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and so to subgroups of order 2. Since there are 3 such subgroups, the quadratic subextensions are: $\mathbb{Q}(\sqrt{6})$, $\mathbb{Q}(\sqrt{10})$ and $\mathbb{Q}(\sqrt{15})$.
We come to the conclusion that if $a = 6$ or $a = 10$ or $a = 15$ up to square factors, then the Galois group of $\mathbb{K}$ is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^2$; otherwise, it is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^3$.

(ii) If the polynomial $f(x) \in \mathbb{Q}[x]$ is irreducible and α is a root of $f(x)$ in $\mathbb{K}$, then $\mathbb{Q}(\alpha) \subseteq \mathbb{K}$ and so $\deg(f(x)) = [\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides $[\mathbb{K} : \mathbb{Q}]$. The possible degrees of $f(x)$ are thus 1, 2, 4 and 8, where the latter is a possibility only if $[\mathbb{K} : \mathbb{Q}] = 8$.
We now intend to show that for $d = 1, 2, 4, 8$ there exists an irreducible polynomial of degree d that has a root in $\mathbb{K}$. The Galois group of $\mathbb{K}/\mathbb{Q}$ is Abelian, so it has a subgroup of order given by each possible divisor of 8; it follows that for each $d = 1, 2, 4, 8$ there is a subextension of $\mathbb{K}$ of degree d over $\mathbb{Q}$. By the Primitive Element Theorem, each of these subextensions is generated over $\mathbb{Q}$ by some element of $\mathbb{K}$: the minimal polynomials of such elements provide the required examples.

(iii) Since $\mathbb{K}/\mathbb{Q}$ is normal, every polynomial that is irreducible over $\mathbb{Q}$ and has a root in $\mathbb{K}$ will split as a product of linear factors in $\mathbb{K}$. So the possible degrees are again 1, 2, 4 and 8.

240 Denote by n the degree of $\mathbb{K}$ over $\mathbb{Q}$; we consider several cases according to whether or not a is a square and whether or not b is a cube.

If a is a square and b is a cube then $\mathbb{K} = \mathbb{Q}$, $n = 1$ and there are no proper subextensions.

If a is a square and b is not a cube then $\mathbb{K} = \mathbb{Q}(\sqrt[3]{b})$ and $n = 3$, as $x^3 - b$ is irreducible in $\mathbb{Q}[x]$ because it has no roots in $\mathbb{Q}$. There are no proper subextensions because the degree is prime.

If a is not a square and b is a cube then $\mathbb{K} = \mathbb{Q}(\sqrt{a})$ and $n = 2$, as $x^2 - a$ is irreducible. Again, there are no proper subextensions.

Finally, assume a is not a square and b is not a cube. The extension $\mathbb{K}/\mathbb{Q}$ has the subextensions $\mathbb{Q}(\sqrt{a})$ of degree 2 and $\mathbb{Q}(\sqrt[3]{b})$ of degree 3. Since $\mathbb{K} = \mathbb{Q}(\sqrt[3]{b})(\sqrt{a})$, we know that $n \leq 6$; but, since 2 and 3 are relatively prime, 6 must divide n, and therefore $n = 6$.

Now let $\mathbb{L} \neq \mathbb{Q}$ be a proper subextension of $\mathbb{K}$ over $\mathbb{Q}$. Since $n = 6$, we have either $[\mathbb{L} : \mathbb{Q}] = 2$ or $[\mathbb{L} : \mathbb{Q}] = 3$. If $[\mathbb{L} : \mathbb{Q}] = 2$ then $\mathbb{L}$ must coincide with $\mathbb{Q}(\sqrt{a})$, or $\mathbb{Q}(\sqrt{a}) \cdot \mathbb{L}$ would be a subextension of degree 4, but 4 does not divide 6.

Suppose $[\mathbb{L} : \mathbb{Q}] = 3$; the polynomial $x^3 - b$ cannot be irreducible in $\mathbb{L}[x]$, or $\mathbb{L}(\sqrt[3]{b})$ would be a subextension of $\mathbb{K}$ whose degree is 9. Thus, letting ζ be a primitive third root of unity in $\mathbb{C}$, one of the three roots $\sqrt[3]{b}$, $\sqrt[3]{b}\zeta$, $\sqrt[3]{b}\zeta^2$ must belong to $\mathbb{L}$; in fact, $\mathbb{L}$ is obtained by extending $\mathbb{Q}$ with this root.

If $\mathbb{L}$ is not $\mathbb{Q}(\sqrt[3]{b})$, then $\sqrt[3]{b}\zeta$ or $\sqrt[3]{b}\zeta^2 = \sqrt[3]{b}\zeta^{-1}$ belong to $\mathbb{K}$; since $\sqrt[3]{b} \in \mathbb{K}$, this implies that ζ is an element of $\mathbb{K}$; consequently, $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{-3})$ is a quadratic subextension of $\mathbb{K}$. But we have already shown that $\mathbb{Q}(\sqrt{a})$ is the only quadratic subextension of $\mathbb{K}$; we can thus conclude that $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{a})$, hence $a = -3a_1^2$, where $a_1 \in \mathbb{Z}$.

In conclusion, if $a = -3$ up to square factors, then we have three subextensions of degree 3: $\mathbb{Q}(\sqrt[3]{b})$, $\mathbb{Q}(\zeta\sqrt[3]{b})$ and $\mathbb{Q}(\zeta^2\sqrt[3]{b})$. Otherwise, $\mathbb{Q}(\sqrt[3]{b})$ is the only subextension of $\mathbb{K}$ whose degree is 3.

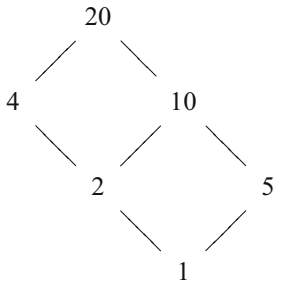
241

(i) Let $\mathbb{L} = \mathbb{Q}(\sqrt[3]{7}, \zeta)$, where $\zeta \in \mathbb{C}$ is a primitive third root of unity, and let $\mathbb{K} = \mathbb{Q}(\sqrt{3})$; we have that $\mathbb{F} = \mathbb{L} \cdot \mathbb{K}$. Clearly, $\mathbb{L}$ is the splitting field of $x^3 - 7$ and its Galois group is isomorphic to S_3, whereas $\mathbb{K}$ is the splitting field of $x^2 - 3$, with Galois group isomorphic to $\mathbb{Z}/2\mathbb{Z}$.
Moreover, $\mathbb{F}$ can be obtained as the composite field of the extensions $\mathbb{Q}(\sqrt[3]{7}, \sqrt{3})$ and $\mathbb{Q}(\sqrt{-3})$, as $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{-3})$. Since the first of the two extensions is real of degree 6 and the second is not real and has degree 2, the degree of $\mathbb{F}$ over $\mathbb{Q}$ must be 12. This implies that $\mathbb{L} \cap \mathbb{K} = \mathbb{Q}$ and thus, since $\mathbb{L}$ and $\mathbb{K}$ are normal extensions that generate $\mathbb{F}$, the Galois group of $\mathbb{F}/\mathbb{Q}$ is isomorphic to the direct product $\mathrm{Gal}(\mathbb{L}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$, that is, to $G = \mathsf{S}_3 \times \mathbb{Z}/2\mathbb{Z}$.

(ii) By the Fundamental Theorem of Galois Theory, non-normal subextensions of $\mathbb{F}$ over $\mathbb{Q}$ correspond to non-normal subgroups of $\mathsf{S}_3 \times \mathbb{Z}/2\mathbb{Z}$. We now wish to find all such subgroups.
A subgroup of G that is not normal has order 2, 3 or 4.

The elements of order 2 in G are $(0, 1)$ and the elements of the form $(\tau, 0)$ or $(\tau, 1)$, where τ is a transposition of S_3. The element $(0, 1)$ generates the subgroup $e \times \mathbb{Z}/2\mathbb{Z} = Z(G)$, whereas all other elements generate subgroups that are not normal, as the transpositions do not belong to the centre of S_3.
No element of G has order 4, so all subgroups of order 4 are isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, that is, they are generated by two elements of order 2 that commute with each other. At least one of these two elements must have a transposition τ as its first coordinate; if both elements have a transposition as their first coordinate, it must be the same transposition τ, because different transpositions do not commute in S_3. Moreover, at least one of the two elements must have 1 as its second coordinate. This leads us to conclude that there are 3 subgroups of order 4 in G, namely, $\langle(\tau, 0), (0, 1)\rangle$ where τ is a transposition of S_3. None of them is normal.
There is a unique subgroup of G whose order is 3, which is generated by $((1, 2, 3), 0)$ and is therefore normal.
We have found 6 non-normal subgroups of order 2, and 3 non-normal subgroups of order 4.
By the Fundamental Theorem of Galois Theory, there are 6 non-normal subextensions of degree 6 and 3 non-normal subextensions of degree 3. We now describe these subextensions.
For the subextensions of degree 3, consider $\mathbb{Q}(\sqrt[3]{7}\zeta^u)$, with $u = 0, 1, 2$. Clearly, these subextensions have degree 3 over $\mathbb{Q}$ and are not normal, as they each contain only one root of the polynomial $x^3 - 7$.
For $u = 0, 1, 2$, let $\mathbb{E}_u$ be the subextension $\mathbb{Q}(\sqrt[3]{7}\zeta^u, \sqrt{3})$. These are the composite fields of $\mathbb{Q}(\sqrt[3]{7}\zeta^u)$ and $\mathbb{Q}(\sqrt{3})$, that is, of extensions of degree 3 and 2, so each $\mathbb{E}_u$ has degree 6 over $\mathbb{Q}$. Moreover, $\mathbb{E}_u$ is not normal over $\mathbb{Q}$: if it were, it would contain all roots of $x^3 - 7$ because it contains at least one; it would then contain all roots of $(x^3 - 7)(x^2 - 3)$ because it contains the roots of $x^2 - 3$, and we would therefore have $\mathbb{E}_u = \mathbb{F}$, but we know that $\mathbb{F}$ has degree 12 over $\mathbb{Q}$.
For $u = 0, 1, 2$, let $\mathbb{E}'_u = \mathbb{Q}(\sqrt[3]{7}\zeta^u, i)$. Similarly to the previous case, $\mathbb{E}'_u$ is the composite field of the two extensions $\mathbb{Q}(\sqrt[3]{7}\zeta^u)$ and $\mathbb{Q}(i)$, whose degrees are 3 and 2, respectively, and whose intersection is $\mathbb{Q}$; it follows that $\mathbb{E}'_u$ has degree 6 over $\mathbb{Q}$. Moreover, it is not a normal extension of $\mathbb{Q}$, as it contains one root of $x^3 - 7$ and, if it were normal, would also contain the root $\sqrt[3]{7}\zeta^{u+1}$; it would therefore contain ζ and finally, since it does contain i, it would have to contain $\sqrt{3}$; we would thus have $\mathbb{E}'_u = \mathbb{F}$, but $\mathbb{F}$ has degree 12, contradiction.

242 Let $\mathbb{K} = \mathbb{F}_q$, where q is some prime power. Since $\alpha\beta \in \mathbb{K}(\alpha, \beta) = \mathbb{F}_{q^{20}}$, the smallest k such that $\alpha\beta \in \mathbb{F}_{q^k}$ is a divisor of 20; therefore, it is a divisor of 4 or a multiple of 5.



In the first case, $\alpha\beta \in \mathbb{F}_{q^4} = \mathbb{K}(\beta)$, hence $\alpha = \alpha\beta/\beta \in \mathbb{K}(\beta)$, which is not possible because 5 does not divide 4. In the second case, we obtain that $\mathbb{K}(\alpha) = \mathbb{F}_{q^5} \subseteq \mathbb{K}(\alpha\beta)$, so $\beta = \alpha\beta/\alpha \in \mathbb{K}(\alpha\beta)$. The latter inclusion implies that 4 must divide k, that is, $k = 20$, as wanted.

243 The field $\mathbb{K} = \mathbb{Q}(\zeta)$ is the cyclotomic extension of order 25 of $\mathbb{Q}$ in $\mathbb{C}$, and its Galois group G is isomorphic to $(\mathbb{Z}/25\mathbb{Z})^*$, that is, to $\mathbb{Z}/20\mathbb{Z}$. The elements of the Galois group can be represented as $\varphi_h : \zeta \longmapsto \zeta^h$, where $0 < h < 25$ is coprime to 5.

(i) Set $\mathbb{F} = \mathbb{Q}(\alpha)$ and remark that, by the Fundamental Theorem of Galois Theory, $[\mathbb{F} : \mathbb{Q}]$ is the index of the subgroup H of elements of G that fix every point of $\mathbb{F}$, that is, the index of the subgroup of elements that fix the generator α of $\mathbb{F}$. We have

$$\begin{aligned}\varphi_7(\alpha) &= \varphi_7(\zeta^7) + \varphi_7(\zeta) + \varphi_7(\zeta^{-1}) + \varphi_7(\zeta^{-7}) \\ &= \zeta^{49} + \zeta^7 + \zeta^{-7} + \zeta^{-49} \\ &= \zeta^{-1} + \zeta^7 + \zeta^{-7} + \zeta \\ &= \alpha\end{aligned}$$

and so $\varphi_7 \in H$. Now, the order of φ_7 in G is given by the order of 7 in $(\mathbb{Z}/25\mathbb{Z})^*$, that is, 4; it follows that $[\mathbb{F} : \mathbb{Q}]$ is a divisor of 5. If we had $[\mathbb{F} : \mathbb{Q}] = 1$ then we would have $\alpha \in \mathbb{Q}$, so the polynomial

$$x^{14} + x^8 + x^6 + 1 - \alpha x^7$$

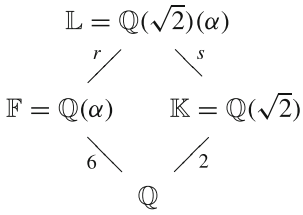
would have rational coefficients. But this is not possible, as this polynomial has degree 14 and has ζ among its roots:

$$\zeta^{14} + \zeta^8 + \zeta^6 + 1 - \alpha\zeta^7 = \zeta^7(\zeta^7 + \zeta + \zeta^{-1} + \zeta^{-7} - \alpha) = 0,$$

which contradicts the fact that $[\mathbb{K} : \mathbb{Q}] = 20$. We have thus shown that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 5$.

(ii) The Galois group of $\mathbb{K}$ is cyclic of order 20, so each divisor of 20 corresponds to a unique subextension. Clearly, the extensions of degree 20 and 1 are $\mathbb{K}$ and $\mathbb{Q}$, respectively, and the extension of degree 5 is $\mathbb{Q}(\alpha)$ by (i).
The real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R} = \mathbb{Q}(\zeta + \zeta^{-1})$ has degree $\phi(25)/2 = 10$. Moreover, ζ^5 is a primitive fifth root of unity, so $\mathbb{Q}(\zeta^5)$ has degree $\phi(5) = 4$ and its real subextension $\mathbb{Q}(\zeta^5) \cap \mathbb{R} = \mathbb{Q}(\zeta^5 + \zeta^{-5})$ has degree $\phi(5)/2 = 2$ over $\mathbb{Q}$. We have thus listed all of the subextensions of $\mathbb{K}$.

244 Let α be a root of $f(x)$ in $\mathbb{C}$, let $\mathbb{F} = \mathbb{Q}(\alpha)$ and $\mathbb{K} = \mathbb{Q}(\sqrt{2})$.



Remark that $[\mathbb{F} : \mathbb{Q}] = 6$, as $f(x)$ is irreducible. Consider the composite field $\mathbb{L} = \mathbb{F} \cdot \mathbb{K} = \mathbb{Q}(\sqrt{2})(\alpha)$; the degree $[\mathbb{L} : \mathbb{K}]$ is the degree of the irreducible factor of $f(x)$ over $\mathbb{Q}(\sqrt{2})$ that has α as a root. Setting $r = [\mathbb{L} : \mathbb{F}]$ and $s = [\mathbb{L} : \mathbb{K}]$, we have two possibilities.

(1) If $\sqrt{2} \notin \mathbb{F}$ then $r = 2$, $s = 6$, and $f(x)$ remains irreducible over $\mathbb{Q}(\sqrt{2})$. We now show that the polynomial $f(x) = x^6 + x^5 + \cdots + x + 1$ does fall into this case. Since $f(x)$ is the seventh cyclotomic polynomial over $\mathbb{Q}$, it is irreducible; $\mathbb{F}$ is the seventh cyclotomic field. If we had $\sqrt{2} \in \mathbb{F}$ then $\mathbb{K}$ would be a subextension of $\mathbb{F} \cap \mathbb{R}$, which, being the one real (nontrivial) subextension of $\mathbb{F}$, has degree $\phi(7)/2 = 3$ over $\mathbb{Q}$; but this yields a contradiction, as $2 = [\mathbb{K} : \mathbb{Q}]$ does not divide 3.
(2) If $\sqrt{2} \in \mathbb{F}$ then $r = 1$, $s = 3$, so the minimal polynomial of α over $\mathbb{K}$ has degree 3. The same holds for each root of $f(x)$, so $f(x)$ is the product of two irreducible factors in $\mathbb{K}[x]$. This is the case, for example, for $f(x) = x^6 - 2$, which is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion. Obviously, over $\mathbb{Q}(\sqrt{2})$ we have $f(x) = (x^3 - \sqrt{2})(x^3 + \sqrt{2})$, and the two factors are irreducible by our discussion above.

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(i) In order to show that $\mathbb{Q}(\alpha)$ has degree 2 over $\mathbb{Q}$, it is enough to prove that α is a root of an irreducible polynomial of degree 2 in $\mathbb{Q}[x]$.
Indeed, developing the square of α and using the fact that $\zeta_7^6 + \cdots + \zeta_7^2 + \zeta_7 + 1 = 0$, one finds immediately that the polynomial $x^2 + x + 2$, which is irreducible over $\mathbb{Q}$ because it has no rational roots, vanishes at α.

(ii) First of all, we show that $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\zeta_5) = \mathbb{Q}$. Thanks to (i), we know that $\mathbb{Q}(\alpha)$ has degree 2 over $\mathbb{Q}$, so the intersection can only be $\mathbb{Q}$ or $\mathbb{Q}(\alpha)$.
The Galois group of the fifth cyclotomic field $\mathbb{Q}(\zeta_5)$ is isomorphic to $(\mathbb{Z}/5\mathbb{Z})^*$, that is, $\mathbb{Z}/4\mathbb{Z}$; it therefore has a unique subgroup of index 2. By the Fundamental Theorem of Galois Theory, there exists only one subfield of $\mathbb{Q}(\zeta_5)$ whose degree is 2 over $\mathbb{Q}$, and this subfield is the real subextension $\mathbb{Q}(\zeta_5) \cap \mathbb{R}$, whose degree is $\phi(5)/2 = 2$.
Thus, if we had $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\zeta_5) \neq \mathbb{Q}$, then the field $\mathbb{Q}(\alpha)$ would be real; but this is not true, as $x^2 + x + 2$ has negative discriminant. This shows that $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\zeta_5) = \mathbb{Q}$.
It follows that $\mathbb{K} = \mathbb{Q}(\alpha, \zeta_5)$ is the composite field of two normal subextensions $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\zeta_5)$ whose intersection is $\mathbb{Q}$ and whose Galois groups are isomorphic to $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$, respectively. We therefore have $\mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$.

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(i) Let r be the order of σ in $\mathrm{Gal}(\mathbb{E}/\mathbb{Q}(\zeta))$. We have $\sigma(\zeta) = \zeta$, so $\alpha = \sigma^r(\alpha) = \zeta^r\alpha$; this proves that n divides r. It follows that n divides the order of $\mathrm{Gal}(\mathbb{E}/\mathbb{Q}(\zeta_n))$, that is, $[\mathbb{E} : \mathbb{Q}(\zeta)]$.

(ii) We have $\sigma(\alpha^n) = \sigma(\alpha)^n = (\zeta\alpha)^n = \alpha^n$, and so α^n is fixed by σ. It follows that α^n is an element of the subfield $\mathbb{F}$ of $\mathbb{E}$ that is fixed by σ, that is, the subfield that corresponds to the cyclic subgroup generated by σ. We cannot have $\mathbb{F} = \mathbb{E}$ because we know that σ does not fix α.

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(i) Letting $\mathbb{E} = \mathbb{Q}(\alpha_n) \cap \mathbb{Q}(\zeta_5)$, notice that $\mathbb{E} \subseteq \mathbb{Q}(\alpha_n) \subseteq \mathbb{R}$, so $\mathbb{E}$ is contained in the real subextension $\mathbb{Q}(\zeta_5) \cap \mathbb{R} = \mathbb{Q}(\zeta_5 + \zeta_5^{-1})$, which is the same as $\mathbb{Q}(\sqrt{5})$ because $\zeta_5 + \zeta_5^{-1}$ is a root of the polynomial $x^2 + x - 1$.
Moreover, it is clear that $[\mathbb{Q}(\alpha_n) : \mathbb{Q}] = n$, as $x^n - 5$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion. It follows that if n is odd then $\mathbb{E} = \mathbb{Q}$, whereas if n is even then $\sqrt{5} = (\alpha_n)^{n/2}$ and thus $\mathbb{E} = \mathbb{Q}(\sqrt{5})$.
Using the fact that $\mathbb{Q}(\alpha_n)/\mathbb{E}$ and $\mathbb{Q}(\sqrt{5})/\mathbb{E}$ are Galois extensions, we obtain that $[\mathbb{Q}(\alpha_n, \zeta_5) : \mathbb{Q}] = [\mathbb{Q}(\alpha_n) : \mathbb{Q}][\mathbb{Q}(\zeta_5) : \mathbb{Q}]/[\mathbb{E} : \mathbb{Q}]$; hence, by our argument above, the degree is $4n$ if n is odd and $2n$ if n is even.

(ii) Suppose $\alpha_n \in \mathbb{Q}(\zeta_m)$. The extension $\mathbb{Q}(\zeta_m)$ is cyclotomic, so its Galois group is Abelian and all of its subextensions are normal. In particular, $\mathbb{Q}(\alpha_n)/\mathbb{Q}$ must be normal; since it contains one root of $x^n - 5$, it must also contain the splitting field of the polynomial. However, for $n > 2$ the splitting field of $x^n - 5$ is not real, whereas $\mathbb{Q}(\alpha_n)$ is a real extension; this implies that we necessarily have $n = 1$ or $n = 2$.
For $n = 1$, we clearly have $\alpha_1 = 5 \in \mathbb{Q}(\zeta_m)$ for all $m \geq 1$. Assume $n = 2$. We have already observed that $\sqrt{5} \in \mathbb{Q}(\zeta_5)$; moreover, since $\mathbb{Q}(\zeta_5) \subseteq \mathbb{Q}(\zeta_{5m})$ for all $m \geq 1$, we obtain that $\alpha_2 = \sqrt{5} \in \mathbb{Q}(\zeta_{5m})$.
Conversely, if m and 5 are relatively prime, then we have $[\mathbb{Q}(\zeta_{5m}) : \mathbb{Q}] = \phi(5m) = 4\phi(m) = [\mathbb{Q}(\zeta_5) : \mathbb{Q}][\mathbb{Q}(\zeta_m) : \mathbb{Q}]$, so the intersection of $\mathbb{Q}(\zeta_5)$ and $\mathbb{Q}(\zeta_m)$ is $\mathbb{Q}$; in particular, $\sqrt{5} \notin \mathbb{Q}(\zeta_m)$.
In conclusion, the pairs that satisfy the desired condition are those of the form $(1, m)$ or $(2, 5m)$, where $m \geq 1$.

248 We show that $[\mathbb{L} : \mathbb{K}] = 5$ is indeed possible. Consider $f(x) = x^5 - 2$, which is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion. We know that, letting $\alpha = \sqrt[5]{2}$ and denoting by ζ a primitive fifth root of unity in $\mathbb{C}$, the roots of $f(x)$ are $\alpha\zeta^k$ for $k = 0, 1, 2, 3, 4$.

The field $\mathbb{L}$ is the composite field of $\mathbb{Q}(\alpha)$, which has degree 5 over $\mathbb{Q}$ because $f(x)$ is irreducible, and $\mathbb{Q}(\zeta)$, which has degree $4 = \phi(5)$ over $\mathbb{Q}$. It follows that $[\mathbb{L} : \mathbb{Q}] = 20$, hence $[\mathbb{L} : \mathbb{Q}(\zeta)] = 5$; in particular, $f(x)$ remains irreducible over $\mathbb{Q}(\zeta)$, as adding its root α to $\mathbb{Q}(\zeta)$ yields an extension of degree 5. We can thus take $\mathbb{K} = \mathbb{Q}(\zeta)$ to obtain the required example.

We now show that $[\mathbb{L} : \mathbb{K}] = 10$ is also possible. Let $f(x)$, α and ζ be defined as above, and consider $\mathbb{K} = \mathbb{Q}(\zeta + \zeta^{-1})$, that is, the real subextension of $\mathbb{Q}(\zeta)$ of

degree 2 over $\mathbb{Q}$. We have $[\mathbb{L} : \mathbb{K}] = 10$. Moreover, the degrees of $\mathbb{Q}(\alpha)$ and $\mathbb{K}$ over $\mathbb{Q}$ are relatively prime, so $\mathbb{K}(\alpha)$ has degree 5 over $\mathbb{K}$. This implies that $f(x)$ is irreducible over $\mathbb{K}$.

Finally, we prove that $[\mathbb{L} : \mathbb{K}] = 15$ is not possible. Indeed, if we had $[\mathbb{L} : \mathbb{K}] = 15$ then the Galois group of $\mathbb{L}/\mathbb{K}$ would be a group of order 15 and thus be cyclic. Such a group, however, would also have to be a subgroup of S_5, which has no elements of order 15.

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(i) Let $\mathbb{F}$ be the splitting field of $f(x)$ over $\mathbb{K}$; note that $\mathbb{F}$ is a Galois extension and that $\mathbb{L}$ is a subextension of $\mathbb{F}$. Any homomorphism of $\mathbb{L}$ to an algebraic closure of $\mathbb{K}$ will extend to $\mathbb{F}$, so it permutes the roots $\alpha_1,\ \alpha_2,\ \alpha_3,\ \alpha_4$. But then it also permutes the elements β, γ, δ because of their expressions in terms of $\alpha_1,\ \alpha_2,\ \alpha_3,\ \alpha_4$. This proves that $\mathbb{L}$ is a normal extension of $\mathbb{K}$.

(ii) Consider the polynomial $g(x) = (x - \beta)(x - \gamma)(x - \delta)$, whose coefficients are clearly in $\mathbb{L}$, and in fact in $\mathbb{K}$, since the elements of the Galois group of $\mathbb{L}/\mathbb{K}$ permute the roots of $g(x)$, so they fix its coefficients. The result follows.

(iii) If $\mathbb{K}$ is a finite field then $[\mathbb{F} : \mathbb{K}] = 4$ and the Galois group of $\mathbb{F}/\mathbb{K}$ is cyclic of order 4. Up to relabelling the roots of the polynomial $f(x)$, which are permuted by the elements of G, we can identify the generator of G with the cycle $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$, since in S_4 the 4-cycles are the only elements of order 4.

Moreover, remark that $\mathbb{L}$ can only have degree 1 or 2 over $\mathbb{K}$, because $\mathbb{L}$ is a subextension of $\mathbb{F}$, which has degree 4, and, by the argument above, its degree divides 3!.

Assume by contradiction that $[\mathbb{L} : \mathbb{K}] = 1$. The elements β, γ, δ are then fixed by the Galois group of $\mathbb{F}/\mathbb{K}$. However, one can easily check that the cycle $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ sends β to δ; the only possibility is therefore $\beta = \delta$, hence

$$\begin{aligned} \alpha_1\alpha_2 + \alpha_3\alpha_4 &= \alpha_1\alpha_4 + \alpha_2\alpha_3 \\ \alpha_1(\alpha_2 - \alpha_4) &= \alpha_3(\alpha_2 - \alpha_4) \\ (\alpha_1 - \alpha_3)(\alpha_2 - \alpha_4) &= 0. \end{aligned}$$

The last equality implies that $\alpha_1 = \alpha_3$ or $\alpha_2 = \alpha_4$. Neither is possible: an irreducible polynomial over a finite field cannot have multiple roots in an algebraic closure. We have thus shown that $[\mathbb{L} : \mathbb{K}] = 2$.

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(i) The field $\mathbb{L}$ is the 24th cyclotomic field, so the Galois group of $\mathbb{L}/\mathbb{Q}$ is isomorphic to $(\mathbb{Z}/24\mathbb{Z})^*$, that is, to $G = (\mathbb{Z}/2\mathbb{Z})^3$. By the Fundamental Theorem of Galois Theory, the subextensions of degree 4 and 2 over $\mathbb{Q}$ correspond to subgroups of G of index 4 and 2, respectively, that is, to subgroups of order 2 and 4.

Note that obviously every element of $G \setminus \{(0, 0, 0)\}$ has order 2 in G. Any subgroup of order 2 contains the neutral element and one element of order 2;

there are thus 7 subgroups of order 2, as many as the elements of order 2 of G. The subgroups of order 4 are all isomorphic to $(\mathbb{Z}/2\mathbb{Z})^2$, so we can enumerate them as follows: we choose an element g of order 2, which we can do in 2^3-1 ways, then a second element h of order 2 that is different from g, which can be done in 2^3-2 ways; finally, we divide the product $7\cdot 6$ by $3\cdot 2$, that is, by the number of possible ordered pairs of generators for a group isomorphic to $(\mathbb{Z}/2\mathbb{Z})^2$. We find that there are 7 subgroups of order 4.
In conclusion, $\mathbb{L}$ has 7 subextensions of degree 4 over $\mathbb{Q}$ and 7 subextensions of degree 2 over $\mathbb{Q}$.

(ii) Since the group G is Abelian, every subextension of $\mathbb{L}$ is normal. Letting $\mathbb{F}$ be a subextension of degree 4, the Fundamental Theorem of Galois Theory implies that $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ is isomorphic to a quotient of G of order 4. Such a quotient can only be isomorphic to $(\mathbb{Z}/2\mathbb{Z})^2$, as every element of G has order at most 2.

(iii) The field $\mathbb{L}$ contains ζ^6, which is a primitive fourth root of unity, ζ^8, which is a primitive third root of unity, and ζ^3, which is a primitive eighth root of unity. It follows that $\mathbb{L}$ also contains $\alpha_1=\sqrt{-1}$, $\alpha_2=\sqrt{2}$ and $\alpha_3=\sqrt{3}$, and in fact $\mathbb{L}=\mathbb{Q}(\alpha_1,\alpha_2,\alpha_3)$, as it is easy to check.
The minimal polynomials of α_1, α_2 and α_3 are x^2+1, x^2-2 and x^3-3, respectively. The automorphisms in $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ are therefore extensions of

$$\varphi_{(h_1,h_2,h_3)}(\alpha_i)=(-1)^{h_i}\alpha_i,\quad \text{for } i=1,2,3,$$

where (h_1,h_2,h_3) ranges in $G=(\mathbb{Z}/2\mathbb{Z})^3$. The 7 subextensions of degree 4 over $\mathbb{Q}$ are thus the subfields fixed by each of the 7 automorphisms $\varphi_{(h_1,h_2,h_3)}$ such that $(h_1,h_2,h_3)\neq(0,0,0)$. These subextensions are

$$\mathbb{Q}(\sqrt{-1},\sqrt{2}),\ \mathbb{Q}(\sqrt{-1},\sqrt{3}),\ \mathbb{Q}(\sqrt{-1},\sqrt{6}),\ \mathbb{Q}(\sqrt{2},\sqrt{3}),$$
$$\mathbb{Q}(\sqrt{2},\sqrt{-3}),\ \mathbb{Q}(\sqrt{-2},\sqrt{3}),\ \mathbb{Q}(\sqrt{-2},\sqrt{-3}).$$

Each of them is such that $\mathbb{Q}(\sqrt{a},\sqrt{b})=\mathbb{Q}(\sqrt{a}+\sqrt{b})$; indeed, one containment is clear and the other one is due to the fact that $\sqrt{a}+\sqrt{b}$ has degree 4 over $\mathbb{Q}$ for all a and b as in the above list.

251 The polynomial $f(x)=x^4+5x+5$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion; we will now determine its splitting field and Galois group over $\mathbb{Q}$. Setting

$$\alpha=\sqrt{\frac{-5+\sqrt{5}}{2}}\quad\text{and}\quad\beta=\sqrt{\frac{-5-\sqrt{5}}{2}},$$

the roots of $f(x)=x^4+5x^2+5$ are $\pm\alpha,\pm\beta$. Note that $\sqrt{5}\in\mathbb{Q}(\alpha)$ and $\alpha\beta=\sqrt{5}$, so $\beta\in\mathbb{Q}(\alpha)$; thus, the splitting field $\mathbb{K}$ of $f(x)$ is $\mathbb{Q}(\alpha)$ and the Galois group $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ has four elements.

Let φ be the unique automorphism of $\mathbb{K}$ that extends the mapping $\alpha \longmapsto \beta$. Since $\sqrt{5} = 2\alpha^2 + 5$ and $-\sqrt{5} = 2\beta^2 + 5$, we have $\varphi(\sqrt{5}) = -\sqrt{5}$, hence $\varphi(\beta) = \varphi(\sqrt{5}/\alpha) = -\sqrt{5}/\beta = -\alpha$. It follows that φ acts on the roots of $f(x)$ as the 4-cycle $(\alpha, \beta, -\alpha, -\beta)$, so $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $\mathbb{Z}/4\mathbb{Z}$, and in particular $\mathbb{Q}(\sqrt{5})$ is the only quadratic subextension of $\mathbb{K}$.

We can therefore conclude that the splitting field of $f(x)(x^2 - a)$ is $\mathbb{K}$ and that the Galois group is isomorphic to $\mathbb{Z}/4\mathbb{Z}$ provided that $a = 5b^2$ or $a = b^2$ with $b \in \mathbb{N}$. If a is not of this form, then $\mathbb{F} = \mathbb{K}(\sqrt{a}) = \mathbb{Q}(\alpha, \sqrt{a})$. In this case, the two subextensions $\mathbb{K}$ and $\mathbb{Q}(\sqrt{a})$ are both normal and have intersection equal to $\mathbb{Q}$, so the Galois group of $\mathbb{F}$ over $\mathbb{Q}$ is isomorphic to $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

252 Let $\alpha = \sqrt[6]{2}$ and let ζ be a primitive sixth root of unity in $\mathbb{C}$. The roots of $f(x) = x^6 - 2$ are $\alpha\zeta^k$ for $k = 0, 1, 2, 3, 4, 5$, and its splitting field is clearly $\mathbb{K} = \mathbb{Q}(\alpha, \zeta)$. The subextension $\mathbb{Q}(\alpha)$ is real, has degree 6 over $\mathbb{Q}$ (because $f(x)$ is irreducible by Eisenstein's Criterion) and is not normal because it only contains the roots $\pm\alpha$ among those of $f(x)$. As for $\mathbb{Q}(\zeta)$, it is the cyclotomic extension of $\mathbb{Q}$ of order 6 and has degree $\phi(6) = 2$ over $\mathbb{Q}$. In particular, $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\zeta) = \mathbb{Q}$, so $[\mathbb{K} : \mathbb{Q}] = 12$.

The field $\mathbb{K}$ is normal over $\mathbb{Q}$, so the subextensions $\mathbb{K}/\mathbb{Q}(\alpha)$ and $\mathbb{K}/\mathbb{Q}(\zeta)$ are also normal. It follows that for the two subgroups $H_1 = \mathrm{Gal}(\mathbb{K}/\mathbb{Q}(\zeta))$ and $H_2 = \mathrm{Gal}(\mathbb{K}/\mathbb{Q}(\alpha))$ of $G = \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ we have: $H_1 \cdot H_2 = G$, because $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\zeta) = \mathbb{Q}$; $H_1 \cap H_2 = \{e\}$, because $|H_1| \cdot |H_2| = 6 \cdot 2 = 12 = |G|$; and H_1 is normal in G. In conclusion, $G \simeq H_1 \rtimes H_2$, and this semidirect product is not a direct product, as G contains the non-normal subgroup H_2 and is therefore not Abelian.

Now, remark that the mapping $\zeta \longmapsto \zeta, \alpha \longmapsto \alpha\zeta$ can be extended in a unique way to an automorphism $\varphi \in H_1$, and that $H_1 = \langle\varphi\rangle \simeq \mathbb{Z}/6\mathbb{Z}$. In conclusion, G is isomorphic to the dihedral group D_6, because there is only one nontrivial homomorphism from $H_2 \simeq \mathbb{Z}/2\mathbb{Z}$ to $\mathrm{Aut}(H_1) \simeq \mathrm{Aut}(\mathbb{Z}/6\mathbb{Z}) \simeq \mathbb{Z}/2\mathbb{Z}$.

A subextension of $\mathbb{F}$ of degree 2 over $\mathbb{Q}$ corresponds to a subgroup of order 6 in D_6. Moreover, it is easy to show that D_6 has exactly three subgroups of order 6, namely, the group of rotations and two subgroups isomorphic to D_3.

The field $\mathbb{K}$ contains the quadratic subextensions $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{2}) = \mathbb{Q}(\alpha^3)$ and $\mathbb{Q}(\sqrt{-6})$; by the argument above, these three are the only quadratic subextensions.

253 The roots of $f(x) = x^4 - 2$ are $\sqrt[4]{2}i^k$ for $k = 0, 1, 2, 3$. Its splitting field is thus $\mathbb{K} = \mathbb{Q}(\sqrt[4]{2}, i)$ and contains the real subextension $\mathbb{Q}(\sqrt[4]{2})$, which is not normal, and the non-real normal subextension $\mathbb{Q}(i)$ of degree 2. It follows that $[\mathbb{K} : \mathbb{Q}] = 8$ and that the Galois group $G = \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to D_4, as it is a subgroup of order 8, and thus a 2-Sylow subgroup, of the group S_4.

Observe that $\mathbb{K}$ has three quadratic subextensions over $\mathbb{Q}$, namely, $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$. Moreover, using the habitual presentation of D_4 as generated by r and s, satisfying the relations $r^4 = s^2 = e$ and $sr = r^{-1}s$, one immediately sees that D_4 has three subgroups of index 2, namely, $\langle r\rangle$, $\langle r^2, s\rangle$ and $\langle r^2, rs\rangle$. By the Fundamental Theorem of Galois Theory, it follows that the ones listed are the only quadratic subextensions of $\mathbb{K}/\mathbb{Q}$.

In particular, since all elements of order 4 in D_4 are contained in $\langle r\rangle$, the latter is the only subgroup of index 2 that is isomorphic to $\mathbb{Z}/4\mathbb{Z}$; the other two are isomorphic to $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$. Finally, up to changing r into r^{-1}, the unique automorphism that extends the mapping $\sqrt[4]{2}\longmapsto i\sqrt[4]{2}$, $i\longmapsto i$ corresponds to r, as its order in G is 4. It follows that the subextension $\mathbb{Q}(i)$, which is fixed by this automorphism, corresponds to the subgroup $\langle r\rangle$ of D_4.

We will now consider the problem for all possible values of a. If $a\in\mathbb{N}^2$, that is, if a is the square of an integer, then $\mathbb{Q}(\sqrt{a})=\mathbb{Q}$ and the Galois group is clearly D_4.

If $a\in -\mathbb{N}^2, 2\mathbb{N}^2, -2\mathbb{N}^2$, then $\mathbb{Q}\subsetneqq\mathbb{Q}(\sqrt{a})\subseteq\mathbb{K}$ and the Galois group has order 4. More specifically, it is isomorphic to $\mathbb{Z}/4\mathbb{Z}$ if $a=-\mathbb{N}^2$, since as discussed $\mathbb{Q}(i)$ corresponds to $\langle r\rangle\simeq\mathbb{Z}/4\mathbb{Z}$, and to $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$ in the other two cases.

Finally, if a is not an element of $\pm\mathbb{N}^2$ or $\pm 2\mathbb{N}^2$, then $\mathbb{Q}(\sqrt{a})\not\subseteq\mathbb{K}$ and so $\mathbb{Q}(\sqrt{a})\cap\mathbb{K}=\mathbb{Q}$. Letting $\mathbb{F}=\mathbb{K}(\sqrt{a})$ denote the composite field of the two extensions $\mathbb{Q}(\sqrt{a})$ and $\mathbb{K}$, we have that $\mathbb{F}$ is the splitting field of the polynomial x^4-2 over $\mathbb{Q}(\sqrt{a})$ and that $\mathbb{F}/\mathbb{Q}$ is a normal extension of $\mathbb{Q}$ whose Galois group is isomorphic to $\mathbb{Z}/2\mathbb{Z}\times\mathsf{D}_4$. Since $\mathbb{Q}(\sqrt{a})$ is fixed by $0\times\mathsf{D}_4$, the Galois group is isomorphic to D_4 in this case, by the Fundamental Theorem of Galois Theory.

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(i) We have $\mathbb{K}_5=\mathbb{Q}(\sqrt[5]{p},\zeta_5)$, $\mathbb{K}_7=\mathbb{Q}(\sqrt[7]{p},\zeta_7)$, $\mathbb{K}_{35}=\mathbb{Q}(\sqrt[35]{p},\zeta_{35})$, where for $m=5,\ 7,\ 35$ we let $\zeta_m=e^{2\pi/m}$ be a primitive mth root of unity in $\mathbb{C}$. Since

$$(\sqrt[35]{p})^7=\sqrt[5]{p},\quad (\sqrt[35]{p})^5=\sqrt[7]{p},\quad \zeta_{35}^7=\zeta_5,\quad \zeta_{35}^5=\zeta_7,$$

it is immediately clear that $\mathbb{K}_5$ and $\mathbb{K}_7$ are contained in $\mathbb{K}_{35}$, so $\mathbb{K}_5\cdot\mathbb{K}_7\subseteq\mathbb{K}_{35}$. On the other hand, if a,b are integers such that $7a+5b=1$, then $(\sqrt[5]{p})^a\cdot(\sqrt[7]{p})^b=\sqrt[35]{p^{7a+5b}}=\sqrt[35]{p}$, and, analogously, $\zeta_5^a\cdot\zeta_7^b=\zeta_{35}$, so $\mathbb{K}_{35}\subseteq\mathbb{K}_5\cdot\mathbb{K}_7$.

(ii) The polynomials x^5-p, x^7-p, $x^{35}-p$ are irreducible by Eisenstein's Criterion. It follows that $[\mathbb{Q}(\sqrt[m]{p}):\mathbb{Q}]=m$ for $m=5,\ 7,\ 35$. Moreover, the cyclotomic extension $\mathbb{Q}(\zeta_m)$ has degree $\phi(m)$ over $\mathbb{Q}$, that is, 4, 6, 24. Since m and $\phi(m)$ are relatively prime for these values of m, we obtain that $[\mathbb{K}_m:\mathbb{Q}]=20,\ 42,\ 840$ for $m=5,\ 7,\ 35$, respectively.
Now, let $\mathbb{K}_0=\mathbb{K}_5\cap\mathbb{K}_7$. We have $[\mathbb{K}_5\cdot\mathbb{K}_7:\mathbb{K}_0]=[\mathbb{K}_5:\mathbb{K}_0][\mathbb{K}_7:\mathbb{K}_0]$, because all extensions whose degrees are involved in the equality are Galois, and $[\mathbb{K}_5\cdot\mathbb{K}_7:\mathbb{Q}]=840=20\cdot 42=[\mathbb{K}_5:\mathbb{Q}][\mathbb{K}_7:\mathbb{Q}]$. It follows, using the fact that $[\mathbb{K}_m:\mathbb{Q}]=[\mathbb{K}_m:\mathbb{K}_0][\mathbb{K}_0:\mathbb{Q}]$ for $m=5,\ 7,\ 35$, that $[\mathbb{K}_0:\mathbb{Q}]=[\mathbb{K}_0:\mathbb{Q}]^2$. Therefore, $[\mathbb{K}_0:\mathbb{Q}]=1$, that is, $\mathbb{K}_0=\mathbb{Q}$.

255 We will first consider the polynomial $f(x)=x^4-49$. Its roots are $\pm\sqrt{7}$ and $\pm\sqrt{-7}$, so $\mathbb{E}=\mathbb{Q}(i,\sqrt{7})$ is the splitting field of the polynomial $f(x)$ over $\mathbb{Q}$. In particular, $\mathbb{E}$ is the composite field of the two Galois extensions $\mathbb{Q}(\sqrt{7})\subseteq\mathbb{R}$ and $\mathbb{Q}(i)$, that are both quadratic and thus have Galois group isomorphic to $\mathbb{Z}/2\mathbb{Z}$. Since the intersection of the two extensions is $\mathbb{Q}$, the Galois group of $\mathbb{E}/\mathbb{Q}$ is isomorphic to the direct product $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$.

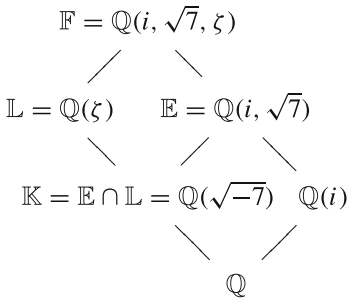
Consider now the polynomial $g(x) = (x^4 - 49)(x^7 - 1)$. Its roots are $\pm\sqrt{7}$, $\pm\sqrt{-7}$ and ζ^h for $0 \le h \le 6$, where ζ is a primitive seventh root of unity in $\mathbb{C}$; so $\mathbb{F} = \mathbb{Q}(i, \sqrt{7}, \zeta)$ is the splitting field of $g(x)$ over $\mathbb{Q}$. The field $\mathbb{F}$ is the compositum of the extension $\mathbb{E}$ from above and the cyclotomic extension $\mathbb{L} = \mathbb{Q}(\zeta)$ whose Galois group over $\mathbb{Q}$ is isomorphic to $(\mathbb{Z}/7\mathbb{Z})^*$, that is, to $\mathbb{Z}/6\mathbb{Z}$.

In order to compute the degree of $\mathbb{F}$ and determine its Galois group, it is useful to consider the field $\mathbb{K} = \mathbb{E} \cap \mathbb{L}$. First of all, remark that the degree of $\mathbb{K}$ over $\mathbb{Q}$ must divide 4 and 6, so it is equal to either one or two.

By the Fundamental Theorem of Galois Theory, the field $\mathbb{L}$ has exactly one quadratic subextension, as its Galois group is cyclic of even order. Moreover, this extension is the fixed field of the unique subgroup of index 2 in $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$. Since 3 generates $(\mathbb{Z}/7\mathbb{Z})^*$, the automorphism given by $\zeta \longmapsto \zeta^3$ generates $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$, and so the automorphism φ given by $\zeta \longmapsto \zeta^9 = \zeta^2$ generates the subgroup of index 2. The element

$$\alpha = \zeta + \zeta^2 + \zeta^4$$

of $\mathbb{L}$ is clearly fixed by φ. If α were in $\mathbb{Q}$ then the rational linear combination $\alpha \cdot 1 - \zeta - \zeta^2 - \zeta^4$ being zero would imply linear dependence between $1, \zeta, \zeta^2, \zeta^4$, whereas we know $1, \zeta, \zeta^2, \ldots, \zeta^5$ to be a basis of $\mathbb{L}/\mathbb{Q}$. It follows that α has degree 2 over $\mathbb{Q}$; computing the value of α^2 immediately yields that α is a root of the polynomial $x^2 + x + 2$, and thus that $\mathbb{Q}(\sqrt{-7})$ is the only quadratic subextension of $\mathbb{L}$.



We can therefore conclude that $\mathbb{K} = \mathbb{E} \cap \mathbb{L} = \mathbb{Q}(\sqrt{-7})$. In particular, we obtain that $\mathbb{F}$ is the composite field of the two Galois extensions $\mathbb{Q}(i)$ and $\mathbb{L}$, whose intersection is $\mathbb{Q}$. The Galois group under consideration is thus the direct product of the Galois groups of the two extensions, that is, it is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$.

256 Consider the splitting field of $x^5 - m$ over $\mathbb{Q}$. Let $\alpha = \sqrt[5]{m}$ and let ζ be a primitive fifth root of unity in $\mathbb{C}$. The roots of $x^5 - m$ are $\alpha\zeta^h$ for $h = 0, 1, 2, 3, 4$, and the splitting field is $\mathbb{K} = \mathbb{Q}(\alpha, \zeta)$.

If $m \in \mathbb{N}^5$, that is, if m is a fifth power, then $\mathbb{K} = \mathbb{Q}(\zeta)$ and the degree of the extension is $\phi(5) = 4$. Now assume that $m \notin \mathbb{N}^5$; we will show that $x^5 - m$ is irreducible over $\mathbb{Q}$. Clearly, $x^5 - m$ does not have any rational roots; moreover, if the polynomial had a factor of degree two, then the product of two of its roots, which is of the form $\alpha\zeta^h \cdot \alpha\zeta^k = \sqrt[5]{m^2}\zeta^{h+k}$, would have to be rational, which implies $k \equiv -h \pmod 5$ and $\sqrt[5]{m^2} \in \mathbb{N}^5$, contradicting the assumption that m is not a fifth power.

This proves that $\mathbb{K}$ contains the subextension $\mathbb{Q}(\sqrt[5]{m})$ of degree 5 over $\mathbb{Q}$. Since $\mathbb{K}$ is the composite field of this subextension and $\mathbb{Q}(\zeta)$, whose degree is $\phi(5) = 4$, and since 5 and 4 are coprime, we obtain that $[\mathbb{K} : \mathbb{Q}] = 20$.

Now consider the splitting field of $x^5 - m$ over $\mathbb{F}_{19}$. Since 5 and $18 = |\mathbb{F}_{19}^*|$ are relatively prime, the map $\mathbb{F}_{19} \ni a \longmapsto a^5 \in \mathbb{F}_{19}$ is bijective. This implies that there are no primitive fifth roots of unity in $\mathbb{F}_{19}$ and that each element of $\mathbb{F}_{19}$ is the fifth power of exactly one element, so $x^5 - m$ has exactly one root in $\mathbb{F}_{19}$ for each m.

Moreover, 5 divides $|\mathbb{F}_{19^2}^*| = 19^2 - 1$, so $\mathbb{F}_{19^2}$ does contain the primitive fifth roots of unity.

These facts imply that, whatever the value of m, $x^5 - m$ does not split as a product of linear factors in $\mathbb{F}_{19}$, but it does in $\mathbb{F}_{19^2}$. The degree of the required splitting field is therefore equal to 2.

257

(i) Since $\mathbb{K}$ is the composite field of the two extensions $\mathbb{Q}(\sqrt[4]{2})$ and $\mathbb{Q}(\sqrt[3]{2})$ of degree 4 and 3, respectively (the polynomials $x^4 - 2$ and $x^3 - 2$ are irreducible over $\mathbb{Q}$ by Eisenstein's Criterion), we obtain that $[\mathbb{K} : \mathbb{Q}] = 12$.

(ii) Remark that $\mathbb{K}$ contains $\sqrt[4]{2}$ and $\sqrt[3]{2}$, so the extension $\mathbb{L}$ must contain the splitting field $\mathbb{L}_4$ of $x^4 - 2$ and the splitting field $\mathbb{L}_3$ of $x^3 - 2$ over $\mathbb{Q}$. But then, since the composite field of normal extensions is a normal extension, we have $\mathbb{L} = \mathbb{L}_4 \cdot \mathbb{L}_3$.

We now proceed to determine the Galois group of $\mathbb{L}_4/\mathbb{Q}$. We have the two subextensions $\mathbb{Q}(\sqrt[4]{2})$ and $\mathbb{Q}(i)$; the former has degree 4 over $\mathbb{Q}$, is real and not normal; the latter has degree 2, is normal and not real. These have $\mathbb{Q}$ as their intersection and $\mathbb{L}_4$ as their composite field. It follows that $\mathrm{Gal}(\mathbb{L}_4/\mathbb{Q})$ has 8 elements and is a subgroup of S_4; in particular, it is a 2-Sylow subgroup of S_4 and thus isomorphic to D_4.

As for the Galois group of $\mathbb{L}_3/\mathbb{Q}$, we know that it has order 6, because of the two subextensions $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\zeta)$, where ζ is a primitive third root of unity in $\mathbb{C}$; this implies that $\mathrm{Gal}(\mathbb{L}_3/\mathbb{Q}) \simeq \mathsf{S}_3$.

We now intend to show that $\mathbb{L}_4 \cap \mathbb{L}_3 = \mathbb{Q}$, which will finally yield that $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{L}_4/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{L}_3/\mathbb{Q})$ and thus to $\mathsf{D}_4 \times \mathsf{S}_3$.

First of all, $[\mathbb{L}_4 \cap \mathbb{L}_3 : \mathbb{Q}]$ must divide both $[\mathbb{L}_4 : \mathbb{Q}] = 8$ and $[\mathbb{L}_3 : \mathbb{Q}] = 6$, so it can only be 1 or 2. If this degree were 2, then $\mathbb{L}_4 \cap \mathbb{L}_3$ would be a common quadratic subextension of $\mathbb{L}_4$ and $\mathbb{L}_3$, which we will now show cannot exist.

With the usual presentation $\langle r, s \mid r^4 = s^2 = e,\ sr = r^{-1}s \rangle$ of D_4, we find that there are only three subgroups of index 2: $\langle r \rangle$, $\langle r^2, s \rangle$ and $\langle r^2, sr \rangle$. But then the only quadratic subextensions of $\mathbb{L}_4$ are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$. The only subgroup of index 2 of S_3 is A_3, so the only quadratic subextension of $\mathbb{L}_3$ is $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{-3})$. We can thus conclude that the two fields have no quadratic subextensions in common.

(iii) We have $\mathbb{L} = \mathbb{K}(i, \zeta) = \mathbb{K}(\omega)$, where $\omega = i\zeta$ is a primitive twelfth root of unity. The automorphisms of $\mathbb{L}/\mathbb{Q}$ that fix $\mathbb{K}$ are determined by the image of ω, which is one of its conjugates over $\mathbb{Q}$. The possibilities for the image of ω are $\omega^{\pm 1}, \omega^{\pm 5}$; moreover, since $[\mathbb{L} : \mathbb{K}] = [\mathbb{L}/\mathbb{Q}]/[\mathbb{K} : \mathbb{Q}] = 4$, every

mapping $\omega \longmapsto \omega^{\pm 1}, \omega^{\pm 5}$ extends to an automorphism of $\mathbb{L}/\mathbb{K}$. In conclusion, $\mathrm{Gal}(\mathbb{L}/\mathbb{K})$ is isomorphic to $(\mathbb{Z}/12\mathbb{Z})^*$, that is, to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(iv) Every Galois subextension of $\mathbb{L}$ of degree 4 over $\mathbb{Q}$ has an Abelian Galois group, so it is fixed by a subgroup containing the derived group G' of $G = \mathrm{Gal}(\mathbb{L}/\mathbb{Q})$. Enumerating such subextensions is thus equivalent to enumerating the subgroups of G of index 4 that contain G'. Since the correspondence between subgroups of G/G' and subgroups of G containing G' preserves the index, we shall instead enumerate the subgroups of index 4 in the Abelianisation G/G'. We have $G \simeq \mathsf{D}_4 \times \mathsf{S}_3$ and $G/G' \simeq (\mathbb{Z}/2\mathbb{Z})^3$; the normal subextensions of $\mathbb{L}$ that have degree 4 over $\mathbb{Q}$ are as many as the subgroups of index 4, or equivalently of order 2, of $(\mathbb{Z}/2\mathbb{Z})^3$, that is, 7.

258 The 15th cyclotomic field $\mathbb{Q}(\zeta)$ has degree $\phi(15) = 8$ over $\mathbb{Q}$; its Galois group is isomorphic to $(\mathbb{Z}/15\mathbb{Z})^*$, that is, to $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. The subextensions of degree 2 over $\mathbb{Q}$ correspond to subgroups of index 2, and thus of order 4, in the Galois group G.

We set out to list these subgroups using the fact that a group of order 4 is either cyclic or isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. The group G has 4 elements of order 4, and thus $4/\phi(4) = 2$ cyclic subgroups of order 4; moreover, G has the subgroup $\mathbb{Z}/2\mathbb{Z} \times 2\mathbb{Z}/4\mathbb{Z}$, which is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and, since G has only 3 elements of order 2, it has no other subgroups of this type. In conclusion, G has 3 subgroups of order 4, so $\mathbb{Q}(\zeta)$ has 3 subextensions of order 2 over $\mathbb{Q}$.

Remark that ζ^5 is a primitive third root of unity and that ζ^3 is a primitive fifth root of unity. It follows that $\mathbb{Q}(\zeta^5) = \mathbb{Q}(\sqrt{-3})$ is a quadratic subextension of $\mathbb{Q}(\zeta)$. Moreover, since $\mathbb{Q}(\zeta^3)/\mathbb{Q}$ is the cyclotomic extension of order 5, we have the quadratic subextension $\mathbb{Q}(\zeta^3 + \zeta^{-3}) = \mathbb{Q}(\sqrt{5})$, as $\zeta^3 + \zeta^{-3}$ is a root of the polynomial $x^2 + x - 1$. We thus have the three quadratic subextensions $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-15})$, which are pairwise distinct.

259

(i) The roots of the polynomial $f(x)$ in $\mathbb{C}$ are $\pm\sqrt[4]{12}, \pm i\sqrt[4]{12}$, and the splitting field of $f(x)$ over $\mathbb{Q}$ is thus $\mathbb{F} = \mathbb{Q}(\sqrt[4]{12}, i)$. We have $[\mathbb{F} : \mathbb{Q}] = [\mathbb{Q}(i, \sqrt[4]{12}) : \mathbb{Q}(\sqrt[4]{12})][\mathbb{Q}(\sqrt[4]{12}) : \mathbb{Q}]$. The degree of $\sqrt[4]{12}$ over $\mathbb{Q}$ is 4, as the polynomial $x^4 - 12$ is irreducible by Eisenstein's Criterion. Moreover, $\mathbb{Q}(\sqrt[4]{12})$ is a real extension whereas $\mathbb{F}$ is not, so $[\mathbb{F} : \mathbb{Q}(\sqrt[4]{12})] = [\mathbb{Q}(i) : \mathbb{Q}] = 2$. In conclusion, $[\mathbb{F} : \mathbb{Q}] = 8$. Remark that $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ is isomorphic to D_4: it is a subgroup of order 8 of S_4, so it is a 2-Sylow subgroup and thus is isomorphic to D_4.

(ii) The splitting field of $f(x)$ over a finite extension $\mathbb{K}$ of $\mathbb{Q}$ is the composite field $\mathbb{K} \cdot \mathbb{F} = \mathbb{K}(\sqrt[4]{12}, i)$; we now show that its degree over $\mathbb{K}$ can only be 1, 2, 4, 8. First of all, we prove that each of these values is actually possible by providing the respective examples $\mathbb{K} = \mathbb{F}$, $\mathbb{K} = \mathbb{Q}(\sqrt[4]{12})$, $\mathbb{K} = \mathbb{Q}(i)$ and $\mathbb{K} = \mathbb{Q}$.
On the other hand, $[\mathbb{K}(i, \sqrt[4]{12}) : \mathbb{K}] = [\mathbb{K}(i, \sqrt[4]{12}) : \mathbb{K}(\sqrt[4]{12})][\mathbb{K}(\sqrt[4]{12}) : \mathbb{K}]$, and we know that $[\mathbb{K}(i, \sqrt[4]{12}) : \mathbb{K}(\sqrt[4]{12})] \leq 2$, as well as $[\mathbb{K}(\sqrt[4]{12}) : \mathbb{K}] \leq 4$. However, we have $[\mathbb{K}(\sqrt[4]{12}) : \mathbb{K}] \neq 3$, as the polynomial $f(x)$ has two pairs

of opposite roots, so if $\mathbb{K}$ contains a root of $f(x)$ then it contains at least two. It follows that the only possible degrees are those listed.

▌More generally, one can show that, since $\mathbb{F}/\mathbb{Q}$ is normal, we have $[\mathbb{F}\cdot\mathbb{K} : \mathbb{K}] = [\mathbb{F} : \mathbb{F}\cap\mathbb{K}]$ and the degree must therefore divide $[\mathbb{F} : \mathbb{Q}]$.▐

260 The polynomial $x^4 - 3$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion, and its roots in $\mathbb{C}$ are $\pm\sqrt[4]{3}$, $\pm i\sqrt[4]{3}$; its splitting field is thus $\mathbb{K} = \mathbb{Q}(\sqrt[4]{3}, i)$. The roots of $x^2 - a$ are $\pm\sqrt{a}$, so the splitting field of the polynomial $(x^4 - 3)(x^2 - a)$ is $\mathbb{F} = \mathbb{K}(\sqrt{a}) = \mathbb{Q}(\sqrt[4]{3}, i, \sqrt{a})$.

Consider the extension $\mathbb{K}/\mathbb{Q}$; we have $[\mathbb{K} : \mathbb{Q}] = [\mathbb{Q}(\sqrt[4]{3})(i) : \mathbb{Q}(\sqrt[4]{3})][\mathbb{Q}(\sqrt[4]{3}) : \mathbb{Q}] = 2 \cdot 4 = 8$, as $\sqrt[4]{3}$ has degree 4 over $\mathbb{Q}$ (because $x^4 - 3$ is irreducible) and because $\mathbb{Q}(\sqrt[4]{3})$ is a real extension whereas the extension $\mathbb{Q}(i)$, of degree 2 over $\mathbb{Q}$, is not.

The group $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to a subgroup of S_4; in particular, since its order is 8, it is isomorphic to a 2-Sylow subgroup of S_4. Remark that D_4, which has itself order 8, is isomorphic to a subgroup of S_4; thus, since all 2-Sylow subgroups are pairwise conjugate and therefore isomorphic, we obtain that $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to D_4.

Now consider the extension $\mathbb{F} = \mathbb{K}(\sqrt{a})$ of $\mathbb{K}$; its degree is 1 if $\sqrt{a} \in \mathbb{K}$ and 2 if $\sqrt{a} \notin \mathbb{K}$. Note that $\sqrt{a} \in \mathbb{K}$ if and only if $\mathbb{Q}(\sqrt{a})$ is $\mathbb{Q}$ or one of the quadratic subextensions of $\mathbb{K}$. The extension $\mathbb{K}$ has three distinct quadratic subextensions, namely, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$, and it cannot have any others, as D_4 only has three subgroups of index 2, namely, $\langle r\rangle$, $\langle r^2, s\rangle$ and $\langle r^2, rs\rangle$.

In conclusion, if $a \in \mathbb{N}^2 \cup (-\mathbb{N}^2) \cup (3\mathbb{N}^2) \cup (-3\mathbb{N}^2)$ then $\sqrt{a} \in \mathbb{K}$, $\mathbb{F} = \mathbb{K}$ and $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ is isomorphic to D_4. Otherwise, $\mathbb{F} = \mathbb{K} \cdot \mathbb{Q}(\sqrt{a})$ and $\mathbb{F} \cap \mathbb{Q}(\sqrt{a}) = \mathbb{Q}$; thus, since these are Galois extensions, $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\sqrt{a})/\mathbb{Q})$, that is, $\mathsf{D}_4 \times \mathbb{Z}/2\mathbb{Z}$.

We now compute the Galois group over $\mathbb{Q}(\sqrt{2})$. The splitting field of the polynomial in the problem statement over $\mathbb{Q}(\sqrt{2})$ is $\mathbb{F}(\sqrt{2}) = \mathbb{Q}(\sqrt{2})(\sqrt[4]{3}, i, \sqrt{a})$. Remark that, by the arguments above, $\sqrt{2} \notin \mathbb{K}$.

If $\sqrt{2} \in \mathbb{F}$ then $\mathbb{F}(\sqrt{2}) = \mathbb{F} = \mathbb{K}(\sqrt{2})$ and, since $\mathbb{K} \cap \mathbb{Q}(\sqrt{2}) = \mathbb{Q}$, we have $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \simeq \mathrm{Gal}(\mathbb{K}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$, which implies that $\mathrm{Gal}(\mathbb{F}(\sqrt{2})/\mathbb{Q}(\sqrt{2}))$ is isomorphic to $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ and thus to D_4.

If $\sqrt{2} \notin \mathbb{F}$ then $\mathbb{F}(\sqrt{2}) = \mathbb{F} \cdot \mathbb{Q}(\sqrt{2})$ and $\mathbb{F} \cap \mathbb{Q}(\sqrt{2}) = \mathbb{Q}$; it follows that $\mathrm{Gal}(\mathbb{F}(\sqrt{2})/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$. Thus, like before, $\mathrm{Gal}(\mathbb{F}(\sqrt{2})/\mathbb{Q}(\sqrt{2}))$ is isomorphic to $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$, which is in turn isomorphic to either D_4 or $\mathsf{D}_4 \times \mathbb{Z}/2\mathbb{Z}$, as shown in the first part of the solution.

What remains to be done is identify the integers a such that $\sqrt{2} \in \mathbb{F} = \mathbb{Q}(\sqrt[4]{3}, i, \sqrt{a})$, that is, such that $\mathbb{Q}(\sqrt{2})$ is a subextension of degree 2 of $\mathbb{F}/\mathbb{Q}$. We proceed to show that this is the case if and only if $a \in (\pm 2\mathbb{N}^2) \cup (\pm 6\mathbb{N}^2)$.

Clearly, for these values of a we have $\sqrt{2} \in \mathbb{F}$. Conversely, the subextensions of degree 2 of $\mathbb{F}$ are Abelian, and thus fixed by the commutator subgroup of $\mathrm{Gal}(\mathbb{F}/\mathbb{Q})$. If $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \simeq \mathsf{D}_4$ then the commutator subgroup is generated by r^2; similarly, if $\mathrm{Gal}(\mathbb{F}/\mathbb{Q}) \simeq \mathsf{D}_4 \times \mathbb{Z}/2\mathbb{Z}$ then the commutator subgroup is generated by $(r^2, 0)$. Moreover, the generator (r^2 or $(r^2, 0)$) corresponds to the

automorphism obtained by extending the mapping $\sqrt[4]{3} \longmapsto -\sqrt[4]{3}$, $i \longmapsto i$ and, in the case of the group $\mathsf{D}_4 \times \mathbb{Z}/2\mathbb{Z}$, $\sqrt{a} \longmapsto \sqrt{a}$. It is then clear that the fixed field of the commutator subgroup is $\mathbb{Q}(i, \sqrt{3}, \sqrt{a})$. Finally, this field only contains the following quadratic subextensions, not necessarily distinct: $\mathbb{Q}(\sqrt{-1})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{a})$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-a})$, $\mathbb{Q}(\sqrt{3a})$, and $\mathbb{Q}(\sqrt{-3a})$; this implies that the conditions for a given above are indeed necessary.

261 The extension $\mathbb{Q}(\zeta)$ is Galois over $\mathbb{Q}$ and the group $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ is isomorphic to $(\mathbb{Z}/36\mathbb{Z})^*$, that is, to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$. By the Fundamental Theorem of Galois Theory, the subextensions of $\mathbb{Q}(\zeta)/\mathbb{Q}$ are in bijection with the subgroups of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$, which are easy to enumerate: other than the group itself, we have the trivial group, three subgroups of order 2, one subgroup of order 3, one subgroup of order 4 and three subgroups of order 6. It follows that, other than the obvious subextensions $\mathbb{Q}$ and $\mathbb{Q}(\zeta)$, we have three subextensions of degree 6 over $\mathbb{Q}$, one of degree 4, one of degree 3 and three of degree 2.

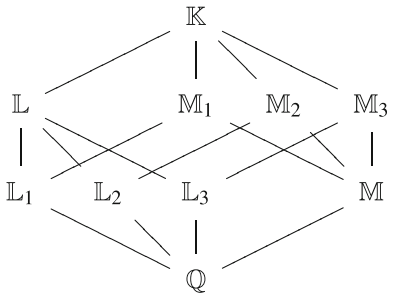
Since ζ^9 and ζ^{12} are a primitive fourth and third root of unity, respectively, we have the two quadratic subextensions $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$, and so the additional subextension $\mathbb{Q}(\sqrt{3})$; the three being all distinct, they are all the quadratic subextensions of $\mathbb{Q}(\zeta)$. Moreover, $\mathbb{Q}(i, \sqrt{3})$ is the unique subextension of degree 4.

We can also obtain the primitive ninth root of unity ζ^4, so $\mathbb{Q}(\zeta^4)$ is a subextension of degree $\phi(9) = 6$. Its real subextension $\mathbb{Q}(\zeta^4 + \zeta^{-4})$ has degree $\phi(9)/2 = 3$, and it is the only one of that degree. Consider now the following subextensions of degree 6: $\mathbb{Q}(\zeta^4 + \zeta^{-4}, i)$, $\mathbb{Q}(\zeta^4 + \zeta^{-4}, \sqrt{-3})$ and $\mathbb{Q}(\zeta^4 + \zeta^{-4}, \sqrt{3})$. Each of them contains a unique quadratic subextension: if this were not the case, it would also contain a subextension of degree 4, but 4 does not divide 6. It follows that the three are distinct, and in particular the second one is $\mathbb{Q}(\zeta^4)$, as the ninth cyclotomic field contains the third. We have thus found all three subextensions of degree 6.

262 Consider the two subextensions $\mathbb{Q}(\zeta_{11})$ and $\mathbb{Q}(\sqrt{11})$, which are both Galois over $\mathbb{Q}$. The group $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ is isomorphic to $(\mathbb{Z}/11\mathbb{Z})^*$, that is, to $\mathbb{Z}/10\mathbb{Z}$, as $\mathbb{Q}(\zeta)$ is a cyclotomic extension; the group $\mathrm{Gal}(\mathbb{Q}(\sqrt{11})/\mathbb{Q})$ is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

We will now prove that $\mathbb{Q}(\zeta) \cap \mathbb{Q}(\sqrt{11}) = \mathbb{Q}$. Indeed, since $\mathbb{Q}(\sqrt{11})$ has degree 2 over $\mathbb{Q}$, it is enough to show that we do not have $\mathbb{Q}(\sqrt{11}) \subseteq \mathbb{Q}(\zeta)$. But, if this were the case, then the fact that $\mathbb{Q}(\sqrt{11})$ is a real extension would imply that it is contained in $\mathbb{Q}(\zeta) \cap \mathbb{R}$, which we know to be $\mathbb{Q}(\zeta_{11} + \zeta_{11}^{-1})$. However, the latter extension has degree $\phi(11)/2 = 5$, so it cannot contain a quadratic subextension.

We therefore conclude that the Galois group of $\mathbb{K} = \mathbb{Q}(\zeta, \sqrt{11})$ over $\mathbb{Q}$ is isomorphic to the direct product $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\sqrt{11})/\mathbb{Q})$ and thus to $\mathbb{Z}/10\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.



By the Fundamental Theorem of Galois Theory, the lattice of all subextensions of $\mathbb{K}/\mathbb{Q}$ corresponds to the lattice of all subgroups of $\mathbb{Z}/10\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Other than the trivial group and the group itself, $\mathbb{Z}/10\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has the following subgroups: three subgroups of order 10, one of order 5, one of order 4 and three of order 2. Obviously, the trivial subgroup corresponds to $\mathbb{Q}$ and the group itself to $\mathbb{K}$.

The subextensions of degree 2 over $\mathbb{Q}$ are fixed by the subgroups of order 10, so there are three, which we denote by $\mathbb{L}_1$, $\mathbb{L}_2$, $\mathbb{L}_3$. There is a unique subextension of degree 4, which is fixed by the subgroup of order 5 and which has to be $\mathbb{L} = \mathbb{L}_1 \cdot \mathbb{L}_2 = \mathbb{L}_1 \cdot \mathbb{L}_3 = \mathbb{L}_2 \cdot \mathbb{L}_3$. There is a unique subextension of degree 5, namely, $\mathbb{M} = \mathbb{Q}(\zeta + \zeta^{-1})$; from it we obtain the three subextensions of degree 10 given by $\mathbb{M} \cdot \mathbb{L}_1$, $\mathbb{M} \cdot \mathbb{L}_2$ and $\mathbb{M} \cdot \mathbb{L}_3$.

▌We can also give an explicit description of the subextensions of degree 2, which yields one for those of degree 6. Setting $\mathbb{L}_1 = \mathbb{Q}(\sqrt{11})$, let $\mathbb{L}_2$ be the unique subextension of degree 2 of $\mathbb{Q}(\zeta)$. The latter is fixed by the subgroup of order 5 of $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$, which is generated by the automorphism induced by $\zeta \longmapsto \zeta^4$. One can check that $\alpha = \zeta + \zeta^4 + \zeta^5 + \zeta^9 + \zeta^3$ is fixed by this automorphism and is not rational, as $1, \zeta, \ldots, \zeta^9$ are linearly independent over $\mathbb{Q}$. It follows that $\mathbb{L}_2 = \mathbb{Q}(\alpha)$; it is easy to see that the minimal polynomial of α is $x^2 + x + 3$, which in turn yields $\mathbb{L}_2 = \mathbb{Q}(\sqrt{-11})$. Finally, this tells us that we can take $\mathbb{L}_3 = \mathbb{Q}(i)$.▐

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(i) The polynomial $f(x) = x^6 - 2$ is irreducible in $\mathbb{Q}[x]$ by Eisenstein's Criterion. It follows that, letting $\alpha = \sqrt[6]{2}$, we have $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 6$. The splitting field of $f(x)$ over $\mathbb{Q}$ is $\mathbb{E} = \mathbb{Q}(\alpha, \zeta)$, where ζ is a primitive sixth root of unity in $\mathbb{C}$. Since $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \phi(6) = 2$ and ζ is not real, the degree $[\mathbb{E} : \mathbb{Q}]$ is $6 \cdot 2 = 12$.

(ii) The extension $\mathbb{Q}(\alpha)/\mathbb{Q}$ is not normal because there exists a homomorphism $\mathbb{Q}(\alpha) \longrightarrow \mathbb{C}$ sending α to $\zeta\alpha \notin \mathbb{Q}(\alpha)$. If $\mathrm{Gal}(\mathbb{E}/\mathbb{Q})$ were Abelian, all of its subgroups would me normal; the extension $\mathbb{Q}(\alpha)$ would thus correspond to a normal subgroup and therefore be a normal extension of $\mathbb{Q}$, which is a contradiction.

Now consider the cyclotomic extension $\mathbb{F} = \mathbb{Q}(\zeta)$. We have $[\mathbb{E} : \mathbb{F}] = 6$ and the group $\mathrm{Gal}(\mathbb{E}/\mathbb{F})$ is generated by the automorphism given by $\alpha \longmapsto \alpha\zeta$, $\zeta \longmapsto \zeta$, which has order 6; it follows that $\mathrm{Gal}(\mathbb{E}/\mathbb{F})$ is a cyclic subgroup of order 6 of $\mathrm{Gal}(\mathbb{E}/\mathbb{Q})$.

(iii) The elements $\sqrt{2}$ and $\zeta = (1 + \sqrt{-3})/2$ belong to $\mathbb{E}$, so $\mathbb{Q}(\sqrt{2}, \sqrt{-3})$ is a subextension of $\mathbb{E}$. Remark that any homomorphism $\mathbb{Q}(\sqrt{2}, \sqrt{-3}) \longrightarrow \mathbb{C}$ must send $\sqrt{2}$ to $\pm\sqrt{2}$ and $\sqrt{-3}$ to $\pm\sqrt{-3}$, as $x^2 - 2$ and $x^2 + 3$ are their

respective minimal polynomials. But then the extension is clearly normal, and its Galois group is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

264 We prove that $\mathbb{K} = \mathbb{Q}(\sqrt{2}, \sqrt{-3}, \sqrt[3]{5})$ is a normal extension using the definition of normality. If $\varphi : \mathbb{K} \longrightarrow \mathbb{C}$ is a homomorphism and $f(x)$ is the minimal polynomial of an element $\alpha \in \mathbb{K}$, then $\varphi(\alpha)$ must be a root of $f(x)$. The minimal polynomials of $\sqrt{2}$, $\sqrt{-3}$, $\sqrt[3]{5}$ over $\mathbb{Q}$ are x^2-2, x^2+3, x^3-5, respectively, and all three are irreducible thanks to Eisenstein's Criterion. It follows that $\varphi(\sqrt{2}) = \pm\sqrt{2}$, $\varphi(\sqrt{-3}) = \pm\sqrt{-3}$, $\varphi(\sqrt[3]{5}) = \sqrt[3]{5}\zeta^h$, where $\zeta = (-1+\sqrt{-3})/2$ is a primitive third root of unity and $h = 0, 1, 2$. Since ζ is in $\mathbb{K}$, we obtain that the image of φ is contained in $\mathbb{K}$, which is therefore a normal extension of $\mathbb{Q}$.

By what we saw above, the subextension $\mathbb{F} = \mathbb{Q}(\sqrt[3]{5}, \zeta)$ is the splitting field of $x^3 - 5$. It has the quadratic subextension $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{-3})$, so its degree over $\mathbb{Q}$ is 6 and its Galois group is isomorphic to S_3.

The field $\mathbb{E} = \mathbb{Q}(\sqrt{2})$ is also a subextension of $\mathbb{K}$, and its Galois group is isomorphic to $\mathbb{Z}/2\mathbb{Z}$; its intersection with $\mathbb{F}$ is equal to $\mathbb{Q}$, because $\mathbb{F}$ has the unique quadratic subextension $\mathbb{Q}(\sqrt{-3})$ corresponding to the subgroup A_3, which is the only subgroup of index 2 in S_3. But then, using the fact that $\mathbb{K} = \mathbb{E} \cdot \mathbb{F}$, we obtain that $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{E}/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{F}/\mathbb{Q})$, that is, to $\mathbb{Z}/2\mathbb{Z} \times \mathsf{S}_3$.

In order to describe the normal subextensions of $\mathbb{K}$, we look for the normal subgroups of the Galois group. It is useful to remark that the mapping $r \longmapsto (1, (1,2,3))$, $s \longmapsto (0, (1,2))$ extends to an isomorphism between D_6, given with the usual presentation $\langle r, s \mid r^6 = s^2 = e, sr = r^{-1}s\rangle$, and $\mathbb{Z}/2\mathbb{Z} \times \mathsf{S}_3$. Moreover, if we label the roots as $\alpha_1 = \sqrt[3]{5}\zeta$, $\alpha_2 = \sqrt[3]{5}\zeta^2$ and $\alpha_3 = \sqrt[3]{5}\zeta^3 = \sqrt[3]{5}$, the elements of D_6 act on $\mathbb{K}$ as follows:

$$\begin{array}{lll} r(\sqrt{2}) = -\sqrt{2}, & r(\sqrt{-3}) = \sqrt{-3}, & r(\sqrt[3]{5}) = \sqrt[3]{5}\zeta, \\ s(\sqrt{2}) = \sqrt{2}, & s(\sqrt{-3}) = -\sqrt{-3}, & s(\sqrt[3]{5}) = \sqrt[3]{5}. \end{array}$$

Excluding the trivial group, each proper normal subgroup of D_6 falls into one of the following two categories.

(1). The subgroups of $\langle r\rangle$, which is a characteristic subgroup of D_6. These are: $\langle r\rangle$, whose fixed field is $\mathbb{Q}(\sqrt{-3})$; $\langle r^2\rangle$, whose fixed field is $\mathbb{Q}(\sqrt{2}, \sqrt{-3})$; $\langle r^3\rangle$, whose fixed field is $\mathbb{Q}(\sqrt[3]{5}, \sqrt{-3})$.

(2). The subgroups containing some symmetry, which must then contain the 3 conjugates of said symmetry. They are: $\langle r^2, s\rangle$, whose fixed field is $\mathbb{Q}(\sqrt{2})$; $\langle r^2, rs\rangle$, whose fixed field is $\mathbb{Q}(\sqrt{-6})$.

In conclusion, the subextensions of $\mathbb{K}$ that are normal over $\mathbb{Q}$ are the following: $\mathbb{K}$, $\mathbb{Q}(\sqrt[3]{5}, \sqrt{-3})$, $\mathbb{Q}(\sqrt{2}, \sqrt{-3})$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-6})$, $\mathbb{Q}$.

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(i) The roots of $x^8 - 2$ are $\alpha\zeta^h$ for $h = 0, 1, \ldots, 7$, so $\mathbb{E} = \mathbb{Q}(\alpha, \zeta)$. The polynomial $x^8 - 2$ is irreducible in $\mathbb{Q}[x]$ by Eisenstein's Criterion, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 8$. Moreover, ζ is a primitive eighth root of unity, so $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \phi(8) = 4$ and we clearly have $\mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{2}, i)$. Now, $\sqrt{2} = \alpha^4 \in \mathbb{Q}(\alpha)$,

hence $\mathbb{E} = \mathbb{Q}(\alpha, i)$. Finally, i does not belong to the real extension $\mathbb{Q}(\alpha)$, so $[\mathbb{E} : \mathbb{Q}] = [\mathbb{E} : \mathbb{Q}(\alpha)][\mathbb{Q}(\alpha) : \mathbb{Q}] = 2 \cdot 8 = 16$.

(ii) Any element of G must send the generators α, i to roots of their respective minimal polynomials over $\mathbb{Q}$, which yields at most $8 \cdot 2 = 16$ possibilities. Since $|G| = 16$, all these possibilities must be realisable, so there do exist automorphisms θ and σ with the desired properties.

(iii) Since $|G| = 16$, if G were a dihedral group it would be isomorphic to D_8. In order to compute the order of θ, remark that the fact that $\sqrt{2} = \alpha^4$ implies

$$\theta(\sqrt{2}) = \theta(\alpha)^4 = (\alpha\zeta)^4 = -\alpha^4 = -\sqrt{2},$$

hence $\theta(\zeta) = -\zeta$. We thus have

$$\theta^2(\alpha) = -\alpha\zeta^2 = -i\alpha, \quad \theta^4(\alpha) = \theta^2(\theta^2(\alpha)) = -\alpha, \quad \theta^8(\alpha) = \theta^4(\theta^4(\alpha)) = \alpha.$$

This proves that θ has order 8 in G.
If G were really isomorphic to D_8, the subgroup generated by θ would have to be the group of rotations, and every element outside this subgroup would be a symmetry: in particular, its order would have to be 2. However, consider the element $\theta \circ \sigma$: we have $\sigma(\sqrt{2}) = \sigma(\alpha^4) = \alpha^4 = \sqrt{2}$, hence $\sigma(\zeta) = \sigma(\sqrt{2}(1+i)/2) = \sqrt{2}(1-i)/2 = \zeta^{-1}$, and finally

$$(\theta\circ\sigma)(\alpha) = \theta(\sigma(\alpha)) = \theta(\alpha) = \alpha\zeta, \ (\theta\circ\sigma)^2(\alpha) = (\theta\circ\sigma)(\alpha\zeta) = \theta(\alpha\zeta^{-1}) = -\alpha.$$

It follows that $\theta \circ \sigma$ does not have order 2 and thus that G is not isomorphic to D_8.

(iv) By the Fundamental Theorem of Galois Theory, a subgroup of order r fixes a subfield of degree $|G|/r = 16/r$ over $\mathbb{Q}$. We have seen that the subgroup generated by θ has order 8, so $[\mathbb{E}^{\langle\theta\rangle} : \mathbb{Q}] = 2$ and, since θ fixes the element i, we find that $\mathbb{E}^{\langle\theta\rangle} = \mathbb{Q}(i)$. Analogously, the fixed fields of θ^2 and θ^4 have degree over $\mathbb{Q}$ given by 4 and 8, respectively. The fact that $\theta^2(\sqrt{2}) = \sqrt{2}$ and $\theta^4(\sqrt[4]{2}) = \sqrt[4]{2}$ yields $\mathbb{E}^{\langle\theta^2\rangle} = \mathbb{Q}(\sqrt{2}, i)$ and $\mathbb{E}^{\langle\theta^4\rangle} = \mathbb{Q}(\sqrt[4]{2}, i)$.
The order of the automorphism σ is 2, so its fixed field has degree 8 over $\mathbb{Q}$. But by definition σ fixes α, and so $\mathbb{E}^{\langle\sigma\rangle} = \mathbb{Q}(\alpha)$. Finally, the fixed field of $\langle\theta^4, \sigma\rangle$ is the intersection of the fixed fields of θ^4 and σ, that is, $\mathbb{Q}(\sqrt[4]{2}, i) \cap \mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt[4]{2})$.

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(i) Let $\mathbb{E}, \mathbb{K}$ be the splitting fields of $x^4 - 3$ and $x^4 - 12$, respectively. It is easy to check that $\mathbb{E} = \mathbb{Q}(\sqrt[4]{3}, i)$ and $\mathbb{K} = \mathbb{Q}(\sqrt[4]{12}, i)$, and thus $\mathbb{F} = \mathbb{E} \cdot \mathbb{K}$; since $\sqrt[4]{12} = \sqrt[4]{3} \cdot \sqrt{2}$, we have $\mathbb{F} = \mathbb{Q}(\sqrt[4]{3}, \sqrt{2}, i)$.
The polynomial $x^4 - 3$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion, so $[\mathbb{Q}(\sqrt[4]{3}) : \mathbb{Q}] = 4$. Since $\mathbb{Q}(\sqrt[4]{3})$ is a real extension, we obtain that $[\mathbb{E} : \mathbb{Q}] = 8$. This implies that $\mathbb{F} = \mathbb{E}(\sqrt{2})$, and thus $[\mathbb{F} : \mathbb{E}] = 1$ if $\sqrt{2} \in \mathbb{E}$ and

$[\mathbb{F} : \mathbb{E}] = 2$ otherwise. In other words, we need to determine whether or not $\mathbb{Q}(\sqrt{2})$ is a subextension of $\mathbb{E}$.
The Galois group of $\mathbb{E}/\mathbb{Q}$ is isomorphic to D_4 because its order is 8 and, since $\mathbb{E}$ is the splitting field of a polynomial of degree 4, it is isomorphic to a 2-Sylow subgroup of S_4. The group D_4 has 3 subgroups of order 4, and therefore $\mathbb{E}$ has three subextensions of degree 2, which are easy to find: they are $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$. Note that $\mathbb{Q}(\sqrt{2})$ is not among them, so $[\mathbb{F} : \mathbb{E}] = 2$ and $[\mathbb{F} : \mathbb{Q}] = 16$.

(ii) One can immediately check that the following 7 fields are quadratic subextensions of $\mathbb{F}$: $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{6})$, $\mathbb{Q}(\sqrt{-2})$, $\mathbb{Q}(\sqrt{-6})$. There are no other quadratic subextensions: if $\mathbb{Q}(\sqrt{d})$ were another one, we would have $\mathbb{F} = \mathbb{Q}(\sqrt{2}, \sqrt{3}, i, \sqrt{d})$. But then the Galois group G of $\mathbb{F}/\mathbb{Q}$ would consist of elements of order 1 or 2, because every square root can only be sent to itself or its opposite, whereas G has the quotient $\mathrm{Gal}(\mathbb{E}/\mathbb{Q})$, which is isomorphic to D_4 and thus not even Abelian.

267 Let $\mathbb{L}$ be the splitting field of $(x^2 + 3)(x^3 - 3x + 1)$. The second factor is irreducible because it has no rational roots, and its discriminant is $-4 \cdot (-3)^3 - 27 \cdot 1^2 = 81 = 9^2$, which is a square in $\mathbb{Q}$. We thus obtain that the splitting field $\mathbb{F}$ of the second factor has degree 3 over $\mathbb{Q}$, and in particular $\mathbb{F}$ does not contain the quadratic subextension $\mathbb{Q}(\sqrt{-3})$. In conclusion, since $\mathbb{L} = \mathbb{Q}(\sqrt{-3}) \cdot \mathbb{L}$ and $\mathbb{Q}(\sqrt{-3}) \cap \mathbb{L} = \mathbb{Q}$, $\mathbb{L}$ has degree 6 over $\mathbb{Q}$ and the Galois group of $\mathbb{L}/\mathbb{Q}$ is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, and thus to $\mathbb{Z}/6\mathbb{Z}$.

Now let $\mathbb{L}'$ be the splitting field of $(x^2 + 3)(x^3 - 5)$. Again, the second factor is irreducible because of its lack of rational roots, but this time it has splitting field $\mathbb{F}' = \mathbb{Q}(\sqrt[3]{5}, \zeta)$, where $\zeta = (-1 + \sqrt{-3})/2$ is a primitive third root of unity in $\mathbb{C}$. Notice that the first factor $x^2 + 3$ splits as a product of linear factors in $\mathbb{F}'$. In conclusion, $\mathbb{L}' = \mathbb{F}'$, and thus the Galois group of $\mathbb{L}'/\mathbb{Q}$ is S_3.

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(i) Let $\zeta \in \mathbb{C}$ be a primitive tenth root of unity. We have $(x^5 - 1)(x^5 + 1) = x^{10} - 1$, so the roots of $x^5 + 1$ are the tenth roots of unity that are not fifth roots of unity, namely, -1 and the primitive tenth roots of unity $\zeta, \zeta^3, \zeta^7, \zeta^9$. The splitting field of $p(x)$ is $\mathbb{Q}(\zeta, \sqrt{a})$ and the minimal polynomial of ζ is $(x^5 + 1)/(x + 1) = x^4 - x^3 + x^2 - x + 1$, the tenth cyclotomic polynomial. The cyclotomic extension $\mathbb{Q}(\zeta)$ has degree $\phi(10) = 4$ over $\mathbb{Q}$, and its Galois group is isomorphic to $(\mathbb{Z}/10\mathbb{Z})^*$, that is, to $\mathbb{Z}/4\mathbb{Z}$. It therefore contains a unique subextension of degree 2 over $\mathbb{Q}$, which is the real subextension $\mathbb{Q}(\alpha)$, where $\alpha = \zeta + \zeta^{-1}$. It follows from the identity $\zeta^4 - \zeta^3 + \zeta^2 - \zeta + 1 = 0$ that the minimal polynomial of α is $x^2 - x - 1$ and thus we have $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{5})$. Consequently, if a is a square in $\mathbb{Q}(\sqrt{5})$, that is, if a is of the form m^2 or $5m^2$ for some $m \in \mathbb{Z}$, then $\mathbb{Q}(\zeta, \sqrt{a}) = \mathbb{Q}(\zeta)$: the degree of the extension is 4 and the Galois group is isomorphic to $\mathbb{Z}/4\mathbb{Z}$.
Otherwise, the degree of the extension is 8. In this case, the Galois group is isomorphic to $\mathrm{Gal}(\mathbb{Q}(\sqrt{a})/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ and thus to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$; this

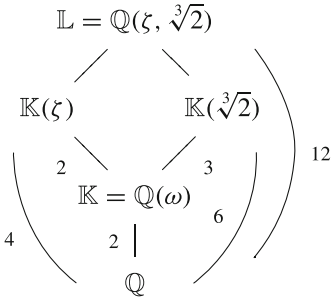
is thanks to the fact that the intersection of $\mathbb{Q}(\sqrt{a})$ and $\mathbb{Q}(\zeta)$ is $\mathbb{Q}$ and that their composite field is the splitting field under consideration.

(ii) By our argument above, when $a = 7$ the Galois group is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, which has three subgroups of index 4, so there are 3 subfields of the splitting field that have degree 2 over $\mathbb{Q}$. Two of them are clearly given by $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{7})$, which are distinct by the arguments above; the third is thus $\mathbb{Q}(\sqrt{35})$.

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(i) Note that ζ^4 is a primitive third root of unity, so $\mathbb{L}$ is the splitting field of the polynomial $(x^{12} - 1)(x^3 - 2)$ and thus a Galois extension of $\mathbb{Q}$. Moreover, the cyclotomic field $\mathbb{Q}(\zeta)$ has degree $\phi(12) = 4$ over $\mathbb{Q}$ and, since $x^3 - 2$ is irreducible by Eisenstein's Criterion and is thus the minimal polynomial of $\sqrt[3]{2}$, the extension $\mathbb{Q}(\sqrt[3]{2})$ has degree 3 over $\mathbb{Q}$. Since the degrees are relatively prime, the intersection of the two extensions is trivial, and so their composite field $\mathbb{L}$ has degree 12 over $\mathbb{Q}$.

(ii) Let $\mathbb{K} = \mathbb{Q}(\omega)$ be the third cyclotomic field; its degree over $\mathbb{Q}$ is $\phi(3) = 2$. We have $\mathbb{K}(\zeta) = \mathbb{Q}(\zeta)$ and $\mathbb{K}(\sqrt[3]{2}) = \mathbb{Q}(\omega, \sqrt[3]{2})$; moreover, we have established that $[\mathbb{K}(\zeta) : \mathbb{Q}] = 4$ and $[\mathbb{K}(\sqrt[3]{2})] = 6$.



It follows that $\mathbb{L}$ has degree 6 over $\mathbb{K}$ and that the two subextensions $\mathbb{K}(\zeta)$ and $\mathbb{K}(\sqrt[3]{2})$ of $\mathbb{L}$ over $\mathbb{K}$ have degree 2 and 3, respectively. Since their degrees are relatively prime, the intersection of the two extensions is $\mathbb{K}$; both extensions are normal, as they are the splitting fields of $x^{12} - 1$ and $x^3 - 2$ over $\mathbb{K}$, so $\mathrm{Gal}(\mathbb{L}/\mathbb{K})$ is isomorphic to $\mathrm{Gal}(\mathbb{K}(\zeta)/\mathbb{K}) \times \mathrm{Gal}(\mathbb{K}(\sqrt[3]{2})/\mathbb{K})$ and thus to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.

(iii) Remark that ζ^3 is a primitive fourth root of unity, so $\mathbb{L}$ contains the two subextensions $\mathbb{L}_1 = \mathbb{Q}(i)$ and $\mathbb{L}_2 = \mathbb{Q}(\omega, \sqrt[3]{2})$; moreover, $\zeta^3 \cdot \omega = \zeta^7$ is a primitive twelfth root of unity, so $\mathbb{L}$ is the composite field of $\mathbb{L}_1$ and $\mathbb{L}_2$. As for the degrees, we have $[\mathbb{L}_1 : \mathbb{Q}] = 2$ and $[\mathbb{L}_2 : \mathbb{Q}] = 6$, so $\mathbb{L}_1 \cap \mathbb{L}_2 = \mathbb{Q}$ because the degree of their composite field $\mathbb{L}$ is the product of their degrees. Finally, both subextensions are normal: $\mathbb{L}_1$ is the splitting field of $x^2 + 1$ and $\mathbb{L}_2$ is the splitting field of $x^3 - 2$. It follows that $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{L}_1/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{L}_2/\mathbb{Q})$. Now, the Galois group of $\mathbb{L}_2/\mathbb{Q}$ is isomorphic to S_3 because $\mathbb{L}_2$ is the splitting field of a cubic polynomial and has degree 6. In conclusion, $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathsf{S}_3$.

We can choose as generators of $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ the automorphisms induced by the mappings

$$\begin{cases} i \longmapsto -i \\ \omega \longmapsto \omega \\ \sqrt[3]{2} \longmapsto \sqrt[3]{2} \end{cases}, \qquad \begin{cases} i \longmapsto i \\ \omega \longmapsto \omega \\ \sqrt[3]{2} \longmapsto \omega\sqrt[3]{2} \end{cases}, \qquad \begin{cases} i \longmapsto i \\ \omega \longmapsto \omega^2 \\ \sqrt[3]{2} \longmapsto \sqrt[3]{2} \end{cases}.$$

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(i) The splitting field of $x^3 - 3$ is $\mathbb{L}_1 = \mathbb{Q}(\sqrt[3]{3}, \zeta)$, where ζ is a primitive third root of unity in $\mathbb{C}$, and the splitting field of $x^4 - 3$ is $\mathbb{L}_2 = \mathbb{Q}(\sqrt[4]{3}, i)$.
The degree $[\mathbb{L}_1 : \mathbb{Q}]$ is 6; indeed, $\sqrt[3]{3}$ has degree 3 over $\mathbb{Q}$, as the polynomial $x^3 - 3$ is irreducible by Eisenstein's Criterion, and the root of unity ζ does not belong to $\mathbb{Q}(\sqrt[3]{3})$ (because the latter field is real) and has degree $\phi(3) = 2$ over $\mathbb{Q}$.
The degree $[\mathbb{L}_2 : \mathbb{Q}]$ is 8 by an argument similar to the one above: $\sqrt[4]{3}$ has degree 4 over $\mathbb{Q}$, as $x^4 - 3$ is irreducible by Eisenstein's Criterion and is thus its minimal polynomial; i does not belong to the real field $\mathbb{Q}(\sqrt[4]{3})$ and has degree $\phi(4) = 2$ over $\mathbb{Q}$.
Remark that $\mathbb{E} = \mathbb{Q}(\zeta) = \mathbb{Q}(i\sqrt{3})$ is a subfield of $\mathbb{L}_1 \cap \mathbb{L}_2$ and that the degree of this intersection is at most 2, as it has to divide both $[\mathbb{L}_1 : \mathbb{Q}]$ and $[\mathbb{L}_2 : \mathbb{Q}]$; it follows that $\mathbb{E} = \mathbb{L}_1 \cap \mathbb{L}_2$ and that $\mathbb{L} = \mathbb{L}_1 \cdot \mathbb{L}_2$ has degree 12 over $\mathbb{E}$, hence degree 24 over $\mathbb{Q}$.

(ii) Note that $\mathbb{L} = \mathbb{Q}(\sqrt[3]{3}, \zeta, \sqrt[4]{3}, i) = \mathbb{Q}(\sqrt[12]{3}, i)$. Each element of its Galois group are therefore uniquely identified by the images of $\sqrt[12]{3}$ and i. The minimal polynomials of these algebraic numbers are $x^{12} - 3$ and $x^2 + 1$, respectively, and their roots are $\omega^h \sqrt[12]{3}$ for $0 \le h \le 11$, where ω is a primitive twelfth root of unity, and $\pm i$. It follows that any automorphism of $\mathbb{L}/\mathbb{Q}$ must satisfy

$$\sqrt[12]{3} \longmapsto \omega^h \sqrt[12]{3}, \quad i \longmapsto i^k,$$

where h is well-defined modulo 12, and b is well-defined modulo 4 and coprime to 4. Since the Galois group $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ has exactly 24 elements, each of the 24 pairs $(h, k) \in \mathbb{Z}/12\mathbb{Z} \times (\mathbb{Z}/4\mathbb{Z})^*$ corresponds to a unique element of $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ that transforms the two generators as described above. In particular,

$$\sqrt[12]{3} \longmapsto -i \sqrt[12]{3}, \quad i \longmapsto i,$$

defines an element τ of $\mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ with the following properties: first of all, we have $\tau(i) = i$; moreover, since $\sqrt[3]{3} = (\sqrt[12]{3})^4$, we have $\tau(\sqrt[3]{3}) = (-i \sqrt[12]{3})^4 = \sqrt[3]{3}$, so τ fixes $\mathbb{Q}(i, \sqrt[3]{3})$. Analogously, it follows from $\sqrt[4]{3} = (\sqrt[12]{3})^3$ that $\tau(\sqrt[4]{3}) = (-i \sqrt[12]{3})^3 = i\sqrt[4]{3}$. Thus the automorphism τ satisfies the question's requirements.

(iii) Let $\mathbb{K}$ be a subextension of $\mathbb{L}$ of degree 4 over $\mathbb{Q}$; we shall first show that $\mathbb{K}$ is contained in $\mathbb{L}_2$. Indeed, assume by contradiction that it isn't; by the Primitive Element Theorem, we have $\alpha \in \mathbb{K}$ such that $\mathbb{K} \cdot \mathbb{L}_2 = \mathbb{Q}(\alpha)$. But then $\mathbb{K} \cdot \mathbb{L}_2 = \mathbb{L}_2(\alpha)$ and, since $\mathbb{K} \cap \mathbb{L}_2$ has degree 1 or 2 over $\mathbb{Q}$, the element α must have degree 4 or 2 over $\mathbb{L}_2$; this, however, would imply that $\mathbb{K} \cdot \mathbb{L}_2$ has degree 16 or 32 over $\mathbb{Q}$, which contradicts the fact that $[\mathbb{L} : \mathbb{Q}] = 24$.
We now determine the Galois group of $\mathbb{L}_2/\mathbb{Q}$. Since $\mathbb{L}_2$ is the splitting field of a polynomial of degree 4 and has degree 8, it is isomorphic to a 2-Sylow subgroup of S_4 and thus to D_4. The subextensions of $\mathbb{K}$ of the desired degree correspond to the subgroups of order 2 of D_4, that is, to the elements of order 2 of D_4. There are 5 such elements in D_4, and we have the following 5 subextensions of $\mathbb{L}$ of degree 4 over $\mathbb{Q}$: $\mathbb{Q}(i, \sqrt{3})$, $\mathbb{Q}(\sqrt[4]{3})$, $\mathbb{Q}((1+i)\sqrt[4]{3})$, $\mathbb{Q}(i\sqrt[4]{3})$ and $\mathbb{Q}((1-i)\sqrt[4]{3})$.

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(i) The roots of the polynomial $x^5 - 2$ are the fifth roots of 2, so its splitting field is $\mathbb{K} = \mathbb{Q}(\sqrt[5]{2}, \zeta)$, where ζ is a primitive fifth root of unity in $\mathbb{C}$. The cyclotomic extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ is normal, has degree $\phi(5) = 4$, and has Galois group isomorphic to $(\mathbb{Z}/5\mathbb{Z})^*$. The polynomial $x^2 - 5$ is irreducible by Eisenstein's Criterion, so $\mathbb{Q}(\sqrt[5]{2})$ is a real extension of degree 5 over $\mathbb{Q}$; it is not normal, as it only contains the real root $\sqrt[5]{2}$ of $x^5 - 2$. The field $\mathbb{K} = \mathbb{Q}(\sqrt[5]{2}) \cdot \mathbb{Q}(\zeta)$ has degree 20 over $\mathbb{Q}$, as the degrees of the two subextensions of $\mathbb{K}$ we have just considered are relatively prime.
Any element of the Galois group G of $\mathbb{K}/\mathbb{Q}$ sends the root $\sqrt[5]{2}$ of $x^5 - 2$ to a root of its minimal polynomial, namely, to one of the five elements $\zeta^h \sqrt[5]{2}$ for $h = 0, 1, 2, 3, 4$; analogously, it sends ζ to a primitive fifth root of unity, that is, to one of the four elements ζ^k for $k = 1, 2, 3, 4$. The complex numbers $\sqrt[5]{2}$ and ζ are generators for $\mathbb{K}/\mathbb{Q}$, so each automorphism is completely determined by the corresponding pair (h, k); we denote the automorphism that corresponds to the pair by $\varphi_{k,h}$. Since the order of G is 20, all pairs (h, k) as above do correspond to an automorphism, and the fact that ζ has multiplicative order 5 implies that we may see h and k as residue classes modulo 5.
By computing the images of the generators ζ, $\sqrt[5]{2}$ of $\mathbb{K}/\mathbb{Q}$ via compositions of automorphisms, we immediately find that the map

$$\mathbb{Z}/5\mathbb{Z} \rtimes (\mathbb{Z}/5\mathbb{Z})^* \ni (h, k) \longmapsto \varphi_{h,k} \in G$$

is a group isomorphism, the semidirect product being defined with respect to the action $(\mathbb{Z}/5\mathbb{Z})^* \ni k \longmapsto (h \longmapsto kh) \in \mathrm{Aut}(\mathbb{Z}/5\mathbb{Z})$.

(ii) First of all, we wish to investigate the quadratic subextensions of $\mathbb{K}$. By the Fundamental Theorem of Galois Theory, we know that they correspond to subgroups of index 2, that is, of order 10, of the group G. Any such subgroup H is the kernel of a homomorphism $G \longrightarrow \mathbb{Z}/2\mathbb{Z}$, so H must contain every element of order 5 of G. In particular, the automorphism $\varphi = \varphi_{1,1}$, defined by

extending the mapping $\sqrt[5]{2} \longmapsto \zeta\sqrt[5]{2}$, $\zeta \longmapsto \zeta$, has to be an element of H since it has order 5.

From this remark we obtain that any subextension $\mathbb{F}$ of degree 2 over $\mathbb{Q}$ is contained in the fixed subfield of the subgroup H' generated by φ. This subfield has degree $[G : H'] = 4$ over $\mathbb{Q}$ and contains $\mathbb{Q}(\zeta)$, so it is in fact equal to $\mathbb{Q}(\zeta)$. It follows that $\mathbb{F}$ is a quadratic subextension of the fifth cyclotomic field $\mathbb{Q}(\zeta)$, but such a subextension is unique: it is the real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R}$ generated by $\zeta + \zeta^{-1}$, which is equal to $\mathbb{Q}(\sqrt{5})$ as $\zeta + \zeta^{-1}$ satisfies $x^2 + x - 1$.

We can now consider the degree r of the splitting field $\mathbb{L}$ of $(x^2 - p)(x^5 - 2)$, where p is a prime number. We have $\mathbb{L} = \mathbb{K}(\sqrt{p})$ and $[\mathbb{Q}(\sqrt{p}) : \mathbb{Q}] = 2$, so $r = 1$ if $\mathbb{Q}(\sqrt{p})$ is a subextension of $\mathbb{K}$ and $r = 2$ otherwise. By our argument above, it follows that $r = 1$ if and only if $p = 5$.

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(i) The ninth cyclotomic field $\mathbb{Q}(\zeta)$ has degree $\phi(9) = 6$ over $\mathbb{Q}$ and its Galois group G is isomorphic to $(\mathbb{Z}/9\mathbb{Z})^*$, thus cyclic of order 6. Moreover, α is a primitive third root of unity, so $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{-3})$ has degree 2 over $\mathbb{Q}$; on the other hand, the extension $\mathbb{Q}(\beta)$ is the real subextension $\mathbb{Q}(\zeta) \cap \mathbb{R}$ of degree $\phi(9)/2 = 3$. Since G is cyclic of order 6, it has a unique subgroup of oder d for each divisor d of 6; by the Fundamental Theorem of Galois Theory, there are no additional subextensions other than the trivial ones.

(ii) Letting $\omega = \zeta \cdot i$, it is clear that ω is a primitive 36th root of unity in $\mathbb{C}$ and that $\mathbb{Q}(\zeta, i) = \mathbb{Q}(\omega)$. In particular, the Galois group of $\mathbb{Q}(\zeta, i)$ over $\mathbb{Q}$ is isomorphic to $G = (\mathbb{Z}/36\mathbb{Z})^* \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. The groups $\mathrm{Gal}\big(\mathbb{Q}(\zeta, i)/\mathbb{Q}(\alpha)\big)$ and $\mathrm{Gal}\big(\mathbb{Q}(\zeta, i)/\mathbb{Q}(\beta)\big)$ are isomorphic to subgroups of G of order 6 and 4, respectively, so to $\mathbb{Z}/2\times\mathbb{Z}/3 \simeq \mathbb{Z}/6\mathbb{Z}$ and to $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$.

(iii) A subextension of $\mathbb{K} = \mathbb{Q}(\zeta, i)$ whose degree is prime has either degree 2 or degree 3. It is easy to check that $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ contains three subgroups of index 2 and only one subgroup of index 3. By the Fundamental Theorem of Galois Theory, $\mathbb{K}$ has three subextensions of degree 2 and one subextension of degree 3 over $\mathbb{Q}$.

We can thus conclude that $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{3})$ are the quadratic subextensions and $\mathbb{Q}(\beta)$ is the only subextension of degree 3.

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(i) The polynomial $p(x) = x^4 - 2x^2 - 10$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion. Letting $\alpha_1 = \sqrt{1 + \sqrt{11}}$ and $\alpha_2 = \sqrt{1 - \sqrt{11}}$, the roots of $p(x)$ are $\pm\alpha_1, \pm\alpha_2$. Note that the extensions

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{11}) \subseteq \mathbb{Q}(\alpha_1) \subseteq \mathbb{Q}(\alpha_1, \alpha_2)$$

are all proper and of degree 2: this is obvious for the first one; for the second one, α_1 is a root of the irreducible polynomial $p(x)$ of degree 4; finally, α_2 is a root of $x^2 - 1 + \sqrt{11}$, which is a polynomial of degree 2 with coefficients in

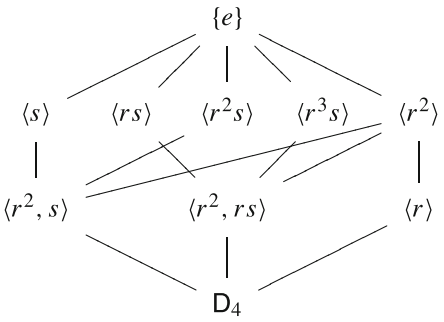
$\mathbb{Q}(\alpha_1)$ and is irreducible, as $\mathbb{Q}(\alpha_1) \subseteq \mathbb{R}$ whereas α_2 is not real. The degree of the extension $\mathbb{E}/\mathbb{Q}$ is therefore 8.

(ii) The Galois group $G = \mathrm{Gal}(\mathbb{E}/\mathbb{Q})$ has order 8 and is a subgroup of S_4, so it is isomorphic to a 2-Sylow subgroup of S_4, and thus to D_4.

(iii) By the Fundamental Theorem of Galois Theory, the subextensions of $\mathbb{E}/\mathbb{Q}$ correspond to the subgroups of D_4. With the usual presentation

$$\langle r, s \mid r^4 = s^2 = e,\ sr = r^{-1}s \rangle$$

of D_4, the lattice of subgroups is represented below.



Among the subgroups of D_4, those that are normal are: $\{e\}$, $\langle r^2 \rangle$, which is the centre, the three subgroups $\langle r^2, s \rangle$, $\langle r^2, rs \rangle$, $\langle r \rangle$, which are normal because they have index 2, and finally D_4 itself. Indeed, the subgroups of order 2 generated by a symmetry cannot be normal, as they contain only one symmetry, whereas in D_4 the symmetries are pairwise conjugate.

Since the normal subfields over $\mathbb{Q}$ correspond to the normal subgroups, we have, other than $\mathbb{E}$ and $\mathbb{Q}$, a subfield of degree 4 and three subfields of degree 2. Observe that $\alpha_1\alpha_2 = \sqrt{-10}$; the quadratic subextensions are thus $\mathbb{Q}(\sqrt{11})$, $\mathbb{Q}(\sqrt{-10})$ and therefore $\mathbb{Q}(\sqrt{-110})$. Finally, the normal extension of degree 4 is the composite field $\mathbb{Q}(\sqrt{11}, \sqrt{-10})$.

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(i) The splitting field $\mathbb{K}$ is clearly obtained from $\mathbb{Q}$ by adding a primitive eleventh root of unity, a primitive third root of unity and the real number $\sqrt[3]{a}$. Since 11 and 3 are relatively prime, this is equivalent to adding a primitive 33rd root of unity ζ and the number $\sqrt[3]{a}$, that is, we have $\mathbb{K} = \mathbb{Q}(\zeta, \sqrt[3]{a})$. We consider two separate cases.

① If a is a perfect cube, that is, if $a \in \mathbb{N}^3$, then $\sqrt[3]{a} \in \mathbb{Q}$ and $\mathbb{K} = \mathbb{Q}(\zeta)$ is the 33rd cyclotomic field, which has degree $\phi(33) = 20$. Its Galois group is isomorphic to $(\mathbb{Z}/33\mathbb{Z})^*$, that is, to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$.

② If a is not a perfect cube, then $[\mathbb{Q}(\sqrt[3]{a}) : \mathbb{Q}] = 3$, because $x^3 - a$ is irreducible over $\mathbb{Q}$ as it has no rational roots, so $\mathbb{K} = \mathbb{Q}(\zeta) \cdot \mathbb{Q}(\sqrt[3]{a})$ has degree $20 \cdot 3$ since 20 and 3 are relatively prime.

Consider the subextension $\mathbb{K}_1 = \mathbb{Q}(\zeta^3)$; this is the eleventh cyclotomic field, and its Galois group is isomorphic to $\mathbb{Z}/10\mathbb{Z}$. Notice that $\mathbb{K}_2 = \mathbb{Q}(\zeta^{11}, \sqrt[3]{a})$ is the splitting field of $x^3 - a$ and has degree 6 over $\mathbb{Q}$ as $\mathbb{Q}(\sqrt[3]{a})$ contains only the real root $\sqrt[3]{a}$. In particular, $\mathrm{Gal}(\mathbb{K}_2/\mathbb{Q})$ is isomorphic to S_3. We have

$\mathbb{K} = \mathbb{K}_1 \cdot \mathbb{K}_2$ and $\mathbb{K}_1 \cap \mathbb{K}_2 = \mathbb{Q}$, as $[\mathbb{K} : \mathbb{Q}] = 60 = [\mathbb{K}_1 : \mathbb{Q}] \cdot [\mathbb{K}_2 : \mathbb{Q}]$. It follows that $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{K}_1/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{K}_2/\mathbb{Q})$ and thus to $\mathbb{Z}/10\mathbb{Z} \times \mathsf{S}_3$.

(ii) By the Fundamental Theorem of Galois Theory, the subextensions $\mathbb{E}$ that are normal over $\mathbb{Q}$ are those fixed by normal subgroups H of $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$; moreover, $\mathrm{Gal}(\mathbb{E}/\mathbb{Q})$, which is isomorphic to $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})/H$, is Abelian if and only if H contains the derived subgroup of $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$. Again, we consider two cases.

① If a is a cube then $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $\mathbb{Z}/10\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, so all of its subgroups are normal and all of its quotients are Abelian. We wish to find the number of subgroups of $\mathbb{Z}/10\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ or, equivalently, of $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. It is easy to see that the subgroups are the following: one subgroup of order 1; three subgroups of order 2; the unique 2-Sylow subgroup, which has order 4; the unique 5-Sylow subgroup, which has order 5; three subgroups of order 10, each of which is cyclic (indeed, we have 12 elements of order 10 and thus $12/\phi(10) = 3$ cyclic subgroups of that order); finally, one subgroup of order 20. We have 10 subgroups in total.

② If a is not a cube, then $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $\mathbb{Z}/10\mathbb{Z} \times \mathsf{S}_3$, whose derived subgroup is $0 \times \mathsf{A}_3$. The subextensions we want are thus in bijection with the subgroups of $(\mathbb{Z}/10\mathbb{Z} \times \mathsf{S}_3)/(0 \times \mathsf{A}_3) \simeq \mathbb{Z}/10\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, of which we know there are 10.

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(i) Setting $\alpha_1 = \sqrt{1+\sqrt{3}}$ and $\alpha_2 = \sqrt{1-\sqrt{3}}$, the roots of $p(x) = x^4 - 2x^2 - 2$ are $\pm\alpha_1$ and $\pm\alpha_2$, so $K = \mathbb{Q}(\alpha_1, \alpha_2)$; since $\alpha_1\alpha_2 = \sqrt{-2}$, we also have $\mathbb{K} = \mathbb{Q}(\alpha_1, \sqrt{-2})$.

Moreover, $\mathbb{Q}(\alpha_1)$ is real of degree 4, as $p(x)$ is irreducible by Eisenstein's Criterion, whereas $\mathbb{Q}(\sqrt{-2})$ is not real, so $[\mathbb{K} : \mathbb{Q}(\alpha_1)] = 2$ and therefore $[\mathbb{K} : \mathbb{Q}] = 8$.

(ii) The Galois group $G = \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ has order 8 and is a subgroup of S_4, so it's a 2-Sylow subgroup and thus is isomorphic to D_4.

⟦ An alternative solution is the following. The Galois group $G = \mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ has order 8 and is not Abelian, as the subextension $\mathbb{Q}(\alpha_1)$ is not normal (it contains the root α_1 of $p(x)$ but is not its splitting field). Moreover, $\mathbb{K}$ contains the two degree 4 subextensions $\mathbb{Q}(\alpha_1)$ and $\mathbb{Q}(\alpha_2)$, which are distinct because the first one is real and the second one is not; it follows by the Fundamental Theorem of Galois Theory that G has two subgroups of index 4, that is, of order 2. We know that the only non-Abelian groups of order 8 are D_4 and the group Q_8 of unit quaternions, but the only element of order two in the latter group is -1; in conclusion, G is isomorphic to D_4. ⟧

(iii) Again by the Fundamental Theorem of Galois Theory, the subextensions of $\mathbb{K}$ correspond to the subgroups of G, that is, of D_4, and normal subextensions correspond to normal subgroups.

The trivial subgroup $\{e\}$ corresponds to the extension $\mathbb{K}$, which is normal over $\mathbb{Q}$.

In D_4 there are 5 subgroups of order 2, namely, the four generated by the symmetries and the one generated by the rotation by π. Only the latter, which is in fact the centre of D_4, is normal. There are thus four subextensions of degree 4 that are not normal over $\mathbb{Q}$ and one that is.
We know that $\mathbb{Q}(\alpha_1)$ and $\mathbb{Q}(\alpha_2)$ have degree 4 and are not normal. In order to construct the other two non-normal extensions, consider the fact that each element of G sends α_1 to a root $\pm\alpha_1, \pm\alpha_2$ of $p(x)$ and $\sqrt{-2}$ to $\pm\sqrt{-2}$; moreover, these 8 cases are all possible because $|G| = 8$. In particular, there exist two automorphisms φ and ψ such that

$$\begin{aligned} \alpha_1 &\overset{\varphi}{\longmapsto} \alpha_2, \quad \sqrt{-2} \overset{\varphi}{\longmapsto} -\sqrt{-2}, \\ \alpha_1 &\overset{\psi}{\longmapsto} \alpha_2, \quad \sqrt{-2} \overset{\psi}{\longmapsto} \sqrt{-2}. \end{aligned}$$

From $\alpha_1\alpha_2 = \sqrt{-2}$ we obtain that the automorphism φ permutes the roots of $p(x)$ as the cycle $(\alpha_1, \alpha_2, -\alpha_1, -\alpha_2)$, whereas ψ exchanges α_1 and α_2. This implies that the orbit of $\alpha_1 + \alpha_2$ for the action of the Galois group consists of the four elements $\pm\alpha_1 \pm \alpha_2$. In conclusion, $\mathbb{Q}(\alpha_1 + \alpha_2)$ and $\mathbb{Q}(\alpha_1 - \alpha_2)$ are two subextensions of degree 4 over $\mathbb{Q}$, neither of which is normal as φ sends the first to the second.
Finally, $\mathbb{Q}(\sqrt{-2}, \sqrt{3})$ is the one normal subextension of degree 4, as it is the composite field of two extensions of degree 2.
In D_4 we also have three subgroups of order 4, which are normal because their index is 2. They correspond to the three quadratic subextensions: $\mathbb{Q}(\sqrt{-2})$, $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{-6})$.
Finally, the whole group D_4 obviously corresponds to the normal extension $\mathbb{Q}$.

276 For ease of notation, set $\mathbb{K} = \mathbb{C}(x)$ and $\mathbb{F} = \mathbb{C}(t)$.

(i) Clearly, the polynomial $z^6 - tz^3 + 1 \in \mathbb{F}[z]$ vanishes at x; we wish to prove it is irreducible in $\mathbb{F}[z] = \mathbb{C}(t)[z]$. By Gauss's Lemma, it is enough to show that it is irreducible in $\mathbb{C}[t][z] = \mathbb{C}[t, z]$. Let $z^6 - tz^3 + 1 = f(t, z)g(t, z)$ be a factorisation in $\mathbb{C}[t, z]$; remark that one of the two polynomials $f(z, t)$ and $g(z, t)$ must have degree 0 in t, as $z^6 - tz^3 + 1$ has degree 1 in t. Assume that $f(z, t)$ has degree 0 in t; since $z^6 - tz^3 + 1$ is a primitive polynomial in the variable t, the polynomial $f(z, t)$ must be a constant and the factorisation is trivial.
This proves that $z^6 - tz^3 + 1$ is the minimal polynomial of x over $\mathbb{F}$, and thus that $[\mathbb{K} : \mathbb{F}] = 6$.

(ii) Letting $\zeta \in \mathbb{C}$ be a primitive third root of unity, observe that $\alpha : x \longmapsto 1/x$ and $\beta : x \longmapsto \zeta x$ induce two automorphisms of $\mathbb{K}/\mathbb{F}$, which we will denote by the same symbols.
The roots of $z^6 - tz^3 + 1$ are $x, \zeta x, \zeta^2 x, x^{-1}, \zeta x^{-1}, \zeta^2 x^{-1}$ and they all belong to $\mathbb{K}$. It follows that $\mathbb{K}$ is the splitting field of $z^6 - tz^3 + 1$ and is thus a Galois extension. Each element of the Galois group is uniquely determined by the image of x, and it is clear that any root of the polynomial can be obtained via

appropriate combinations of α and β. Therefore, $G = \mathrm{Gal}(\mathbb{K}/\mathbb{F}) = \langle \alpha, \beta \rangle$. Since α has order 2 and β has order 3 and since $\alpha\beta\alpha = \beta^2$, the group G is isomorphic to S_3.

(iii) The subextension that is invariant under $\langle \beta \rangle$ is $\mathbb{F}(x^3) = \mathbb{C}(x^3)$; indeed, $\mathbb{C}(x^3)$ is β-invariant and is a proper subextension of $\mathbb{F}$, as it is not invariant under α. The subextensions $\mathbb{F}(x + x^{-1}) = \mathbb{C}(x + x^{-1})$, $\mathbb{F}(x + \zeta x^{-1}) = \mathbb{C}(x + \zeta x^{-1})$ and $\mathbb{F}(x + \zeta^2 x^{-1}) = \mathbb{C}(x + \zeta^2 x^{-1})$ are invariant under $\langle \alpha \rangle$, $\langle \alpha\beta \rangle$ and $\langle \alpha\beta^2 \rangle$, respectively. Moreover, they are proper extensions of $\mathbb{F}$, as they are not β-invariant and thus are the fixed fields of the listed subgroups.
By the Fundamental Theorem of Galois Theory, the list above includes all proper subextensions of $\mathbb{K}/\mathbb{F}$, as we have listed all proper subgroups of S_3.

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(i) It is clear that $\mathbb{Q}(\alpha)$ is contained in $\mathbb{Q}(\sqrt[3]{3}, \sqrt{5})$. In order to obtain the opposite inclusion, note that $\mathbb{Q}(\sqrt[3]{3}, \sqrt{5}) = \mathbb{Q}(\alpha)(\sqrt[3]{3})$. The minimal polynomial of $\sqrt[3]{3}$ over $\mathbb{Q}$ is $x^3 - 3$, which is irreducible by Eisenstein's Criterion; its roots are $\sqrt[3]{3}$ and two non-real complex numbers. Thus, if $\sqrt[3]{3}$ does not belong to $\mathbb{Q}(\alpha)$ then $x^3 - 3$ has no real roots in the real extension $\mathbb{Q}(\alpha)$ and therefore $[\mathbb{Q}(\alpha)(\sqrt[3]{3}) : \mathbb{Q}(\alpha)] = 3$. Moreover, since $\mathbb{Q}(\sqrt[3]{3}, \sqrt{5}) = \mathbb{Q}(\alpha)(\sqrt{5})$, the degree $[\mathbb{Q}(\sqrt[3]{3}, \sqrt{5}) : \mathbb{Q}(\alpha)]$ is at most 2. The only possible value for it is therefore 1, and we have shown that $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt[3]{3}, \sqrt{5})$. In particular, it follows that α has degree 6 over $\mathbb{Q}$.

(ii) We show that $\mathbb{K}$ is the smallest normal extension of $\mathbb{Q}$ that contains $\mathbb{Q}(\alpha)$. Indeed, if a normal extension contains $\mathbb{Q}(\alpha)$ then it contains α and thus the splitting field of α; conversely, $\mathbb{K}$ is a normal extension containing α, so it also contains $\mathbb{Q}(\alpha)$.
Since $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt[3]{3}, \sqrt{5})$ and $x^3 - 3$, $x^2 - 5$ are the minimal polynomials of the two generators, we immediately find that $\mathbb{K} = \mathbb{Q}(\sqrt[3]{3}, \zeta, \sqrt{5})$, where ζ is a primitive third root of unity in $\mathbb{C}$.
It follows that $\mathbb{K}$ is the composite field of the two extensions $\mathbb{K}_1 = \mathbb{Q}(\sqrt[3]{3}, \zeta)$ and $\mathbb{K}_2 = \mathbb{Q}(\sqrt{5})$. The first is the splitting field of $x^3 - 3$ and has thus degree 6 and Galois group isomorphic to S_3, since it contains the real subextension $\mathbb{Q}(\sqrt[3]{3})$ but is not real itself. As for $\mathbb{K}_2$, it is the splitting field of $x^2 - 5$; it has degree 2 and Galois group isomorphic to $\mathbb{Z}/2\mathbb{Z}$.
In conclusion, since the degree of $\mathbb{K}$ over $\mathbb{Q}$ is the product of the degrees of these two subextensions, we find that their intersection is $\mathbb{Q}$, and thus that $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$ is isomorphic to $\mathrm{Gal}(\mathbb{K}_1/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{K}_2/\mathbb{Q})$, that is, to $\mathsf{S}_3 \times \mathbb{Z}/2\mathbb{Z}$.

(iii) By the Fundamental Theorem of Galois Theory, a subextension $\mathbb{F}$ of $\mathbb{K}$ is normal and has an Abelian Galois group if and only if it is fixed by the derived group G' of G, that is, if and only if it is contained in $\mathbb{K}^{G'}$. We know that G is isomorphic to $\mathsf{S}_3 \times \mathbb{Z}/2\mathbb{Z}$, so the group G' corresponds to the subgroup $\mathsf{A}_3 \times 0$ and is therefore generated by the automorphism φ such that $\varphi(\sqrt[3]{3}) = \zeta\sqrt[3]{3}$, $\varphi(\zeta) = \zeta$ and $\varphi(\sqrt{5}) = \sqrt{5}$. So $\mathbb{K}^{G'}$ is $\mathbb{Q}(\zeta, \sqrt{5}) = \mathbb{Q}(\sqrt{-3}, \sqrt{5})$.

In conclusions, the desired subextensions $\mathbb{F}$ are: $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$, which is the only one of degree 4, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-15})$, which have degree 2, and $\mathbb{Q}$.

278 Letting $\alpha = \sqrt[5]{5}$, the roots of the polynomial $f(x) = x^5 - 5$ are $\alpha\zeta^h$ for some fixed primitive fifth root of unity ζ in $\mathbb{C}$ and $h = 0, 1, 2, 3, 4$. The splitting field $\mathbb{K}$ is thus $\mathbb{Q}(\alpha, \zeta)$: it is the composite field of $\mathbb{Q}(\alpha)$, whose degree over $\mathbb{Q}$ is 5 because $f(x)$ is irreducible by Eisenstein's Criterion, and $\mathbb{Q}(\zeta)$, which is the fifth cyclotomic field and has degree $\phi(5) = 4$ over $\mathbb{Q}$. The fact that 5 and 4 are relatively prime implies that $[\mathbb{K} : \mathbb{Q}] = 20$ and that $\mathbb{Q}(\alpha) \cap \mathbb{Q}(\zeta) = \mathbb{Q}$.

Let G be the Galois group $\mathrm{Gal}(\mathbb{K}/\mathbb{Q})$; by the Fundamental Theorem of Galois Theory, G has a normal subgroup N of order 5 which fixes $\mathbb{Q}(\zeta)$; N is the only 5-Sylow subgroup of G, and the quotient G/N is isomorphic to $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$, that is, to $(\mathbb{Z}/5\mathbb{Z})^*$, the cyclic group with 4 elements. Since the quotient G/N contains elements of order 4 and 4 is the greatest power of 2 that divides $|G|$, the group G also has elements of order 4; in particular, the 2-Sylow subgroups are isomorphic to $\mathbb{Z}/4\mathbb{Z}$. In conclusion, G is isomorphic to a semidirect product $\mathbb{Z}/5\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$.

We now analyse the subextensions of $\mathbb{K}$ by considering the corresponding subgroups of G, which we list according to their degree.

(1) Obviously, $\mathbb{Q}$ is the only extension of degree 1; it is normal and it corresponds to G.

(2) A subextension of degree 2 corresponds to a subgroup of order 10 in G. There is only one such extension, as any subgroup of order 10 must contain the 5-Sylow subgroup N, and so the extension must be contained in $\mathbb{K}^N = \mathbb{Q}(\zeta)$; but the only subextension of degree 2 of the fifth cyclotomic field is the real extension $\mathbb{Q}(\zeta + \zeta^{-1}) = \mathbb{Q}(\sqrt{5})$, which is obviously normal over $\mathbb{Q}$.

(3) The subextensions of degree 4 are those fixed by the 5-Sylow subgroups of G; but we have seen that N is the only 5-Sylow subgroup, so $\mathbb{Q}(\zeta)$ is the only subextension of degree 4 and it is normal.

(4) It is easy to check that there are five 2-Sylow subgroups of G, so $\mathbb{K}$ has five subextensions of degree 5 over $\mathbb{Q}$. They are therefore $\mathbb{Q}(\alpha\zeta^h)$ for $h = 0, 1, 2, 3, 4$. Indeed, each of these extensions are degree 5 because it is generated by a root of $f(x)$; moreover, if two were the same, then they would also contain ζ and would therefore coincide with $\mathbb{K}$, contradicting the fact that $[\mathbb{K} : \mathbb{Q}] = 20$. These extensions are not normal, as they only contain one root of $f(x)$.

(5) The subextensions $\mathbb{Q}(\sqrt[5]{5}\zeta^h, \zeta + \zeta^{-1})$, for $h = 0, 1, 2, 3, 4$ have degree 10 over $\mathbb{Q}$ and are all distinct by the same argument as above. By our arguments so far, it is easy to see that G contains: one element of order 1, four elements of order 5, two elements of order 4 for each 2-Sylow subgroup, and thus ten overall; we are therefore left with five elements of order 2, because the group is not cyclic and therefore cannot have any elements of order 20. This proves that we have found all subextensions of degree 10; they are not normal because they contain only one root of $f(x)$.

279 Letting $\alpha = \sqrt[6]{2}$, it is clear that $\mathbb{K} = \mathbb{Q}(\sqrt[3]{2}, \sqrt{2}) = \mathbb{Q}(\alpha)$, as 2 and 3 are relatively prime; it follows that $\mathbb{K}$ has degree 6 over $\mathbb{Q}$ and that $x^6 - 2$ is the minimal polynomial of α, as it is irreducible by Eisenstein's Criterion.

The smallest extension $\mathbb{L}$ of $\mathbb{K}$ that is normal over $\mathbb{Q}$ coincides with the splitting field of α: indeed, any normal extension must contain the splitting field of α; conversely, a splitting field is a normal extension. This implies that $\mathbb{L} = \mathbb{Q}(\alpha, \zeta)$, where ζ is a primitive sixth root of unity in $\mathbb{C}$. Moreover, $\mathbb{Q}(\alpha)$ is a real extension, whereas $\mathbb{Q}(\zeta)$ is not, so $[\mathbb{L} : \mathbb{Q}] = 12$ because $\mathbb{Q}(\zeta)$ has degree $\phi(6) = 2$ over $\mathbb{Q}$.

Any automorphism of $\mathbb{L}$ is uniquely determined by the images of the generators α and ζ. Moreover, α must be sent to a root of $f(x)$, that is, to $\alpha\zeta^h$, where h is well-defined modulo 6; analogously, ζ is sent to a primitive sixth root of unity, that is, to $\zeta^{\pm 1}$, or in other words to ζ^k for some k that is well-defined modulo 6 and coprime to 6. Letting $\varphi_{h,k}$ denote the unique automorphism induced by the images corresponding to h and k as above, we have a map $\varphi_{h,k} \longmapsto (h, k)$ from the group $G = \mathrm{Gal}(\mathbb{L}/\mathbb{Q})$ to the set of pairs (h, k) such that $h \in \mathbb{Z}/6\mathbb{Z}$ and $k \in (\mathbb{Z}/6\mathbb{Z})^*$.

By computing the images of α and ζ via compositions of automorphisms we obtain that the map we have defined is an isomorphism between G and the semidirect product $\mathbb{Z}/6\mathbb{Z} \rtimes (\mathbb{Z}/6\mathbb{Z})^*$ with respect to the action $(\mathbb{Z}/6\mathbb{Z})^* \ni k \longmapsto (h \longmapsto kh) \in \mathrm{Aut}(\mathbb{Z}/6\mathbb{Z})$. It is easy to see that this group is isomorphic to the dihedral group D_6.

280 The Galois group G of $\mathbb{L}/\mathbb{Q}$ is isomorphic to $(\mathbb{Z}/35\mathbb{Z})^*$, that is, to $(\mathbb{Z}/5\mathbb{Z})^* \times (\mathbb{Z}/7\mathbb{Z})^*$, which is to say $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$. We denote by ζ a primitive 35th root of unity in $\mathbb{C}$ and let $\alpha = \zeta^7$ be a primitive fifth root of unity and $\beta = \zeta^5$ a primitive seventh root of unity. The extension $\mathbb{L}$ is the composite field of the two cyclotomic fields $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta)$, whose intersection is $\mathbb{Q}$.

The group G has three elements of order 2 and therefore three subgroups of order 2; by the Fundamental Theorem of Galois Theory, this implies that there are three subextensions $\mathbb{K}$ such that $[\mathbb{L} : \mathbb{K}] = 2$, which are those fixed by these subgroups.

Now remark that $\mathbb{Q}(\alpha)$, $\mathbb{Q}(\beta)$ and $\mathbb{L} = \mathbb{Q}(\zeta)$ all have degree 2 over their respective real subextensions $\mathbb{Q}(\alpha) \cap \mathbb{R} = \mathbb{Q}(\alpha + \alpha^{-1})$, $\mathbb{Q}(\beta) \cap \mathbb{R} = \mathbb{Q}(\beta + \beta^{-1})$ and $\mathbb{K}_1 = \mathbb{Q}(\zeta) \cap \mathbb{R} = \mathbb{Q}(\zeta + \zeta^{-1})$. It follows that $\mathbb{L}$ has degree 2 over $\mathbb{K}_2 = Q(\alpha, \beta + \beta^{-1})$ and $\mathbb{K}_3 = \mathbb{Q}(\beta, \alpha + \alpha^{-1})$. Indeed, as was the case for $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta)$, the extensions $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(\beta + \beta^{-1})$ have intersection equal to $\mathbb{Q}$, so the degree of the composite field $\mathbb{K}_2$ is the product of their degrees, and thus $[\mathbb{L} : \mathbb{K}_2] = 2$. Similarly, one shows that $[\mathbb{L} : \mathbb{K}_3] = 2$.

Finally, notice that α belongs to $\mathbb{K}_2$ but not to $\mathbb{K}_3$, as otherwise we would have $\mathbb{K}_3 = \mathbb{L}$; moreover, $\mathbb{K}_1$ is real but $\mathbb{K}_2$ and $\mathbb{K}_3$ are not. This proves that $\mathbb{K}_1$, $\mathbb{K}_2$ and $\mathbb{K}_3$ are distinct and are thus the three extensions we were after.

281 Remark that $f(x) = (x^2 - \sqrt{6}x + 3)(x^2 + \sqrt{6}x + 3)$ is the factorisation of $f(x)$ in $\mathbb{R}[x]$ and therefore $f(x)$ has no roots in $\mathbb{Q}$ and is not the product of two polynomials of degree 2 in $\mathbb{Q}[x]$, because $\mathbb{R}[x]$ is a unique factorisation domain and $\sqrt{6} \notin \mathbb{Q}$. This proves that $f(x)$ is irreducible in $\mathbb{Q}[x]$.

The roots of $f(x)$ are $\sqrt{3}\zeta i^k$ for $k = 0, 1, 2, 3$, where we have denoted by ζ the primitive eighth root of unity $\sqrt{2}(1 + i)/2$. Now, setting $\alpha = \sqrt{3}\zeta$, we have $\alpha^2 = 3\zeta^2 = 3i$, so the splitting field $\mathbb{K} = \mathbb{Q}(\alpha, i)$ of $f(x)$ over $\mathbb{Q}$ is equal to $\mathbb{Q}(\alpha)$. In particular, the Galois group G of $f(x)$ has 4 elements.

We have $\alpha+3\alpha^{-1} = \sqrt{6}$, so $\mathbb{K}$ contains the three distinct quadratic subextensions $\mathbb{Q}(\sqrt{d})$, where $d = -1, 6, -6$. In conclusion, G is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

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